CH32V407/V467 Reference Manual

V1.3

https://wch-ic.com

Overview

CH32V407 is an industrial-grade interconnected microcontroller based on the QingKe RISC-V core. It supports zero-wait operation at a 200MHz clock speed and the V-vector instruction set, making it suitable for applications such as accelerated computing and parallel processing. The CH32V407 integrates 2 sets of USB2.0 high-speed PHY transceivers (480Mbps), an Ethernet MAC controller, and 10/100M physical layer transceivers; It features integrated analog resources such as dual ADCs, dual DACs, and OPAs; it also provides a wealth of peripheral resources, including 2 sets of DMA controllers (totaling 24 channels), SDIO, DVP interfaces, RNG, ARGB, LTDC, FSMC, multiple timer units, 10 USARTs, I2C interfaces, I3C interfaces, 3 SPIs, 2 I2Ss, and CAN interfaces.

The CH32V467 adds 4MB or 8MB of on-chip extended PSRAM memory to the CH32V407 and supports DMA.

This manual provides detailed usage information for the CH32V407, V467 chips to support user application development.

For device specifications of the CH32V407 series chips, please refer to the *CH32V407DS0*.

RISC-V core version comparison overview

Features / Core versions	Instruction set	Hardware stack levels	Interrupt nesting levels	Fast interrupt channels	Pipeline	Vector table model	Extended instruction	Interface debug
QingKeV3V	IMACBV-X	2	2	4	3	Address or command	Supported	1/2-wire

For information regarding the RISC-V core, refer to *QingKeV3_Processor_Manual*.

Abbreviated description of the bit attribute in the register:

Register bit attributes	Property description
RF	Read-only attribute, read a fixed value.
RO	Read-only attribute, changed by hardware.
RZ	Read-only attribute, auto bit clear 0 after read operation.
WO	Write only attribute (not readable, read value uncertain)
WA	Write-only attribute, writable in Safe mode.
WZ	Write only attribute, auto bit clear 0 after write operation.
RW	Readable and writable.
RWA	Readable, writable in Safe mode.
RW1	Readable, write 1 valid, write 0 invalid.
RW0	Readable, write 0 valid, write 1 invalid.
RW1T	Readable, write 0 invalid, write 1 flipped.
RW1Z	Readable, write 1 to clear, write 0 invalid.
SC	Auto cleared.

Chapter 1 Memory and Bus Architecture

1.1 Bus Architecture

This series of chips is an industrial general-purpose microcontroller based on QingKe RISC-V3V core. Within this architecture, components such as the cores, arbitration unit, DMA module, and SRAM storage interact via multiple bus groups. The system block diagram is shown in Figure 1-1-1 and Figure 1-1-2.

Figure 1-1-1 CH32V407 System block diagram

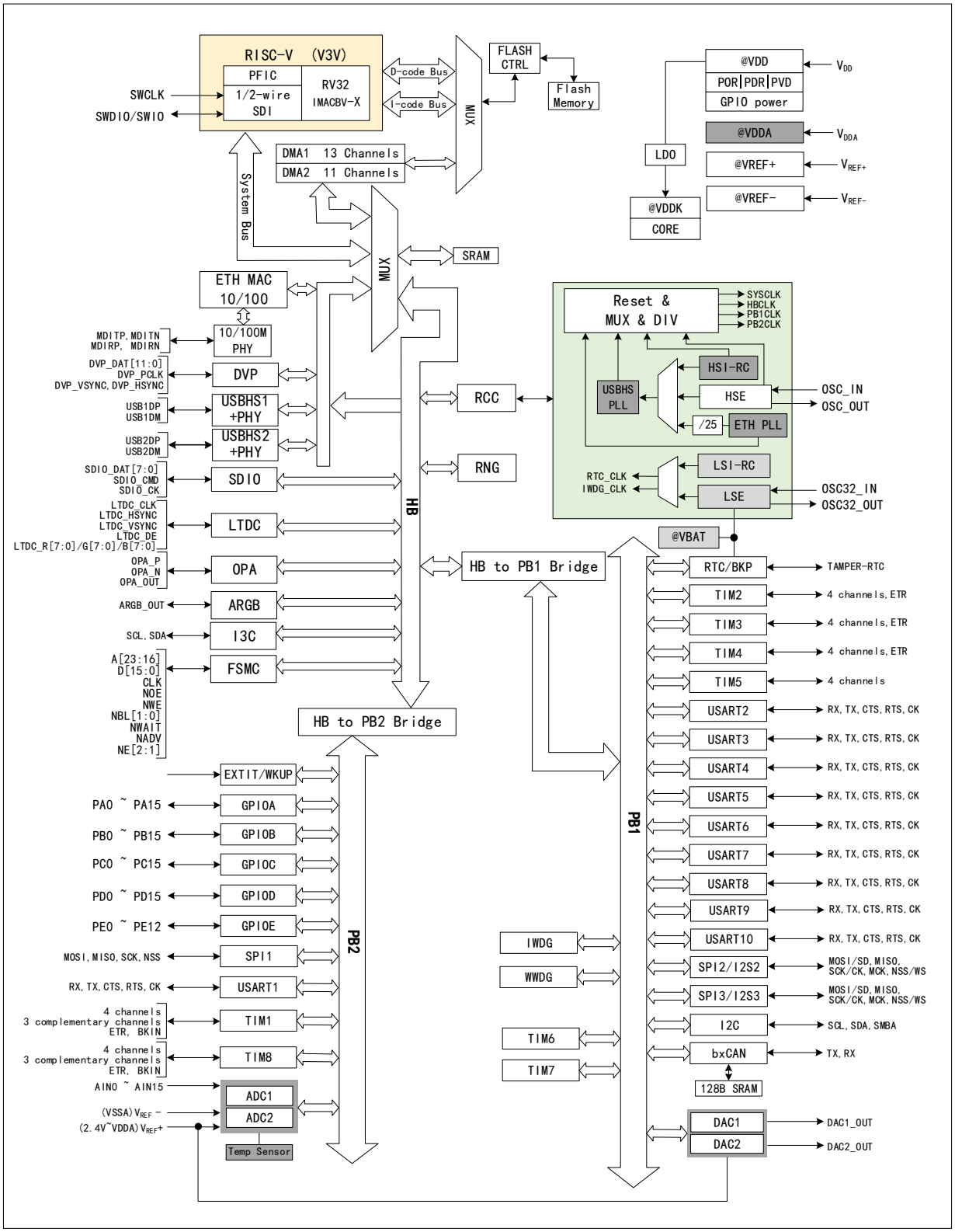
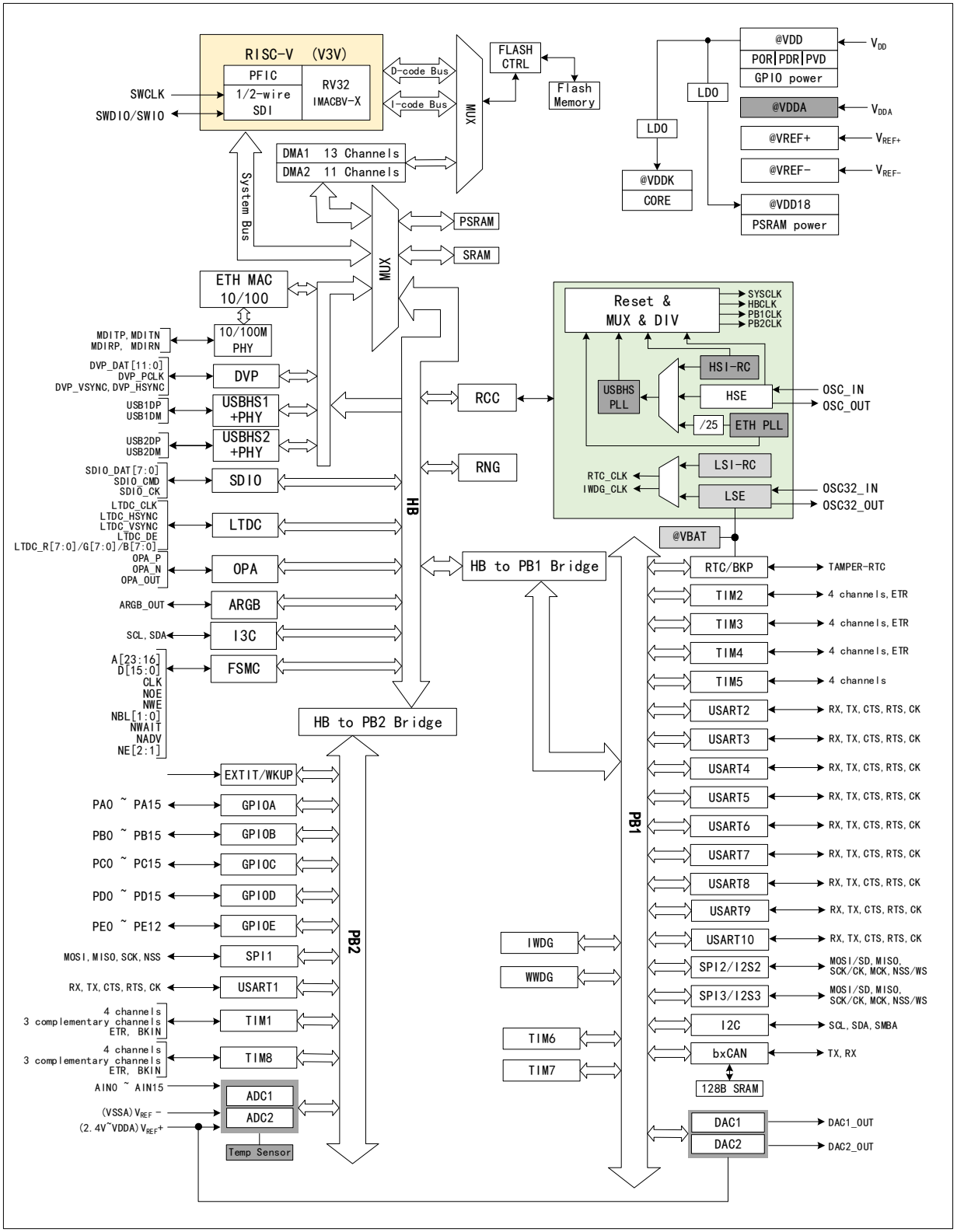


Figure 1-1-2 CH32V467 System block diagram



The system is equipped with: General-purpose DMA controller to reduce the CPU burden and improve efficiency; clock tree hierarchy management to reduce the total power consumption of peripherals, as well as data protection mechanisms, clock security system protection mechanisms and other measures to increase system stability.

- The instruction bus (I-Code) connects the core to the FLASH instruction interface, where instruction prefetching occurs.

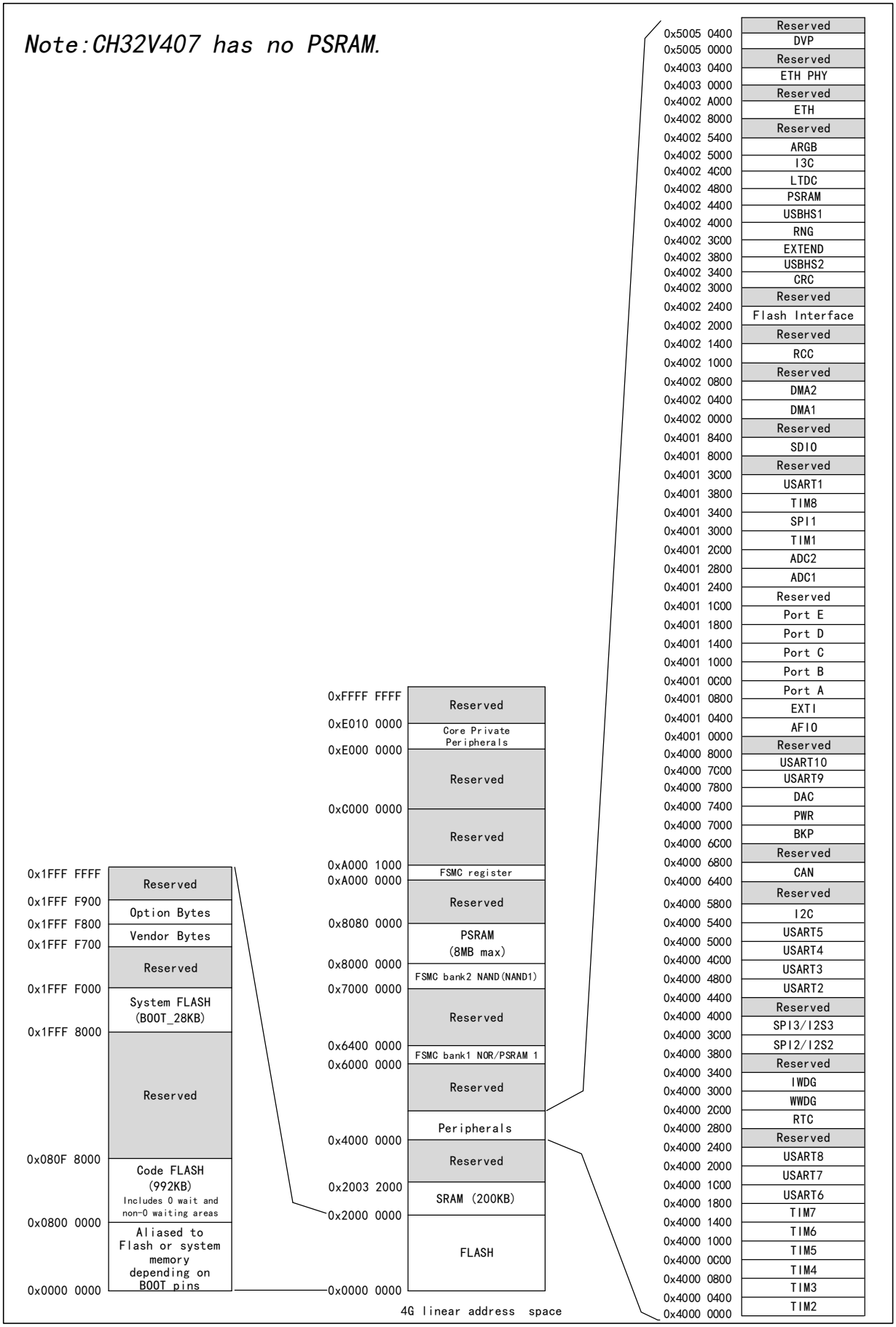
- The data bus (D-Code) connects the core to the FLASH data interface, used for constant loading and debugging.
- The system bus connects the core to the bus matrix, coordinating access between the core, DMA, SRAM, and peripherals.
- The DMA bus connects the HB controller interface for DMA operations to the bus matrix. This bus accesses FLASH data, SRAM, and peripherals.
- The bus matrix coordinates access between the system bus, data bus, DMA bus, SRAM and HB/PB bridge.
- The HB/PB bridge provides a synchronous connection between the HB bus and 2 PB buses. Different peripherals are connected to different PB buses, and the bus clocks can be configured according to actual requirements to optimize performance.

1.2 Memory Map

The chip contains program memory, data memory, core registers, peripheral registers, and more, all addressed in a 4GB linear space.

System storage stores data in little-endian format, i.e., low bytes are stored at the low address and high bytes are stored at the high address.

Figure 1-2 Storage image



1.2.1 Memory Allocation

Built-in SRAM with zero waiting 200K bytes is used to store data, and the data will be lost after power failure. The starting address is 0x20000000, which supports byte, half word (2 bytes) and whole word (4 bytes) access.

Built-in 992K bytes program flash storage area (Code FLASH), the user area, is used for user application and constant data storage. Including 480K bytes of non-zero wait program area and 512K bytes of zero wait program running area.

There are 64KB of zero-wait data SRAM that can be configured for zero-wait programs, namely, 416KB of non-zero-wait program area, 576KB of zero-wait program running area and 136KB of zero-wait data SRAM area.

Built-in 28K byte System FLASH, namely the BOOT area, is used for storing system boot programs (the bootloader is solidified by the manufacturer).

128 bytes are used in the nonvolatile configuration information storage area of the system, and are used to store the manufacturer's configuration words. They are solidified before leaving the factory and cannot be modified by users.

128 bytes are used for user-defined information storage area and for user option byte storage.

Note: For chips whose fifth digit from the bottom of the lot number is greater than 0, their non-zero waiting area supports both RVV instructions and DMA 64-bit access.

1.3 Boot Configuration

The system can select 3 different boot modes through the BOOT0 and BOOT1 pins.

Table 1-1 Boot mode

BOOT0	BOOT1	Boot mode
0	X	Boot from program flash memory
1	0	Boot from system memory
1	1	Boot from internal SRAM

The user selects the BOOT mode after reset by setting the state value of the boot pin. After system reset or power reset, the value of the BOOT pin will be re-latched.

With different startup modes, program flash memory, system memory and internal SRAM have different access modes:

- When booting from program flash memory, the program flash memory addresses are mapped to the 0x00000000 address space, and can also be accessed in the original address space at 0x08000000.
- When booting from system memory, the system memory address is mapped to the 0x00000000 address space, and can also be accessed in the original address space at 0x1FFF0000.
- When booting from internal SRAM, access is limited to the 0x20000000 address space.

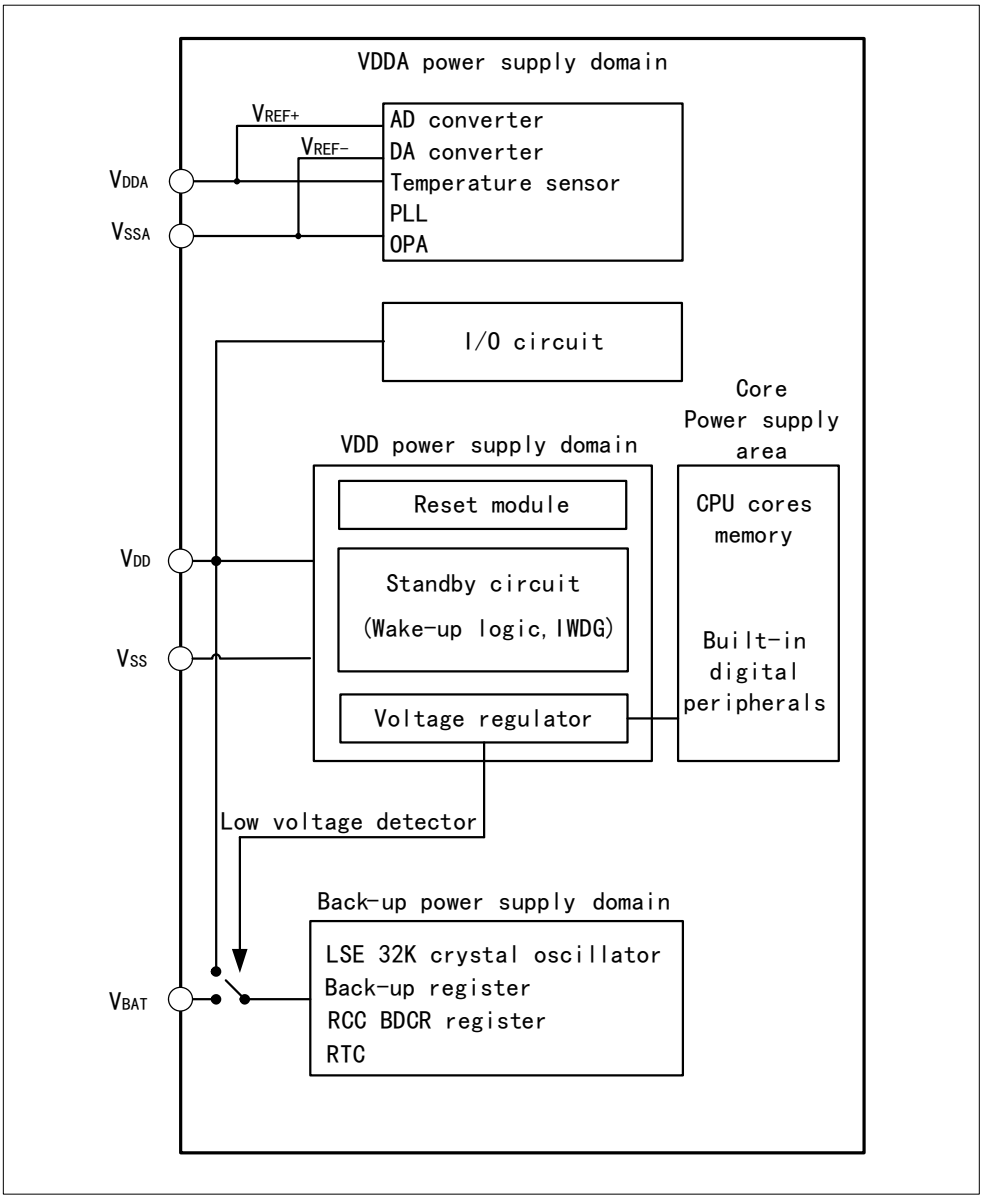
Chapter 2 Power Control (PWR)

2.1 Overview

The operating voltage V_{DD} ranges from 2.9 to 3.6 V. The built-in voltage regulator provides the low-voltage power supply required by the core. When the main power supply, V_{DD}, is powered down, a backup power source such as a battery can provide power to the real-time clock (RTC) and backup registers through the V_{BAT} pin. If no backup power is required, it is recommended that V_{DD} be connected directly to the V_{BAT} pin.

The V_{DDA} and V_{SSA} pins are specifically designed to supply power to analog-related circuits in the system, including ADCs, DACs, temperature sensors, and others. V_{REF+} and V_{REF-} serve as reference points for certain analog circuits and are internally connected to V_{DDA} and V_{SSA}.

Figure 2-1 Block diagram of power supply structure



The following parameters also apply to the CH32V467 product (refer to the *CH32V407DS0* for specific values):

V_{DD18}: Supplies power to the on-chip PSRAM module.

When the main power supply (V_{DD}) is turned off, the analog switch switches to V_{BAT}, and the backup area is powered by the V_{BAT} pin. At this point, pins PC13 through PC15 cannot be used as GPIOs and are limited to the following functions:

- PC13 can be used as a TAMPER pin, RTC alarm, or second output.
- PC14 and PC15 can only be used as LSE pins.

Once the main power supply (V_{DD}) has stabilized, the system automatically switches to V_{DD} to power the backup domain, and pins PC13~PC15 can be used as GPIOs.

When pins PC13~PC15 are used as GPIO outputs, the clock speed must be limited to 2MHz or less, the maximum load capacitance is 30pF, and they must not be used in applications that continuously draw or supply current, such as LED drivers.

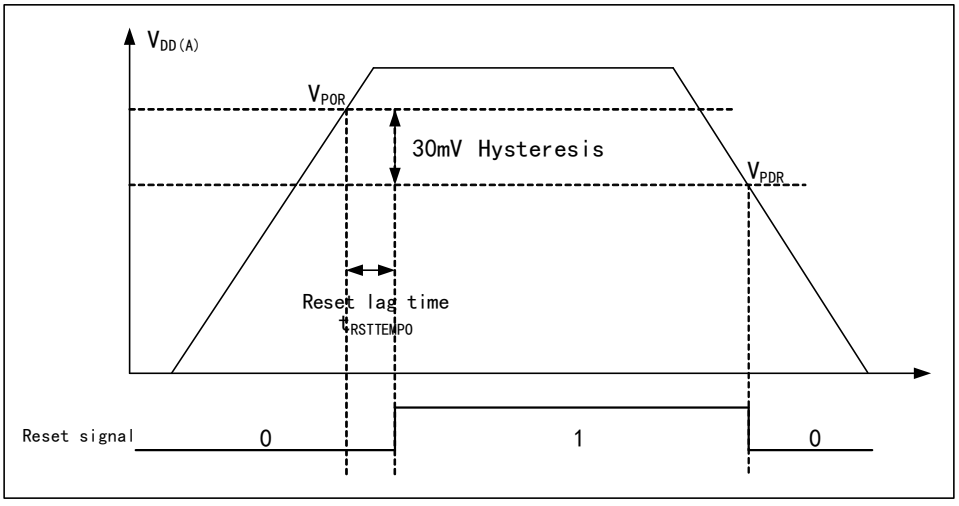
Note: During the restoration of the main power supply V_{DD}, the internal V_{BAT} power supply remains connected to the external backup power source via the corresponding V_{BAT} pin. If V_{DD} stabilizes within a time shorter than the reset delay time $t_{RSTTEMPO}$ and is more than 0.6V higher than V_{BAT}, there may be a brief moment when current may flow through the diode between V_{DD} and V_{BAT} into V_{BAT}, and subsequently into the backup power source (such as a battery) via the V_{BAT} pin. If the backup power source cannot withstand such an instantaneous current surge, it is recommended to install a forward-biased, low-dropout diode between the backup power source and the V_{BAT} pin.

2.2 Power Management

2.2.1 Power-on Reset and Power-down Reset

The system integrated a power-on reset POR and a power-down reset PDR circuit. When the chip supply voltage V_{DD} and V_{DDA} falls below the corresponding threshold voltage, the system is reset by the relevant circuit, and no additional external reset circuit is required. Please refer to the corresponding datasheet for the parameters of the power-on threshold voltage V_{POR} and the power-down threshold voltage V_{PDR}.

Figure 2-2 Schematic diagram of the operation of POR and PDR



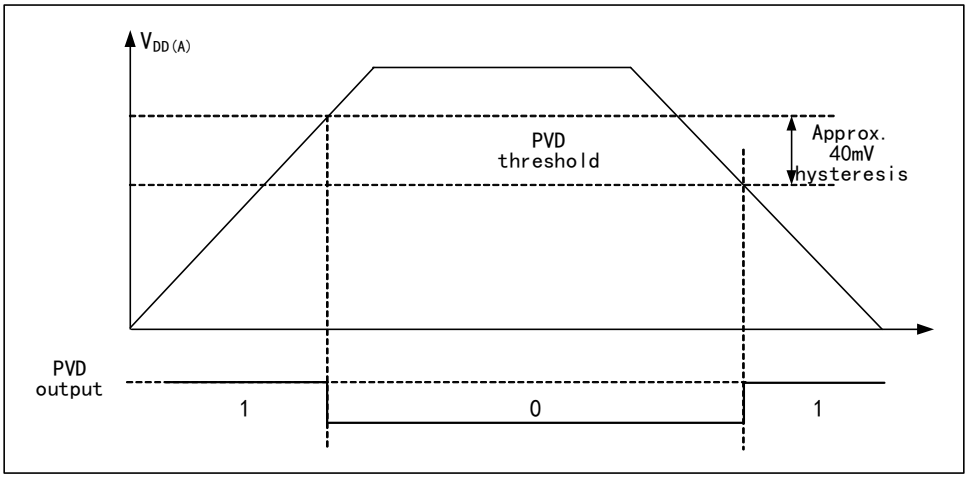
2.2.2 Programmable Voltage Detector

The programmable voltage detector, PVD, is mainly used to monitor the change of the main power supply of the system, compared with the threshold voltage set by PLS[1:0] of the power control register PWR_CTLR, and together with the setting of the external interrupt register (EXTI), the relevant interrupt can be generated in order to notify the system to carry out the pre-dropout operation, such as the data saving, in time.

The specific configuration is as follows:

1) Set the PLS[1:0] fields of the PWR_CTLR register to select the voltage threshold to be monitored.
2) Optional interrupt handling. The PVD function is internally linked to the rising/falling edge trigger setting on pin 16 of the EXTI module. Enabling this interrupt (by configuring EXTI) causes a PVD interrupt to be generated when V_{DD} drops below the PVD threshold or rises above it.
3) Set the PVDE bit of the PWR_CTLR register to enable the PVD function.
4) Read the PVDO bit of PWR_CSR status register to obtain the relationship between the current system main power supply and the threshold value set by PLS[1:0], and perform the corresponding soft processing. When the V_{DD} voltage is higher than the PLS[1:0] setting threshold, the PVDO position is 0; when the V_{DD} voltage is lower than the PLS[1:0] setting threshold, the PVDO position is 1.

Figure 2-3 Schematic diagram of PVD operation



2.3 Low-power Modes

After a system reset, the microcontroller is in a normal operating state (run mode), where system power can be saved by reducing the system main frequency or turning off the unused peripheral clock or reducing the operating peripheral clock. If the system does not need to work, you can set the system to enter low-power mode and let the system jump out of this state by specific events.

Microcontrollers currently offer 3 low-power modes, divided in terms of operating differences between processors, peripherals, voltage regulators, etc.

- Sleep mode: The core stops running and all peripherals (including core private peripherals) are still running.
- Stop mode: All clocks are stopped; The system continues to run after waking up.
- Standby mode: All clocks are stopped; The microcontroller resets after waking up (power reset).

Table 2-1 Low-power mode list

Mode	Entry	Wake-up source	Effect on clock	Voltage regulator
SLEEP	WFI	Any interrupt	Core clock OFF, no effect on other clocks	Normal
	WFE	Wake-up event		

STOP	Set SLEEPDEEP to 1 Clear PDDS to 0 WFI or WFE	Any external interrupt/event (EXTI signal), an external reset signal on NRST, or an IWDG reset; the EXTI signal includes one of the 77 external I/O pins, the PVD output, the RTC alarm, the I3C wake-up signal, and the USB wake-up signal.	HSE, HSI, PLL, and peripheral clocks are disabled.	Normal: LPDS=0 or Low power: LPDS=1
STANDBY	Set SLEEPDEEP to 1 Set PDDS to 1 WFI or WFE	Any external event (EXTI signal), an external reset signal on NRST, an IWDG reset, or a rising edge on the WKUP pin, where the EXTI signal includes one of the 77 external I/O ports or the RTC alarm.	HSE, HSI, PLL and peripheral clocks are disabled	Disabled

Note: The SLEEPDEEP bit is a core-private peripheral control bit; refer to the PFIC_SCTLR register.

2.3.1 Low-power Configuration Options

- WFI and WFE

WFI: The microcontroller is woken up by an interrupt source with interrupt controller response, and the interrupt service function will be executed first after the system wakes up (except for microcontroller reset).

WFE: The wakeup event triggers the microcontroller to exit low-power mode. Wake-up events include:

1) Configure an external or internal EXTI line to event mode, when no interrupt controller needs to be configured.
2) Or configure an interrupt source, equivalent to a WFI wakeup, where the system prioritizes the execution of the interrupt service function.
3) Or configure the SEVONPEND bit to turn on peripheral interrupt enable, but not interrupt enable in the interrupt controller, and the interrupt pending bit needs to be cleared after the system wakes up.

- SLEEPONEXIT

  Enable: After executing the WFI or WFE instruction, the microcontroller ensures that all pending interrupt services are exited and then enters low-power mode.

  Disable: The microcontroller enters low-power mode immediately after executing the WFI or WFE command.

- SEVONPEND

  Enable: All interrupts or wake-up events can wake up the low-power consumption entered by executing WFE.

  Disable: Only interrupts or wake-up events enabled in the interrupt controller can wake up the low-power consumption entered by executing WFE.

2.3.2 Sleep Mode

In this mode, all IO pins maintain their status in running mode, and all peripheral clocks are normal. Therefore,

before entering sleep mode, try to turn off useless peripheral clocks to reduce power consumption. This mode takes the shortest time to wake up.

Enter: Configure the core register control bit SLEEPDEEP=0, the power control register PDDS=0, LPDS determines the internal voltage regulator status, executes WFI or WFE, optional SEVONPEND and SLEEPONEXIT.

Exit: Any interrupt or wakeup event.

2.3.3 Stop Mode

Stop mode combines the clock control mechanisms of the peripherals with the fact that the core enters deep sleep (SLEEPDEEP). The high-frequency clock (HSE/HSI/PLL) domains are disabled, SRAM and register contents are preserved, and I/O pin states are maintained. Upon waking from this mode, the system can resume operation, with HSI serving as the default system clock.

If flash programming is in progress, the system does not enter stop mode until access to memory is complete; if access to the PB is in progress, the system does not enter stop mode until access to the PB is complete.

Modules that can operate in Stop mode: Independent Watchdog (IWDG), Real Time Clock (RTC), Low Frequency Clock (LSI/LSE).

Entry: Configuration of the core register control bit SLEEPDEEP=1, set PDDS to 0, optional LPDS bit, execute WFI or WFE, optional SEVONPEND and SLEEPONEXIT.

Exit: Any external interrupt/event, external reset on the NRST pin, IWDG reset. The EXTI signal includes one of the 77 external I/O pins, the PVD output, the RTC alarm, the I3C wake-up signal, and the USB wake-up signal.

2.3.4 Standby Mode

In standby mode, the main LDO of the system is turned off, and the low-power LDO supplies power to the wake-up circuit. All other digital circuits are powered off, and FLASH is in a power-off state. Waking up the system from standby mode will generate a reset and SBF (PWR_CSR) will be set. After waking up, query the SBF status to know the low power consumption mode before waking up. SBF is cleared by the CSBF (PWR_CR) bit. In standby mode, the contents of the 32KB SRAM can be retained (depending on the planning configuration before sleep), and the contents of the back-up register are retained.

Modules that operate in standby mode: Independent Watchdog (IWDG), Real-Time Clock (RTC), and Low-Frequency Clock (LSI/LSE).

Enter: configure the core register control bit SLEEPDEEP=1, the power control register PDDS=1, execute WFI or WFE, optional SEVONPEND and SLEEPONEXIT.

Exit: any external event (EXTI signal), external reset signal on NRST, IWDG reset, a rising edge on the WKUP pin, where the EXTI signal includes one of the 77 external I/O ports, RTC alarm clock.

In standby mode, when normal power is supplied, the 32K-byte RAM is kept powered by configuring R32KSTY=1 in the PWR_CTLR register; When powered by VBAT, the 32K-byte RAM is kept non-volatile by configuring R32KVBAT=1 in the PWR_CTLR register. On this basis, RAM low-voltage mode can be enabled by configuring RAMLV=1 in the PWR_CTLR register to achieve the lowest power consumption.

Note: (1) In debug mode, if the microprocessor enters stop or standby mode, the debug connection will be lost.

(2) For chips whose fifth digit from the bottom of the lot number is 0, the I/O pins not used for wake-up function must be configured as analog inputs before entering standby mode.

(3) R32KSTY=1 controls the address range of the 32K-byte RAM: 0x20000000—0x20000000+32K.

2.3.5 RTC Auto-wakeup

RTC can automatically wake up without external interruption. By programming the time base, it can be awakened periodically from Stop or Standby mode.

The accurate external low frequency 32.768kHz crystal oscillator LSE can be selected as the RTC clock source, or the internal LSI oscillator can be selected as the RTC clock source. The accuracy and power consumption index of LSI is worse than that of LSE.

The RTC alarm clock event can wake the MCU from stop mode. In order to achieve this function, the EXTI line 17 needs to be configured, and the RTC needs to be set to generate alarm clock events. To wake up from standby mode, simply set the RTC to generate an alarm event.

2.4 Register Description

Table 2-2 PWR-related registers list

Name	Access address	Description	Reset value
R32_PWR_CTLR	0x40007000	Power control register	0x1B000200
R32_PWR_CSR	0x40007004	Power control/status register	0x00000000

2.4.1 Power Control Register (PWR_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		LDO_VDD12A[2:0]			LDO_VDDK[2:0]			Reserved			RAMLV	Reserved	R32KVBAT	Reserved	R32KSTY

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						LDO18_EN	DBP	Reserved	PLS[1:0]		PVDE	CSBF	CWUF	PDDS	LPDS

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:27]	LDO_VDD12A[2:0]	RW	Adjust the voltage value of the internal analog power supply.	0x3
[26:24]	LDO_VDDK[2:0]	RW	Adjust the digital core voltage value and VDDK voltage value.	0x3
[23:21]	Reserved	RO	Reserved	0
20	RAMLV	RW	RAM low-voltage mode enable control bit (relatively lower power consumption): 1: Enabled; 0: Disabled.	0

			Note: Valid when the LPDS bit in the PWR_CTLR register is set to 1.	
19	Reserved	RO	Reserved	0
18	R32KVBAT	RW	When powered by VBAT, whether the 32K RAM is powered in Standby mode: 1: Powered; 0: Not powered.	0
17	Reserved	RO	Reserved	0
16	R32KSTY	RW	Whether the 32K RAM is powered in Standby mode: 1: Powered; 0: Not powered.	0
[15:10]	Reserved	RO	Reserved	0
9	LDO18_EN	RW	VDD18 power enable bit: 1: Enables the PSRAM power supply; 0: Disables the PSRAM power supply.	1
8	DBP	RW	Write enable for the backup domain: In the reset state, all registers in the backup domain are protected against write access. This bit must be set to 1 to enable write access to these registers. 1: Enables access to registers in the backup domain; 0: Disables access to registers in the backup domain.	0
7	Reserved	RO	Reserved	0
[6:5]	PLS[1:0]	RW	PVD voltage monitoring threshold settings. For detailed information, refer to the Electrical Characteristics section of the data sheet. 00: 2.85V (rising edge)/2.81V (falling edge); 01: 2.94V (rising edge)/2.90V (falling edge); 10: 3.04V (rising edge)/3.00V (falling edge); 11: 3.14V (rising edge)/3.10V (falling edge);	0
4	PVDE	RW	Power supply voltage monitoring enable flag: 1: Enable power supply voltage monitoring; 0: Disable power supply voltage monitoring.	0
3	CSBF	RW1	Clear the standby status flag; a read operation always returns 0. 1: Setting this bit clears the SBF standby status flag; 0: Clearing this bit has no effect.	0
2	CWUF	RW1	Clear the wake-up status flag; a read operation always returns 0. 1: Clear the WUF flag 2 system clock cycles after setting this bit to 1; 0: Clearing this bit has no effect.	0
1	PDDS	RW	Standby/Stop mode selection bit for deep sleep scenarios during power loss. 1: Enter Standby mode; 0: Enter Stop mode; the voltage regulator status is controlled by the LPDS.	0
0	LPDS	RW	In Stop mode, this bit selects the operating mode of the voltage regulator. When PDDS=0, this bit is active.	0

			1: The voltage regulator operates in low-power mode; 0: The voltage regulator operates in normal mode.	

Note: BIT16, BIT18, and BIT20 of this register can only be reset by the backup command; the other BIT are reset when the device is awakened from standby mode.

2.4.2 Power Control/Status Register (PWR_CSR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							EWUP	Reserved					PVDO	SBF	WUF

Bit	Name	Access	Description	Reset value
[31:9]	Reserved	RO	Reserved	0
8	EWUP	RW	WKUP pin enable settings: 1: The WKUP pin is forced into an input pull-down state, used to wake the MCU from standby mode; 0: The WKUP pin can be used as a general-purpose I/O pin, without standby wake-up functionality.	0
[7:3]	Reserved	RO	Reserved	0
2	PVDO	RO	PVD output status flag. This bit is active when PVDE in the PWR_CTLR register is set to 1. 1: V_{DD} and V_{DDA} are below the PVD thresholds set by PLS[1:0]; 0: V_{DD} and V_{DDA} are above the PVD thresholds set by PLS[1:0].	0
1	SBF	RO	Standby status flag; can be cleared via CSBF bit 1. 1: The MCU enters standby mode; 0: The MCU is not in standby mode.	0
0	WUF	RO	The wake-up event status flag can be cleared via CWUF bit 1. 1: A wake-up event or RTC alarm event has been detected on the WKUP pin; 0: No wake-up event has occurred.	0

Note: This register remains unchanged after waking up from Standby mode.

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Chapter 3 Reset and Clock Control (RCC)

The controller provides different forms of resets and configurable clock tree structures based on the division of power areas and peripheral power management considerations in the application. This section describes the scope of each clock in the system.

3.1 Main Features

- Multiple reset forms
- Multiple clock sources, bus clock management
- Built-in external crystal oscillation monitoring and clock security system
- Independent management of each peripheral clock: Reset, On, Off
- Supports internal clock output

3.2 Reset

The controller provides 3 forms of reset: Power Reset, System Reset, and Backup Domain Reset.

3.2.1 Power Reset

When a power reset occurs, all registers except the backup domain (which is powered by V_{BAT}) will be reset.

The conditions for its production include:

- Power-on/power-off reset (POR/PDR reset)
- Wake up from standby mode

3.2.2 System Reset

When a system reset occurs, all registers except the reset flag and the backup domain in the control / status register RCC_RSTSCKR are reset. The source of the reset event is identified by looking at the reset status flag bit in the RCC_RSTSCKR register.

The conditions for its production include:

- Low level signal on NRST pin (External reset)
- Window watchdog count terminated (WWDG reset)
- Independent watchdog count terminated (IWDG reset)
- Software reset (SW reset)
- Low power management reset
- Core deadlock reset

Window/independent watchdog reset: Triggered by window/independent watchdog peripheral timer count cycle overflow, see the corresponding chapter for a detailed description.

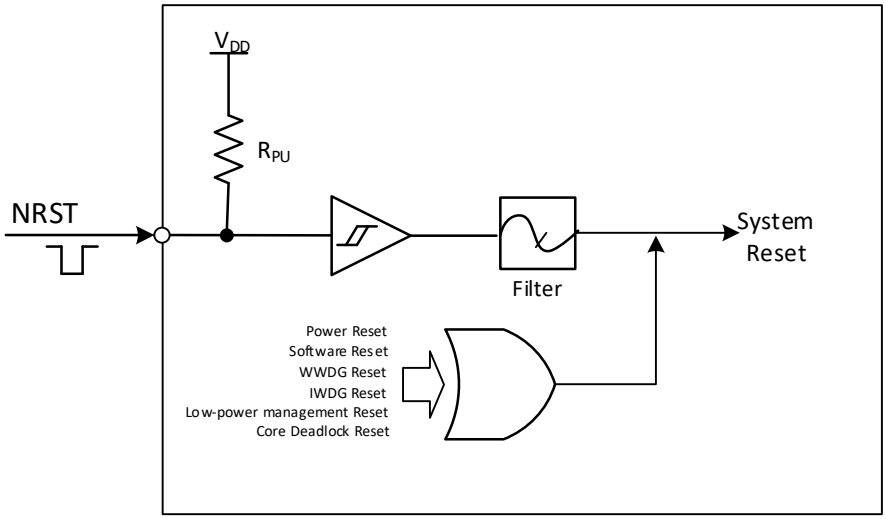
Software reset: This product resets the system through the SYSRST position 1 of the interrupt configuration register PFIC_CFGR in the programmable interrupt controller PFIC or the RSTSYS position 1 of the configuration register PFIC_SCTLR, refer to the corresponding chapter.

Low Power Management Reset: Standby mode reset is enabled by setting the STANDYRST bit in the user-selected byte to 0. After executing the process of entering standby mode at this time, a system reset will be executed instead of entering standby mode. Stop mode reset is enabled by setting the STOPRST bit in the user-selected byte to 0. After executing the process of entering stop mode at this time, a system reset will be

performed instead of entering stop mode.

Core deadlock reset: When the LKUPEN bit of the EXTEN_CTR register is enabled, if the core runs abnormally and cannot be recovered and a deadlock occurs, the system will be reset.

Figure 3-1 System reset structure



3.2.3 Backup Domain Reset

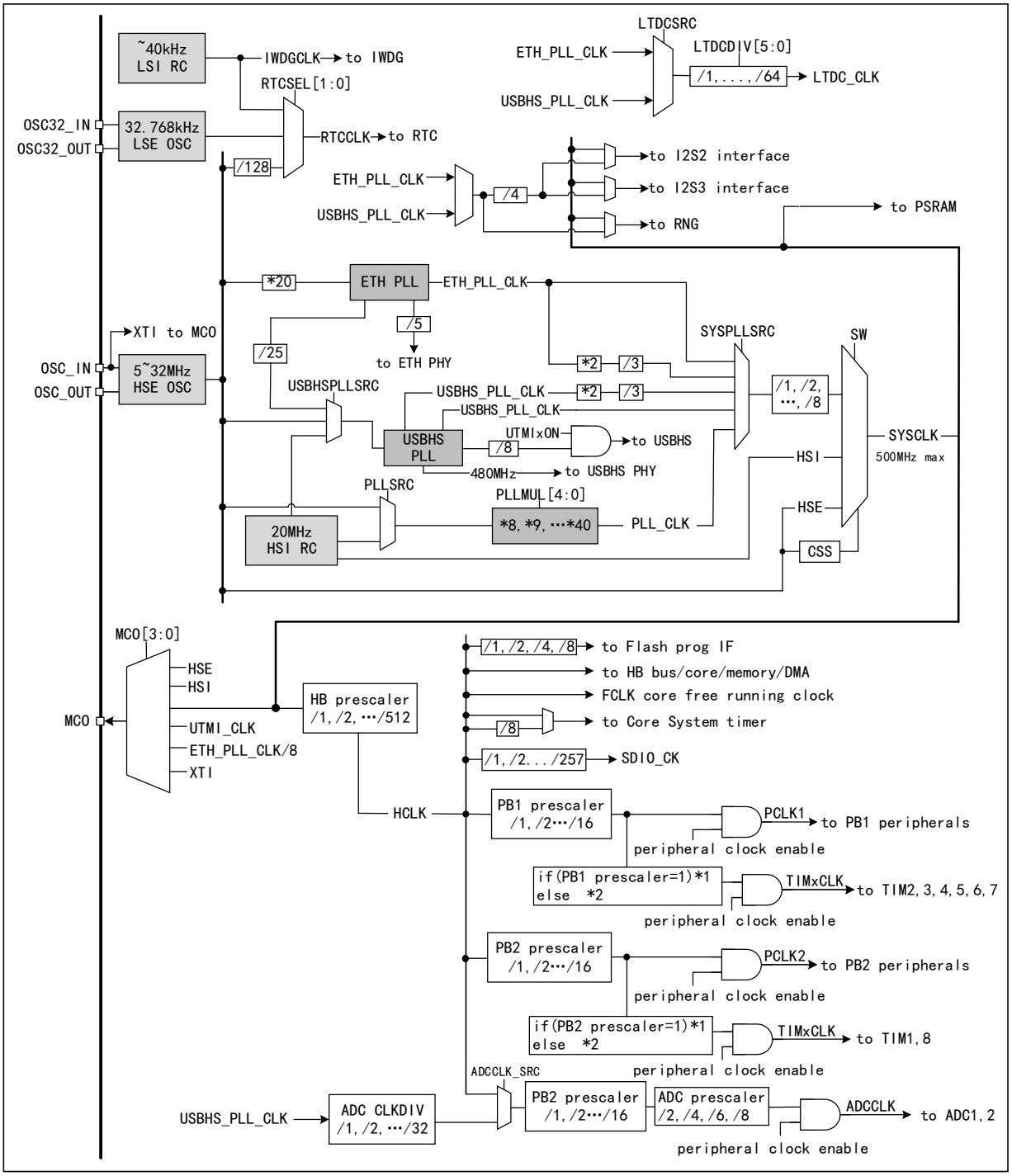
When a backup domain reset occurs, only the backup domain registers are reset, including the backup registers, the RCC_BDCTLR registers (RTC enable and LSE oscillator). The conditions for its generation include:

- Caused by V_{DD} or V_{BAT} power-up with both V_{DD} and V_{BAT} powered down
- Set BDRST to 1 of the RCC_BDCTLR register
- Set BKPRST to 1 of the RCC_PB1PRSTR register

3.3 Clock

3.3.1 System Clock Structure

Figure 3-2 Clock tree block diagram



3.3.2 High-speed Clock (HSI/HSE)

HSI is a high-speed clock signal generated by the RC oscillator of 20MHz in the system. The HSI RC oscillator can provide the system clock without any external devices. Its start-up time is very short, but the clock frequency accuracy is poor. The HSI is turned on and off by setting the HSION bit in the RCC_CTLR register, and the HSIRDY bit indicates whether the HSIRC oscillator is stable. The system defaults to HSION and HSIRDY setting 1 (it is recommended that you do not turn it off). If the HSIRDYIE bit of the RCC_INTR

register is set, the corresponding interrupt will be generated.

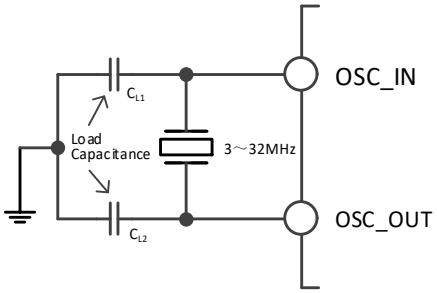
- Factory calibration: differences in manufacturing processes will lead to different RC oscillation frequencies of each chip, so HSI calibration is performed for each chip before it leaves the factory. After the system is reset, the factory calibration value is loaded into the HSICAL[7:0] in the RCC_CTLR register.
- User adjustment: depending on the voltage or ambient temperature, the application can adjust the HSI frequency through the HSITRIM[4:0] bit in the RCC_CTLR register.

Note: If the HSE crystal oscillator fails, the HSI clock will be used as a backup clock source (clock security system).

HSE is an external high-speed clock signal, including external crystal/ceramic resonator generation or external high-speed clock input.

- External Crystal/Ceramic Resonator (HSE Crystal): An external 3-32MHz oscillator provides a more accurate clock source for the system. Further information can be found in the Electrical Characteristics section of the datasheet. The HSE crystal can be turned on and off by setting the HSEON bit in the RCC_CTLR register. The HSERDY bit indicates whether the HSE crystal oscillation is stable or not, and the hardware sends the clock into the system only after the HSERDY bit is set to one. If the HSERDYIE bit in the RCC_INTR register is set, the appropriate interrupt will be generated.

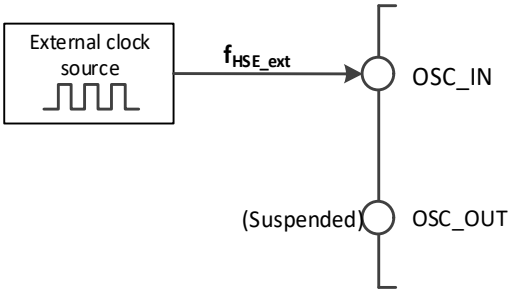
Figure 3-3 High-speed external crystal circuit



Note: The load capacitance needs to be as close as possible to the oscillator pins and the capacitance value selected according to the crystal manufacturer's parameters.

- External High Speed Clock Source (HSE Bypass): this mode feeds the clock source directly from the external source to the OSC_IN pin, with the OSC_OUT pin dangling. A maximum frequency of 32MHz is supported. The application program needs to set the HSEBYP bit to turn on the HSE bypass function with the HSEON bit at 0, and then set the HSEON bit again.

Figure 3-4 High-speed clock source circuit



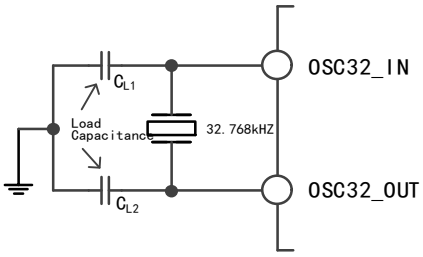
3.3.3 Low-speed Clock (LSI/LSE)

LSI is a low-speed clock signal generated by a RC oscillator in the system. It can be kept running in stop and standby mode, providing clock references for RTC clocks, independent watchdogs, and wake-up units. For further information, please refer to the electrical characteristics section of the data manual. LSI can be turned on and off by setting the LSION bit in the RCC_RSTSCKR register, and then check whether the LSIRC oscillation is stable by querying the LSIRDY bit, and the hardware sends the clock in after LSIRDY position 1. If the LSIRDYIE bit of the RCC_INTR register is set, the corresponding interrupt will be generated.

LSE is an external low-speed clock signal, including external crystal/ceramic resonator generation or external low-speed clock input. It provides a low-power and accurate clock source for RTC clocks or other timing functions.

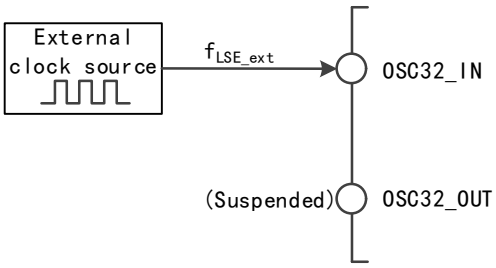
- External crystal/ceramic resonator (LSE crystal): External low-speed oscillator with external 32.768kHz. LSE is turned on and off by setting the LSEON bit in the RCC_BDCTLR register, and the LSERDY bit indicates whether the LSE crystal oscillation is stable or not, and the hardware sends the clock into the system after LSERDY position 1. If the LSERDYIE bit of the RCC_INTR register is set, the corresponding interrupt will be generated.

Figure 3-5 Low-speed external crystal circuit



- External low-speed clock source (LSE bypass): this mode feeds the clock source directly from the outside to the OSC32_IN pin, and the OSC32_OUT pin is suspended. The application needs to set the LSEBYP bit when the LSEON bit is 0, turn on the LSE bypass function, and then set the LSEON bit.

Figure 3-6 Low-speed clock source circuit



3.3.4 System PLL Clock (SYSPLL)

By configuring the RCC_CFGR0 register and RCC_CFGR2 register, the system PLL clock can select 5 clock sources and frequency division factors. These settings must be completed before each PLL is turned on. Once the PLL is started, these parameters cannot be changed.

The PLLON bit in the RCC_CTLR register is set to be turned on and off. The PLLRDY bit indicates whether the PLL clock is stable. The hardware only sends the clock to the system after the PLLRDY bit is set to 1.

It is enabled and disabled by configuring the ETH_PLLON bit in the RCC_CTLR register. The ETH_PLLRDY bit indicates whether the ETH_PLL clock is stable. The hardware only sends the clock to the system after the

ETH_PLLRDY bit is set to 1.

The USBHS_PLLON bit in the RCC_CTLR register is set to be enabled and disabled. The USBHS_PLLRDY bit indicates whether the USBHS_PLL clock is stable. The hardware only sends the clock to the system after the USBHS_PLLRDY bit is set to 1.

SYSPLL clock source:

- PLL_CLK clock input
- USBHS_PLL (480MHz) clock input
- ETH_PLL (500MHz) clock input
- USBHS_PLL (320MHz) frequency division clock input
- ETH_PLL(333MHz) frequency division clock input

3.3.5 Bus/Peripheral Clock

3.3.5.1 System Clock (SYSCLK)

By configuring the RCC_CFGR0 register SW[1:0] bit to configure the system clock source, SWS[1:0] indicates the current system clock source.

- HSI as the system clock.
- HSE as the system clock.
- SYSPLL clock as the system clock.

After the controller is reset, the default HSI clock is selected as the system clock source. The switching between clock sources does not occur until the target clock source is ready.

3.3.5.2 HB/PB1/PB2 Bus Peripheral Clock (HCLK/PCLK1/PCLK2)

HB stands for High Performance Bus, and its peripheral clock is HCLK; PB1 stands for Peripheral Bus 1, and its peripheral clock is PCLK1; PB2 stands for Peripheral Bus 2, and its peripheral clock is PCLK2.

By configuring the HPRE[3:0], PPRE1[2:0], and PPRE2[2:0] bits in the RCC_CFGR0 register, you can configure the clocks for the HB, PB1, and PB2 buses, respectively. These bus clocks determine the clock reference for accessing the peripheral interfaces connected to them. Applications can adjust these values to reduce the power consumption of certain peripherals during operation.

Different peripheral modules can be reset and restored to their initial state through each bit in the RCC_HBRSTR, RCC_PB1PRSTR, and RCC_PB2PRSTR registers.

Different peripheral module communication clock interfaces can be individually turned on or off through each bit in the RCC_HBPCENR, RCC_PB1PCENR, and RCC_PB2PCENR registers. When using a peripheral, you first need to turn on its clock enable bit before you can access its registers.

3.3.5.3 RTC Clock (RTCCLK)

By setting the RTCSEL[1:0] bit of the RCC_BDCTLR register, the RTCCLK clock source can be provided by the HSE/128, LSE, or LSI clock. Before modifying this bit, make sure that the DBP position 1 in the power control register (PWR_CTLR) can be reset only if the backup domain is reset.

- LSE acts as the RTC clock: Since LSE is powered by V_{BAT} in the backup domain, RTC continues to work as long as V_{BAT} maintains power, even though V_{DD} power is cut off.
- LSI as the RTC clock: If the V_{DD} power is cut off, RTC automatic wake up is not guaranteed.
- HSE/128 as RTC clock: If the V_{DD} power supply is cut off or the internal voltage regulator is turned off

(the power supply to the 1.8V domain is cut off), the RTC status is undefined.

3.3.5.4 Independent Watchdog Clock

If the independent watchdog has been started by the hardware configuration or software, the LSI oscillator will be forced to open and cannot be turned off. After the LSI oscillator is stabilized, the clock is supplied to the IWDG.

3.3.5.5 Microcontroller Clock Output (MCO)

The microcontroller allows clock signals to be output to MCO pins. The alternate push-pull output mode is configured in the corresponding GPIO port register. By configuring the RCC_CFGR0 register MCO[3:0] bit, the following 6 clock signals can be selected as MCO clock output:

- System clock (SYSCLK) output.
- HSI clock output.
- HSE clock output.
- UTMI clock output.
- ETH_PLL clock output through 8 frequency division
- XT1 external oscillator clock output

3.3.6 Clock Security System

The clock security system is a running protection mechanism of the controller, which can switch to the HSI clock in the case of HSE clock transmission failure, and generate interrupt notification, allowing application software to complete the rescue operation.

Activate the clock security system by setting CSSON position 1 of the RCC_CTLR register. At this point, the clock monitor will be enabled after the HSE oscillator startup (HSERDY=1) delay and turned off after the HSE clock is turned off. Once the HSE clock fails during the operation of the system, the HSE oscillator will be turned off, the clock failure event will be sent to the brake input of the advanced timer (TIM1 and TIM8), and the clock security interrupt will be generated, CSSF bit is set to 1, and the application will enter the NMI unshielded interrupt. By setting the CSSC bit, the CSSF bit flag can be cleared and the NMI interrupt suspension bit can be revoked.

If the current HSE is used as the system clock, or the current HSE is used as the PLL input clock, and the PLL is used as the system clock, the clock safety system will automatically switch the system clock to the HSI oscillator when the HSE fails, and shut down the HSE oscillator and PLL.

3.4 Register Description

Table 3-1 RCC-related registers list

Name	Access address	Description	Reset value
R32_RCC_CTLR	0x40021000	Clock control register	0x0000xx83
R32_RCC_CFGR0	0x40021004	Clock configuration register 0	0x20000000
R32_RCC_INTR	0x40021008	Clock interrupt register	0x00000000
R32_RCC_PB2PRSTR	0x4002100C	PB2 peripheral reset register	0x00000000
R32_RCC_PB1PRSTR	0x40021010	PB1 peripheral reset register	0x00000000
R32_RCC_HBPCENR	0x40021014	HB peripheral clock enable register	0x00000004

R32_RCC_PB2PCENR	0x40021018	PB2 peripheral clock enable register	0x00000000
R32_RCC_PB1PCENR	0x4002101C	PB1 peripheral clock enable register	0x00000000
R32_RCC_BDCTLR	0x40021020	Backup domain control register	0x00000000
R32_RCC_RSTSCKR	0x40021024	Control/status register	0x08000000
R32_RCC_HBRSTR	0x40021028	HB peripheral reset register	0x00000000
R32_RCC_CFGR2	0x4002102C	Clock configuration register 2	0x00000000
R32_RCC_CFGR3	0x40021030	Clock configuration register 3	0x00000000

3.4.1 Clock Control Register (RCC_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved						PLLRDY	PLLON	ETH_PLLRDY	ETH_PLLON	USBHS_PLLRDY	USBHS_PLLON	CSSON	HSEBYP	HSERDY	HSEON

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSICAL[7:0]								HSITRIM[4:0]					Reserved	HSIRDY	HSION

Bit	Name	Access	Description	Reset value
[31:26]	Reserved	RO	Reserved	0
25	PLLRDY	RO	PLL clock ready flag: 1: PLL locked; 0: PLL unlocked	0
24	PLLON	RW	PLL enable: 1: Enable PLL clock; 0: Disable PLL clock. *Note: Cleared by hardware when entering Stop or Standby mode.*	0
23	ETH_PLLRDY	RO	ETH_PLL clock ready locked flag: 1: ETH_PLL clock locked; 0: ETH_PLL clock not locked.	0
22	ETH_PLLON	RW	ETH_PLL enable control bit: 1: Enable ETH_PLL clock; 0: Disable ETH_PLL clock.	0
21	USBHS_PLLRDY	RO	USBHS_PLL clock ready locked flag: 1：USBHS_PLL clock locked; 0：USBHS_PLL clock not locked.	0
20	USBHS_PLLON	RW	USBHS_PLL clock enable control bit: 1: Enable USBHS_PLL clock; 0: Disable USBHS_PLL clock. Hardware automatically resets upon entering deep sleep.	0
19	CSSON	RW	Clock security system enable:	0

			1: Enable clock security system. When HSE is ready (HSERDY is set to 1), the hardware turns on the clock monitoring function for HSE and finds that HSE abnormal triggers the CSSF flag and NMI interrupt; when HSE is not ready, the hardware turns off the clock monitoring function for HSE. 0: Disable clock security system.	
18	HSEBYP	RW	External high-speed clock bypass: 1: Bypassing external high-speed crystal/ceramic resonator (using external clock source); 0: No bypass high-speed external crystal/ceramic resonator. *Note: This bit needs to be written when HSEON is 0.*	0
17	HSERDY	RO	External high-speed crystal oscillation stability ready flag (set by hardware): 1: External high-speed crystal oscillation is stable. 0: External high-speed crystal oscillation is not stable. *Note: After the HSEON bit is cleared, it needs 6 HSE cycles to clear 0.*	0
16	HSEON	RW	HSE clock enable: 1: Enable HSE oscillator; 0: Disable HSE oscillator. *Note: Clear by hardware to stop HSE oscillator when entering Stop or Standby mode.*	0
[15:8]	HSICAL[7:0]	RO	Internal high-speed clock calibration value These bits are automatically initialized at system startup.	x
[7:3]	HSITRIM[4:0]	RW	Internal high-speed clock adjustment: The user can enter an adjustment value that is superimposed on the HSICAL[7:0] value to adjust the frequency of the internal HSI RC oscillator according to changes in voltage and temperature. The default value is 16, which can adjust the HSI to 20MHz±0.25%; the change of HSICAL is adjusted about 40kHz per step.	10000b
2	Reserved	RO	Reserved	0
1	HSIRDY	RO	Internal high-speed HSI (20MHz) stable ready flag (set by hardware) 1: Internal high-speed HSI (20MHz) is stable; 0 Internal high-speed HSI (20MHz) is not stable. *Note: After the HSION bit is cleared, it needs 6 HSI cycles to clear 0.*	1
0	HSION	RW	Internal high-speed clock HSI (20MHz) enable control bit: 1: Enable the HSI oscillator; 0: Disable the HSI oscillator. *Note: This bit is set by hardware to start the internal*	1

			20MHz RC oscillator when returning from standby and stop modes or when the external oscillator HSE used as the system clock fails.	

3.4.2 Clock Configuration Register 0 (RCC_CFGR0)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	ADC_DUTY_SEL		ADCSRC	MCO[3:0]				Reserved		PLLMUL[4:0]					PLLSRC

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADCPRE[1:0]		PPRE2[2:0]			PPRE1[2:0]			HPRE[3:0]				SWS[1:0]		SW[1:0]	

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
[30:29]	ADC_DUTY_SEL[1:0]	RW	ADC clock duty cycle selection: 00: ADC clock duty cycle is 50%; 01: ADC clock duty cycle is 62.5%; 10: ADC clock duty cycle is 75%; 11: ADC clock duty cycle is 62.5%.	01b
28	ADCSRC	RW	ADC input clock source selection: 1: Clock from the USBHS_PLL after division; 0: HCLK.	0
[27:24]	MCO[3:0]	RW	Microcontroller MCO pin clock output control: 00xx: No clock output; 0100: System clock (SYSCLK) output; 0101: Internal 20MHz RC oscillator clock (HSI) output; 0110: External oscillator clock (HSE) output; 0111: UTMI clock output; 1000: ETH_PLL clock output after 8 frequency division; 1010: XT1 external oscillator clock output; Others: Reserved. *Note: When starting or switching the MCO clock, there may be a few cycles of clock loss.*	0000b
[23:22]	Reserved	RO	Reserved	0
[21:17]	PLLMUL[4:0]	RW	System PLL frequency multiplication coefficient: 00000: 8x frequency; 00001: 9x frequency; 00010: 10x frequency; 00011: 11x frequency; 00100: 12x frequency; 00101: 12.5x frequency; 00110: 13x frequency; 00111: 13.5x frequency; 01000: 14x frequency; 01001: 14.5x frequency; 01010: 15x frequency; 01011: 15.5x frequency; 01100: 16x frequency; 01101: 16.5x frequency; 01110: 17x frequency; 01111: 17.5x frequency;	0

			10000: 18x frequency; 10001: 18.5x frequency; 10010: 19x frequency; 10011: 19.5x frequency; 10100: 20x frequency; 10101: 21x frequency; 10110: 22x frequency; 10111: 24x frequency; 11000: 26x frequency; 11001: 28x frequency; 11010: 30x frequency; 11011: 32x frequency; 11100: 34x frequency; 11101: 36x frequency; 11110: 38x frequency; 11111: 40x frequency.	
16	PLLSRC	RW	System PLL source: 1: HSE; 0: HSI.	0
[15:14]	ADCPRE[1:0]	RW	ADC clock source prescaler control: 00: PCLK2 divided by 2 and used as the ADC clock; 01: PCLK2 divided by 4 and used as the ADC clock; 10: PCLK2 divided by 6 and used as the ADC clock; 11: PCLK2 divided by 8 and used as the ADC clock. *Note: The ADC clock should not exceed 30MHz.*	00b
[13:11]	PPRE2[2:0]	RW	PB2 clock source prescaler control: 0xx: HCLK no prescaling; 100: HCLK divided by 2; 101: HCLK divided by 4; 110: HCLK divided by 8; 111: HCLK divided by 16.	000b
[10:8]	PPRE1[2:0]	RW	PB1 clock source prescaler control: 0xx: HCLK no prescaling; 100: HCLK divided by 2; 101: HCLK divided by 4; 110: HCLK divided by 8; 111: HCLK divided by 16.	000b
[7:4]	HPRE[3:0]	RW	HB clock source prescaler control: 0xxx: SYSCLK no prescaling; 1000: SYSCLK divided by 2; 1001: SYSCLK divided by 4; 1010: SYSCLK divided by 8; 1011: SYSCLK divided by 16; 1100: SYSCLK divided by 64; 1101: SYSCLK divided by 128; 1110: SYSCLK divided by 256; 1111: SYSCLK divided by 512.	0000b
[3:2]	SWS[1:0]	RO	System clock (SYSCLK) status (set by hardware): 00: System clock source is HSI; 01: System clock source is HSE; 10: System clock source is SYSPLL; 11: Not available.	00b
[1:0]	SW[1:0]	RW	Select the system clock source: 00: HSI as system clock;	00b

			01: HSE as system clock; 10: SYSPLL output as system clock; 11: Not available. *Note: When the clock safety system is enabled (CSSON=1), when returning from standby and stop modes or the external oscillator HSE used as the system clock fails, HSI is forced to be selected by hardware as the system clock.*	

3.4.3 Clock Interrupt Register (RCC_INTR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CSSC	Reserved			HSERDYC	HSIRDYC	LSERDYC	LSIRDYC

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				HSERDYIE	HSIRDYIE	LSERDYIE	LSIRDYIE	CSSF	Reserved			HSERDYF	HSIRDYF	LSERDYF	LSIRDYF

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
23	CSSC	WO	Clear clock security system interrupt flag (CSSF): 1: Clear CSSF interrupt flag; 0: No effect.	0
[22:20]	Reserved	RO	Reserved	0
19	HSERDYC	WO	Clear HSE oscillator ready interrupt flag: 1: Clear HSERDYF interrupt flag; 0: No action.	0
18	HSIRDYC	WO	Clear HSI oscillator ready interrupt flag: 1: Clear HSIRDYF interrupt flag; 0: No action.	0
17	LSERDYC	WO	Clear LSE oscillator ready interrupt flag: 1: Clear LSERDYF interrupt flag; 0: No action.	0
16	LSIRDYC	WO	Clear LSI oscillator ready interrupt flag: 1: Clear LSIRDYF interrupt flag; 0: No action.	0
[15:12]	Reserved	RO	Reserved	0
11	HSERDYIE	RW	HSE ready interrupt enable: 1: Enable HSE ready interrupt; 0: Disable HSE ready interrupt.	0
10	HSIRDYIE	RW	HSI ready interrupt enable: 1: Enable HSI ready interrupt; 0: Disable HSI ready interrupt.	0

9	LSERDYIE	RW	LSE ready interrupt enable: 1: Enable LSE ready interrupt; 0: Disable LSE ready interrupt.	0
8	LSIRDYIE	RW	LSI ready interrupt enable: 1: Enable LSI ready interrupt; 0: Disable LSI ready interrupt.	0
7	CSSF	RO	Clock security system interrupt flag bit: 1: HSE clock failure, resulting in clock security interrupt CSSI. 0: No clock security system interrupt. Set by hardware, clear when software write CSSC bit 1.	0
[6:4]	Reserved	RO	Reserved	0
3	HSERDYF	RO	HSE clock ready interrupt flag: 1: HSE clock ready generates an interrupt; 0: No HSE clock ready interrupt. Set by hardware, clear when software write HSERDYC bit 1.	0
2	HSIRDYF	RO	HSI clock ready interrupt flag: 1: HSI clock ready generates an interrupt; 0: No HSI clock ready interrupt. Set by hardware, clear when software write HSIRDYC bit 1.	0
1	LSERDYF	RO	LSE clock ready interrupt flag: 1: LSE clock ready generates an interrupt; 0: No LSE clock ready interrupt. Set by hardware, clear when software write LSERDYC bit 1.	0
0	LSIRDYF	RO	LSI clock ready interrupt flag: 1: LSI clock ready generates an interrupt; 0: No LSI clock ready interrupt. Set by hardware, clear when software write LSIRDYC bit 1.	0

3.4.4 PB2 Peripheral Reset Register (RCC_PB2PRSTR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	USART1RST	TIM8RST	SPI1RST	TIM1RST	ADC2RST	ADC1RST	Reserved		IOPERST	IOPDRST	IOPCRST	IOPBRST	IOPARST	Reserved	AFIORST

Bit	Name	Access	Description	Reset value
[31:15]	Reserved	RO	Reserved	0
14	USART1RST	RW	USART1 module reset control:	0

			1: Reset module; 0: No effect.	
13	TIM8RST	RW	TIM8 module reset control: 1: Reset module; 0: No effect.	0
12	SPI1RST	RW	SPI1 module reset control: 1: Reset module; 0: No effect.	0
11	TIM1RST	RW	TIM1 module reset control: 1: Reset module; 0: No effect.	0
10	ADC2RST	RW	ADC2 module reset control: 1: Reset module; 0: No effect.	0
9	ADC1RST	RW	ADC1 module reset control: 1: Reset module; 0: No effect.	0
[8:7]	Reserved	RO	Reserved	0
6	IOPERST	RW	IO's PE port module reset control: 1: Reset module; 0: No effect.	0
5	IOPDRST	RW	IO's PD port module reset control: 1: Reset module; 0: No effect.	0
4	IOPCRST	RW	IO's PC port module reset control: 1: Reset module; 0: No effect.	0
3	IOPBRST	RW	IO's PB port module reset control: 1: Reset module; 0: No effect.	0
2	IOPARST	RW	IO's PA port module reset control: 1: Reset module; 0: No effect.	0
1	Reserved	RO	Reserved	0
0	AFIORST	RW	IO auxiliary function module reset control: 1: Reset module; 0: No effect.	0

3.4.5 PB1 Peripheral Reset Register (RCC_PB1PRSTR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		DACRST	PWRRST	BKPRST	Reserved	CAN1RST	Reserved			I2CRST	USART5RST	USART4RST	USART3RST	USART2RST	Reserved

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3RST	SPI2RST	Reserved		WWDGRST	USART10RST	USART9RST	USART8RST	USART7RST	USART6RST	TIM7RST	TIM6RST	TIM5RST	TIM4RST	TIM3RST	TIM2RST

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
29	DACRST	RW	DAC module reset control: 1: Reset module; 0: No effect.	0
28	PWRRST	RW	Power interface module reset control: 1: Reset module; 0: No effect.	0
27	BKPRST	RW	Backup unit reset control: 1: Reset module; 0: No effect.	0

26	Reserved	RO	Reserved	0
25	CAN1RST	RW	CAN1 module reset control: 1: Reset module; 0: No effect.	0
[24:22]	Reserved	RO	Reserved	0
21	I2CRST	RW	I2C module reset control: 1: Reset module; 0: No effect.	0
20	USART5RST	RW	USART5 module reset control: 1: Reset module; 0: No effect.	0
19	USART4RST	RW	USART4 module reset control: 1: Reset module; 0: No effect.	0
18	USART3RST	RW	USART3 module reset control: 1: Reset module; 0: No effect.	0
17	USART2RST	RW	USART2 module reset control: 1: Reset module; 0: No effect.	0
16	Reserved	RO	Reserved	0
15	SPI3RST	RW	SPI3 module reset control: 1: Reset module; 0: No effect.	0
14	SPI2RST	RW	SPI2 module reset control: 1: Reset module; 0: No effect.	0
[13:12]	Reserved	RO	Reserved	0
11	WWDGRST	RW	Window watchdog reset control: 1: Reset module; 0: No effect.	0
10	USART10RST	RW	USART10 module reset control: 1: Reset module; 0: No effect.	0
9	USART9RST	RW	USART9 module reset control: 1: Reset module; 0: No effect.	0
8	USART8RST	RW	USART8 module reset control: 1: Reset module; 0: No effect.	0
7	USART7RST	RW	USART7 module reset control: 1: Reset module; 0: No effect.	0
6	USART6RST	RW	USART6 module reset control: 1: Reset module; 0: No effect.	0
5	TIM7RST	RW	Timer 7 module reset control: 1: Reset module; 0: No effect.	0
4	TIM6RST	RW	Timer 6 module reset control: 1: Reset module; 0: No effect.	0
3	TIM5RST	RW	Timer 5 module reset control: 1: Reset module; 0: No effect.	0
2	TIM4RST	RW	Timer 4 module reset control: 1: Reset module; 0: No effect.	0
1	TIM3RST	RW	Timer 3 module reset control: 1: Reset module; 0: No effect.	0
0	TIM2RST	RW	Timer 2 module reset control: 1: Reset module; 0: No effect.	0

3.4.6 HB Peripheral Clock Enable Register (RCC_HBPCENR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	ETHMACEN	DVPEN	USBHS1EN	USBHS2EN	SDIOEN	RNGEN	FSMCEN	ARGBEN	CRCEN	I3CEN	PSRAMEN	LTDCEN	SRAMEN	DMA2EN	DMA1EN

Bit	Name	Access	Description	Reset value
[31:15]	Reserved	RO	Reserved	0
14	ETHMACEN	RW	Ethernet MAC Clock Enable: 1: Ethernet MAC clock enabled; 0: Ethernet MAC clock disabled.	0
13	DVPEN	RW	DVP module clock enable: 1: Module clock on; 0: Module clock off.	0
12	USBHS1EN	RW	USBHS1 module clock enable: 1: Module clock on; 0: Module clock off.	0
11	USBHS2EN	RW	USBHS2 module clock enable: 1: Module clock on; 0: Module clock off.	0
10	SDIOEN	RW	SDIO module clock enable: 1: Module clock on; 0: Module clock off.	0
9	RNGEN	RW	RNG module clock enable: 1: Module clock on; 0: Module clock off.	0
8	FSMCEN	RW	FSMC module clock enable: 1: Module clock on; 0: Module clock off.	0
7	ARGBEN	RW	ARGB module clock enable: 1: Module clock on; 0: Module clock off.	0
6	CRCEN	RW	CRC module clock enable: 1: Module clock on; 0: Module clock off.	0
5	I3CEN	RW	I3C module clock enable: 1: Module clock on; 0: Module clock off.	0
4	PSRAMEN	RW	PSRAM module clock enable: 1: Module clock on; 0: Module clock off.	0
3	LTDCEN	RW	LTDC module clock enable: 1: Module clock on; 0: Module clock off.	0
2	SRAMEN	RW	SRAM interface module clock enable bit: 1: In sleep mode, the SRAM interface module clock is on; 0: In sleep mode, the SRAM interface module clock is off.	1
1	DMA2EN	RW	DMA2 module clock enable: 1: Module clock on; 0: Module clock off.	0
0	DMA1EN	RW	DMA1 module clock enable: 1: Module clock on; 0: Module clock off.	0

3.4.7 PB2 Peripheral Clock Enable Register (RCC_PB2PCENR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	USART1EN	TIM8EN	SPI1EN	TIM1EN	ADC2EN	ADC1EN	Reserved		IOPEEN	IOPDEN	IOPCEN	IOPBEN	IOPAEN	Reserved	AFIOEN

Bit	Name	Access	Description	Reset value
[31:15]	Reserved	RO	Reserved	0
14	USART1EN	RW	USART1 module clock enable: 1: Module clock on; 0: Module clock off.	0
13	TIM8EN	RW	TIM8 module clock enable: 1: Module clock on; 0: Module clock off.	0
12	SPI1EN	RW	SPI1 module clock enable: 1: Module clock on; 0: Module clock off.	0
11	TIM1EN	RW	TIM1 module clock enable: 1: Module clock on; 0: Module clock off.	0
10	ADC2EN	RW	ADC2 module clock enable: 1: Module clock on; 0: Module clock off.	0
9	ADC1EN	RW	ADC1 module clock enable: 1: Module clock on; 0: Module clock off.	0
[8:7]	Reserved	RO	Reserved	0
6	IOPEEN	RW	IO's PE port module clock enable: 1: Module clock on; 0: Module clock off.	0
5	IOPDEN	RW	IO's PD port module clock enable: 1: Module clock on; 0: Module clock off.	0
4	IOPCEN	RW	IO's PC port module clock enable: 1: Module clock on; 0: Module clock off.	0
3	IOPBEN	RW	IO's PB port module clock enable: 1: Module clock on; 0: Module clock off.	0
2	IOPAEN	RW	IO's PA port module clock enable: 1: Module clock on; 0: Module clock off.	0
1	Reserved	RO	Reserved	0
0	AFIOEN	RW	IO auxiliary function module clock enable: 1: Module clock on; 0: Module clock off.	0

3.4.8 PB1 Peripheral Clock Enable Register (RCC_PB1PCENR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		DACEN	PWREN	BKPEN	Reserved	CAN1EN	Reserved			I2CEN	USART5EN	USART4EN	USART3EN	USART2EN	Reserved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

SPI3 EN	SPI2 EN	Reserved	WW DG EN	USA RT10 EN	USA RT9E N	USA RT8E N	USA RT7E N	USA RT6E N	TIM7 EN	TIM6 EN	TIM5 EN	TIM4 EN	TIM3 EN	TIM2 EN

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
29	DACEN	RW	DAC module clock enable: 1: Module clock on; 0: Module clock off.	0
28	PWREN	RW	Power interface module clock enable: 1: Module clock on; 0: Module clock off.	0
27	BKPEN	RW	Backup unit clock enable: 1: Module clock on; 0: Module clock off.	0
26	Reserved	RO	Reserved	0
25	CAN1EN	RW	CAN1 module clock enable: 1: Module clock on; 0: Module clock off.	0
[24:22]	Reserved	RO	Reserved	0
21	I2CEN	RW	I2C module clock enable: 1: Module clock on; 0: Module clock off.	0
20	USART5EN	RW	USART5 module clock enable: 1: Module clock on; 0: Module clock off.	0
19	USART4EN	RW	USART4 module clock enable: 1: Module clock on; 0: Module clock off.	0
18	USART3EN	RW	USART3 module clock enable: 1: Module clock on; 0: Module clock off.	0
17	USART2EN	RW	USART2 module clock enable: 1: Module clock on; 0: Module clock off.	0
16	Reserved	RO	Reserved	0
15	SPI3EN	RW	SPI3 module clock enable: 1: Module clock on; 0: Module clock off.	0
14	SPI2EN	RW	SPI2 module clock enable: 1: Module clock on; 0: Module clock off.	0
[13:12]	Reserved	RO	Reserved	0
11	WWDGEN	RW	Window watchdog clock enable: 1: Module clock on; 0: Module clock off.	0
10	USART10EN	RW	USART10 module clock enable: 1: Module clock on; 0: Module clock off.	0
9	USART9EN	RW	USART9 module clock enable: 1: Module clock on; 0: Module clock off.	0
8	USART8EN	RW	USART8 module clock enable: 1: Module clock on; 0: Module clock off.	0
7	USART7EN	RW	USART7 module clock enable: 1: Module clock on; 0: Module clock off.	0
6	USART6EN	RW	USART6 module clock enable: 1: Module clock on; 0: Module clock off.	0
5	TIM7EN	RW	Timer 7 module clock enable:	0

			1: Module clock on; 0: Module clock off.	
4	TIM6EN	RW	Timer 6 module clock enable: 1: Module clock on; 0: Module clock off.	0
3	TIM5EN	RW	Timer 5 module clock enable: 1: Module clock on; 0: Module clock off.	0
2	TIM4EN	RW	Timer 4 module clock enable: 1: Module clock on; 0: Module clock off.	0
1	TIM3EN	RW	Timer 3 module clock enable: 1: Module clock on; 0: Module clock off.	0
0	TIM2EN	RW	Timer 2 module clock enable: 1: Module clock on; 0: Module clock off.	0

3.4.9 Backup Domain Control Register (RCC_BDCTLR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															BDRST

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RTCEN	Reserved					RTCSEL[1:0]		Reserved					LSEBYP	LSERDY	LSEON

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	BDRST	RW	Backup domain software reset control: 1: Reset the entire backup domain. 0: Undo the reset.	0
15	RTCEN	RW	RTC Clock Enable Control: 1: Enable the RTC clock; 0: Disable the RTC clock. *Note: The RTC clock can only be enabled when RTCSEL!=0; otherwise, the hardware forces it to 0.*	0
[14:10]	Reserved	RO	Reserved	0
[9:8]	RTCSEL[1:0]	RW	RTC clock source selection: 00: No clock; 01: LSE oscillator as RTC clock; 10: LSI oscillator as RTC clock; 11: HSE oscillator, divided by 128, as RTC clock. *Note: Once the RTC clock source is selected (RTCEN=1), it cannot be changed until the backup domain is reset. The default setting can be restored by setting the BDRST bit.*	0
[7:3]	Reserved	RO	Reserved	0
2	LSEBYP	RW	External Low-speed crystal (LSE) bypass control bit: 1: Bypass external low-speed crystal/ceramic resonator (using external clock source);	0

			0: Do not bypass low-speed external crystal/ceramic resonator. *Note: This bit needs to be written when LSEON is 0.*	
1	LSERDY	RO	External low-speed crystal oscillation stabilization ready flag bit (set by hardware): 1: External low-speed crystal oscillation is stabilized; 0: External low-speed crystal oscillation is not stabilized. *Note: After the LSEON bit is cleared to 0, this bit takes 6 LSE cycles to clear to 0.*	0
0	LSEON	RW	External low-speed crystal oscillation enable control bit: 1: Enable the LSE oscillator; 0: Disable the LSE oscillator.	0

Note: The LSEON, LSEBYP, RTCSEL and RTCEN bits in the backup domain control register (RCC_BDCTLR) are in the backup domain. Therefore, these bits are write-protected after reset and can only be changed after the DBP bit in the power control register (PWR_CTLR) is set to 1. These bits can only be cleared by a backup domain reset. These bits are not affected by any internal or external reset.

3.4.10 Control/Status Register (RCC_RSTSCKR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LPWRRSTF	WWDGRSTF	IWDGRSTF	SFTRSTF	PORRSTF	PINRSTF	Reserved	RMVF	Reserved							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														LSIRDY	LSION

Bit	Name	Access	Description	Reset value
31	LPWRRSTF	RO	Low power reset flag: 1: Low power reset occurs; 0: No low power reset occurs. Set by hardware when a low-power management reset occurs; software writes the RMVF bit to clear it. *Note: This function is only applicable to chips whose fifth digit from the bottom of the lot number is greater than 0.*	0
30	WWDGRSTF	RO	Window watchdog reset flag: 1: Occurrence of a window watchdog reset. 0: No window watchdog reset occurs. Set to 1 by hardware when a window watchdog reset occurs; cleared by software writing of the RMVF bit.	0
29	IWDGRSTF	RO	Independent watchdog reset flag: 1: Occurrence of an independent watchdog reset. 0: No independent watchdog reset occurs. Set to 1 by hardware when an independent watchdog reset occurs; cleared by software writing of the RMVF bit.	0

28	SFTRSTF	RO	Software reset flag: 1: Software reset occurs. 0: No software reset occurs. Set to 1 by hardware when a software reset occurs; software write RMVF bit cleared.	0
27	PORRSTF	RO	Power-up/power-down reset flag: 1: Power-up/power-down reset occurs. 0: No power-up/power-down reset occurs. Set to 1 by hardware when power-up/power-down reset occurs; cleared by software writing of RMVF bit.	1
26	PINRSTF	RO	External manual reset (NRST pin) flag: 1: Occurrence of NRST pin reset. 0: No NRST pin reset occurs. Set to 1 by hardware when NRST pin reset occurs; cleared by software writing of RMVF bit.	0
25	Reserved	RO	Reserved	0
24	RMVF	RW	Clear reset flag control: 1: Clear the reset flag. 0: No effect.	0
[23:2]	Reserved	RO	Reserved	0
1	LSIRDY	RO	Internal low-speed clock (LSI) stabilization ready flag bit (set by hardware): 1: The internal low-speed clock (40kHz) is stabilized; 0: The internal low-speed clock (40kHz) is not stabilized. *Note: After the LSION bit is cleared to 0, it takes 3 LSI cycles for the bit to be cleared.*	0
0	LSION	RW	Internal low-speed clock (LSI) enable control bits: 1: Enables the LSI (40kHz) oscillator; 0: Disables the LSI (40kHz) oscillator.	0

Note: Except for BIT1, which is cleared by power-on reset, other write-clear reset flags can be cleared.

3.4.11 HB Peripheral Reset Register (RCC_HBRSTR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	ETHMACRST	DVPRST	USBHS1RST	USBHS2RST	Reserved			ARGBRST	Reserved	I3CRST	PSRAMRST	LTDCRST	Reserved		

Bit	Name	Access	Description	Reset value
[31:15]	Reserved	RO	Reserved	0
14	ETHMACRST	RW	Ethernet MAC module reset control: 1: Reset module; 0: No effect.	0

13	DVPRST	RW	DVP module reset control: 1: Reset module; 0: No effect.	0
12	USBHS1RST	RW	USBHS1 module reset control: 1: Reset module; 0: No effect.	0
11	USBHS2RST	RW	USBHS2 module reset control: 1: Reset module; 0: No effect.	0
[10:8]	Reserved	RO	Reserved	0
7	ARGBRST	RW	ARGB module reset control: 1: Reset module; 0: No effect.	0
6	Reserved	RO	Reserved	0
5	I3CRST	RW	I3C module reset control: 1: Reset module; 0: No effect.	0
4	PSRAMRST	RW	PSRAM module reset control: 1: Reset module; 0: No effect.	0
3	LTDCRST	RW	LTDC module reset control: 1: Reset module; 0: No effect.	0
[2:0]	Reserved	RO	Reserved	0

3.4.12 Clock Configuration Register 2 (RCC_CFGR2)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UTMI2ON	UTMI1ON	USBHSPLLCLK[1:0]		USBHSPLLSRC[1:0]		Reserved						RNGSRC	I2S3SRC	I2S2SRC	Reserved

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SYSPLLSRC[2:0]			SYSPLL_GATE	Reserved	SYSPLLDIV[2:0]			I2SPLLSRC	LTDCSRC	LTDCDIV[5:0]					

Bit	Name	Access	Description	Reset value
31	UTMI2ON	RW	UTMI2 clock gating enable: 1: Enable; 0: Disable.	0
30	UTMI1ON	RW	UTMI1 clock gating enable: 1: Enable; 0: Disable.	0
[29:28]	USBHSPLLCLK[1:0]	RW	USBHSPLL reference clock frequency selection: 00: 25MHz; 01: 20MHz; 10: 24MHz; 11: 32MHz; Writing is only allowed when USBHS_PLLON is 0.	01b
[27:26]	USBHSPLLSRC[1:0]	RW	USBHSPLL reference source selection: 00: HSE;	0

			01: HSI; 10: ETHCLK_20M. Writing is only allowed when USBHS_PLLON is 0.	
[25:20]	Reserved	RO	Reserved	0
19	RNGSRC	RW	RNG clock source selection: 1: PLL clock used after selection by I2SPLLSRC bit; 0: System clock (SYSCLK).	0
18	I2S3SRC	RW	I2S3 clock source selection: 1: PLL clock used after selection by I2SPLLSRC bit; 0: System clock (SYSCLK).	0
17	I2S2SRC	RW	I2S2 clock source selection: 1: PLL clock used after selection by I2SPLLSRC bit; 0: System clock (SYSCLK).	0
16	Reserved	RO	Reserved	0
[15:13]	SYSPLLSRC[2:0]	RW	The system clock switches to PLL selection, and writing is only allowed when SYSPLL_GATE is 0: 000: USBHS_PLL(480MHz); 001: ETH_PLL(500MHz); 010: USBHS_PLL*2/3(320MHz); 011: ETH_PLL*2/3(333MHz); 110: PLL_CLK; Others: Reserved.	0
12	SYSPLL_GATE	RW	This bit needs to be set before the system clock switches to the PLL.	0
11	Reserved	RO	Reserved	0
[10:8]	SYSPLLDIV[2:0]	RW	System PLL frequency division configuration: 000: No frequency division; 001: 2 frequency divider; …… 111: 8 frequency divider.	0
7	I2SPLLSRC	RW	I2S/RNG PLL source selection: 1: ETH_PLL; 0: USBHS_PLL.	0
6	LTDCSRC	RW	LTDC clock source selection: 1: ETH_PLL clock; 0: USBHS_PLL clock.	0
[5:0]	LTDCDIV[5:0]	RW	LTDC clock source prescaler control: 000000: No frequency division; 000001: 2 frequency divider; 000010: 3 frequency divider; 000011: 4 frequency divider; 000100: 5 frequency divider; …… 111110: 63 frequency divider; 111111: 64 frequency divider.	0

3.4.13 Clock Configuration Register 3 (RCC_CFGR3)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											ADC_CLKDIV[4:0]				

Bit	Name	Access	Description	Reset value
[31:5]	Reserved	RO	Reserved	0
[4:0]	ADC_CLKDIV [4:0]	RW	When the ADC clock source is selected as USBHS_PLL, the clock frequency division configuration is: 00000: No frequency division; 00001: 2 frequency divider; 00010: 3 frequency divider; …… 11111: 32 frequency divider;	00000b

Chapter 4 Backup Register (BKP)

Backup Register (BKP) provides up to 42×16-bit backup data registers, which can be used to store up to 84 bytes of user data. After the main power supply (V_{DD}) switched off, these data can still be maintained by the power supply of V_{BAT} without being affected by the standby status, system reset or power reset. In addition, the BKP unit also provides tamper detection management, RTC clock calibration and pulse output functions.

4.1 Main Features

- Provide 84 bytes of backing data register
- RTC clock calibration function
- Output RTC clock divided by 64, alarm pulse or second pulse on PC13 pin
- Tamper detection (TAMPER) function

4.2 Function Description

After the microcontroller reset, the access to the backup register and RTC will be disabled, and the access to the backup register needs to be enabled by the following operations:
1) Set the PWREN bit and BKPEN bit in the RCC_PB1PCENR register to enable the power supply and the operating clock of the backup interface;
2) Set the DBP bit in the power control register (PWR_CTLR) to enable access to the backup register and RTC register.

4.2.1 Backup Data Register

The backup data register can be used as a general data buffer. Because of its feature of saving data by V_{BAT} power when V_{DD} is powered off, it can be used to store some important or sensitive data. But these data will be completely cleared after the tamper event occurs.

4.2.2 Tamper Detection

When an external signal (Rising or falling edge) is provided, a tamper event is generated, and the important information retained in the current system is cleared automatically by hardware. Tamper detection increases the security of system information.

A tamper event is generated when a transition edge is detected on the tamper detection pin (Depending on the TPAL bit). If the tamper detection interrupt is enabled, a tamper detection interrupt is also generated at the same time. As long as a tamper event occurs, all backup data registers are cleared. In addition, the hardware detection adopts the memory mode. Even if the tamper detection function is not enabled (TPE=0), the system still samples and checks whether there is a transition edge, and if the TPAL bit selection is met, the tamper event is locked in advance, and the TPE bit is set to 1 to trigger a tamper event.

For example: When TPAL=0, if it is not enabled (TPE=0) but the TAMPER pin is already high, an additional tamper event is generated (The system locks the rising edge in advance) once TPE=1. When TPAL=1, if it is not enabled (TPE=0) but the TAMPER pin is already low, an additional tamper event is generated (The system locks the falling edge in advance) once TPE=1.

Therefore, in order to avoid unnecessary tamper events from causing the backup registers to be cleared, it is recommended to set the CTE bit in the BKP_TPCSR register to 1 at the beginning of the hardware detection

of the tamper pin, to firstly clear the tamper events that may have been remembered by hardware, and ensure that the current tamper detection pin is invalid.

Note: When the V_{DD} power supply is disconnected, the tamper detection function is still valid. In order to avoid unnecessary backup registers of resetting data, the TAMPER pin shall be connected to the correct level correctly.

4.2.3 RTC Calibration

For this function, the tamper detection pin must be selected as a common IO port. Clear the TPE bit in the BKP_TPCTLR register.

- Pulse output

Configure the ASOE bit in the BKP_OCTLR register, enable RTC pulse output, set the ASOS bit, and select either the second pulse output or the alarm pulse output.

- RTC calibration

After the CCO bit in the BKP_OCTLR register is configured, the internal RTC clock divided by 64 is output to the tamper detection pin (TAMPER). Through the actual test, adjust the clock and calibrate the RTC by configuring the CAL[6:0] bits by software.

4.2.4 BKP Interface Reset

The BKP domain can be independently powered-on by V_{BAT} when the V_{DD} power is switched off. The application code controls the reset of BKP domain register, the BKP_DATAR1-10, ASOS bit and ASOE bit in the backup data register are reset by setting the BDRST bit in the RCC_BDCTLR register by software, which is not affected by the RCC peripheral interface control BKPRST bit.

4.3 Register Description

Table 4-1 BKP related registers

Name	Access address	Description	Reset value
R16_BKP_DATAR1	0x40006C04	Backup data register 1	0x0000
R16_BKP_DATAR2	0x40006C08	Backup data register 2	0x0000
R16_BKP_DATAR3	0x40006C0C	Backup data register 3	0x0000
R16_BKP_DATAR4	0x40006C10	Backup data register 4	0x0000
R16_BKP_DATAR5	0x40006C14	Backup data register 5	0x0000
R16_BKP_DATAR6	0x40006C18	Backup data register 6	0x0000
R16_BKP_DATAR7	0x40006C1C	Backup data register 7	0x0000
R16_BKP_DATAR8	0x40006C20	Backup data register 8	0x0000
R16_BKP_DATAR9	0x40006C24	Backup data register 9	0x0000
R16_BKP_DATAR10	0x40006C28	Backup data register 10	0x0000
R16_BKP_OCTLR	0x40006C2C	RTC calibration register	0x0000
R16_BKP_TPCTLR	0x40006C30	Tamper detection control register	0x0000
R16_BKP_TPCSR	0x40006C34	Tamper detection status register	0x0000
R16_BKP_DATAR11	0x40006C40	Backup data register 11	0x0000
R16_BKP_DATAR12	0x40006C44	Backup data register 12	0x0000
R16_BKP_DATAR13	0x40006C48	Backup data register 13	0x0000
R16_BKP_DATAR14	0x40006C4C	Backup data register 14	0x0000

R16_BKP_DATAR15	0x40006C50	Backup data register 15	0x0000
R16_BKP_DATAR16	0x40006C54	Backup data register 16	0x0000
R16_BKP_DATAR17	0x40006C58	Backup data register 17	0x0000
R16_BKP_DATAR18	0x40006C5C	Backup data register 18	0x0000
R16_BKP_DATAR19	0x40006C60	Backup data register 19	0x0000
R16_BKP_DATAR20	0x40006C64	Backup data register 20	0x0000
R16_BKP_DATAR21	0x40006C68	Backup data register 21	0x0000
R16_BKP_DATAR22	0x40006C6C	Backup data register 22	0x0000
R16_BKP_DATAR23	0x40006C70	Backup data register 23	0x0000
R16_BKP_DATAR24	0x40006C74	Backup data register 24	0x0000
R16_BKP_DATAR25	0x40006C78	Backup data register 25	0x0000
R16_BKP_DATAR26	0x40006C7C	Backup data register 26	0x0000
R16_BKP_DATAR27	0x40006C80	Backup data register 27	0x0000
R16_BKP_DATAR28	0x40006C84	Backup data register 28	0x0000
R16_BKP_DATAR29	0x40006C88	Backup data register 29	0x0000
R16_BKP_DATAR30	0x40006C8C	Backup data register 30	0x0000
R16_BKP_DATAR31	0x40006C90	Backup data register 31	0x0000
R16_BKP_DATAR32	0x40006C94	Backup data register 32	0x0000
R16_BKP_DATAR33	0x40006C98	Backup data register 33	0x0000
R16_BKP_DATAR34	0x40006C9C	Backup data register 34	0x0000
R16_BKP_DATAR35	0x40006CA0	Backup data register 35	0x0000
R16_BKP_DATAR36	0x40006CA4	Backup data register 36	0x0000
R16_BKP_DATAR37	0x40006CA8	Backup data register 37	0x0000
R16_BKP_DATAR38	0x40006CAC	Backup data register 38	0x0000
R16_BKP_DATAR39	0x40006CB0	Backup data register 39	0x0000
R16_BKP_DATAR40	0x40006CB4	Backup data register 40	0x0000
R16_BKP_DATAR41	0x40006CB8	Backup data register 41	0x0000
R16_BKP_DATAR42	0x40006CBC	Backup data register 42	0x0000

4.3.1 Backup Data Register (BKP_DATARx) (x=1-42)

Offset address: 0x04~0x28, 0x40~0xBC

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
D[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	D[15:0]	RW	Backup data, which can be written with user data. *Note: They are only reset by backup domain reset (BDRST) or by a TAMPER pin event (If the TAMPER pin function is activated).*	0

4.3.2 RTC Calibration Register (BKP_OCTLR)

Offset address: 0x2C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						ASO	ASO	CCO	CAL[6:0]						

	S	E		

Bit	Name	Access	Description	Reset value
[15:10]	Reserved	RO	Reserved.	0
9	ASOS	RW	TAMPER pin alarm/second output selection: 0: Alarm pulse output selected; 1: Second pulse output selected. *Note: This bit can only be reset by backup domain reset (BDRST).*	0
8	ASOE	RW	TAMPER pin alarm/second output enable: 0: Alarm/second pulse output disabled; 1: Alarm/second pulse output enabled. *Note: This bit can only be reset by backup domain reset (BDRST).*	0
7	CCO	RW	Calibration clock output selection: 0: No calibration clock output; 1: TAMPER pin output RTC clock divided by 64. *Note 1: The tamper detection function must be switched off to enable this function.* *Note2: This bit is cleared when V_{DD} power supply is switched off.*	0
[6:0]	CAL[6:0]	RW	Calibration value register. The value of this register indicates the number of clock pulses that will be skipped every 2^{20} clock pulses. This allows the calibration of the RTC clock. The clock can be slowed down from 0 to 121ppm.	0

4.3.3 Tamper Detection Control Register (BKP_TPCTLR)

Offset address: 0x30

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														TPAL	TPE

Bit	Name	Access	Description	Reset value
[15:2]	Reserved	RO	Reserved.	0
1	TPAL	RW	TAMPER pin active level: 0: High level on the TAMPER pin cause all backup data registers cleared (hardware lock rising edge); 1: Low level on the TAMPER pin cause all backup data registers cleared (hardware lock falling edge);	0
0	TPE	RW	TAMPER pin enable: 0: TAMPER pin used as common IO port; 1: TAMPER pin used for the tamper detection.	0

Note: When the TPAL and TPE bits are cleared at the same time, a false tamper event occurs. It is recommended to change the status of the TPAL bit only when TPE is 0.

4.3.4 Tamper Detection Status Register (BKP_TPCSR)

Offset address: 0x34

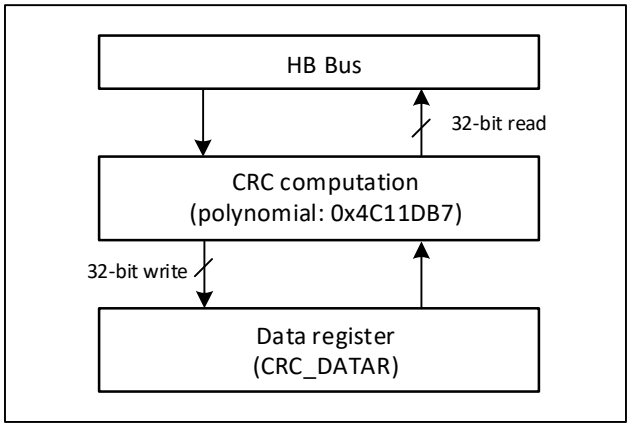
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						TIF	TEF	Reserved					TPIE	CTI	CTE

Bit	Name	Access	Description	Reset value
[15:10]	Reserved	RO	Reserved.	0
9	TIF	RO	Tamper interrupt flag. When a tamper event is detected and the TPIE bit is set to 1, this bit is set. Cleared by writing 1 to the CTI bit. If the TPIE bit is reset, this bit is reset at the same time. *Note: This bit is reset only when the system is reset or woken up from Standby mode.*	0
8	TEF	RO	Tamper event flag. When a tamper event is detected, this bit is set. Cleared by writing 1 to the CTE bit. *Note: When this bit is 1, all BKP_DATARx registers are cleared, and all write operations to the BKP_DATARx register are invalid before this bit is not reset.*	0
[7:3]	Reserved	RO	Reserved.	0
2	TPIE	RW	Tamper interrupt enable: 0: Tamper detection interrupt disabled; 1: Tamper detection interrupt enabled (TPE needs to be set to 1). *Note 1: The tamper detection interrupt cannot wake up the core from low-power mode.* *Note 2: This bit is reset only when the system is reset or woken up from standby mode.*	0
1	CTI	WO	Clear Tamper detection interrupt. Write 1 to clear it, and the value read out is invalid.	0
0	CTE	WO	Clear Tamper detection event. Write 1 to clear it, and the value read out is invalid.	0

Chapter 5 Cyclic Redundancy Check (CRC)

The cyclic redundancy check (CRC) computation unit is used to obtain the result of the CRC computation for any 32-bit data based on a fixed generating polynomial. It is generally used in the field of data storage and data communication to verify the correctness of data. The system provides hardware CRC calculation unit which can greatly save CPU and RAM resources to improve efficiency.

Figure 5-1 CRC structure diagram



5.1 Main Features

- CRC32 polynomial (0x4C11DB7): $X^{32}+X^{26}+X^{23}+X^{22}+X^{16}+X^{12}+X^{11}+X^{10}+X^{8}+X^{7}+X^{5}+X^{4}+X^{2}+X+1$;
- Same 32-bit register as input for data and output for CRC32 calculation
- Single conversion time: 4 HB clock cycles (HCLK)

5.2 Function Description

- CRC unit reset

To start a CRC calculation for a new data set, the CRC calculation unit needs to be reset. Writing a 1 to the RST bit of the control register CRC_CTLR will reset the data register by the hardware, restoring the initial value 0xFFFFFFFF.

- CRC calculation

The CRC unit calculates the CRC result of the previous CRC calculation and the CRC result of the newly involved data. The CRC_DATAR data register, for which a write operation will feed new data to the hardware calculation unit; and a read operation will get the value of the latest round of CRC calculation. The hardware calculation interrupts the system write operation, so new values can be written continuously.

Note: The CRC unit calculates the entire 32-bit data, not byte-by-byte.

- Independent data buffer

The CRC unit provides an 8-bit independent data register, CRC_IDATAR, which is used for the application code to temporarily store 1 byte of data independent of the CRC unit reset.

5.3 Register Description

Table 5-1 CRC-related registers list

Name	Access address	Description	Reset value
R32_CRC_DATAR	0x40023000	Data register	0xFFFFFFFF
R8_CRC_IDATAR	0x40023004	Independent data buffer	0x00
R32_CRC_CTLR	0x40023008	Control register	0x00000000

5.3.1 Data Register (CRC_DATAR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	DR[31:0]	RW	Write raw data; read calculations.	0xFFFFFFFF

5.3.2 Independent Data Buffer (CRC_IDATAR)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								IDR[7:0]							

Bit	Name	Access	Description	Reset value
[7:0]	IDR[7:0]	RW	An 8-bit general-purpose register that can be used as a data cache, this register is not affected by the RST field of the control register.	0

5.3.3 Control Register (CRC_CTLR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															RST

Bit	Name	Access	Description	Reset value
[31:1]	Reserved	RO	Reserved	0
0	RST	WO	CRC calculation unit reset control, write 1 execution, hardware auto clear, after execution, data register is 0xFFFFFFFF.	0

Chapter 6 Real Time Clock (RTC)

The Real Time Clock (RTC) is a standalone timer module with a programmable counter up to 32 bits, which can be used with software to realize the real time clock function, and the counter value can be modified to reconfigure the current time and date of the system. The RTC module is in the backup power supply area, and the system reset and wake-up from Standby mode do not have any effect on it.

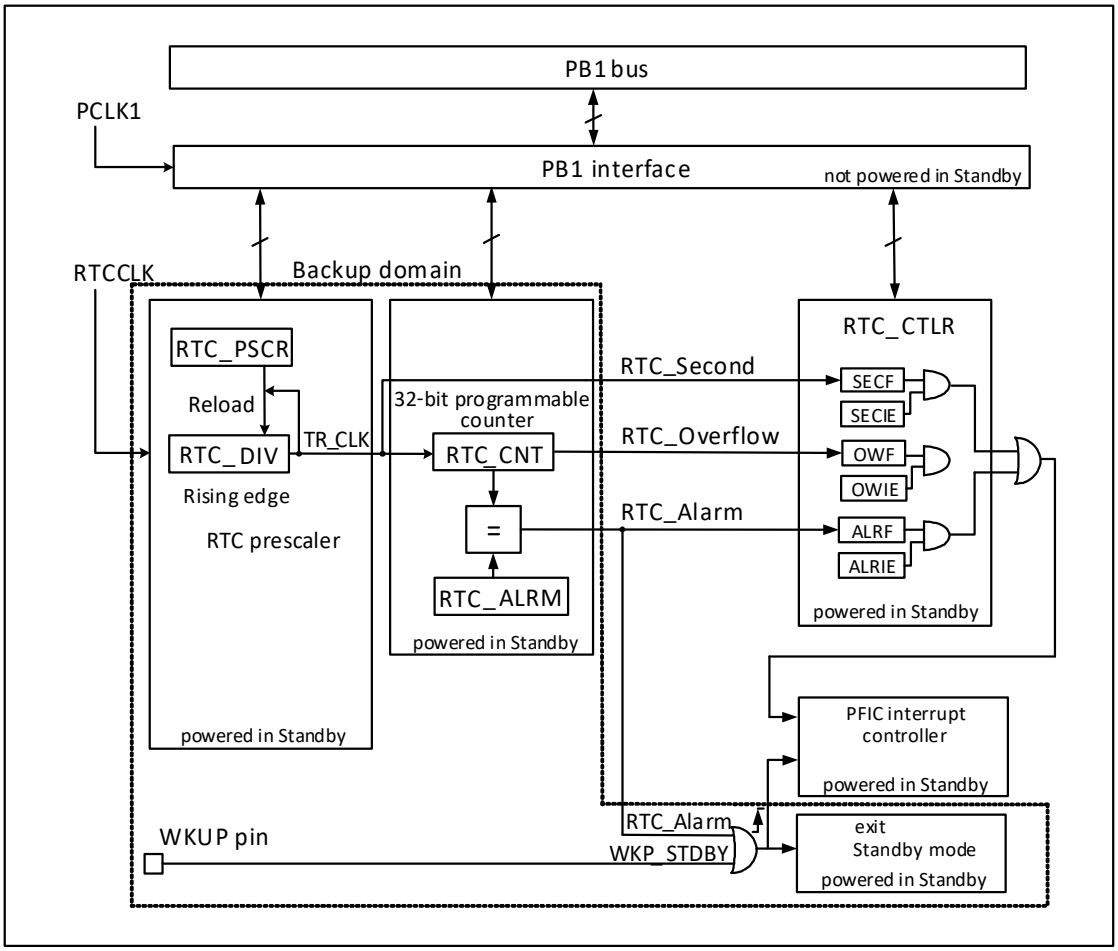
6.1 Main Features

- Prescaler coefficients up to 2^{20}
- 32-bit programmable counter
- Multiple clock sources, interrupts
- Independent reset

6.2 Function Description

6.2.1 Overview

Figure 6-1 RTC structure diagram



As shown in figure 6-1, the RTC module is mainly composed of 3 parts: PB1 bus interface, frequency divider and counter, control and status register, in which the frequency divider and counter are in the backup area and can be powered by V_{BAT}. After inputting the frequency divider (RTC_DIV), the RTCCK is divided into TR_CLK. It is worth noting that the inside of the frequency divider (RTC_DIV) is a self-subtractive counter,

which will output a TR_CLK from the overrun, then take the default value from the reload value register (RTC_PSCR) and reinstall it into the frequency divider, the read divider actually reads its real-time value (read only), and the write division factor should be written to the reload value register (RTC_PSCR). Generally, when the period of TR_CLK is set to 1 second, TR_CLK will trigger the second event and increase the main counter (RTC_CNT) by 1; when the main counter is increased to the same value as the alarm clock register, the alarm clock event will be triggered; when the master counter increases to overflow, the overflow event will be triggered. All of the above three events can trigger interrupts and correspond to the corresponding interrupts to enable level control.

6.2.2 Reset

Due to the special purpose of the real-time clock, the four sets of registers that are in the backup domain: the prescaler, prescaler reload value, master counter, and alarm clock, can only be reset by the reset signal in the backup domain, refer to the RCC's Backup Domain Reset chapter. The control registers of the real-time clock are controlled by a system reset or a power reset.

6.2.3 Special Read-Write Register Operation

Due to the special use of the real-time clock, RTC and PB1 buses are independent, and the reading of RTC by PB1 is not necessarily real-time. Reading the register of RTC through PB1 must be after PB1 startup and passing through a RTC rising edge. This situation may occur after system reset and power reset, after waking up from Standby or Stop mode. It is convenient to wait for the RSF bit of the control register (CTLR) to be set high. The write operation for RTC must wait for the end of the last write operation and must enter configuration mode. The specific steps are as follows:

1) Query the RTOFF bit until it becomes 1.
2) Set the CNF bit and enter the configuration mode.
3) Write to one or more RTC registers.
4) Clear the CNF bit, exit the configuration mode, and the PB1 interface starts to write the RTC register.
5) Query the RTOFF bit until it becomes 1.

6.3 Register Description

Table 6-1 RTC-related registers list

Name	Access address	Description	Reset value
R16_RTC_CTLRH	0x40002800	RTC control register high	0x0000
R16_RTC_CTLRL	0x40002804	RTC control register low	0x0020
R16_RTC_PSCRH	0x40002808	Prescaler reload register high	0x000X
R16_RTC_PSCRL	0x4000280C	Prescaler reload register low	0xXXXX
R16_RTC_DIVH	0x40002810	Divider register high	0x000X
R16_RTC_DIVL	0x40002814	Divider register low	0xXXXX
R16_RTC_CNTH	0x40002818	RTC counter high	0xXXXX
R16_RTC_CNTL	0x4000281C	RTC counter low	0xXXXX
R16_RTC_ALRMH	0x40002820	Alarm clock register high	0xXXXX
R16_RTC_ALRML	0x40002824	Alarm clock register low	0xXXXX

6.3.1 RTC Control Register High (RTC_CTLRH)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													OWIE	ALRIE	SECIE

Bit	Name	Access	Description	Reset value
[15:3]	Reserved	RO	Reserved	0
2	OWIE	RW	Overflow interrupt enable: 1: Enable; 0: Disable.	0
1	ALRIE	RW	Alarm interrupt enable: 1: Enable; 0: Disable.	0
0	SECIE	RW	Second interrupt enable: 1: Enable; 0: Disable.	0

6.3.2 RTC Control Register Low (RTC_CTLRL)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RTOFF	CNF	RSF	OWF	ALRF	SECF

Bit	Name	Access	Description	Reset value
[15:6]	Reserved	RO	Reserved	0
5	RTOFF	RO	RTC operation status indication bit, indicates the execution status of the last operation to the RTC, the operation to the RTC must wait for this bit to be 1. 1: The last operation on the RTC has been completed; 0: The last operation on the RTC is still in progress.	1
4	CNF	RW	Configuration flag bit, writing this bit by 1 enters the configuration mode, thereby allowing values to be written to the Counter (R16_RTC_CNTx), the Alarm Clock Register (R16_RTC_ALRMx) and the Prescaler Reload Value Register (R16_RTC_PSCRx). The write operation is performed only after the bit is written 1 and re-cleared 0 by the software: 1: Enter configuration mode; 0: Exit configuration mode and start updating the RTC registers.	0
3	RSF	RW0	Register synchronization flag bit, before reading and writing registers such as prescaler (PSCRx), alarm clock (ALRMx) and counter (CNTx) of RTC module, make sure that this bit has been set by hardware to make sure that these registers have been	0

			synchronized; when reading and writing these registers, or after PB1 reset or PB1 clock stops, the first step should reset this bit. 1: The register is synchronized; 0: The register is not synchronized.	
2	OWF	RW0	Counter overflow flag, this bit is set by hardware when the 32-bit counter overflows. An overflow interrupt is also generated if the OWIE bit is set. This bit can only be cleared by software and cannot be set by software.	0
1	ALRF	RW0	Alarm Clock Flag, this bit is set by hardware when the counter value reaches the value of the Alarm Clock Register (ALRMx), and an alarm clock interrupt is also generated if the Alarm Clock Interrupt Enable Bit (ALRIE) is set. This bit can only be cleared by software and cannot be set by software.	0
0	SECF	RW0	The second event flag, when the clock is divided by the prescaler after each falling edge, will cause the counter to increment by one, and at the same time generate a second event, this bit will be set, if the second interrupt is enabled (SECIE is set), and at the same time will also generate a second interrupt. This bit can only be cleared by software and cannot be set by software.	0

6.3.3 Prescaler Reload Register High (RTC_PSCRH)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												PRL[19:16]			

Bit	Name	Access	Description	Reset value
[15:4]	Reserved	RO	Reserved	0
[3:0]	PRL[19:16]	WO	Reload value high	x

6.3.4 Prescaler Reload Register Low (RTC_PSCRL)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRL[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	PRL[15:0]	WO	The reloading value is low. The actual frequency division factor is (PRL[19:0] + 1). If the RTC input frequency is 32768Hz, then this value is set to 0x7fff, you can divide the signal with a period	X

			of 1 second.	

6.3.5 Divider Register High (RTC_DIVH)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												DIV[19:16]			

Bit	Name	Access	Description	Reset value
[15:4]	Reserved	RO	Reserved	0
[3:0]	DIV[19:16]	RO	Divider register high.	x

6.3.6 Divider Register Low (RTC_DIVL)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIV[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	DIV[15:0]	RO	Divider register low. DIV is actually a self-subtractive counter. Every time a clock comes to RTC_CLK, the DIV counter is subtracted by 1. After the overflow, a TR_CLK is output and the value is reloaded from the PSCR. DIV can only read and read the remaining value of the counter of the current frequency divider.	X

6.3.7 RTC Counter High (RTC_CNTH)

Offset address: 0x18

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[31:16]															

Bit	Name	Access	Description	Reset value
[15:0]	CNT[31:16]	RW	Counter High	X

6.3.8 RTC Counter Low (RTC_CNTL)

Offset address: 0x1C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CNT[15:0]	RW	Counter low, the core device of the RTC timer, the clock is provided by the TRCLK (the period is generally set to 1 second). Calculate the current	X

			time by reading CNT [31:0]. To write this value, you need to enter configuration mode.	

6.3.9 Alarm Register High (RTC_ALRMH)

Offset address: 0x20

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ALR[31:16]															

Bit	Name	Access	Description	Reset value
[15:0]	ALR[31:16]	WO	Alarm register high	X

6.3.10 Alarm Register Low (RTC_ALRML)

Offset address: 0x24

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ALR[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	ALR[15:0]	WO	Alarm clock register is low. An alarm clock event occurs when the value of the alarm clock register ALRM[31:0] is the same as that of the counter CNT[31:0]. Changing this value requires you to enter configuration mode.	X

Chapter 7 Independent Watchdog (IWDG)

The system is equipped with an independent watchdog (IWDG) to detect software failures caused by logic errors and external environment interference. The IWDG clock source comes from LSI and can be run independently of the main program, so it is suitable for situations with low precision requirements.

7.1 Main Features

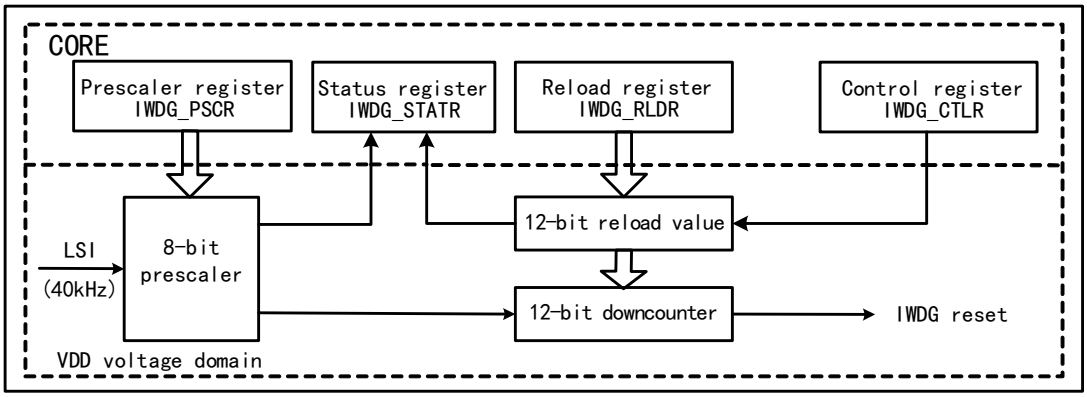
- 12-bit self-subtractive counter.
- Clock source LSI frequency division, can run in low-power mode.
- Reset condition: counter value reduced to 0

7.2 Functional Description

7.2.1 Principle and Application

The independent watchdog is clocked from the LSI clock and functions in stop and standby modes. When the watchdog counter decrements itself to 0, a system reset will be generated, so the timeout is (reload value + 1) clock.

Figure 7-1 Structure block diagram of Independent Watchdog



- Enable independent watchdog

After the system reset, the watchdog is OFF. Write 0xCCCC to the IWDG_CTLR register to enable the watchdog, and then it can no longer be disabled unless a reset occurs.

If the hardware independent watchdog enable bit (IWDGSW) is enabled in User Option Bytes, the IWDG is permanently enabled after the microcontroller reset.

- Watchdog configuration

Inside the watchdog is a 12-bit counter running progressively. When the value of the counter is reduced to 0, a system reset will occur. To enable the IWDG function, you need to perform the following actions:

1) Count time base: IWDG clock source LSI, through the IWDG_PSCR register to set the LSI frequency division value clock as the IWDG count time base. The operation method first writes 0x5555 to the IWDG_CTLR register, and then modifies the frequency division value in the IWDG_PSCR register. The PVU

bit in the IWDG_STATR register indicates the update status of the frequency division value, and the frequency division value can only be modified and read out when the update is completed.

2) Reload value: used to update the current value of the counter in the independent watchdog, and the counter is decremented by this value. The operation method first writes 0x5555 to the IWDG_CTLR register, and then modifies the IWDG_RLDR register to set the target reload value. The RVU bit in the IWDG_STATR register indicates the update status of the reload value, and the IWDG_RLDR register can be modified and read out only after the update is completed.

3) Watchdog enable: write 0xCCCC to the IWDG_CTLR register to turn on the watchdog function.

4) Feeding the dog: that is, before the watchdog counter decreases to 0, refresh the current counter value to prevent system reset. Write 0xAAAA to the IWDG_CTLR register and have the hardware update the IWDG_RLDR register value to the watchdog counter. This action needs to be performed regularly after the watchdog function is turned on, otherwise the watchdog reset action will occur.

7.2.2 Debug Mode

When the system enters the debug mode, the IWDG counter can be configured by the debug module register to continue working or stop.

7.3 Register Description

Table 7-1 IWDG-related registers

Name	Access address	Description	Reset value
R16_IWDG_CTLR	0x40003000	IWDG Control register	0x0000
R16_IWDG_PSCR	0x40003004	IWDG Prescaler register	0x0000
R16_IWDG_RLDR	0x40003008	IWDG Reload Value register	0x0FFF
R16_IWDG_STATR	0x4000300C	IWDG Status register	0x0000

7.3.1 IWDG Control Register (IWDG_CTLR)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KEY[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	KEY[15:0]	WO	Operate the key value lock. 0xAAAA: Feed the dog. Load the IWDG_RLDR register value into the independent watchdog counter. 0x5555: Allow modification of R16_IWDG_PSCR and R16_IWDG_RLDR registers. 0xCCCC: Start the watchdog, which is not subject to this restriction if the hardware watchdog is enabled (user chooses word configuration).	0

7.3.2 IWDG Prescaler Register (IWDG_PSCR)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													PR[2:0]		

Bit	Name	Access	Description	Reset value
[15:3]	Reserved	RO	Reserved	0
[2:0]	PR[2:0]	RW	IWDG clock division factor, write 0x5555 to KEY before modifying this field. 000: Divided by 4; 001: Divided by 8; 010: Divided by 16; 011: Divided by 32; 100: Divided by 64; 101: Divided by 128; 110: Divided by 256; 111: Reserved. IWDG count time base = LSI/division factor. *Note: Before reading the value of this field, make sure that the PVU bit in the IWDG_STATR register is 0, otherwise the read value is invalid.*	000b

7.3.3 IWDG Reload Value Register (IWDG_RLDR)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				RL[11:0]											

Bit	Name	Access	Description	Reset value
[15:12]	Reserved	RO	Reserved	0
[11:0]	RL[11:0]	RW	Counter reload value. Write 0x5555 to KEY before modifying this field. When 0xAAAA is written to KEY, the value of this field is loaded into the counter by the hardware, and the counter is then decremented from that value. *Note: Before reading and writing the field value, make sure that the RVU bit in the IWDG_STATR register is 0, otherwise it is invalid to read and write this field.*	0xFFF

Note: This register will be reset in standby mode.

7.3.4 IWDG Status Register (IWDG_STATR)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														RVU	PVU

Bit	Name	Access	Description	Reset value

[15:2]	Reserved	RO	Reserved	0
1	RVU	RO	The reinstall value updates the flag bit. Hardware setting or clearing 0. 1: Reload value update is in progress. 0: Reload update ends (up to 5 LSI cycles). *Note: The reload value register IWDG_RLDR can be read and written only after the RVU bit has been cleared.*	0
0	PVU	RO	Clock frequency division coefficient updates flag bits. Hardware setting or clearing 0. 1: Clock division value update is in progress. 0: Clock division value update ends (up to 5 LSI cycles). *Note: The frequency division factor register IWDG_PSCR can be read and written only after the PVU bit has been cleared.*	0

Note: After the pre-division or reinstallation value is updated, you do not have to wait for RVU or PVU to reset, you can continue to execute the following code. (This write operation continues to be completed even in low power mode.)

Chapter 8 Window Watchdog (WWDG)

The window watchdog is generally used to monitor the software failures of the system, such as external interference, unforeseen logic errors and so on. It needs to refresh the counter (feed the dog) within a specific window time (with upper and lower limits), otherwise the watchdog circuit will produce a system reset earlier or later than this window time.

8.1 Main Features

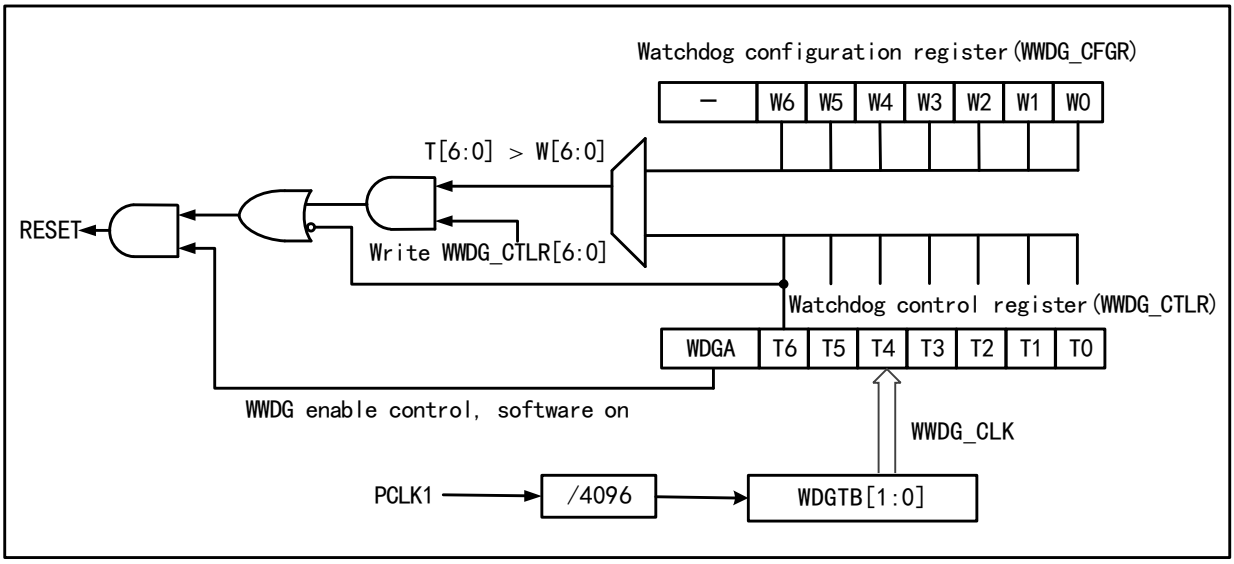
- Programmable 7-bit self-subtractive counter.
- Double conditional reset: the current counter value is less than 0x40, or the counter value is reloaded outside the window time.
- Early wakeup interrupt (EWI), which is used to feed the dog in time to prevent system reset

8.2 Function Description

8.2.1 Principle and Application

The window watchdog runs based on a 7-bit decrement counter, which is mounted on the PB1 bus to count the frequency division of the time-based WWDG_CLK source (PCLK1/4096) clock, and the frequency division factor is set in the WDGTB[1:0] field in the configuration register WWDG_CFGR. The decrement counter is in a state of free operation, regardless of whether the watchdog function is turned on or not, the counter has been cyclically decreasing counting. As shown in figure 8-1, the internal structure block diagram of the window watchdog.

Figure 8-1 Window Watchdog structure diagram



- Enable window watchdog

After the system reset, the watchdog is disabled. Set the WDGA bit in the WWDG_CTLR register to switch on the watchdog, and then it can no longer be disabled unless a reset occurs.

Note: WWDG clock source can be disabled by setting the RCC_PB1PCENR register to suspend WWDG_CLK counting and indirectly stop the watchdog function. Or reset the WWDG module by setting the RCC_PB1PRSTR register, which is equivalent to the function of reset.

- Watchdog configuration

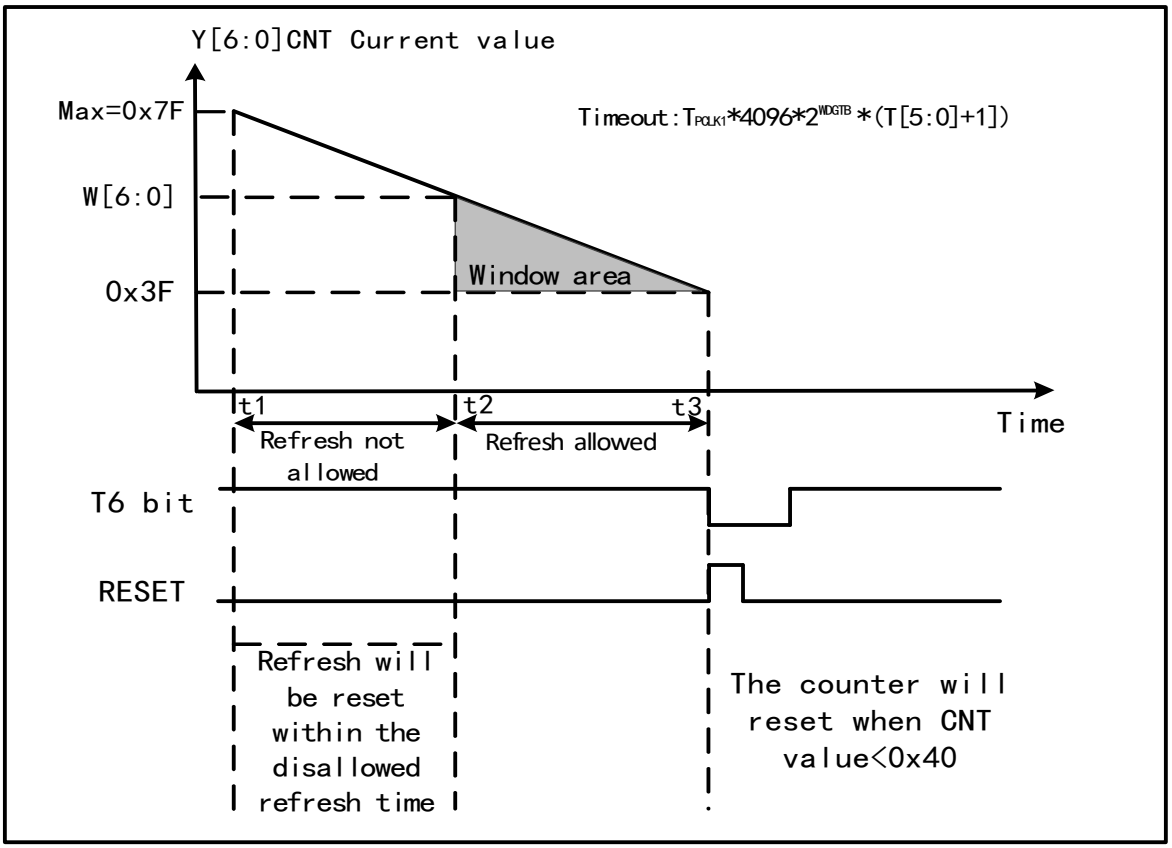
Inside the watchdog is a 7-bit counter running in a continuous cycle, which supports read and write access. To use the watchdog reset function, you need to perform the following actions:

1) Counter time base: The WDGTB[1:0] bits in the WWDG_CFGR register. Note to switch on the WWDG module clock of the RCC unit.
2) Window counter: Set the W[6:0] bits in the WWDG_CFGR register. This counter is used to be compared with the current counter by hardware, the value is configured by the user software and will not change. It serves as the maximum value of window time.
3) Watchdog enable: The WDGA bit in the WWDG_CTLR register is set to 1 by software, and the watchdog function is enabled to reset the system.
4) Feed dog: Refresh the current counter value and configure the T[6:0] bits in the WWDG_CTLR register. This action needs to be executed in the periodic window time after the watchdog function is enabled. Otherwise, the watchdog reset action occurs.

- Feed dog window time

As shown in Figure 8-2, the gray area is the detector window area of the window watchdog. Its maximum timeout (t_2) corresponds to the time point when the current counter value reaches the window value W[6:0]. Its minimum timeout (t_3) corresponds to the time point when the current counter value reaches 0x3F. Within this area time ($t_2<t<t_3$), the feed dog operation can be performed (write T[6:0]) to refresh the current counter value.

Figure 8-2 Counting mode of window watchdog



- Watchdog reset:

1) When the feed dog operation is not performed in time, the value of the T[6:0] counter changes from 0x40

to 0x3F, a "Window Watchdog Reset" occurs, and a system reset occurs. I.e., when T6-bit is detected as 0 by hardware, the system reset occurs.

Note: The application program can write 0 to the T6-bit by software to implement system reset, which is equivalent to software reset function.

2) When the counter refresh action is executed when the feed dog operation is disabled, i.e., when write operation is performed on the T[6:0] bits when $t_1 \leq t \leq t_2$, a "Window Watchdog Reset" occurs, and a system reset occurs.

- Early wake-up

To prevent the system from resetting due to failure to refresh the counter in time, the watchdog module provides early wake-up interrupt (EWI) notification. When the counter is reduced to 0x40, an early wake-up signal is generated, and the EWIF flag is set to 1. If the EWI bit is set, the window watchdog will be triggered to interrupt at the same time. At this point, there is a counter clock cycle (self-reduced to 0x3F) from the hardware reset, during which the application can immediately feed the dog.

8.2.2 Debug Mode

When the system enters debug mode, the WWDG counter can either continues to work normally or stops, depending on the debugging module register.

8.3 Register Description

Table 8-1 WWDG-related registers

Name	Access address	Description	Reset value
R16_WWDG_CTLR	0x40002C00	WWDG Control Register	0x007F
R16_WWDG_CFGR	0x40002C04	WWDG Configuration Register	0x007F
R16_WWDG_STATR	0x40002C08	WWDG Status Register	0x0000

8.3.1 WWDG Control Register (WWDG_CTLR)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								WDGA	T[6:0]						

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
7	WDGA	RW1	The window watchdog resets the enable position. 1: Enable the watchdog function (can generate reset signal). 0: Disable the watchdog function. Software write 1 is on, but only hardware is allowed to clear 0 after reset.	0
[6:0]	T[6:0]	RW	7-bit self-subtractive counter, minus 1 per $4096*2^{WDGTB}$ PCLK1 cycle. When the counter is reduced from 0x40 to 0x3F, that is, when T6 jumps	0x7F

			to 0, a watchdog reset is generated.	

8.3.2 WWDG Configuration Register (WWDG_CFGR)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						EWI	WDGTB[1:0]		W[6:0]						

Bit	Name	Access	Description	Reset value
[15:10]	Reserved	RO	Reserved	0
9	EWI	RW1	Early wakeup interrupt: If it set to 1, interrupt is generated when the counter reaches 0x40. It can only be cleared by hardware after reset.	0
[8:7]	WDGTB[1:0]	RW	Window watchdog time base: 00: Divided by 1, counter time base =PCLK1/4096; 01: Divided by 2, counter time base =PCLK1/4096/2; 10: Divided by 4, counter time base = PCLK1/4096/4; 11: Divided by 8, counter time base = PCLK1/4096/8.	0
[6:0]	W[6:0]	RW	Window watchdog 7-bit window value. It is used to be compared with the counter value. The feed dog operation can be performed only when the counter value is less than the window value and is greater than 0x3F.	0x7F

8.3.3 WWDG Status Register (WWDG_STATR)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															EWIF

Bit	Name	Access	Description	Reset value
[15:1]	Reserved	RO	Reserved	0
0	EWIF	RW0	Early wakeup interrupt flag. When the counter reaches 0x40, this bit is set by hardware, and it must be cleared by software. User setting is invalid. Even if the EWI is not set, this bit is still set as usual when the event occurs.	0

Chapter 9 Central Processing Unit (CPU)

CH32V407 uses the QingKe V3V core. RISC-V3V supports the IMACV-X instruction set. It adopts a 2-3 level pipeline internally and a sequential execution structure. The core supports RISC-V V1.0 vector instruction set and zvbb subset, and supports EEW bit width up to 64 bits.

The internal management is modular and includes fast programmable interrupt controller (PFIC), memory protection, branch prediction mode, extended instruction support and other units. External multiple sets of buses are connected to external unit modules to realize the interaction between external functional modules and the core.

9.1 Main Features

- Sequential single-issue architecture
- 2~3 stage pipeline
- Provide one non-maskable interrupt (NMI)
- 2-level interrupt nesting
- Programmable Fast Interrupt Controller (PFIC)
- Custom extendable instructions
- Hardware stack
- Vector operations/parallel processing

9.2 Function Description

9.2.1 Instruction Set

The QingKe V3V processor supports vector integer and fixed-point arithmetic instructions in the RISC-VV1.0 instruction set, but does not support vector floating-point instructions. Vector operations use 32 sets of 64-bit wide vector registers. The maximum EEW of vector instructions is 64 bits. The configuration complies with the vector embedded extension Zve64x extension set, and supports vector instruction encoding check and instruction execution exception check.

9.2.1.1 RISC-V V1.0

Vector operations use 32 groups of 64-bit wide vector registers as operation registers, fully supporting the vector embedded extension Zve64x extension set.

Note: 1. The zve64x extended instruction set is all instructions defined in the RVV1.0 manual. For specific instruction functions, please refer to RISC-V 'V' vector Extension1.0;

2. All memory access instructions support eew=64 and iew=64 operations, but the scalar address bit width XLEN=32 bits and does not support 64-bit addressing, which complies with the manual Zve64x specification.

9.2.1.2 RISC-V zvbb

zvbb is an extended bit-based operation subset of the vector instruction set. Full support for Zve64x extension set.

9.2.1.3 Custom Delay Instructions

The custom delay instruction is a precise delay instruction at the expense of a single instruction overhead. It can delay the specified matching instruction to the accuracy of the clock cycle, avoiding the uncertainty of instruction jump execution in timers, delay and other operations. The instruction can automatically match the instruction encoding, configure the delay period, and clock division coefficient. It is flexible to use and has obvious advantages in applications such as precise control, process management, and low-overhead delay.

Bit	Name	Description
[31:20]	imm[11:0]	Delay immediate value, which can represent the number of delayed clock cycles or the number of divided cycles according to the configuration.
[19:15]	rs1[4:0]	Matching rs1 register encoding.
[14:12]	func3[2:0]	Fixed value 001b.
[11:9]	match[2:0]	Matching instruction type. The matching instruction type cannot be executed before the delay is over. 000: Reserved; 001: Matching load instruction; 010: Matching store instruction; 011: Matching load/store instruction; 100: Matching delay instruction; 101: Rerserved; 110: Match all instructions (pipeline paused); 111: Match instructions specifying func3 and opcode (The matching value is configured by the csr register).
8	div	Use the main frequency clock or divided clock to count. 1: Use the frequency division cycle counting (the frequency division coefficient is configured by the CSR register); 0: Use the main frequency clock cycle count.
7	sel	Delay number selection: 1: rs1 + rs2; 0: rs1 + imm.
[6:0]	func7[6:0]	Fixed value 0001011b.

Note: The execution of read and write instructions with the sp register as the base address is not restricted by matching load/store instructions (match=001/010/011).

9.2.1.4 Custom mcpy instructions

The mcpy instruction is used to replace the memcpy function to implement continuous memory transfer functions. The instruction register stores the address information of the start address, target address, and end address. All addresses have no alignment requirements.

Bit	Name	Description
[31:27]	rs3[4:0]	End address register encoding.
[26:25]	Reserved	Fixed value 00b.
[24:20]	rs2[4:0]	Starting address register encoding.
[19:15]	rs1[4:0]	Destination address register encoding.

[14:12]	func3[2:0]	Fixed value 111b.
[11:7]	func5[4:0]	Fixed value 00000b.
[6:0]	func7[6:0]	Fixed value 0001111b.

9.2.2 Vector Privileged Register (CSR)

Table 9-1 CSR related register list

Name	Access address	Description	Reset value
VXSAT	0x009	Fixed-point saturation status register	0x00000000
VXRM	0x00A	Fixed-point rounding register	0x00000000
VCSR	0x00F	Vector control and status registers	0x00000000
VL	0xC20	Element body length register	0x00000000
VTYPE	0xC21	Vector data type register	0x00000000
VLENB	0xC22	Vector register byte length register	0x00000008
VCONTROL	0x805	Custom vector control register	0x00000X0C
VPPADDR	0x806	Custom vector register stack base address register	0x20000000
VCAUSE	0x808	Custom vector exception source register	0x00000000
VTVAL	0x809	Custom vector exception trap information register	0x00000000

9.2.2.1 Fixed-point Saturation Status Register (VXSAT)

CSR address: 0x009

Bit	Name	Access	Description	Reset value
[31:1]	Reserved	URW	Reserved	0
0	VXSAT	URW	Fixed-point saturation status register, set to 1 when the result of the fixed-point operation must be output as a saturated value.	0

9.2.2.2 Fixed-point Rounding Register (VXRM)

CSR address: 0x00A

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	URW	Reserved	0
[1:0]	VXRM	URW	Fixed-point operation rounding register, used to control the rounding mode of fixed-point operations: 00: Round to the nearest value (the intermediate value is rounded up); 01: Round to the nearest value (the intermediate value is rounded to the even number); 10: Round down; 11: Round to the odd number.	0

9.2.2.3 Vector Control and Status Register (VCSR)

CSR address: 0x00F

Bit	Name	Access	Description	Reset value
[31:3]	Reserved	URW	Reserved	0
[2:1]	VXRM	URW	Fixed-point operation rounding register, used to control the rounding mode of fixed-point operations: 00: Round to the nearest value (the intermediate value is rounded up); 01: Round to the nearest value (the intermediate value is rounded to the even number); 10: Round down; 11: Round to the odd number.	0
0	VXSAT	URW	Fixed-point saturation status register, set to 1 when the result of the executed fixed-point operation must be output as a saturated value.	0

9.2.2.4 Element Body Length Register (VL)

CSR address: 0XC20

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	URW	Reserved	0
[6:0]	VL	URO	The vector length register is used to store the length of elements used in operations. The register value can only be modified by the vset instruction. For assignment rules, see the *RISC-V 'V' vector Extension 1.0*.	0

9.2.2.5 Vector Data Type Register (VTYPE)

CSR address: 0XC21

Bit	Name	Access	Description	Reset value
[31:6]	Reserved	URW	Reserved	0
[5:3]	VSEW[2:0]	URO	Vector width register, used to select the width of vectors and elements, only configurable by the VSET instruction: 000: 8bit; 001: 16bit; 010: 32bit; 011: 64bit; 1xx: Reserved. *Note: Configuring eew to 64bit requires enabling VCONTROL related configurations.*	0

[2:0]	VLMUL[2:0]	URO	The vector group allocation register is used to select grouping information in vector operations. Only the VSET instruction can be configured: 000: 1; 001: 2; 010: 4; 011: 8; 100: Reserved; 101: 1/8; 110: 1/4; 111: 1/2.	0

9.2.2.6 Vector Register Byte Length Register (VLENB)

CSR address: 0XC22

Bit	Name	Access	Description	Reset value
[31:0]	VLENB[31:0]	URO	Vector register group byte length register, this core vector register length is 64 bits, byte length is 8, fixed value.	0x8

9.2.2.7 Custom Vector Control Register (VCONTROL)

CSR address: 0x805

Bit	Name	Access	Description	Reset value
31	VEC_ILL	URW1C	Vector instruction exception flag, set when a vector instruction generates an exception. When the NMI_EN register is enabled, the vector exception will enter the NMI exception.	0
[30:24]	Reserved	URO	Reserved	0
[23:22]	NEST_INT_PTR[1:0]	URO	Interrupt nesting depth: 00: No interrupt; 01: 1-level interrupt nesting; 10: 2-level interrupt nesting; 11: 3-level interrupt nesting.	0
[21:20]	VEC_INT_PTR[1:0]	URW	Vector pipeline interrupt nesting depth: 00: No interrupt; 01: 1-level interrupt nesting; 10: 2-level interrupt nesting; 11: 3-level interrupt nesting. When vector instructions are not used in the interrupt service function, the vector pipeline interrupt nesting depth retains the previous value.	0
[19:16]	VEC_REG_STATE[3:0]	URW	The vector register status register has a total of 4 bits corresponding to the maximum 4-	0

			layer vector register status depth. A value of 1 indicates that the vector register status corresponding to the interrupt depth is a dirty state, that is, the vector register value changes due to the execution of vector instructions.	
[15:10]	Reserved	URO	Reserved	0
9	VEC_ILL_INFO	URW	Vector exception information saving is enabled. After being set to 1, the VCAUSE and VTVAL registers will save vector exception information.	1
8	Reserved	URO	Reserved, cannot write 0.	x
[7:4]	Reserved	URO	Reserved	0
3	LSU_WAX_ON	URW	Forced write order is enabled, forcing write operations to be executed in instruction order.	1
2	LSU_RAW_ON	URW	Forced read order is enabled, forcing read operations to be performed in instruction order.	1
1	VEC_HW_PP_ABI	URW	The vector register hardware stack protects the ABI configuration register: 1: Call variant ABI, hardware stack protects V0, V8-V23 registers; 0: Call standard ABI, hardware stack protects V0-V31 registers. *Note: For ABI calling, please refer to 'RISC-VABISSPECIFICATIONV1.1'*	0
0	VEC_HW_PP_EN	URW	Vector register hardware stack protection enable: 1: Turn on the vector stack protection function; 0: Turn off the vector stack protection function.	0

9.2.2.8 Custom Vector Register Stack Base Address Register (VPPADDR)

CSR address: 0x806

Bit	Name	Access	Description	Reset value
[31:0]	VPPADDR[31:0]	URW	The vector register stack protection memory address register is used to configure the vector register stack protection address. The lower 10 bits are fixed to 0 and are aligned to 1KB in the memory.	0x20000000

9.2.2.9 Custom Vector Exception Source Register (VCAUSE)

CSR address: 0x808

Bit	Name	Access	Description	Reset value
[31:0]	VCAUSE[31:0]	URO	Vector instruction exception cause register, when VEC_ILL_INFO is enabled, is used to save the exception cause of vector exceptions. *Note: Debug mode will cause the exception cause information to be lost and cannot be used in debug mode.*	0

9.2.2.10 Custom Vector Exception Source Register (VTVAL)

CSR address: 0x809

Bit	Name	Access	Description	Reset value
[31:0]	VTVAL[31:0]	URO	The vector instruction trap information register is used to save the PC value of the vector exception when VEC_ILL_INFO is enabled. *Note: Debug mode will cause the exception cause information to be lost and cannot be used in debug mode.*	0

9.2.3 General Privilege Register

The RISC-V architecture defines a set of Control and Status Registers (CSRs) used to configure, identify, or record operational states. CSR registers are internal to the core and utilize a dedicated 12-bit address space.

Note: Registers marked with the "MRW, MRO, MRW1" attributes require the system to be in machine mode for access.

Table 9-2 General privilege register list

Name	CSR address	Description	Reset
MVENDORID	0xF11	Chip manufacturer ID register	0x00000000
MARCHID	0xF12	Chip structure ID register	0xDC68D876
MIMPID	0xF13	Machine implementation ID register	0xDC688003
MSTATUS	0x300	Machine status register	0x00000000
MISA	0x301	Machine instruction set register	0x40B01107
MTVEC	0x305	Machine mode exception base address register	0x00000000
MSCRATCH	0x340	Machine mode temporary register	X
MEPC	0x341	Machine abnormal program counter register	X
MCAUSE	0x342	Machine exception register	0xX00000XX
MTVAL	0x343	Machine self-trapping value register	X
TINFO	0x7A4	Trigger information register	0x00000004
DPC	0x7B1	Debugging PC address register	X
DSCRATCH0	0x7B2	Debug mode temporary register 0	X
DSCRATCH1	0x7B3	Debug mode temporary register 1	X

DBGMCU_0	0x7C0	Debug register 0	0x00000000
UACCES_MSTATUS	0x800	User access to machine status register	0x00000000
HW_POPDM_CTLR	0x804	Hardware stack control register	0x00000004
TSELECT	0x7A0	Trigger channel selection register	0x0000000X
TDATA1	0x7A1	Trace trigger data register	0x23E00048
TDATA2	0x7A2	Save trigger data register	X
MCOUNT_INHIBIT	0x320	Counter mask register	0x00000000
MCYCLE	0xB00	Machine cycle counter register	0x00000000
MINSTRET	0xB02	Machine instruction counter register	0x00000000
UCYCLE	0xC00	User mode cycle counter register	0x00000000
UINSTRET	0xC02	User mode instruction counter register	0x00000000

9.2.3.1 Chip Manufacturer ID Register (MVENDORID)

CSR address: 0xF11

Bit	Name	Access	Description	Reset value
[31:0]	mvendorid[31:0]	MRO	Chip manufacturer ID, fixed value is 0.	0

9.2.3.2 Chip Structure ID Register (MARCHID)

CSR address: 0xF12

Bit	Name	Access	Description	Reset value
[31:0]	marchid[31:0]	MRO	Chip structure ID, fixed value cannot be modified.	0xDC68D876

9.2.3.3 Machine Implementation ID Register (MIMPID)

CSR address: 0xF13

Bit	Name	Access	Description	Reset value
[31:0]	mimpid[31:0]	MRO	Machine implementation ID, fixed value cannot be modified.	0xDC688003

9.2.3.4 Machine Status Register (MSTATUS)

CSR address: 0x300

Bit	Name	Access	Description	Reset value
[31:13]	Reserved	MRO	Reserved	0
[12:11]	MPP[1:0]	MRW	Save/restore the machine state before entering abnormality, and update it to the machine state before entering abnormality: 00: Set machine status to U mode upon exit exception; 01: Reserved; 10: Reserved; 11: Set machine status to M mode upon exit exception.	0
[10:9]	VS[1:0]	MRW	Vector instruction set status, vector instructions (V) are allowed to execute	0

			when the value is not 0: 00: OFF; 01: Initial; 10: Clean; 11: Dirty.	
8	Reserved	MRW	Reserved	0
7	MPIE	MRW	Preserve the interrupt enable status upon entering an interrupt: 1: Enable global interrupts after exiting the interrupt; 0: Disable global interrupts after exiting the interrupt.	0
[6:4]	Reserved	MRO	Reserved	0
3	MIE	MRW	Global interrupts enabled; updates to MPIE value upon interrupt exit: 1: Global interrupts enabled, allowing the processor to respond to interrupt requests; 0: Global interrupts disabled, preventing the processor from responding to interrupt requests.	0
[2:0]	Reserved	MRO	Reserved	0

9.2.3.5 Machine Instruction Set Register (MISA)

CSR address: 0x301

Bit	Name	Access	Description	Reset value
[31:30]	XLEN[1:0]	MRO	Machine basic integer bit width and basic instruction bit width information, 32-bit wide core, fixed value is 1.	01b
[29:24]	Reserved	MRO	Reserved	0
23	X	MRO	This core supports the execution of custom instruction sets.	1
[22:21]	Reserved	MRO	Reserved	01b
20	U	MRO	Support user mode operation.	1
[19:13]	Reserved	MRO	Reserved	0
12	M	MRO	This core supports the execution of multiplication instruction sets.	1
[11:9]	Reserved	MRO	Reserved	0
8	I	MRO	This core supports the execution of basic integer instruction sets.	1
[7:6]	Reserved	MRO	Reserved	0
5	F	MRO	This core supports the execution of single-precision floating-point instruction sets.	0
[4:3]	Reserved	MRO	Reserved	0
2	C	MRO	This core supports the execution of compressed instruction sets.	1

1	B	MRO	This core supports the execution of bit operation instruction sets.	1
0	A	MRO	This core supports the execution of atomic instruction sets.	1

9.2.3.6 Machine Mode Exception Base Address Register (MTVEC)

CSR address: 0x305

Bit	Name	Access	Description	Reset value
[31:10]	mtvec_base[21:0]	MRW	The high-order bits of the base address for jump instructions when processor exceptions occur, aligned to 1kB. The actual 32-bit base address BASE = {mtvec_base, 10'h0}.	0
[9:2]	Reserved	MRO	Reserved	0
1	mode1	RW	Vector Table Mode 1: 1: The interrupt vector table stores jump addresses; 0: The interrupt vector table stores jump instructions.	0
0	mode0	RW	Vector Table Mode 0: 1: All interrupts/exceptions have independent entry points, with jump addresses set to BASE + 4 × IRQ/EXC ID; 0: All interrupts/exceptions share a unified entry point, all jumping to the exception jump base address.	0

9.2.3.7 Machine Mode Temporary Register (MSCRATCH)

CSR address: 0x340

Bit	Name	Access	Description	Reset value
[31:0]	mscratch[31:0]	MRW	Machine mode temporary register.	x

9.2.3.8 Machine Exception Program Counter Register (MEPC)

CSR address: 0x341

Bit	Name	Access	Description	Reset value
[31:0]	mepc[31:0]	MRW	Save the PC address recovered when exiting the interrupt/exception, and mepc[0] is fixed to 0.	x

9.2.3.9 Machine Exception Register (MCAUSE)

CSR address: 0x342

Bit	Name	Access	Description	Reset value
31	Interrupt	RW	Interrupt/Exception Flag: 1: Exception/interrupt generated externally;	x

			0: Exception/interrupt generated internally by the core.	
[30:8]	Reserved	RO	Reserved	0
[7:0]	Exception Code[7:0]	RW	Save the source information that caused the exception or interrupt.	x

Interrupt exception source code:

Interrupt/Exception	Exception Code	Interrupt/exception source
1	2	Keep NMI
1	3	Machine mode software interrupt
1	4	User mode timer interrupt
1	5	Monitor mode timer interrupt
1	6	Reserved
1	7	Machine mode timer interrupt
1	8	User mode external interrupt
1	11	Machine mode external interrupt
1	12	System timer interrupt
1	≥16	Reserved platform interrupt id
0	0	Instruction address misalignment
0	1	Instruction access error
0	2	Illegal instruction
0	3	Breakpoint
0	4	Read address misalignment
0	5	Read access error
0	6	Write/atomic operation address misalignment
0	7	Write/atomic operation access error
0	8	User mode ecall
0	11	Machine mode ecall
0	≥16	Reserved

9.2.3.10 Machine Self-trapping Value Register (MTVAL)

CSR address: 0x343

Bit	Name	Access	Description	Reset value
[31:0]	mtval[31:0]	RW	Save additional information about self-trapping.	x

Additional information saved by mtval register in trap:

Abnormal type	Mcause	Save information
Instruction address misalignment	0	Instruction PC
Instruction access failed	1	Instruction PC
Illegal instruction	2	Instruction machine code
Read address misalignment	4	Read data address

Read access failed	5	Read data address
Write/AMO address misaligned	6	Write data address
Write/ AMO access failed	7	Write data address

9.2.3.11 Trigger Information Register (TINFO)

CSR address: 0x7A4

Bit	Name	Access	Description	Reset value
[31:0]	tinfo[31:0]	MRO	Trigger information register, corresponding to the type value in the TDATA1 register. The type value is 2, and the second bit of tinfo is set to 1.	0x4

9.2.3.12 Debug PC Address Register (DPC)

CSR address: 0x7B1

Bit	Name	Access	Description	Reset value
[31:0]	dpc[31:0]	RW	Save the PC address recovered when exiting debug mode.	x

The addresses saved by dpc for jumping to debugging mode caused by different debugging request sources are as follows:

Cause	Save address
ebreak	The instruction address of Ebreak.
One step	The address of the next instruction to be executed when saved in non-debug mode.
Trigger module	tdata1.timing=0: Instruction address that triggers the breakpoint.
Pause request	The address of the next instruction to be executed when entering debug mode.

9.2.3.13 Debug Mode Temporary Register 0 (DSCRATCH0)

CSR address: 0x7B2

Bit	Name	Access	Description	Reset value
[31:0]	Dscratch0[31:0]	RW	Debug mode temporary register 0.	x

9.2.3.14Debug Mode Temporary Register 1 (DSCRATCH1)

CSR address: 0x7B3

Bit	Name	Access	Description	Reset value
[31:0]	Dscratch0[31:0]	RW	Debug mode temporary register 1.	x

9.2.3.15 Debug Register 0 (DBGMCU_0)

CSR address: 0x7C0

Bit	Name	Access	Description	Reset value
[31:8]	dbg_mode_stop	MRW	Registered modules cease operation when entering debug mode:	0

			1: Registered peripherals cease operation when the core enters debug mode; 0: Registered peripherals continue normal operation when the core enters debug mode.	
[7:0]	Reserved	MRO	Reserved	0

9.2.3.16 User Access Machine Status Register (UACCES_MSTATUS)

CSR address: 0x800

Bit	Name	Access	Description	Reset value
31	SD	RO	Used to mark whether a floating-point instruction is in a dirty state.	0
[30:15]	Reserved	RO	Reserved	0
[14:13]	FS[1:0]	RO	Floating-point instruction set status, when the value is not 0, single-precision floating-point instruction (F) is allowed to execute: 00: OFF; 01: Initial; 10: Clean; 11: Dirty.	0
[12:11]	MPP[1:0]	RO	Preserve/restore the machine state prior to entering an abnormal condition; upon exiting the abnormal state, update to the pre-abnormal machine state: 00: Set machine state to U mode upon exiting abnormal state; 01: Reserved 10: Reserved 11: Set machine state to M mode upon exiting abnormal state.	0
[10:8]	Reserved	RO	Reserved	0
7	MPIE	RW	Preserve the interrupt enable status upon entering an interrupt: 1: Enable global interrupts after exiting the interrupt; 0: Disable global interrupts after exiting the interrupt.	0
[6:4]	Reserved	RO	Reserved	0
3	MIE	RW	Global interrupts enabled; updates to MPIE value upon interrupt exit: 1: Global interrupts enabled, allowing the processor to respond to interrupt requests; 0: Global interrupts disabled, preventing the processor from responding to interrupt requests.	0

[2:0]	Reserved	RO	Reserved	0

Note: This register is used by the core to configure the MSTATUS register when running in user mode. This address becomes accessible when the non_usta bit in the CPU_RUN_CTLR register is enabled. The register settings are identical to those of the MSTATUS (0x300) register.

9.2.3.17 Hardware Stack Control Register (HW_POPDM_CTLR)

CSR address: 0x804

Bit	Name	Access	Description	Reset value
31	lock	RW	User Mode Lock Flag When locked, user mode cannot rewrite this register group configuration: 1: Locked; configurable only in machine mode. 0: Unlocked; configurable in machine/user mode.	0
[30:6]	Reserved	RO	Reserved	0
5	hw_pop_off	RW	Disable hardware push upon exit from interrupt, automatically reset: 1: Disable hardware push upon next interrupt exit; 0: Do not disable hardware push.	0
4	Reserved	RO	Reserved	0
[3:2]	preempt[1:0]	RW	The preemption priority bit width configuration register is used for configuring the preemption priority bit width in the interrupt priority: 00: The preemption bit width is 0, and interrupts with any priority cannot be nested; 01: The preemption bit width is 1, that is, bit [7] of the interrupt priority register is the preemption priority; 10: The preemption bit width is 2, that is, [7:6] bits of the interrupt priority register are preemption priority; 11: The preemption bit width is 3, that is, [7:5] bits of the interrupt priority register are preemption priority. *Note: The CH32V4x7 series only supports two levels of interrupt nesting; this field may only be configured as 01b or 00b.*	0x1
1	nest_en	RW	Interrupt Nesting Enable Register: 1: Enable interrupt nesting 0: Disable interrupt nesting	0
0	hw_stk_en	RW	Hardware stack protection enable:	0

			1: Use hardware stack protection when entering and exiting interrupts; 0: Do not use hardware stack protection when entering and exiting interrupts.	

9.2.3.18 Trigger Channel Selection Register (TSELECT)

CSR address: 0x7A0

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	MRO	Reserved	0
[1:0]	tselect[1:0]	MRW	Trigger channel selection register.	x

Note: Both Cores implement 4-channel triggers. The trigger channel is configured via the TSELECT register, where the lower two bits of TSELECT are active, while the remaining bits are permanently set to 0.

9.2.3.19 Trace Trigger Data Register (TDATA1)

CSR address: 0x7A1

Bit	Name	Access	Description	Reset value
[31:28]	type[3:0]	MRO	Trigger type definition.	0x2
27	dmode	MRO	Both machine mode and debug mode can modify the trigger-related registers.	0
[26:21]	maskmax	MRO	When macth=1, the maximum exponential power range of matching is allowed.	0x1F
[20:19]	Reserved	MRO	Reserved	0
18	timing	MRO	Triggered before the instruction/operation is executed.	0
[17:13]	Reserved	MRO	Reserved	0
12	action	MRW	Set the processing mode when triggering: 1: When triggered, enter the debugging mode, and dpc saves the instruction address that needs to be restored when exiting debugging; 0: When triggered, an exception is generated that leads the thread to M mode.	0
[11:8]	Reserved	MRO	Reserved	0
7	match	MRO	Matching policy configuration: Allow address trigger with comparison value equal to tdata2 value.	0
6	m	MRO	Trigger enable in M mode: 1: Enable; 0: Disable.	1
[5:4]	Reserved	MRO	Reserved	0
3	u	MRO	Trigger enable in U mode: 1: Enable; 0: Disable.	1
2	execute	MRW	Triggered by the virtual address of the executed instruction	0

			1: The trigger compares the instruction address; 0: The trigger does not compare the instruction address.	
1	store	MRW	Triggered based on the virtual address of stored data 1: Trigger compares the write data address; 0: Trigger does not compare the write data address.	0
0	load	MRW	Triggered based on the virtual address of the read data 1: Trigger compares the read data address; 0: Trigger does not compare the read data address	0

Note: The processor trigger configuration complies with the RISC-V debugging standard. The trigger configuration type is 2, matching based on the least significant bit of the accessed virtual address. The trigger operation executes before the matched instruction or data read/write.

9.2.3.20 Save Trigger Data Register (TDATA2)

CSR address: 0x7A2

Bit	Name	Access	Description	Reset value
[31:0]	tdata2[31:0]	MRW	Used to save trigger data (instruction PC or data address).	x

9.2.3.21 Counter Mask Register (MCOUNT_INHIBIT)

CSR address: 0x320

Bit	Name	Access	Description	Reset value
[31:3]	Reserved	MRO	Reserved	0
2	IR	MRW	Instruction counter mask register: 1: Mask the instruction counter, MINSTRET and UNSTRET stop counting; 0: Do not mask the instruction counter, MINSTRET and UNSTRET count normally.	0
1	Reserved	MRO	Reserved	0
0	CY	MRW	Cycle counter mask register: 1: Mask the cycle counter, MCYCLE and UCYCLE stop counting; 0: Do not mask the cycle counter, MCYCLE and UCYCLE count normally.	0

9.2.3.22 Machine Cycle Counter Register (MCYCLE)

CSR address: 0xB00

Bit	Name	Access	Description	Reset value
[31:0]	MCYLE[31:0]	MRW	The machine cycle counter is used to count the clock cycles of the machine. It counts when the	0

			CY bit of the MCOUNT_INHIBIT register is 0. It does not count when the counter stops in debug mode.	

9.2.3.23 Machine Instruction Counter Register (MINSTRET)

CSR address: 0xB02

Bit	Name	Access	Description	Reset value
[31:0]	MINSTRET[31:0]	MRW	The instruction counter is used to count the number of instructions correctly executed by the machine. It counts when the IR bit of the MCOUNT_INHIBIT register is 0. It does not count when the counter stops in debug mode.	0

9.2.3.24 User Mode cycle Counter Register (UCYCLE)

CSR address: 0xC00

Bit	Name	Access	Description	Reset value
[31:0]	UCYLE[31:0]	RO	The user mode cycle counter is used to count the clock cycles of the machine. It counts when the CY bit of the MCOUNT_INHIBIT register is 0. It does not count when the counter stops in debug mode.	0

Note: The user mode cycle counter has the same function as the MCYCLE register and is used to read the number of machine operating cycles in user mode.

9.2.3.25 User Mode Instruction Counter Register (UINSTRET)

CSR address: 0xC02

Bit	Name	Access	Description	Reset value
[31:0]	UINSTRET[31:0]	RO	The instruction counter is used to count the number of instructions correctly executed by the machine. It counts when the IR bit of the MCOUNT_INHIBIT register is 0. It does not count when the counter stops in debug mode.	0

Note: The user mode instruction counter has the same function as the MINSTRET register and is used to read the number of instructions correctly executed by the machine in user mode.

9.2.4 RISC-V3V Core Custom Register

Table 9-3 RISC-V3V core custom registers list

Name	CSR address	Description	Reset value
CPU_RUN_CTLR	0xBC0	Processor run control register	0x00000000
INEST_CTLR	0xBC1	Interrupt nested control register	0x00000000
MIE	0xBC8	Machine interrupt register	0x00000000
DCSR	0x7B0	Debug control and status register	0x40000000
U_NONS_DLY_0	0x8C0	Delay command control register	0xXXXX0XXX

9.2.4.1 Processor Run Control Register (CPU_RUN_CTLR)

CSR address: 0xBC0

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	MRO	Reserved	0
7	nmi_ie	MRW	Non-maskable interrupt enable: 1: Triggers NMI interrupt when an exception causes nested overflow; 0: Does not trigger NMI interrupt when an exception causes nested overflow.	0
6	int_fence	MRW	fence interrupt enable: 1: Clear interrupt requests when executing fence instructions; 0: Do not clear interrupt requests when executing fence instructions.	0
5	non_usta	MRW	UACCES_MSTATUS register enable: 1: Bits 3 and 7 of CSR address 0x800 map to MIE and MPIE bits of the MSTATUS register, respectively; 0: CSR address 0x800 is a read-only register, returning the value of MSTATUS.	0
[4:2]	Reserved	MRO	Reserved	0
[1:0]	pipe_acc[1:0]	MRW	Fetching mode: 00: Prefetch disabled. Disables instruction prefetching to avoid invalid fetches, allowing a maximum of one valid instruction in the CPU pipeline. This mode consumes the least power but reduces performance by approximately 2 to 3 times. 01: Prefetch enabled. Instruction prefetching is enabled. The CPU continuously accesses the instruction memory until the number of pending instructions in the internal instruction buffer exceeds a certain threshold, or until the buffer is full, at which point instruction fetching pauses. This mode consumes high power but delivers strong performance. (CPU prediction failures may introduce redundant fetch operations; in some cases, the execution unit may incur an additional 0~2 cycles of bubble, though performance degradation is generally negligible for most programs.) Other: Reserved.	0

9.2.4.2 Interrupt Nesting Control Register (INEST_CTLR)

CSR address: 0xBC1

Bit	Name	Access	Description	Reset value

31	Reserved	RO	Reserved	0
30	nest_ovr	MRW	Interrupt/Exception nesting overflow flag: Set to 1 when nesting depth exceeds maximum depth; write 1 to clear. *Note: Interrupt overflow occurs only during instruction exceptions generated by secondary interrupt service functions and NMI interrupts. In these cases, the exception and NMI interrupts proceed normally, but the CPU stack overflows, preventing exit from these exceptions and NMI interrupts.*	0
[29:12]	Reserved	MRO	Reserved	0
[11:8]	nest_sta[3:0]	MRO	Nested status flag: 0000: No interrupt; 0001: Level 1 interrupt; 0011: Level 2 interrupt nested; 0111: Level 3 interrupt nested, overflow; Others: Reserved.	0
[7:2]	Reserved	MRO	Reserved	0
[1:0]	nest_max[1:0]	MRW	Maximum nesting level: 00: Nesting disabled (nesting function off); 01: 2-level nesting (enable nesting function); 10: Unwritable; 11: Unwritable. *Note: (1) Writing 10b or 11b to this field sets the register to 01b;* *(2) When 11b is written, reading this register returns the chip's maximum nesting level.*	0

9.2.4.3 Machine Interrupt Register (MIE)

CSR address: 0xBC8

Bit	Name	Access	Description	Reset value
[31:5]	Reserved	RO	Reserved	0
[4:0]	nest_mie[4:0]	RW	Machine interrupt register, storing the MIE information for each level of nested interrupts: status.mie=nest_mie[0]; status.mpie=nest_mie[1].	0

9.2.4.4 Debug Control and Status Register (DCSR)

CSR address: 0x7B0

Bit	Name	Access	Description	Reset value
[31:28]	exdebug[3:0]	MRO	There is an external debugger that conforms to the debugger protocol.	0x4
[27:16]	Reserved	MRO	Reserved	0

15	ebreakm	MRW	Enable debug mode via ebreak instruction in machine mode: 1: Execute ebreak instruction in machine mode to enter debug mode; 0: Execute ebreak instruction in machine mode to trigger an exception.	0
[14:13]	Reserved	MRO	Reserved	0
12	ebreaku	MRW	Enable debug mode via ebreak instruction in user mode: 1: Execute ebreak instruction in user mode to enter debug mode; 0: Execute ebreak instruction in user mode to trigger an exception.	0
[11:10]	Reserved	MRO	Reserved	0
9	stoptime	MRW	Timer pause enable: 1: Counter stops counting in debug mode; 0: Counter operates normally in debug mode.	0
[8:6]	dcause[2:0]	MRW	Debug source register, used to mark the specific event triggering entry into debug mode 000: Reset value; 001: ebreak instruction in program triggers debugging (priority 3); 010: Trigger module causes a breakpoint exception (priority 4); 011: Debugger generates a pause request (priority 1); 100: Pause in single-step mode (priority 0, lowest); 101: Not implemented; 110: Not implemented; 111: Reserved.	0
[5:4]	Reserved	MRO	Reserved	0
3	nmip	MRO	Core NMI interrupt pending flag.	0
2	step	MRW	Single-step mode: 1: Single-step mode; 0: Non-single-step mode.	0
[1:0]	prv[1:0]	MRW	Record the machine state before entering debug mode: 00: Transition from user mode to debug mode; upon recovery, the processor returns to user mode. 01, 10: Reserved. 11: Transition from machine mode to debug mode; upon recovery, the processor	0

			returns to machine mode.	

9.2.4.5 Delay Command Control Register (U_NONS_DLY_0)

CSR address: 0x8C0

Bit	Name	Access	Description	Reset value
[31:24]	dly_mask[7:0]	URW	The matching command parameter mask when the dly command match=111.	x
[23:16]	dly_data[7:0]	URW	The {func, opcode} pairs matched when the dly command is set to match=111.	x
15	CPU_dly_busy_sta	URO	Core is in delayed state flag.	0
14	dly_tout	RW1Z	Timeout information, there is no matching command request within the delay range.	0
13	dly_ie	URW	Delayed interrupt request, if a timeout occurs after being enabled, a software interrupt request is generated.	0
12	dly_priv	URW	1: Matching is allowed in both machine mode and user mode. 0: The delay operation performed in user mode does not match the instruction in machine mode.	0
[11:10]	Reserved	URO	Reserved	0
[9:0]	dly_freq[9:0]	URW	Delay count division coefficient configuration, if configured as the main clock frequency, us can achieve unit delay.	x

9.3 System Counter (SysTick)

Table 9-4 STK registers list

Name	Access	Description	Reset value
R32_STK_CTLR	0xE000F000	System count control register 0	0x00000000
R32_STK_SR	0xE000F004	System counter interrupt status register	0x00000000
R32_STK_CNTL	0xE000F008	System counter low register	0xXXXXXXXX
R32_STK_CMPLR	0xE000F010	Count comparison low register	0xXXXXXXXX

Note: Applicable to general-purpose microcontrollers designed based on 32-bit RISC-V instruction set and architecture.

9.3.1 System Count Control Register 0 (STK_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										INIT	MODE	STRE	STCLK	STIE	STE

Bit	Name	Access	Description	Reset value
[31:6]	Reserved	RO	Reserved	0
5	INIT	RW1	The initial value of the counter is updated: 1: Updated to 0 when counting upwards, and updated to the comparison value when counting downwards; 0: Invalid.	0
4	MODE	RW	Counting mode: 1: Count down; 0: Count up.	0
3	STRE	RW	Automatic reload counting enable bit: 1: Count upwards to the comparison value and then count again from 0; After counting downwards to 0, count again from the comparison value; 0: Continue counting up/down.	0
2	STCLK	RW	Counter clock source selection bit: 1: HCLK is used as the time base; 0: HCLK/8 is used as the time base.	0
1	STIE	RW	Counter interrupt enable control bit: 1: Enable counter interrupt; 0: Disable counter interrupt.	0
0	STE	RW	System counter enable control bit: 1: Start the system counter STK; 0: Turn off the system counter STK, and the counter stops counting.	0

9.3.2 System Counter Status Register (STK_SR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SWIE	Reserved														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															CNTIF

Bit	Name	Access	Description	Reset value
31	SWIE	RW	Software interrupt trigger enable (SW): 1: Trigger software interrupt; 0: Turn off trigger. *Note: This bit must be cleared to 0 after entering the software interrupt, otherwise it will always trigger.*	0
[30:1]	Reserved	RO	Reserved	0
0	CNTIF	RW	Count value comparison flag, clear by writing 0,	0

			invalid by writing 1: 1: Counting up to reach the comparison value, counting down to 0; 0: Not reaching the comparison value.	

9.3.3 System Counter Low Register (STK_CNTL)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNT[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	CNT[31:0]	RW	The lower 32 bits of the current counter value.	X

9.3.4 Count Comparison Low Register (STK_CMPLR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CMP[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMP[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	CMP[31:0]	RW	Set the lower 32 bits of the compare counter value. The STK interrupt will be triggered when the up-counting CMP register value is equal to the CNT counter value or the down-counting CNT counter value is 0.	X

9.4 Interrupt and Event (PFIC)

9.4.1 PFIC Controller

- Each interrupt request has independent trigger and mask control bits, and a dedicated status bit
- Support up to configurable 2-level interrupt nesting
- Unique fast interrupt entry and exit mechanism, automatic hardware push and restore, maximum hardware push depth of 2 levels, no instruction overhead
- Vector Table Free (VTF) interrupt response mechanism, 4 programmable direct interrupt vector addresses

9.4.2 Vector Table of Interrupt and Exception

Table 9-5 Vector table

No.	Priority	Type	Name	Description	Entrance address
0	-6	Fixed	RESET	Reset interrupt	0x00000000

1	-	-	-	-	0x00000004
2	-5	Fixed	NMI	Non-maskable interrupt	0x00000008
3	-4	Fixed	HardFault	Exception	0x0000000C
4	-	-	-	-	0x00000010
5	-3	Fixed	Ecall-M	Machine mode callback interrupt	0x00000014
6-7	-	-	-	-	0x00000018- 0x0000001C
8	-2	Fixed	Ecall-U	User mode callback interrupt	0x00000020
9	-1	Fixed	BreakPoint	Breakpoint callback interrupt	0x00000024
10-11	-	-	-	-	0x00000028- 0x0000002C
12	0	Programmable	SysTick0	System timer interrupt 0	0x00000030
13	-	-	-	-	0x00000034
14	1	Programmable	SW	Software interrupt	0x00000038
15	-	-	-	-	0x0000003C
16	2	Programmable	WWDG	Window timer interrupt	0x00000040
17	3	Programmable	PVD	Supply voltage detection interrupt (EXTI)	0x00000044
18	4	Programmable	TAMPER	Intrusion detection interrupt	0x00000048
19	5	Programmable	RTC	Real time clock interrupt	0x0000004C
20	6	Programmable	FLASH	Flash global interrupt	0x00000050
21	7	Programmable	RCC	Reset and clock control interrupts	0x00000054
22	8	Programmable	EXTI0	EXTI line 0 interrupt	0x00000058
23	9	Programmable	EXTI1	EXTI line 1 interrupt	0x0000005C
24	10	Programmable	EXTI2	EXTI line 2 interrupt	0x00000060
25	11	Programmable	EXTI3	EXTI line 3 interrupt	0x00000064
26	12	Programmable	EXTI4	EXTI line 4 interrupt	0x00000068
27	13	Programmable	DMA1_CH1	DMA channel 1 global interrupt	0x0000006C
28	14	Programmable	DMA1_CH2	DMA channel 2 global interrupt	0x00000070
29	15	Programmable	DMA1_CH3	DMA channel 3 global interrupt	0x00000074
30	16	Programmable	DMA1_CH4	DMA channel 4 global interrupt	0x00000078
31	17	Programmable	DMA1_CH5	DMA channel 5 global interrupt	0x0000007C
32	18	Programmable	DMA1_CH6	DMA channel 6 global interrupt	0x00000080
33	19	Programmable	DMA1_CH7	DMA channel 7 global interrupt	0x00000084
34	20	Programmable	ADC1_2	ADC1_2 global interrupt	0x00000088
35	21	Programmable	CAN1_TX	CAN1_TX global interrupt	0x0000008C
36	22	Programmable	CAN1_RX0	CAN1_RX0 global interrupt	0x00000090
37	23	Programmable	CAN1_RX1	CAN1_RX1 global interrupt	0x00000094
38	24	Programmable	CAN1_SCE	CAN1_SCE global interrupt	0x00000098
39	25	Programmable	EXTI9_5	EXTI line [9:5] interrupt	0x0000009C
40	26	Programmable	TIM1_BRK	TIM1 brake interrupt	0x000000A0
41	27	Programmable	TIM1_UP	TIM1 update interrupt	0x000000A4
42	28	Programmable	TIM1_TRG_ COM	TIM1 trigger and communication interrupt	0x000000A8
43	29	Programmable	TIM1_CC	TIM1 capture compare interrupt	0x000000AC

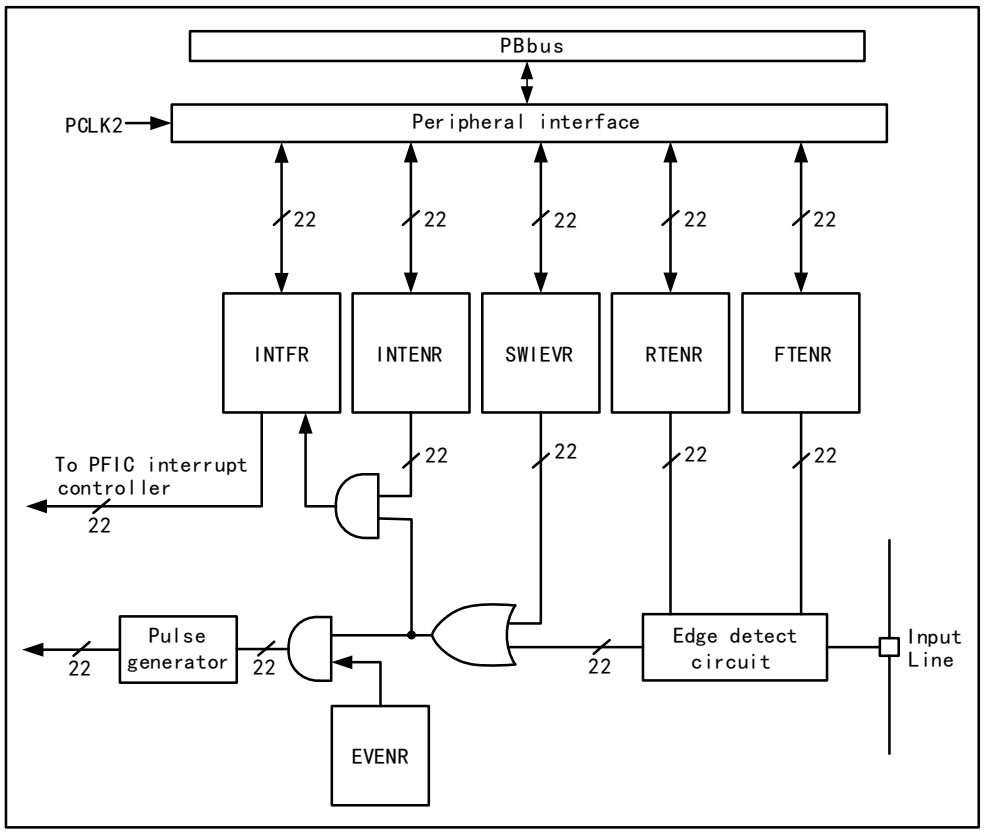
44	30	Programmable	TIM2	TIM2 global interrupt	0x000000B0
45	31	Programmable	TIM3	TIM3 global interrupt	0x000000B4
46	32	Programmable	TIM4	TIM4 global interrupt	0x000000B8
47	33	Programmable	I2C_EV	I^2C event interrupt	0x000000BC
48	34	Programmable	I2C_ER	I^2C error interrupt	0x000000C0
49	35	Programmable	I3C_EV	I^3C event interrupt	0x000000C4
50	36	Programmable	I3C_ER	I^3C error interrupt	0x000000C8
51	37	Programmable	SPI1	SPI1 global interrupt	0x000000CC
52	38	Programmable	SPI2	SPI2 global interrupt	0x000000D0
53	39	Programmable	USART1	USART1 global interrupt	0x000000D4
54	40	Programmable	USART2	USART2 global interrupt	0x000000D8
55	41	Programmable	USART3	USART3 global interrupt	0x000000DC
56	42	Programmable	EXTI15_10	EXTI line [15:10] interrupt	0x000000E0
57	43	Programmable	RTCAlarm	RTC alarm interrupt (EXTI)	0x000000E4
58	44	Programmable	USBHS1WakeUP	USBHS1 wakeup interrupt	0x000000E8
59	45	Programmable	TIM8_BRK	TIM8 brake interrupt	0x000000EC
60	46	Programmable	TIM8_UP	TIM8 update interrupt	0x000000F0
61	47	Programmable	TIM8_TRG_COM	TIM8 trigger and communication interrupt	0x000000F4
62	48	Programmable	TIM8_CC	TIM8 capture compare interrupt	0x000000F8
63	49	Programmable	RNG	RNG global interrupt	0x000000FC
64	-	-	-	-	0x00000100
65	50	Programmable	SDIO	SDIO global interrupt	0x00000104
66	51	Programmable	TIM5	TIM5 global interrupt	0x00000108
67	52	Programmable	SPI3	SPI3 global interrupt	0x0000010C
68	53	Programmable	USART4	USART4 global interrupt	0x00000110
69	54	Programmable	USART5	USART5 global interrupt	0x00000114
70	55	Programmable	TIM6	TIM6 global interrupt	0x00000118
71	56	Programmable	TIM7	TIM7 global interrupt	0x0000011C
72	57	Programmable	DMA2_CH1	DMA2 channel 1 global interrupt	0x00000120
73	58	Programmable	DMA2_CH2	DMA2 channel 2 global interrupt	0x00000124
74	59	Programmable	DMA2_CH3	DMA2 channel 3 global interrupt	0x00000128
75	60	Programmable	DMA2_CH4	DMA2 channel 4 global interrupt	0x0000012C
76	61	Programmable	DMA2_CH5	DMA2 channel 5 global interrupt	0x00000130
77	62	Programmable	ETH	ETH global interrupt	0x00000134
78	63	Programmable	ETH_WKUP	ETH wakeup interrupt	0x00000138
79	64	Programmable	LTDC	LTDC global interrupt	0x0000013C
80	65	Programmable	PSRAM	PSRAM global interrupt	0x00000140
81	66	Programmable	USART9	USART9 global interrupt	0x00000144
82	67	Programmable	USART10	USART10 global interrupt	0x00000148
83	68	Programmable	USBHS1	USBHS1 global interrupt	0x0000014C
84	69	Programmable	USBHS2WakeUP	USBHS2 wakeup interrupt	0x00000150
85	70	Programmable	USBHS2	USBHS2 global interrupt	0x00000154

86	71	Programmable	DVP	DVP global interrupt	0x00000158
87	72	Programmable	USART6	USART6 global interrupt	0x0000015C
88	73	Programmable	USART7	USART7 global interrupt	0x00000160
89	74	Programmable	USART8	USART8 global interrupt	0x00000164
90	75	Programmable	DMA1_CH8	DMA1 channel 8 global interrupt	0x00000168
91	76	Programmable	DMA1_CH9	DMA1 channel 9 global interrupt	0x0000016C
92	77	Programmable	DMA1_CH10	DMA1 channel 10 global interrupt	0x00000170
93	78	Programmable	DMA1_CH11	DMA1 channel 11 global interrupt	0x00000174
94	79	Programmable	DMA1_CH12	DMA1 channel 12 global interrupt	0x00000178
95	80	Programmable	DMA1_CH13	DMA1 channel 13 global interrupt	0x0000017C
96	81	Programmable	I3CWakeUP	I3C wakeup interrupt	0x00000180
97	82	Programmable	ARGB	ARGB global interrupt	0x00000184
98	83	Programmable	DMA2_CH6	DMA2 channel 6 global interrupt	0x00000188
99	84	Programmable	DMA2_CH7	DMA2 channel 7 global interrupt	0x0000018C
100	85	Programmable	DMA2_CH8	DMA2 channel 8 global interrupt	0x00000190
101	86	Programmable	DMA2_CH9	DMA2 channel 9 global interrupt	0x00000194
102	87	Programmable	DMA2_CH10	DMA2 channel 10 global interrupt	0x00000198
103	88	Programmable	DMA2_CH11	DMA2 channel 11 global interrupt	0x0000019C

9.4.3 External Interrupt and Event Controller (EXTI)

9.4.3.1 Overview

Figure 9-1 External interrupt (EXTI) interface block diagram



As can be seen from figure 9-1, the trigger source of the external interrupt can be either a software interrupt (SWIEVR) or an actual external interrupt channel, and the signal of the external interrupt channel is first screened by the edge detection circuit. As long as a software interrupt or an external interrupt signal is

generated, it will be output to the event enable and interrupt enable AND gate circuits through the OR gate circuit in the figure. As long as an interrupt is enabled or an event is enabled, an interrupt or event will be generated. The six registers of EXTI are accessed by the processor through the PB2 interface.

9.4.3.2 Wakeup Event

The system can wake up sleep patterns caused by WFE instructions through wake-up events. Wake-up events are generated through the following 2 configurations:

- Enable an interrupt in the register of the peripheral, but not in the PFIC of the core, as well as the SEVONPEND bit in the core. Reflected in EXTI, it enables EXTI interrupts, but does not enable EXTI interrupts in PFIC, while enabling SEVONPEND bits. After the CPU wakes up from WFE, it must clear the EXTI interrupt flag and the PFIC suspend bit.
- Enable an EXTI channel to be an event channel, and after CPU wakes up from WFE, there is no need to clear the operation of the interrupt flag bit and PFIC hang bit.

9.4.3.3 Description

The use of external interrupt needs to configure the corresponding external interrupt channel, that is, select the appropriate trigger edge to enable the corresponding interrupt. When a set trigger edge appears on the external interrupt channel, an interrupt request will be generated and the corresponding interrupt flag bit will be set. Write 1 to the flag bit to clear it.

Use external hardware interrupt steps:
1) Configure GPIO operation.
2) Configure the interrupt enable bit (EXTI_INTENR) of the corresponding external interrupt channel.
3) Configure trigger edge (EXTI_RTENR or EXTI_FTENR), select rising edge trigger, falling edge trigger or double edge trigger.
4) Configure EXTI interrupts in the PFIC of the core to ensure that they respond correctly.

To use external hardware events:
1) Configure GPIO operation.
2) Configure the event enable bit (EXTI_EVENR) of the corresponding external interrupt channel.
3) Configure trigger edge (EXTI_RTENR or EXTI_FTENR), select rising edge trigger, falling edge trigger or double edge trigger.

Use software interrupt/event steps:
1) Enable external interrupt (EXTI_INTENR) or external event (EXTI_EVENR).
2) If you use the interrupt service function, you need to set the EXTI interrupt in the PFIC of the core.
3) Set the software interrupt trigger (EXTI_SWIEVR), that is, an interrupt will occur.

9.4.3.4 External Event Map

Table 9-6 EXTI interrupt map

External interrupt/event line	Mapping event description
EXTI0~EXTI15	Px0~Px15 (x=A/B/C/D/E), any of the IO ports can be enabled for external interrupt/event functionality, configured by the AFIO_EXTICRx register.
EXTI16	PVD event: Voltage monitoring threshold exceeded

EXTI17	RTC alarm event
EXTI18	USBHS1 wakeup event
EXTI19	ETH wakeup event
EXTI20	USBHS2 wakeup event
EXTI21	I3C wakeup event

9.4.4 EXTI Register

Table 9-7 EXTI registers list

Name	Access	Description	Reset value
R32_EXTI_INTENR	0x40010400	Interrupt enable register	0x00000000
R32_EXTI_EVENR	0x40010404	Event enable register	0x00000000
R32_EXTI_RTENR	0x40010408	Rising edge trigger enable register	0x00000000
R32_EXTI_FTENR	0x4001040C	Falling edge trigger enable register	0x00000000
R32_EXTI_SWIEVR	0x40010410	Software interrupt event register	0x00000000
R32_EXTI_INTFR	0x40010414	Interrupt flag register	0x0000XXXX

9.4.4.1 Interrupt Enable Register (EXTI_INTENR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										MR21	MR20	MR19	MR18	MR17	MR16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
[21:0]	MRx	RW	Enable the interrupt request signal for external interrupt channel x: 1: Enable interrupt for this channel; 0: Mask interrupt for this channel.	0

9.4.4.2 Event Enable Register (EXTI_EVENR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										MR21	MR20	MR19	MR18	MR17	MR16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
[21:0]	MRx	RW	Enable the event request signal for external interrupt channel x:	0

			1: Enable event for this channel; 0: Mask event for this channel.	

9.4.4.3 Rising Edge Trigger Enable Register (EXTI_RTENR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										TR21	TR20	TR19	TR18	TR17	TR16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
[21:0]	TRx	RW	Enable rising edge triggering of external interrupt channel x: 1: Enable rising edge triggering for this channel; 0: Disable rising edge triggering for this channel.	0

9.4.4.4 Falling Edge Trigger Enable Register (EXTI_FTENR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										TR21	TR20	TR19	TR18	TR17	TR16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
[21:0]	TRx	RW	Enable falling edge triggering of external interrupt channel x: 1: Enable falling edge triggering for this channel; 0: Disable falling edge triggering for this channel.	0

9.4.4.5 Software Interrupt Event Register (EXTI_SWIEVR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										SWIER 21	SWIER 20	SWIER 19	SWIER 18	SWIER 17	SWIER 16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWIER15	SWIER 14	SWIER13	SWIER12	SWIER11	SWIER 10	SWIER 9	SWIER 8	SWIER 7	SWIER 6	SWIER 5	SWIER 4	SWIER 3	SWIER 2	SWIER 1	SWIER 0

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
[21:0]	SWIERx	RW	A software interrupt is set on the corresponding externally triggered interrupt channel. Setting it here causes the interrupt flag bit (EXTI_INTFR) to be set to the corresponding position bit, and an interrupt or event is generated if the interrupt enable (EXTI_INTENR) or event enable (EXTI_EVENR) is turned on.	0

9.4.4.6 Interrupt Flag Register (EXTI_INTFR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										IF21	IF20	IF19	IF18	IF17	IF16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IF15	IF14	IF13	IF12	IF11	IF10	IF9	IF8	IF7	IF6	IF5	IF4	IF3	IF2	IF1	IF0

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
[21:0]	IFx	RW1	Interrupt Flag Bit, this location bit flag indicates that the corresponding external interrupt has occurred. Writing 1 clears this bit.	0

9.4.5 PFIC Registers

Table 9-8 PFIC-related registers

Name	Access address	Description	Reset value
R32_PFIC_ISR1	0xE000E000	PFIC interrupt enable status register 1	0x0000032C
R32_PFIC_ISR2	0xE000E004	PFIC interrupt enable status register 2	0x00000000
R32_PFIC_ISR3	0xE000E008	PFIC interrupt enable status register 3	0x00000000
R32_PFIC_ISR4	0xE000E00C	PFIC interrupt enable status register4	0x00000000
R32_PFIC_IPR1	0xE000E020	PFIC interrupt pending status register 1	0x00000000
R32_PFIC_IPR2	0xE000E024	PFIC interrupt pending status register 2	0x00000000
R32_PFIC_IPR3	0xE000E028	PFIC interrupt pending status register 3	0x00000000
R32_PFIC_IPR4	0xE000E02C	PFIC interrupt pending status register 4	0x00000000
R32_PFIC_ITHRESDR	0xE000E040	PFIC interrupt priority threshold configuration register	0x00000000
R32_PFIC_CFGR	0xE000E048	PFIC interrupt configuration register	0x00000000
R32_PFIC_GISR	0xE000E04C	PFIC interrupt global status register	0x00000000
R32_PFIC_VTFIDR	0xE000E050	PFIC VTF interrupt ID configuration register	0xXXXXXXXX
R32_PFIC_VTFADDRR0	0xE000E060	PFIC VTF interrupt 0 offset address register	0xXXXXXXXX
R32_PFIC_VTFADDRR1	0xE000E064	PFIC VTF interrupt 1 offset address	0xXXXXXXXX

		register	
R32_PFIC_VTFADDRR2	0xE000E068	PFIC VTF interrupt 2 offset address register	0xXXXXXXXX
R32_PFIC_VTFADDRR3	0xE000E06C	PFIC VTF interrupt 3 offset address register	0xXXXXXXXX
R32_PFIC_IENR1	0xE000E100	PFIC interrupt enable set register 1	0x00000000
R32_PFIC_IENR2	0xE000E104	PFIC interrupt enable set register 2	0x00000000
R32_PFIC_IENR3	0xE000E108	PFIC interrupt enable set register 3	0x00000000
R32_PFIC_IENR4	0xE000E10C	PFIC interrupt enable set register 4	0x00000000
R32_PFIC_IRER1	0xE000E180	PFIC interrupt enable clear register 1	0x00000000
R32_PFIC_IRER2	0xE000E184	PFIC interrupt enable clear register 2	0x00000000
R32_PFIC_IRER3	0xE000E188	PFIC interrupt enable clear register 3	0x00000000
R32_PFIC_IRER4	0xE000E18C	PFIC interrupt enable clear register 4	0x00000000
R32_PFIC_IPSR1	0xE000E200	PFIC interrupt pending set register 1	0x00000000
R32_PFIC_IPSR2	0xE000E204	PFIC interrupt pending set register 2	0x00000000
R32_PFIC_IPSR3	0xE000E208	PFIC interrupt pending set register 3	0x00000000
R32_PFIC_IPSR4	0xE000E20C	PFIC interrupt pending set register 4	0x00000000
R32_PFIC_IPRR1	0xE000E280	PFIC interrupt pending clear register 1	0x00000000
R32_PFIC_IPRR2	0xE000E284	PFIC interrupt pending clear register 2	0x00000000
R32_PFIC_IPRR3	0xE000E288	PFIC interrupt pending clear register 3	0x00000000
R32_PFIC_IPRR4	0xE000E28C	PFIC interrupt pending clear register 4	0x00000000
R32_PFIC_IACTR1	0xE000E300	PFIC interrupt activation register 1	0x00000000
R32_PFIC_IACTR2	0xE000E304	PFIC interrupt activation register 2	0x00000000
R32_PFIC_IACTR3	0xE000E308	PFIC interrupt activation register 3	0x00000000
R32_PFIC_IACTR4	0xE000E30C	PFIC interrupt activation register 4	0x00000000
R32_PFIC_IPRIORx	0xE000E400	PFIC interrupt priority configuration register	0x00000000
R32_PFIC_WAKEIP0	0xE000E720	PFIC wakeup instruction pointer register 0	0x00000001
R32_PFIC_CSTAR0	0xE000E780	PFIC core status register 0	0x00000000
R32_PFIC_EENR	0xE000EC80	PFIC event enable register	0xFFFFFFFF
R32_PFIC_EPR	0xE000EC84	PFIC event pending register	0x00000000
R32_PFIC_EWUPR	0xE000EC88	PFIC event wakeup register	0x00000000
R32_PFIC_SCTLR	0xE000ED10	PFIC system control register	0x00000000

Note: 1. NMI, EXC, ECALL-M, ECALL-U, BREAKPOINT interrupts are always enabled by default.

2. ECALL-M, ECALL-U, and BREAKPOINT are all EXC cases, and the status is indicated by EXC status bit 3.

3. NMI and EXC support interrupt suspend clear and set operations, but not interrupt enable clear and set operations.

4. ECALL-M, ECALL-U, and BREAKPOINT do not support interrupt hang clear and set, interrupt enable clear and set operations.

9.4.5.1 PFIC Interrupt Enable Status Register 1 (PFIC_ISR1)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTENSTA[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	INTENSTA14	Reserved	INTENSTA12	Reserved		INTENSTA9	INTENSTA8	Reserved		INTENSTA5	Reserved	INTENSTA3	INTENSTA2	Reserved	

Bit	Name	Access	Description	Reset value
[31:16]	INTENSTA	RO	16#-31#Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0
15	Reserved	RO	Reserved	0
14	INTENSTA	RO	14# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0
13	Reserved	RO	Reserved	0
12	INTENSTA	RO	12# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0
[11:10]	Reserved	RO	Reserved	0
[9:8]	INTENSTA	RO	8#-9# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0x3
[7:6]	Reserved	RO	Reserved	0
5	INTENSTA	RO	5# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	1
4	Reserved	RO	Reserved	0
[3:2]	INTENSTA	RO	2#-3# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0x3
[1:0]	Reserved	RO	Reserved	0

9.4.5.2 PFIC Interrupt Enable Status Register 2 (PFIC_ISR2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTENSTA[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTENSTA[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	INTENSTA	RO	32#-63# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0

9.4.5.3 PFIC Interrupt Enable Status Register 3 (PFIC_ISR3)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTENSTA[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTENSTA[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	INTENSTA	RO	64#-95# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0

9.4.5.4 PFIC Interrupt Enable Status Register 4 (PFIC_ISR4)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								INTENSTA[103:96]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[31:0]	INTENSTA	RO	96#-103# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0

9.4.5.5 PFIC Interrupt Pending Status Register 1 (PFIC_IPR1)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSTA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSTA15	PENDSTA14	PENDSTA13	PENDSTA12	Reserved		PENDSTA9	PENDSTA8	Reserved		PENDSTA5	Reserved	PENDSTA3	PENDSTA2	Reserved	

Bit	Name	Access	Description	Reset value
[31:16]	PENDSTA	RO	16#-31#interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0
15	Reserved	RO	Reserved	0
14	PENDSTA	RO	14# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0
13	Reserved	RO	Reserved	0
12	PENDSTA	RO	12# interrupts currently pending status.	0

			1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	
[11:10]	Reserved	RO	Reserved	0
[9:8]	PENDSTA	RO	8#-9#interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0
[7:6]	Reserved	RO	Reserved	0
5	PENDSTA	RO	5# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0
4	Reserved	RO	Reserved	0
[3:2]	PENDSTA	RO	2#-3# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0
[1:0]	Reserved	RO	Reserved	0

9.4.5.6 PFIC Interrupt Pending Status Register 2 (PFIC_IPR2)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSTA[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSTA[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDSTA	RO	32#-63# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0

9.4.5.7 PFIC Interrupt Pending Status Register 3 (PFIC_IPR3)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSTA[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSTA[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDSTA	RO	64#-95# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0

9.4.5.8 PFIC Interrupt Pending Status Register 4 (PFIC_IPR4)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								PENDSTA[103:96]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	PENDSTA	RO	96#-103# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0

9.4.5.9 PFIC Interrupt Priority Threshold Configuration Register (PFIC_ITHRESDR)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved																								THRESHOLD[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	THRESHOLD	RW	Interrupt priority threshold setting value. Interrupt priority values lower than the current set value do not perform interrupt service when hung; a 0 in this register indicates that the threshold register function is invalid. [7:4]: Priority threshold value. [3:0]: Reserved, fixed to 0, write invalid.	0

9.4.5.10 PFIC Interrupt Configuration Register (PFIC_CFGR)

Offset address: 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
KEYCODE[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								RSTSYS	Reserved						

Bit	Name	Access	Description	Reset value
[31:16]	KEYCODE[15:0]	WO	The "key" control keyword is valid when writing to "beef"; otherwise, the operation is invalid.	0
[15:8]	Reserved	RO	Reserved	0
7	RSTSYS	WO	Reset register: System reset occurs when 0xBEEF is written to register 0080. Other values are invalid. *Note: Same function as PFIC_SCTLR register SYSRST bit.*	0
[6:0]	Reserved	RO	Reserved	0

9.4.5.11 PFIC Interrupt Global Status Register (PFIC_GISR)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EX_STATE[1:0]		LOCK_UP	DBG_MODE	GLOBL_IE	Reserved	GPENDSTA	GACTSTA	NESTSTA[7:0]							

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:14]	EX_STATE[1:0]	RO	Core status register, used to obtain the core's operational state 00: Running; 01: Light sleep; 10: Deep sleep; 11: Deep sleep (Locked).	0
13	LOCK_UP	RO	Core lock status register, used to determine whether the core is in a locked state. 1: Locked; 0: Unlocked.	0
12	DBG_MODE	RO	Core debug mode register, used to determine whether the core is in debug mode 1: Core is in debug mode; 0: Core is in non-debug mode.	0
11	GLOBL_IE	RO	Core global interrupt enable register, used to query whether core global interrupts are enabled 1: Core global interrupts enabled; 0: Core global interrupts disabled.	0
10	Reserved	RO	Reserved	0
9	GPENDSTA	RO	Whether an interrupt is currently pending: 1: Yes; 0: No.	0
8	GACTSTA	RO	Whether the interrupt is executed currently: 1: Yes; 0: No.	0
[7:0]	NESTSTA	RO	Current interrupt nesting state, used to query the core interrupt nesting state.	0

9.4.5.12 PFIC VTF Interrupt ID Configuration Register (PFIC_VTFIDR)

Offset address: 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VTFID3								VTFID2							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VTFID1								VTFID0							

Bit	Name	Access	Description	Reset value
[31:24]	VTFID3	RW	Configure VTF interrupt 3 ID	X
[23:16]	VTFID2	RW	Configure VTF interrupt 2 ID	X
[15:8]	VTFID1	RW	Configure VTF interrupt 1 ID	X
[7:0]	VTFID0	RW	Configure VTF interrupt 0 ID	X

9.4.5.13 PFIC VTF Interrupt 0 Address Register (PFIC_VTFADDRR0)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR0[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR0[15:1]															VTF0EN

Bit	Name	Access	Description	Reset value
[31:1]	ADDR0	RW	VTF interrupt 0 service program address bit[31:1], bit0 is 0.	X
0	VTF0EN	RW	VTF interrupt 0 enable bit: 1: Enabled VTF interrupt 0 channel; 0: Disabled.	0

9.4.5.14 PFIC VTF Interrupt 1 Address Register (PFIC_VTFADDRR1)

Offset address: 0x64

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR1[15:1]															VTF1EN

Bit	Name	Access	Description	Reset value
[31:1]	ADDR1	RW	VTF interrupt 1 service program address bit[31:1], bit0 is 0.	X
0	VTF1EN	RW	VTF interrupt 1 enable bit: 1: Enabled VTF interrupt 1 channel; 0: Disabled.	0

9.4.5.15 PFIC VTF Interrupt 2 Address Register (PFIC_VTFADDRR2)

Offset address: 0x68

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR2[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

ADDR2[15:1]	VTF2EN

Bit	Name	Access	Description	Reset value
[31:1]	ADDR2	RW	VTF interrupt 2 service program address bit[31:1], bit0 is 0.	X
0	VTF2EN	RW	VTF interrupt 2 enable bit: 1: Enabled VTF interrupt 2 channel; 0: Disabled.	0

9.4.5.16 PFIC VTF Interrupt 3 Address Register (PFIC_VTFADDRR3)

Offset address: 0x6C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR3[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR3[15:1]															VTF3EN

Bit	Name	Access	Description	Reset value
[31:1]	ADDR3	RW	VTF interrupt 3 service program address bit[31:1], bit0 is 0.	X
0	VTF3EN	RW	VTF interrupt 3 enable bit: 1: Enabled VTF interrupt 3 channel; 0: Disabled.	0

9.4.5.17 PFIC Interrupt Enable Set Register 1 (PFIC_IENR1)

Offset address: 0x100

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTEN[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	INTEN14	Reserved	INTEN12	Reserved											

Bit	Name	Access	Description	Reset value
[31:12]	INTEN	WO	16#-31# interrupt enable control. 1: Current numbered interrupt enable; 0: No effect.	0
15	Reserved	RO	Reserved	0
14	INTEN	WO	14# interrupt enable control. 1: Current numbered interrupt enable; 0: No effect.	0
13	Reserved	RO	Reserved	0
12	INTEN	WO	12# interrupt enable control.	0

			1: Current numbered interrupt enable; 0: No effect.	
[11:0]	Reserved	RO	Reserved	0

9.4.5.18 PFIC Interrupt Enable Set Register 2 (PFIC_IENR2)

Offset address: 0x104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTEN[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTEN[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	INTEN	WO	32#-63# interrupt enable control. 1: Current numbered interrupt enable; 0: No effect.	0

9.4.5.19 PFIC Interrupt Enable Set Register 3 (PFIC_IENR3)

Offset address: 0x108

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTEN[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTEN[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	INTEN	WO	64#-95# interrupt enable control. 1: Current numbered interrupt enable; 0: No effect.	0

9.4.5.20 PFIC Interrupt Enable Set Register 4 (PFIC_IENR4)

Offset address: 0x10C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								INTEN[103:96]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	INTEN	WO	96#-103# interrupt enable control. 1: Current numbered interrupt enable; 0: No effect.	0

9.4.5.21 PFIC Interrupt Enable Clear Register 1 (PFIC_IRER1)

Offset address: 0x180

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTRSET[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTRSET15	INTRSET14	INTRSET13	INTRSET12	Reserved											

Bit	Name	Access	Description	Reset value
[31:16]	INTRSET	WO	16#-31# interrupt shutdown control. 1: Current numbered interrupt shutdown; 0: No effect.	0
15	Reserved	RO	Reserved	0
14	INTRSET	WO	14# interrupt shutdown control. 1: Current numbered interrupt shutdown; 0: No effect.	0
13	Reserved	RO	Reserved	0
12	INTRSET	WO	12# interrupt shutdown control. 1: Current numbered interrupt shutdown; 0: No effect.	0
[11:0]	Reserved	RO	Reserved	0

9.4.5.22 PFIC Interrupt Enable Clear Register 2 (PFIC_IRER2)

Offset address: 0x184

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTRSET[63:48]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTRSET[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	INTRSET	WO	32#-63# interrupt shutdown control. 1: Current numbered interrupt shutdown; 0: No effect.	0

9.4.5.23 PFIC Interrupt Enable Clear Register 3 (PFIC_IRER3)

Offset address: 0x188

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTRSET[95:80]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTRSET[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	INTRSET	WO	64#-95# interrupt shutdown control. 1: Current numbered interrupt shutdown; 0: No effect.	0

9.4.5.24 PFIC Interrupt Enable Clear Register 4 (PFIC_IRER4)

Offset address: 0x18C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								INTRSET[103:96]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	INTRSET	WO	96#-103# interrupt shutdown control. 1: Current numbered interrupt shutdown; 0: No effect.	0

9.4.5.25 PFIC Interrupt Pending Set Register 1 (PFIC_IPSR1)

Offset address: 0x200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSET[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSET15	PENDSET14	PENDSET13	PENDSET12	Reserved		PEND SET9	PEND SET8	Reserved		PEND SET5	Reserved	PEND SET3	PEND SET2	Reserved	

Bit	Name	Access	Description	Reset value
[31:16]	PENDSET	WO	16#-31# interrupt pending settings. 13# and 15# reserved. 1: Current numbered interrupt pending; 0: No effect.	0
15	Reserved	RO	Reserved	0
14	PENDSET	WO	14# interrupt pending settings. 13# and 15# reserved. 1: Current numbered interrupt pending; 0: No effect.	
13	Reserved	RO	Reserved	0
12	PENDSET	WO	12# interrupt pending settings. 13# and 15# reserved. 1: Current numbered interrupt pending; 0: No effect.	

[11:10]	Reserved	RO	Reserved	0
[9:8]	PENDSET	WO	8#-9# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0
[7:6]	Reserved	RO	Reserved	0
5	PENDSET	WO	5# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0
4	Reserved	RO	Reserved	0
[3:2]	PENDSET	WO	2#-3# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0
[1:0]	Reserved	RO	Reserved	0

9.4.5.26 PFIC Interrupt Pending Set Register 2 (PFIC_IPSR2)

Offset address: 0x204

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSET[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSET[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDSET	WO	32#-63# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0

9.4.5.27 PFIC Interrupt Pending Set Register 3 (PFIC_IPSR3)

Offset address: 0x208

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSET[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSET[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDSET	WO	64#-95# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0

9.4.5.28 PFIC Interrupt Pending Set Register 4 (PFIC_IPSR4)

Offset address: 0x20C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								PENDSET[103:96]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	PENDSET	WO	96#-103# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0

9.4.5.29 PFIC Interrupt Pending Clear Register 1 (PFIC_IPRR1)

Offset address: 0x280

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDRST[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDRST15	PENDRST14	PENDRST13	PENDRST12	Reserved		PEND RST9	PEND RST8	Reserved		PEND RST5	Reserved	PEND RST3	PEND RST2	Reserved	

Bit	Name	Access	Description	Reset value
[31:16]	PENDRST	WO	16#-31# interrupt pending clear. 1: Current numbered interrupt pending; 0: No effect.	0
15	Reserved	RO	Reserved	0
14	PENDRST	WO	14# interrupt pending clear. 13# and 15# reserved. 1: Current numbered interrupt pending; 0: No effect.	0
13	Reserved	RO	Reserved	0
12	PENDRST	WO	12# interrupt pending clear. 1: Current numbered interrupt pending; 0: No effect.	0
[11:10]	Reserved	RO	Reserved	0
[9:8]	PENDRST	WO	8#-9# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0
[7:6]	Reserved	RO	Reserved	0
5	PENDRST	WO	5# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0
4	Reserved	RO	Reserved	0

[3:2]	PENDRST	WO	2#-3# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0
[1:0]	Reserved	RO	Reserved	0

9.4.5.30 PFIC Interrupt Pending Clear Register 2 (PFIC_IPRR2)

Offset address: 0x284

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDRST[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDRST[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDRST	WO	32#-63# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0

9.4.5.31 PFIC Interrupt Pending Clear Register 3 (PFIC_IPRR3)

Offset address: 0x288

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDRST[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDRST[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDRST	WO	64#-95# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0

9.4.5.32 PFIC Interrupt Pending Clear Register 4 (PFIC_IPRR4)

Offset address: 0x28C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								PENDRST[103:96]							

Bit	Name	Access	Description	Reset value

[31:8]	Reserved	RO	Reserved	0
[31:0]	PENDRST	WO	96#-103# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0

9.4.5.33 PFIC Interrupt Activation Register 1 (PFIC_IACTR1)

Offset address: 0x300

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IACTS[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IACTS15	IACTS14	IACTS13	IACTS12	Reserved								IACTS3	IACTS2	Reserved	

Bit	Name	Access	Description	Reset value
[31:16]	IACTS	RO	16#-31# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0
15	Reserved	RO	Reserved	0
14	IACTS	RO	14# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0
13	Reserved	RO	Reserved	0
12	IACTS	RO	12# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0
[11:4]	Reserved	RO	Reserved	0
[3:2]	IACTS	RO	2#-3# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0
[1:0]	Reserved	RO	Reserved	0

9.4.5.34 PFIC Interrupt Activation Register 2 (PFIC_IACTR2)

Offset address: 0x304

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IACTS[63:48]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IACTS[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	IACTS	RO	32#-63# interrupt execution status.	0

			1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	

9.4.5.35 PFIC Interrupt Activation Register 3 (PFIC_IACTR3)

Offset address: 0x308

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IACTS[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IACTS[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	IACTS	RO	64#-95# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0

9.4.5.36 PFIC Interrupt Activation Register 4 (PFIC_IACTR4)

Offset address: 0x30C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								IACTS[103:96]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	IACTS	RO	96#-103# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0

9.4.5.37 PFIC Interrupt Priority Configuration Register (PFIC_IPRIORx) (x=0-63)

Offset address: 0x400 - 0x4FF

The controller supports 256 interrupts (0-255), and the priority of each interrupt is controlled by 8bits.

	31 24	23 16	15 8	7 0
IPRIOR63	PRIO_255	PRIO_254	PRIO_253	PRIO_252
…	…	…	…	…
IPRIORx	PRIO_（4x+3）	PRIO_（4x+2）	PRIO_（4x+1）	PRIO_（4x）
…	…	…	…	…
IPRIOR0	PRIO_3	PRIO_2	PRIO_1	PRIO_0

Bit	Name	Access	Description	Reset value
[2047:2040]	IP_255	RW	Same as IP_0 description.	0
…	…	…	…	…
[31:24]	IP_3	RW	Same as IP_0 description.	0
[23:16]	IP_2	RW	Same as IP_0 description.	0
[15:8]	IP_1	RW	Same as IP_0 description.	0
[7:0]	IP_0	RW	No. 0 Interrupt Priority Configuration: [7:4]: Priority control bits: the lower the value, the higher the priority. The preemption bit is controlled by the CSR register's preempt flag. When the preempt register value is 0, there is no preemption bit. When the preempt register value is 1, bit 7 is the preemption bit. When the preempt register value is 2, bits 7 through 6 are the preemption bits. When the preempt register value is 3, bits 7 through 5 are the preemption bits. The core supports up to 2 levels of nesting. When the nesting is full, preemption cannot continue. [3:0]: fixed to 0.	0

9.4.5.38 PFIC Wakeup Instruction Pointer Register 0 (PFIC_WAKEIP0)

Offset address: 0x720

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IP_RELOAD0[30:15]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IP_RELOAD0[14:0]															SHUTDOWN0

Bit	Name	Access	Description	Reset value
[31:1]	IP_RELOAD0	RW	PC address register during core C0 wake-up, used for reloading the PC value after power-on wake-up and reset wake-up.	0
0	SHUTDOWN0	RW	Core C0 Deep Sleep (Lock) Unlock Register: Used to unlock a core that entered deep sleep (lock) state after power-on, allowing it to be woken by the timer: 1: Locked; 0: Unlocked.	1

9.4.5.39 PFIC Core Status Register 0 (PFIC_CSTAR0)

Offset address: 0x780

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CPU_EX_STATE_0		CPU_LOCK_UP_0	CPU_DBG_MODE_0	CPU_GLOBL_IE_0	RESERVED	CPU_IRQ_PEND_0	CPU_IRQ_ACTIVE_0	CPU_NEST_STA_0							

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:14]	CPU_EX_STATE_0	RO	Core C0 Status Register, used to obtain the operational status of core C0: 00: Running; 01: Light Sleep; 10: Deep Sleep; 11: Deep Sleep (Locked).	0
13	CPU_LOCK_UP_0	RO	Core C0 Lock Status Register, used to determine whether core C0 is in a locked state: 1: Locked; 0: Unlocked.	0
12	CPU_DBG_MODE_0	RO	Core C0 Debug Mode Register, used to determine whether the core C0 is in debug mode: 1: Core is in debug mode; 0: Core is in non-debug mode.	0
11	CPU_GLOBL_IE_0	RO	Core C0 Global Interrupt Enable Register, used to query whether the core C0 global interrupt is enabled: 1: Core global interrupt enable is active; 0: Core global interrupt enable is inactive.	0
10	Reserved	RO	Reserved	0
9	CPU_IRQ_PEND_0	RO	Core C0 Interrupt Pending Flag Register, used to check whether the core C0 has any unprocessed interrupts: 1: The core has unprocessed interrupts; 0: The core has no unprocessed interrupts.	0
8	CPU_IRQ_ACTIVE_0	RO	Core C0 Interrupt Active Flag Register, used to determine whether the core is currently handling the C0 interrupt 1: Core is handling the interrupt; 0: Core is not handling the interrupt.	0
[7:0]	CPU_NEST_STA_0	RO	The core C0 Interrupt Nesting Status Register, used to query the nesting status of	0

			core C0 interrupts. It corresponds to the nest_sta bit in the INEST_CTLR register of the CSR register.	

9.4.5.40 PFIC Event Enable Register (PFIC_EENR)

Offset address: 0xC80

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EVENTEN[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EVENTEN[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	EVENTEN	RW	31#-0# event wake-up enable: 1: Allow the corresponding event to wake up the core in sleep; 0: Mask the corresponding event, and it is not allowed to wake up the core when this event occurs. Up to 32 event sources are supported, of which 0-7 is level triggered (high active) and 8-31 is pulse triggered: 0#: Core pause request; 1#: Core interrupt request; 7#-2#: Reserved; 8#: Core enters interrupt; 9#: Core exit interrupt; 10#: A new interrupt is pending; (For shared interrupts, it is not distinguished whether the interrupt is routed to the local core.) 11#: The core sends an event; (Send event of local core) 12#: The core receives an event; (Send events from other cores) 13#: The system receives the event; (System-configured events, such as GPIO flip events ...) 31#-14#: Reserved.	0xFFFFFFFF

9.4.5.41 PFIC Event Pending Register (PFIC_EPR)

Offset address: 0xC84

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EVENT_PEND[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EVENT_PEND[15:8]								EVENT_PEND[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	EVENT_PEND	RW1Z	31#-8# event pending status, write 1 to clear it: 1: The occurrence of the corresponding event was not handled; 0: The corresponding event did not occur.	0
[7:0]	EVENT_PEND	RO	7#-0# event pending status, cannot be cleared: 1: The occurrence of the corresponding event was not handled; 0: The corresponding event did not occur.	0

9.4.5.42 PFIC Event Wakeup Register (PFIC_EWUPR)

Offset address: 0xC88

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EVENT_WUP[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EVENT_WUP[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	EVENT_WUP	RO	31#-0# event wake-up register: 1: Corresponding events wake the core from sleep; 0: Corresponding event is not the event that wakes up the core this time.	0

9.4.5.43 PFIC System Control Register (PFIC_SCTLR)

Offset address: 0xD10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SYS RST	Reserved							HART_ID							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										SET EVENT	SEV ONPEND	WFITO WFE	SLEEP DEEP	SLEEP ONEXIT	Reserved

Bit	Name	Access	Description	Reset value
31	SYSRST	WO	System reset, auto clear 0. Write 1 is valid, write 0 is invalid, same effect as PFIC_CFGR register.	0
[30:24]	Reserved	RO	Reserved	0
[23:16]	HART_ID	RO	Core ID Register, used to obtain the core ID of the register being read:	0
[15:6]	Reserved	RO	Reserved	0
5	SETEVENT	WO	Set up events that can wake up the WFE situation.	0
4	SEVONPEND	RW	When an event occurs or interrupts the pending state, the system can be woken up from the WFE instruction, and if the WFE instruction is not executed, the system will be awakened immediately after the next execution of the instruction. 1: Enabled events and all interrupts (including unopened interrupts) can wake up the system. 0: Only enabled events and enabled interrupts can wake up the system.	0
3	WFITOWFE	RW	The WFI instruction executes as WFE, which is a custom instruction. When this register is set to 1, the core enters a low-power state after executing the WFI instruction and waits for an external event to wake up the core. 1: After the WFI instruction, wait for an event to wake up the core; 0: After the WFI instruction, wait for an interrupt to wake up the core.	0
2	SLEEPDEEP	RW	Control the low-power mode of the system: 1: deepsleep 0: sleep	0
1	SLEEPONEXIT	RW	Controls the state of the system after leaving the interrupt service program: 1: The system enters low-power mode; 0: The system enters the main program.	0
0	Reserved	RO	Reserved	0

Chapter 10 GPIO and Alternate Functions (GPIO/AFIO)

The GPIO port can be configured into a variety of input or output modes, has a built-in pull-up or pull-down resistor that can be turned off, and can be configured as a push-pull or open-drain function. The GPIO port can also be reused for other functions.

10.1 Main Features

Each pin of the port can be configured into one of the following modes:

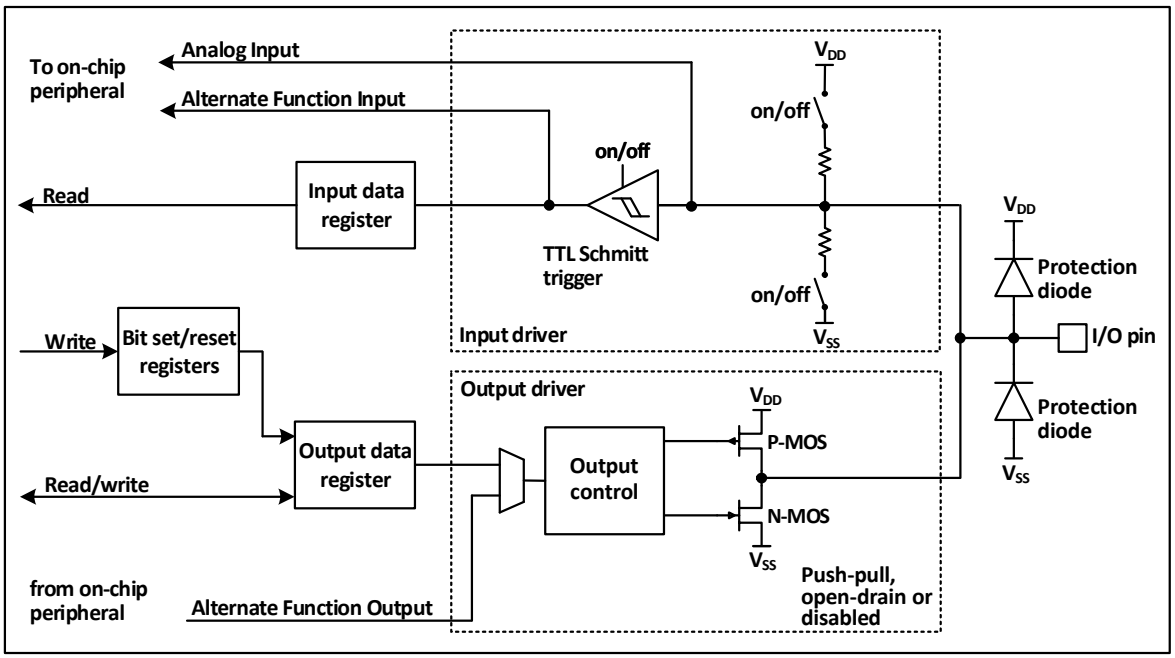
- Floating input
- Pull-up input
- Pull-down input
- Analog input
- Open-drain output
- Push-pull output
- Input and output of alternate function

Many pins have alternate functions, and many other peripherals map their own output and input channels to these pins. The specific application of these alternate pins needs to be with reference to each peripheral, and this chapter shall specify whether these pins are alternate and remapped.

10.2 Function Description

10.2.1 Overview

Figure 10-1 Basic structure of GPIO module



As shown in figure 10-1, the IO port structure, each pin has 2 protection diodes inside the chip, and the IO port can be divided into input and output drive modules. The input driver has a weak pull-up resistor, which can be connected to AD and other analog input peripherals; if you input to a digital peripheral, you need to go through a TTL Schmitt trigger, and then connect to the GPIO input register or other multiplexed peripherals. The output driver has a pair of MOS tubes, and the IO port can be configured to open leakage or push-pull

output by configuring whether the upper and lower MOS tubes are enabled or not; the output driver can also be configured to be controlled by GPIO or by other peripherals that are reused.

10.2.2 GPIO Initialization

Just after the reset, the GPIO port is running in the initial state, at this time, most IO ports are running in the floating input state, but there are also HSE and other peripheral-related pins that run on the peripheral reuse function. For specific initialization functions, please refer to the relevant sections of the pin description.

10.2.3 External Interrupt

All GPIO ports can be configured as external interrupt input channels, but each external interrupt input channel can only be mapped to one GPIO pin. Furthermore, the external interrupt channel number must correspond to the bit number of the GPIO port. For example, PA1 (or PB1, PC1, PD1, PE1, etc.) can only be mapped to EXTI1, and EXTI1 can only accept the mapping of either PA1, PB1, PC1, PD1, or PE1. Both sides maintain a one-to-one relationship.

10.2.4 Alternate Function

When using alternate functions, you must pay attention to:

- To use the alternate function of the input direction, the port must be configured in the multiplexing input mode, and the pull-down settings can be set according to actual needs.
- To use the alternate function in the output direction, the port must be configured in multiplexed output mode. Push-pull or open-drain can be set according to the actual situation.
- For the bidirectional alternate function, the port must be configured to multiplex the output mode, and then the driver is configured to float the input mode.

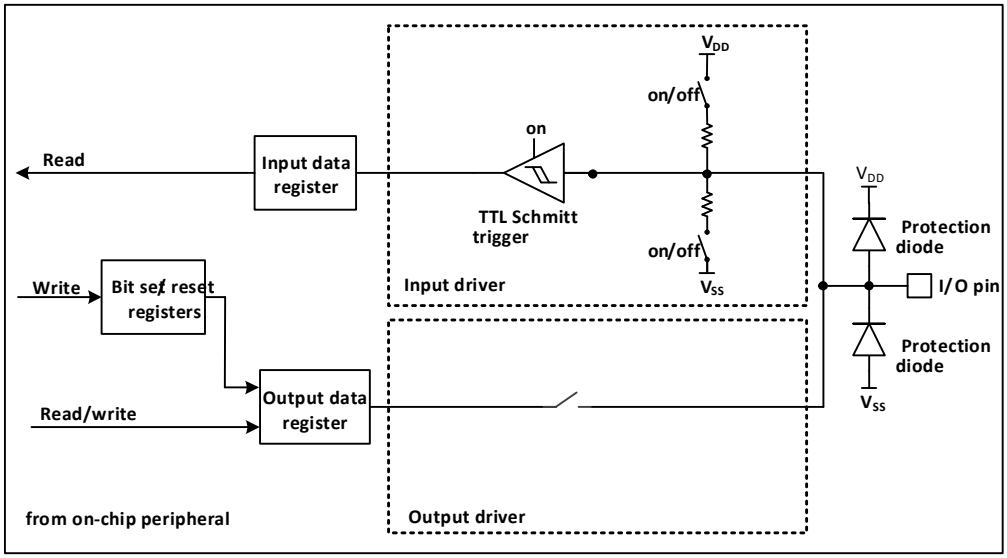
There may be multiple peripherals multiplexed to this pin in the same IO port, so in order to maximize the space for each peripheral, in addition to the default multiplexed pin, the multiplexed pin of the peripheral can be remapped to other pins to avoid the occupied pin.

10.2.5 Locking Mechanism

The locking mechanism can lock the configuration of the IO port. After a specific write sequence, the selected IO pin configuration is locked and cannot be changed until the next reset.

10.2.6 Input Configuration

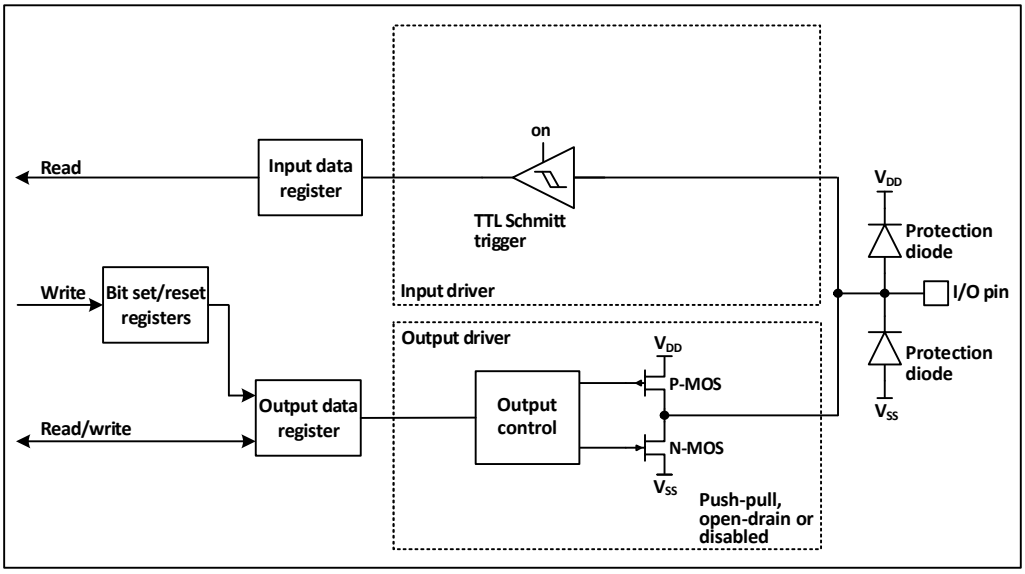
Figure 10-2 Input configuration structure block diagram of GPIO module



When the IO port is configured in input mode, the output driver is disconnected, the input up and down is optional, and the alternate function and analog input are not connected. The data on each IO port is sampled to the input data register at each PB2 clock, and the level state of the corresponding pin is obtained by reading the corresponding bit of the input data register.

10.2.7 Output Configuration

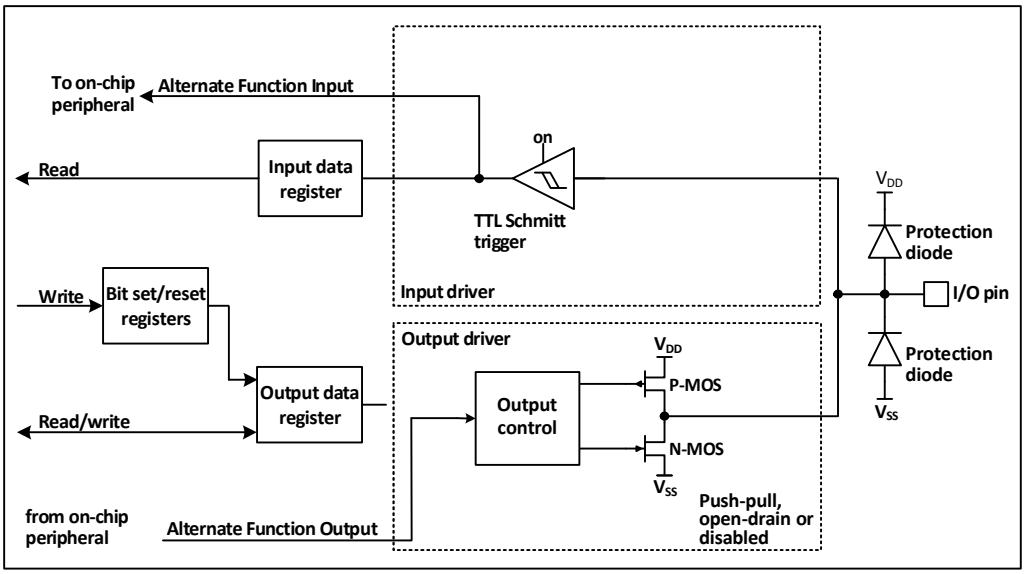
Figure 10-3 Output configuration structure block diagram of GPIO module



When the IO port is configured in output mode, a pair of MOS in the output driver can be configured in push-pull or open-drain mode as needed, without using the alternate function. The input-driven pull-up resistor is disabled, the TTL Schmitt trigger is activated, and the level that appears on the IO pin will be sampled to the input data register at each PB2 clock, so reading the input data register will get the IO state, and in push-pull output mode, access to the output data register will get the last written value.

10.2.8 Alternate Function Configuration

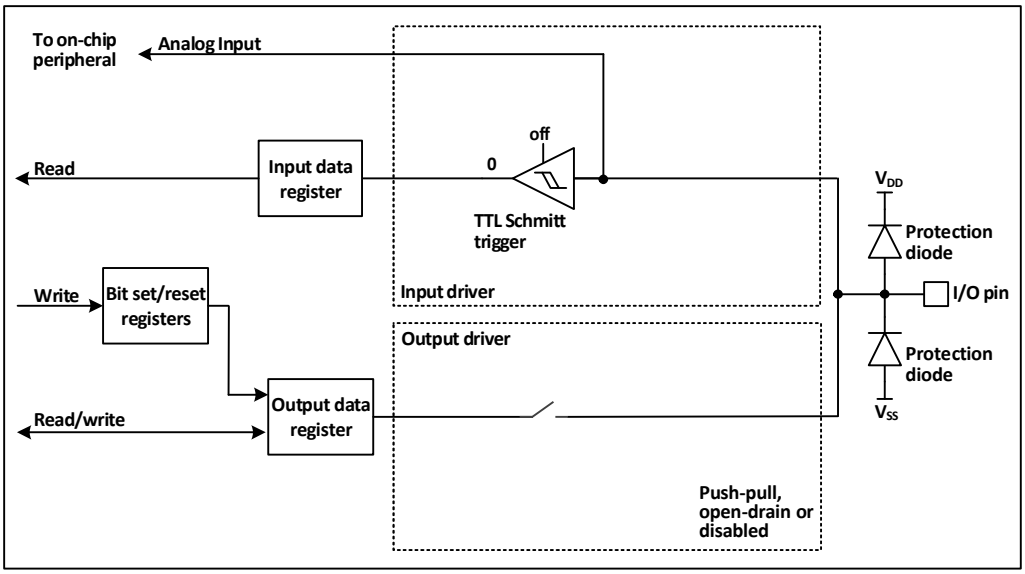
Figure 10-4 Structure block diagram when GPIO module is multiplexed by other peripherals



When the alternate function is enabled, the output driver is enabled and can be configured in open-drain or push-pull mode as needed, the Schmitt trigger is also turned on, the input and output lines of the alternate function are connected, but the output data register is disconnected, and the level that appears on the IO pin will be sampled to the input data register at each PB2 clock. In open-drain mode, reading the input data register will get the current state of the IO port. In push-pull mode, reading the output data register will get the last written value.

10.2.9 Analog Input Configuration

Figure 10-5 Configuration structure block diagram of GPIO module as analog input



When the analog input is enabled, the output buffer is disconnected, the input of the Schmitt trigger in the input driver is disabled to prevent consumption on the IO port, the pull-up resistor is prohibited, and the read input data register will always be 0.

10.2.10 Peripheral GPIO Setting

The following table recommends the corresponding GPIO port configuration of each peripheral pin.

Table 10-1 Advanced-control timer (TIM1/8)

TIM1/8	Configuration	GPIO configuration
TIM1/8_CHx	Input capture channel x	Floating input
	Output compare channel x	Push-pull alternate output
TIM1/8_CHxN	Complementary output channel x	Push-pull alternate output
TIM1/8_BKIN	Break input	Floating input
TIM1/8_ETR	External trigger clock input	Floating input

Table 10-2 General-purpose timer (TIM2/3/4/5)

TIM2/3/4/5 pin	Configuration	GPIO configuration
TIM2/3/4/5_CHx	Input capture channel x	Floating input
	Output compare channel x	Push-pull alternate output
TIM2/3/4_ETR	External trigger clock input	Floating input

Table 10-3 Universal Synchronous/Asynchronous Receiver Transmitter (USART1/2/3/4/5/6/7/8/9/10)

USART pin	Configuration	GPIO configuration
USARTx_TX	Full-duplex mode	Push-pull alternate output
	Half-duplex synchronous mode	Open-drain alternate output
USARTx_RX	Full-duplex mode	Floating input or pull-up input
	Half-duplex synchronous mode	Not used
USARTx_CK	Synchronous mode	Push-pull alternate output
USARTx_RTS	Hardware flow control	Push-pull alternate output
USARTx_CTS	Hardware flow control	Floating input or pull-up input

Table 10-4 Serial Peripheral Interface (SPI1/2/3) module

SPI pin	Configuration	GPIO configuration
SPIx_SCK	Master mode	Push-pull alternate output
	Slave mode	Floating input
SPIx_MOSI	Full-duplex master mode	Push-pull alternate output
	Full-duplex slave mode	Floating input or pull-up input
	Simplex bidirectional data line/master mode	Push-pull alternate output
	Simplex bidirectional data line/slave mode	Not used
SPIx_MISO	Full duplex master mode	Floating input or pull-up input
	Full duplex slave mode	Push-pull alternate output
	Simplex bidirectional data line/master mode	Not used
	Simplex bidirectional data line/slave mode	Push-pull alternate output
SPIx_NSS	Hardware master or slave mode	Floating input or pull-up or pull-down input

	Hardware master mode/NSS output enable	Push-pull alternate output
	Software mode	Not used

Table 10-5 Built-in audio bus (I2S2/3) module

I2Sx pin	Configuration	GPIO configuration
I2Sx_WS	Master mode	Push-pull alternate output
	Slave mode	Floating input
I2Sx_CK	Master mode	Push-pull alternate output
	Slave mode	Floating input
I2Sx_SD	Transmitter	Push-pull alternate output
	Receiver	Floating input or pull-up or pull-down input
I2Sx_MCK	Master mode	Push-pull alternate output
	Slave mode	Not used

Table 10-6 Inter IC Sound (I2C) module

I2C pin	Configuration	GPIO configuration
I2C_SCL	I2C clock	Open-drain alternate output
I2C_SDA	I2C data	Open-drain alternate output

Table 10-7 Inter IC Sound (I3C) module

I3C pin	Configuration	GPIO configuration
I3C_SCL	I3C clock	Push-pull alternate output
I3C_SDA	I3C data	Push-pull alternate output

Table 10-8 Controller LAN (CAN) module

CAN pin	GPIO configuration
CAN1_TX	Push-pull alternate output
CAN1_RX	Floating input or pull-up input

Table 10-9 Digital image interface (DVP) module

DVP pin	Configuration	GPIO configuration
PCLK	Pixel clock input	Floating input or pull-up input
DATA[11:0]	Pixel data input	Floating input or pull-up input
VSYNC	Frame synchronization signal input	Floating input or pull-up input
HSYNC	Line synchronization signal input	Floating input or pull-up input

Table 10-10 LCD-TFT interface (LTDC) module

LTDC pin	Configuration	GPIO configuration
LCD_CLK	Clock output	Push-pull alternate output
LCD_VSYNC	Vertical synchronous output	Push-pull alternate output
LCD_HSYNC	Horizontal synchronous output	Push-pull alternate output

LCD_DE	Non-data enabled output	Push-pull alternate output
LCD_R[7:0]	Data: 8-bit red data output	Push-pull alternate output
LCD_G[7:0]	Data: 8-bit green data output	Push-pull alternate output
LCD_B[7:0]	Data: 8-bit blue data output	Push-pull alternate output

Note: The LCD-TFT controller pin must be configured through the user program. Unused pins can be used for other functions.

For LTDC outputs up to 24 bits (RGB888), when outputting RGB565 or RGB666 to 16-bit or 18-bit displays using a pixel depth lower than 8bpp, the RGB display data lines must be connected to the MSB of the LCD-TFT controller's RGB data lines. For example, when the LCD-TFT controller is connected to an RGB565 16-bit display, the LCD display's R[4:0], G[4:0], and B[4:0] data line pins must connect to the LCD-TFT controller's LCD_R[7:3], LCD_G[7:3], and LCD_B[7:3] pins.

Table 10-11 SDIO controller module

SDIO pin	Configuration	GPIO configuration
SDIO_CK	Clock	Push-pull alternate output
SDIO_CMD	Command	Push-pull alternate output
SDIO[D7:D0]	Data	Push-pull alternate output

Table 10-12 Flexible Static Memory Controller (FSMC) Controller

FSMC pin	GPIO configuration
FSMC_A[23:16] FSMC_D[15:0]	Push-pull alternate output
FSMC_CK	Push-pull alternate output
FSMC_NOE FSMC_NWE	Push-pull alternate output
FSMC_NE1 FSMC_NCE2	Push-pull alternate output
FSMC_NWAIT	Floating input or pull-up input
FSMC_NBL[1:0]	Push-pull alternate output

Table 10-13 Analog-to-digital converter (ADC) and digital-to-analog converter (DAC)

ADC/DAC pin	GPIO configuration
ADC/DAC	Analog input

Table 10-14 Other IO function settings

Pin	Configuration function	GPIO configuration
TAMPER_RTC	RTC output	Hardware automatic settings
	Intrusion event input	
MCO	Clock output	Push-pull alternate output
EXTI	External interrupt input	Floating input or pull-up or pull-down input

10.2.11 Alternate Function Remapping GPIO Settings

10.2.11.1 OSC32_IN/OSC32_OUT as GPIO Port PC14/PC15

When LSEON=0, the LSE oscillator pin OSC32_IN/OSC32_OUT can be used as PC14/PC15 of GPIO respectively.

When LSEON=1, it is used as LSE pin.

10.2.11.2 OSC_IN/OSC_OUT as GPIO port PD0/PD1

OSC_IN/OSC_OUT can be used as GPIO PD0/PD1, which is realized by setting Remap Register 1 (AFIO_PCFR1).

10.2.11.3 TIM Alternate Function Remapping

Table 10-15 TIM1 alternate function remapping

Alternate function	TIM1_RM=00 Default mapping	TIM1_RM=01 Remapping	TIM1_RM=11 Remapping
TIM1_ETR	PA12	PA12	PE7
TIM1_CH1	PA8	PA8	PE9
TIM1_CH2	PA9	PA9	PE11
TIM1_CH3	PA10	PA10	PB8
TIM1_CH4	PA11	PA11	PA7
TIM1_BKIN	PB12	PA6	PB10
TIM1_CH1N	PB13	PA7	PE8
TIM1_CH2N	PB14	PB0	PE10
TIM1_CH3N	PB15	PB1	PE12

Table 10-16 TIM2 alternate function remapping

Alternate function	TIM2_RM=00 Default mapping	TIM2_RM=01 Remapping	TIM2_RM=10 Remapping	TIM2_RM=11 Remapping
TIM2_ETR	PA0	PA15	PA0	PA15
TIM2_CH1	PA0	PA15	PA0	PA15
TIM2_CH2	PA1	PB3	PA1	PB3
TIM2_CH3	PA2	PA2	PB10	PB10
TIM2_CH4	PA3	PA3	PB11	PB11

Table 10-17 TIM3 alternate function remapping

Alternate function	TIM3_RM=00 Default mapping	TIM3_RM=10 Remapping	TIM3_RM=11 Remapping
TIM3_ETR	PD2	PD2	PD2
TIM3_CH1	PA6	PB4	PC6
TIM3_CH2	PA7	PB5	PC7
TIM3_CH3	PB0	PB0	PC8
TIM3_CH4	PB1	PB1	PC9

Table 10-18 TIM4 alternate function remapping

Alternate	TIM4_RM=0	TIM4_RM=1

function	Default mapping	Remapping
TIM4_ETR	PE0	PE0
TIM4_CH1	PB6	PD12
TIM4_CH2	PB7	PD13
TIM4_CH3	PB8	PD14
TIM4_CH4	PB9	PD15

Table 10-19 TIM5 alternate function remapping

Alternate function	Default mapping	
TIM5_CH1	PA0	
TIM5_CH2	PA1	
TIM5_CH3	PA2	
Alternate function	TIM5CH4_RM=0 Default mapping	TIM5CH4_RM=1 Remapping
TIM5_CH4	PA3	LSI internal clock

Table 10-20 TIM8 alternate function remapping

Alternate function	TIM8_RM=0 Default mapping	TIM8_RM=1 Remapping
TIM8_ETR	PA0	PA0
TIM8_CH1	PC6	PB6
TIM8_CH2	PC7	PB7
TIM8_CH3	PC8	PB8
TIM8_CH4	PC9	PC13
TIM8_BKIN	PA6	PB9
TIM8_CH1N	PA7	PA13
TIM8_CH2N	PB0	PA14
TIM8_CH3N	PB1	PA15

10.2.11.4 USART Alternate Function Remapping

Table 10-21 USART1 alternate function remapping

Alternate function	USART1_RM1=0(1) USART1_RM=0(2) Default mapping	USART1_RM1=0(1) USART1_RM=1(2) Remapping	USART1_RM1=1(1) USART1_RM=0(2) Remapping	USART1_RM1=1(1) USART1_RM=1(2) Remapping
USART1_CK	PA8	PA8	PA10	PA5
USART1_TX	PA9	PB6	PB15	PA6
USART1_RX	PA10	PB7	PA8	PA7
USART1_CTS	PA11	PA11	PA5	PC4
USART1_RTS	PA12	PA12	PA9	PC5

Note: (1) USART1_RM is bit 2 of the AFIO_PCFR1 register, which is the low-order bit of the mapping configuration.

(2) USART1_RM1 is bit 26 of the AFIO_PCFR2 register, which is the high-order bit of the mapping

configuration.

Table 10-22 USART2 alternate function remapping

Alternate function	USART2_RM=0 Default mapping	USART2_RM=1 Remapping
USART2_CTS	PA0	PD3
USART2_RTS	PA1	PD4
USART2_TX	PA2	PD5
USART2_RX	PA3	PD6
USART2_CK	PA4	PD7

Table 10-23 USART3 alternate function remapping

Alternate function	USART3_RM=00 Default mapping	USART3_RM=01 Remapping	USART3_RM=10 Remapping	USART3_RM=11 Remapping
USART3_TX	PB10	PC10	PA13	PD8
USART3_RX	PB11	PC11	PA14	PD9
USART3_CK	PB12	PC12	PD10	PD10
USART3_CTS	PB13	PB13	PD11	PD11
USART3_RTS	PB14	PB14	PD12	PD12

Table 10-24 USART4 alternate function remapping

Alternate function	USART4_RM=00 Default mapping	USART4_RM=01 Remapping	USART4_RM=1x Remapping
USART4_TX	PC10	PB0	PE0
USART4_RX	PC11	PB1	PE1
USART4_CK	PC12	PB2	PE12
USART4_CTS	PA13	PC4	PB8
USART4_RTS	PD2	PC5	PB9

Table 10-25 USART5 alternate function remapping

Alternate function	USART5_RM=00 Default mapping	USART5_RM=01 Remapping	USART5_RM=1x Remapping
USART5_TX	PC12	PB4	PE8
USART5_RX	PD2	PB5	PE9
USART5_CK	PB5	PB3	PB2
USART5_CTS	PB3	PB8	PB0
USART5_RTS	PB4	PB9	PB1

Table 10-26 USART6 alternate function remapping

Alternate function	USART6_RM=00 Default mapping	USART6_RM=01 Remapping	USART6_RM=10 Remapping	USART6_RM=11 Remapping

USART6_TX	PC0	PB8	PE10	PC0
USART6_RX	PC1	PB9	PE11	PC1
USART6_CK	PB0	PB5	PA13	PA5
USART6_CTS	PC2	PE0	PA14	PC2
USART6_RTS	PC3	PE1	PA15	PC3

Table 10-27 USART7 alternate function remapping

Alternate function	USART7_RM=00 Default mapping	USART7_RM=01 Remapping	USART7_RM=10 Remapping	USART7_RM=11 Remapping
USART7_TX	PC2	PA6	PA9	PA9
USART7_RX	PC3	PA7	PA10	PA10
USART7_CK	PC1	PB0	PA8	PA8
USART7_CTS	PC4	PC4	PB6	PC8
USART7_RTS	PC5	PC5	PB7	PC9

Table 10-28 USART8 alternate function remapping

Alternate function	USART8_RM=00 Default mapping	USART8_RM=01 Remapping	USART8_RM=1x Remapping
USART8_TX	PC4	PA14	PB15
USART8_RX	PC5	PA15	PB14
USART8_CK	PA7	PB15	PB11
USART8_CTS	PB0	PE10	PB13
USART8_RTS	PB1	PE11	PB12

Table 10-29 USART9 alternate function remapping

Alternate function	USART9_RM=00 Default mapping	USART9_RM=01 Remapping	USART9_RM=1x Remapping
USART9_TX	PB1	PA4	PE3
USART9_RX	PB2	PA5	PE4
USART9_CK	PD8	PD8	PE2
USART9_CTS	PE8	PE8	PE5
USART9_RTS	PE9	PE9	PE6

Table 10-30 USART10 alternate function remapping

Alternate function	USART10_RM=00 Default mapping	USART10_RM=01 Remapping	USART10_RM=1x Remapping
USART10_TX	PE0	PB8	PD11
USART10_RX	PE1	PB9	PD12
USART10_CK	PE12	PE12	PD10
USART10_CTS	PC14	PC14	PD13
USART10_RTS	PC15	PC15	PD14

10.2.11.5 SPI Alternate Function Remapping

Table 10-31 SPI1 alternate function remapping

Alternate function	SPI1_RM=0 Default mapping	SPI1_RM=1 Remapping
SPI1_NSS	PA4	PA15
SPI1_SCK	PA5	PB3
SPI1_MISO	PA6	PB4
SPI1_MOSI	PA7	PB5

Table 10-32 SPI3 alternate function remapping

Alternate function	SPI3_RM=0 Default mapping	SPI3_RM=1 Remapping
SPI3_NSS	PA15	PA4
SPI3_SCK	PB3	PC10
SPI3_MISO	PB4	PC11
SPI3_MOSI	PB5	PC12

Table 10-33 I2S3 alternate function remapping

Alternate function	SPI3_RM=0 Default mapping	SPI3_RM=1 Remapping
I2S3_WS	PA15	PA4
I2S3_CK	PB3	PC10
I2S3_MCK	PC7	PC7
I2S3_SD	PB5	PC12

Note: I2S mode is selected when bit 1 of the I2SMOD field in the SPI_I2S_CFGR register is set.

10.2.11.6 I2C Alternate Function Remapping

Table 10-34 I2C alternate function remapping

Alternate function	I2C_RM=0 Default mapping	I2C_RM=1 Remapping
I2C_SCL	PB6	PB8
I2C_SDA	PB7	PB9

10.2.11.7 CAN Alternate Function Remapping

Table 10-35 CAN1 alternate function remapping

Alternate function	CAN1_RM=00 Default mapping	CAN1_RM=10 Remapping	CAN1_RM=11 Remapping
CAN1_RX	PA11	PB8	PD0
CAN1_TX	PA12	PB9	PD1

10.2.11.8 ADC Alternate Function Remapping

Table 10-36 ADC1 external trigger injection conversion alternate function remapping

Alternate function	ADC1_ETRGINJ_RM=0 Default mapping	ADC1_ETRGINJ_RM=1 Remapping
ADC1 external trigger injection conversion	ADC1 external trigger injection conversion is connected to EXTI15	ADC1 external trigger injection conversion is connected to TIM8_CH4

Table 10-37 ADC1 external trigger rule conversion alternate function remapping

Alternate function	ADC1_ETRGREG_RM=0 Default mapping	ADC1_ETRGREG_RM=1 Remapping
ADC1 external trigger rule conversion	ADC1 external trigger rule conversion is connected to EXTI11	ADC1 external trigger rule conversion is connected to TIM8_TRGO

Table 10-38 ADC2 external trigger injection conversion alternate function remapping

Alternate function	ADC2_ETRGINJ_RM=0 Default mapping	ADC2_ETRGINJ_RM=1 Remapping
ADC2 external trigger injection conversion	ADC1 external trigger injection conversion is connected to EXTI15	ADC1 external trigger injection conversion is connected to TIM8_CH4

Table 10-39 ADC2 external trigger rule conversion alternate function remapping

Alternate function	ADC2_ETRGREG_RM=0 Default mapping	ADC2_ETRGREG_RM=1 Remapping
ADC2 external trigger rule conversion	ADC2 external trigger rule conversion is connected to EXTI11	ADC2 external trigger rule conversion is connected to TIM8_TRGO

Note: ADC1 remapping only supports products with TIM8. For details, see the relevant datasheet.

10.2.11.9 FSMC Alternate Function Remapping

Table 10-40 FSMC alternate function remapping

Alternate function	FSMCEN=0 Default mapping	FSMCEN=1 Remapping
NBL[0]	PE0	PE0
NBL[1]	PE1	PE1
CLK	PD3	PD3
NOE	PD4	PD4
NWE	PD5	PD5
NWAIT	PD6	PD6
NE1/NCE2	PD7	PD7

NADV	PB7	PB7
A16	PD11	PD11
A17	PD12	PD12
A18	PD13	PD13
A19	PE3	PE3
A20	PE4	PE4
A21	PE5	PE5
A22	PE6	PE6
A23	PE2	PE2
D0	PD14	PD14
D1	PD15	PD15
D2	PD0	PD0
D3	PD1	PD1
D4	PE7	PC4
D5	PE8	PE8
D6	PE9	PE9
D7	PE10	PE10
D8	PE11	PE11
D9	PB11	PB11
D10	PB12	PB12
D11	PB13	PB13
D12	PB14	PB14
D13	PD8	PD8
D14	PD9	PD9
D15	PD10	PD10

10.2.11.10 FSMC_NADV Alternate Function Remapping

Table 10-41 FSMC_NADV alternate function remapping

Alternate function(1)	FSMCEN=1 Default mapping	FSMCEN=1&USBHS2EN=1 &RB_UD_RST_SIE=0
FSMC_NADV	PB7	PD2

Note: (1) When FSMC_NADV=1, the FSMC_NADV output is disabled.

10.2.11.11 DVP Alternate Function Remapping

Table 10-42 DVP alternate function remapping

Alternate function	DVPEN=1 DVP_RM=0 Default mapping	DVPEN=1&USBHS1EN=1 &RB_UD_RST_SIE=0	DVP_RM=1 Remapping
PCLK	PA6	PA6	PA6

HSYNC	PA4	PA4	PA4
VSYNC	PA5	PA5	PA5
D0	PA9	PA9	PA9
D1	PA10	PA10	PA10
D2	PC8	PC8	PA8
D3	PC9	PC9	PB4
D4	PC11	PC11	PC11
D5	PB6	PB3	PB3
D6	PB8	PB8	PB8
D7	PB9	PB9	PB9
D8	PC10	PC10	PC10
D9	PC12	PC12	PC12
D10	PD6	PD6	PB5
D11	PD2	PD2	PD2

10.2.11.12 SDIO Alternate Function Remapping

Table 10-43 SDIO alternate function remapping

Alternate function	SDIO_RM=0 Default mapping	SDIOEN=1&ETHMACEN=1	SDIO_RM=1 Remapping
CK	PC12	PC12	PC12
CMD	PD2	PD2	PD2
SD0	PC8	PB14	PB14
SD1	PC9	PB15	PB15
SD2	PC10	PC10	PC10
SD3	PC11	PC11	PC11
SD4	PB8	PB8	PB8
SD5	PB9	PB9	PB9
SD6	PC6	PC6	PA9
SD7	PC7	PC7	PA10

10.2.11.13 I3C Alternate Function Remapping

Table 10-44 I3C alternate function remapping

Alternate function	I3C_RM=0 Default mapping	I3C_RM=1 Remapping
I3C_SCL	PB10	PB4
I3C_SDA	PB11	PB5

10.2.11.14 LTDC Alternate Function Remapping

Table 10-45 LTDC alternate function remapping

Alternate function	LTDC_RM=0 Default mapping	LTDC_RM=1 Remapping
CLK	PE1	PA10
HSYNC	PB4	PB8
VSYNC	PB5	PB9
DE	PE0	PA9
R[0]	PC0	PC0
R[1]	PC1	PC1
R[2]	PC2	PC2
R[3]	PC3	PC3
R[4]	PA1	PA1
R[5]	PA2	PA2
R[6]	PA3	PA3
R[7]	PA7	PA7
G[0]	PC4	PC4
G[1]	PC5	PC5
G[2]	PB0	PB0
G[3]	PB1	PB1
G[4]	PB2	PB3
G[5]	PD8	PA4
G[6]	PB12	PB12
G[7]	PB13	PB13
B[0]	PB14	PB14
B[1]	PB15	PB15
B[2]	PB11	PB11
B[3]	PB10	PB10
B[4]	PA8	PA8
B[5]	PA13	PA13
B[6]	PA14	PA14
B[7]	PA15	PA15

10.3 Register Description

10.3.1 GPIO Registers

Unless otherwise specified, the GPIO registers must be operated in words (operate these registers in 32 bits).

Table 10-46 GPIO-related registers

Name	Access address	Description	Reset value
R32_GPIOA_CFGLR	0x40010800	PA port configuration register low	0x44444444
R32_GPIOB_CFGLR	0x40010C00	PB port configuration register low	0x44444444
R32_GPIOC_CFGLR	0x40011000	PC port configuration register low	0x44444444
R32_GPIOD_CFGLR	0x40011400	PD port configuration register low	0x44444444
R32_GPIOE_CFGLR	0x40011800	PE port configuration register low	0x44444444
R32_GPIOA_CFGHR	0x40010804	PA port configuration register high	0x44444444

R32_GPIOB_CFGHR	0x40010C04	PB port configuration register high	0x44444444
R32_GPIOC_CFGHR	0x40011004	PC port configuration register high	0x44444444
R32_GPIOD_CFGHR	0x40011404	PD port configuration register high	0x44444444
R32_GPIOE_CFGHR	0x40011804	PE port configuration register high	0x44444444
R32_GPIOA_INDR	0x40010808	PA port input data register	0x0000XXXX
R32_GPIOB_INDR	0x40010C08	PB port input data register	0x0000XXXX
R32_GPIOC_INDR	0x40011008	PC port input data register	0x0000XXXX
R32_GPIOD_INDR	0x40011408	PD port input data register	0x0000XXXX
R32_GPIOE_INDR	0x40011808	PE port input data register	0x0000XXXX
R32_GPIOA_OUTDR	0x4001080C	PA port output data register	0x00000000
R32_GPIOB_OUTDR	0x40010C0C	PB port output data register	0x00000000
R32_GPIOC_OUTDR	0x4001100C	PC port output data register	0x00000000
R32_GPIOD_OUTDR	0x4001140C	PD port output data register	0x00000000
R32_GPIOE_OUTDR	0x4001180C	PE port output data register	0x00000000
R32_GPIOA_BSHR	0x40010810	PA port set/reset register	0x00000000
R32_GPIOB_BSHR	0x40010C10	PB port set/reset register	0x00000000
R32_GPIOC_BSHR	0x40011010	PC port set/reset register	0x00000000
R32_GPIOD_BSHR	0x40011410	PD port set/reset register	0x00000000
R32_GPIOE_BSHR	0x40011810	PE port set/reset register	0x00000000
R32_GPIOA_BCR	0x40010814	PA port reset register	0x00000000
R32_GPIOB_BCR	0x40010C14	PB port reset register	0x00000000
R32_GPIOC_BCR	0x40011014	PC port reset register	0x00000000
R32_GPIOD_BCR	0x40011414	PD port reset register	0x00000000
R32_GPIOE_BCR	0x40011814	PE port reset register	0x00000000
R32_GPIOA_LCKR	0x40010818	PA port lock configuration register	0x00000000
R32_GPIOB_LCKR	0x40010C18	PB port lock configuration register	0x00000000
R32_GPIOC_LCKR	0x40011018	PC port lock configuration register	0x00000000
R32_GPIOD_LCKR	0x40011418	PD port lock configuration register	0x00000000
R32_GPIOE_LCKR	0x40011818	PE port lock configuration register	0x00000000

10.3.1.1 GPIO Configuration Register Low (GPIOx_CFGLR) (x=A/B/C/D/E)

Offset address: 0x00

31 30	29 28	27 26	25 24	23 22	21 20	19 18	17 16
CNF7[1:0]	MODE7[1:0]	CNF6[1:0]	MODE6[1:0]	CNF5[1:0]	MODE5[1:0]	CNF4[1:0]	MODE4[1:0]
15 14	**13 12**	**11 10**	**9 8**	**7 6**	**5 4**	**3 2**	**1 0**
CNF3[1:0]	MODE3[1:0]	CNF2[1:0]	MODE2[1:0]	CNF1[1:0]	MODE1[1:0]	CNF0[1:0]	MODE0[1:0]

Bit	Name	Access	Description	Reset value
[31:30] [27:26] [23:22] [19:18]	CNFy[1:0]	RW	(y=0-7), the configuration bits of port x, through which the corresponding ports are configured. When entering a mode (MODE=00b): 00: Analog input mode.	01b

[15:14] [11:10] [7:6] [3:2]			01: Floating input mode. 10: With pull-up mode. 11: Reserved. In output mode (MODE > 00b): 00: Universal push-pull output mode. 01: Universal open-drain output mode. 10: Alternate function push-pull output mode. 11: Alternate function open-leak output mode.	
[29:28] [25:24] [21:20] [17:16] [13:12] [9:8] [5:4] [1:0]	MODEy[1:0]	RW	(y=0-7), port x mode selection, configure the corresponding port through these bits. 00: Input mode. 01: Output mode, maximum speed 10MHz. 10: Output mode, maximum speed 2MHz. 11: Output mode, maximum speed 50MHz.	00b

10.3.1.2 GPIO Configuration Register High (GPIOx_CFGHR) (x=A/B/C/D/E)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNF15[1:0]		MODE15[1:0]		CNF14[1:0]		MODE14[1:0]		CNF13[1:0]		MODE13[1:0]		CNF12[1:0]		MODE12[1:0]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNF11[1:0]		MODE11[1:0]		CNF10[1:0]		MODE10[1:0]		CNF9[1:0]		MODE9[1:0]		CNF8[1:0]		MODE8[1:0]	

Bit	Name	Access	Description	Reset value
[31:30] [27:26] [23:22] [19:18] [15:14] [11:10] [7:6] [3:2]	CNFy[1:0]	RW	(y=8-15), the configuration bits for port x, by which the corresponding port is configured. When in input mode (MODE=00b): 00: Analog input mode; 01: Float input mode; 10: With pull-up and pull-down modes; 11: Reserved. In output mode (MODE>00b): 00: General-purpose push-pull output mode; 01: General-purpose open-drain output mode; 10: Alternate function push-pull output mode; 11: Alternate function open-drain output mode.	01b
[29:28] [25:24] [21:20] [17:16] [13:12] [9:8] [5:4] [1:0]	MODEy[1:0]	RW	(y=8-15), the mode bits of port x, by which the corresponding port is configured. 00: Input mode; 01: Output mode, maximum speed 10MHz. 10: Output mode, maximum speed 2MHz; 11: Output mode, maximum speed 50MHz.	00b

10.3.1.3 Port Input Data Register (GPIOx_INDR) (x=A/B/C/D/E)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDR15	IDR14	IDR13	IDR12	IDR11	IDR10	IDR9	IDR8	IDR7	IDR6	IDR5	IDR4	IDR3	IDR2	IDR1	IDR0

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	IDRy	RO	(y=0-15), port input data. These bits are read-only and can only be read out in 16-bit form. The value read out is the high and low state of the corresponding bit.	X

10.3.1.4 Port Output Data Register (GPIOx_OUTDR) (x=A/B/C/D/E)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ODR15	ODR14	ODR13	ODR12	ODR11	ODR10	ODR9	ODR8	ODR7	ODR6	ODR5	ODR4	ODR3	ODR2	ODR1	ODR0

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	ODRy	RW	For output mode: (y=0-15). the port outputs data. This data can only be manipulated in 16-bit form. the IO port externally outputs the values of these registers. For input modes with pull-down: 0: Pull-down input; 1: Pull-up input.	0

10.3.1.5 Port Set/Reset Register (GPIOx_BSHR) (x=A/B/C/D/E)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0

Bit	Name	Access	Description	Reset value
[31:16]	BRy	WO	(y=0-15), the corresponding OUTDR bits are	0

			cleared for these location bits; writing 0 has no effect. These bits can only be accessed in 16-bit format. If both BR and BS bits are set, the BS bit acts.	
[15:0]	BSy	WO	(y=0-15), for which positional bits cause the corresponding OUTDR positional bits, and writing 0 has no effect. These bits can only be accessed in 16-bit format. If both BR and BS bits are set, the BS bit acts.	0

10.3.1.6 Port Reset Register (GPIOx_BCR) (x=A/B/C/D/E)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	BRy	WO	(y=0-15), the corresponding OUTDR bits are cleared for these location bits; writing 0 has no effect. These bits can only be accessed in 16-bit form.	0

10.3.1.7 Lock Configuration Register (GPIOx_LCKR) (x=A/B/C/D/E)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															LCKK
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LCK15	LCK14	LCK13	LCK12	LCK11	LCK10	LCK9	LCK8	LCK7	LCK6	LCK5	LCK4	LCK3	LCK2	LCK1	LCK0

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	LCKK	RW	Lock key, which can be locked by writing a specific sequence, but it can be read out at any time. When it reads 0, it means that the lock is not in effect, and read 1 means that the lock is in effect. The write sequence of the lock key is: write 1-write 0-write 1-read 0-read 1, the last step is not necessary, but can be used to confirm that the lock key has been activated. Any error when writing a sequence does not activate a lock, and the value of LCK[15:0] cannot be changed	0

			when writing a sequence. After the lock is in effect, the configuration of the port can be changed only after the next reset.	
[15:0]	LCKy	RW	(y=0-15), where 1 indicates that the configuration of the corresponding port is locked. These bits can only be changed before the LCKK is locked. Locked configurations refer to the configuration registers GPIOx_CFGLR and GPIOx_CFGHR.	0

Note: After the LOCK sequence is performed on the corresponding port bit, the configuration of the port bit cannot be changed until the next system reset.

10.3.2 AFIO Registers

Unless otherwise specified, the AFIO registers must be operated in words (operate these registers in 32 bits).

Table 10-47 AFIO-related registers

Name	Access address	Description	Reset value
R32_AFIO_ECR	0x40010000	Event control register	0x00000000
R32_AFIO_PCFR1	0x40010004	Remap register 1	0x00000000
R32_AFIO_EXTICR1	0x40010008	External interrupt configuration register 1	0x00000000
R32_AFIO_EXTICR2	0x4001000C	External interrupt configuration register 2	0x00000000
R32_AFIO_EXTICR3	0x40010010	External interrupt configuration register 3	0x00000000
R32_AFIO_EXTICR4	0x40010014	External interrupt configuration register 4	0x00000000
R32_AFIO_PCFR2	0x4001001C	Remap register 2	0x00000000

10.3.2.1 Event Control Register (AFIO_ECR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								EVOE	PORT[2:0]			PIN[3:0]			

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
7	EVOE	RW	Allow event output bit, setting this bit will connect the EVENTOUT of the core to the IO port selected by PORT and PIN.	0
[6:4]	PORT[2:0]	RW	Used to select the port for core output EVENTOUT: 000: Select PA port; 001: Select PB port; 010: Select PC port; 011: Select PD port; 100: Select PE port; Others: Reserved.	000b
[3:0]	PIN[3:0]	RW	The value of this bit is used to determine the specific pin number that selects the core output EVENTOUT to the port. Values 0-15 correspond	0

			to pins 0-15 of the Px selected in PORT.	

10.3.2.2 Remap Register 1 (AFIO_PCFR1)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ETH PHY _LED _RM	PTP_ PPS_ RM	TIM2 ITR1 _RM	SPI3_ RM	Reser ved	SW_CFG[2:0]			Reserved			ADC 2_ET RGR EG_R M	ADC 2_ET RGIN J_RM	ADC 1_ET RGR EG_R M	ADC 1_ET RGIN J_RM	TIM5 CH4_ RM

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PDP D1 _RM	CAN1_RM [1:0]		TIM4 _RM	TIM3_RM [1:0]		TIM2_RM [1:0]		TIM1_RM [1:0]		USART3_RM [1:0]		USAR T2 _RM	USAR T1 _RM	I2C _RM	SPI1 _RM

Bit	Name	Access	Description	Reset value
31	ETHPHY_LED_RM	RW	ETHPHY LED remapping: 1：LED0:PD14 LED1:PD15; 0：LED0:PE8 LED1:PE9.	0
30	PTP_PPS_RM	RW	Ethernet PTP PPS remapping: 1: PTP PPS is output to pin PB5; 0: PTP PPS is not output to pin PB5.	0
29	TIM2ITR1_RM	RW	TIM2 internal trigger 1 remapping: 1: Reserved; 0: Internally connect TIM2_ITR1 to the Ethernet PTP output.	0
28	SPI3_RM	RW	SPI3 remapping: 1: Remapping (NSS/PA4, SCK/PC10, MISO/PC11, MOSI/PC12); 0: Default mapping (NSS/PA15, SCK/PB3, MISO/PB4, MOSI/PB5).	0
27	Reserved	RO	Reserved	0
[26:24]	SW_CFG[2:0]	RW	These bits are used to configure the IO port of the SW function and tracking function. SWD (SDI) is the debug interface for accessing the core. After system reset, it always acts as SWD port. 0xx: Enable SWD (SDI); 100: Disable SWD (SDI), as GPIO function. Others: Reserved.	0
[23:21]	Reserved	RO	Reserved	0
20	ADC2_ETRGREG_ RM	RW	Remap bit for ADC2 external trigger rule conversion. 1: ADC2 external trigger rule conversion is connected to TIM8_TRGO; 0: ADC2 external trigger rule conversion is connected to EXTI11.	0
19	ADC2_ETRGINJ_R	RW	ADC2 external trigger injects the remapped bit into	0

	M		the conversion. 1: ADC2 external trigger injection conversion is connected to TIM8_CH4; 0: ADC2 external trigger injection conversion is connected to EXTI15.	
18	ADC1_ETRGREG_RM	RW	ADC1 external trigger rule conversion remap bit. 1: ADC1 external trigger rule conversion is connected to TIM8_TRGO; 0: ADC1 external trigger rule conversion is connected to EXTI11.	0
17	ADC1_ETRGINJ_RM	RW	The ADC1 external trigger injects the remapped bit of the conversion. 1: ADC1 external trigger injection conversion is connected to TIM8_CH4; 0: ADC1 external trigger injection conversion is connected to EXTI15.	0
16	TIM5CH4_RM	RW	Remapping of timer 5 channel 4: 1: Remapping, timer 5 channel 4 is mapped to the LSI internal clock; 0: Default mapping, timer 5 channel 4 is mapped to PA3 by default.	0
15	PD0PD1_RM	RW	Pin PD0&PD1 remap bit, this bit can be read and written by the user. It controls whether the GPIO functions of PD0 and PD1 are remapped, that is, PD0&PD1 is mapped to OSC_IN&OSC_OUT. 1: The pin is used as a GPIO port; 0: The pin is used as a crystal oscillator pin. *Note: For the LQFP100 package, since PD0 and PD1 are inherent function pins, there is no need to reimage settings by software.*	0
[14:13]	CAN1_RM[1:0]	RW	The CAN1 alternate function remaps the bits and these bits can be read and written by the user. Control remapping of CAN1_RX and CAN1_TX: 00: CAN1_RX is mapped to PA11, and CAN1_TX is mapped to PA12; 10: CAN1_RX is mapped to PB8, and CAN1_TX is mapped to PB9; 01: Reserved; 11: CAN1_RX is mapped to PD0, and CAN1_TX is mapped to PD1.	0
12	TIM4_RM	RW	The remapped bits of Timer 4, this bit can be read and written by the user. It controls the remapping of channels 1 to 4 of timer 4 on the GPIO port: 1: Remapping (ETR/PE0, CH1/PD12, CH2/PD13, CH3/PD14, CH4/PD15); 0: Default mapping (ETR/PE0, CH1/PB6, CH2/PB7,	0

			CH3/PB8, CH4/PB9).	
[11:10]	TIM3_RM[1:0]	RW	The remapped bits of Timer 3, these bits can be read and written by the user. It controls the remapping of channels 1 to 4 of timer 3 on the GPIO port: 00: Default mapping (ETR/PD2, CH1/PA6, CH2/PA7, CH3/PB0, CH4/PB1); 01: Reserved; 10: Remapping (ETR/PD2, CH1/PB4, CH2/PB5, CH3/PB0, CH4/PB1); 11: Remapping (ETR/PD2, CH1/PC6, CH2/PC7, CH3/PC8, CH4/PC9); *Note: Remapping does not affect TIM3_ETR on PD2.*	0
[9:8]	TIM2_RM[1:0]	RW	The remapped bits of Timer 2. These bits can be read and written by the user. It controls the mapping of channels 1 to 4 of Timer 2 and the external trigger (ETR) on the GPIO port: 00: Default mapping (ETR/PA0, CH1/PA0, CH2/PA1, CH3/PA2, CH4/PA3); 01: Remapping (ETR/PA15, CH1/PA15, CH2/PB3, CH3/PA2, CH4/PA3); 10: Remapping (ETR/PA0, CH1/PA0, CH2/PA1, CH3/PB10, CH4/PB11); 11: Remapping (ETR/PA15, CH1/PA15, CH2/PB3, CH3/PB10, CH4/PB11).	0
[7:6]	TIM1_RM[1:0]	RW	The remapped bits of Timer 1. These bits can be read and written by the user. It controls the mapping of channels 1 to 4, 1N to 3N, external trigger (ETR) and brake input (BKIN) of timer 1 on the GPIO port: 00: Default mapping (ETR/PA12, CH1/PA8, CH2/PA9, CH3/PA10, CH4/PA11, BKIN/PB12, CH1N/PB13, CH2N/PB14, CH3N/PB15); 01: Remapping (ETR/PA12, CH1/PA8, CH2/PA9, CH3/PA10, CH4/PA11, BKIN/PA6, CH1N/PA7, CH2N/PB0, CH3N/PB1); 10: Reserved; 11: Remapping (ETR/PE7, CH1/PE9, CH2/PE11 CH3/PB8, CH4/PA7, BKIN/PB10, CH1N/PE8, CH2N/PE10, CH3N/PE12).	0
[5:4]	USART3_RM[1:0]	RW	The remapped bits of USART3, these bits can be read and written by the user. It controls the mapping of USART3's CTS, RTS, CK, TX and RX alternate functions on the GPIO port: 00: Default mapping (TX/PB10, RX/PB11, CK/PB12 CTS/PB13, RTS/PB14); 01: Remapping (TX/PC10, RX/PC11, CK/PC12,	0

			CTS/PB13, RTS/PB14); 10: Remapping (TX/PA13, RX/PA14, CK/PD10, CTS/PD11, RTS/PD12); 11: Remapping (TX/PD8, RX/PD9, CK/PD10, CTS/PD11, RTS/PD12).	
3	USART2_RM	RW	The remapped bits of USART2. This bit is readable and writable by the user. It controls the mapping of USART2's CTS, RTS, CK, TX and RX multiplexing functions on the GPIO port: 1: Remapping (CTS/PD3, RTS/PD4, TX/PD5, RX/PD6, CK/PD7); 0: Default mapping (CTS/PA0, RTS/PA1, TX/PA2, RX/PA3, CK/PA4).	0
2	USART1_RM	RW	USART1 low-bit mapping configuration (used in conjunction with bit 26 (USART1_RM1) of the AFIO_PCFR2 register). 00: Default mapping (CK/PA8, TX/PA9, RX/PA10, CTS/PA11, RTS/PA12); 01: Remapping (CK/PA8, TX/PB6, RX/PB7, CTS/PA11, RTS/PA12); 10: Remapping (CK/PA10, TX/PB15, RX/PA8, CTS/PA5, RTS/PA9); 11: Remapping (CK/PA5, TX/PA6, RX/PA7, CTS/PC4, RTS/PC5).	0
1	I2C_RM	RW	I2C remapping. This bit is readable and writable by the user. It controls the mapping of I2C's SCL and SDA multiplexing functions on the GPIO port: 1: Remapping (SCL/PB8, SDA/PB9); 0: Default mapping (SCL/PB6, SDA/PB7).	0
0	SPI1_RM	RW	Remapping of SPI1. This bit is readable and writable by the user. It controls the mapping of SPI1's NSS, SCK, MISO and MOSI multiplexing functions on the GPIO port: 1: Remapping (NSS/PA15, SCK/PB3, MISO/PB4, MOSI/PB5); 0: Default mapping (NSS/PA4, SCK/PA5, MISO/PA6, MOSI/PA7).	0

10.3.2.3 External Interrupt Configuration Register 1 (AFIO_EXTICR1)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI3[3:0]				EXTI2[3:0]				EXTI1[3:0]				EXTI0[3:0]			

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:12] [11:8] [7:4] [3:0]	EXTIx[3:0]	RW	(x=0-3), External interrupt input pin configuration bit. Used to determine which port the external interrupt pin is mapped to: 0000: x pin of PA pin; 0001: x pin of PB pin; 0010: x pin of PC pin; 0011: x pin of PD pin; 0100: x pin of PE pin; Other: Reserved.	0000b

10.3.2.4 External Interrupt Configuration Register 2 (AFIO_EXTICR2)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI7[3:0]				EXTI6[3:0]				EXTI5[3:0]				EXTI4[3:0]			

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:12] [11:8] [7:4] [3:0]	EXTIx[3:0]	RW	(x=4-7), External interrupt input pin configuration bit. Used to determine which port the external interrupt pin is mapped to: 0000: x pin of PA pin; 0001: x pin of PB pin; 0010: x pin of PC pin; 0011: x pin of PD pin; 0100: x pin of PE pin; Other: Reserved.	0000b

10.3.2.5 External Interrupt Configuration Register 3 (AFIO_EXTICR3)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI11[3:0]				EXTI10[3:0]				EXTI9[3:0]				EXTI8[3:0]			

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:12] [11:8] [7:4] [3:0]	EXTIx[3:0]	RW	(x=8-11), External interrupt input pin configuration bit. Used to determine which port the external interrupt pin is mapped to: 0000: x pin of PA pin;	0000b

			0001: x pin of PB pin; 0010: x pin of PC pin; 0011: x pin of PD pin; 0100: x pin of PE pin; Other: Reserved.	

10.3.2.6 External Interrupt Configuration Register 4 (AFIO_EXTICR4)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI15[3:0]				EXTI14[3:0]				EXTI13[3:0]				EXTI12[3:0]			

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:12] [11:8] [7:4] [3:0]	EXTIx[3:0]	RW	(x=12-15), External interrupt input pin configuration bit. Used to determine which port the external interrupt pin is mapped to: 0000: x pin of PA pin; 0001: x pin of PB pin; 0010: x pin of PC pin; 0011: x pin of PD pin; 0100: x pin of PE pin; Other: Reserved.	0000b

10.3.2.7 Remap Register 2 (AFIO_PCFR2)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
USART10_RM		USART9_RM		Reserved	USART1_RM1	USART8_RM [1:0]		USART7_RM [1:0]		USART6_RM [1:0]		USART5_RM [1:0]		USART4_RM [1:0]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	I3C_RM	DVP_RM	SDIO_RM	LTDC_RM	FSMC_NADV	Reserved							TIM8_RM	Reserved	FSMC_RM

Bit	Name	Access	Description	Reset value
[31:30]	USART10_RM	RW	USART10 remapping: 00: Default mapping (CK/PE12, TX/PE0, RX/PE1, CTS/PC14, RTS/PC15); 01: Remapping (CK/PE12, TX/PB8, RX/PB9, CTS/PC14, RTS/PC15); 1x: Remapping (CK/PD10, TX/PD11, RX/PD12, CTS/PD13, RTS/PD14).	0

[29:28]	USART9_RM	RW	USART9 remapping: 00: Default mapping (CK/PD8, TX/PB1, RX/PB2, CTS/PE8, RTS/PE9); 01: Remapping (CK/PD8, TX/PA4, RX/PA5, CTS/PE8, RTS/PE9); 1x: Remapping (CK/PE2, TX/PE3, RX/PE4, CTS/PE5, RTS/PE6).	0
27	Reserved	RO	Reserved	0
26	USART1_RM1	RW	USART1 map configuration high bits (Used in conjunction with bit 2 of the AFIO_PCFR1 register, USART1_RM). 00: Default mapping (CK/PA8, TX/PA9, RX/PA10, CTS/PA11, RTS/PA12); 01: Remapping (CK/PA8, TX/PB6, RX/PB7, CTS/PA11, RTS/PA12); 10: Remapping (CK/PA10, TX/PB15, RX/PA8, CTS/PA5, RTS/PA9); 11: Remapping (CK/PA5, TX/PA6, RX/PA7, CTS/PC4, RTS/PC5).	0
[25:24]	USART8_RM[1:0]	RW	USART8 remapping: 00: Default mapping (TX/PC4, RX/PC5, CK/PA7, CTS/PB0, RTS/PB1); 01: Remapping (TX/PA14, RX/PA15, CK/PB15, CTS/PE10, RTS/PE11); 1x: Remapping (TX/PB15, RX/PB14, CK/PB11, CTS/PB13, RTS/PB12).。	0
[23:22]	USART7_RM[1:0]	RW	USART7 remapping: 00: Default mapping (TX/PC2, RX/PC3, CK/PC1, CTS/PC4, RTS/PC5); 01: Remapping (TX/PA6, RX/PA7, CK/PB0, CTS/PC4, RTS/PC5); 10: Remapping (TX/PA9, RX/PA10, CK/PA8, CTS/PB6, RTS/PB7); 11: Remapping (TX/PA9, RX/PA10, CK/PA8, CTS/PC8, RTS/PC9).	0
[21:20]	USART6_RM[1:0]	RW	USART6 remapping: 00: Default mapping (TX/PC0, RX/PC1, CK/PB0, CTS/PC2, RTS/PC3); 01: Remapping (TX/PB8, RX/PB9, CK/PB5, CTS/PE0, RTS/PE1); 10: Remapping (TX/PE10, RX/PE11, CK/PA13, CTS/PA14, RTS/PA15); 11: Remapping (TX/PC0, RX/PC1, CK/PA5, CTS/PC2, RTS/PC3).	0
[19:18]	USART5_RM[1:0]	RW	USART5 remapping: 00: Default mapping (TX/PC12, RX/PD2,	0

			CK/PB5, CTS/PB3, RTS/PB4); 01: Remapping (TX/PB4, RX/PB5, CK/PB3, CTS/PB8, RTS/PB9); 1x: Remapping (TX/PE8, RX/PE9, CK/PB2, CTS/PB0, RTS/PB1).	
[17:16]	USART4_RM[1:0]	RW	UART4 remapping: 00: Default mapping (TX/PC10, RX/PC11, CK/PC12, CTS/PA13, RTS/PD2); 01: Remapping (TX/PB0, RX/PB1, CK/PB2, CTS/PC4, RTS/PC5); 1x: Remapping (TX/PE0, RX/PE1, CK/PE12, CTS/PB8, RTS/PB9).	0
15	Reserved	RW	Reserved	0
14	I3C_RM	RW	I3C remapping: 1: Remapping (SCL/PB4, SDA/PB5); 0: Default mapping (SCL/PB10, SDA/PB11).	0
13	DVP_RM	RW	DVP remapping: 1: Remapping (PCLK/PE0, HSYNC/PE1, VSYNC/PD3, D0/PD4, D1/PD5, D2/PD6, D3/PD7, D4/PB7, D5/PD11, D6/PD12, D7/PD13, D8/PE3, D9/PE4, D10/PE5, D11/PE6); 0: Default mapping (PCLK/PA6, HSYNC/PA4, VSYNC/PA5, D0/PA9, D1/PA10, D2/PC8, D3/PC9, D4/PC11, D5/PB6, D6/PB8, D7/PB9, D8/PC10, D9/PC12, D10/PD6, D11/PD2).	0
12	SDIO_RM	RW	SDIO remapping: 1: Remapping (K/PC12, CMD/PD2, SD0/PB14, SD1/PB15, SD2/PC10, SD3/PC11, SD4/PB8, SD5/PB9, SD6/PA9, SD7/PA10); 0: Default mapping (CK/PC12, CMD/PD2, SD0/PC8, SD1/PC9, SD2/PC10, SD3/PC11, SD4/PB8, SD5/PB9, SD6/PC6, SD7/PC7).	0
11	LTDC_RM	RW	LTDC remapping: 1: Remapping (CLK/PA10, HSYNC/PB8, VSYNC/PB9, DE/PA9, R[0]/PC0, R[1]/PC1, R[2]/PC2, R[3]/PC3, R[4]/PA1, R[5]/PA2 R[6]/PA3, R[7]/PA7, G[0]/PC4, G[1]/PC5, G[2]/PB0, G[3]/PB1, G[4]/PB3, G[5]/PA4, G[6]/PB12, G[7]/PB13, B[0]/PB14, B[1]/PB15, B[2]/PB11, B[3]/PB10, B[4]/PA8, B[5]/PA13, B[6]/PA14, B[7]/PA15); 0: Default mapping (CLK/PE1, HSYNC/PB4, VSYNC/PB5, DE/PE0, R[0]/PC0, R[1]/PC1, R[2]/PC2, R[3]/PC3, R[4]/PA1, R[5]/PA2, R[6]/PA3, R[7]/PA7, G[0]/PC4, G[1]/PC5 G[2]/PB0, G[3]/PB1, G[4]/PB2, G[5]/PD8,	0

			G[6]/PB12, G[7]/PB13, B[0]/PB14, B[1]/PB15, B[2]/PB11, B[3]/PB10, B[4]/PA8, B[5]/PA13, B[6]/PA14, B[7]/PA15).	
10	FSMC_NADV	RW	FSMC_NADV remapping: 1: Disable FSMC NADV output; 0: Map FSMC NADV to PB7.	0
[9:3]	Reserved	RO	Reserved	0
2	TIM8_RM	RW	Remapping bits of TIM8. 1: Remapping (ETR/PA0, CH1/PB6, CH2/PB7, CH3/PB8, CH4/PC13, BKIN/PB9, CH1N/PA13, CH2N/PA14, CH3N/PA15); 0: Default mapping (ETR/PA0, CH1/PC6, CH2/PC7, CH3/PC8, CH4/PC9, BKIN/PA6, CH1N/PA7, CH2N/PB0, CH3N/PB1).	0
1	Reserved	RW	Reserved	0
0	FSMC_RM	RW	FSMC remapping: 1: Remapping (NBL[0]/PE0, NBL[1]/PE1, CLK/PD3, NOE/PD4, NWE/PD5, NWAIT/PD6, NE1/NCE2/PD7, NADV/PB7, A16/PD11, A17/PD12, A18/PD13, A19/PE3, A20/PE4, A21/PE5, A22/PE6, A23/PE2, D0/PD14, D1/PD15, D2/PD0, D3/PD1, D4/PC4, D5/PE8, D6/PE9, D7/PE10, D8/PE11, D9/PB11, D10/PB12, D11/PB13, D12/PB14, D13/PD8, D14/PD9, D15/PD10). 0: Default mapping (NBL[0]/PE0, NBL[1]/PE1, CLK/PD3, NOE/PD4, NWE/PD5, NWAIT/PD6, NE1/NCE2/PD7, NADV/PB7, A16/PD11, A17/PD12, A18/PD13, A19/PE3, A20/PE4, A21/PE5, A22/PE6, A23/PE2, D0/PD14, D1/PD15, D2/PD0, D3/PD1, D4/PE7, D5/PE8, D6/PE9, D7/PE10, D8/PE11, D9/PB11 D10/PB12, D11/PB13，D12/PB14，D13/PD8，D14/PD9，D15/PD10).	0

Chapter 11 Direct Memory Access Control (DMA)

Direct Memory Access (DMA) controllers provide a high-speed means of transferring data between a peripheral and memory or between memory and memory without CPU intervention, and data can be moved quickly through the DMA to conserve CPU resources for other operations.

The DMA controller supports the reconfiguration of DMA request lines between peripherals and DMA. There is also an arbiter to coordinate the priorities between channels.

11.1 Main Features

- Multiple independent configurable channels
- Each channel is directly connected to a dedicated hardware DMA request and supports software triggering
- Support cyclic buffer management
- The priority of requests between multiple channels can be set by software programming (very high, high, medium and low level). When the priority settings are equal, it is determined by the channel number (the lower the channel number, the higher the priority)
- Support transmission from peripheral to memory, memory to peripheral, and memory to memory
- Flash memory, SRAM, peripheral SRAM, PB1, PB2 and HB peripherals can all be used as the sources and targets of access
- Number of programmable data transfer: 65535 at most

11.2 Functional Description

11.2.1 DMA Channel Processing

1) Arbitration priority

DMA requests generated by multiple independent channels are logically or structurally input to the DMA controller, and currently only one channel request is answered. The arbitrator within the module selects the access of the peripheral / memory to be initiated according to the priority of the channel request.

In software management, applications can independently configure priority levels for each channel, including the very high, high, medium and low levels, by setting the PL[1:0] bit of the DMA_CFGRx register. When the software settings between the channels are the same, the modules will be selected according to the fixed hardware priority, and the lower channel number will have higher priority than the higher one.

2) DMA configuration

When the DMA controller receives a request signal, it accesses the requesting peripheral or memory to establish the peripheral or data transfer between the memory and the memory. It mainly includes the following 3 steps:

1) Data is taken from the memory address indicated by the peripheral data register or the current peripheral / memory address register, and the starting address of the first transmission is the peripheral base address or memory address specified by the DMAy_PADDRx or DMAy_MADDRx register.
2) Data is stored in the peripheral data register or the memory address indicated by the current peripheral / memory address register, and the starting address of the first transmission is the peripheral base address or memory address specified in the DMAy_PADDRx or DMAy_MADDRx register.

3) Perform a decrement operation of the values in the DMAy_CNTRx register, which indicates the current number of outstanding operations.

Each channel has 3 DMA data transfer modes:

- Peripheral to memory (MEM2MEM=0, DIR=0)
- Memory to peripheral (MEM2MEM=0, DIR=1)
- Memory to memory (MEM2MEM=1)

Note: Memory-to-memory mode does not require a peripheral request signal, after configuring this mode (MEM2MEM=1), the channel is turned on (EN=1) to start data transfer. This method does not support cyclic mode.

The configuration process is as follows:

1) Set the first address of the peripheral register or the memory data address in the memory-to-memory mode (MEM2MEM=1) in the DMAy_PADDRx register. When an DMA request occurs, this address will be the source or destination address of the data transfer.

2) The memory data address is set in the DMAy_MADDRx register. When a DMA request occurs, data will be read from or written to this address.

3) Set the amount of data to be transferred in the DMAy_CNTRx register. After each data transmission, this value decreases.

4) Set the channel priority in the PL[1:0] bit of the DMAy_CFGRx register.

5) Set the data transfer direction, loop mode, incremental mode of peripherals and memory, data width of peripherals and memory, transmission halfway, transmission completion, transmission error interrupt enable level in DMAy_CFGRx register.

6) Set the EN bit of the DMAy_CFGRx register and start the channel x.

Note: The control bits in the DMAy_PADDRx/DMAy_MADDRx/DMAy_CNTRx register and the data transfer direction (DIR), loop mode (CIRC), peripheral and memory incremental mode (MINC/PINC) in the DMAy_CFGRx register can be configured to write only when the DMA channel is turned off.

3) Cycle mode

Set the CIRC location 1 of the DMAy_CFGRx register to enable the circular mode function of channel data transfer. In circular mode, when the number of data transfers becomes 0, the contents of the DMAy_CNTRx register are automatically reloaded to their initial values, the internal peripherals and memory address registers are also reloaded to the initial address values set by the DMAy_PADDRx and DMAy_MADDRx registers, and the DMA operation continues until the channel is closed or DMA mode is turned off.

4) DMA processing status

- Transfer more than half: corresponding to the HTIFx bit hardware setting in the DMAy_INTFR register. When the number of DMA transmissions is reduced to less than half of the initial setting value, more than half of the DMA transfer flag will be generated, and if HTIE is set in the DMAy_CFGRx register, an interrupt will occur. The hardware uses this flag to remind the application that it can prepare for a new round of data transfer.
- Transfer completed: corresponding to the TCIFx bit hardware setting in the DMA_INTFR register. When the number of DMA transmissions is reduced to 0, the DMA transfer completion flag will be generated, and if TCIE is set in the DMAy_CFGRx register, an interrupt will occur.
- Transfer error: corresponding to the TEIFx bit hardware setting in the DMAy_INTFR register. Reading and writing a reserved address area will result in a DMA transmission error. At the same time, the module

hardware will automatically clear the EN bits of the DMAy_CFGRx register corresponding to the channel where the error occurred, and the channel is closed. If TEIE is set in the DMAy_CFGRx register, an interrupt will occur.

When querying the status of the DMA channel, the application can first access the GIFx bit of the DMAy_INTFR register, determine which channel has the DMA event, and then deal with the specific DAM event content of the channel.

11.2.2 Programmable Data Transmission Total Size/Data Bit Width/Alignment

The total amount of data transferred in one round of each DMA channel is programmable, up to 65535 times. The number to be transferred is indicated in the DMAy_CNTRx register. When EN=0, write the setting value. After EN=1 turns on the DMA transmission channel, this register becomes a read-only attribute, and the value decreases after each transmission.

The transmission data value of peripherals and memory supports the automatic increment function of the address pointer, and the pointer increment is programmable. The first transferred data address they access is stored in the DMAy_PADDRx and DMAy_MADDRx registers. By setting the PINC bit or MINC bit of the DMAy_CFGRx register to 1, the peripheral address auto-increment mode can be turned on respectively. Or the memory address auto-increment mode, PSIZE[1:0] sets the data size of the peripheral address and the address auto-increment size, MSIZE[1:0] sets the memory address data size and the address auto-increment size, including 4 options: 8-bit, 16-bit, 32-bit, 64-bit. The specific data transfer methods are as follows:
Note: For chips whose fifth digit from the bottom of the lot number is greater than 0, their non-zero waiting area supports DMA access with a data width of 64 bits.

Table 11-1 DMA transfer under different data bit width (PINC=MINC=1)

Source bit width	Target bit width	Transfer data Number	Source: Address/data	Target: Address/data	Transfer operation
8	8	4	0x00/B0 0x01/B1 0x02/B2 0x03/B3	0x00/B0 0x01/B1 0x02/B2 0x03/B3	● The source address increment is aligned with the data bit width set by the source, and the value is equal to the source data bit width. ● The target address increment is aligned with the bit width of the target setting data, and the value size is equal to the target data bit width. ● The data transferred to the target by DMA is based on the principle: if the data size is insufficient, the high bits are filled with 0, and if the data size exceeds, the high bits are removed. ● Data storage method: little
8	16	4	0x00/B0 0x01/B1 0x02/B2 0x03/B3	0x00/00B0 0x02/00B1 0x04/00B2 0x06/00B3	
8	32	4	0x00/B0 0x01/B1 0x02/B2 0x03/B3	0x00/000000B0 0x04/000000B1 0x08/000000B2 0x0C/000000B3	
8	64	4	0x00/B0 0x01/B1 0x02/B2 0x03/B3	0x00/00…0000B0 0x08/00…0000B1 0x10/00…0000B2 0x18/00…0000B3	
16	8	4	0x00/B1B0 0x02/B3B2 0x04/B5B4 0x06/B7B6	0x00/B0 0x01/B2 0x02/B4 0x03/B6	

16	16	4	0x00/B1B0 0x02/B3B2 0x04/B5B4 0x06/B7B6	0x00/B1B0 0x02/B3B2 0x04/B5B4 0x06/B7B6	endian mode, low address stores low byte, high address stores high byte
16	32	4	0x00/B1B0 0x02/B3B2 0x04/B5B4 0x06/B7B6	0x00/0000B1B0 0x04/0000B3B2 0x08/0000B5B4 0x0C/0000B7B6	
16	64	4	0x00/B1B0 0x02/B3B2 0x04/B5B4 0x06/B7B6	0x00/00…00B1B0 0x08/00…00B3B2 0x10/00…00B5B4 0x18/00…00B7B6	
32	8	4	0x00/B3B2B1B0 0x04/B7B6B5B4 0x08/BBBAB9B8 0x0C/BFBEBDBC	0x00/B0 0x01/B4 0x02/B8 0x03/BC	
32	16	4	0x00/B3B2B1B0 0x04/B7B6B5B4 0x08/BBBAB9B8 0x0C/BFBEBDBC	0x00/B1B0 0x02/B5B4 0x04/B9B8 0x06/BDBC	
32	32	4	0x00/B3B2B1B0 0x04/B7B6B5B4 0x08/BBBAB9B8 0x0C/BFBEBDBC	0x00/B3B2B1B0 0x04/B7B6B5B4 0x08/BBBAB9B8 0x0C/BFBEBDBC	
32	64	4	0x00/B3B2B1B0 0x04/B7B6B5B4 0x08/BBBAB9B8 0x0C/BFBEBDBC	0x00/00…B3B2B1B0 0x08/00…B7B6B5B4 0x10/00…BBBAB9B8 0x18/00…BFBEBDBC	
64	8	4	0x00/1716…10 0x08/1F1E…18 0x10/2726…20 0x18/2F2E…28	0x00/10 0x01/18 0x02/20 0x03/28	
64	16	4	0x00/1716…10 0x08/1F1E…18 0x10/2726…20 0x18/2F2E…28	0x00/1110 0x02/1918 0x04/2120 0x06/2928	
64	32	4	0x00/1716…10 0x08/1F1E…18 0x10/2726…20 0x18/2F2E…28	0x00/13121110 0x04/1B1A1918 0x08/23222120 0x0C/2B2A2928	
64	64	4	0x00/1716…10 0x08/1F1E…18 0x10/2726…20 0x18/2F2E…28	0x00/1716…10 0x08/1F1E…18 0x10/2726…20 0x18/2F2E…28	

11.2.3 DMA Request Mapping

The DMA controller provides 24 channels, of which DMA1 contains 13 channels and DMA2 contains 11 channels. Each channel corresponds to multiple peripheral requests. By setting the corresponding DMA control bit in the corresponding peripheral register, the DMA function of each peripheral can be independently turned on or off. The specific correspondence is as follows.

Figure 11-1 DMA1 request mapping

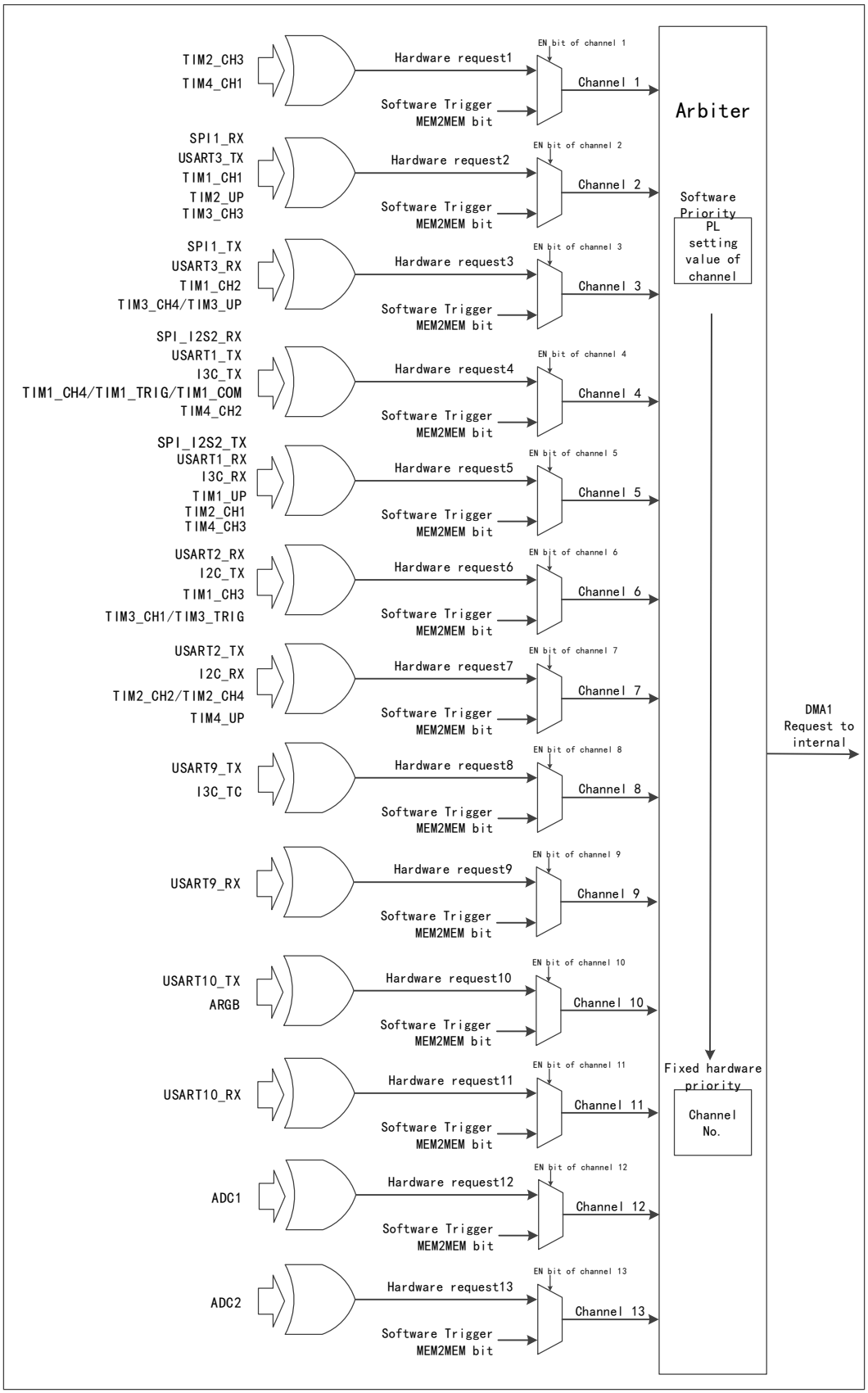


Figure 11-2 DMA2 request mapping

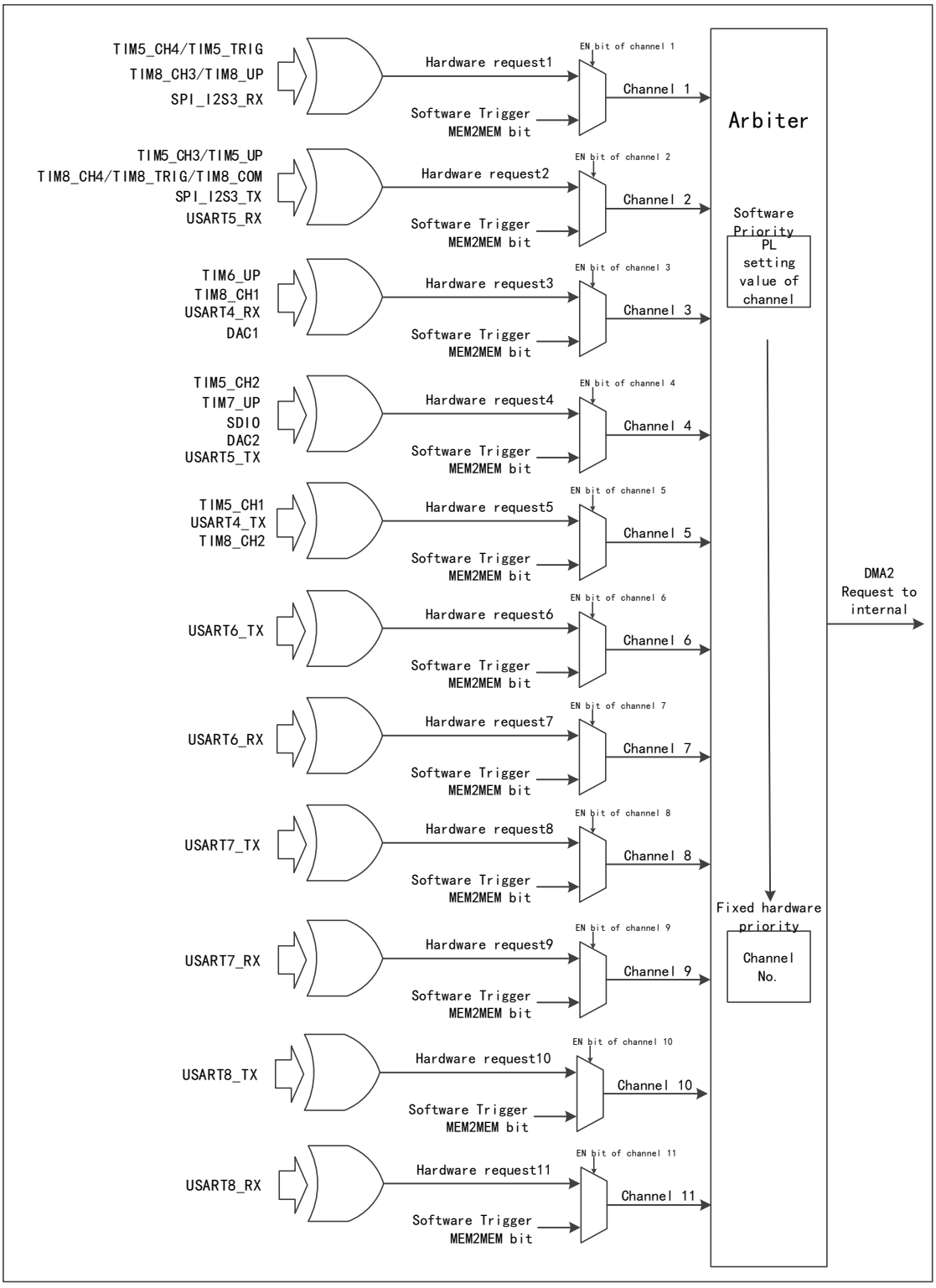


Figure 11-2 DMA1 peripheral mapping table for each channel

Peripheral	Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7
SPI1		SPI1_RX	SPI1_TX				
SPI/I2S2				SPI_I2S2_RX	SPI_I2S2_TX		
USART1				USART1_TX	USART1_RX		
USART2						USART2_RX	USART2_TX

USART3		USART3_TX	USART3_RX				
I2C						I2C_TX	I2C_RX
I3C				I3C_TX	I3C_RX		
TIM1		TIM1_CH1	TIM1_CH2	TIM1_CH4 TIM1_TRIG TIM1_COM	TIM1_UP	TIM1_CH3	
TIM2	TIM2_CH3	TIM2_UP			TIM2_CH1		TIM2_CH2 TIM2_CH4
TIM3		TIM3_CH3	TIM3_UP TIM3_CH4			TIM3_CH1 TIM3_TRIG	
TIM4	TIM4_CH1			TIM4_CH2	TIM4_CH3		TIM4_UP

Peripheral	Channel 8	Channel 9	Channel 10	Channel 11	Channel 12	Channel 13
ADC1					ADC1	
ADC2						ADC2
USART9	USART9_TX	USART9_RX				
USART10			USART10_TX	USART10_RX		
I3C	I3C_TC					
ARGB			ARGB			

Figure 11-3 DMA2 peripheral mapping table for each channel

Peripheral	Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7
TIM5	TIM5_CH4 TIM5_TRIG	TIM5_CH3 TIM5_UP		TIM5_CH2	TIM5_CH1		
TIM6			TIM6_UP				
TIM7				TIM7_UP			
TIM8	TIM8_CH3 TIM8_UP	TIM8_CH4 TIM8_TRIG TIM8_COM	TIM8_CH1		TIM8_CH2		
SPI/I2S3	SPI_I2S3_RX	SPI_I2S3_TX					
SDIO				SDIO			
DAC1			DAC1				
DAC2				DAC2			
USART4			USART4_RX		USART4_TX		
USART5		USART5_RX		USART5_TX			
USART6						USART6_TX	USART6_RX

Peripheral	Channel 8	Channel 9	Channel 10	Channel 11
USART7	USART7_TX	USART7_RX		
USART8			USART8_TX	USART8_RX

11.3 Register Description

Table 11-4 DMA1 registers

Name	Access address	Description	Reset value
R32_DMA1_INTFR	0x40020000	DMA1 interrupt status register	0x00000000
R32_DMA1_INTFCR	0x40020004	DMA1 interrupt flag clear register	0x00000000
R32_DMA1_CFGR1	0x40020008	DMA1 channel 1 configuration register	0x00000000
R32_DMA1_CNTR1	0x4002000C	DMA1 channel 1 transfer data number register	0x00000000
R32_DMA1_PADDR1	0x40020010	DMA1 channel 1 peripheral address register	0x00000000
R32_DMA1_MADDR1	0x40020014	DMA1 channel 1 memory address register	0x00000000
R32_DMA1_CFGR2	0x4002001C	DMA1 channel 2 configuration register	0x00000000
R32_DMA1_CNTR2	0x40020020	DMA1 channel 2 transfer data number register	0x00000000
R32_DMA1_PADDR2	0x40020024	DMA1 channel 2 peripheral address register	0x00000000
R32_DMA1_MADDR2	0x40020028	DMA1 channel 2 memory address register	0x00000000
R32_DMA1_CFGR3	0x40020030	DMA1 channel 3 configuration register	0x00000000
R32_DMA1_CNTR3	0x40020034	DMA1 channel 3 transfer data number register	0x00000000
R32_DMA1_PADDR3	0x40020038	DMA1 channel 3 peripheral address register	0x00000000
R32_DMA1_MADDR3	0x4002003C	DMA1 channel 3 memory address register	0x00000000
R32_DMA1_CFGR4	0x40020044	DMA1 channel 4 configuration register	0x00000000
R32_DMA1_CNTR4	0x40020048	DMA1 channel 4 transfer data number register	0x00000000
R32_DMA1_PADDR4	0x4002004C	DMA1 channel 4 peripheral address register	0x00000000
R32_DMA1_MADDR4	0x40020050	DMA1 channel 4 memory address register	0x00000000
R32_DMA1_CFGR5	0x40020058	DMA1 channel 5 configuration register	0x00000000
R32_DMA1_CNTR5	0x4002005C	DMA1 channel 5 transfer data number register	0x00000000
R32_DMA1_PADDR5	0x40020060	DMA1 channel 5 peripheral address register	0x00000000
R32_DMA1_MADDR5	0x40020064	DMA1 channel 5 memory address register	0x00000000
R32_DMA1_CFGR6	0x4002006C	DMA1 channel 6 configuration register	0x00000000
R32_DMA1_CNTR6	0x40020070	DMA1 channel 6 transfer data number register	0x00000000
R32_DMA1_PADDR6	0x40020074	DMA1 channel 6 peripheral address register	0x00000000
R32_DMA1_MADDR6	0x40020078	DMA1 channel 6 memory address register	0x00000000
R32_DMA1_CFGR7	0x40020080	DMA1 channel 7 configuration register	0x00000000
R32_DMA1_CNTR7	0x40020084	DMA1 channel 7 transfer data number register	0x00000000
R32_DMA1_PADDR7	0x40020088	DMA1 channel 7 peripheral address register	0x00000000
R32_DMA1_MADDR7	0x4002008C	DMA1 channel 7 memory address register	0x00000000
R32_DMA1_CFGR8	0x40020090	DMA1 channel 8 configuration register	0x00000000
R32_DMA1_CNTR8	0x40020094	DMA1 channel 8 transfer data number register	0x00000000
R32_DMA1_PADDR8	0x40020098	DMA1 channel 8 peripheral address register	0x00000000
R32_DMA1_MADDR8	0x4002009C	DMA1 channel 8 memory address register	0x00000000
R32_DMA1_CFGR9	0x400200A0	DMA1 channel 9 configuration register	0x00000000

R32_DMA1_CNTR9	0x400200A4	DMA1 channel 9 transfer data number register	0x00000000
R32_DMA1_PADDR9	0x400200A8	DMA1 channel 9 peripheral address register	0x00000000
R32_DMA1_MADDR9	0x400200AC	DMA1 channel 9 memory address register	0x00000000
R32_DMA1_CFGR10	0x400200B0	DMA1 channel 10 configuration register	0x00000000
R32_DMA1_CNTR10	0x400200B4	DMA1 channel 10 transfer data number register	0x00000000
R32_DMA1_PADDR10	0x400200B8	DMA1 channel 10 peripheral address register	0x00000000
R32_DMA1_MADDR10	0x400200BC	DMA1 channel 10 memory address register	0x00000000
R32_DMA1_CFGR11	0x400200C0	DMA1 channel 11 configuration register	0x00000000
R32_DMA1_CNTR11	0x400200C4	DMA1 channel 11 transfer data number register	0x00000000
R32_DMA1_PADDR11	0x400200C8	DMA1 channel 11 peripheral address register	0x00000000
R32_DMA1_MADDR11	0x400200CC	DMA1 channel 11 memory address register	0x00000000
R32_DMA1_CFGR12	0x400200D0	DMA1 channel 12 configuration register	0x00000000
R32_DMA1_CNTR12	0x400200D4	DMA1 channel 12 transfer data number register	0x00000000
R32_DMA1_PADDR12	0x400200D8	DMA1 channel 12 peripheral address register	0x00000000
R32_DMA1_MADDR12	0x400200DC	DMA1 channel 12 memory address register	0x00000000
R32_DMA1_CFGR13	0x400200E0	DMA1 channel 13 configuration register	0x00000000
R32_DMA1_CNTR13	0x400200E4	DMA1 channel 13 transfer data number register	0x00000000
R32_DMA1_PADDR13	0x400200E8	DMA1 channel 13 peripheral address register	0x00000000
R32_DMA1_MADDR13	0x400200EC	DMA1 channel 13 memory address register	0x00000000
R32_DMA1_EXTEM_INTFR	0x400200F0	DMA1 extended interrupt status register	0x00000000
R32_DMA1_EXTEM_INTFCR	0x400200F4	DMA1 extended interrupt flag clear register	0x00000000

Table 11-5 DMA2 registers

Name	Access address	Description	Reset value
R32_DMA2_INTFR	0x40020400	DMA2 interrupt status register	0x00000000
R32_DMA2_INTFCR	0x40020404	DMA2 interrupt flag clear register	0x00000000
R32_DMA2_CFGR1	0x40020408	DMA2 channel 1 configuration register	0x00000000
R32_DMA2_CNTR1	0x4002040C	DMA2 channel 1 transfer data number register	0x00000000
R32_DMA2_PADDR1	0x40020410	DMA2 channel 1 peripheral address register	0x00000000
R32_DMA2_MADDR1	0x40020414	DMA2 channel 1 memory address register	0x00000000
R32_DMA2_CFGR2	0x4002041C	DMA2 channel 2 configuration register	0x00000000
R32_DMA2_CNTR2	0x40020420	DMA2 channel 2 transfer data number register	0x00000000
R32_DMA2_PADDR2	0x40020424	DMA2 channel 2 peripheral address register	0x00000000
R32_DMA2_MADDR2	0x40020428	DMA2 channel 2 memory address register	0x00000000
R32_DMA2_CFGR3	0x40020430	DMA2 channel 3 configuration register	0x00000000
R32_DMA2_CNTR3	0x40020434	DMA2 channel 3 transfer data number	0x00000000

		register	
R32_DMA2_PADDR3	0x40020438	DMA2 channel 3 peripheral address register	0x00000000
R32_DMA2_MADDR3	0x4002043C	DMA2 channel 3 memory address register	0x00000000
R32_DMA2_CFGR4	0x40020444	DMA2 channel 4 configuration register	0x00000000
R32_DMA2_CNTR4	0x40020448	DMA2 channel 4 transfer data number register	0x00000000
R32_DMA2_PADDR4	0x4002044C	DMA2 channel 4 peripheral address register	0x00000000
R32_DMA2_MADDR4	0x40020450	DMA2 channel 4 memory address register	0x00000000
R32_DMA2_CFGR5	0x40020458	DMA2 channel 5 configuration register	0x00000000
R32_DMA2_CNTR5	0x4002045C	DMA2 channel 5 transfer data number register	0x00000000
R32_DMA2_PADDR5	0x40020460	DMA2 channel 5 peripheral address register	0x00000000
R32_DMA2_MADDR5	0x40020464	DMA2 channel 5 memory address register	0x00000000
R32_DMA2_CFGR6	0x4002046C	DMA2 channel 6 configuration register	0x00000000
R32_DMA2_CNTR6	0x40020470	DMA2 channel 6 transfer data number register	0x00000000
R32_DMA2_PADDR6	0x40020474	DMA2 channel 6 peripheral address register	0x00000000
R32_DMA2_MADDR6	0x40020478	DMA2 channel 6 memory address register	0x00000000
R32_DMA2_CFGR7	0x40020480	DMA2 channel 7 configuration register	0x00000000
R32_DMA2_CNTR7	0x40020484	DMA2 channel 7 transfer data number register	0x00000000
R32_DMA2_PADDR7	0x40020488	DMA2 channel 7 peripheral address register	0x00000000
R32_DMA2_MADDR7	0x4002048C	DMA2 channel 7 memory address register	0x00000000
R32_DMA2_CFGR8	0x40020490	DMA2 channel 8 configuration register	0x00000000
R32_DMA2_CNTR8	0x40020494	DMA2 channel 8 transfer data number register	0x00000000
R32_DMA2_PADDR8	0x40020498	DMA2 channel 8 peripheral address register	0x00000000
R32_DMA2_MADDR8	0x4002049C	DMA2 channel 8 memory address register	0x00000000
R32_DMA2_CFGR9	0x400204A0	DMA2 channel 9 configuration register	0x00000000
R32_DMA2_CNTR9	0x400204A4	DMA2 channel 9 transfer data number register	0x00000000
R32_DMA2_PADDR9	0x400204A8	DMA2 channel 9 peripheral address register	0x00000000
R32_DMA2_MADDR9	0x400204AC	DMA2 channel 9 memory address register	0x00000000
R32_DMA2_CFGR10	0x400204B0	DMA2 channel 10 configuration register	0x00000000
R32_DMA2_CNTR10	0x400204B4	DMA2 channel 10 transfer data number register	0x00000000
R32_DMA2_PADDR10	0x400204B8	DMA2 channel 10 peripheral address register	0x00000000
R32_DMA2_MADDR10	0x400204BC	DMA2 channel 10 memory address register	0x00000000
R32_DMA2_CFGR11	0x400204C0	DMA2 channel 11 configuration register	0x00000000
R32_DMA2_CNTR11	0x400204C4	DMA2 channel 11 transfer data number register	0x00000000
R32_DMA2_PADDR11	0x400204C8	DMA2 channel 11 peripheral address register	0x00000000
R32_DMA2_MADDR11	0x400204CC	DMA2 channel 11 memory address register	0x00000000
R32_DMA2_EXTEM_INTFR	0x400204D0	DMA2 extended interrupt status register	0x00000000
R32_DMA2_EXTEM_INTFC	0x400204D4	DMA2 extended interrupt flag clear register	0x00000000

R			

11.3.1 DMAy Interrupt Status Register (DMAy_INTFR) (y=1/2)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				TEIF7	HTIF7	TCIF7	GIF7	TEIF6	HTIF6	TCIF6	GIF6	TEIF5	HTIF5	TCIF5	GIF5

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TEIF4	HTIF4	TCIF4	GIF4	TEIF3	HTIF3	TCIF3	GIF3	TEIF2	HTIF2	TCIF2	GIF2	TEIF1	HTIF1	TCIF1	GIF1

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
27/23/19/15/11/7/3	TEIFx	RO	Transmission error flag for channel x (x= 1/2/3/4/5/6/7): 1: A transmission error occurred on channel x. 0: There is no transmission error on channel x. The hardware is set, and the software writes the CTEIFx bit to clear this flag.	0
26/22/18/14/10/6/2	HTIFx	RO	Transmission more than half flag of channel x (x= 1/2/3/4/5/6/7): 1: More than half of the transmission events occurred on channel x. 0: No more than half of the transmission is on channel x. The hardware is set, and the software writes the CHTIFx bit to clear this flag.	0
25/21/17/13/9/5/1	TCIFx	RO	Transmission completion flags for channel x (x=1/2/3/4/5/6/7): 1: A transmission completion event was generated on channel x; 0: No transmission completion event on channel x. Hardware sets and software writes the CTCIFx bit to clear this flag.	0
24/20/16/12/8/4/0	GIFx	RO	Global interrupt flag for channel x (x=1/2/3/4/5/6/7): 1: TEIFx or HTIFx or TCIFx was generated on channel x; 0: No TEIFx or HTIFx or TCIFx has been generated on channel x. Hardware sets and software writes the CGIFx bit to clear this flag.	0

11.3.2 DMA Interrupt Flag Clear Register (DMAy_INTFCR) (y=1/2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				CTEIF7	CHTIF7	CTCIF7	CGIF7	CTEIF6	CHTIF6	CTCIF6	CGIF6	CTEIF5	CHTIF5	CTCIF5	CGIF5

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CTEIF4	CHTIF4	CTCIF4	CGIF4	CTEIF3	CHTIF3	CTCIF3	CGIF3	CTEIF2	CHTIF2	CTCIF2	CGIF2	CTEIF1	CHTIF1	CTCIF1	CGIF1

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
27/23/19/15/11/7/3	CTEIFx	WO	Clear the transmission error flag for channel x (x=1/2/3/4/5/6/): 1: Clear the TEIFx flag in the DMA_INTFR register; 0: No effect.	0
26/22/18/14/10/6/2	CHTIFx	WO	Clear the transmit half flag for channel x (x=1/2/3/4/5/6/7): 1: Clear the HTIFx flag in the DMA_INTFR register; 0: No effect.	0
25/21/17/13/9/5/1	CTCIFx	WO	Clear the transmission completion flag for channel x (x=1/2/3/4/5/6/7): 1: Clear the TCIFx flag in the DMA_INTFR register; 0: No effect.	0
24/20/16/12/8/4/0	CGIFx	WO	Clear the global interrupt flag for channel x (x=1/2/3/4/5/6/7): 1: Clears the TEIFx/HTIFx/TCIFx/ GIFx flags in the DMA_INTFR register; 0: No effect.	0

11.3.3 DMAy Channel x Configuration Register (DMAy_CFGRx) (x=1-13, y=1) (x=1-11, y=2)

Offset address: 0x08 + (x-1)*20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	MEM2MEM	PL[1:0]		MSIZE[1:0]		PSIZE[1:0]		MINC	PINC	CIRC	DIR	TEIE	HTIE	TCIE	EN

Bit	Name	Access	Description	Reset value
[31:15]	Reserved	RO	Reserved	0
14	MEM2MEM	RW	Memory to memory mode enable: 1: Enable memory to memory mode; 0: Disable memory to memory mode.	0
[13:12]	PL[1:0]	RW	Channel priority level setting: 00: Low; 01: Medium; 10: High; 11: Very high.	00b
[11:10]	MSIZE[1:0]	RW	Memory address data width setting: 00: 8-bit; 01: 16-bit; 10: 32-bit; 11: 64-bit.	00b

			Note: For chips whose fifth digit from the bottom of the lot number is greater than 0, their non-zero waiting area supports DMA access with a data width of 64 bits.	
[9:8]	PSIZE[1:0]	RW	Peripheral address data width setting: 00: 8-bit; 01: 16-bit; 10: 32 bits; 11: 64-bit. *Note: For chips whose fifth digit from the bottom of the lot number is greater than 0, their non-zero waiting area supports DMA access with a data width of 64 bits.*	00b
7	MINC	RW	Memory address increment mode enables: 1: Enable memory address increment operation; 0: Memory address remains unchanged operation.	0
6	PINC	RW	Peripheral address increment mode enables: 1: Enable memory address increment operation; 0: Peripheral address remains unchanged operation.	0
5	CIRC	RW	DMA channel cyclic mode enable: 1: Enable cyclic operation; 0: Perform a single operation.	0
4	DIR	RW	Data transmission direction: 1: Read from memory. 0: Read from the peripheral.	0
3	TEIE	RW	Transmission error interrupt enable control: 1: Enable transmission error interrupt. 0: Disable transmission error interrupt.	0
2	HTIE	RW	More than half of the transmission interruption enables control: 1: Enable transmission for more than half an interruption. 0: Disable transmission for more than half an interruption.	0
1	TCIE	RW	Transfer complete interrupt enable control: 1: Enable transmission completion interrupt; 0: Disable transmission completion interrupt.	0
0	EN	RW	Channel enable control: 1: Channel enabled; 0: Channel disabled. When a DMA transmission error occurs, it will be cleared to 0 automatically by hardware, and channel is disabled.	0

11.3.4 DMAy Channel x Transfer Data Number Register (DMAy_CNTRx) (x=1-13, y=1) (x=1-11, y=2)

Offset address: 0x0C + (x-1)*20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NDT[15:0]															

Bit	Name	Access	Description	Reset value

[31:16]	Reserved	RO	Reserved	0
[15:0]	NDT[15:0]	RW	Number of data transfers, range 0-65535. Indicating the remaining number of transfers (the register contents are decremented after each DMA transfer). When the channel is in loop mode, the contents of the register are automatically reloaded to the previously configured value.	0

Note: Indicating the current number of transfers. When the register content is 0, no data transfer occurs regardless of whether the channel is open or not.

11.3.5 DMAy Channel x Peripheral Address Register (DMAy_PADDRx) (x=1-13, y=1) (x=1-11, y=2)

Offset address: 0x10 + (x-1)*20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PA[31:0]																															

Bit	Name	Access	Description	Reset value
[31:0]	PA[31:0]	RW	The base address of the peripheral, which is used as the source or destination address of the peripheral data transmission. When PSIZE [1:0] = '01' (16 bits), the module automatically ignores bit0, and the operation address automatically aligns with 2 bytes; when PSIZE [1:0] ='10'(32 bits), the module automatically ignores bit [1:0], and the operation address automatically aligns with 4 bytes.	0

Note: This register can only be changed when EN=0, and cannot be written when EN=1.

11.3.6 DMAy Channel x Memory Address Register (DMAy_MADDRx) (x=1-13, y=1) (x=1-11, y=2)

Offset address: 0x14 + (x-1)*20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MA[31:0]																															

Bit	Name	Access	Description	Reset value
[31:0]	MA[31:0]	RW	Memory data address, as the source or destination address for data transmission. When MSIZE[1:0] = '01' (16 bits), the module automatically ignores bit0, and the operation address automatically aligns with 2 bytes; when MSIZE[1:0] ='10'(32 bits), the module automatically ignores bit[1:0], and the operation address automatically aligns with 4 bytes.	0

Note: This register can only be changed when EN=0, and cannot be written when EN=1.

11.3.7 DMAy Extended Interrupt Status Register (DMAy_EXTEM_INTFR) (y=1/2)

Offset address: DMA1:0xF0; DMA2:0xD0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								TEIF13	HTIF13	TCIF13	GIF13	TEIF12	HTIF12	TCIF12	GIF12

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TEIF11	HTIF11	TCIF11	GIF11	TEIF10	HTIF10	TCIF10	GIF10	TEIF9	HTIF9	TCIF9	GIF9	TEIF8	HTIF8	TCIF8	GIF8

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
23/19/15/11/7/3	TEIFx	RO	Transmission error flag for channel x (x=8/9/10/11/12/13): 1: A transmission error occurred on channel x; 0: No transmission error on channel x. Set by hardware, software writes the CTEIFx bit to clear this flag.	0
22/18/14/10/6/2	HTIFx	RO	Transmission halfway mark of channel x (x=8/9/10/11/12/13): 0: There is no more than half transmission on channel x; 1: A more than half transmission event occurs on channel x. Set by hardware, software writes the CHTIFx bit to clear this flag.	0
21/17/13/9/5/1	TCIFx	RO	Transmission completion flag of channel x (x=8/9/10/11/12/13): 1: A transmission completion event is generated on channel x; 0: There is no transmission completion event on channel x. Set by hardware, software clears this flag by writing the CTCIFx bit.	0
20/16/12/8/4/0	GIFx	RO	Global interrupt flag for channel x (x=8/9/10/11/12/13): 1: TEIFx or HTIFx or TCIFx occurred on channel x; 0: No TEIFx or HTIFx or TCIFx occurred on channel x. Set by hardware, software clears this flag by writing to the CGIFx bit.	0

11.3.8 DMAy Extended Interrupt Flag Clear Register (DMAy_EXTEM_INTFR) (y=1/2)

Offset address: DMA1:0xF4; DMA2:0xD4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CTEIF13	CHTIF13	CTCIF13	CGIF13	CTEIF12	CHTIF12	CTCIF12	CGIF12

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CTEIF11	CHTIF11	CTCIF11	CGIF11	CTEIF10	CHTIF10	CTCIF10	CGIF10	CTEIF9	CHTIF9	CTCIF9	CGIF9	CTEIF8	CHTIF8	CTCIF8	CGIF8

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
23/19/15/11/7/3	CTEIFx	WO	Clear the transmission error flag of channel x (x=8/9/10/11/12/13): 1: Clear the TEIFx flag in the DMA_INTFR register; 0: No effect.	0
22/18/14/10/6/2	CHTIFx	WO	Clear the half-transmission flag of channel x (x=8/9/10/11/12/13): 1: Clear the HTIFx flag in the DMA_INTFR register; 0: No effect.	0
21/17/13/9/5/1	CTCIFx	WO	Clear the transfer completion flag of channel x (x=8/9/10/11/12/13): 1: Clear the TCIFx flag in the DMA_INTFR register; 0: No effect.	0
20/16/12/8/4/0	CGIFx	WO	Clear the global interrupt flag of channel x (x=8/9/10/11/12/13): 1: Clear TEIFx/HTIFx/TCIFx/GIFx flag in the DMA_INTFR register; 0: No effect.	0

Chapter 12 Analog-to-digital Converter (ADC)

The ADC module contains two 12-bit n analog-to-digital converters with the highest 30MHz input clock. Support 16 external channels and 2 internal signal source sampling sources. It can complete the functions of single conversion, continuous conversion, automatic scanning mode, discontinuous mode, external trigger mode and so on. You can monitor whether the channel voltage is within the threshold range by simulating the watchdog function.

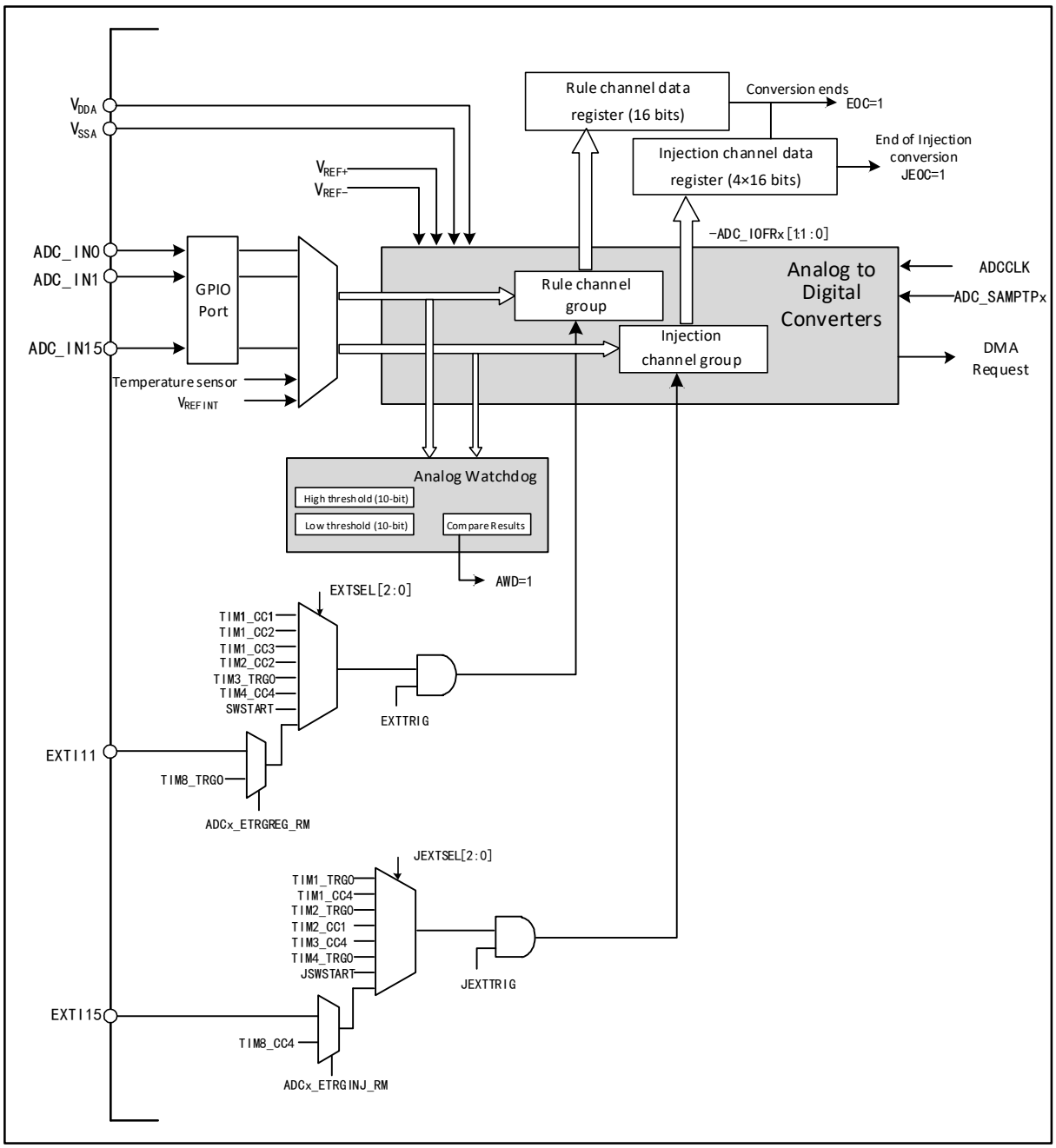
12.1 Main Features

- 12-bit resolution
- 16 external channels and 2 internal signal sources for sample
- Multiple sampling conversion modes for multiple channels: single, continuous, scan, trigger, discontinuous, etc.
- Data alignment mode: Left alignment, right alignment
- Sampling time can be programmed separately per channel
- Both regular conversion and injected conversion support external trigger
- Analog watchdog monitors the channel voltage
- ADC channel input range: $0 \leq V_{IN} \leq V_{DDA}$
- Adjustable input gain, to implement small signal amplification and sampling

12.2 Functional Description

12.2.1 Module Structure

Figure 12-1 ADC module block diagram



12.2.2 ADC Configuration

1) Module power on

An ADON bit of 1 in the ADCx_CTLR2 register indicates that the ADC module is powered on. When the ADC module enters the power-on state (ADON=1) from the power-off mode (ADON=0), it needs to be delayed for a period of time for t_{STAB} to stabilize the module. After that, the ADON bit is written again as 1, which is used as the startup signal for the software to start the ADC conversion. By clearing the ADON bit to 0, you can terminate the current conversion and put the ADC module in power-off mode, where ADC

consumes almost no power.

2) Sample clock

The register operation of the module is based on the PCLK2 (PB2 bus) clock. The clock reference ADCCLK of the conversion unit is synchronized with the PCLK2. The frequency division is configured by the ADCPRE[1:0] domain of the RCC_CFGR0 register, and the maximum cannot exceed 30MHz.

3) Channel configuration

The ADC module provides 18 channel sampling sources, including 16 external channels and 2 internal channels. They can be configured into 2 translation groups: rule groups and injection groups. To realize the group conversion consisting of a series of transformations in any order on any number of channels.

Conversion group:

- Rule group: consists of up to 16 transformations. Regular channels and their conversion order are set in the ADCx_RSQRx register. The total number of translations in the rule group should be written to L[3:0] in the ADCx_RSQR1 register.
- Injection group: composed of up to 4 transformations. The injection channels and their conversion order are set in the ADCx_ISQR register. The total number of translations in the injection group should be written to JL[1:0] in the ADCx_ISQR register.

Note: If the ADCx_RSQRx or ADCx_ISQR register is changed during the conversion, the current conversion is terminated and a new startup signal will be sent to ADC to convert the newly selected group.

2 internal channels:

- Temperature sensor: Connect the ADC_IN16 channel to measure the internal temperature(TA) of the chip.
- V_{REFINT} internal reference voltage: Connect the ADC_IN17 channel.

4) Calibration

Initialize the calibration register by writing the RSTCAL bit of the ADCx_CTLR2 register to 1, and wait for the RSTCAL hardware to clear to 0 to indicate that the initialization is complete. Set the CAL bit to start the calibration function. Once the calibration is completed, the hardware will automatically clear the CAL bit. Perform offset calibration. After that, the normal conversion function can be started, and the offset value is added after each ADC conversion. It is recommended to perform an ADC calibration when the ADC module is powered on.

Note: Before starting the calibration, you must ensure that the ADC module is powered on (ADON=1) for at least 2 ADC clock cycles.

5) Programmable sample time

ADC uses several ADCCLK cycles to sample the input voltage, and the number of sampling cycles of the channels can be changed by the SMPx[2:0] bit in the ADCx_SAMPTR1 and ADCx_SAMPTR2 registers. Each channel can be sampled at different times.

The total conversion time is calculated as follows:

T_{CONV} = Sampling time + 12.5T_{ADCCLK}

The sampling time can be determined based on SMPx[2:0].

The regular channel transformation of ADC supports the DMA feature. The value of regular channel conversion is stored in a single data register ADCx_RDATAR. In order to prevent the continuous conversion

of multiple regular channels from not taking the data from the ADCx_RDATAR register in time, the DMA function of ADC can be turned on. The hardware generates an DMA request at the end of the conversion of the regular channel (EOC setting) and transfers the translated data from the ADCx_RDATAR register to the destination address specified by the user.

After the channel configuration of the DMA controller module is completed, write the DMA location 1 of the ADCx_CTLR2 register and turn on the DMA function of ADC.

Note: the injection group transformation does not support the DMA feature.

6) Data alignment

The ALIGN bit in the ADCx_CTLR2 register selects the data storage alignment after ADC conversion. 12-bit data supports left and right alignment modes.

The data register ADCx_RDATAR of the regular group channel stores the actual converted 12-bit digital value, while the data register ADCx_IDATARx of the injected group channel is the value written after the actual converted data minus the offset of the ADCx_IOFRx register, so there are positive and negative SIGNB.

Figure 12-2 Data left alignment

Rule group data register

D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	0	0	0	0

Injected group data register

SIGNB	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	0	0	0

Figure 12-3 Data right alignment

Rule group data register

0	0	0	0	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

Injected group data register

SIGNB	SIGNB	SIGNB	SIGNB	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

12.2.3 External Trigger Source

The start event of an ADC transformation can be triggered by an external event. If the EXTTRIG or JEXTTRIG bit of the ADCx_CTLR2 register is set, the conversion of the rule group or injection group channel can be triggered by external events, respectively. At this point, the configuration of the EXTSEL[2:0] and JEXTSEL[2:0] bits determines the external event sources of the rule group and the injection group.

Table 12-1 External trigger sources of regular group channel

EXTSEL[2:0]	Trigger source	Type
000	CC1 event of timer 1	Internal signal from on-chip timers
001	CC2 event of timer 1	
010	CC3 event of timer 1	
011	CC2 event of timer 2	
100	TRGO event of timer 3	
101	CC4 event of timer 4	
110	EXTI line 11/ TIM8_TRGO	From external pin/internal timer signal
111	SWSTART position 1 software trigger	Software control bit

Table 12-2 External trigger sources of injected group channel

JEXTSEL[2:0]	Trigger source	Type
000	TRGO event of timer1	Internal signal from on-chip timers
001	CC4 event of timer1	
010	TRGO event of timer2	
011	CC1 event of timer2	
100	CC4 event of timer3	
101	TRGO event of timer4	
110	EXTI line 15/TIM8_CC4	From external pin/Internal timer signal
111	JSWSTART position 1 software trigger	Software control bit

12.2.4 Conversion Mode

Table 12-3 Conversion mode combination

ADCx_CTLR1 and ADCx_CTLR2 register control bits					ADC conversion mode
CONT	SCAN	DISCEN/JDISCEN	IAUTO	Start event	
0	0	0	0	ADON bit set to 1	Single channel mode: A regular channel performs a single conversion.
				External trigger mode	Single channel mode: A regular channel or a channel of an injection channel performs a single conversion.
	1	0	0	ADON bit set to 1 or external trigger mode	Single scan mode: performs a single conversion on all selected regular group channels (ADCx_RSQRx) or all injection group channels (ADCx_ISQR) one by one in sequence. Trigger injection method: during the rule group channel conversion, all the transformations of the injection group channel can be inserted, and then the rule group channel conversion can be continued; however, the rule group channel transformation will not be inserted when the conversion is injected into the group channel.
			1	ADON bit set to 1 or external trigger mode	Single scan mode: performs a single conversion on all selected regular group channels (ADCx_RSQRx) or all injection group channels (ADCx_ISQR) one by one in sequence. Automatic injection mode: after the rule group channel is converted, the injection group channel is automatically converted. *Note: the external trigger signal of the injection channel is not allowed during*

					the conversion process.
	0	1 (DISCEN and JDISCEN cannot be 1 at the same time)	0	External trigger mode	Single break mode: each time an event is started, a short sequence of channel number conversions (the number defined by DISCNUM[2:0]) is performed until all selected channel conversions are completed. *Note: Rule group and injection group select this mode control bit as DISCEN and JDISCEN respectively. Discontinuous mode cannot be configured for both rule group and injection group. Discontinuous mode can only be used for one set of transformations.*
			1	-	Disable such mode.
	1	1	X	-	No such mode.
1	0	0	0	ADON bit set to 1 or external trigger mode	Continuous single channel/scan mode: repeat a new round of conversion at the end of each round until the CONT is cleared 0.
	1	0	0		
			1		

Note: The external trigger events of the rule group and the injection group are different, and the "ADON" bit can only start the rule group channel transformation, so the start events of the rule group and injection group channel transformation are independent.

1) Single channel conversion mode

In this mode, only one conversion is performed for the current 1 channel. This mode performs conversion on the channels sorted first in the rule group or injection group, where it can be started by setting ADON position 1 of the ADCx_CTLR2 register (for regular channels only) or by external triggering (for regular channels or injection channels). Once the conversion of the selected channel is completed:

If the rule group channel is converted, the conversion data is stored in the 16-bit ADCx_RDATAR register, the EOC flag is set, and if the EOCIE bit is set, the ADC interrupt will be triggered.

If the injection group channel is converted, the conversion data is stored in the 16-bit ADCx_IDATAR1 register, the EOC and JEOC flags are set, and if the JEOCIE or EOCIE bit is set, the ADC interrupt will be triggered.

2) Single scan mode conversion

Enter ADC scan mode by setting the scan bit of the ADCx_CTLR1 register to 1. This mode is used to scan a set of analog channels and perform a single conversion one by one for all channels selected by the ADCx_RSQRx register (for regular channels) or ADCx_ISQR (for injection channels). When the current channel conversion ends, the next channel of the same group is automatically converted.

In the scanning mode, according to the state of IAUTO bits, it can be divided into trigger injection mode and

automatic injection mode.

- Trigger injection

The IAUTO bit is 0, when the trigger event of the injection group channel conversion occurs in the process of scanning the rule group channel, the current conversion is reset, and the sequence of the injection channel is carried out in a single scan mode. After all the selected injection group channel scan conversion is completed, the last interrupted rule group channel conversion is restored.

If the start event of the regular channel occurs when scanning the channel sequence of the injection group, the conversion of the injection group will not be interrupted, but the conversion of the rule sequence will be performed after the conversion of the injection sequence is completed.

Note: When using triggered injection transformations, you must ensure that the interval between triggered events is longer than the injection sequence. For example, if it takes 28 ADCCLK to complete the conversion of the injection sequence, the minimum time between events to trigger the injection channel is 29 ADCCLK.

- Automatic injection

The JAUTO bit is 1, and after scanning all the channel translations selected by the rule group, the channel selected by the injection group is converted automatically. This method can be used to convert up to 20 conversion sequences in ADCx_RSQRx and ADCx_ISQR registers.

In this mode, the external trigger (JEXTTRIG=0) of the injection channel must be disabled.

Note: When the ADC clock pre-division factor (ADCPRE[1:0]) is 4 to 8, 1 ADCCLK interval is automatically inserted when switching from regular conversion to injection sequence or from injection conversion to regular sequence; when the ADC clock pre-division factor is 2, there is a delay of 2 ADCCLK intervals.

3) Single discontinuous mode conversion

Enter the break mode of the rule group or injection group by setting the DISCEN or JDISCEN bit of the ADCx_CTLR1 register to 1. This mode distinguishes scanning a complete set of channels in the scan mode, but divides a group of channels into multiple short sequences, and each external trigger event will perform a scan conversion of a short sequence.

The length of the short sequence n (n < = 8) is defined in the DISCNUM[2:0] of the ADCx_CTLR1 register. When DISCEN is 1, it is the discontinuous mode of the rule group, and the total length to be converted is defined in the L[3:0] of the ADCx_RSQR1 register. When JDISCEN is 1, it is the discontinuous mode of the injection group, and the total length to be converted is defined in the JL[1:0] of the ADCx_ISQR register. Both rule group and injection group cannot be set to discontinuous mode at the same time.

Example of rule group discontinuity mode:
DISCEN=1, DISCNU [2:0] = 3, L[3:0] = 8, to be switched channel = 1, 3, 2, 5, 8, 4, 7, 6.
The first external trigger: the conversion sequence is: 1, 3, 2.
The second external trigger: the conversion sequence is: 5, 8, 4.
The third external trigger: the conversion sequence is: 7, 6, and EOC events are generated at the same time.
The 4th external trigger: the conversion sequence is: 1, 3, 2.

Example of injection group discontinuity mode:
JDISCEN=1, DISCNUM[2:0]=1, JL[1:0]=3, Channel to be changed = 1, 3, 2.
The first external trigger: the conversion sequence is: 1.
The second external trigger: conversion sequence is: 3.
The third external trigger: the conversion sequence is: 2, and both EOC and JEOC events are generated.

The 4th external trigger: conversion sequence is: 1.

Note:
1. When a rule group or injection group is converted in discontinuous mode, the conversion sequence does not automatically start from scratch at the end of the conversion sequence. When all subgroups are converted, the next trigger event initiates the conversion of the first subgroup.
2. You cannot use both automatic injection (IAUTO=1) and discontinuous mode.
3. Discontinuous mode cannot be set for both rule groups and injection groups, and can only be used for one set of transformations.
4. In the intermittent mode of the injection group, the number of injection channels to be switched after an external trigger is 1.

4) Continuous conversion

In continuous conversion mode, another conversion is started as soon as the current ADC conversion is completed, and the conversion does not stop on the last channel of the selection group, but continues again from the first channel of the selection group. Boot events in this mode include external trigger events and ADON position 1. After setting boot, CONT position 1 needs to be set.

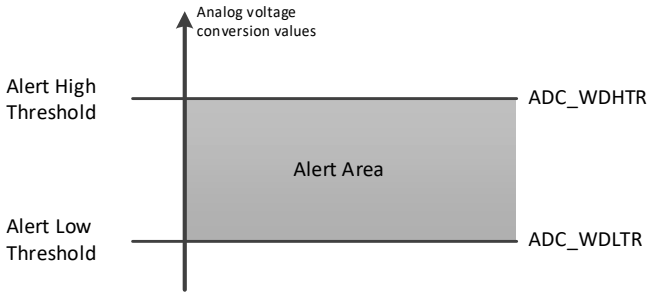
If a regular channel is converted, the converted data is stored in the ADCx_RDATAR register, the end of the conversion indicates that the EOC is set, and if EOCIE is set, an interrupt is generated.

If an injection channel is converted, the converted data is stored in the ADCx_IDATARx register, the end of injection conversion flag JEOC is set, and an interrupt is generated if JEOCIE is set.

12.2.5 Analog Watchdog

If the analog voltage being converted by the ADC is below the low threshold or above the high threshold, the AWD analog watchdog status bit is set. The threshold setting is located in the lowest 12 valid bits of the ADCx_WDHTR and ADCx_WDLTR registers. The corresponding interrupt is allowed to be generated by setting the AWDIE bit in the ADCx_CTLR1 register. The analog watchdog reset can be enabled by setting the AWDRST_EN bit in the ADCx_CFG register.

Figure 12-4 Analog watchdog threshold area



Configure the AWDSGL, AWDEN, JAWDEN, and AWDCH[4:0] bits of the ADCx_CTLR1 register to select the channel for analog watchdog alert, see the following table for the relationship:

Table 12-4 Analog Watchdog channel selection

Analog Watchdog alert channel	ADCx_CTLR1 register control bit			
	AWDSGL	AWDEN	JAWDEN	AWDCH[4:0]
No alert	Ignore	0	0	Ignore
All injection channels	0	0	1	Ignore

All rule channels	0	1	0	Ignore
All injection and rule channels	0	1	1	Ignore
Single injected channel	1	0	1	Determine channel No.
Single regular channel	1	1	0	Determine channel No.
Single injected and regular channel	1	1	1	Determine channel No.

12.2.6 Temperature Sensor

Chip built-in temperature sensor, connected to the ADC_IN16 channel, through ADC to convert the output voltage of the sensor into digital value to feedback the internal temperature of the chip, it is recommended to set the sampling time is 17.1μs. The output voltage of the temperature sensor varies linearly with the temperature. Due to the manufacturing discreteness, the slope and offset of the linear curve are different, so the internal temperature sensor is more suitable for detecting the change of temperature rather than measuring the absolute temperature. If you need to measure the temperature accurately, you should use an external temperature sensor.

By setting the TSVREFE position 1 of the ADCx_CTLR2 register, awakening the ADC internal sampling channel, software startup or external trigger starts the temperature sensor channel conversion of the ADC, and reads the data result (mV). Among them, the conversion formula of numerical value and temperature (℃) is as follows:

Temperature (℃) = ((V_{SENSE}-V_{25})/Avg_Slope)+25

V25: The voltage value of the temperature sensor at 25℃

Avg_Slope: Average slope of temperature and V_{SENSE} curve (mV/℃)

Refer to the actual values of V_{25} and Avg_Slope in the electrical characteristics section of the datasheet.

Note: it takes a setup time for the internal temperature sensor to power on (the TSVREFE bit is changed from 0 to 1), and the ADC module also needs a setup time (the ADON bit is changed from 0 to 1), so in order to shorten the waiting time, both the ADON and TSVREFE bits can be set.

12.2.7 Dual ADC Mode

In ADC modules with 2 chips, both ADCs can be used together to enable dual ADC mode. In dual ADC mode, ADC1 serves as the primary ADC and ADC2 as the secondary ADC. By configuring DUALMOD[3:0] in ADC1_CTLR1 to select the mode, ADC1 and ADC2 can be triggered alternately or synchronously for conversion.

Note: In dual ADC mode, when selecting external trigger events for activation, users must enable external triggering on both the master and slave ADCs. The master ADC must be configured for the corresponding trigger, while the slave ADC should be set to software triggering. This prevents unnecessary triggering that could cause the slave ADC to convert.

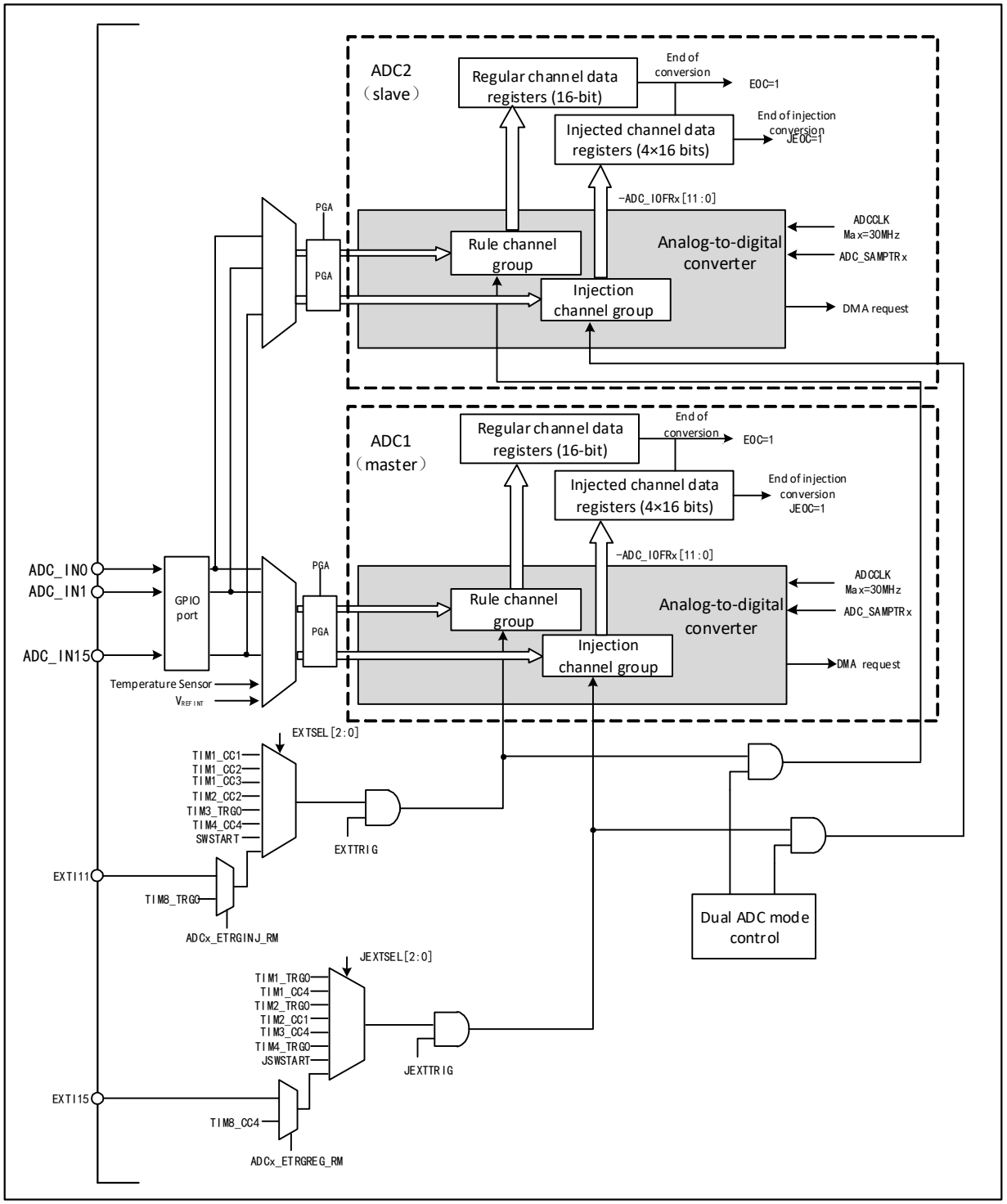
The following possible modes can be achieved through configuration:

- Independent Mode
- Synchronous Injection Mode
- Synchronous Rule Mode
- Fast Alternating Mode
- Slow Alternating Mode
- Alternating Trigger Mode
- Synchronous Rule Mode + Synchronous Injection Mode
- Synchronous Rule Mode + Alternating Trigger Mode

- Synchronous Injection Mode + Fast Alternation Mode
- Synchronous Injection Mode + Slow Alternation Mode

Note: 1. In dual ADC mode, the DMA bit must be enabled to allow the main data register to read conversion data from the slave ADC.

Figure 12-5 Dual ADC block diagram



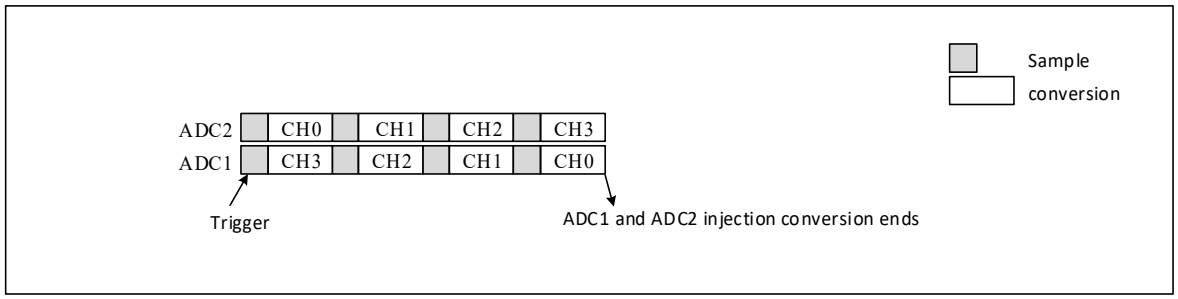
1) Independent Mode

In this mode, the dual ADCs work asynchronously and independently.

2) Synchronous injection mode

In this mode, it is used to convert a channel group. Set JEXTSEL[2:0] in ADC1_CTLR2 to select the trigger source, which will also synchronize the trigger for ADC2. Upon conversion completion, the converted data is stored in ADCx_IDATARx for each ADCx. If any ADC interrupt is enabled, a JEOC interrupt will be generated after conversion ends.

Figure 12-6 4-channel synchronous injection conversion



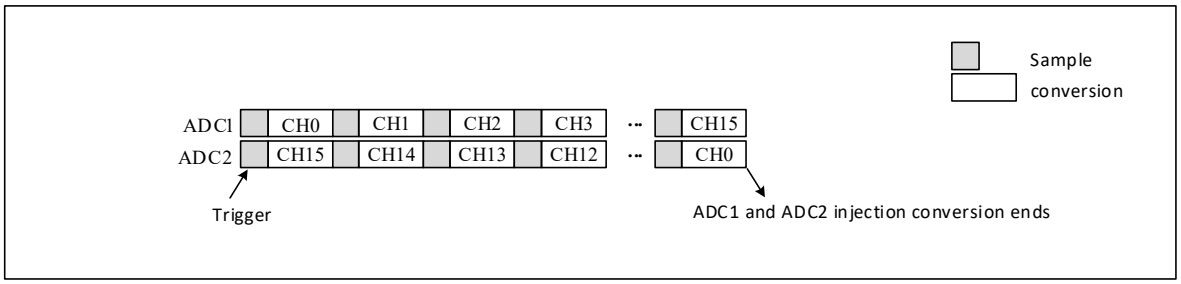
Note: 1. The conversion channels of ADC1 and ADC2 should not overlap at the same moment.

2. In synchronous mode, ADC1 and ADC2 should have conversion sequences of equal duration, or the duration of the longer sequence should be shorter than the trigger interval to ensure both sequences complete conversion with each trigger.

3) Synchronous rule mode

In this mode, the conversion rule channel sequence is used to configure EXTSEL[2:0] in ADC1_CTLR2 to select the trigger source, which also synchronizes the trigger for ADC2. Upon conversion completion, a 32-bit DMA transfer request is generated to transfer the contents of the ADC1_RDATAR data register to SRAM. The upper 16 bits contain ADC2 conversion data, while the lower 16 bits contain ADC1 conversion data. Enabling interrupts for either ADC will generate an EOC interrupt.

Figure 12-7 16-channel synchronous rule conversion



Note: 1. The conversion channels of ADC1 and ADC2 should not overlap at the same moment.

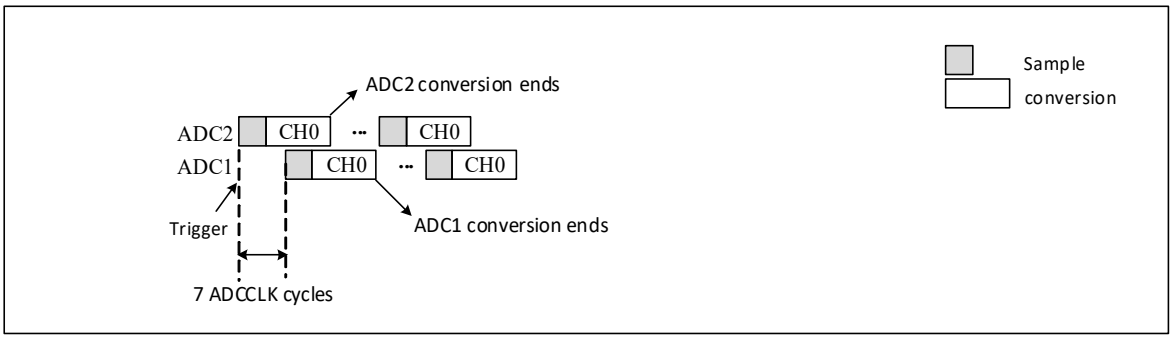
2. In synchronous mode, ADC1 and ADC2 should have conversion sequences of equal duration, or the duration of the longer sequence should be shorter than the trigger interval to ensure both sequences complete conversion with each trigger.

4) Fast alternating mode

This mode applies only to the regular channel (typically the sole channel). Configure EXTSEL[2:0] in ADC1_CTLR2 to select the trigger source. Upon trigger generation, ADC2 initiates conversion immediately, while ADC1 begins conversion after a delay of 7 ADC clock cycles. If both ADCs are enabled in continuous mode (CONT set), they will perform alternating conversions on the regular channel. If interrupts are enabled, ADC1 will generate an EOC interrupt. If DMA is also enabled, this will trigger a 32-bit DMA transfer request, transferring the contents of the ADC1_RDATAR data register to SRAM. The upper 16 bits contain ADC2

conversion data, while the lower 16 bits contain ADC1 conversion data.

Figure 12-8 Single channel fast alternating continuous conversion

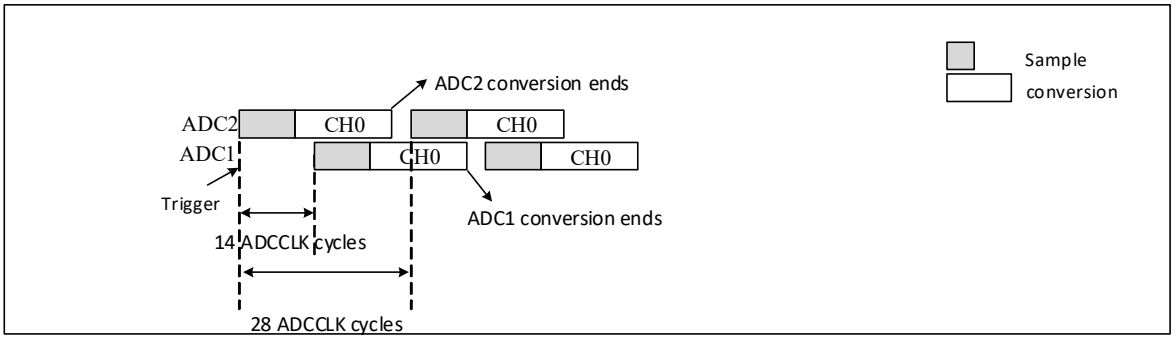


Note: The sampling time should be less than 7 ADC clock cycles, so as to avoid the problem of overlapping sampling periods when ADC1 and ADC2 sample the same channel.

5) Slow alternating mode

This mode is only applicable to the regular channel and supports only one channel. Configure EXTSEL[2:0] in ADC1_CTLR2 to select the trigger source. Upon trigger generation, ADC2 initiates conversion immediately, while ADC1 starts conversion after a delay of 14 ADC clock cycles. After another delay of 14 ADC clock cycles, ADC2 initiates conversion again, repeating this cycle. If interrupts are enabled, ADC1 will generate an EOC interrupt. If DMA is simultaneously enabled, a 32-bit DMA transfer request will be generated, transferring the contents of the ADC1_RDATAR data register to SRAM. The upper 16 bits contain the ADC2 conversion data, while the lower 16 bits contain the ADC1 conversion data.

Figure 12-9 Single channel slow alternating continuous conversion



Note: 1. The sampling time should be less than 14 ADC clock cycles to avoid overlapping with the next sampling cycle;

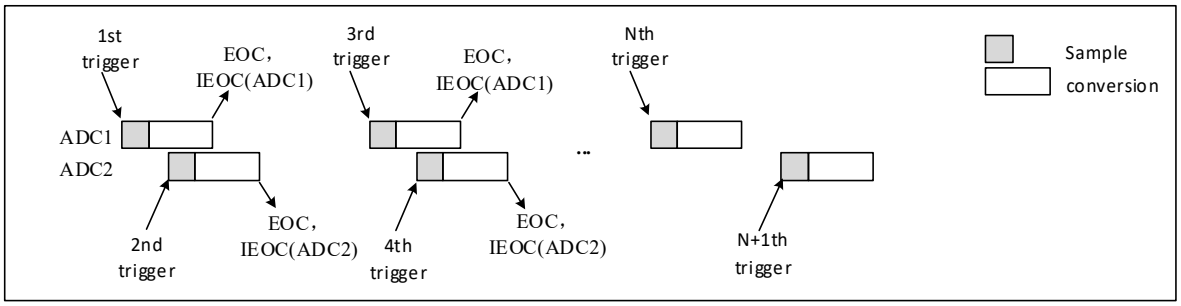
2. A new ADC2 conversion will automatically start after 28 ADC clock cycles;

3. The CONT bit does not need to be set.

6) Alternate trigger mode

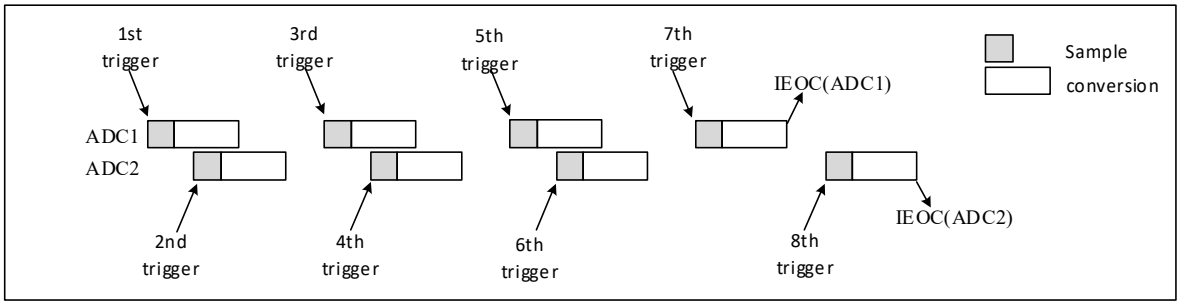
This mode applies only to injection channel groups. Configure JEXTSEL[2:0] in ADC1_CTLR2 to select the trigger source. Upon the first trigger event, all injection channels on ADC1 undergo conversion. Upon the second trigger event, all injection channels on ADC2 undergo conversion. This cycle repeats sequentially. If interrupts are enabled, a JEOC interrupt is generated when all injection channels on ADC1 complete conversion, and another JEOC interrupt is generated when all injection channels on ADC2 complete conversion.

Figure 12-10 Alternate trigger conversion for each ADC injection channel group



If injection break mode is used simultaneously, the first injection channel on ADC1 is converted when the first trigger event occurs, the first injection channel on ADC2 is converted when the second trigger event occurs, and so on. If interrupt is enabled, a JEOC interrupt is generated when all injection channels on ADC1 are converted, and a JEOC interrupt is generated when all injection channels on ADC2 are converted.

Figure 12-11 Alternating trigger transitions for each ADC injection channel in discontinuous mode



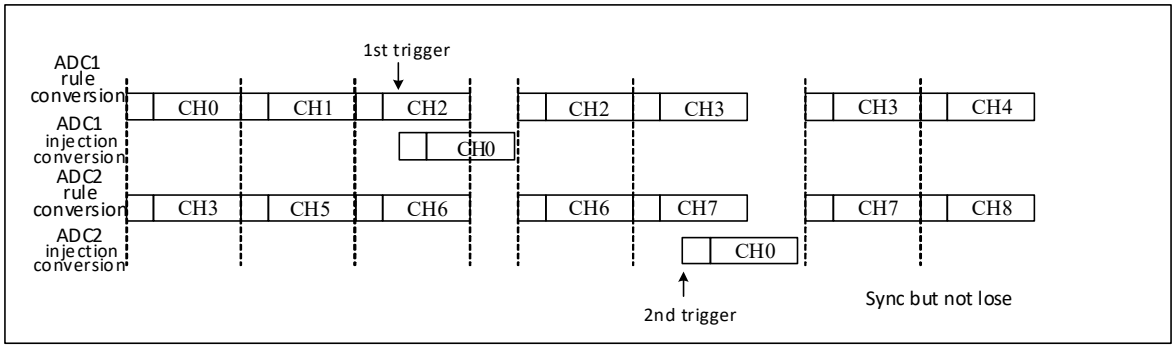
7) Synchronous rule mode + synchronous injection mode

In this mode, the synchronous transition of a rule group can be interrupted to initiate the synchronous transition of an injection group. At the same time, it is necessary to ensure that the transitions have the same length of time, or that the trigger time interval is longer than the longer of the two sequences.

8) Synchronous rule mode + alternating trigger mode

In this mode, synchronous conversion of rule groups can be interrupted to initiate alternate trigger conversion of injection groups. When an injection event occurs, alternate trigger conversion is initiated immediately. If synchronous rule conversion is ongoing, rule conversion of all ADCs is stopped, and synchronization resumes after injection conversion.

Figure 12-12 Alternating trigger injection channel switching in synchronous regular mode

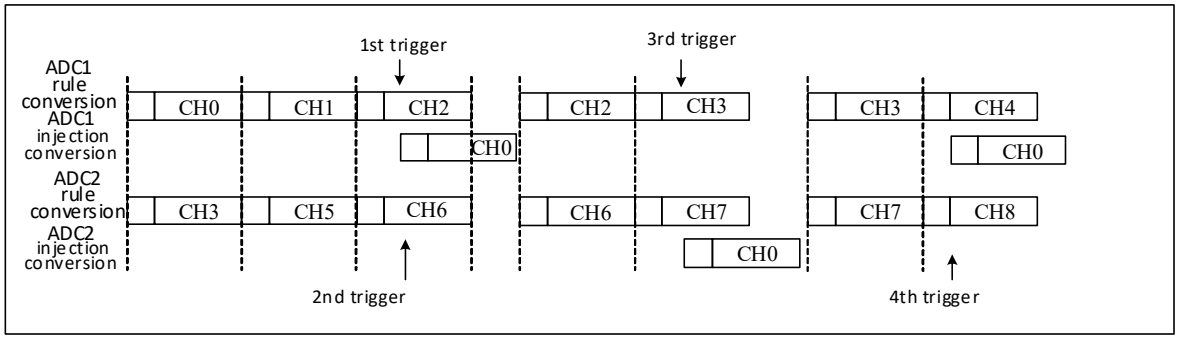


Note: In this mode, you need to ensure that sequences with the same time length are converted, or that the

trigger interval is longer than the longer of the two sequences.

If an injection trigger event occurs during an injection transition that interrupts a rule transition, the trigger event is ignored, such as in the case of the second trigger depicted in the figure below.

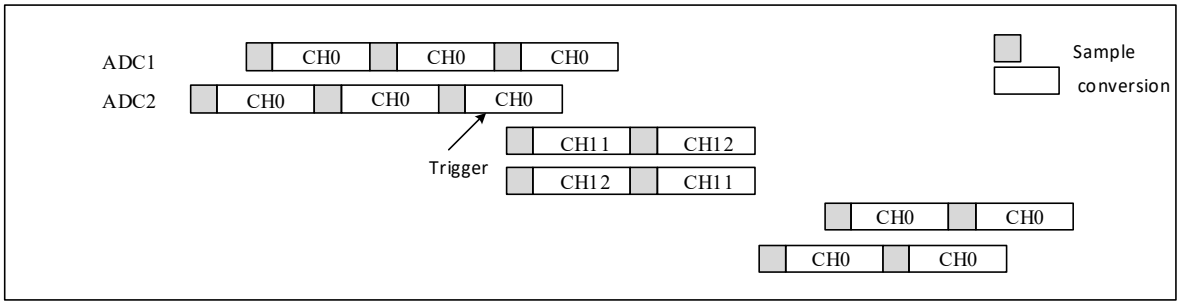
Figure 12-13 Trigger event occurs during injection transition



9) Synchronous injection mode + alternating mode

In this mode, injection transitions can interrupt alternating transitions. When an injection event occurs, the alternating transition is interrupted, the injection transition begins, and after the injection transition completes, the alternating transition resumes.

Figure 12-14 Trigger injection group switching under alternate switching



Note: When the prescaler of the ADC is 4, the sampling interval after alternate conversion recovery is no longer uniform at 7 ADC clocks, but alternating between 8 and 6 clock cycles.

12.3 Register Description

Table 12-5 ADC1-related registers

Name	Access address	Description	Reset value
R32_ADC1_STATR	0x40012400	ADC1 status register	0x00000000
R32_ADC1_CTLR1	0x40012404	ADC1 control register 1	0x00000000
R32_ADC1_CTLR2	0x40012408	ADC1 control register 2	0x00000000
R32_ADC1_SAMPTR1	0x4001240C	ADC1 sample time configuration register 1	0x00000000
R32_ADC1_SAMPTR2	0x40012410	ADC1 sample time configuration register 2	0x00000000
R32_ADC1_IOFR1	0x40012414	ADC1 injected channel data offset register 1	0x00000000
R32_ADC1_IOFR2	0x40012418	ADC1 injected channel data offset register 2	0x00000000
R32_ADC1_IOFR3	0x4001241C	ADC1 injected channel data offset register 3	0x00000000
R32_ADC1_IOFR4	0x40012420	ADC1 injected channel data offset register 4	0x00000000

R32_ADC1_WDHTR	0x40012424	ADC1 watchdog high threshold register	0x00000FFF
R32_ADC1_WDLTR	0x40012428	ADC1 watchdog high threshold register	0x00000000
R32_ADC1_RSQR1	0x4001242C	ADC1 regular channel sequence register 1	0x00000000
R32_ADC1_RSQR2	0x40012430	ADC1 regular channel sequence register 2	0x00000000
R32_ADC1_RSQR3	0x40012434	ADC1 regular channel sequence register 3	0x00000000
R32_ADC1_ISQR	0x40012438	ADC1 injected channel sequence register	0x00000000
R32_ADC1_IDATAR1	0x4001243C	ADC1 injected data register 1	0x00000000
R32_ADC1_IDATAR2	0x40012440	ADC1 injected data register 2	0x00000000
R32_ADC1_IDATAR3	0x40012444	ADC1 injected data register 3	0x00000000
R32_ADC1_IDATAR4	0x40012448	ADC1 injected data register 4	0x00000000
R32_ADC1_RDATAR	0x4001244C	ADC1 regular data register	0x00000000
R32_ADC1_AUX	0x40012454	ADC1 sample time register	0x00000000

Table 12-6 ADC2-related registers

Name	Access address	Description	Reset value
R32_ADC2_STATR	0x40012800	ADC2 status register	0x00000000
R32_ADC2_CTLR1	0x40012804	ADC2 control register 1	0x00000000
R32_ADC2_CTLR2	0x40012808	ADC2 control register 2	0x00000000
R32_ADC2_SAMPTR1	0x4001280C	ADC2 sample time configuration register 1	0x00000000
R32_ADC2_SAMPTR2	0x40012810	ADC2 sample time configuration register 2	0x00000000
R32_ADC2_IOFR1	0x40012814	ADC2 injected channel data offset register 1	0x00000000
R32_ADC2_IOFR2	0x40012818	ADC2 injected channel data offset register 2	0x00000000
R32_ADC2_IOFR3	0x4001281C	ADC2 injected channel data offset register 3	0x00000000
R32_ADC2_IOFR4	0x40012820	ADC2 injected channel data offset register 4	0x00000000
R32_ADC2_WDHTR	0x40012824	ADC2 watchdog high threshold register	0x00000FFF
R32_ADC2_WDLTR	0x40012828	ADC2 watchdog high threshold register	0x00000000
R32_ADC2_RSQR1	0x4001282C	ADC2 regular channel sequence register 1	0x00000000
R32_ADC2_RSQR2	0x40012830	ADC2 regular channel sequence register 2	0x00000000
R32_ADC2_RSQR3	0x40012834	ADC2 regular channel sequence register 3	0x00000000
R32_ADC2_ISQR	0x40012838	ADC2 injected channel sequence register	0x00000000
R32_ADC2_IDATAR1	0x4001283C	ADC2 injected data register 1	0x00000000
R32_ADC2_IDATAR2	0x40012840	ADC2 injected data register 2	0x00000000
R32_ADC2_IDATAR3	0x40012844	ADC2 injected data register 3	0x00000000
R32_ADC2_IDATAR4	0x40012848	ADC2 injected data register 4	0x00000000
R32_ADC2_RDATAR	0x4001284C	ADC2 regular data register	0x00000000
R32_ADC2_AUX	0x40012854	ADC2 sample time register	0x00000000

12.3.1 ADCx Status Register (ADCx_STATR) (x=1/2)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RSTN	Reserved														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Reserved	STRT	JSTRT	JEOC	EOC	AWD

Bit	Name	Access	Description	Reset value
31	RSTN	RO	ADC module reset flag: 1: ADC module is in reset release state; 0: ADC module is in reset state.	0
[30:5]	Reserved	RO	Reserved	0
4	STRT	RW0	Regular channel conversion start status: 1: The regular channel conversion has started; 0: The regular channel conversion has not started. This bit is set by hardware and cleared by software (invalid if writing 1).	0
3	JSTRT	RW0	Injected channel conversion start status: 1: The injected channel conversion has started; 0: The injected channel conversion has not started; This bit is set by hardware and cleared by software (invalid if writing 1).	0
2	JEOC	RW0	Injected channel group conversion completion status: 1: The conversion has completed; 0: The conversion has not completed. This bit is set to 1 by hardware (the conversion of all injected channels is completed), and cleared by software (invalid if writing 1).	0
1	EOC	RW0	Conversion completion status: 1: The conversion has completed; 0: The conversion has not completed. This bit is set to 1 by hardware (the regular or injected channel group conversion ends), and is cleared by software (invalid if writing 1) or clearing when ADCx_RDATAR is read.	0
0	AWD	RW0	Analog watchdog flag bit: 1: The analog watchdog event occurs; 0: No analog watchdog event occurs. This bit is set to 1 by hardware (the conversion value is out of the ADCx_WDHTR and ADCx_WDLTR register range), and is cleared by software (invalid if writing 1).	0

12.3.2 ADC Control Register 1 (ADCx_CTLR1) (x=1/2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SW_PRE	ANA_RST	Reserved						AWDEN	JAWDEN	Reserved		DUALMOD[3:0]			

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DISCNUM[2:0]			JDISC	DISC	JAUT	AWD	SCAN	JEOCI	AWDIE	EOCI	AWDCH[4:0]				

	EN	EN	O	SGL		E		E	

Bit	Name	Access	Description	Reset value
31	SW_PRE	RW	Channel pre-switching off: 1: Channel pre-switching disabled; 0: Channel pre-switching enabled.	0
30	ANA_RST	RW	Analog module reset enable: 1: Enable; 0: Disable. *Note: Resetting the ADC analog circuit during an ADC conversion will corrupt the conversion result; this bit can be used in conjunction with the RCC's ADCxRST.*	0
[29:24]	Reserved	RO	Reserved	0
23	AWDEN	RW	Analog watchdog enable bit on regular channels: 1: Enable analog watchdog on regular channels; 0: Disable analog watchdog on regular channels.	0
22	JAWDEN	RW	Analog watchdog enable bit on injected channels: 1: Enable analog watchdog on injected channels; 0: Disable analog watchdog on injected channels.	0
[21:20]	Reserved	RO	Reserved	0
[19:16]	DUALMOD[3:0]	RW	Dual mode selection: 0000: Independent mode; 0001: Synchronous rules + synchronous injection mode; 0010: Synchronous rules + alternating trigger mode; 0011: Synchronous injection + rapid alternating mode; 0100: Synchronous injection + slow alternating mode; 0101: Synchronous injection mode; 0110: Synchronous rule mode; 0111: Fast alternating mode; 1000: Slow alternating mode; 1001: Alternating trigger mode. *Note: These bits in ADC2 are reserved. Any configuration bit modifications should be performed with dual mode disabled.*	0000b
[15:13]	DISCNUM[2:0]	RW	In discontinuous mode, the number of regular channels to be converted after external trigger: 000: 1 channel; … 111: 8 channels.	000b
12	JDISCEN	RW	Discontinuous mode enable bit on injected channel: 1: Enable discontinuous mode on the injected channel; 0: Disable discontinuous mode on the injected channel.	0
11	DISCEN	RW	Discontinuous mode enable level on the regular channel: 1: Enable discontinuous mode on regular channel. 0: Disable discontinuous mode on regular channel.	0

10	JAUTO	RW	After opening the regular channel, automatically convert the enable bit of the injected channel group: 1: Enable automatic injection channel group conversion. 0: Disable automatic injection channel group conversion. *Note: This mode needs to disable the external trigger function of the injection channel.*	0
9	AWDSGL	RW	In scan mode, analog watchdog enable bit on a single channel: 1: Enable analog watchdog on single channel (AWDCH[4:0] selection); 0: Disable analog watchdog on all channels.	0
8	SCAN	RW	Scan mode enable bit: 1: Enable scan mode (Sequentially convert all channels selected by ADCx_RSQRx (rule group) and ADCx_ISQR (injection group)); 0: Disable scan mode.	0
7	JEOCIE	RW	Injected channel group conversion completion interrupt enable bit: 1: Enable injected channel group transfer completion interrupt (JEOC flag); 0: Disable injected channel group transfer completion interrupt.	0
6	AWDIE	RW	Analog watchdog interrupt enable bit: 1: Enable analog watchdog interrupt; 0: Disable analog watchdog interrupt. *Note: In scan mode, if this interrupt occurs, the scan will be aborted.*	0
5	EOCIE	RW	Conversion completion (regular or injected channel group) interrupt enable bit; 1: Enable the transfer completion bit (EOC flag): 0: Disable the transfer completion interrupt.	0
[4:0]	AWDCH[4:0]	RW	Analog watchdog channel selection bit: 00000: Analog input channel 0; 00001: Analog input channel 1; … 10000: Analog input channel 16; 10001: Analog input channel 17. Others: Reserved.	00000b

12.3.3 ADCx Control Register 2 (ADCx_CTLR2) (x=1/2)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								TSVREFE	RSWSTART	JSWSTART	EXTTRIG	EXTSEL[2:0]			Reserved

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JEXTTRIG	JEXTSEL[2:0]			ALIGN	Reserved		DMA	Reserved				RSTCAL	CAL	CONT	ADON

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
23	TSVREFE	RW	Temperature sensor and internal voltage (V_{REFINT}) channel enable: 1: Enable the temperature sensor and V_{REFINT} channel; 0: Disable the temperature sensor and V_{REFINT} channel. *Note: This bit is only available for ADC1.*	0
22	RSWSTART	RW	To start a regular channel conversion, you need to set the software trigger: 1: Start regular channel conversion. 0: Reset state. This bit is set by the software, and the hardware is cleared 0 after the conversion starts.	0
21	JSWSTART	RW	To start an injection channel conversion, you need to set the software trigger: 1: Start the injection channel conversion. 0: Reset state. This bit is set by the software, and the hardware or software clears 0 after the conversion starts.	0
20	EXTTRIG	RW	External trigger conversion mode enable for regular channels: 1: Enable conversion on external event; 0: Disable conversion on external event.	0
[19:17]	EXTSEL[2:0]	RW	External trigger event selection for regular channel: 000: CC1 event of timer 1; 001: CC2 event of timer 1; 010: CC3 event of timer 1; 011: CC2 event of timer 2; 100: TRGO event of timer 3; 101: CC4 event of timer 4; 110: EXTI line 11/TRGO event of Timer 8; 111: SWSTART software trigger.	000b
16	Reserved	RO	Reserved	0
15	JEXTTRIG	RW	External trigger conversion mode enable for injected channels: 1: Enable conversion on external event; 0: Disable conversion on external event.	0
[14:12]	JEXTSEL[2:0]	RW	External trigger event select for injected channels: 000: TRGO event of timer 1; 001: CC4 event of timer 1; 010: TRGO event of timer 2;	000b

			011: CC1 event of timer 2; 100: CC4 event of timer 3; 101: TRGO event of timer 4; 110: EXTI line 15/CC4 event of Timer 8; 111: JSWSTART software trigger.	
11	ALIGN	RW	Data alignment: 1: Left alignment; 0: Right alignment.	0
[10:9]	Reserved	RO	Reserved	0
8	DMA	RW	Direct memory access (DMA) mode enable: 1: Enable DMA mode; 0: Disable DMA mode.	0
[7:4]	Reserved	RO	Reserved	0
3	RSTCAL	RW	Reset calibration, this bit is set by software, and cleared by hardware after reset: 1: Initialize calibration register; 0: The calibration register initialized. *Note: If RSTCAL is set while the conversion is in progress, it takes extra cycles to clear the calibration register.*	0
2	CAL	RW	A/D calibration, set by software and cleared by hardware when the calibration is completed. 1: Enable the calibration: 0: Calibration completed.	0
1	CONT	RW	Continuous conversion enable: 1: Continuous conversion mode; 0: Single conversion mode. If this bit is set, the conversion will continue until the bit is cleared.	0
0	ADON	RW	A/D converter ON/OFF When this bit is 0, writing 1 will wake up the ADC from power-down mode; When this bit is 1, writing a 1 will start the conversion, and if any other bit changes, a new conversion will not be started. 1: Enable ADC and to start conversion; 0: Disable ADC conversion/calibration, and go to power down mode.	0

12.3.4 ADCx Sample Time Configuration Register 1 (ADCx_SAMPTR1) (x=1/2)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					SMP18[2:0]			SMP17[2:0]			SMP16[2:0]			SMP15[2:1]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMP15[0]		SMP14[2:0]		SMP13[2:0]			SMP12[2:0]			SMP11[2:0]			SMP10[2:0]		

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:0]	SMPx[2:0]	RW	SMPx[2:0]: Sample time configuration of channel x: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 41.5 cycles; 101: 55.5 cycles; 110: 71.5 cycles; 111: 239.5 cycles; SMPx[2:0]: When the ADC_SMP_SELx bit is set, the sampling interval configuration of channel x is: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 2.5 cycles; 101: 3.5 cycles; 110: 4.5 cycles; 111: 5.5 cycles; These bits are used to independently select the sample time of each channel, and the channel configuration value must remain unchanged during the sampling period.	000b

12.3.5 ADCx Sample Time Configuration Register 2 (ADCx_SAMPTR2) (x=1/2)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		SMP9[2:0]			SMP8[2:0]			SMP7[2:0]			SMP6[2:0]			SMP5[2:1]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMP5[0]		SMP4[2:0]			SMP3[2:0]			SMP2[2:0]			SMP1[2:0]			SMP0[2:0]	

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:0]	SMPx[2:0]	RW	SMPx [2:0]: sampling time configuration for channel x: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 41.5 cycles; 101: 55.5 cycles; 110: 71.5 cycles; 111: 239.5 cycles; SMPx[2:0]: When the ADC_SMP_SELx bit is set, the sampling interval configuration of channel X is: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 2.5 cycles; 101: 3.5 cycles; 110: 4.5 cycles; 111: 5.5 cycles; These bits are used to independently select the sample time of each channel, and the channel configuration value must remain unchanged during the sampling period.	000b

12.3.6 ADCy Injected Channel Data Offset Register x (ADCy_IOFRx) (x=1/2/3/4, y=1/2)

Offset address: 0x14 + (x-1)*4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				JOFFSETx[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
[11:0]	JOFFSETx[11:0]	RW	Data offset for injected channel x. When the injected channel is converted, these bits define the value to be subtracted from the original conversion data. The result of the conversion can be read in the ADCx_IDATARx register.	0

12.3.7 ADCx Watchdog High Threshold Register (ADCx_WDHTR) (x=1/2)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				HT[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
[11:0]	HT[11:0]	RW	Analog watchdog high threshold set bits.	0xFFF

Note: The values of WDHTR and LTR can be changed during the conversion, but they will take effect in the next conversion.

12.3.8 ADCx Watchdog Low Threshold Register (ADCx_WDLTR) (x=1/2)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				LT[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
[11:0]	LT[11:0]	RW	Analog watchdog low threshold set bits.	0

Note: The values of WDHTR and LTR can be changed during the conversion, but they will take effect in the next conversion.

12.3.9 ADCx Regular Channel Sequence Register1 (ADCx_RSQR1) (x=1/2)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								L[3:0]				RSQ16[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ16[0]	SQ15[4:0]					SQ14[4:0]					SQ13[4:0]				

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[23:20]	L[3:0]	RW	The number of channels to be converted in a regular channel conversion sequence: 0000-1111: 1-16 conversions.	0
[19:15]	SQ16[4:0]	RW	Number of the 16th conversion channel in the regular sequence (0-17).	0
[14:10]	SQ15[4:0]	RW	Number of the 15th conversion channel in the regular sequence (0-17).	0
[9:5]	SQ14[4:0]	RW	Numbering of the 14th conversion channel in the regular sequence (0-17).	0
[4:0]	SQ13[4:0]	RW	Numbering of the 13th conversion channel in the regular sequence (0-17).	0

12.3.10 ADCx Regular Channel Sequence Register 2 (ADCx_RSQR2) (x=1/2)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		SQ12[4:0]					SQ11[4:0]					SQ10[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ10[0]	SQ9[4:0]					SQ8[4:0]					SQ7[4:0]				

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:25]	SQ12[4:0]	RW	Number of the 12th conversion channel in the regular sequence (0-17).	0
[24:20]	SQ11[4:0]	RW	Number of the 11th conversion channel in the regular sequence (0-17).	0
[19:15]	SQ10[4:0]	RW	Number of the 10th conversion channel in the regular sequence (0-17).	0
[14:10]	SQ9[4:0]	RW	Number of the 9th conversion channel in the regular sequence (0-17).	0
[9:5]	SQ8[4:0]	RW	Number of the 8th conversion channel in the regular sequence (0-17).	0
[4:0]	SQ7[4:0]	RW	Number of the 7th conversion channel in the regular sequence (0-7).	0

12.3.11 ADCx Regular Channel Sequence Register 3 (ADCx_RSQR3) (x=1/2)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		SQ6[4:0]					SQ5[4:0]					SQ4[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ4[0]		SQ3[4:0]					SQ2[4:0]					SQ1[4:0]			

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:25]	SQ6[4:0]	RW	Number of the 6th conversion channel in the regular sequence (0-17).	0
[24:20]	SQ5[4:0]	RW	Number of the 5th conversion channel in the regular sequence (0-17).	0
[19:15]	SQ4[4:0]	RW	Number of the 4th conversion channel in the regular sequence (0-17).	0
[14:10]	SQ3[4:0]	RW	Number of the 3th conversion channel in the regular sequence (0-17).	0
[9:5]	SQ2[4:0]	RW	Number of the 2nh conversion channel in the regular sequence (0-17).	0
[4:0]	SQ1[4:0]	RW	Number of the 1st conversion channel in the regular sequence (0-17).	0

12.3.12 ADCx Injected Channel Sequence Register (ADCx_ISQR) (x=1/2)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										JL[1:0]		JSQ4[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JSQ4[0]		JSQ3[4:0]					JSQ2[4:0]					JSQ1[4:0]			

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
[21:20]	JL[1:0]	RW	Injects the number of channels to be converted in the channel conversion sequence: 00-11: 1-4 conversions.	0
[19:15]	JSQ4[4:0]	RW	Inject the number of the 4th conversion channel in the sequence (0-17). *Note: The software writes and assigns the channel number (0-17) as the 4th in the sequence to be converted.*	0
[14:10]	JSQ3[4:0]	RW	Inject the number of the 3rd conversion channel in the sequence (0-17).	0
[9:5]	JSQ2[4:0]	RW	Inject the number of the 2nd conversion channel in the sequence (0-17).	0
[4:0]	JSQ1[4:0]	RW	Inject the number of the 1st conversion channel in the sequence (0-17).	0

Note: Unlike regular conversion sequences, if the length of JL[1:0] is less than 4, the sequence order of

conversions starts with (4-JL).
For example, when JL[1:0]=3 (4 injected transitions in the sequencer), the ADC will convert channels in the following order: JSQ1[4:0], JSQ2[4:0], JSQ3[4:0], and JSQ4[4:0];
When JL[1:0]=2 (3 injected transitions in the sequencer), the ADC will convert the channels in the following order: JSQ2[4:0], JSQ3[4:0] and JSQ4[4:0];
When JL[1:0]=1 (2 injected conversions in the sequencer), the ADC converts the channels in the following order: first JSQ3[4:0], then JSQ4[4:0];
When JL[1:0] = 0 (1 injection conversion in the sequencer), the ADC will convert only the JSQ4[4:0] channels.
If ADCx_ISQR[21:0]=10 00111 00011 00111 00010, the ADC will convert channels in the following order: JSQ2[4:0], JSQ3[4:0], and JSQ4[4:0], indicating that the scan conversions are performed in the following channel order: 7, 3, 7.

12.3.13 ADCy Injected Data Register (ADCy_IDATARx) (x=1/2/3/4, y=1/2)

Offset address: 0x3C + (x-1)*4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JDATA[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved.	0
[15:0]	JDATA[15:0]	RO	Injected channel converted data (data left alignment or right alignment).	0

12.3.14 ADCx Regular Data Register (ADCx_RDATAR) (x=1/2)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADC2DATA[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	ADC2DATA[15:0]	RO	Data converted by ADC2: In ADC1: In dual-mode operation, these bits contain the regular channel data converted by ADC2. In ADC2: These bits are not used.	0
[15:0]	DATA[15:0]	RO	Regular channel converted data (data left alignment or right alignment).	0

12.3.15 ADCx Sampling Time Register (ADCx_AUX) (x=1/2)

Offset address: 0x54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														ADC	ADC

																SMP_SEL17	SMP_SEL16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
ADC_SMP_SEL15	ADC_SMP_SEL14	ADC_SMP_SEL13	ADC_SMP_SEL12	ADC_SMP_SEL11	ADC_SMP_SEL10	ADC_SMP_SEL9	ADC_SMP_SEL8	ADC_SMP_SEL7	ADC_SMP_SEL6	ADC_SMP_SEL5	ADC_SMP_SEL4	ADC_SMP_SEL3	ADC_SMP_SEL2	ADC_SMP_SEL1	ADC_SMP_SEL0		

Bit	Name	Access	Description	Reset value
[31:18]	Reserved	RO	Reserved	0
[17:0]	ADC_SMP_SELx	RW	x=0-17, optional enabling bits for sampling time of channel x (100-111 only in SMPx): 1: SMPx=100: The sampling time is 2.5 cycles; SMPx=101: The sampling time is 3.5 cycles; SMPx=110: The sampling time is 4.5 cycles; SMPx=111: The sampling time is 5.5 cycles. 0: Sampling time is the corresponding SMPx configuration cycle.	0

Chapter 13 Digital-to-Analog Converter (DAC)

Digital/analog conversion module (DAC), including two 12-bit voltage output digital/analog converters (DAC), converts 2 digital signals into 2 analog voltage signals and outputs them, supports independent or synchronous conversion of dual DAC channels, supports 12-bit data left or right alignment, supports 12-bit or 8-bit data, and supports external event-triggered conversion. Triangular wave and noise generation can be achieved. Support DMA function.

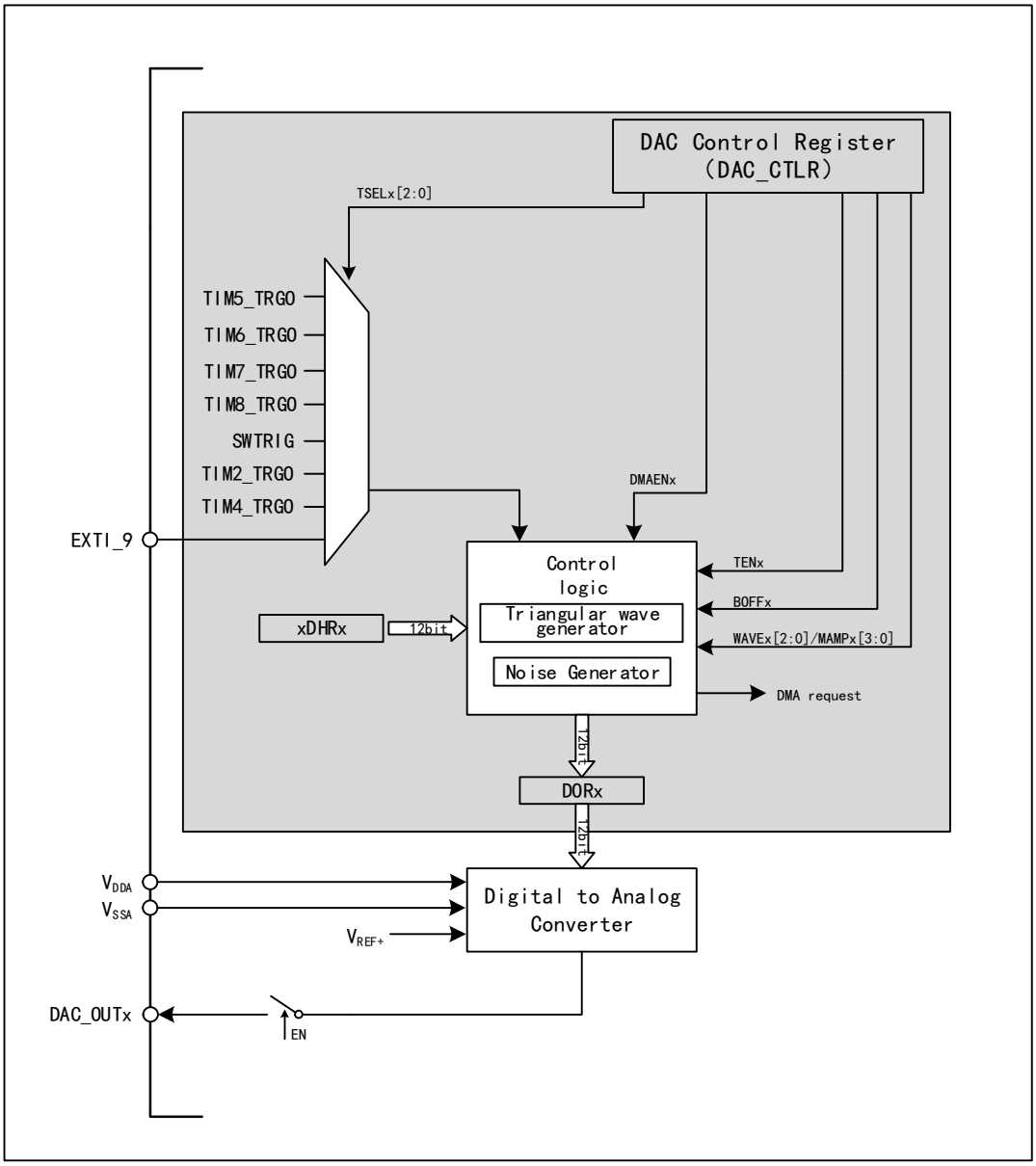
13.1 Main Features

- 2 DAC converters, and each has single output channel
- Triangular-wave generator and noise-wave generator
- Configurable 8-bit or 12-bit output
- Left or right data alignment in 12-bit mode
- Dual DAC channel for independent or simultaneous conversions
- DMA capability
- Multiple trigger events

13.2 Functional Description

13.2.1 DAC Structure

Figure 13-1 DAC block diagram



13.2.2 DAC Channel Configuration

13.2.2.1 Enable DAC:

Set the ENx bit in the DAC_CTLR register to 1, to enable the analog power to DAC channel x. After a period of start-up time, DAC channel x can be enabled. The DAC contains 2 analog output channels which can output independently or simultaneously.

Note: In order to avoid parasitic interference and additional power consumption, the corresponding pins of the DAC channel need to be set to analog input (AIN) mode in advance.

13.2.2.2 Enable Output Buffer:

DAC integrates output buffers, which can be used to reduce the output impedance and increase the drive capacity to directly drive the external loads. Each DAC channel output buffer can be enabled or disabled by setting the BOFFx bit in the DAC_CTLR register.

13.2.2.3 Data Format:

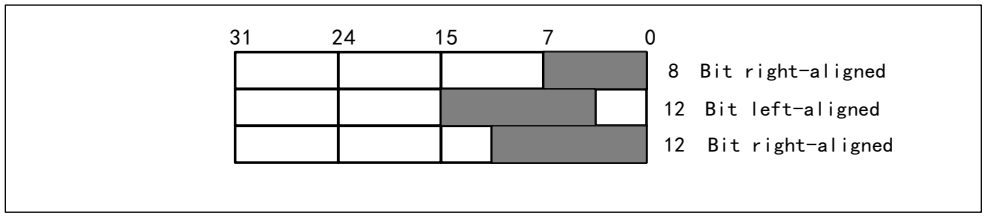
In single DAC channel mode, data formats include 8-bit right alignment, 12-bit left alignment and 12-bit right alignment.

When 8-bit data is right-aligned, write data to DAC_R8BDHRx[7:0], and the module will load (after 1 PB1 clock cycle) its left-shifted data to the data output register DAC_DORx[11:4].

When 12-bit data is right-aligned, write data to DAC_R12BDHRx[11:0], and the module will load (after 1 PB1 clock cycle) right-aligned data into the data output register DAC_DORx[11:0].

When 12 data are left-aligned, write the data to DAC_L12BDHRx[15:4]. After the corresponding shift, the module will load (after 1 PB1 clock cycle) the left-aligned data to the data output register DAC_DORx[11:0].

Figure 13-2 Data formats in single channel mode



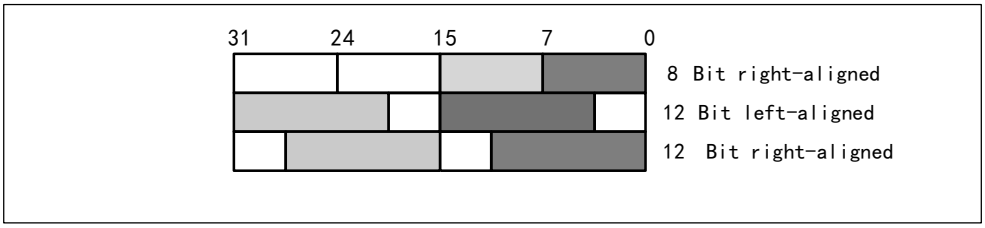
In dual DAC channel mode, there are also 3 methods: 8-bit data right alignment, 12-bit data left alignment, and 12-bit data right alignment.

8-bit data right-aligned, write data to DAC_RD8BDHR[7:0], the module will load (after 1 HB clock cycle) bits [7:0] shifted to DAC_DOR1[11:4] and bits [15:8] shifted to DAC_DOR2[11:4].

12 data left-aligned, write data to DAC_LD12BDHR[31:0], the module will load (after 1 HB clock cycle) bit[15:4] data shifted to DAC_DOR1[11:0] and bit[31:20] data shifted to DAC_DOR2[11:0].

12-bit data right-aligned, write data to DAC_RD12BDHR[31:0], the module will load (after 1 HB clock cycle) bit[11:0] data to DAC_DOR1[11:0] and bit[27:16] data to DAC_DOR2[11:0].

Figure 13-3 Data formats in dual channel mode



13.2.2.4 DMA Capability:

DAC channel has the DMA capability. Set the DMAENx bit in the DAC_CTLR register to 1, to enable DMA of the corresponding channel. When a trigger event occurs (Software trigger not included), a DMA request is generated. And then the data in the DAC_DORx register can be updated.

13.2.2.5 Trigger Event Selection:

DAC conversion can be triggered by the following events. When the TENx bit in the DAC_CTLR register is set to 1, a trigger event to trigger DAC conversion can be selected by configuring the TSELx[2:0] bits.

Table 13-1 Trigger events

Trigger source	Type	TSELx[2:0]
Timer 6 TRGO event	Internal signal from on-chip timer	000

Timer 8 TRGO event		001
Timer 7 TRGO event		010
Timer 5 TRGO event		011
Timer 2 TRGO event		100
Timer 4 TRGO event		101
EXTI line9	External pin	110
SWTRIG (Software trigger)	Software control bit	111

Every time a DAC interface detects a rising edge on the selected timer TRGO output or on the selected external interrupt line 9, the DAC_DORx register is updated 3 HB clock cycles after the trigger occurs.

If the software trigger mode is configured, once the SWTRIG bit is set to 1, a conversion is started. The DAC_DORx register is updated single PB1 clock cycle after the trigger occurs, and the SWTRIG bit can be automatically cleared by hardware.

Note: The TSELx[2:0] bits cannot be changed when ENx is 1.

13.2.3 DAC Conversion

The data of the DAC channel comes from the DAC_DORx register, but data cannot be directly written to the register DAC_DORx. Any data output to the DAC channel x must be written into the DAC_R12BDHR1, DAC_L12BDHR1, DAC_R12BDHR2, DAC_L12BDHR2, DAC_RD12BDHR, DAC_LD12BDHR and DAC_RD8BDHR registers. The internal DAC_DHRx register obtains the value of above registers and transfers it to the DAC_DORx register after the corresponding time.

In the non-trigger mode, the data written into the DAC_xDHRx register is shifted into the DAC_DORx register in a HB clock cycle.

In software trigger mode, the DAC_DORx register is automatically updated in a HB clock cycle after the rising edge of the event trigger.

In hardware trigger mode (Timer TRGO event or external interrupt line 9 rising edge), the DAC_DORx register is automatically updated in 3 PB1 clock cycles after the trigger event.

Load the DAC_DORx register data. After the time of $t_{SETTLING}$, the output can be valid, and the length of this period of time will vary depending on the supply voltage and the analog output load.

The digital input is linearly converted to analog voltage output by the DAC, and it ranges from 0 to V_{DDA}. The output voltage on any DAC channel pin shall meet the following relationship:

DAC output voltage = V_{DDA}* (DAC_DORx/4096).

13.2.4 DAC Triangular-wave Generator

The module has a built-in triangular-wave generator, which can add a small amplitude triangle-wave to the reference signal. Set WAVEx[1:0] bit as '10b' and select the triangular-wave generation function of DAC. Set MAMPx[3:0] bit in the DAC_CTLR register to select the amplitude of triangular-wave.

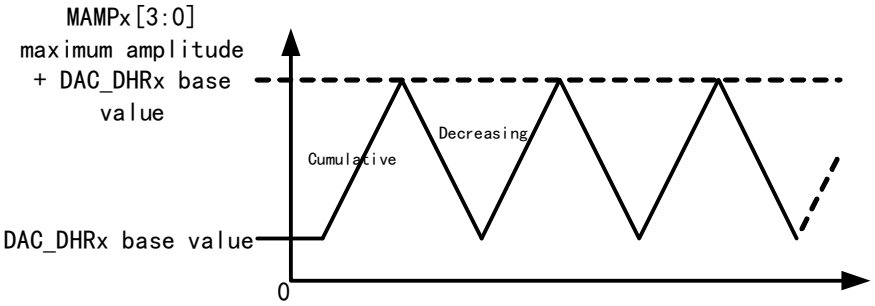
The system contains a triangular-wave counter starting from 0, which accumulates by 1 in 3 HB clock cycles after each trigger event. The value of the counter is added to the value of the DAC_DHRx register and the overflow bit is discarded and then written to the DAC_DORx register. When the value transmitted into the

DAC_DORx register is smaller than the maximum amplitude defined by the MAMPx[3:0] bits, the triangular-wave counter will gradually accumulate. Once it reaches the set maximum amplitude, the counter will begin to decrease progressively, and then start to accumulate after reaching 0. Repeat this cycle. Set WAVEx[1:0] bits to '00' to reset the generation of triangle-waves.

Note: 1. To generate a triangular-wave, DAC trigger must be enabled, i.e., setting the TENx bit in the DAC_CTLR register to 1.

2. MAMPx[3:0] bits must be set before enable the DAC. Otherwise, its value cannot be modified.

Figure 13-4 Triangular wave generation



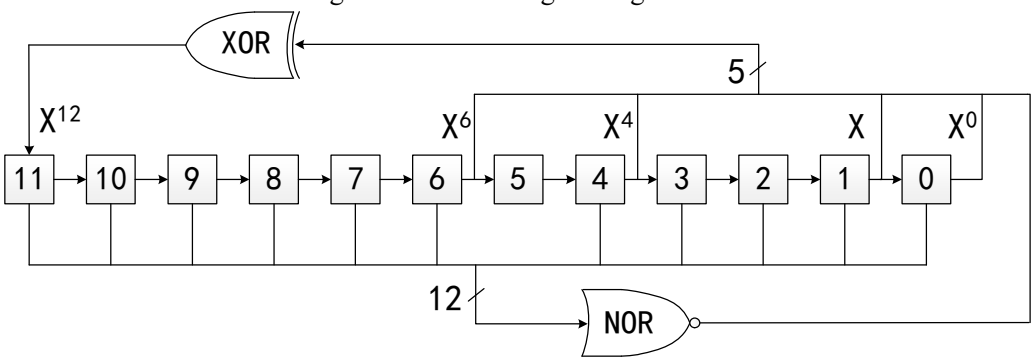
13.2.5 DAC Noise-wave Generator

The module has a built-in noise-generator that uses the Linear Feedback Shift Register (LFSR) to generate pseudo noise with varying amplitude. Set WAVE[1:0] bits to '01b' to select the DAC noise generation function. Set the MAMPx[3:0] bits in the DAC_CTLR register to select the data of the masked part of the LFSR.

The preload value of the register LFSR is 0xAAA. According to a specific algorithm, the value of this register is updated in 3 HB clock cycles after each trigger event. Setting the MAMPx[3:0] bits in the DAC_CTLR register can mask part or all of the LFSR data, so that the LSFR value obtained is added to the value of DAC_DHRx, and the overflow bit is removed and then written into the DAC_DORx register. If the register LFSR value is 0x000, it will inject '1' (Anti-lock mechanism). Set WAVEx[1:0] bit to '00b' to reset the generation algorithm of LFSR waveform.

Note: To generate a noise-wave, DAC trigger must be enabled, i.e., setting the TENx bit in the DAC_CTLR register to 1.

Figure 13-5 LFSR register algorithm



13.3 Dual DAC Conversion

When the 2 DAC channels are required at the same time, the module integrates 3 data registers in dual DAC mode (DAC_RD8BDHR, DAC_LD12BDHR, DAC_RD12BDHR), in order to efficiently operate the DAC module. The conversion value of the 2 DACs can be updated when only one of the 3 registers is operated.

For dual DAC conversion, other registers of the module can be used to implement 11 types of conversion modes with different combinations. The value of data to be converted in the 2 channels needs to be written to one of the 3 data registers.

13.3.1 Independent Trigger with the Same LFSR Generation

Set the TENx bits. TSELx can be set as different values. WAVEx is set to 0b01. MAMPx is set as the same LFSR mask value. When a channel1 trigger event occurs, add the counter (LFSR1) with the same mask value to the channel1 data register (DAC_DHR1), and the sum is transferred to DAC_DOR1 after delay of 3 PB1 clocks, for conversion. Then the LFSR1 counter is updated. When a channel2 trigger event occurs, add the counter (LFSR2) with the same mask value to the channel 2 data register (DAC_DHR2), and the sum is transferred to DAC_DOR2 for conversion after delay of 3 PB1 clocks. Then the LFSR2 counter is updated.

13.3.2 Independent Trigger with Different LFSR Generation

Set the TENx bits. TSELx can be set as different values. WAVEx is set to 0b01. MAMPx can be set as different mask values of LFSR. When a channel1 trigger event occurs, add the LFSR1 counter, with the mask value configured by the MAMP1[3:0] bits, to the channel 1 data register (DAC_DHR1), and the sum is transferred to DAC_DOR1 for conversion after delay of 3 PB1 clocks. Then the LFSR1 counter is updated. When a channel 2 trigger event occurs, add the LFSR2 counter, with the mask value configured by the MAMP2[3:0] bits, to the channel 2 data register (DAC_DHR2), and the sum is transferred to DAC_DOR2 for conversion after delay of 3 PB1 clocks. Then the LFSR2 counter is updated.

13.3.3 Independent Trigger with the Same Triangle Generation

Set the TENx bits. TSELx can be set as different values. WAVEx is set to 0b1x. MAMPx is set as the same triangle amplitude value. When a channel1 trigger event occurs, add the channel1 triangle counter, with the same triangle amplitude, to the channel1 data register (DAC_DHR1), and the sum is transferred to DAC_DOR1 for conversion after delay of 3 PB1 clocks. Then the channel1 triangle counter is updated. When a channel2 trigger event occurs, add the channel2 triangle counter, with the same triangle amplitude, to the channel2 data register (DAC_DHR2), and the sum is transferred to DAC_DOR2 for conversion after delay of 3 PB1 clocks. Then the channel2 triangle counter is updated.

13.3.4 Independent Trigger with Different Triangle Generation

Set the TENx bits. TSELx can be set as different values. WAVEx is set to 0b1x. MAMPx can be set as different triangle amplitude values. When a channel1 trigger event occurs, add the channel1 triangle counter, with a triangle amplitude configured by MAMP1, to the channel1 data register (DAC_DHR1), and the sum is transferred to DAC_DOR1 for conversion after delay of 3 PB1 clocks. Then the channel1 triangle counter is updated. When a channel2 trigger event occurs, add the channel2 triangle counter, with a triangle amplitude configured by MAMP2, to the channel2 data register (DAC_DHR2), and the sum is transferred to DAC_DOR2 for conversion after delay of 3 PB1 clocks. Then the channel2 triangle counter is updated.

13.3.5 Independent Trigger with No Wave Generation

Set the TENx bits. TSELx can be set as different values to select different trigger sources. When a channel1 trigger event occurs, the value of the channel1 data register (DAC_DHR1) delayed 3 PB1 clocks is transferred to DAC_DOR1 for conversion. When a channel2 trigger event occurs, the value of the channel2 data register (DAC_DHR2) delayed 3 PB1 clocks is transferred to DAC_DOR2 for conversion.

13.3.6 Simultaneous Software Trigger

In this conversion mode, write the dual DAC channel data to the desired register. A PB1 clock cycle later, the data in DAC_DHR1 and DAC_DHR2 is respectively transferred to DAC_DOR1 and DAC_DOR2 for conversion.

13.3.7 Simultaneous Trigger with the Same LFSR Generation

Set the TENx bits. TSELx is set as the same value. WAVEx is set to 0b01. MAMPx is set as the same LFSR mask value. When a trigger event occurs, add the LFDR1 counter, with the same mask value, to the DAC_DHR1 register, and the sum is transferred to DAC_DOR1 after delay of 3 PB1 clocks for conversion. Then the LFSR1 counter is updated. At the same time, add the LFDR2 counter, with the same mask value, to the DAC_DHR2 register, and the sum is transferred to DAC_DOR2 after delay of 3 PB1 clocks for conversion. Then the LFSR2 counter is updated.

13.3.8 Simultaneous Trigger with Different LFSR Generation

Set the TENx bits. TSELx is set as the same value. WAVEx is set to 0b01. MAMPx can be set as different LFSR mask values. When a trigger event occurs, add the LFSR1 counter, with different mask values, to the DAC_DHR1 register, and the sum is transferred to DAC_DOR1 after delay of 3 PB1 clocks for conversion. Then the LFSR1 counter is updated. At the same time, add the LFSR2 counter, with different mask values, to the DAC_DHR2 register, and the sum is transferred to DAC_DOR2 after delay of 3 PB1 clocks for conversion. Then the LFSR2 counter is updated.

13.3.9 Simultaneous Trigger with the Same Triangle Generation

Set the TENx bits. TSELx is set as the same value. WAVEx is set to 0b1x. MAMPx is set as the same triangle amplitude value. When a trigger event occurs, add the counter, with the same triangle amplitude value, to the DAC_DHR1 register, and the sum is transferred to DAC_DOR1 after delay of 3 PB1 clocks for conversion. Then the channel1 triangle counter is updated. At the same time, add the counter, with the same triangle amplitude value, to the DAC_DHR2 register, and the sum is transferred to DAC_DOR2 after delay of 3 PB1 clocks for conversion. Then the channel2 triangle counter is updated.

13.3.10 Simultaneous Trigger with Different Triangle Generation

Set the TENx bits. TSELx is set as the same value. WAVEx is set to 0b1x. MAMPx can be set as different triangle amplitude values. When a trigger event occurs, add the counter, with a triangle amplitude configured by MAMP1[3:0], to the DAC_DHR1 register, and the sum is transferred to DAC_DOR1 after delay of 3 PB1 clocks for conversion. Then the channel1 triangle counter is updated. At the same time, add the counter, with a triangle amplitude configured by MAMP2[3:0], to the DAC_DHR2 register, and the sum is transferred to DAC_DOR2 after delay of 3 PB1 clocks for conversion. Then the channel2 triangle counter is updated.

13.3.11 Simultaneous Trigger with No Wave Generation

Set the TENx bits. TSELx is set as the same value. In this conversion mode, when a trigger event occurs, the values of the DAC_DHR1 register and the DAC_DHR2 register delayed 3 PB1 clocks are respectively transferred to DAC_DOR1 and DAC_DOR2 for DAC conversion.

13.4 Register Description

Table 13-2 DAC registers

Name	Access address	Description	Reset value
R32_DAC_CTLR	0x40007400	DAC control register	0x00000000
R32_DAC_SWTR	0x40007404	DAC software trigger register	0x00000000
R32_DAC_R12BDHR1	0x40007408	DAC channel1 right-aligned 12-bit data hold register	0x00000000
R32_DAC_L12BDHR1	0x4000740C	DAC channel1 left-aligned 12-bit data hold register	0x00000000
R32_DAC_R8BDHR1	0x40007410	DAC channel1 right-aligned 8-bit data hold register	0x00000000
R32_DAC_R12BDHR2	0x40007414	DAC channel2 right-aligned 12-bit data hold register	0x00000000
R32_DAC_L12BDHR2	0x40007418	DAC channel2 left-aligned 12-bit data hold register	0x00000000
R32_DAC_R8BDHR2	0x4000741C	DAC channel2 right-aligned 8-bit data hold register	0x00000000
R32_DAC_RD12BDHR	0x40007420	Dual channel right-aligned 12-bit data hold register	0x00000000
R32_DAC_LD12BDHR	0x40007424	Dual channel left-aligned 12-bit data hold register	0x00000000
R32_DAC_RD8BDHR	0x40007428	Dual channel right-aligned 8-bit data hold register	0x00000000
R32_DAC_DOR1	0x4000742C	DAC channel1 data output register	0x00000000
R32_DAC_DOR2	0x40007430	DAC channel2 data output register	0x00000000

13.4.1 DAC Control Register (DAC_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			DMAEN2	MAMP2[3:0]				WAVE2[1:0]		TSEL2[2:0]			TEN2	BOFF2	EN2
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DMAEN1	MAMP1[3:0]				WAVE1[1:0]		TSEL1[2:0]			TEN1	BOFF1	EN1

Bit	Name	Access	Description	Reset value
[31:29]	Reserved	RO	Reserved.	0
28	DMAEN2	RW	DMA enable of DAC channel2: 0: DMA of DAC channel2 disabled; 1: DMA of DAC channel2 enabled.	0
[27:24]	MAMP2[3:0]	RW	DAC channel2 mask/amplitude setting. The software sets this area to select the LFSR data mask bit in the noise generation mode, and select the waveform amplitude in the triangle waveform	0000b

			generation mode: 0000: Unmask bit0 of LFSR/Triangle amplitude equal to 1; 0001: Unmask bits[1:0] of LFSR / Triangle amplitude equal to 3; 0010: Unmask bits[2:0] of LFSR / Triangle amplitude equal to 7; 0011: Unmask bits[3:0] of LFSR / Triangle amplitude equal to 15; 0100: Unmask bits[4:0] of LFSR / Triangle amplitude equal to 31; 0101: Unmask bits[5:0] of LFSR / Triangle amplitude equal to 63; 0110: Unmask bits[6:0] of LFSR / Triangle amplitude equal to 127; 0111: Unmask bits[7:0] of LFSR / Triangle amplitude equal to 255; 1000: Unmask bits[8:0] of LFSR / Triangle amplitude equal to 511; 1001: Unmask bits[9:0] of LFSR / Triangle amplitude equal to 1023; 1010: Unmask bits[10:0] of LFSR / Triangle amplitude equal to 2047; ≥1011: Unmask bits[11:0] of LFSR / Triangle amplitude equal to 4095.	
[23:22]	WAVE2 [1:0]	RW	Noise/triangular wave generation enable of DAC channel2 00: Wave generator disabled; 01: Noise-wave generator enabled; 1x: Triangular-wave generator enabled.	00b
[21:19]	TSEL2[2:0]	RW	Trigger event selection of DAC channel2: 000: TIM6 TRGO event; 001: TIM8 TRGO event; 010: TIM7 TRGO event; 011: TIM5 TRGO event; 100: TIM2 TRGO event; 101: TIM4 TRGO event; 110: External interrupt line 9; 111: Software trigger; Others: Reserved.	000b
18	TEN2	RW	External trigger mode enable of DAC channel2: 0: Trigger of DAC channel2 disabled. The data written into the DAC_xDHR register is sent to the DAC_DOR2 register in 1 PB1 clock cycle. 1: Trigger of DAC channel2 enabled. The data written into the DAC_xDHR register is sent to the DAC_DOR2 register in 3 PB1 clock cycles. *Note: If software trigger is selected, the data in DAC_xDHR only needs to be sent to the DAC_DOR2 register in an PB1 clock cycle.*	0
17	BOFF2	RW	DAC channel2 output buffer disable control (recommended to be enabled): 0: DAC channel2 output buffer enabled; 1: DAC channel2 output buffer disabled.	0
16	EN2	RW	DAC channel2 enable: 0: DAC channel2 disabled; 1: DAC channel2 enabled.	0
[15:13]	Reserved	RO	Reserved.	0
12	DMAEN1	RW	DMA enable of DAC channel1:	0

			0: DMA function of DAC channel1 disabled; 1: DMA function of DAC channel1 enabled.	
[11:8]	MAMP1 [3:0]	RW	DAC channel 1 mask/amplitude setting. The software sets these bits to select the LFSR data mask bit in the noise generation mode, and select the wave amplitude in the triangle waveform generation mode: 0000: Unmask bit0 of LFSR / Triangle amplitude equal to 1; 0001: Unmask bits[1:0] of LFSR / Triangle amplitude equal to 3; 0010: Unmask bits[2:0] of LFSR / Triangle amplitude equal to 7; 0011: Unmask bits[3:0] of LFSR / Triangle amplitude equal to 15; 0100: Unmask bits[4:0] of LFSR / Triangle amplitude equal to 31; 0101: Unmask bits[5:0] of LFSR / Triangle amplitude equal to 63; 0110: Unmask bits[6:0] of LFSR / Triangle amplitude equal to 127; 0111: Unmask bits[7:0] of LFSR / Triangle amplitude equal to 255; 1000: Unmask bits[8:0] of LFSR / Triangle amplitude equal to 511; 1001: Unmask bits[9:0] of LFSR / Triangle amplitude equal to 1023; 1010: Unmask bits[10:0] of LFSR / Triangle amplitude equal to 2047; ≥1011: Unmask bits[11:0] of LFSR / Triangle amplitude equal to 4095.	0000b
[7:6]	WAVE1 [1:0]	RW	Noise/triangular-wave generation enable of DAC channel1. 00: Wave generator disabled; 01: Noise wave generator enabled; 1x: Triangular wave generator enabled.	00b
[5:3]	TSEL1 [2:0]	RW	Trigger event selection of DAC channel1: 000: TIM6 TRGO event; 001: TIM8 TRGO event; 010: TIM7 TRGO event; 011: TIM5 TRGO event; 100: TIM2 TRGO event; 101: TIM4 TRGO event; 110: External interrupt line9; 111: Software trigger; Others: Reserved.	000b
2	TEN1	RW	External trigger mode enable of DAC channel1: 0: Trigger of DAC channel1 disabled. The data written into the DAC_xDHR register is sent to the DAC_DOR1 register in 1 PB1 clock cycle. 1: Trigger of DAC channel1 enabled. The data written into the DAC_xDHR register is sent to the DAC_DOR1 register in 3 PB1 clock cycles. *Note: If software trigger is selected, the data in DAC_xDHR only needs to be sent to the DAC_DOR1 register in an PB1 clock cycle.*	0
1	BOFF1	RW	DAC channel1 output buffer disable control (Recommended to be enabled): 0: DAC channel1 output buffer enabled;	0

			1: DAC channel1 output buffer disabled.	
0	EN1	RW	DAC channel1 enable: 0: DAC channel1 disabled; 1: DAC channel1 enabled.	0

13.4.2 DAC Software Trigger Register (DAC_SWTR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														SW TRIG 2	SW TRIG 1

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	RO	Reserved.	0
1	SWTRIG2	WO	Software trigger control bit of DAC channel2: 0: Software trigger of DAC channel2 disabled; 1: Software trigger of DAC channel2 enabled. *Note: Once the data in DAC_xDHR (After 1 PB1 clock cycle) is sent to the DAC_DOR2 register, this bit can be cleared by hardware.*	0
0	SWTRIG1	WO	Software trigger control bit of DAC channel1: 0: Software trigger of DAC channel1 disabled; 1: Software trigger of DAC channel1 enabled. *Note: Once the data in DAC_xDHR (After 1 PB1 clock cycle) is sent to the DAC_DOR1 register, this bit can be cleared by hardware.*	0

13.4.3 DAC Channel 1 Right-aligned 12-bit Data Hold Register (DAC_R12BDHR1)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DACC1DHR[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved.	0
[11:0]	DACC1DHR[11:0]	RW	12-bit right-aligned data of DAC channel1.	0

13.4.4 DAC Channel 1 Left-aligned 12-bit Data Hold Register (DAC_L12BDHR1)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC1DHR[11:0]												Reserved			

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved.	0
[15:4]	DACC1DHR[11:0]	RW	12-bit left-aligned data of DAC channel1.	0
[3:0]	Reserved	RO	Reserved.	0

13.4.5 DAC Channel 1 Right-aligned 8-bit Data Hold Register (DAC_R8BDHR1)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DACC1DHR[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	DACC1DHR[7:0]	RW	8-bit right-aligned data of DAC channel 1.	0

13.4.6 DAC Channel 2 Right-aligned 12-bit Data Hold Register (DAC_R12BDHR2)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DACC2DHR[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
[11:0]	DACC2DHR[11:0]	RW	12-bit right-aligned data of DAC channel 2.	0

13.4.7 DAC Channel 2 Left-aligned 12-bit Data Hold Register (DAC_L12BDHR2)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC2DHR[11:0]												Reserved			

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:4]	DACC2DHR[11:0]	RW	12-bit left-aligned data of DAC channel2.	0

[3:0]	Reserved	RO	Reserved	0

13.4.8 DAC Channel 2 Right-aligned 8-bit Data Hold Register (DAC_R8BDHR2)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DACC2DHR[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	DACC2DHR[7:0]	RW	8-bit right-aligned data of DAC channel2.	0

13.4.9 DAC Dual Channel Right-aligned 12-bit Data Hold Register (DAC_RD12BDHR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				DACC2DHR[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DACC1DHR[11:0]											

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	DACC2DHR[11:0]	RW	12-bit right-aligned data of DAC channel 2.	0
[15:12]	Reserved	RO	Reserved	0
[11:0]	DACC1DHR[11:0]	RW	12-bit right-aligned data of DAC channel 1.	0

13.4.10 DAC Dual Channel Left-aligned 12-bit Data Hold Register (DAC_LD12BDHR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DACC2DHR[11:0]												Reserved			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC1DHR[11:0]												Reserved			

Bit	Name	Access	Description	Reset value
[31:20]	DACC2DHR[11:0]	RW	12-bit left-aligned data of DAC channel 2.	0
[19:16]	Reserved	RO	Reserved	0
[15:4]	DACC1DHR[11:0]	RW	12-bit left-aligned data of DAC channel 1.	0
[3:0]	Reserved	RO	Reserved	0

13.4.11 DAC Dual Channel Right-aligned 8-bit Data Hold Register (DAC_RD8BDHR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC2DHR[7:0]								DACC1DHR[7:0]							

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:8]	DACC2DHR[7:0]	RW	8-bit right-aligned data of DAC channel 2.	0
[7:0]	DACC1DHR[7:0]	RW	8-bit right-aligned data of DAC channel 1.	0

13.4.12 DAC Channel 1 Data Output Register (DAC_DOR1)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DACC1DOR[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved.	0
[11:0]	DACC1DOR[11:0]	RO	Output data of DAC channel 1	0

13.4.13 DAC Channel 2 Data Output Register (DAC_DOR2)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DACC2DOR[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved.	0
[11:0]	DACC2DOR[11:0]	RO	Output data of DAC channel 2	0

Chapter 14 Advanced-control Timer (ADTM)

The advanced-control timer module contains a powerful 16-bit auto-reload timer (TIM1, TIM8), which can be used to measure pulse width or generate pulses, PWM waves, etc. Used in motor control, power supply and other fields.

14.1 Main Features

Advanced-control timer (TIM1/8) features include:

- 16-bit auto-reload counter, supports up count, down count and up/down count
- 16-bit prescaler; the frequency division factor is dynamically adjustable from 1 to 65536
- 4 independent compare/capture channels
- Each compare/capture channel supports multiple working modes, such as: input capture, output compare, PWM generation and single pulse output
- Complementary outputs with programmable dead-time
- External signal to control timer
- Repetition counter to update the timer after the determination of the cycle
- Break signal input to put the timer's output signals in reset status or in a known status
- DMA generation in multiple modes
- Incremental encoder
- Cascade connection and synchronization between timers

14.2 Principle and Structure

This section describes the internal structure of the advanced-control timer.

14.2.1 Overview

As shown in Figure 14-1, the structure of the advanced-control timer can be roughly divided into 3 parts: Input clock part, core counterpart and compare/capture channel part.

The advanced-control timer clock can come from PB bus clock (CK_INT), external clock input pin (TIMx_ETR), other timers with clock output function (ITRx), or the input end of compare capture channel (TIMx_CHx). These input clock signals will become CK_PSC clocks after various set filtering and frequency division operations, and will output to the core counterpart. In addition, these complex clock sources can also be output as TRGO to other timers, ADC and other peripherals.

The core of the advanced-control timer is a 16-bit counter (CNT). After CK_PSC is divided by the prescaler (PSC), it becomes CK_CNT and is output to CNT. CNT supports up counting mode, down counting mode and up and down counting mode, and has an automatic reload value register (ATRLR) to reload the initial value for CNT after the end of each counting cycle. There is also an auxiliary counter that counts the number of times ATRLR reloads the initial value for CNT. When the number reaches the number set in the repetition count value register (RPTCR), a specific event can be generated.

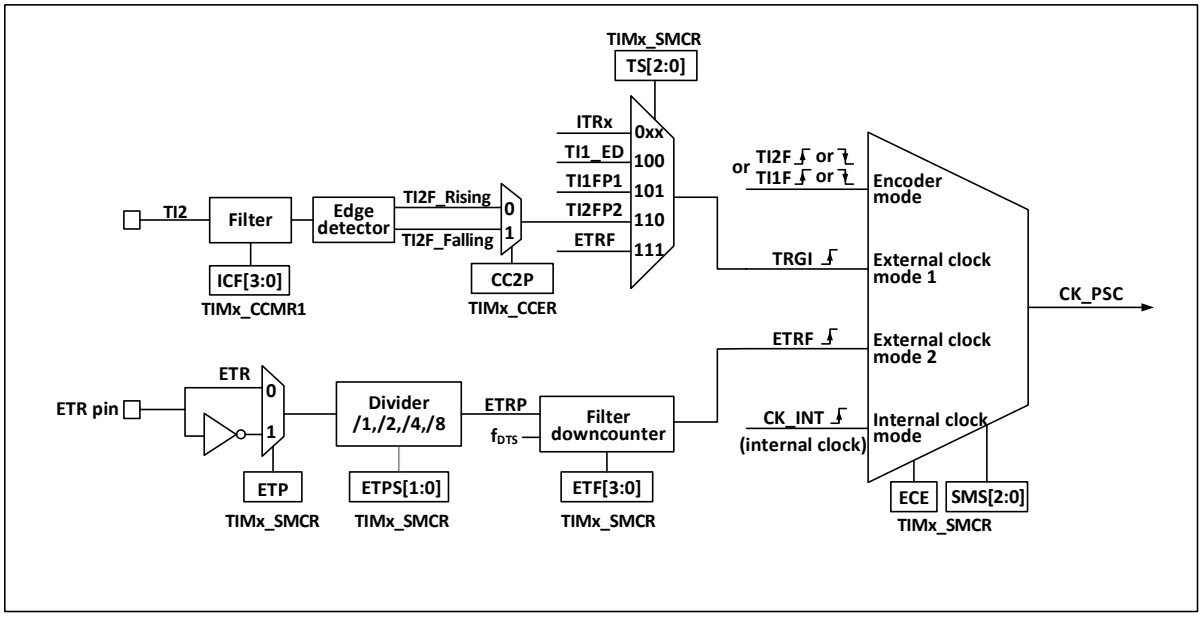
The advanced-control timer has 4 groups of compare/capture channels. On each group of compare/capture channel, pulses can be inputted from its dedicated pins or output waveforms to the pins, i.e., the compare/capture channels support input and output modes. The input of each channel of the compare/capture register supports operations such as filtering, frequency division and edge detection, and supports mutual

triggering between channels, and can also provide a clock for the core counter CNT. Each compare/capture channel has a set of compare/capture register (CHxCVR), which supports comparison with the main counter (CNT) so as to output pulse.

Figure 14-1 Structure block diagram of advanced-control timer

14.2.2 Clock Input

Figure 14-2 CK_PSC source block diagram of advanced-control timer



There are many clock sources for advanced-control timer CK_PSC, which can be divided into 4 categories:

1) The route of external clock pin (ETR) input clock: ETR→ETRP→ETRF;
2) Internal PB clock input route: CK_INT;
3) The route from the compare/capture channel pin (TIMx_CHx): TIMx_CHx→TIx→TIxFPx; this route is also used in encoder mode;
4) Input from other internal timers: ITRx;

The actual operation can be divided into 4 categories by determining the input pulse selection of the SMS from the CK_PSC source:

1) Select the internal clock source (CK_INT);
2) External clock source mode 1;
3) External clock source mode 2;
4) Encoder mode;

The 4 clock sources mentioned above can be selected by these 4 operations.

14.2.2.1 Internal Clock Source (CK_INT)

If the advanced-control timer is started when the SMS field is kept at 000b, then the internal clock source (CK_INT) is selected as the clock. At this moment, CK_INT is CK_PSC.

14.2.2.2 External Clock Source Mode 1

If SMS is set to 111b, the external clock source mode1 is enabled. When external clock source 1 is enabled, TRGI is selected as the source of CK_PSC. It is worth noting that you need to configure TS to select the source of TRGI. For TS, the following pulses can be used as the clock sources:

1) Internal Trigger (ITRx, x is 0,1,2,3);
2) Signal of compare/capture 1 after passing through the edge detector (TI1F_ED);
3) Signals TI1FP1 and TI2FP2 of compare/capture channel;
4) Signal ETRF from external clock pin.

14.2.2.3 External Clock Source Mode 2

Use external trigger mode 2 to count on every rising or falling edge of the external clock pin input. When the ECE bit is set, the external clock source mode 2 is used. When the external clock source mode2 is used, ETRF is selected as CK_PSC. The ETR pin passes through the optional inverter (ETP) and frequency divider (ETPS) to become ETRP, and then passes through the filter (ETF) to become ETRF.

When ECE bit is set and the SMS is set to 111b, it means that the TS selects ETRF as the input.

14.2.2.4 Encoder Mode

Set SMS as 001b, 010b and 011b to enable the encoder mode. After enabling the encoder mode, you may choose to use another transition edge as a signal for signal output at a certain level in TI1FP1 and TI2FP2. This mode is used when the external encoder is used. Refer to Section 14.3.10 for specific functions.

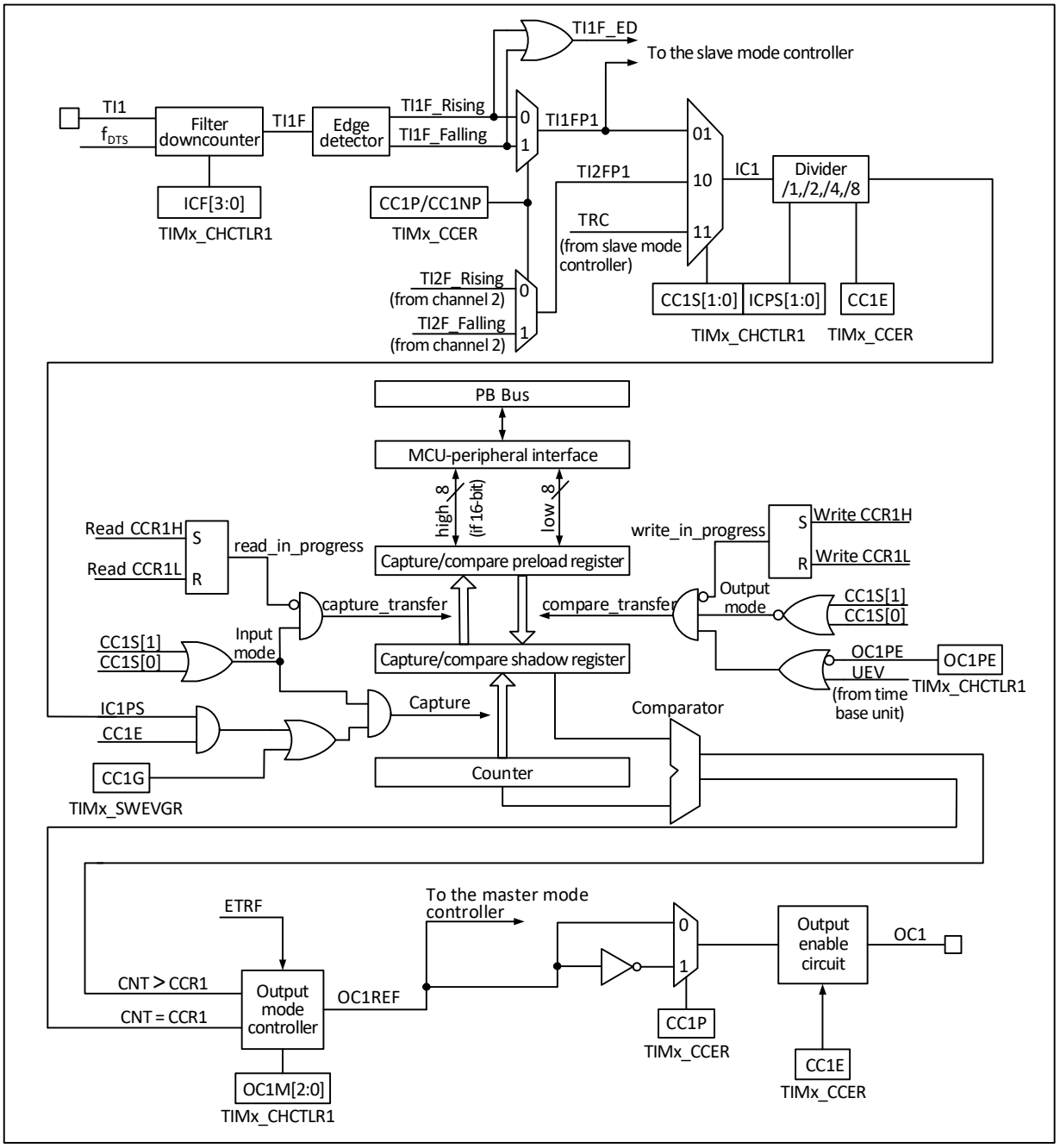
14.2.3 Counter and Periphery

CK_PSC inputs to the prescaler (PSC) for frequency division. PSC is 16 bits, and the actual frequency division factor is equivalent to the value of R16_TIMx_PSC+1. CK_PSC will become CK_CNT after PSC. The changed value of R16_TIMx_PSC will not take effect in real time, but will be updated to the PSC after the update event. Update events include clearing and resetting the UG bit. The core of the timer is a 16-bit counter (CNT). CK_CNT will eventually be inputted to CNT. CNT supports up-counting mode, down-counting mode and up/down counting mode, and there is an automatic reload value register (ATRLR) which re-loads the initial value for CNT after each counting cycle is completed. In addition, there is an auxiliary counter that records the number of times that ATRLR reloads the initial value for CNT. When the number of times reaches the number set in the repeat count register (RPTCR), a specific event can be generated.

14.2.4 Compare/Capture Channel and Periphery

The compare/capture channel is the main component of the timer to achieve complex functions. Its core is the compare/capture register, supplemented by the digital filtering of the peripheral input part, frequency division and channel multiplexing, the output part comparator and output control.

Figure 14-3 Structure block diagram of compare/capture channel



The block diagram of the compare/capture channel is as shown in Figure 14-3. After the signal is inputted from the channel x pin, it can be selected as TIx (the source of TI1 may be more than CH1. See the timer structure block diagram 14-1). Take TI1 as example, TI1 passes through the filter (ICF[3:0]) to generate TI1F, and then is divided into TI1F_Rising and TI1F_Falling after passing through the edge detector. These 2 signals are selected (CC1P) to generate TI1FP1, TI1FP1 and TI2FP1 from channel 2 are sent to CC1S together to be selected as IC1, and then sent to the compare/capture register after going through the ICPS frequency division.

The compare/capture register is composed of a preload register and a shadow register, and only the preload register is operated during reading and writing. In the capture mode, the capture occurs on the shadow register, and then copied to the preload register; in the comparison mode, the content of the preload register is copied to the shadow register, and then the content of the shadow register is compared with the core counter (CNT).

14.3 Function and Implementation

The advanced-control timer complex functions are implemented by the operation of compare/capture channel, clock input circuit, counter and peripheral parts of the timer. The timer's clock input can come from multiple clock sources including the input of the compare/capture channel. The operation of compare/capture channel and the clock source selection directly determines its function. The compare/capture channel is bidirectional and can work in input and output modes.

14.3.1 Counter Mode

Up counting mode

In up-counting mode, the counter counts from 0 to the auto-reload value (contents of the R16_TIMx_ATRLR register), then restarts counting from 0 and generates a counter overflow event.

If a repeat counter is used, an update event (UEV) is generated when the number of incremented repeats reaches the number programmed in the repeat counter register plus one (R16_TIMx_RPTCR+1). Otherwise, an update event will be generated every time the counter overflows.

An update event will also be generated when the UG bit of the TIMx_SWEVGR register is set (either by software or using a slave mode controller).

UEV events can be disabled by software by setting the UDIS bit in the R16_TIMx_CTLR1 register. This avoids updating the shadow register when a new value is written to the preload register. No update event will be generated until the UDIS bit is written to 0. However, both the counter and the prescaler counter will restart counting from 0 (while the prescaler ratio remains unchanged). Additionally, if the URS bit (update request selection) in the R16_TIMx_CTLR1 register is set, setting the UG bit generates an update event UEV but does not set the UIF flag (therefore, no interrupt or DMA request is sent). In this way, if the counter is cleared when a capture event occurs, the update interrupt and capture interrupt will not be generated at the same time.

When an update event occurs, all registers are updated and the update flag (UIF bit in the R16_TIMx_INTFR register) is set (depending on the URS bit):

1) The repetition counter will be reloaded with the contents of the R16_TIMx_RPTCR register
2) The auto-reload shadow register will be updated with the preload value (R16_TIMx_ATRLR)
3) The prescaler buffer will be reloaded with the preload value (contents of the R16_TIMx_PSC register)

Down counting mode

In down-counting mode, the counter starts counting down from the auto-reload value (contents of the R16_TIMx_ATRLR register) to 0, then starts counting again from the auto-reload value and generates a counter underflow event.

If a repeat counter is used, an update event (UEV) is generated when the number of count-down repeats reaches the number programmed in the repeat counter register plus one (R16_TIMx_RPTCR+1). Otherwise, an update event will be generated every time the counter underflows.

An update event will also be generated when the UG bit of the R16_TIMx_EGR register is set (either by software or using a slave mode controller).

UEV update events can be disabled by software by setting the UDIS bit in the R16_TIMx_CTLR1 register. This avoids updating the shadow register when a new value is written to the preload register. No update event will be generated until the UDIS bit is written to 0. However, the counter will restart from the current auto-reload value and the prescaler counter will restart from 0 (but the prescaler ratio remains unchanged).

Additionally, if the URS bit (update request selection) in the R16_TIMx_CTLR1 register is set, setting the UG bit generates an update event UEV but does not set the UIF flag (therefore, no interrupt or DMA request is sent). In this way, if the counter is cleared when a capture event occurs, the update interrupt and capture interrupt will not be generated at the same time.

When an update event occurs, all registers are updated and the update flag (UIF bit in the R16_TIMx_INTFR register) is set (depending on the URS bit):

1) The repetition counter will be reloaded with the contents of the R16_TIMx_RPTCR register
2) The prescaler buffer will be reloaded with the preload value (contents of the R16_TIMx_PSC register)
3) The auto-reload activity register will be updated with the preload value (contents of the R16_TIMx_ATRLR register).

Note: The auto-reload register is updated before the counter is reloaded, so the next count cycle is the desired new cycle length.

Center-aligned mode (up/down counting)

In center-aligned mode, the counter counts from 0 to the auto-reload value (contents of the R16_TIMx_ATRLR register) - 1, generating a counter overflow event; then it counts down from the auto-reload value to 1 and generates a counter underflow event. Then the count starts again from 0.

Center-aligned mode is valid when the CMS bit in the R16_TIMx_CTLR1 register is not '00'. When a channel is configured in output mode, its output compare interrupt flag will be set in the following modes: counter counting down (center-aligned mode 1, CMS='01'), counter counting up (center-aligned mode 2, CMS='10'), and counter up/down counting (center-aligned mode 3, CMS='11').

In this mode, the DIR direction bit of the R16_TIMx_CTLR1 register cannot be written to a value, but is updated by hardware and indicates the current counter direction.

An update event is generated each time a counter overflow and underflow occurs, or can be generated by setting the UG bit in the R16_TIMx_SWEVGR register (either through software or using a slave mode controller). In this case, the counter and the prescaler counter will start counting from 0 again.

UEV update events can be disabled by software by setting the UDIS bit in the R16_TIMx_CTLR1 register. This avoids updating the shadow register when a new value is written to the preload register. No update event will be generated until the UDIS bit is written to 0. However, the counter will still increment and decrement based on the current auto-reload value.

Additionally, if the URS bit (update request selection) in the R16_TIMx_CTLR1 register is set, setting the UG bit generates a UEV update event but does not set the UIF flag (therefore, no interrupt or DMA request is sent). In this way, if the counter is cleared when a capture event occurs, the update interrupt and capture interrupt will not be generated at the same time.

When an update event occurs, all registers are updated and the update flag (UIF bit in the R16_TIMx_INTFR register) is set (depending on the URS bit):

1) The repetition counter will be reloaded with the contents of the R16_TIMx_RPTCR register;
2) The prescaler buffer will be reloaded with the preload value (contents of the R16_TIMx_PSC register);
3) The auto-reload activity register will be updated with the preload value (contents of the R16_TIMx_ATRLR register). Note that if the update operation is triggered by a counter overflow, the auto-reload register is updated before the counter is reloaded, so the next counting period is the desired new period length (the counter is reloaded with the new value).

14.3.2 Input Capture Mode

The input capture mode is one of basic functions of timer. The principle of the input capture mode is that when a certain edge on the ICxPS signal is detected, a capture event will occur, and the current value of the counter will be latched into the compare/capture register (R16_TIMx_CHCTLRx). When a capture event occurs, CCxIF (In R16_TIMx_INTFR) bit will be set. If an interrupt or DMA is enabled, a corresponding interrupt or DMA will be generated. If CCxIF is already set when a capture event occurs, then the CCxOF bit will be set. CCxIF can be cleared by software or by hardware through reading the compare/capture register. CCxOF is cleared by the software.

Take an example of channel 1 to illustrate the steps to use the input capture mode, as follows:
1) Configure CC1S and select the source of IC1 signal. For example, it is set to 10b, and TI1FP1 is selected as the source of IC1, and the default setting cannot be used. CC1S defaults to use the compare capture module as the output channel;
2) Configure IC1F domain, and these bits set the sampling frequency of TI1 input and the length of digital filter. The digital filter consists of an event counter, which will produce an output jump after recording n events;
3) Configure CC1P bit and set the polarity of TI1FP1. For example, maintain CC1P bit to be low and select the jump of rising edge;
4) Configure IC1PS and set IC1 signal as the frequency division factor between IC1PS. For example, maintain the IC1PS as 00b without frequency division;
5) Configure the CC1E bit to allow to capture the core counter (CNT) value to the compare/capture register.
6) The CC1IE and CC1DE bits are configured according to whether an interrupt or DMA is needed.

When TI1 inputs a captured pulse, the value of the core counter (CNT) will be recorded in the compare/capture register, and CC1IF will be set. When CC1IF has been set before, the CCIOF bit will also be set. If CC1IE is set, then an interrupt will be generated; if CC1DE is set, a DMA request will be generated. An input capture event can be generated by software through writing the event generation register (TIMx_SWEVGR).

14.3.3 Compare Output Mode

The compare output mode is one of basic functions of timer. The principle of the compare output mode is to output a specific change or waveform when the value of the core counter (CNT) is consistent with the value of the compare/capture register. OCxM (in R16_TIMx_CHCTLRx) and the CCxP bit (in R16_TIMx_CCER) determine whether the output is determined high or low level or level inversion. When a comparison consistent event is generated, the CCxIF bit will be also set. If the CCxIE bit is preset, an interrupt will be generated; if the CCxDE bit is preset, a DMA request will be generated.

The procedure of compare output mode configuration is as follows:
1) Configure the clock source and auto-reload value of the core counter (CNT);
2) Set the count value to be compared to the compare/capture register (R16_TIMx_CHxCVR);
3) If an interrupt needs to be generated, set the CCxIE bit;
4) Keep OCxPE as 0 and disable the preload register of the compare register;
5) Set the output mode, and set OCxM and CCxP bit;
6) Enable the output and set the CCxE bit;
7) Set the CEN bit to start the timer.

14.3.4 Forced Output Mode

The output mode of the compare/capture channel of the timer can be forced to output a certain level by software, instead of relying on the shadow register and the core counter of the compare/capture register.

The specific means is to set OCxM to 100b, which means to force OCxREF to be low; or to set OCxM to 101b, which means setting OCxREF to a high value by force.

It should be noted that if OCxM is set to 100b or 101b by force, the comparison process between the internal core counter and the compare/capture register will be still in progress, the corresponding flag bit will be still set, and interrupts and DMA request will still be generated.

14.3.5 PWM Input Mode

The PWM input mode is used to measure the duty cycle and frequency of the PWM, which is a special case of the input capture mode. The operation is the same as the input capture mode except for the following differences: PWM occupies 2 compare/capture channels, and the input polarity of the 2 channels is set to opposite. One of the signals is set to trigger input, and SMS is set to reset mode.

For example, to measure the cycle and frequency of the PWM wave input from TI1, the following operations are required:
1) Set TI1 (TI1FP1) as the input of IC1 signal. Set CC1S to 01b;
2) Set TI1FP1 as the rising edge valid. Keep CC1P to 0;
3) Set TI1 (TI1FP2) as the input of IC2 signal. Set CC2S to 10b;
2) Set TI1FP2 as the falling edge valid. Set CC2P to 1;
5) The source of the clock source is TI1FP1. Set TS to 101b;
6) Set SMS to reset mode, i.e., 100b;
7) Enable the input capture. Set CC1E and CC2E bits;
In this way, the value of the compare/capture register 1 is the cycle of PWM, and the value of the compare/capture register 2 is its duty cycle.

Note: Since only TI1FP1 and TI2FP2 are connected to the slave mode controller, only TIM1_CH1/TIM1_CH2 can be used in PWM input mode.

14.3.6 PWM Output Mode

The PWM output mode is one of basic functions of timer. The most common method of PWM output mode is to use the reload value to determine the PWM frequency, and to use the capture comparison register to determine the duty cycle. Set 110b or 111b in OCxM to use PWM mode 1 or mode 2, set the OCxPE bit to enable the preload register, and finally set the ARPE bit. Since the value of the preload register can be sent to the shadow register when an update event occurs, it is necessary to set the UG bit to initialize all registers before the core counter starts counting. In the PWM mode, the core counter and the compare/capture register are always being compared. According to the CMS bit, the timer can output edge-aligned or center-aligned PWM signals.

- Edge alignment

When the edge alignment is used, the core counter counts up or down. In the scenario of PWM mode 1, when the value of the core counter is greater than that of the compare/capture register, OCxREF will be low; when the value of the core counter is less than the compare capture register (such as when the core counter increases to the value of R16_TIMx_ATRLR and returns to all 0), OCxREF drops to high.

- Central alignment

When the center-aligned mode is used, the core counter will run in a mode where up counting and down counting are performed alternately, and OCxREF performs rising and falling jumps when the values of the core counter and the compare/capture register are consistent. However, in 3 types of central alignment mode

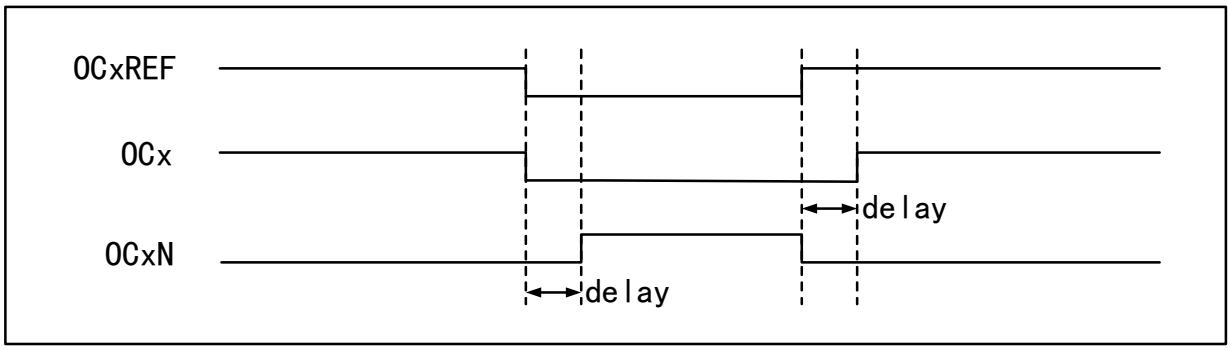
of comparison flag, the bit setting timing is different somewhat. When the center-alignment mode is used, it is the best to generate a software update flag (setting the UG bit) before starting the core counter.

14.3.7 Complementary Output and Dead-time

The compare/capture channel generally has 2 output pins (compare/capture channel 4 has only one output pin), to output 2 complementary signals (OCx and OCxN). OCx and OCxN can be independently set by the CCxP and CCxNP bits. The output enable is set independently through CCxE and CCxNE, and the deadband and other controls are performed through the MOE, OIS, OISN, OSSI and OSSR bits. Meanwhile, OCx and OCxN outputs are enabled to insert into the deadband, each channel has a 10-bit deadband generator. If there is a break circuit, set the MOE bit. OCx and OCxN are generated by OCxREF in association. If OCx and OCxN are both high and effective, then OCx will be the same as OCxREF, but the rising edge of OCx is equivalent to OCxREF with a delay. OCxN is opposite to OCxREF, and its rising edge has a delay relative to the falling edge of the reference signal, and if the delay is greater than the effective output width, the corresponding pulse will not be generated.

Figure 14-4 shows the relationship between OCx, OCxN and OCxREF, and shows the deadband.

Figure 14-4 Complementary output and deadband



14.3.8 Break Signal

When the break signal is generated, the output enable signal and the invalid level will be modified according to the MOE, OIS, OISN, OSSI and OSSR bits. But OCx and OCxN will not be at the effective level at any time. The break event source can come from the break input pin, or it can be a clock failure event, and the clock failure event will be generated by CSS (Clock Security System).

After the system reset, the break function will be disabled by default (MOE bit is low). Setting the BKE bit can enable the break function. The polarity of the input break signal can be set by setting BKP. The BKE and BKP signals can be written at the same time. There will be an APB clock delay before the actual write, so you need to wait for an PB cycle to read the written value correctly.

When the selected level appears on the break pin, the system will generate the following actions:

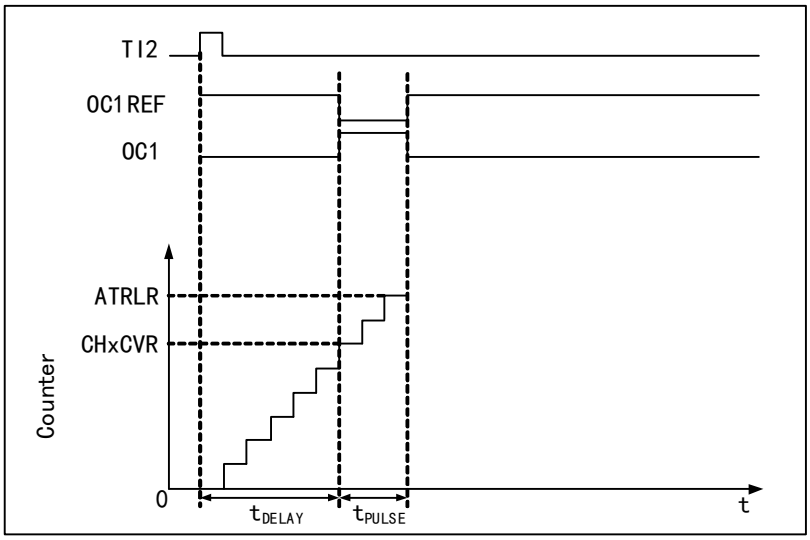
1) The MOE bit is asynchronously cleared, and the output is set to the invalid status, idle status or reset status according to the setting of the OSSI bit;
2) After MOE is cleared, each output channel will output the level determined by OISx;
3) During the supplementary output: the output will be in an invalid status, depending on the polarity;
4) If BIE is set, an interrupt will be generated when BIF is set; if the BDE bit is set, a DMA request will be generated;
5) If AOE is set, the MOE bit will be automatically set during the next update of event UEV.

14.3.9 Single Pulse Mode

The single pulse mode can be used to allow the microcontroller to respond to a specific event to generate a pulse after a delay. The delay and pulse width are programmable. Setting the OPM bit can make the core counter stop when the next update event UEV is generated (the counter turns over to 0).

As shown in Figure 14-5, it is necessary to detect the beginning of a rising edge on the TI2 input pin. After delaying Tdelay, a positive pulse of length Tpulse will be generated on OC1:

Figure 14-5 Single pulse generation



1) Set TI2 as trigger. Set CC2S to 01b and map TI2FP2 to TI2; set CC2P bit to 0b and set TI2FP2 to rising edge detection; set TS to 110b and set TI2FP2 as the trigger source; set SMS to 110b, and TI2FP2 is used to start the counter;
2) Tdelay is determined by the value of the compare/capture register, and Tpulse is determined by the value of the auto-reload value register and the value of the compare/capture register.

14.3.10 Encoder Mode

The encoder mode is a typical application of the timer. It can be used to access the dual-phase output of the encoder. The counting direction of the core counter is synchronized with the rotating shaft of the encoder. Each pulse outputted by the encoder will increase the core counter by adding one or subtracting one. The steps to use the encoder are: set the SMS field to 001b (counting only on TI2 edge), 010b (counting only on TI1 edge) or 011b (counting on both TI1 and TI2 edges), and connect the encoder to compare/capture channel 1, 2 input terminals, set a value for the reload value register and this value can be set to be greater. In the encoder mode, the internal compare/capture register of timer, prescaler, repeat count register, etc. all work normally. The following table shows the relationship between the counting direction and the encoder signal.

Table 14-1 Relationship between counting direction of timer encoder mode and encoder signal

Counting active edge	Relative signal level	TI1FP1 signal edge		TI2FP2 signal	
		Rising edge	Falling edge	Rising edge	Falling edge
Only count at TI1 edge	High	Downcount	Upcount	Not count	
	Low	Upcount	Downcount		
Only count at TI2 edge	High	Not count		Upcount	Downcount
	Low			Downcount	Upcount

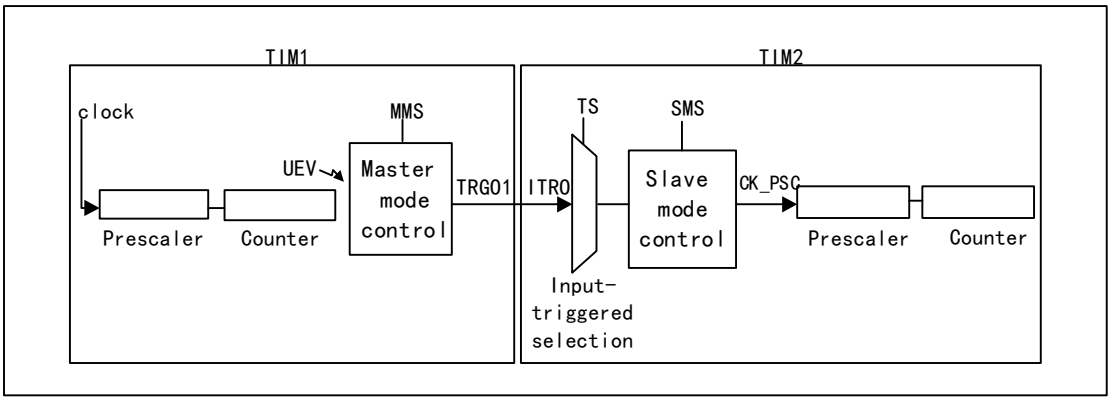
Count on both edges of TI1 and TI2	High	Downcount	Upcount	Upcount	Downcount
	Low	Upcount	Downcount	Downcount	Upcount

14.3.11 Timer Synchronization Mode

TIMx timers are internally wired together for timer synchronization or cascading. When a timer is configured in master mode, it can reset, start, stop, or provide a clock to the counter of another timer configured in slave mode.

Using one timer as a prescaler for another timer

Figure 14-10 Master/Slave timer example



For example, Timer 1 can be configured as a prescaler for Timer 2. To this end:

1) Configure timer 1 as the main mode and output a periodic trigger signal every time an update event UEV occurs. If MMS=010 is written to R16_TIM1_CTLR2, TRGO1 will output a rising edge every time an update event is generated.
2) To connect the TRGO1 output of Timer 1 to Timer 2, Timer 2 must be configured in slave mode, using ITR0 as the internal trigger. This is selected via the TS bit in the R16_TIM2_SMCFGR register (write TS=000).
3) Then set the slave mode controller to external clock mode 1 (write SMS=111 in R16_TIM2_SMCFGR register). In this way, the clock of Timer 2 will be provided by the rising edge of the periodic trigger signal of Timer 1 (corresponding to the counter overflow of Timer 1).
4) Finally, both timers must be enabled simultaneously by setting their corresponding CEN bits (R16_TIMx_CTLR1 register) to 1.

Note: If the OCx signal of timer 1 is selected as the trigger output (MMS=1xx), the rising edge of this signal will be used to drive the counter of timer 2

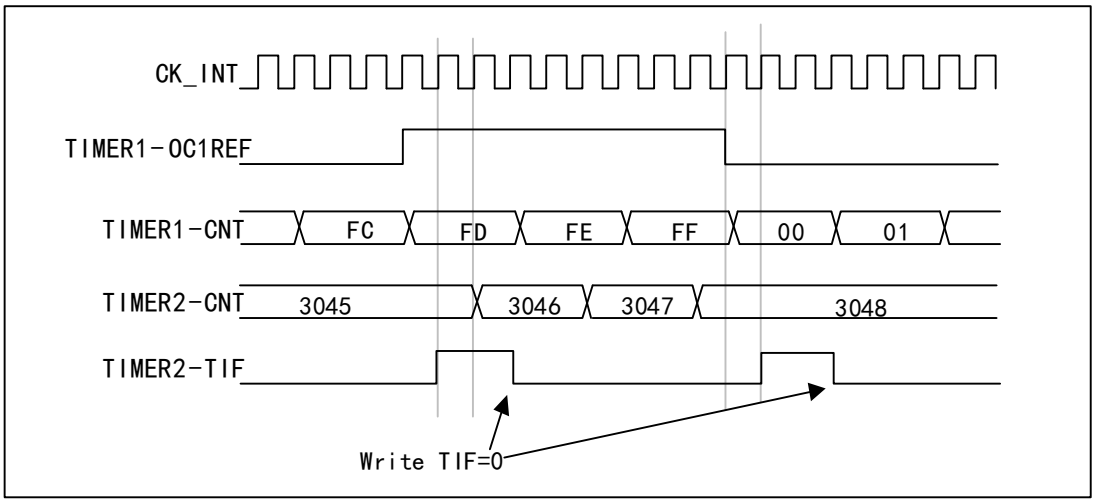
Use one timer to enable another timer

In this example, Timer 2 is enabled by comparing 1 to the output of Timer 1. Only when OC1REF of Timer 1 is high, Timer 2 counts based on the divided internal clock. The clock frequency of both counters is divided by 3 through the prescaler based on CK_INT (fCK_CNT=fCK_INT/3).

1) Configure timer 1 in master mode and send its output compare 1 reference signal (OC1REF) as the trigger output (MMS=100 in the R16_TIM1_CR2 register).
2) Configure the OC1REF waveform for Timer 1 (R16_TIM1_CHCTLR1 register).
3) Configure Timer 2 to receive an input trigger from Timer 1 (TS=000 in the R16_TIM2_SMCFGR register).
4) Configure Timer 2 in gated mode (SMS=101 in R16_TIM2_SMCFGR register).
5) Enable Timer 2 by writing '1' to the CEN bit (R16_TIM2_CTLR1 register).
6) Start Timer 1 by writing '1' to the CEN bit (R16_TIM1_CTLR1 register).

Note: The clock of counter 2 is not synchronized with counter 1. This mode only affects the counter enable signal of timer 2.

Figure 14-11 Use OC1REF of timer 1 to implement gating control of timer 2

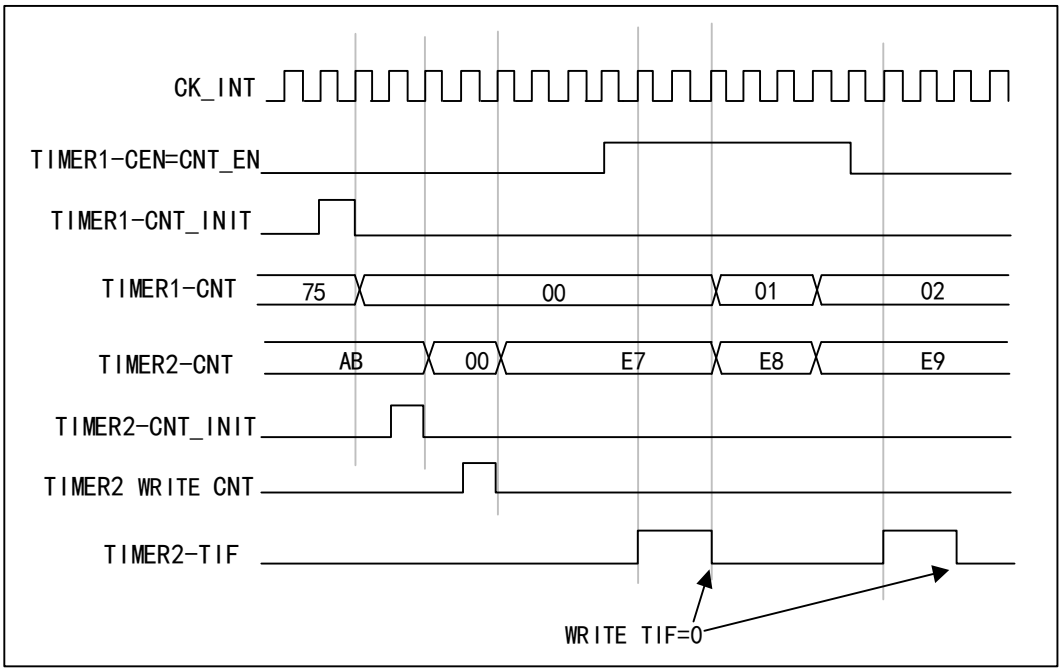


The Timer 2 counter and prescaler are not initialized before starting. So counting starts from the respective current value. Both timers can be reset from a specified value before starting Timer 1. This allows you to write any desired value into the timer counter. Both timers can be easily reset by software using the UG bit in the R16_TIMx_SWEVGR register.

In the next example, Timer1 is synchronized with Timer2. Timer 1 is in master mode and starts counting from 0. Timer 2 is in slave mode and starts counting from 0xE7. The prescaler ratios of both timers are the same. Timer 2 will be stopped when Timer 1 is disabled by writing '0' to the CEN bit in the R16_TIM1_CTLR1 register:

1) Configure timer 1 in master mode and send its output compare 1 reference signal (OC1REF) as a trigger output (MMS=100 in the R16_TIM1_CTLR2 register).
2) Configure the OC1REF waveform for Timer 1 (R16_TIM1_CHCTLR1 register).
3) Configure Timer 2 to receive an input trigger from Timer 1 (TS=000 in the R16_TIM2_SMCFGR register).
4) Configure Timer 2 in gated mode (SMS=101 in R16_TIM2_SMCFGR register).
5) Reset Timer 1 by writing '1' to the UG bit (R16_TIM1_SWEVGR register).
6) Reset Timer 2 by writing '1' to the UG bit (R16_TIM2_SWEVGR register).
7) Initialize Timer 2 to 0xE7 by writing '0xE7' in the timer 2 counter (R16_TIM2_CNTL).
8) Enable Timer 2 by writing '1' to the CEN bit (R16_TIM2_CTLR1 register).
9) Start Timer 1 by writing '1' to the CEN bit (R16_TIM1_CTLR1 register).
10) Stop Timer 1 by writing '0' to the CEN bit (R16_TIM1_CTLR1 register).

Figure 14-12 Use the enable signal of timer 1 to implement gating control on timer 2

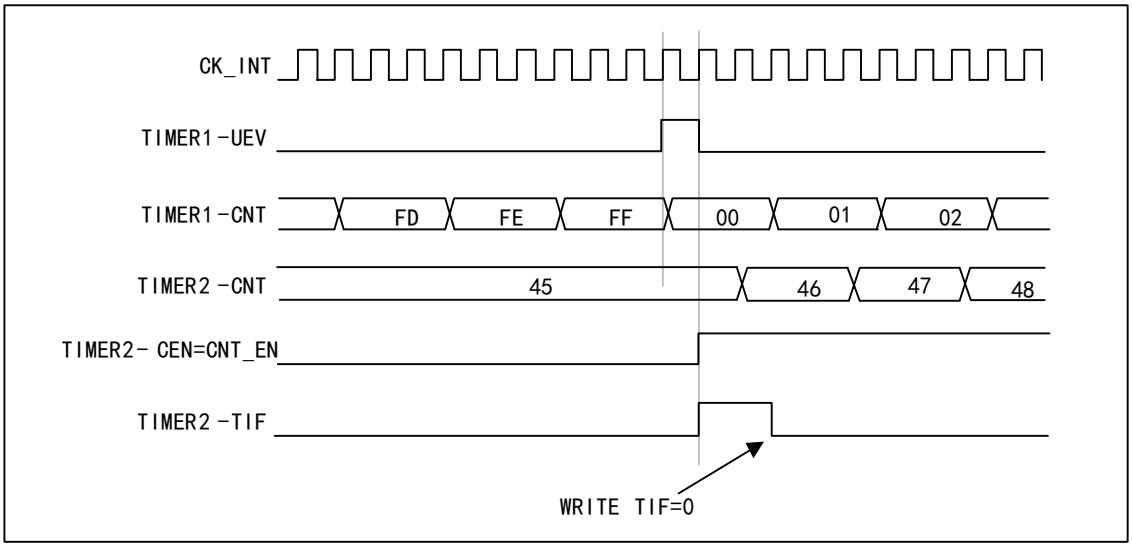


Use one timer to start another timer

In this example, the update event of timer 1 is used to enable timer 2. Whenever Timer 1 generates an update event, Timer 2 starts counting from the current value (which may not be 0) based on the divided internal clock. When timer 2 receives the trigger signal, its CEN bit is automatically set to 1, and the counter starts counting until "0" is written to the CEN bit of the R16_TIM2_CTLR1 register to stop counting. The clock frequency of both counters is divided by 3 through the prescaler based on CK_INT (fCK_CNT=fCK_INT/3).

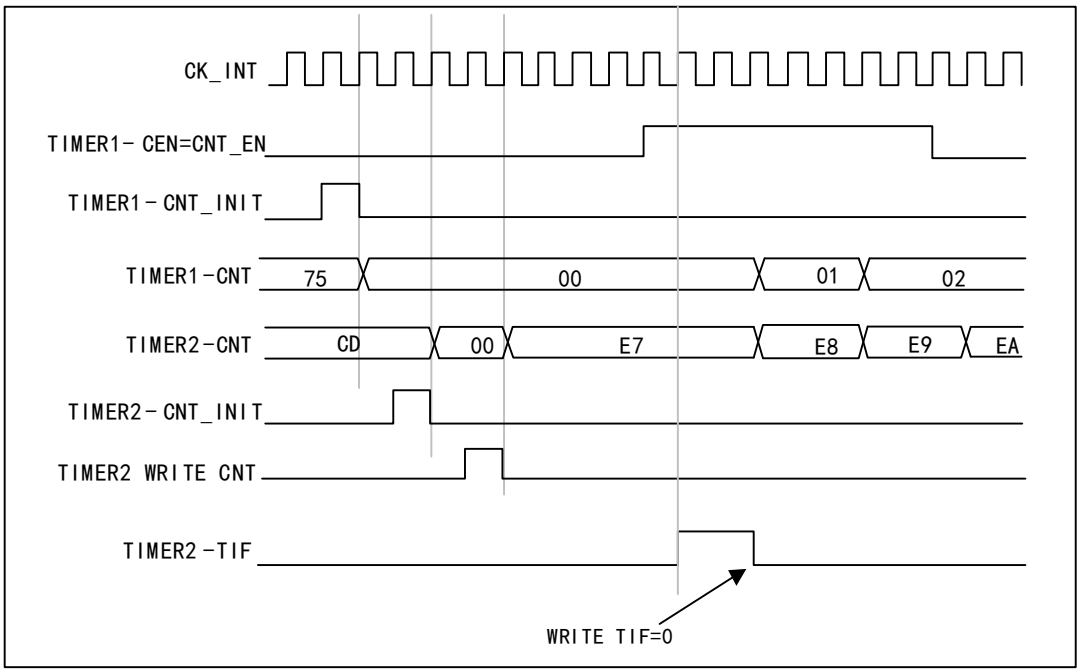
1) Configure timer 1 in master mode and send its update event (UEV) as a trigger output (MMS=010 in the R16_TIM1_CTLR2 register).
2) Configure the period of timer 1 (R16_TIM1_ATRLR register).
3) Configure Timer 2 to receive an input trigger from Timer 1 (TS=000 in the R16_TIM2_SMCFGR register).
4) Configure timer 2 in trigger mode (SMS=110 in R16_TIM2_SMCFGR register).
5) Start Timer 1 by writing '1' to the CEN bit (R16_TIM1_CTLR1 register).

Figure 14-13 Using the update event of timer 1 to trigger timer 2



As shown in the above example, the user can initialize both counters before starting counting. Figure 14-14 shows the counting behavior with the same configuration as Figure 14-13, but in trigger mode (SMS=110 in the R16_TIM2_SMCFGR register) instead of gated mode.

Figure 14-14 Use the enable signal of timer 1 to trigger timer 2



Using one timer as a prescaler for another timer

For example, Timer 1 can be configured as a prescaler for Timer 2. To this end:

1) Configure timer 1 in master mode and send its update event (UEV) as a trigger output (MMS=010 in the R16_TIM1_CTLR2 register). This results in a periodic signal being output each time the counter overflows.
2) Configure the period of timer 1 (R16_TIM1_ATRLR register).
3) Configure Timer 2 to receive an input trigger from Timer 1 (TS=000 in the R16_TIM2_SMCFGR register).
4) Configure Timer 2 in external clock mode (SMS=111 in R16_TIM2_SMCFGR register).
5) Start Timer 2 by writing '1' to the CEN bit (R16_TIM2_CTLR1 register).

6) Start Timer 1 by writing '1' to the CEN bit (R16_TIM1_CTLR1 register).

Use an external trigger to start 2 timers synchronously

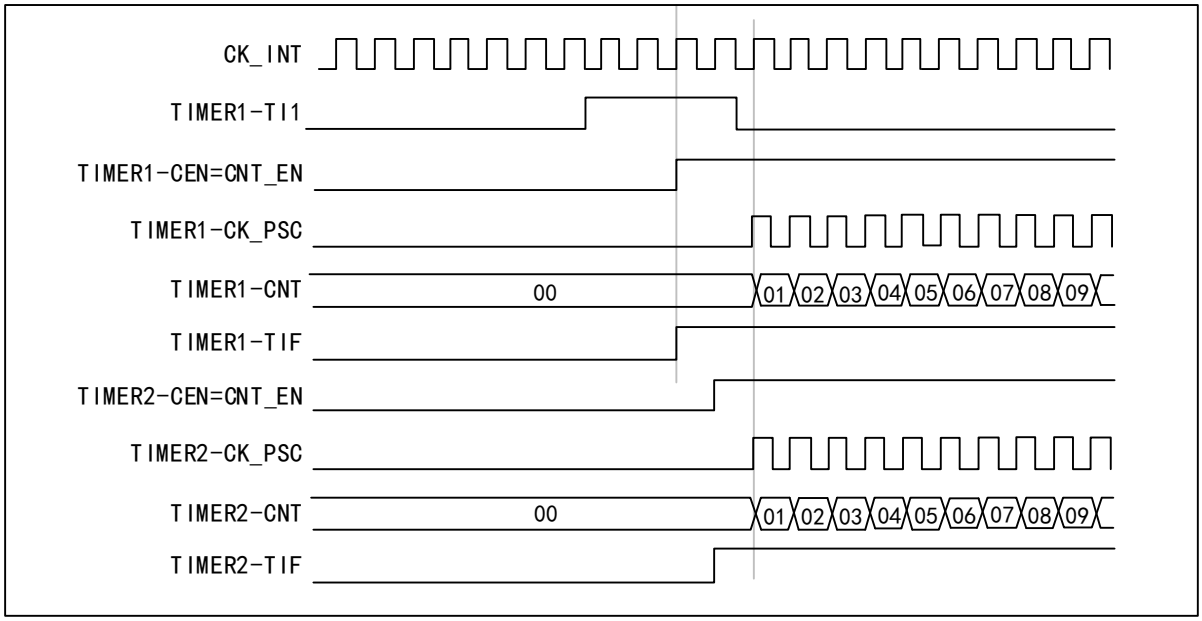
In this example, timer 1 is enabled when a rising edge occurs at the TI1 input of timer 1. When timer 1 is enabled, timer 2 is enabled at the same time. To ensure that the two counters are aligned, Timer 1 must be configured in master/slave mode (corresponding to TI1 as slave and corresponding to Timer 2 as master):

1) Configure timer 1 in master mode and send its enable signal as a trigger output (MMS=001 in the R16_TIM1_CTLR2 register).
2) Configure Timer 1 in slave mode to receive input triggers from TI1 (TS=100 in R16_TIM1_SMCFGR register).
3) Configure timer 1 in trigger mode (SMS=110 in R16_TIM1_SMCFGR register).
4) Configure Timer 1 in master/slave mode by writing MSM=1 (R16_TIMx_SMCFGR register).
5) Configure Timer 2 to receive an input trigger from Timer 1 (TS=000 in the R16_TIM2_SMCFGR register).
6) Configure timer 2 in trigger mode (SMS=110 in R16_TIM2_SMCFGR register).

When a rising edge occurs on TI1 (Timer 1), both counters start counting synchronously according to the internal clock, and both TIF flags are set to 1.

Note: In this example, both timers are initialized before starting (by setting their respective UG bits to 1). Both counters start counting at 0, but an offset between the two can be easily inserted by writing to either counter register (R16_TIMx_CNT). It can be noted that the master/slave mode creates a delay between CNT_EN and CK_PSC of timer 1.

Figure 14-15 Trigger Timer 1 and Timer 2 using the TI1 input of Timer 1



The timer can output clock pulses (TRGO) and also receive inputs from other timers (ITRx). The source of ITRx for different timers (the TRGO from other timers) varies. The internal trigger connections within the timer are shown in Table 14-2.

Table 14-2 Relationship between counting direction of timer encoder mode and encoder signal

Slave timer	ITR0(TS=000)	ITR1(TS=001)	ITR2(TS=010)	ITR3(TS=011)
TIM1	-	TIM2	TIM3	TIM4

14.3.12 Debug Mode

When the system enters debug mode, the timer continues to run or stops according to the settings of the DBG module.

14.3.13 Dual Edge Capture Mode

The pulses can be measured via the TIMx_AUX register.

For example: To capture pulse width using Channel 2, configure the capture function by selecting the IC2 signal source (CC2S bit in TIMx_CHCTLR1 register set to 11b) and enabling Channel 2's dual-edge capture (CAP_ED_CH2 bit in TIMx_AUX register set to 1). This completes the dual-edge capture configuration.

The captured dual-edge pulse width value can be read through the CCR2 bit of the TIMx_CH2CVR register.

14.4 Register Description

Table 14-3 TIM1-related registers list

Name	Access address	Description	Reset value
R16_TIM1_CTLR1	0x40012C00	Control register 1	0x0000
R16_TIM1_CTLR2	0x40012C04	Control register 2	0x0000
R16_TIM1_SMCFGR	0x40012C08	Slave mode control register	0x0000
R16_TIM1_DMAINTENR	0x40012C0C	DMA/interrupt enable register	0x0000
R16_TIM1_INTFR	0x40012C10	Interrupt status register	0x0000
R16_TIM1_SWEVGR	0x40012C14	Event generation register	0x0000
R16_TIM1_CHCTLR1	0x40012C18	Compare/capture control register 1	0x0000
R16_TIM1_CHCTLR2	0x40012C1C	Compare/capture control register 2	0x0000
R16_TIM1_CCER	0x40012C20	Compare/capture enable register	0x0000
R16_TIM1_CNT	0x40012C24	Counters	0x0000
R16_TIM1_PSC	0x40012C28	Counting clock prescaler	0x0000
R16_TIM1_ATRLR	0x40012C2C	Auto-reload value register	0xFFFF
R16_TIM1_RPTCR	0x40012C30	Repeat Count Register	0x0000
R32_TIM1_CH1CVR	0x40012C34	Compare/capture register 1	0x0000
R32_TIM1_CH2CVR	0x40012C38	Compare/capture register 2	0x0000
R32_TIM1_CH3CVR	0x40012C3C	Compare/capture register 3	0x0000
R32_TIM1_CH4CVR	0x40012C40	Compare/capture register 4	0x0000
R16_TIM1_BDTR	0x40012C44	Brake and deadband registers	0x0000
R16_TIM1_DMACFGR	0x40012C48	DMA control register	0x0000
R16_TIM1_DMAADR	0x40012C4C	DMA address register for continuous mode	0x0000
R16_TIM1_AUX	0x40012C50	Dual edge capture register	0x0000

Table 14-4 TIM8-related registers list

Name	Access address	Description	Reset value
R16_TIM8_CTLR1	0x40013400	Control register 1	0x0000
R16_TIM8_CTLR2	0x40013404	Control register 2	0x0000
R16_TIM8_SMCFGR	0x40013408	Slave mode control register	0x0000

R16_TIM8_DMAINTENR	0x4001340C	DMA/interrupt enable register	0x0000
R16_TIM8_INTFR	0x40013410	Interrupt status register	0x0000
R16_TIM8_SWEVGR	0x40013414	Event generation register	0x0000
R16_TIM8_CHCTLR1	0x40013418	Compare/capture control register 1	0x0000
R16_TIM8_CHCTLR2	0x4001341C	Compare/capture control register 2	0x0000
R16_TIM8_CCER	0x40013420	Compare/capture enable register	0x0000
R16_TIM8_CNT	0x40013424	Counters	0x0000
R16_TIM8_PSC	0x40013428	Counting clock prescaler	0x0000
R16_TIM8_ATRLR	0x4001342C	Auto-reload value register	0xFFFF
R16_TIM8_RPTCR	0x40013430	Repeat Count Register	0x0000
R32_TIM8_CH1CVR	0x40013434	Compare/capture register 1	0x0000
R32_TIM8_CH2CVR	0x40013438	Compare/capture register 2	0x0000
R32_TIM8_CH3CVR	0x4001343C	Compare/capture register 3	0x0000
R32_TIM8_CH4CVR	0x40013440	Compare/capture register 4	0x0000
R16_TIM8_BDTR	0x40013444	Brake and deadband registers	0x0000
R16_TIM8_DMACFGR	0x40013448	DMA control register	0x0000
R16_TIM8_DMAADR	0x4001344C	DMA address register for continuous mode	0x0000
R16_TIM8_AUX	0x40013450	Dual edge capture register	0x0000

14.4.1 Control Register 1 (TIMx_CTLR1) (x=1/8)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CKD[1:0]		ARPE	CMS[1:0]		DIR	OPM	URS	UDIS	CEN

Bit	Name	Access	Description	Reset value
[15:10]	Reserved	RO	Reserved	0
[9:8]	CKD[1:0]	RW	These 2 bits define the division ratio between the timer clock (CK_INT) frequency, the dead time and the sampling clock used by the dead time generator and the digital filter (ETR,TIx). 00: Tdts=Tck_int 01: Tdts = 2 × Tck_int 10: Tdts = 4 × Tck_int 11: Reserved.	00b
7	ARPE	RW	Auto-reload preload enable bit. 0: Disable Auto-reload Value Register (ATRLR). 1: Enable Auto-reload Value Register (ATRLR).	0
[6:5]	CMS[1:0]	RW	Central alignment mode selection. 00: Edge-aligned mode. The counter counts up or down based on the direction bit (DIR). 01: Central alignment mode 1. The counter counts up and down alternately. The output compare interrupt flag bit of the channel configured as output	00b

			(CCxS=00 in the CHCTLRx register) is set only when the counter counts down. 10: Central alignment mode 2. The counter counts up and down alternately. The output compare interrupt flag bit of the channel configured as output (CCxS=00 in the CHCTLRx register) is set only when the counter counts up. 11: Central alignment mode 3. The counter counts up and down alternately. The output compare interrupt flag bit of the channel configured as output (CCxS=00 in the CHCTLRx register) is set when the counter counts both up and down. *Note: When the counter is enabled (CEN=1), the transition from edge-aligned mode to center-aligned mode is not allowed.*	
4	DIR	RW	Counter direction: 0: The counting mode of the counter is increment. 1: The counting mode of the counter is subtractive. *Note: This bit is not valid when the counter is configured in central alignment mode or encoder mode.*	0
3	OPM	RW	Single pulse mode. 0: The counter does not stop when the next update event occurs. 1: The counter stops (clear the CEN bit) when the next update event occurs.	0
2	URS	RW	Update request source, by which the software selects the source of the UEV event. 0: If an update interrupt or DMA request is enabled, an update interrupt or DMA request is generated by any of the following events. -Counter overflow/underflow -Setting the UG position -Updates generated by the slave mode controller 1: If an update interrupt or DMA request is enabled, only an update interrupt or DMA request is generated if the counter overflows/underflows.	0
1	UDIS	RW	Updates are prohibited, and the software allows / disables the generation of UEV events through this bit. 0: UEV is allowed. The UEV event is generated by any of the following events: -Counter overflow / underflow. -Set the UG bit. Updates generated from the mode controller with cached registers are loaded into their preload values.	0

			1: UEV is prohibited. No update events are generated, and the registers (ARR, PSC, CCRx) hold their values. If the UG bit is set or a hardware reset is issued from the mode controller, the counter and prescaler are reinitialized.	
0	CEN	RW	Enables the counter. 1: Enable the counter; 0: Disable the counter. *Note: The external clock, gated mode and encoder mode will not work until the CEN bit is set in software. Trigger mode can automatically set the CEN bit in hardware.*	0

14.4.2 Control Register 2 (TIMx_CTLR2) (x=1/8)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6 5 4	3	2	1	0
Reserved	OIS4	OIS3N	OIS3	OIS2N	OIS2	OIS1N	OIS1	TI1S	MMS[2:0]	CCDS	CCUS	Reserved	CCPC

Bit	Name	Access	Description	Reset value
15	Reserved	RO	Reserved	0
14	OIS4	RW	Output idle state 4. 1: When MOE=0, if OC4N is implemented, OC4=1 after deadband; 0: When MOE=0, if OC4N is implemented, OC4=0 after deadband. *Note: This bit cannot be modified after LOCK (TIMx_BDTR register) level 1, 2 or 3 has been set.*	0
13	OIS3N	RW	Output idle state 3. 1: OC1N = 1 after the dead zone when MOE = 0. 0: When MOE=0, OC1N=0 after dead zone. *Note: This bit cannot be modified after the LOCK (TIMx_BDTR register) level 1, 2 or 3 has been set.*	0
12	OIS3	RW	Output idle state 3, see OIS4.	0
11	OIS2N	RW	Output idle state 2, see OIS3N.	0
10	OIS2	RW	Output idle state 2, see OIS4.	0
9	OIS1N	RW	Output idle state 1, see OIS3N.	0
8	OIS1	RW	Output idle state 1, see OIS4.	0
7	TI1S	RW	TI1 selection. 1: TIMx_CH1, TIMx_CH2 and TIMx_CH3 pins connected to TI1 input after heterodyning. 0: TIMx_CH1 pin is connected directly to TI1 input.	0
[6:4]	MMS[2:0]	RW	Master mode selection: These 3 bits are used to select the synchronization information (TRGO) sent to the slave timer in master mode. The possible combinations are as follows.	000b

			000: The UG bit of the Reset-TIMx_EGR register is used as the trigger output (TRGO). In the case of a reset generated by a trigger input (from a mode controller in reset mode), there is a delay in the signal on TRGO relative to the actual reset. 001: Enable - The counter enable signal CNT_EN is used as a trigger output (TRGO). Sometimes it is necessary to start multiple timers at the same time or to control the enable from timers over a period of time. The counter enable signal is generated by the logical or of the trigger input signal in CEN control bit and gated mode. When the counter enable signal is controlled by a trigger input, there is a delay on TRGO unless master/slave mode is selected (see the description of the MSM bit in the TIMx_SMCFGR register). 010: Update - The update event is selected as a trigger output (TRGO). For example, the clock of a master timer may be used as a prescaler for a slave timer. 011: Comparison pulse - on the occurrence of a capture or a successful comparison, when the CC1IF flag is to be set (even if it is already high), the trigger output sends a positive pulse (TRGO). 100: The comparison-OC1REF signal is used as a trigger output (TRGO). 101: The comparison-OC2REF signal is used as a trigger output (TRGO). 110: The comparison-OC3REF signal is used as a trigger output (TRGO). 111: The comparison -OC4REF signal is used as the trigger output (TRGO).	
3	CCDS	RW	Capture the DMA selection for comparison. 1: Send a DMA request for CHxCVR when an update event occurs. 0: Generate a DMA request for CHxCVR when CHxCVR occurs.	0
2	CCUS	RW	Compare capture control update selection bits. 1: If CCPC is set, they can be updated by setting the COM bit or a rising edge on TRGI. 0: If the CCPC is set, they can only be updated by setting the COM bit. *Note: This bit only works for channels with complementary outputs.*	0
1	Reserved	RO	Reserved	0
0	CCPC	RW	Compare capture preload control bits. 1: CCxE, CCxNE and OCxM bits are preloaded and when this bit is set they are only updated when the	0

			COM bit is set. 0: CCxE, CCxNE and OCxM bits are not preloaded. *Note: This bit only works for channels with complementary outputs.*	

14.4.3 Slave Mode Control Register (TIMx_SMCFGR) (x=1/8)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETP	ECE	ETPS[1:0]		ETF[3:0]				MSM	TS[2:0]			Reserved	SMS[2:0]		

Bit	Name	Access	Description	Reset value
15	ETP	RO	ETR trigger polarity selection, this bit selects whether to input ETR directly or to input the inverse of ETR. 1: Invert ETR, low or falling edge active; 0: ETR, active high or rising edge.	0
14	ECE	RW	External clock mode 2 enable selection. 1: Enables external clock mode 2. 0: Disable external clock mode 2. *Note 1: Slave mode can be used simultaneously with external clock mode 2: reset mode, gated mode and trigger mode; however, TRGI cannot be connected to ETRF in this case (TS bit cannot be '111').* *Note 2: When both external clock mode 1 and external clock mode 2 are enabled, the external clock input is ETRF.*	0
[13:12]	ETPS[1:0]	RW	The external trigger signal (ETRP) divides the frequency of this signal, which cannot exceed a maximum of 1/4 of the TIMxCLK frequency, and can be downconverted through this domain. 00: Prescaler off. 01: ETRP frequency divided by 2. 10: ETRP frequency divided by 4. 11: ETRP frequency divided by 8.	00b
[11:8]	ETF[3:0]	RW	Externally triggered filtering, in fact, the digital filter is an event counter, which uses a certain sampling frequency to record up to N events and then produces a jump in the output. 0001: Sampling frequency Fsampling=Fck_int, N=2. 0010: Sampling frequency Fsampling=Fck_int, N=4. 0011: Sampling frequency Fsampling=Fck_int, N=8. 0100: Sampling frequency Fsampling = Fdts/2, N = 6. 0101: Sampling frequency Fsampling = Fdts/2, N = 8. 0110: Sampling frequency Fsampling = Fdts/4, N = 6. 0111: Sampling frequency Fsampling = Fdts/4, N = 8. 1000: Sampling frequency Fsampling = Fdts/8, N = 6.	0000b

			1001: Sampling frequency Fsampling = Fdts/8, N = 8. 1010: Sampling frequency Fsampling = Fdts/16, N = 5. 1011: Sampling frequency Fsampling = Fdts/16, N = 6. 1100: Sampling frequency Fsampling = Fdts/16, N = 8. 1101: Sampling frequency Fsampling = Fdts/32, N = 5. 1110: Sampling frequency Fsampling = Fdts/32, N = 6. 1111: Sampling frequency Fsampling=Fdts/32, N=8.	
7	MSM	RW	Master/slave mode selection. 1: The event on the trigger input (TRGI) is delayed to allow perfect synchronization between the current timer (via TRGO) and its slave timer. This is useful when the synchronization of several timers to a single external event is required. 0: Does not function.	0
[6:4]	TS[2:0]	RW	Trigger selection field, these 3 bits select the trigger input source used to synchronize the counter. 000: Internal trigger 0 (ITR0). 001: Internal trigger 1 (ITR1). 010: Internal trigger 2 (ITR2). 011: Internal trigger 3 (ITR3). 100: Edge detector of TI1 (TI1F_ED). 101: Filtered timer input 1 (TI1FP1). 110: Filtered timer input 2 (TI2FP2). 111: External trigger input (ETRF). The above only changes when SMS is 0. *Note: See Table 14-2 for details.*	000b
3	Reserved	RO	Reserved.	0
[2:0]	SMS[2:0]	RW	Input mode selection field. Selects the clock and trigger mode of the core counter. 000: driven by the internal clock CK_INT. 001: Encoder mode 1, where the core counter increments or decrements the count at the edge of TI2FP2 depending on the level of TI1FP1. 010: Encoder mode 2, where the core counter increments or decrements the count at the edge of TI1FP1, depending on the level of TI2FP2. 011: Encoder mode 3, where the core counter increments and decrements the count on the edges of TI1FP1 and TI2FP2 depending on the input level of another signal; 100: reset mode, where the rising edge of the trigger input (TRGI) will initialize the counter and generate a signal to update the registers. 101: Gated mode, when the trigger input (TRGI) is high, the counter clock is turned on; at the trigger input becomes low, the counter is stopped, and the counter	000b

			starts and stops are controlled. 110: Trigger mode, where the counter is started on the rising edge of the trigger input TRGI and only the start of the counter is controlled. 111: External clock mode 1, rising edge of the selected trigger input (TRGI) drives the counter.	

14.4.4 DMA/Interrupt Enable Register (TIMx_DMAINTENR) (x=1/8)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	TDE	COMDE	CC4DE	CC3DE	CC2DE	CC1DE	UDE	BIE	TIE	COMIE	CC4IE	CC3IE	CC2IE	CC1IE	UIE

Bit	Name	Access	Description	Reset value
15	Reserved	RO	Reserved	0
14	TDE	RW	Trigger the DMA request enable bit. 1: Allow DMA requests to be triggered. 0: Triggering of DMA requests is disabled.	0
13	COMDE	RW	DMA request enable bit of COM. 1: Allow DMA requests for COM. 0: DMA request for COM is disabled.	0
12	CC4DE	RW	Compare the DMA request enable bit of capture channel 4. 1: Allow comparison of DMA requests for capture channel 4. 0: Disable comparison of DMA requests for capture channel 4.	0
11	CC3DE	RW	Compare the DMA request enable bit of capture channel 3. 1: Allow comparison of DMA requests for capture channel 3. 0: Disable comparison of DMA requests for capture channel 3.	0
10	CC2DE	RW	Compare the DMA request enable bit of capture channel 2. 1: Allow comparison of DMA requests for capture channel 2. 0: Disable comparison of DMA requests for capture channel 2.	0
9	CC1DE	RW	Compare the DMA request enable bit of capture channel 1. 1: Allow comparison of DMA requests for capture channel 1. 0: Disable comparison of DMA requests for capture channel 1.	0

8	UDE	RW	Updated DMA request enable bit. 1: DMA requests that allow updates. 0: DMA requests for updates are disabled.	0
7	BIE	RW	Brake interrupt enable bit. 1: Allow brakes to be interrupted. 0: Brake interruption is prohibited.	0
6	TIE	RW	Trigger the interrupt enable bit. 1: Enable triggering of interrupts. 0: Trigger interrupt is disabled.	0
5	COMIE	RW	COM interrupt allow bit. 1: Allow COM interrupts. 0: COM interrupt is disabled.	0
4	CC4IE	RW	Compare capture channel 4 interrupt enable bit. 1: Allow comparison of capture channel 4 interrupts. 0: Disable compare capture channel 4 interrupt.	0
3	CC3IE	RW	Compare capture channel 3 interrupt enable bit. 1: Allow comparison of capture channel 3 interrupts. 0: Disable compare capture channel 3 interrupt.	0
2	CC2IE	RW	Compare capture channel 2 interrupt enable bit. 1: Allow comparison of capture channel 2 interrupts. 0: Disable compare capture channel 2 interrupt.	0
1	CC1IE	RW	Compare capture channel 1 interrupt enable bit. 1: Allow comparison of capture channel 1 interrupts. 0: Disable compare capture channel 1 interrupt.	0
0	UIE	RW	Update the interrupt enable bit. 1: Allow updates to be interrupted. 0: Disable update interruption.	0

14.4.5 Interrupt Status Register (TIMx_INTFR) (x=1/8)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CC4OF	CC3OF	CC2OF	CC1OF	Reserved	BIF	TIF	COMIF	CC4IF	CC3IF	CC2IF	CC1IF	UIF

Bit	Name	Access	Description	Reset value
[15:13]	Reserved	RO	Reserved	0
12	CC4OF	RW0	Compare capture channel 4 repeat capture flag bit.	0
11	CC3OF	RW0	Compare capture channel 3 repeat capture flag bit.	0
10	CC2OF	RW0	Compare capture channel 2 repeat capture flag bit.	0
9	CC1OF	RW0	Compare capture channel 1 repeat capture flag bit is used only when the compare capture channel is configured for input capture mode. This flag is set by hardware and a software write of 0 clears this bit.	0

			1: The value of the counter is captured into the capture comparison register when the status of CC1IF has been set. 0: No duplicate captures are generated.	
8	Reserved	RO	Reserved	0
7	BIF	RW0	The brake interrupt flag bit, once the brake input is valid, by hardware for this position bit, can be cleared by software. 1: A set valid level is detected on the brake pin input. 0: No braking event is generated.	0
6	TIF	RW0	Trigger interrupt flag bit, when a trigger event occurs by hardware to this location bit, by software to clear. Trigger events include the detection of a valid edge at the TRGI input from a mode other than gated, or any edge in gated mode. 1: Trigger event generation. 0: No trigger event is generated.	0
5	COMIF	RW0	COM interrupt flag bit, this bit is set by hardware and cleared by software once a COM event is generated. com events including CCxE, CCxNE, OCxM are updated. 1: COM event generation. 0: No COM event is generated.	0
4	CC4IF	RW0	Compare capture channel 4 interrupt flag bit.	0
3	CC3IF	RW0	Compare capture channel 3 interrupt flag bit.	0
2	CC2IF	RW0	Compare capture channel 2 interrupt flag bit.	0
1	CC1IF	RW0	Compare capture channel 1 interrupt flag bit. If the compare capture channel is configured in output mode. This bit is set by hardware when the counter value matches the comparison value, except in centrosymmetric mode. This bit is cleared by software. 1: The value of the core counter matches the value of compare capture register 1; 0: No match occurs. If compare capture channel 1 is configured as input mode. This bit is set by hardware when a capture event occurs, and it is cleared by software or by reading the compare capture register. 1: The counter value has been captured compare capture register 1. 0: No input capture is generated.	0
0	UIF	RW0	Update interrupt flag bit, this bit is set by hardware when an update event is generated and cleared by software. 1: Update interrupt generation.	0

			0: No update event is generated. The following scenarios generate update events. If UDIS = 0, when the repeat counter value overflows or underflows. If URS = 0, UDIS = 0, when the UG bit is set, or when the counter core counter is reinitialized by software. If URS = 0, UDIS = 0, when the counter CNT is reinitialized by a trigger event.	

14.4.6 Event Generation Register (TIMx_SWEVGR) (x=1/8)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								BG	TG	COMG	CC4G	CC3G	CC2G	CC1G	UG

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved.	0
7	BG	WO	The brake event generation bit, which is set and cleared by software, is used to generate a brake event. 1: Generate a brake event. At this point, MOE=0, BIF=1, if the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA are generated. 0: No action.	0
6	TG	WO	The trigger event generation bit, which is set by software and cleared by hardware, is used to generate a trigger event. 1: Generate a trigger event, TIF is set, and the corresponding interrupts and DMAs are generated if enabled. 0: No action.	0
5	COMG	WO	Compare capture control update generation bit. Generates a compare capture control update event. This bit is set by software and automatically cleared by hardware. 1: when CCPC = 1, allow updating of CCxE, CCxNE, OCxM bits. 0: No action. *Note: This bit is only valid for channels with complementary outputs (channels 1, 2, 3).*	0
4	CC4G	WO	Compare capture event generation bit 4. generates compare capture event 4.	0
3	CC3G	WO	Compare capture event generation bit 3. generates compare capture event 3.	0
2	CC2G	WO	Compare capture event generation bit 2. generates compare capture event 2.	0

1	CC1G	WO	Compare capture event generation bit 1. generates compare capture event 1. This bit is set by software and cleared by hardware. It is used to generate a compare capture event. 1: Generate a compare capture event on compare capture channel 1. If compare capture channel 1 is configured as output. Set the CC1IF bit. Generate the corresponding interrupts and DMAs if they are enabled. If compare capture channel 1 is configured as input. The current core counter value is captured to compare capture register 1; set the CC1IF bit to generate the corresponding interrupts and DMAs if they are enabled; if CC1IF is already set, set the CC1OF bit. 0: No action.	0
0	UG	WO	Update event generation bit to generate an update event. This bit is set by software and is automatically cleared by hardware. 1: Initialize the counter and generate an update event. 0: No action. *Note: The prescaler counter is also cleared to zero, but the prescaler factor remains unchanged. The core counter is cleared if in centrosymmetric mode or incremental counting mode; if in decremental counting mode, the core counter takes the value of the reload value register.*	0

14.4.7 Compare/Capture Control Register 1 (TIMx_CHCTLR1) (x=1/8)

Offset address: 0x18

The channel can be used in input (capture mode) or output (compare mode), and the direction of the channel is defined by the corresponding CCxS bit. The other bits of this register have different roles in input and output modes. OCxx describes the function of the channel in output mode and ICxx describes the function of the channel in input mode.

15	14 13 12	11	10	9 8	7	6 5 4	3	2	1 0
OC2CE	OC2M[2:0]	OC2PE	OC2FE	CC2S[1:0]	OC1CE	OC1M[2:0]	OC1PE	OC1FE	CC1S[1:0]
IC2F[3:0]		IC2PSC[1:0]			IC1F[3:0]		IC1PSC[1:0]		

Compare mode (pin direction is output).

Bit	Name	Access	Description	Reset value
15	OC2CE	RW	Compare capture channel 2 clear enable bit. 1: Clear OC2REF bit zero once ETRF input is detected high; 0: OC2REF is not affected by ETRF input.	0
[14:12]	OC2M[2:0]	RW	Compare Capture Channel 2 mode setting field. The 3 bits define the action of the output reference	000b

			signal OC2REF, which determines the values of OC2, OC2N. OC2REF is active high, while the active levels of OC2 and OC2N depend on the CC2P, CC2NP bits. 000: Freeze. Comparison of the value of the capture register with the value of the comparison between the core counters does not work for OC2REF. 001: force to set to valid level. Forcing OC2REF high when the core counter has the same value as the comparison capture register 2. 010: Force to set to invalid level. Forcing OC2REF low when the value of the core counter is the same as the comparison capture register 2. 011: Flip. Flips the level of OC2REF when the core counter is the same as the value of compare capture register 2. 100: Forced to invalid level. Forces OC2REF to low. 101: Forced to valid level. Force OC2REF to high. 110: PWM Mode 1: In up count, once the core counter is smaller than the value of the compare capture register, channel 2 is valid, otherwise it is invalid; in downward counting, once the core counter is larger than the value of the compare capture register, channel 2 is invalid (OC2REF=0), otherwise it is valid (OC2REF=1) 111: PWM mode 2: In up count, once the core counter is smaller than the value of the compare capture register, channel 2 is invalid level, otherwise it is valid level; in down count, once the core counter is larger than the value of the compare capture register, channel 2 is valid level (OC2REF=1), otherwise it is invalid level (OC2REF=0). *Note: This bit cannot be modified once the LOCK level is set to 3 and CC2S=00b. In PWM mode 1 or PWM mode 2, the OC2REF level is changed only when the comparison result is changed or when switching from freeze mode to PWM mode in the output comparison mode.*	
11	OC2PE	RW	Compare Capture Register 2 preload enable bit. 1: Enable the preload function of compare capture register 2, read and write operations only operate on the preload registers, the preload value of compare capture register 2 is loaded into the current shadow register when the update event comes; 0: Disable the preload function of compare capture register 2, compare capture register 2 can be written at any time, and the newly written value takes effect	0

			immediately. *Note: Once the LOCK level is set to 3 and CC2S=00, this bit cannot be modified; PWM mode can be used only in single pulse mode (OPM=1) without confirming the pre-load register, otherwise its action is not determined.*	
10	OC2FE	RW	Compare Capture Channel 2 fast enable bit, this bit is used to speed up the response of the compare capture channel output to a trigger input event. 1: The active edge of the input to the flipflop acts as if a comparison match has occurred. Therefore, the OC is set to the comparison level independent of the comparison result. The delay between the valid edge of the sample trigger and the output of the compare capture channel 2 is reduced to 3 clock cycles. 0: Based on the value of the counter and compare capture register 1, compare capture channel 2 operates normally, even if the flip-flop is open. The minimum delay to activate the compare capture channel 2 output is 5 clock cycles when the input of the flipflop has a valid edge. OC2FE only works when the channel is configured to PWM1 or PWM2 mode.	0
[9:8]	CC2S[1:0]	RW	Compare capture channel 2 input selection fields. 00: comparison capture channel 2 is configured as an output. 01: comparison capture channel 2 is configured as an input and IC2 is mapped on TI2. 10: comparison capture channel 2 is configured as an input and IC2 is mapped on TI1. 11: Compare Capture Channel 2 is configured as an input and IC2 is mapped on TRC. This mode works only when the internal trigger input is selected (by the TS bit). *Note: Compare Capture Channel 2 is writable only when the channel is off (when CC2E is zero).*	00b
7	OC1CE	RW	Compare capture channel 1 clear enable bit.	0
[6:4]	OC1M[2:0]	RW	Compare capture channel 1 mode setting field.	0
3	OC1PE	RW	Compare capture register 1 preload enable bit.	0
2	OC1FE	RW	Compare capture channel 1 fast enable bit.	0
[1:0]	CC1S[1:0]	RW	Compare capture channel 1 input selection fields.	0

Capture mode (pin direction is input).

Bit	Name	Access	Description	Reset value
[15:12]	IC2F[3:0]	RW	The input capture filter 2 configuration field, these bits set the sampling frequency of the TI1 input and the	0000b

			digital filter length. The digital filter consists of an event counter, which records N events and then generates a jump in the output. 0000: no filter, sampled at fDTS. 1000: sampling frequency Fsampling = Fdts/8, N = 6. 0001: sampling frequency Fsampling=Fck_int, N=2. 1001: sampling frequency Fsampling = Fdts/8, N = 8. 0010: sampling frequency Fsampling=Fck_int, N=4. 1010: sampling frequency Fsampling = Fdts/16, N = 5. 0011: sampling frequency Fsampling=f=Fck_int, N=8. 1011: sampling frequency Fsampling = Fdts/16, N = 6. 0100: sampling frequency Fsampling = Fdts/2, N = 6. 1100: sampling frequency Fsampling = Fdts/16, N = 8. 0101: sampling frequency Fsampling = Fdts/2, N = 8. 1101: sampling frequency Fsampling = Fdts/32, N = 5. 0110: sampling frequency Fsampling = Fdts/4, N = 6. 1110: sampling frequency Fsampling = Fdts/32, N = 6. 0111: sampling frequency Fsampling = Fdts/4, N = 8. 1111: Sampling frequency Fsampling=Fdts/32, N=8.	
[11:10]	IC2PSC[1:0]	RW	Compare capture channel 2 prescaler configuration field, these 2 bits define the prescaler coefficient for compare capture channel 2. Once CC1E = 0, the prescaler is reset. 00: Without prescaler, one capture is triggered for each edge detected on the capture input. 01: Capture triggered every 2 events. 10: Capture triggered every 4 events. 11: Capture is triggered every 8 events.	00b
[9:8]	CC2S[1:0]	RW	Compare the capture channel 2 input selection field, these 2 bits define the direction of the channel (input/output), and the selection of the input pin. 00: Compare capture channel 2 channel is configured as an output. 01: Compare capture channel 2 channel is configured as an input and IC1 is mapped on TI1. 10: Compare capture channel 2 channel is configured as an input and IC1 is mapped on TI2. 11: Compare capture channel 2 channel is configured as an input and IC1 is mapped on TRC. This mode works only when the internal trigger input is selected (by the TS bit). *Note: CC2S is writable only when the channel is off (CC2E is 0).*	00b
[7:4]	IC1F[3:0]	RW	Input capture filter 1 configuration field.	0
[3:2]	IC1PSC[1:0]	RW	Compare the capture channel 1 prescaler configuration field.	0

[1:0]	CC1S[1:0]	RW	Compare capture channel 1 input selection fields.	0

14.4.8 Compare/Capture Control Register 2 (TIMx_CHCTLR2) (x=1/8)

Offset address: 0x1C

The channel can be used in input (capture mode) or output (compare mode), and the direction of the channel is defined by the corresponding CCxS bit. The other bits of this register serve different purposes in input and output modes. OCxx describes the function of the channel in output mode and ICxx describes the function of the channel in input mode.

15	14 13 12	11	10	9 8	7	6 5 4	3	2	1 0
OC4CE	OC4M[2:0]	OC4PE	OC4FE	CC4S[1:0]	OC3CE	OC3M[2:0]	OC3PE	OC3FE	CC3S[1:0]
IC4F[3:0]		IC4PSC[1:0]			IC3F[3:0]		IC3PSC[1:0]		

Compare mode (Pin direction is output):

Bit	Name	Access	Description	Reset value
15	OC4CE	RW	Compare capture channel 4 clear enable bit.	0
[14:12]	OC4M[2:0]	RW	Compare capture channel 4 mode setting field.	0
11	OC4PE	RW	Compare capture register 4 preload enable bit.	0
10	OC4FE	RW	Compare capture channel 4 fast enable bit.	0
[9:8]	CC4S[1:0]	RW	Compare capture channel 4 input selection fields.	0
7	OC3CE	RW	Compare capture channel 3 clear enable bit.	0
[6:4]	OC3M[2:0]	RW	Compare capture channel 3 mode setting field.	0
3	OC3PE	RW	Compare capture register 3 preload enable bit.	0
2	OC3FE	RW	Compare capture channel 3 fast enable bit.	0
[1:0]	CC3S[1:0]	RW	Compare capture channel 3 input selection fields.	0

Capture mode (pin direction is input).

Bit	Name	Access	Description	Reset value
[15:12]	IC4F[3:0]	RW	Input capture filter 4 configuration field.	0
[11:10]	IC4PSC[1:0]	RW	Compare capture channel 4 prescaler configuration field.	0
[9:8]	CC4S[1:0]	RW	Compare capture channel 4 input selection fields.	0
[7:4]	IC3F[3:0]	RW	Input capture filter 3 configuration field.	0
[3:2]	IC3PSC[1:0]	RW	Compare capture channel 3 prescaler configuration fields.	0
[1:0]	CC3S[1:0]	RW	Compare capture channel 3 input selection fields.	0

14.4.9 Compare/Capture Enable Register (TIMx_CCER) (x=1/8)

Offset address: 0x20

15 14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	CC4P	CC4E	CC3NP	CC3NE	CC3P	CC3E	CC2NP	CC2NE	CC2P	CC2E	CC1NP	CC1NE	CC1P	CC1E

Bit	Name	Access	Description	Reset value
[15:14]	Reserved	RO	Reserved	0
13	CC4P	RW	Compare the capture channel 4 output polarity setting bit.	0
12	CC4E	RW	Compare capture channel 4 output enable bit.	0

11	CC3NP	RW	Compare capture channel 3 complementary output polarity setting bit.	0
10	CC3NE	RW	Compare capture channel 3 complementary output enable bits.	0
9	CC3P	RW	Compare capture channel 3 output polarity setting bit.	0
8	CC3E	RW	Compare capture channel 3 output enable bit.	0
7	CC2NP	RW	Compare capture channel 2 complementary output polarity setting bit.	0
6	CC2NE	RW	Compare capture channel 2 complementary output enable bits.	0
5	CC2P	RW	Compare capture channel 2 output polarity setting bit.	0
4	CC2E	RW	Compare capture channel 2 output enable bit.	0
3	CC1NP	RW	Compare capture channel 1 complementary output polarity setting bit.	0
2	CC1NE	RW	Compare capture channel 1 complementary output enable bit.	0
1	CC1P	RW	Compare the output polarity setting bits of capture channel 1. The CC1 channel is configured to output: 1: OC1 low level effective. 0: OC1 high level is effective. The CC1 channel is configured to enter: The bit selects whether the inverse signal of IC1 or IC1 is used as the trigger or capture signal. 1: inversion: capture occurs at the falling edge of the IC1; when used as an external trigger, the IC1 is inverted. 0: no inversion: capture occurs on the rising edge of the IC1; when used as an external trigger, the IC1 is not inverted. *Note: Once the LOCK level (the LOCK bit in the TIMx_BDTR register) is set to 3 or 2, the bit cannot be modified.*	0
0	CC1E	RW	Compare the output enable bits of capture channel 1. The CC1 channel is configured to output: 1: On. The OC1 signal is output to the corresponding output pin, and its output level depends on the values of MOE, OSSI, OSSR, OIS1, OIS1N and CC1NE bits. 0: Off. OC1 forbids output, so the output level of OC1 depends on the values of MOE, OSSI, OSSR, OIS1, OIS1N, and CC1NE bits. The CC1 channel is configured to enter: This bit determines whether the value of the counter can be captured into the TIMx_CCR1 register. 1: Capture enable. 0: Capture is prohibited.	0

14.4.10 Counter of Advanced-control Timer (TIMx_CNT) (x=1/8)

Offset address: 0x24

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CNT[15:0]	RW	The real-time value of the timer's counter.	0

14.4.11 Counting Clock Prescaler (TIMx_PSC) (x=1/8)

Offset address: 0x28

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	PSC[15:0]	RW	The dividing factor of the prescaler of the timer; the clock frequency of the counter is equal to the input frequency of the divider/(PSC+1).	0

14.4.12 Auto-reload Value Register (TIMx_ATRLR) (x=1/8)

Offset address: 0x2C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	ARR[15:0]	RW	The value of this field will be loaded into the counter, see section 14.2.3 for when the ATRLR acts and updates; the counter stops when the ATRLR is empty.	FFFFh

14.4.13 Repeat Count Value Register (TIMx_RPTCR) (x=1/8)

Offset address: 0x30

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								REP[7:0]							

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:0]	REP[7:0]	RW	The value of the repeat counter.	0

14.4.14 Compare/Capture Register 1 (TIMx_CH1CVR) (x=1/8)

Offset address: 0x34

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CCR1[15:0]	RW	The value compare/capture register channel 1.	0

14.4.15 Compare/Capture Register 2 (TIMx_CH2CVR) (x=1/8)

Offset address: 0x38

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CCR2[15:0]	RW	The value compare/capture register channel 2.	0

14.4.16 Compare/Capture Register 3 (TIMx_CH3CVR) (x=1/8)

Offset address: 0x3C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR3[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CCR3[15:0]	RW	The value compare/capture register channel 3.	0

14.4.17 Compare/Capture Register 4 (TIMx_CH4CVR) (x=1/8)

Offset address: 0x40

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CCR4[15:0]	RW	The value compare/capture register channel 4.	0

14.4.18 Brake and Dead-time Register (TIMx_BDTR) (x=1/8)

Offset address: 0x44

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK[1:0]		DTG[7:0]							

Bit	Name	Access	Description	Reset value
15	MOE	RW	Main output enable bit. Once the brake signal is active, it will be cleared asynchronously. 1: Allow OCx and OCxN to be set as outputs. 0: Disable the output of OCx and OCxN or force to idle state.	0
14	AOE	RW	Auto output enable. 1: MOE can be set by software or set in the next update event. 0: MOE can only be set by software.	0
13	BKP	RW	The brake input polarity setting bit. 1: Brake input active high. 0: Brake input is active low. *Note: When LOCK level 1 is set, this bit cannot be*	0

			modified. A write to this bit requires an PB clock before it can take effect.	
12	BKE	RW	Brake function enable bit. 1: Turn on the brake input. 0: Brake input is disabled. *Note: When LOCK level 1 is set, this bit cannot be modified. A write to this bit requires an PB clock before it can take effect.*	0
11	OSSR	RW	1: When the timer is not working, once CCxE=1 or CCxNE=1, first turn on OC/OCN and outputinvalid level, then set OCx, OCxN enable output signal=1. 0: When the timer is not operating, OC/OCN output is disabled. *Note: When LOCK level 1 is set, this bit cannot be modified.*	0
10	OSSI	RW	1: when the timer is not operating, once CCxE = 1 or CCxNE = 1, OC/OCN first outputs its idle level, then OCx, OCxN enable output signal = 1. 0: When the timer is not operating, OC/OCN output is disabled. *Note: When LOCK level 1 is set, this bit cannot be modified.*	0
[9:8]	LOCK[1:0]	RW	Lock the function setting field. 00: Disable the locking function. 01: Lock level 1, no DTG, BKE, BKP, AOE, OISx and OISxN bits can be written. 10: Lock level 2, where the bits in lock level 1 cannot be written, nor the CC polarity bits, nor the OSSR and OSSI bits. 11: Lock level 3, cannot write to the bits in lock level 2, and cannot write to the CC control bits. *Note: After system reset, the LOCK bit can only be written once and cannot be modified again until reset.*	00b
[7:0]	DTG[7:0]	RW	Deadband setting bits that define the duration of the deadband between complementary outputs. Assume that DT denotes its duration. DTG[7:5]=0xx=>DT=DTG[7:0]*Tdtg, Tdtg=TDTS; DTG[7:5]=10x=>DT=(64+DTG[5:0])*Tdtg, Tdtg=2*TDTS; DTG[7:5]=110=>DT=(32+DTG[4:0])*Tdtg, Tdtg =8×TDTS; DTG[7:5]=111=>DT=(32+DTG[4:0])*Tdtg, Tdtg =16*TDTS. *Note: Once the LOCK level (the LOCK[1:0] bit in the TIM1_BDTR register) is set to 1, 2, or 3, these bits cannot be modified.*	0

14.4.19 DMA Control Register (TIMx_DMACFGR) (x=1/8)

Offset address: 0x48

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DBL[4:0]					Reserved			DBA[4:0]				

Bit	Name	Access	Description	Reset value
[15:13]	Reserved	RO	Reserved	0
[12:8]	DBL[4:0]	RW	The length of the DMA continuous transmission, the actual value of which is the value of this field + 1. When reading or writing to the TIMx_DMAADR register, the timer performs a continuous transfer), that is, it defines the number of transfers, which can be half words (double bytes) or bytes: 00000: 1 transmission; 00001: 2 transmissions; 00010: 3 transmissions; …… …… 10001: 18 transmissions. Suppose we perform the following transmission: DBL=7, DBA=TIM2_CTLR1. If DBL=7 and DBA=TIM2_CTLR1 represents the address of the data to be transferred, then the transfer address is given by the following formula: (TIMx_CTLR1 address) + DBA + (DMA index), where DMA index = DBL. where (TIMx_CTLR1 address) + DBA plus 7 provides the address where data will be written or read. Thus, data transfer occurs across 7 registers starting from the slave address (TIMx_CTLR1 address) + DBA. Depending on the configured DMA data length, the following scenarios may occur: 1. If the data is set as a half-word (16 bits), it will be transferred to all 7 registers. 2. If the data is set as a byte, it will still be transferred to all 7 registers: the first register contains the first MSB byte, the second register contains the first LSB byte, and so on. Therefore, for timers, the user must specify the data width transferred by the DMA.	0
[7:5]	Reserved	RO	Reserved	0
[4:0]	DBA[4:0]	RW	These bits define the offset of the DMA in continuous mode from the address where control register 1 is located. 00000: TIMx_CTLR1; 00001: TIMx_CTLR2;	0

			00010: TIMx_SMCFGR; ……	

14.4.20 DMA Address Register in Continuous Mode (TIMx_DMAADR) (x=1/8)

Offset address: 0x4C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAB[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	DMAB [31:0]	RW	The address of the DMA in continuous mode. Reading or writing the TIMx_DMAADR register results in accessing the register located at the address: TIMx_CTLR1 Address + DBA + DMA Index, where: 'TIMx_CTLR1 Address' is the address of Control Register 1 (TIMx_CTLR1); "DBA" is the base address defined in the TIMx_DMACFGR register; "DMA Index" is an offset automatically controlled by the DMA, determined by the DBL defined in the TIMx_DMACFGR register.	0

14.4.21 Dual-edge Capture Register (TIMx_AUX) (x=1/8)

Offset address: 0x50

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													CAP_ED_CH4	CAP_ED_CH3	CAP_ED_CH2

Bit	Name	Access	Description	Reset value
[15:3]	Reserved	R0	Reserved	0
2	CAP_ED_CH4	RW	Edge capture enable for Channel 4: 0: Disable edge capture for Channel 4; 1: Enable edge capture for Channel 4.	0
1	CAP_ED_CH3	RW	Edge capture enable for Channel 3: 0: Disable edge capture for Channel 3; 1: Enable edge capture for Channel 3.	0
0	CAP_ED_CH2	RW	Edge capture enable for Channel 2: 0: Disable edge capture for Channel 2; 1: Enable edge capture for Channel 2.	0

Chapter 15 General-Purpose Timer (GPTM)

The general timer module consists of four 16-bit automatic reassembling timers (TIM2, TIM3, TIM4 and TIM5), which are used to measure pulse width or generate pulses and PWM waves with specific frequencies. It can be used in automatic control, power supply and other fields.

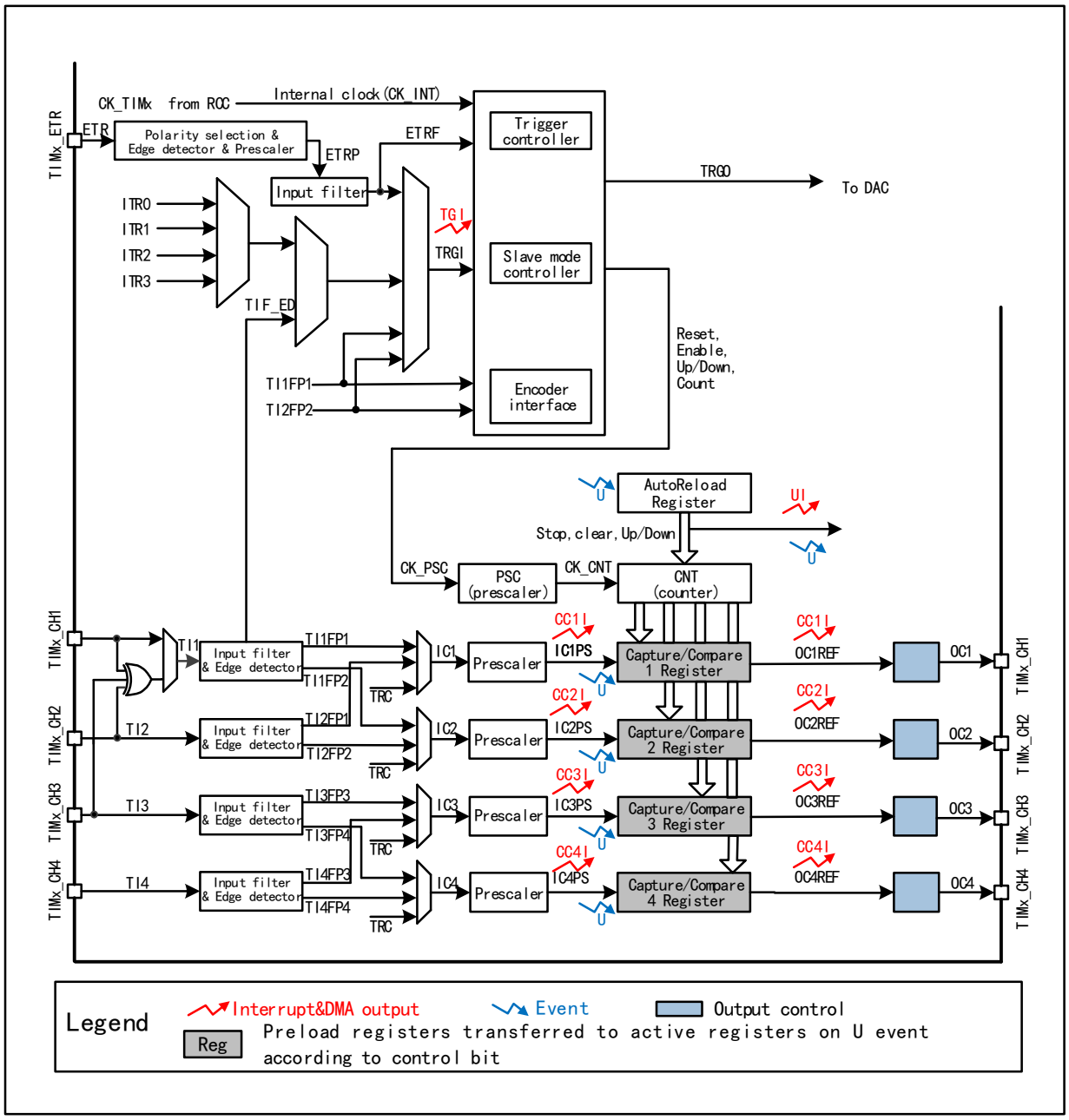
15.1 Main Features

The main features of a 16-bit general-purpose timer include:

- 16-bit automatic reinstall counter, supporting increasing counting mode, decreasing counting mode and increasing and decreasing counting mode.
- 16-bit prescaler, the frequency division coefficient is dynamically adjustable from 1 to 65536.
- Support 4 independent comparison capture channels.
- Each comparison capture channel supports multiple operating modes, such as input capture, output comparison, PWM generation and single pulse output.
- Support external signal control timer.
- Support using DMA in multiple modes.
- Support incremental coding, cascading and synchronization between timers.

15.2 Principle and Structure

Figure 15-1 Block diagram of the structure of the general-purpose timer



15.2.1 Overview

As shown in Figure 15-1, the structure of the general-purpose timer can be roughly divided into three parts, namely the input clock part, the core counter part and the compare capture channel part.h

The clock for the general-purpose timer can come from the PB1 bus clock (CK_INT), from the external clock input pin (TIMx_ETR), from other timers with clock output (ITRx), and from the input of the compare capture channel (TIMx_CHx). These input clock signals become CK_PSC clocks after various set filtering and dividing operations, etc., and are output to the core counter section. In addition, these complex clock sources can also be output as TRGO to other peripherals such as timers, ADCs.

The core of the general-purpose timer is a 16-bit counter (CNT). After CK_PSC is divided by the prescaler (PSC), it becomes CK_CNT and is finally lost to CNT. CNT supports up counting mode, down counting mode

and up and down counting mode, and has an automatic reload value register (ATRLR) to reload the initialization value for CNT after the end of each counting cycle.

The general-purpose timer has 4 sets of compare capture channels, each of which can input pulses from exclusive pins or output waveforms to pins, i.e., the compare capture channels support both input and output modes. The input of each channel of the compare capture register supports filtering, dividing, edge detection, and other operations, and supports mutual triggering between channels, and can also provide clock for the core counter CNT. Each compare capture channel has a set of compare capture registers (CHxCVR) that support comparison with the main counter (CNT) to output pulses.

15.2.2 Difference between General-purpose Timer and Advanced-control Timer

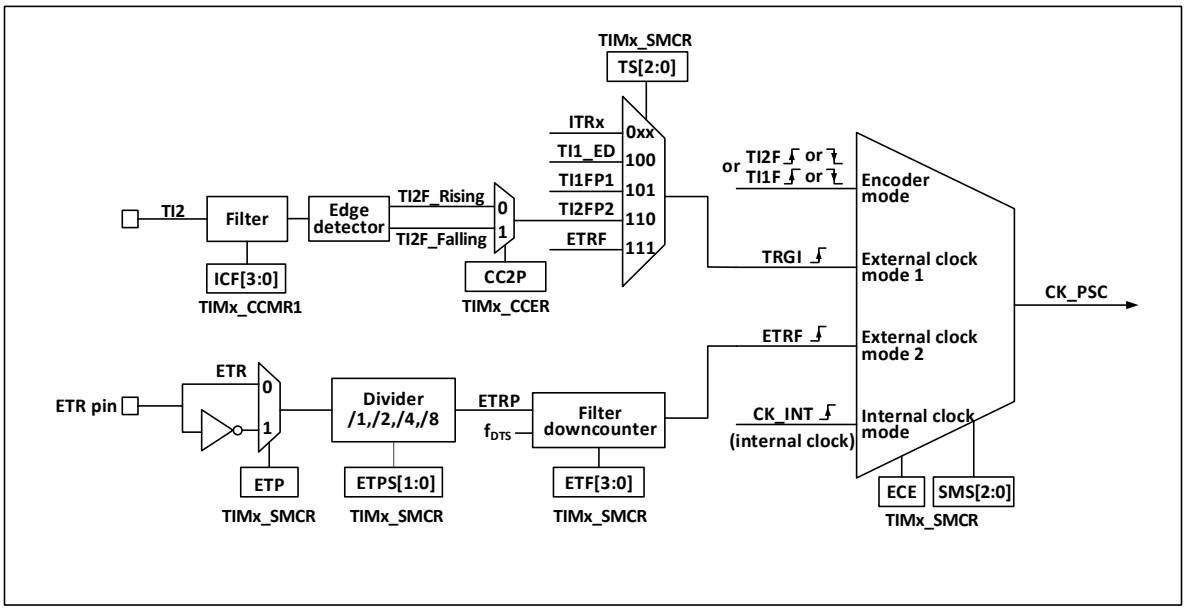
Compared with the advanced-control timer, the general-purpose timer is lack of the following functions:

1) The general-purpose timer lacks a repeated counting register that counts the count cycle of core counter.
2) The compare/capture of general-purpose timer lacks deadband generation and has no complementary output.
3) The general-purpose timer has no break signal mechanism.
4) The default clock CK_INT of general timers comes from PB1, while the CK_INT of advanced timers comes from PB2.

15.2.3 Clock Input

This section describes the source of CK_PSC. The clock source part of the general structure block diagram of the general-purpose timer is abstracted here.

Figure 15-2 Block diagram of general-purpose timer source



The available input clocks can be divided into 4 categories:

1) External clock pin (ETR) input: ETR→ETRP→ETRF;
2) Internal PB clock input: CK_INT;
3) From the compare/capture pin (TIMx_CHx): TIMx_CHx→TIx→TIxFPx; it is also used in encoder mode;
4) Input from other internal timers: ITRx.

The actual operation can be divided into 3 categories by determining the input pulse selection of the SMS from the CK_PSC source:

1) Select the internal clock source (CK_INT);
2) External clock source mode 1;

3) External clock source mode 2;
4) Encoder code.
The 4 clock sources mentioned above can be selected by these 4 operations.

15.2.3.1 Internal Clock Source (CK_INT)

If the general-purpose timer is started when the SMS domain is kept at 000b, then the internal clock source (CK_INT) is selected as the clock. At this moment, CK_INT is CK_PSC.

15.2.3.2 External Clock Source Mode 1

If SMS is set to 111b, the external clock source mode1 is enabled. When external clock source mode1 is enabled, TRGI is selected as the source of CK_PSC. It is worth noting that the user needs to configure TS to select the source of TRGI. For TS, the following pulses can be used as the clock source:
1) Internal Trigger (ITRx, x is 0, 1, 2, 3);
2) Signal of compare/capture1 after passing through the edge detector (TI1F_ED);
3) Signals TI1FP1 and TI2FP2 of compare/capture;
4) Signal ETRF from external clock pin.

15.2.3.3 External Clock Source Mode 2

Use external trigger mode2 to count on every rising or falling edge of the external clock pin input. When the ECE bit is set, the external clock source mode2 is enabled. When the external clock source mode2 is enabled, ETRF is selected as CK_PSC. The ETR pin passes through the optional inverter (ETP) and frequency divider (ETPS) to become ETRP, and then passes through the filter (ETF) to become ETRF.

When the ECE bit is set and the SMS is set to 111b, it means that the TS selects ETRF as the input.

15.2.3.4 Encoder Mode

Set SMS to 001b, 010b and 011b to enable the encoder mode. After enable the encoder mode, you may choose to use another transition edge as a signal for signal output at a certain level in TI1FP1 and TI2FP2. This mode is used when the external encoder is used. Refer to Section15.3.7 for the specific functions.

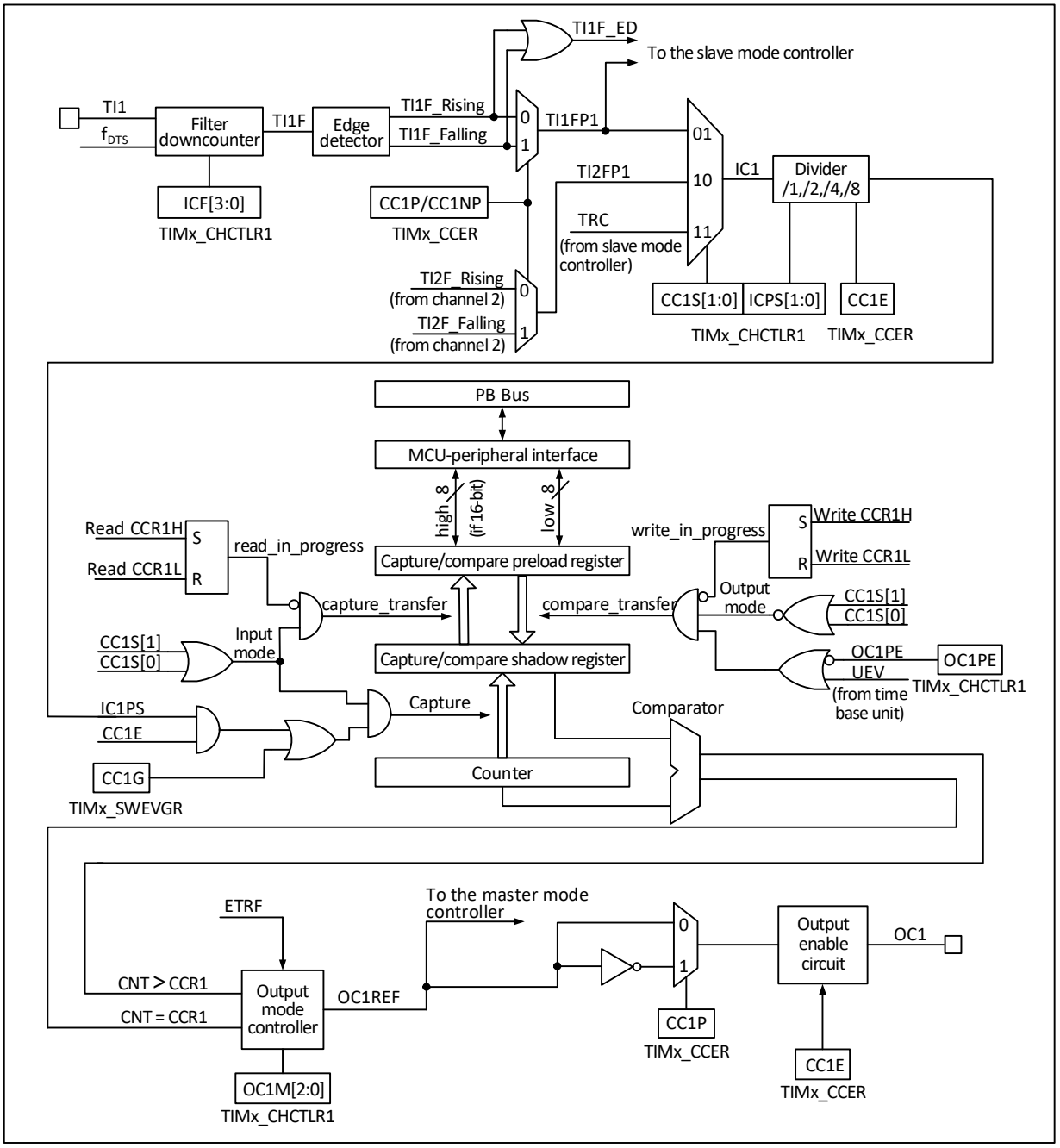
15.2.4 Counter and Periphery

CK_PSC inputs to the prescaler (PSC) for frequency division. PSC has 16 bits, and the actual frequency division factor is equivalent to the value of R16_TIMx_PSC+1. CK_PSC becomes CK_CNT through PSC. The changed value of R16_TIMx_PSC does not take effect in real time, but can be updated to the PSC after the update event. Update events include clearing and resetting the UG bit.

15.2.5 Compare/capture Channel

The compare/capture is the core of the timer to achieve complex functions. Its core is the compare/capture register, supplemented by the digital filtering of the peripheral input part, frequency division and channel multiplexing, the output part comparator and output control. The block diagram of the compare/capture is as shown in Figure 15-3.

Figure 15-3 Structure block diagram of compare/capture channel



After the signal is input from the channel x pin, it can be selected as TIx (the source of TI1 may be more than CH1. See Figure 15-1 timer block diagram). Take T1 as example, TI1 passes through the filter (ICF[3:0]) to generate TI1F, and then is divided into TI1F_Rising and TI1F_Falling after passing through the edge detector. These 2 signals are selected (CC1P) to generate TI1FP1, and TI1FP1 and TI2FP1 from channel 2 are sent to CC1S together to be selected as IC1, and then sent to the compare/capture register after going through the ICPS frequency division.

The compare/capture register is composed of preload register and shadow register, and only the preload register is operated during reading and writing. In the capture mode, the capture occurs on the shadow register, and then copied to the preload register; in the comparison mode, the content of the preload register is copied to the shadow register, and then the content of the shadow register is compared with the core counter (CNT).

15.3 Function and Implementation

The general-purpose timer complex functions are implemented by the operation of compare/capture channel, clock input circuit, counter and peripheral parts of the timer. The timer's clock input can come from multiple clock sources including the input of the compare/capture. The operation of compare/capture register channel and the clock source selection directly determines its function. The compare/capture is bidirectional and can work in input and output modes.

15.3.1 Input Capture Mode

Input capture mode is one of the basic functions of timer. The principle of the input capture mode is that when a determined edge on the ICxPS signal is detected, a capture event is generated and the current value of the counter is locked in the comparison capture register (R16_TIMx_CHxCVR). When a capture event occurs, the CCxIF (in R16_TIMx_INTFR) is set, and if the interrupt or DMA is enabled, the corresponding interrupt or DMA is generated. If the CCxIF is already set when the capture event occurs, the CCxOF bit will be set. CCxIF can be cleared by software or by hardware by reading comparison capture registers. The CCxOF is cleared by the software.

Give an example of channel 1 to illustrate the steps to use input capture mode, as follows:
1) Configure the IC1 domain and select the source of the CC1S signal. For example, if you set it to 10b and select TI1FP1 as the source of the IC1, you cannot use the default setting. The CC1S domain defaults to using the comparison capture module as the output channel.
2) Configure the IC1F field. These bits set the sampling frequency for TI1 input and the length of the digital filter. The digital filter consists of an event counter that generates an output transition after recording N events.
3) Configure the CC1P bit and set the polarity of TI1FP1. For example, keep the CC1P bit low and choose the rising edge jump.
4) Configure the IC1PS domain and set the IC1 signal to become the frequency division coefficient between IC1PS. For example, keep the IC1PS at 00b without frequency division.
5) Configure the CC1E bit to allow the value of the core counter (CNT) to be captured into the comparison capture register.
6) Configure the CCxIE and CCxDE bits according to whether interrupts or DMA are required.

When TI1 inputs a captured pulse, the value of the core counter (CNT) will be recorded in the compare capture register, CC1IF is set, and when CC1IF has been set before, the CCIOF bit will also be set. If CC1IE is set, an interrupt will be generated; if CC1DE is set, a DMA request will be generated. An input capture event can be generated by software by writing to the event generation register (R16_TIMx_SWEVGR).

15.3.2 Compare Output Mode

The compare output mode is one of basic functions of timer. The principle of the compare output mode is to output a specific change or waveform when the value of the core counter (CNT) is consistent with the value of the compare/capture register. OCxM (in R16_TIMx_CHCTLRx) and the CCxP bit (in R16_TIMx_CCER) determine whether the output is determined high or low level or level inversion. When a compare consistent event is generated, the CCxIF bit will be also set. If the CCxIE bit is preset, an interrupt will be generated; if the CCxDE bit is preset, a DMA request will be generated.

The procedure of compare output mode configuration is as follows:
1) Configure the clock source and auto-reload value of the core counter (CNT);
2) Set the count value to be compared to the compare/capture register (R16_TIMx_CHxCVR);
3) If an interrupt needs to be generated, set the CCxIE bit;

4) Keep OCxPE as 0 and disable the preload register of the compare/capture register;
5) Set the output mode, and set OCxM and CCxP bit;
6) Enable the output and set the CCxE bit;
7) Set the CEN bit and start the timer;

15.3.3 Forced Output Mode

The output mode of the compare/capture of the timer can be forced to output a certain level by software, instead of relying on the shadow register and the core counter of the compare/capture register.

The specific means is to set OCxM to 100b, which means to force OCxREF to be low; or to set OCxM to 101b, which means setting OCxREF to a high value by force.

It shall be noted that if OCxM is set to 100b or 101b by force, the compare process between the internal main counter and the compare/capture register will be still in progress, the corresponding flag bit will be still set, and interrupts and DMA request will still be generated.

15.3.4 PWM Input Mode

The PWM input mode is used to measure the duty cycle and frequency of the PWM, which is a special case of the input capture mode. The operation is the same as the input capture mode except for the following differences: PWM occupies 2 compare/captures, and the input polarity of the 2 channels is set to opposite. One of the signals is set to trigger input, and SMS is set to reset mode.

For example, to measure the cycle and frequency of the PWM wave input from TI1, the following operations are required:
1) Set TI1 (TI1FP1) as the input of IC1 signal. Set CC1S to 01b;
2) Set TI1FP1 as the rising edge valid. Keep CC1P as 0;
3) Set TI1 (TI1FP2) as the input of IC2 signal. Set CC2S to 10b;
2) Set TI1FP2 as the falling edge valid. Set CC2P to 1;
5) The source of the clock source is TI1FP1. Set TS to 101b;
6) Set SMS to reset mode, i.e., 100b;
7) Enable the input capture. Set CC1E and CC2E bits.

Note: Since only TI1FP1 and TI2FP2 are connected to the slave mode controller, only TIMx_CH1/TIMx_CH2 can be used in PWM input mode.

15.3.5 PWM Output Mode

The PWM output mode is one of basic functions of timer. The most common method of PWM output mode is to use the reload value to determine the PWM frequency, and to use the capture comparison register to determine the duty cycle. Set 110b or 111b in OCxM to use PWM mode 1 or mode 2, set the OCxPE bit to enable the preload register, and finally set the ARPE bit. Since the value of the preload register can be sent to the shadow register when an update event occurs, it is necessary to set the UG bit to initialize all registers before the core counter starts counting. In the PWM mode, the core counter and the compare/capture register are always being compared. According to the CMS bit, the timer can output edge-aligned or center-aligned PWM signals.

- Edge alignment

When the edge alignment is used, the core counter counts up or down. In the scenario of PWM mode 1, when the value of the core counter is greater than that of the compare/capture register, OCxREF will rise to be low;

when the value of the core counter is less than the compare/capture register (such as When the core counter increases to the value of R16_TIMx_ATRLR and returns to all 0s), OCxREF drops to high.

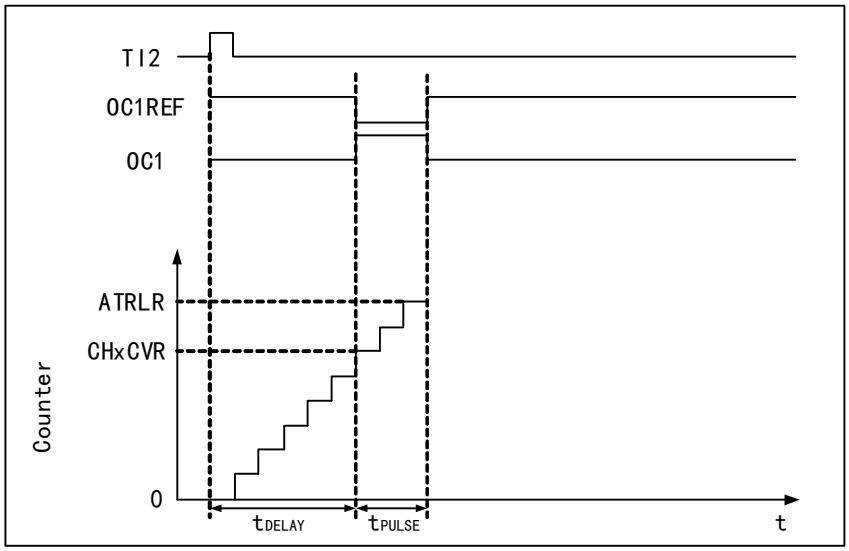
- Central alignment

When the center-aligned mode is used, the core counter will run in a mode where up counting and down counting are performed alternately, and OCxREF performs rising and falling jumps when the values of the core counter and the compare/capture register are consistent. However, in 3 types of central alignment mode, the bit setting timing of comparison flag is different somewhat. When the center-alignment mode is used, it is the best to generate a software update flag (set the UG bit) before starting the core counter.

15.3.6 Single Pulse Mode

The single pulse mode can be used to respond to a specific event to generate a pulse after a delay. The delay and pulse width are programmable. Setting the OPM bit can make the core counter stop when the next update event UEV is generated (the counter turns over to 0).

Figure 15-4 Event generation and pulse response



As shown in Figure 15-4, it is necessary to detect the beginning of a rising edge on the TI2 input pin. After delaying Tdelay, a positive pulse of length Tpulse will be generated on OC1:

1) Set TI2 as trigger. Set the CC2S field to 01b and map TI2FP2 to TI2; set the CC2P bit to 0b and set TI2FP2 to rising edge detection; set the TS field to 110b and set TI2FP2 as the trigger source; set the SMS field to 110b, and TI2FP2 is used to start the counter;

2) Tdelay is defined by the value of the compare/capture register, and Tpulse is determined by the value of the auto-reload value register and the value of the compare/capture register.

15.3.7 Encoder Mode

The encoder mode is a typical application of the timer. It can be used to access the dual-phase output of the encoder. The count direction of the core counter is synchronized with the rotating shaft of the encoder. Each pulse output by the encoder will increase the core counter by adding one or subtracting one. The steps to use the encoder are: set the SMS field to 001b (count only on TI2 edge), 010b (count only on TI1 edge) or 011b (count on both TI1 and TI2 edges), and connect the encoder to compare/capture 1, 2 inputs, set a value for the reload value register and this value can be set to be greater. In the encoder mode, the internal compare/capture register of timer, prescaler, repeat count register and other registers all work normally. The following table shows the relationship between the counting direction and the encoder signal.

Table 15-1 Relationship between count direction of timer in encoder mode and encoder signal

Count active edge	Relative signal level	TI1FP1 signal edge		TI2FP2 signal edge	
		Rising edge	Falling edge	Rising edge	Falling edge
Only count at TI1 edge	High	Downcount	Upcount	Not count	
	Low	Upcount	Downcount		
Only count at TI2 edge	High	Not count		Upcount	Downcount
	Low			Downcount	Upcount
Count on both edges of TI1 and TI2	High	Downcount	Upcount	Upcount	Downcount
	Low	Upcount	Downcount	Downcount	Upcount

15.3.8 Timer Synchronous Mode

The timer can output clock pulses (TRGO) and can also receive input from other timers (ITRx). The sources of ITRx of different timers (TRGO of other timers) are different. Table 15-2 shows the internal trigger connection of timers.

Figure 15-2 GTPM internal trigger connection

Slave timer	ITR0(TS=000)	ITR1(TS=001)	ITR2(TS=010)	ITR3(TS=011)
TIM2	TIM1	USB	TIM3	TIM4
TIM3	TIM1	TIM2	-	TIM4
TIM4	TIM1	TIM2	TIM3	-
TIM5	TIM2	TIM3	TIM4	-

15.3.9 Dual Edge Capture Mode

The pulses can be measured via the TIMx_AUX register.

For example: capture the pulse width through channel 2. In the capture function configuration, select the source of the IC2 signal (the CC2S bit of the TIMx_CHCTLR1 register is 11b), and enable the dual-edge capture function of channel 2 (the CAP_ED_CH2 bit of the TIMx_AUX register is 1). At this point, the dual edge capture configuration has been completed.

The captured dual-edge pulse width value can be read through the CCR2 bit of the TIMx_CH2CVR register.

15.3.10 Debug Mode

When the system enters debug mode, the timer continues to run or stops according to the setting of the DBG module.

15.4 Register Description

Table 15-3 TIM2 registers

Name	Offset address	Description	Reset value
R16_TIM2_CTLR1	0x40000000	TIM2 control register1	0x0000
R16_TIM2_CTLR2	0x40000004	TIM2 control register2	0x0000

R16_TIM2_SMCFGR	0x40000008	TIM2 slave mode configuration register	0x0000
R16_TIM2_DMAINTENR	0x4000000C	TIM2 DMA/interrupt enable register	0x0000
R16_TIM2_INTFR	0x40000010	TIM2 interrupt flag register	0x0000
R16_TIM2_SWEVGR	0x40000014	TIM2 event generation register	0x0000
R16_TIM2_CHCTLR1	0x40000018	TIM2 compare/capture control register 1	0x0000
R16_TIM2_CHCTLR2	0x4000001C	TIM2 compare/capture control register 2	0x0000
R16_TIM2_CCER	0x40000020	TIM2 compare/capture enable register	0x0000
R16_TIM2_CNT	0x40000024	TIM2 counter	0x0000
R16_TIM2_PSC	0x40000028	TIM2 prescaler	0x0000
R16_TIM2_ATRLR	0x4000002C	TIM2 auto-reload register	0xFFFF
R32_TIM2_CH1CVR	0x40000034	TIM2 compare/capture register 1	0x0000
R32_TIM2_CH2CVR	0x40000038	TIM2 compare/capture register 2	0x0000
R32_TIM2_CH3CVR	0x4000003C	TIM2 compare/capture register 3	0x0000
R32_TIM2_CH4CVR	0x40000040	TIM2 compare/capture register 4	0x0000
R16_TIM2_DMACFGR	0x40000048	TIM2 DMA configuration register	0x0000
R16_TIM2_DMAADR	0x4000004C	TIM2 DMA address register in continuous mode	0x0000
R16_TIM2_AUX	0x40000050	TIM2 dual edge capture register	0x0000

Table 15-4 TIM3 registers

Name	Offset address	Description	Reset value
R16_TIM3_CTLR1	0x40000400	TIM3 control register 1	0x0000
R16_TIM3_CTLR2	0x40000404	TIM3 control register 2	0x0000
R16_TIM3_SMCFGR	0x40000408	TIM3 slave mode configuration register	0x0000
R16_TIM3_DMAINTENR	0x4000040C	TIM3 DMA/interrupt enable register	0x0000
R16_TIM3_INTFR	0x40000410	TIM3 interrupt flag register	0x0000
R16_TIM3_SWEVGR	0x40000414	TIM3 event generation register	0x0000
R16_TIM3_CHCTLR1	0x40000418	TIM3 compare/capture control register 1	0x0000
R16_TIM3_CHCTLR2	0x4000041C	TIM3 compare/capture control register 2	0x0000
R16_TIM3_CCER	0x40000420	TIM3 compare/capture enable register	0x0000
R16_TIM3_CNT	0x40000424	TIM3 counter	0x0000
R16_TIM3_PSC	0x40000428	TIM3 prescaler	0x0000
R16_TIM3_ATRLR	0x4000042C	TIM3 auto reload register	0xFFFF
R32_TIM3_CH1CVR	0x40000434	TIM3 compare/capture register 1	0x0000
R32_TIM3_CH2CVR	0x40000438	TIM3 compare/capture register 2	0x0000
R32_TIM3_CH3CVR	0x4000043C	TIM3 compare/capture register 3	0x0000
R32_TIM3_CH4CVR	0x40000440	TIM3 compare/capture register 4	0x0000
R16_TIM3_DMACFGR	0x40000448	TIM3 DMA configuration register	0x0000
R16_TIM3_DMAADR	0x4000044C	TIM3 DMA address register in continuous mode	0x0000
R16_TIM3_AUX	0x40000450	TIM3 dual edge capture register	0x0000

Table 15-5 TIM4 registers

Name	Offset address	Description	Reset value
R16_TIM4_CTLR1	0x40000800	TIM4 control register1	0x0000
R16_TIM4_CTLR2	0x40000804	TIM4 control register2	0x0000
R16_TIM4_SMCFGR	0x40000808	TIM4 slave mode configuration register	0x0000
R16_TIM4_DMAINTENR	0x4000080C	TIM4 DMA/interrupt enable register	0x0000
R16_TIM4_INTFR	0x40000810	TIM4 interrupt flag register	0x0000
R16_TIM4_SWEVGR	0x40000814	TIM4 event generation register	0x0000
R16_TIM4_CHCTLR1	0x40000818	TIM4 compare/capture control register1	0x0000
R16_TIM4_CHCTLR2	0x4000081C	TIM4 compare/capture control register2	0x0000
R16_TIM4_CCER	0x40000820	TIM4 compare/capture enable register	0x0000
R16_TIM4_CNT	0x40000824	TIM4 counter	0x0000
R16_TIM4_PSC	0x40000828	TIM4 prescaler	0x0000
R16_TIM4_ATRLR	0x4000082C	TIM4 auto-reload register	0xFFFF
R16_TIM4_CH1CVR	0x40000834	TIM4 compare/capture register1	0x0000
R16_TIM4_CH2CVR	0x40000838	TIM4 compare/capture register2	0x0000
R16_TIM4_CH3CVR	0x4000083C	TIM4 compare/capture register3	0x0000
R16_TIM4_CH4CVR	0x40000840	TIM4 compare/capture register4	0x0000
R16_TIM4_DMACFGR	0x40000848	TIM4 DMA configuration register	0x0000
R16_TIM4_DMAADR	0x4000084C	TIM4 DMA address register in continuous mode	0x0000
R16_TIM4_AUX	0x40000850	TIM4 dual edge capture register	0x0000

Table 15-6 TIM5 registers

Name	Offset address	Description	Reset value
R16_TIM5_CTLR1	0x40000C00	TIM5 control register1	0x0000
R16_TIM5_CTLR2	0x40000C04	TIM5 control register2	0x0000
R16_TIM5_SMCFGR	0x40000C08	TIM5 slave mode configuration register	0x0000
R16_TIM5_DMAINTENR	0x40000C0C	TIM5 DMA/interrupt enable register	0x0000
R16_TIM5_INTFR	0x40000C10	TIM5 interrupt flag register	0x0000
R16_TIM5_SWEVGR	0x40000C14	TIM5 event generation register	0x0000
R16_TIM5_CHCTLR1	0x40000C18	TIM5 compare/capture control register1	0x0000
R16_TIM5_CHCTLR2	0x40000C1C	TIM5 compare/capture control register2	0x0000
R16_TIM5_CCER	0x40000C20	TIM5 compare/capture enable register	0x0000
R16_TIM5_CNT	0x40000C24	TIM5 counter	0x0000
R16_TIM5_PSC	0x40000C28	TIM5 prescaler	0x0000
R16_TIM5_ATRLR	0x40000C2C	TIM5 auto-reload register	0xFFFF
R32_TIM5_CH1CVR	0x40000C34	TIM5 compare/capture register1	0x0000
R32_TIM5_CH2CVR	0x40000C38	TIM5 compare/capture register2	0x0000

R32_TIM5_CH3CVR	0x40000C3C	TIM5 compare/capture register3	0x0000
R32_TIM5_CH4CVR	0x40000C40	TIM5 compare/capture register4	0x0000
R16_TIM5_DMACFGR	0x40000C48	TIM5 DMA configuration register	0x0000
R16_TIM5_DMAADR	0x40000C4C	TIM5 DMA address register in continuous mode	0x0000
R16_TIM5_AUX	0x40000C50	TIM5 dual edge capture register	0x0000

15.4.1 Control Register 1 (TIMx_CTLR1) (x=2/3/4/5)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CKD[1:0]		ARPE	CMS[1:0]		DIR	OPM	URS	UDIS	CEN

Bit	Name	Access	Description	Reset value
[15:10]	Reserved	RO	Reserved	0
[9:8]	CKD[1:0]	RW	These 2 bits define the frequency division ratio of timer clock (CK_INT) frequency and sampling clock used for the digital filter: 00: Tdts=Tck_int; 01: Tdts= 2×Tck_int; 10: Tdts= 4×Tck_int; 11: Reserved.	00b
7	ARPE	RW	Auto-reload and preload enable: 1: Auto-reload value register (ATRLR) enabled; 0: Auto-reload value register (ATRLR) disabled.	0
[6:5]	CMS[1:0]	RW	Central alignment mode selection: 00: Edge alignment mode. The counter counts up or down according to the direction bit (DIR). 01: Center alignment mode 1. The counter counts up and down alternately. The output comparison interrupt flag bit of the channel configured as an output (CCxS=00 in the CHCTLRx register) is only set when the counter counts down. 10: Center alignment mode 2. The counter counts up and down alternately. The output comparison interrupt flag bit of the channel configured as an output (CCxS=00 in the CHCTLRx register) is only set when the counter counts up. 11: Center alignment mode 3. The counter counts up and down alternately. The output comparison interrupt flag bit of the channel configured as an output (CCxS=00 in the CHCTLRx register) is only set when the counter counts up and down. *Note: When the counter is enabled (CEN=1), it is not allowed to switch from edge alignment mode to center*	00b

			alignment mode.	
4	DIR	RW	Counter direction: 1: Downcount; 0: Upcount. *Note: When the counter is configured in the center alignment mode or encoder mode, this bit will be invalid.*	0
3	OPM	RW	Single pulse mode. 1: The counter stops when the next update event (clearing the CEN bit) occurs; 0: The counter does not stop when the next update event occurs.	0
2	URS	RW	Update Request Source: The software uses this bit to select the source of UEV events. 1: If update interrupts or DMA requests are enabled, only counter overflows/underflows generate update interrupts or DMA requests; 0: If update interrupts or DMA requests are enabled, any of the following events generate update interrupts or DMA requests: -Counter overflow/underflow. -Set the UG bit. -Updates generated from the mode controller.	0
1	UDIS	RW	Updates are prohibited, and the software allows/ disables the generation of UEV events through this bit. 1: UEV is prohibited. No update events are generated, and the registers (ATRLR, PSC, CHxCVR) hold their values. If the UG bit is set or a hardware reset is issued from the mode controller, the counter and prescaler are reinitialized. 0: UEV is allowed. The UEV event is generated by any of the following events: -Counter overflow / underflow. -Set the UG bit. -Updates generated from the mode controller. Registers with caches are loaded into their preloaded values.	0
0	CEN	RW	Counter enable: 1: Enable counter. 0: Disables counters. *Note: the external clock, gating mode and encoder mode will not work until the CEN bit is set in the software. The trigger mode automatically sets the CEN bit through the hardware.*	0

15.4.2 Control Register 2 (TIMx_CTLR2) (x=2/3/4/5)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								TI1S	MMS[2:0]			CCDS	CCUS	Reserved	CCPC

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
7	TI1S	RW	TI1 selection: 1: TIMx_CH1, TIMx_CH2 and TIMx_CH3 pins are connected to TI1 input through XOR; 0: TIMx_CH1 pin is directly connected to TI1 input.	0
[6:4]	MMS[2:0]	RW	Master mode selection: These 3 bits are used to select the synchronization information (TRGO) sent to the slave timer in the master mode. The possible combination is as follows: 000: Reset – The UG bit is used as a trigger output (TRGO). If it is a reset generated by a trigger input (the slave mode controller is in reset mode), the signal on TRGO will have a delay relative to the actual reset; 001: Enable-the counter enables signal CNT_EN to be used as a trigger output (TRGO). Sometimes, it is necessary to start multiple timers at the same time or control to enable slave timers within a period of time. The counter enable signal is generated by the logical OR of the CEN control bit and the trigger input signal in the gating mode. When the counter enable signal is controlled by the trigger input, there will be a delay on TRGO, unless the master/slave mode is selected (see the description of the MSM bit in the TIMx_SMCFGR register); 010: An update event is selected as the trigger output (TRGO). For example, the clock of a master timer can be used as a prescaler for a slave timer; 011: Comparison pulse, when a capture occurs or a comparison is successful, and the CC1IF flag is to be set (even if it is already high), the trigger output will send a positive pulse (TRGO); 100: OC1REF signal is used as trigger output (TRGO); 101: OC2REF signal is used as trigger output (TRGO); 110: OC3REF signal is used as trigger output (TRGO); 111: OC4REF signal is used as trigger output (TRGO).	000b
3	CCDS	RW	1: When an update event occurs, send a DMA request of CHxCVR; 0: When CHxCVR occurs, a DMA request of CHxCVR will be generated.	0
[2:1]	Reserved	RO	Reserved.	0

0	CCPC	RW	Compare/capture preload control bit: 1: The CCxE, CCxNE, and OCxM bits are preloaded. When this bit is set, they are updated only after the COM bit is set. 0: The CCxE, CCxNE, and OCxM bits are not preloaded. *Note: This bit affects only channels with complementary outputs.*	0

15.4.3 Slave Mode Control Register (TIMx_SMCFGR) (x=2/3/4/5)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETP	ECE	ETPS[1:0]		ETF[3:0]				MSM	TS[2:0]			Reserved	SMS[2:0]		

Bit	Name	Access	Description	Reset value
15	ETP	RO	ETR trigger polarity selection; this bit selects whether to directly input ETR or input inverted ETR. 1: ETR inverted, active at low level or falling edge; 0: ETR, active at high level or rising edge.	0
14	ECE	RW	External clock mode 2 enable. 1: External clock mode 2 enabled; 0: External clock mode 2 disabled. *Note 1: Slave mode can be used simultaneously with external clock mode 2: reset mode, gating mode and trigger mode; however, TRGI cannot be connected to ETRF at this time (TS bit cannot be 111b).* *Note 2: When both external clock mode 1 and external clock mode 2 are enabled at the same time, the input of the external clock will be ETRF.*	0
[13:12]	ETPS[1:0]	RW	External trigger prescaler (ETRP); the frequency must be at most 1/4 of TIMxCLK frequency, and the frequency can be reduced through this domain. 00: Prescale off; 01: ETRP frequency divided by 2; 10: ETRP frequency divided by 4; 11: ETRP frequency divided by 8.	00b
[11:8]	ETF[3:0]	RW	External trigger filter. In fact, the digital filter is an event counter. N events are needed to validate a transition on the output. 0000: No filter, sampling is done at Fdts; 0001: Fsampling=Fck_int, N=2; 0010: Fsampling=Fck_int，N=4; 0011: Fsampling=Fck_int, N=8; 0100: Fsampling=Fdts/2, N=6; 0101: Fsampling=Fdts/2, N=8;	0000b

			0110: Fsampling=Fdts/4, N=6; 0111: Fsampling=Fdts/4, N=8; 1000: Fsampling=Fdts/8, N=6; 1001: Fsampling=Fdts/8, N=8; 1010: Fsampling=Fdts/16, N=5; 1011: Fsampling=Fdts/16, N=6; 1100: Fsampling=Fdts/16, N=8; 1101: Fsampling=Fdts/32, N=5; 1110: Fsampling=Fdts/32, N=6; 1111: Fsampling=Fdts/32, N=8.	
7	MSM	RW	Master/Slave mode selection: 1: The event on the trigger input (TRGI) is delayed to allow perfect synchronization between the current timer (via TRGO) and its slave timer. This is very useful when it is required to synchronize several timers to a single external event; 0: Not action.	0
[6:4]	TS[2:0]	RW	Trigger selection; these 3 bits select the trigger input source used to synchronize the counter. 000: Internal trigger 0 (ITR0); 100: Edge detector of TI1 (TI1F_ED); 001: Internal trigger 1 (ITR1); 101: Timer input 1 after filtering (TI1FP1); 010: Internal trigger 2 (ITR2); 110: Timer input 2 after filtering (TI12FP2); 011: Internal trigger 3 (ITR3); 111: External trigger input (ETRF); The values can be changed only when SMS is 0.	000b
3	Reserved	RO	Reserved	0
[2:0]	SMS[2:0]	RW	Input mode selection. Select the clock and trigger mode of the core counter. 000: Driven by the internal clock CK_INT; 001: Encoder mode1; depending on TI1FP1 level, the core counter counts up or down on edge of TI2FP2; 010: Encoder mode2; depending on TI2FP2 level, the core counter counts up or down on edge of TI1FP1; 011: Encoder mode3; depending on the input level of another signal, the core counter counts up and down on the edge of TI1FP1 and TI2FP2; 100: Reset mode; the rising edge of the trigger input (TRGI) will initialize the counter and generate a signal for updating the register; 101: Gating mode; when the trigger input (TRGI) is high, the clock of the counter will be turned on; when the trigger input becomes low, the counter will stop, and the start and stop of the counter will be controlled;	000b

			110: Trigger mode; the counter starts on the rising edge of the trigger input TRGI, and only the start of the counter is controlled; 111: External clock mode1; the rising edge of the selected trigger input (TRGI) drives the counter.	

15.4.4 DMA/Interrupt Enable Register (TIMx_DMAINTENR) (x=2/3/4/5)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	TDE	COMDE	CC4DE	CC3DE	CC2DE	CC1DE	UDE	Reserved	TIE	Reserved	CC4IE	CC3IE	CC2IE	CC1IE	UIE

Bit	Name	Access	Description	Reset value
15	Reserved	RO	Reserved	0
14	TDE	RW	Trigger DMA request enable. 1: Trigger DMA request enabled; 0: Trigger DMA request disabled.	0
13	Reserved	RW	Reserved	0
12	CC4DE	RW	DMA request enable of compare/capture 4. 1: DMA request of compare/capture 4 enabled; 0: DMA request of compare/capture 4 disabled.	0
11	CC3DE	RW	DMA request enable of compare/capture 3. 1: DMA request of compare/capture 3 enabled; 0: DMA request of compare/capture 3 disabled.	0
10	CC2DE	RW	DMA request enable of compare/capture 2. 1: DMA request of compare/capture 2 enabled; 0: DMA request of compare/capture 2 disabled.	0
9	CC1DE	RW	DMA request enable of compare/capture 1. 1: DMA request of compare/capture 1 enabled; 0: DMA request of compare/capture 1 disabled.	0
8	UDE	RW	Update DMA request enable. 1: Update DMA request enabled; 0: Update DMA request disabled.	0
7	Reserved	RO	Reserved	0
6	TIE	RW	Trigger interrupt enable. 1: Trigger interrupt enabled; 0: Trigger interrupt disabled.	0
5	Reserved	RO	Reserved	0
4	CC4IE	RW	Interrupt enable of compare/capture 4. 1: Interrupt of compare/capture 4 enabled; 0: Interrupt of compare/capture 4 disabled.	0
3	CC3IE	RW	Interrupt enable of compare/capture 3. 1: Interrupt of compare/capture 3 enabled; 0: Interrupt of compare/capture 3 disabled.	0

2	CC2IE	RW	Interrupt enable of compare/capture 2. 1: Interrupt of compare/capture 2 enabled; 0: Interrupt of compare/capture 2 disabled.	0
1	CC1IE	RW	Interrupt enable of compare/capture 1. 1: Interrupt of compare/capture 1 enabled; 0: Interrupt of compare/capture 1 disabled.	0
0	UIE	RW	Update interrupt enable. 1: Update interrupt enabled; 0: Update interrupt disabled.	0

15.4.5 Interrupt Flag Register (R16_TIMx_INTFR) (x=2/3/4/5)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CC4OF	CC3OF	CC2OF	CC1OF	Reserved		TIF	Reserved	CC4IF	CC3IF	CC2IF	CC1IF	UIF

Bit	Name	Access	Description	Reset value
[15:13]	Reserved	RO	Reserved	0
12	CC4OF	RW0	Compare/capture4 recapture flag.	0
11	CC3OF	RW0	Compare/capture3 recapture flag.	0
10	CC2OF	RW0	Compare/capture2 recapture flag.	0
9	CC1OF	RW0	Compare/capture1 recapture flag is only used when the compare/capture is configured in the input capture mode. This flag bit is set by the hardware, write 0 by software to clear the bit. 1: When the value of the counter is captured into the capture comparison register, the status of CC1IF has been set; 0: No recapture is generated.	0
[8:7]	Reserved	RO	Reserved	0
6	TIF	RW0	Trigger interrupt flag. When a trigger event occurs, set by hardware and cleared by software. Trigger events include the detection of a valid edge at the TRGI input terminal from modes other than gating mode, or any edge in gating mode. 1: Trigger event occurs; 0: No trigger event occurs.	0
5	Reserved	RO	Reserved.	0
4	CC4IF	RW0	Compare/capture 4 interrupt flag.	0
3	CC3IF	RW0	Compare/capture 3 interrupt flag.	0
2	CC2IF	RW0	Compare/capture 2 interrupt flag.	0
1	CC1IF	RW0	Compare/capture 1 interrupt flag. If the compare/capture is configured as the output mode, this bit is set by hardware when the counter value matches the compare value, except in center-aligned mode. This bit	0

			is cleared by software. 1: The value of core counter matches the value of compare/capture register 1; 0: No. If the compare/capture is configured as the output mode, this bit is set by hardware when a capture event occurs, and it is cleared by software or cleared by reading the compare/capture register. 1: The counter value has been captured by the compare/capture register 1; 0: No input capture is generated.	
0	UIF	RW0	Update interrupt flag. When an update event occurs, it is set by hardware and cleared by software. 1: Update interrupt generated; 0: No update interrupt generated. The update event generates in case of the following circumstances: For UDIS=0, when the repeated counter value overflows or underflows; For URS=0, UDIS=0, when the UG bit is set, or when the counter core is reinitialized by software; For URS=0, UDIS=0, when the counter CNT is reinitialized by a trigger event.	0

15.4.6 Event Generation Register (TIMx_SWEVGR) (x=2/3/4/5)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									TG	Reserved	CC4G	CC3G	CC2G	CC1G	UG

Bit	Name	Access	Description	Reset value
[15:7]	Reserved	RO	Reserved.	0
6	TG	WO	Trigger event generation. Set by software, and cleared by hardware to generate a trigger event. 1: A trigger event generated; if TIF is set and the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA will be generated; 0: No effect.	0
5	Reserved	RO	Reserved.	0
4	CC4G	WO	Compare/capture 4 generation.	0
3	CC3G	WO	Compare/capture 3 generation.	0
2	CC2G	WO	Compare/capture 2 generation.	0
1	CC1G	WO	Compare/capture1 generation. This bit is set by software and cleared by hardware. It is used to generate a compare/capture event.	0

			1: Generate compare/capture event on channel 1: If compare/capture 1 is configured as output: Set the CC1IF bit. If the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA will be generated; If compare/capture 1 is configured as input, the current core counter value is captured to compare/capture register 1; set the CC1IF bit, if the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA will be generated. If the CC1IF bit has been set, set the CC1OF bit. 0: No action.	
0	UG	WO	Update event generation. This bit is set by software and cleared automatically by hardware. 1: Initialize the counter and generate an update event; 0: No action. *Note: The counter of the prescaler is also cleared, but the prescaler factor remains unchanged. In Centro symmetric mode or up-counting mode, the core counter will be cleared; in the down-counting mode, the core counter will take the value of the reload value register.*	0

15.4.7 Compare/Capture Control Register 1 (TIMx_CHCTLR1) (x=2/3/4/5)

Offset address: 0x18

The channel can be used for input (capture mode) or output (compare mode), and the direction of the channel is defined by the corresponding CCxS bit. The functions of other bits of this register are different in input and output modes. OCxx describes the function of the channel in output mode, and ICxx describes the function of the channel in input mode.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		OC1CE	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
IC2F[3:0]				IC2PSC[1:0]				IC1F[3:0]				IC1PSC[1:0]			

Compare mode (pin direction is output):

Bit	Name	Access	Description	Reset value
15	OC2CE	RW	Compare/capture 2 clear enable. 1: Once the ETRF input high level is detected, clear the OC2REF bit to zero; 0: OC2REF is not affected by the ETRF input.	0
[14:12]	OC2M[2:0]	RW	Compare/capture2 mode setting. These 3 bits define the action of the output reference signal OC2REF, and OC2REF determines the value of OC2 and OC2N. OC2REF is active at high level, while the active level of OC2 and OC2N depends on the	000b

			CC2P and CC2NP bits. 000: Frozen. The comparison value between the value of the compare/capture register and the core counter has no effect on OC2REF; 001: Active by force. When the core counter and compare/capture register1 have the same value, force OC2REF to be high; 010: Set as inactive level by force. When the value of the core counter is the same as compare/capture register 1, force OC2REF to be low; 011: Overturn. When the core counter and compare/capture register1 have the same value, overturn the level of OC2REF; 100: Inactive by force. Force OC2REF to be low. 101: Force to be active level. Force OC2REF to be high. 110: PWM Mode 1: In up count, once the core counter is less than the value of the compare capture register, channel 2 is valid level, otherwise it is invalid level; in downward counting, once the core counter is greater than the value of the compare capture register, channel 2 is invalid level (OC2REF=0), otherwise it is valid level (OC2REF=1). 111: PWM mode 2: In up count, once the core counter is smaller than the value of the compare capture register, channel 2 is invalid level, otherwise it is valid level; in down count, once the core counter is larger than the value of the compare capture register, channel 2 is valid level (OC2REF=1), otherwise it is invalid level (OC2REF=0). *Note: Once the LOCK level is set to 3 and CC2S=00b, this bit cannot be modified. In PWM mode1 or PWM mode2, the OC2REF level changes only when the comparison result changes or when switching from freezing mode to PWM mode in output compare mode.*	
11	OC2PE	RW	Compare/capture register 2 preload enable. 1: Enable the preload function of the compare/capture register. Read and write operations are only made on the preload register. The preload value of the compare/capture register 2 is loaded into the current shadow register when the update event arrives; 0: Disable the pre-loading function of compare/capture register 2. Compare/capture register 2 can be written at any time, and the newly written value takes effect immediately. *Note: Once the LOCK level is set to 3 and CC2S=00,*	0

			this bit cannot be modified; only in single pulse mode (OPM=1) you can use PWM mode without confirming the preload register; otherwise its action is uncertain.	
10	OC2FE	RW	Compare/capture 2 fast enable. It is used to speed up the response of the compare/capture output to the trigger input event. 1: The effect of the inactive edge inputted to the trigger is like a comparison match. Therefore, OC is set to the comparison level regardless of the comparison result. The delay between the valid edge of the sampling trigger and the output of compare/capture 2 is shortened to 3 clock cycles; 0: According to the value of counter and compare/capture register 2, compare/capture 2 operates normally, even if the trigger is turned on. When the input of the trigger has a valid edge, the minimum delay for activating the output of the compare/capture 2 is 5 clock cycles. OC2FE only works when the channel is configured in PWM1 or PWM2 mode.	0
[9:8]	CC2S[1:0]	RW	Compare/capture 2 input selection. 00: The compare/capture 2 is configured as output; 01: Compare/capture 2 is configured as input, and IC2 is mapped on TI2; 10: Compare/capture 2 is configured as input, and IC2 is mapped on TI1; 11: Compare/capture 2 is configured as an input, and IC2 is mapped on TRC. This mode only works when the internal trigger input is selected (selected by the TS bit). *Note: Compare/capture 2 is only writable when the channel is switched off (CC2E is zero).*	00b
7	OC1CE	RW	Compare/capture channel 1 clear enable.	0
[6:4]	OC1M[2:0]	RW	Compare/capture channel 1 mode setting.	0
3	OC1PE	RW	Compare/capture register 1 preload enable.	0
2	OC1FE	RW	Compare/capture channel 1 fast enable.	0
[1:0]	CC1S[1:0]	RW	Compare/capture channel 1 input selection.	0

Capture mode (pin direction is input):

Bit	Name	Access	Description	Reset value
[15:12]	IC2F[3:0]	RW	Input capture2 filter configuration. These bits set the sampling frequency and digital filter length of TI1 input. The digital filter is composed of an event counter, in which N events are needed to validate a transition on the output. 0000: No filter, sampling is done at Fdts;	0000b

			1000: Fsampling=Fdts/8, N=6; 0001: Fsampling=Fck_int, N=2; 1001: Fsampling=Fdts/8, N=8; 0010: Fsampling=Fck_int， N=4; 1010: Fsampling=Fdts/16, N=5; 0011: Fsampling=f=Fck_int, N=8; 1011: Fsampling=Fdts/16, N=6; 0100: Fsampling=Fdts/2, N=6; 1100: Fsampling=Fdts/16, N=8; 0101: Fsampling=Fdts/2, N=8; 1101: Fsampling=Fdts/32, N=5; 0110: Fsampling=Fdts/4, N=6; 1110: Fsampling=Fdts/32, N=6; 0111: Fsampling=Fdts/4, N=8; 1111: Fsampling=Fdts/32, N=8.	
[11:10]	IC2PSC[1:0]	RW	Compare/capture channel 2 prescaler configuration. These 2 bits define the prescaler factor of compare/capture 2. Once CC1E=0, the prescaler will be reset. 00: Prescaler OFF, each edge detected on the capture input port triggers a capture; 01: Trigger a capture every 2 events; 10: Trigger a capture every 4 events; 11: Trigger a capture every 8 events.	00b
[9:8]	CC2S[1:0]	RW	Compare/capture channel 2 input selection. These 2 bits define the direction of the channel (input/output) and selection of input pins. 00: Compare/capture channel 1 is configured as output; 01: Compare/capture channel 1 is configured as input, and IC1 is mapped on TI1; 10: Compare/capture channel 1 is configured as input, and IC1 is mapped on TI2; 11: Compare/capture channel 1 is configured as an input, and IC1 is mapped on TRC. This mode only works when the internal trigger input is selected (selected by the TS bit). *Note: CC2S can be written only when the channel is closed (CC2E is 0).*	00b
[7:4]	IC1F[3:0]	RW	Input capture1 filter configuration.	0
[3:2]	IC1PSC[1:0]	RW	Compare/capture1 prescaler configuration.	0
[1:0]	CC1S[1:0]	RW	Compare/capture1 input selection.	0

15.4.8 Compare/Capture Control Register 2 (TIMx_CHCTLR2) (x=2/3/4/5)

Offset address: 0x1C

The channel can be used for input (capture mode) or output (comparison mode), and the direction of the channel is defined by the corresponding CCxS bit. The functions of other bits of this register are different in

input and output modes. OCxx describes the function of the channel in output mode, and ICxx describes the function of the channel in input mode.

15	14 13 12	11	10	9 8	7	6 5 4	3	2	1 0
OC4CE	OC4M[2:0]	OC4PE	OC4FE	CC4S[1:0]	OC3CE	OC3M[2:0]	OC3PE	OC3FE	CC3S[1:0]
IC4F[3:0]		IC4PSC[1:0]			IC3F[3:0]		IC3PSC[1:0]		

Compare mode (pin direction is output):

Bit	Name	Access	Description	Reset value
15	OC4CE	RW	Compare/capture channel 4 clear enable bit.	0
[14:12]	OC4M[2:0]	RW	Compare/capture channel 4 mode setting fields.	0
11	OC4PE	RW	Compare/capture register 4 preload enable bit.	0
10	OC4FE	RW	Compare/capture channel 4 fast enable bit.	0
[9:8]	CC4S[1:0]	RW	Compare/capture channel 4 input selection fields.	0
7	OC3CE	RW	Compare/capture channel 3 clear enable bit.	0
[6:4]	OC3M[2:0]	RW	Compare/capture channel 3 mode setting fields.	0
3	OC3PE	RW	Compare/capture register 3 preload enable bit.	0
2	OC3FE	RW	Compare/capture channel 3 fast enable bit.	0
[1:0]	CC3S[1:0]	RW	Compare/capture channel 3 input selection fields.	0

Capture mode (pin direction is input):

Bit	Name	Access	Description	Reset value
[15:12]	IC4F[3:0]	RW	Input capture filter 4 configuration fields.	0
[11:10]	IC4PSC[1:0]	RW	Compare/capture channel 4 prescale configuration fields.	0
[9:8]	CC4S[1:0]	RW	Compare/capture channel 4 input selection fields.	0
[7:4]	IC3F[3:0]	RW	Input capture filter 3 configuration fields.	0
[3:2]	IC3PSC[1:0]	RW	Compare/capture channel 3 prescale configuration fields.	0
[1:0]	CC3S[1:0]	RW	Compare/capture channel 3 input selection fields.	0

15.4.9 Compare/Capture Enable Register (TIMx_CCER) (x=2/3/4/5)

Offset address: 0x20

15 14	13	12	11 10	9	8	7 6	5	4	3 2	1	0
Reserved	CC4P	CC4E	Reserved	CC3P	CC3E	Reserved	CC2P	CC2E	Reserved	CC1P	CC1E

Bit	Name	Access	Description	Reset value
[15:14]	Reserved	RO	Reserved	0
13	CC4P	RW	Compare/capture channel 4 output polarity	0
12	CC4E	RW	Compare/capture channel 4 output enable	0
[11:10]	Reserved	RO	Reserved	0
9	CC3P	RW	Compare/capture channel 3 output polarity	0
8	CC3E	RW	Compare/capture channel 3 output enable	0

[7:6]	Reserved	RO	Reserved	0
5	CC2P	RW	Compare/capture channel 2 output polarity	0
4	CC2E	RW	Compare/capture channel 2 output enable	0
[3:2]	Reserved	RO	Reserved	0
1	CC1P	RW	Compare capture channel 1 output polarity setting bit. CC1 channel is configured as output: 1: OC1 active low; 0: OC1 active high. CC1 channel is configured as input: This bit selects whether IC1 or the inverted signal of IC1 is used as the trigger or capture signal. 1: Inverting: Capture occurs on the falling edge of IC1; IC1 is inverted when used as an external trigger. 0: Non-inverting: Capture occurs on the rising edge of IC1; IC1 is non-inverted when used as an external trigger.	0
0	CC1E	RW	Compare/capture channel 1 output enable bit CC1 channel is configured as output: 1: Enable: OC1 signal output to the corresponding output pin; 0: Disable: OC1 output is inhibited. CC1 channel is configured as input: This bit determines whether the counter value can be captured into the TIMx_CH1CVR register. 1: Capture enabled; 0: Capture disabled.	0

15.4.10 Counter of General-purpose Timer (TIMx_CNT) (x= 2/3/4/5)

Offset address: 0x24

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CNT[15:0]	RW	Real-time value of timer counter.	0

15.4.11 Count Clock Prescaler (TIMx_PSC) (x=2/3/4/5)

Offset address: 0x28

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	PSC[15:0]	RW	The frequency division factor of the timer's prescaler; the clock frequency of the counter is equal to the input frequency of the frequency divider/(PSC+1).	0

15.4.12 Auto-reload Value Register (TIMx_ATRLR) (x=2/3/4/5)

Offset address: 0x2C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	ARR[15:0]	RW	Counter thresholds, when ATRLR is active and updated read Section 15.2.4; when ATRLR is empty, the counter stops.	0xFFFF

15.4.13 Compare/Capture Register 1 (TIMx_CH1CVR) (x=2/3/4/5)

Offset address: 0x34

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CCR1[15:0]	RW	The value of compare capture register channel 1.	0

15.4.14 Compare/Capture Register 2 (TIMx_CH2CVR) (x=2/3/4/5)

Offset address: 0x38

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CCR2[15:0]	RW	The value of compare capture register channel 2.	0

15.4.15 Compare/Capture Register 3 (TIMx_CH3CVR) (x=2/3/4/5)

Offset address: 0x3C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR3[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CCR3[15:0]	RW	The value of compare capture register channel 3.	0

15.4.16 Compare/Capture Register 4 (TIMx_CH4CVR) (x=2/3/4/5)

Offset address: 0x40

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	CCR4[15:0]	RW	The value of compare capture register channel 4.	0

15.4.17 DMA Control Register (TIMx_DMACFGR) (x=2/3/4/5)

Offset address: 0x48

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DBL[4:0]					Reserved			DBA[4:0]				

Bit	Name	Access	Description	Reset value
[15:13]	Reserved	RO	Reserved	0
[12:8]	DBL[4:0]	RW	Length of data that DMA continuously transfers; the actual value is the value of this domain + 1.	0
[7:5]	Reserved	RO	Reserved.	0
[4:0]	DBA[4:0]	RW	These bits define the offset of DMA from the address of control register 1 in continuous mode.	0

15.4.18 DMA Address Register in Continuous Mode (TIMx_DMAADR) (x=2/3/4/5)

Offset address: 0x4C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAB[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	DMAB[15:0]	RW	DMA address in continuous mode.	0

15.4.19 Dual-edge Capture Register (TIMx_AUX) (x=2/3/4/5)

Offset address: 0x50

Bit	Name	Access	Description	Reset value
[15:3]	Reserved	R0	Reserved	0
2	CAP_ED_CH4	RW	Dual-edge capture enable for Channel 4: 1: Enable dual-edge capture for Channel 4; 0: Disable dual-edge capture for Channel 4.	0
1	CAP_ED_CH3	RW	Dual-edge capture enable for Channel 3: 1: Enable dual-edge capture for Channel 3; 0: Disable dual-edge capture for Channel 3.	0
0	CAP_ED_CH2	RW	Dual-edge capture enable for Channel 2: 1: Enable dual-edge capture for Channel 2; 0: Disable dual-edge capture for Channel 2.	0

Chapter 16 Basic Timer (BCTM)

The basic timer module comprises two 16-bit auto-reloadable timers (TIM6 and TIM7) for counting and generating interrupts upon update events or DMA requests. TIM6 and TIM7 support 16-bit programmable prescalers. They can provide clocks for digital-to-analog conversion (DAC) and trigger DAC synchronization circuits. The basic timers operate independently and do not share any resources.

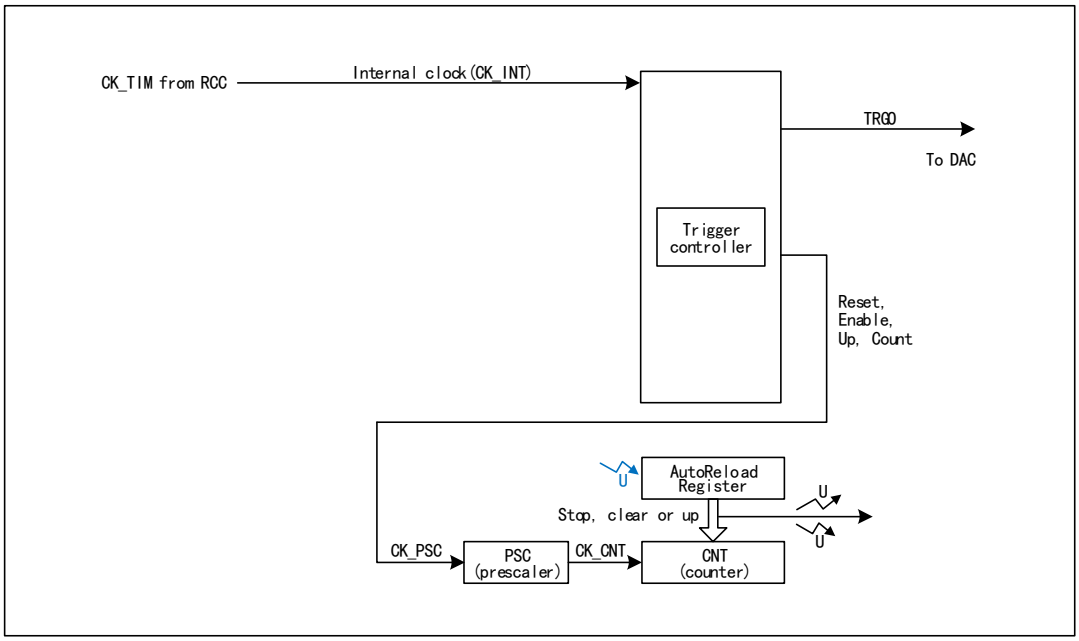
16.1 Main Features

Main features of the basic timer include:

- 16-bit auto-reload counter supporting increment mode
- 16-bit prescaler with dynamically adjustable division factor from 1 to 65536
- Triggered DAC synchronization circuit
- Generates interrupts or DMA requests upon update events

16.2 Principle and Structure

Figure 16-1 Block diagram of the structure of the basic timer



16.2.1 Overview

As shown in Figure 16-1, the structure of the basic timer can be broadly divided into 2 parts: the input clock section and the core counter section.

The clock for the basic timer originates from the HB bus clock (CK_INT). These input clock signals undergo various operations such as filtering and frequency division to become the CK_PSC clock, which is then output to the core counter section. Additionally, these complex clock sources can also be output as TRGO to the DAC peripheral.

The core of the basic timer is a 16-bit counter (CNT). After CK_PSC is divided by the prescaler (PSC), it becomes CK_CNT and is finally lost to CNT. CNT supports increment counting mode and has an automatic

reload value register (ATRLR) to reload the initialization value for CNT after the end of each counting cycle.

16.2.2 Difference between Basic Timer and General-purpose Timer

Compared with the general timer, the basic timer lacks the following functions:

1) The basic timer lacks down-counting mode and up-down counting mode.
2) The basic timer lacks 4-channel independent compare capture channels.
3) The basic timer does not support external signal control timer.
4) The basic timer does not support incremental coding, cascade and synchronization between timers.

16.2.3 Clock Input

The clock of the basic timer is provided by the internal clock CK_INT.

16.2.4 Counter and Periphery

CK_PSC is input to the prescaler (PSC) for frequency division. PSC is 16 bits, and the actual frequency division coefficient is equivalent to the value of R16_TIMx_PSC + 1. CK_PSC will become CK_CNT after passing through PSC. Changing the value of R16_TIMx_PSC will not take effect in real time, but will be updated to PSC after the update event. Update events include clearing and resetting the UG bit.

16.3 Debug Mode

When the system enters debug mode, the timer continues to run or stops according to the setting of the DBG module.

16.4 Register Description

Table 16-1 TIM6-related registers

Name	Offset address	Description	Reset value
R16_TIM6_CTLR1	0x40001000	TIM6 control register 1	0x0000
R16_TIM6_CTLR2	0x40001004	TIM6 control register 2	0x0000
R16_TIM6_DMAINTENR	0x4000100C	TIM6 DMA/interrupt enable register	0x0000
R16_TIM6_INTFR	0x40001010	TIM6 interrupt status register	0x0000
R16_TIM6_SWEVGR	0x40001014	TIM6 event generation register	0x0000
R16_TIM6_CNT	0x40001024	TIM6 counter	0x0000
R16_TIM6_PSC	0x40001028	TIM6 count clock prescaler	0x0000
R16_TIM6_ATRLR	0x4000102C	TIM6 auto-reload value register	0xFFFF

Table 16-2 TIM7-related registers

Name	Offset address	Description	Reset value
R16_TIM7_CTLR1	0x40001400	TIM7 control register 1	0x0000
R16_TIM7_CTLR2	0x40001404	TIM7 control register 2	0x0000
R16_TIM7_DMAINTENR	0x4000140C	TIM7 DMA/interrupt enable register	0x0000
R16_TIM7_INTFR	0x40001410	TIM7 interrupt status register	0x0000
R16_TIM7_CNT	0x40001424	TIM7 counter	0x0000

R16_TIM7_PSC	0x40001428	TIM7 count clock prescaler	0x0000
R16_TIM7_ATRLR	0x4000142C	TIM7 auto-reload value register	0xFFFF

16.4.1 Control Register 1 (TIMx_CTLR1) (x=6/7)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
7	ARPE	RW	Auto-reload and preload enable: 1: Auto-reload value register (ATRLR) enabled; 0: Auto-reload value register (ATRLR) disabled.	0
[6:4]	Reserved	RO	Reserved	0
3	OPM	RW	Single pulse mode. 1: The counter stops when the next update event (clearing the CEN bit) occurs; 0: The counter does not stop when the next update event occurs.	0
2	URS	RW	Update Request Source: The software uses this bit to select the source of UEV events. 1: If update interrupts or DMA requests are enabled, only counter overflows/underflows generate update interrupts or DMA requests; 0: If update interrupts or DMA requests are enabled, any of the following events generate update interrupts or DMA requests: -Counter overflow/underflow. -Set the UG bit. -Updates generated from the mode controller.	0
1	UDIS	RW	Updates are prohibited, and the software allows/ disables the generation of UEV events through this bit. 1: UEV is prohibited. No update events are generated, and the registers (ATRLR, PSC, CHCTLRx) hold their values. If the UG bit is set or a hardware reset is issued from the mode controller, the counter and prescaler are reinitialized. 0: UEV is allowed. The UEV event is generated by any of the following events: -Counter overflow/underflow. -Set the UG bit. -Updates generated from the mode controller. Registers with caches are loaded into their preloaded values.	0
0	CEN	RW	Counter enable: 1: Enable counter. 0: Disables counters.	0

16.4.2 Control Register 2 (TIMx_CTLR2) (x=6/7)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[15:7]	Reserved	RO	Reserved	0
[6:4]	MMS[2:0]	RW	Master mode selection: These 3 bits are used to select the synchronization information (TRGO) sent to the slave timer in the master mode. The possible combination is as follows: 000: Reset – The UG bit is used as a trigger output (TRGO). If it is a reset generated by a trigger input (the slave mode controller is in reset mode), the signal on TRGO will have a delay relative to the actual reset; 001: Enable-the counter enables signal CNT_EN to be used as a trigger output (TRGO). Sometimes, it is necessary to start multiple timers at the same time or control to enable slave timers within a period of time. The counter enable signal is generated by the logical OR of the CEN control bit and the trigger input signal in the gating mode. When the counter enable signal is controlled by the trigger input, there will be a delay on TRGO, unless the master/slave mode is selected (see the description of the MSM bit in the TIMx_SMCFGR register); 010: An update event is selected as the trigger output (TRGO). For example, the clock of a master timer can be used as a prescaler for a slave timer.	0
[3:0]	Reserved	RO	Reserved.	0

16.4.3 DMA/Interrupt Enable Register (TIMx_DMAINTENR) (x=6/7)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[15:9]	Reserved	RO	Reserved	0
8	UDE	RW	Update DMA request enable. 1: Update DMA request enabled; 0: Update DMA request disabled.	0
[7:1]	Reserved	RO	Reserved	0
0	UIE	RW	Update interrupt enable. 1: Update interrupt enabled; 0: Update interrupt disabled.	0

16.4.4 Interrupt Flag Register (R16_TIMx_INTFR) (x=6/7)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[15:1]	Reserved	RO	Reserved	0
0	UIF	RW0	Update interrupt flag. When an update event occurs, it is	0

			set by hardware and cleared by software. 1: Update interrupt generated; The update event generates in case of the following circumstances: For UDIS=0, when the repeated counter value overflows or underflows; For URS=0, UDIS=0, when the UG bit is set, or when the counter core is reinitialized by software; 0: No update interrupt generated.	

16.4.5 Event Generation Register (TIMx_SWEVGR) (x=6/7)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[15:1]	Reserved	RO	Reserved.	0
0	UG	WO	Update event generation. This bit is set by software and cleared automatically by hardware. 1: Initialize the counter and generate an update event; 0: No action. *Note: The counter of the prescaler is also cleared, but the prescaler factor remains unchanged. In up-counting mode, the core counter will be cleared.*	0

16.4.6 Counter of General-purpose Register (TIMx_CNT) (x=6/7)

Offset address: 0x24

Bit	Name	Access	Description	Reset value
[15:0]	CNT[15:0]	RW	The real-time value of the counter of the timer.	0

16.4.7 Count Clock Prescaler (TIMx_PSC) (x=6/7)

Offset address: 0x28

Bit	Name	Access	Description	Reset value
[15:0]	PSC[15:0]	RW	The frequency division factor of the timer's prescaler; the clock frequency of the counter is equal to the input frequency of the frequency divider/(PSC+1).	0

16.4.8 Auto-reload Value Register (TIMx_ATRLR) (x=6/7)

Offset address: 0x2C

Bit	Name	Access	Description	Reset value
[15:0]	ARR[15:0]	RW	The value of ATRLR[15:0] will be loaded into the counter. Please read Section 16.2.4 when ATRLR acts and updates; when ATRLR is empty, the counter stops.	0xFFFF

Chapter 17 Universal Synchronous Asynchronous Receiver Transmitter (USART)

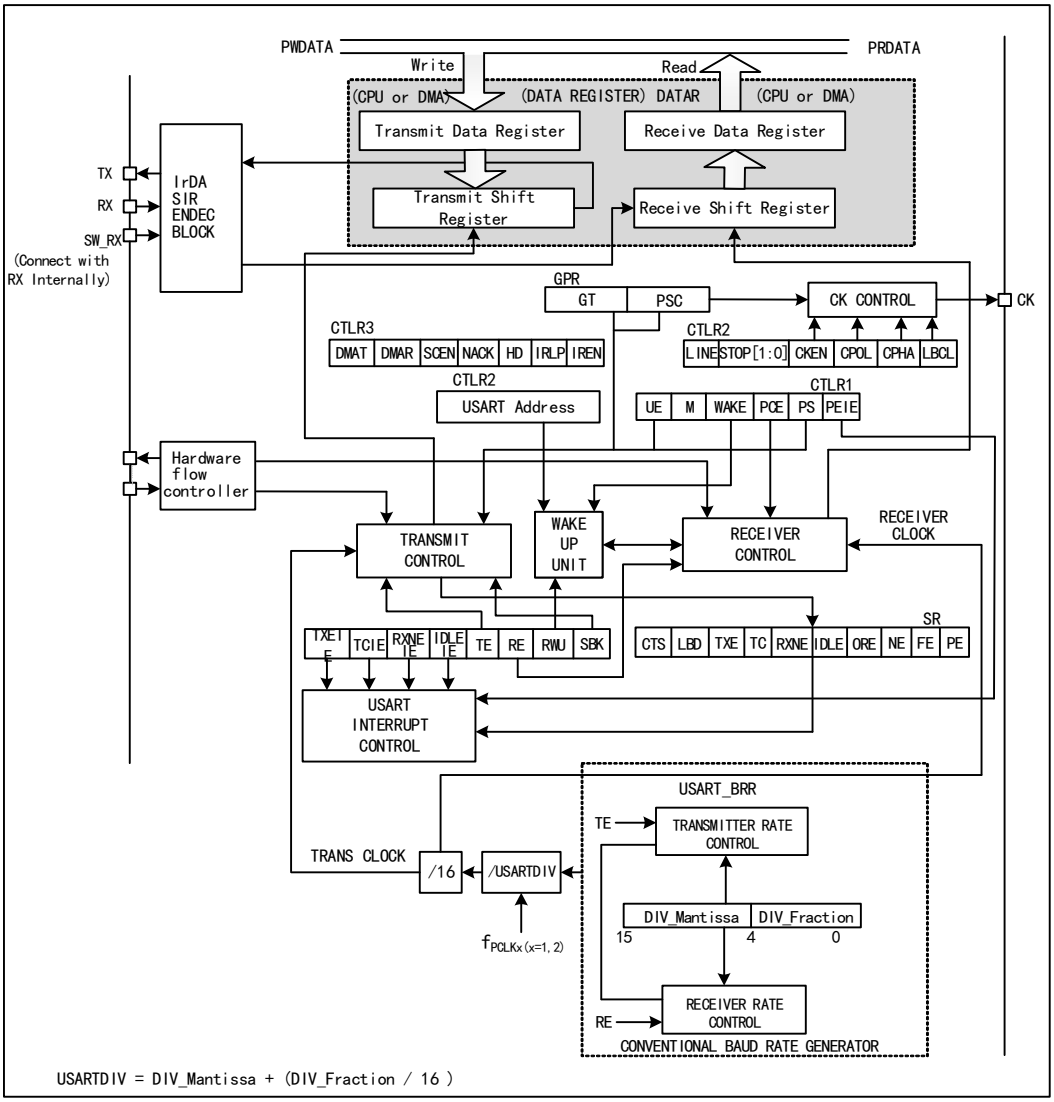
The module contains 8 universal synchronous asynchronous transceivers (USART1/2/3/4/5/6/7/8/9/10).

17.1 Main Features

- Full-duplex or half-duplex synchronous or asynchronous communication
- NRZ data format
- Fractional baud rate generator, highest 12.5Mbps
- Programmable data length
- Configurable stop bit
- Support LIN, IrDA encoders, smart cards
- Support DMA
- Multiple interrupt sources

17.2 Overview

Figure 17-1 USART block diagram



When TE (transmission enable bit) is set, the data in the transmitter shift register will be outputted on the TX pin, and the clock will be outputted on the CK pin. During transmission, the lowest significant bit is the first to be shifted out. Each data frame starts with a low-level start bit, and then the transmitter sends 8-bit data or 9-bit data according to the setting of the M (word length) bit, and finally a configurable number of stop bits. If there is a parity check bit, the last bit of the data word is the check bit. After TE is set, an idle frame is sent. The idle frame is 10-bit or 11-bit high level, including the stop bit. The break frame is a 10-bit or 11-bit low level, followed by a stop bit.

17.3 Baud Rate Generator

The baud rate of the transceiver = F_{CLK}/(16*USARTDIV); F_{CLK} is the clock of PBx, i.e., PCLK1 or PCLK2, PCLK2 is used for the USART1 module, and PCLK1 shall be used for the rest. The value of USARTDIV is determined according to the 2 domains: DIV_M and DIV_F in USART_BRR. The specific calculation formula is:

$$USARTDIV = DIV_M+(DIV_F/16)$$

It should be noted that the baud rate generated by the baud rate generator may not always generate just the baud rate required by the user, which may be biased. In addition to taking the value as close as possible, the method to reduce the deviation can also be to increase the HB clock. For example, when setting the baud rate to 9600bps, the USARTDIV value is set to 78.125. At 12MHz, this yields a baud rate of 9600 bps. However, if you require a baud rate of 115200bps, the calculated USARTDIV value is 6.51. but the closest value that can be entered into USART_BRR is 6.5. The resulting baud rate is 115384bps, resulting in an error of 0.16%.

When the serial port waveform sent by the transmitter is transmitted to the receiver, there is a certain error in the baud rate between the receiver and the sender. The error mainly comes from 3 aspects: the actual baud rate of the receiver and the sender are inconsistent; the clocks of the receiver and the sender have errors; the waveform changes in the circuit. The receiver of the peripheral module has a certain tolerance for receiving. When the sum of the total deviations generated in the above 3 aspects is less than the tolerance limit of the module, the total deviation will not affect the receiving and sending. The tolerance limit of the module is affected by the use of fractional baud rate and M bit (data field word length) or not. The use of fractional baud rate and the use of 9-bit data field length will reduce the tolerance limit, but it shall not be less than 3%.

17.4 Synchronous Mode

The synchronous mode enables the system to output clock signals when the USART module is used. When the synchronous mode is enabled to send data externally, the CK pin will output clock externally at the same time.

To enable synchronous mode, set CLKEN bit in the control register2 (R16_USARTx_CTLR2), but you need to switch off the LIN mode, smart card mode, infrared mode and half-duplex mode at the same time, i.e., to ensure that the LINEN(R16_USARTx_CTLR2), SCEN, HDSEL and IREN bits are in the reset status. These 3 bits are in the control register 3 (R16_USARTx_CTLR3).

The main point of the synchronous mode is the output control of the clock. Attention shall be paid to the following:
The synchronous mode of the USART module only works in the master mode, i.e., the CK pin only outputs the clock and does not receive input;
The clock signal is outputted only when TX pin outputs data;

The LBCL bit determines whether the clock is outputted when the last data bit is sent. The CPOL bit determines the polarity of the clock, and the CPHA determines the phase position of the clock. These 3 bits are in the control register 2 (R16_USARTx_CTLR2). These 3 bits need to be set when TE and RE are not enabled. The specific difference is shown in Figure 17-2.

In the synchronous mode, the receiver will only sample when outputting the clock, and the slave needs to maintain a certain signal setup time and hold time, specifically as shown in Figure 17-3.

Figure 17-2 Example of USART clock timing (M=0)

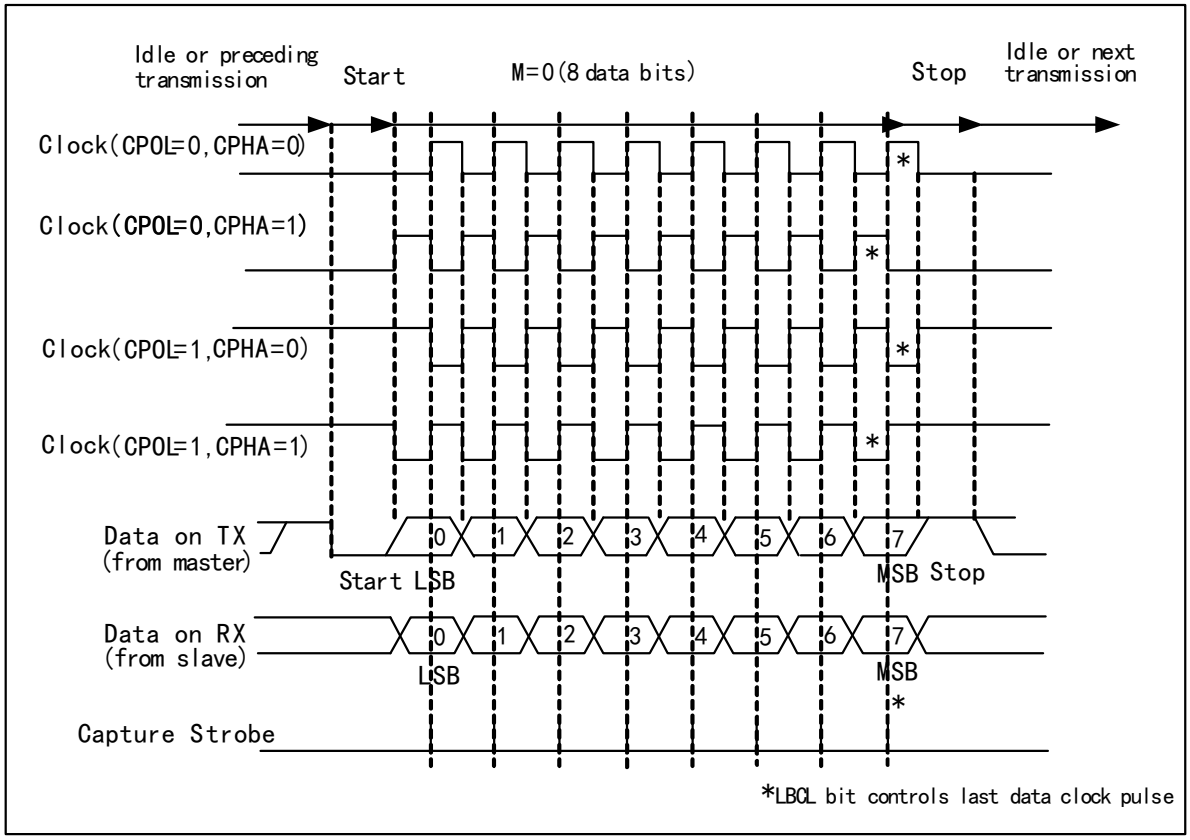
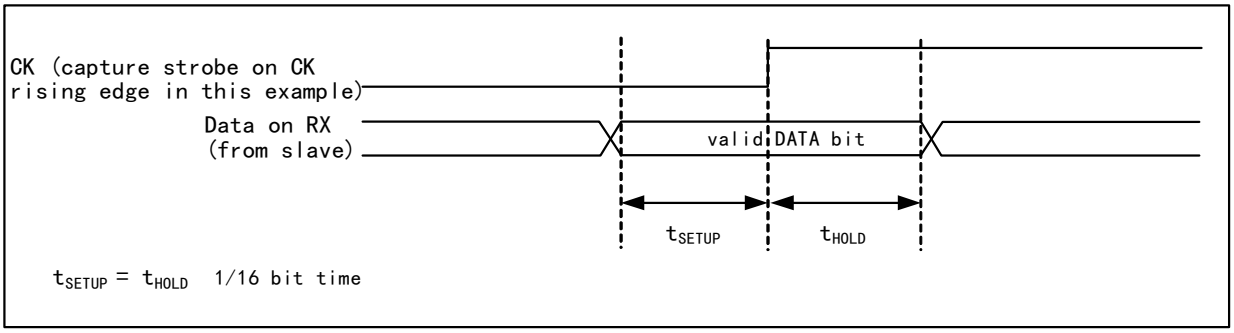


Figure 17-3 Data sample hold time



17.5 1-wire Half-duplex Mode

The half-duplex mode supports the use of a single pin (only TX pin) to receive and transmit, and the TX pin and RX pin are connected inside the chip.

To enable half-duplex mode, set HDSEL bit in the control register 3 (R16_USARTx_CTLR3), but you need to disable LIN mode, smartcard mode, infrared mode and synchronous mode at the same time, i.e., to ensure that the SCEN, CLKEN and IREN bits are in the reset status. These 3 bits are in the control register 2 and the control register 3 (R16_USARTx_CTLR2 and R16_USARTx_CTLR3).

After setting to half-duplex mode, it is needed to set the TX IO port to open-drain output high mode. When TE bit is set, the data will be sent out as long as the data is written to the data register. Special attention shall be paid to the fact that bus conflicts may occur when multiple devices use a single bus to transmit and receive in half-duplex mode. This requires users to avoid it by software.

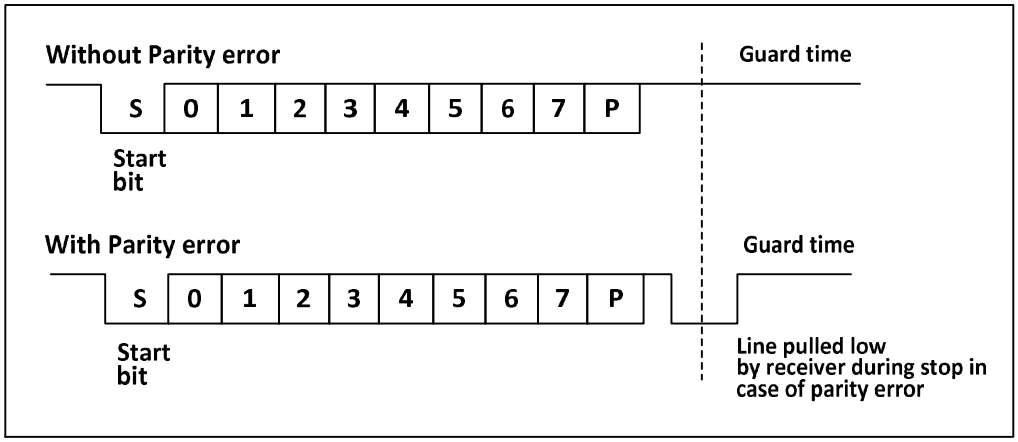
17.6 Smart Card

The smartcard mode supports ISO7816-3 protocol to access the smart card controller.

To enable smartcard mode, set the SCEN bit in the control register 3 (R16_USARTx_CTLR3), but it is needed to disable LIN mode, half-duplex mode and infrared mode at the same time, i.e., to ensure that the LINEN, HDSEL and IREN bits are in the reset status, but CLKEN can be switched on to output the clock. These 3 bits are in the control register 2 and the control register 3 (R16_USARTx_CTLR2 and R16_USARTx_CTLR3).

In order to support smartcard mode, USART shall be set to 8 data bits plus 1 check bit. It is recommended that the stop bit be configured to 1.5 bits for both sending and receiving. The smart card mode is a 1-wire half-duplex protocol, which uses TX line as the data communication and shall be configured as open drain output plus pull-up. When the receiver receives a frame of data and detects a parity check error, it will send a NACK signal at the stop bit, i.e., actively reducing a cycle of TX during the stop bit. After the sender detects the NACK signal, a frame error will be generated, and the application can resend accordingly. Figure 17-4 shows the waveforms on the TX pin under correct conditions and in the event of parity check errors. The TC flag (transmission completion flag) of the USART can delay the generation of GT (protection time) clocks, and the receiver will not recognize the NACK signal set by itself as the start bit.

Figure 17-4 (No) parity check error



In smartcard mode, the output waveform after the CK pin is enabled has nothing to do with the communication. It only provides the clock for the smart card. Its value is the PB clock and then the 5-bit settable clock frequency division (the frequency division value is double of PSC, and the highest is frequency division 62).

17.7 IrDA

USART module supports control IrDA infrared transceiver for physical layer communication. To use IrDA,

the LINEN, STOP, CLKEN, SCEN and HDSEL bits must be cleared. NRZ (non-return-to-zero) coding is used between the USART module and the SIR physical layer (infrared transceiver), and the maximum support rate is 115200bps.

IrDA is a half-duplex protocol. If UASRT sends data to the SIR physical layer, the IrDA decoder ignores the newly sent infrared signal. If the USART receives data from SIR, then SIR does not accept USART signal. The level logic sent by USART to SIR and SIR to USART is different. In SIR receive logic, '1' represents high level and '0' represents low level. However, in the SIR transmit logic, '0' represents high level and '1' represents low level.

17.8 DMA

The USART module supports DMA, and can use DMA to implement fast continuous reception and transmission. When DMA is enabled and the TXE bit is set, DMA writes data to the transmit buffer from the set memory space. When DMA is used for reception, DMA transfers the data in the receive buffer to a specific memory space each time the RXNE bit is set.

17.9 Interrupt

The USART module supports multiple interrupt sources, including transmit data register empty (TXE), CTS, transmission complete (TC), received data ready (RXNE), data overrun error (ORE), idle line (IDLE), parity check error (PE), break flag (LBD), noise (NE), multi-buffer communication overrun (ORT) and framing error (FE).

Table 17-1 Interrupts and the corresponding enable bits

Interrupt source	Enable bit
Transmit data register empty (TXE)	TXEIE
Transmission allowed (CTS)	CTSIE
Transmission complete (TC)	TCIE
Received data ready (RXNE)	RXNEIE
Transmit data register empty (TXE)	
Idle line (IDLE)	IDLEIE
Parity error (PE)	PEIE
Break flag (LBD)	LBDIE
Noise (NE)	EIE
Overrun error in multi-buffer communication (ORT)	
Framing error (FE) in multi-buffer communication	

17.10 Register Description

Table 17-2 USART1-related registers

Name	Access address	Description	Reset value
R32_USART1_STATR	0x40013800	UASRT1 status register	0x000000C0
R32_USART1_DATAR	0x40013804	UASRT1 data register	0x000000XX

R32_USART1_BRR	0x40013808	UASRT1 baud rate register	0x00000000
R32_USART1_CTLR1	0x4001380C	UASRT1 control register1	0x00000000
R32_USART1_CTLR2	0x40013810	UASRT1 control register2	0x00000000
R32_USART1_CTLR3	0x40013814	UASRT1 control register3	0x00000000
R32_USART1_GPR	0x40013818	UASRT1 guard time and prescaler register	0x00000000
R32_USART1_CTLR4	0x4001381C	UASRT1 control register4	0x00000000

Table 17-3 USART2-related registers

Name	Access address	Description	Reset value
R32_USART2_STATR	0x40004400	UASRT2 status register	0x000000C0
R32_USART2_DATAR	0x40004404	UASRT2 data register	0x000000XX
R32_USART2_BRR	0x40004408	UASRT2 baud rate register	0x00000000
R32_USART2_CTLR1	0x4000440C	UASRT2 control register1	0x00000000
R32_USART2_CTLR2	0x40004410	UASRT2 control register2	0x00000000
R32_USART2_CTLR3	0x40004414	UASRT2 control register3	0x00000000
R32_USART2_GPR	0x40004418	UASRT2 guard time and prescaler register	0x00000000
R32_USART2_CTLR4	0x4000441C	UASRT2 control register4	0x00000000

Table 17-4 USART3-related registers

Name	Access address	Description	Reset value
R32_USART3_STATR	0x40004800	UASRT3 status register	0x000000C0
R32_USART3_DATAR	0x40004804	UASRT3 data register	0x000000XX
R32_USART3_BRR	0x40004808	UASRT3 baud rate register	0x00000000
R32_USART3_CTLR1	0x4000480C	UASRT3 control register1	0x00000000
R32_USART3_CTLR2	0x40004810	UASRT3 control register2	0x00000000
R32_USART3_CTLR3	0x40004814	UASRT3 control register3	0x00000000
R32_USART3_GPR	0x40004818	UASRT3 guard time and prescaler register	0x00000000
R32_USART3_CTLR4	0x4000481C	UASRT3 control register4	0x00000000

Table 17-5 USART4-related registers

Name	Access address	Description	Reset value
R32_USART4_STATR	0x40004C00	UASRT4 status register	0x000000C0
R32_USART4_DATAR	0x40004C04	UASRT4 data register	0x000000XX
R32_USART4_BRR	0x40004C08	UASRT4 baud rate register	0x00000000
R32_USART4_CTLR1	0x40004C0C	UASRT4 control register1	0x00000000
R32_USART4_CTLR2	0x40004C10	UASRT4 control register2	0x00000000
R32_USART4_CTLR3	0x40004C14	UASRT4 control register3	0x00000000
R32_USART4_GPR	0x40004C18	UASRT4 guard time and prescaler register	0x00000000
R32_USART4_CTLR4	0x40004C1C	UASRT4 control register4	0x00000000

Table 17-6 USART5-related registers

Name	Access address	Description	Reset value
R32_USART5_STATR	0x40005000	UASRT5 status register	0x000000C0
R32_USART5_DATAR	0x40005004	UASRT5 data register	0x00000000
R32_USART5_BRR	0x40005008	UASRT5 baud rate register	0x00000000
R32_USART5_CTLR1	0x4000500C	UASRT5 control register1	0x00000000
R32_USART5_CTLR2	0x40005010	UASRT5 control register2	0x00000000
R32_USART5_CTLR3	0x40005014	UASRT5 control register3	0x00000000
R32_USART5_GPR	0x40005018	UASRT5 guard time and prescaler register	0x00000000
R32_USART5_CTLR4	0x4000501C	UASRT5 control register4	0x00000000

Table 17-7 USART6-related registers

Name	Access address	Description	Reset value
R32_USART6_STATR	0x40001800	UASRT6 status register	0x000000C0
R32_USART6_DATAR	0x40001804	UASRT6 data register	0x00000000
R32_USART6_BRR	0x40001808	UASRT6 baud rate register	0x00000000
R32_USART6_CTLR1	0x4000180C	UASRT6 control register1	0x00000000
R32_USART6_CTLR2	0x40001810	UASRT6 control register2	0x00000000
R32_USART6_CTLR3	0x40001814	UASRT6 control register3	0x00000000
R32_USART6_GPR	0x40001818	UASRT6 guard time and prescaler register	0x00000000
R32_USART6_CTLR4	0x4000181C	UASRT6 control register4	0x00000000

Table 17-8 USART7-related registers

Name	Access address	Description	Reset value
R32_USART7_STATR	0x40001C00	UASRT7 status register	0x000000C0
R32_USART7_DATAR	0x40001C04	UASRT7 data register	0x00000000
R32_USART7_BRR	0x40001C08	UASRT7 baud rate register	0x00000000
R32_USART7_CTLR1	0x40001C0C	UASRT7 control register1	0x00000000
R32_USART7_CTLR2	0x40001C10	UASRT7 control register2	0x00000000
R32_USART7_CTLR3	0x40001C14	UASRT7 control register3	0x00000000
R32_USART7_GPR	0x40001C18	UASRT7 guard time and prescaler register	0x00000000
R32_USART7_CTLR4	0x40001C1C	UASRT7 control register4	0x00000000

Table 17-9 USART8-related registers

Name	Access address	Description	Reset value
R32_USART8_STATR	0x40002000	UASRT8 status register	0x000000C0
R32_USART8_DATAR	0x40002004	UASRT8 data register	0x00000000
R32_USART8_BRR	0x40002008	UASRT8 baud rate register	0x00000000

R32_USART8_CTLR1	0x4000200C	UASRT8 control register1	0x00000000
R32_USART8_CTLR2	0x40002010	UASRT8 control register2	0x00000000
R32_USART8_CTLR3	0x40002014	UASRT8 control register3	0x00000000
R32_USART8_GPR	0x40002018	UASRT8 guard time and prescaler register	0x00000000
R32_USART8_CTLR4	0x4000201C	UASRT8 control register4	0x00000000

Table 17-10 USART9-related registers

Name	Access address	Description	Reset value
R32_USART9_STATR	0x40007800	USART9 status register	0x000000C0
R32_USART9_DATAR	0x40007804	UASRT9 data register	0x00000000
R32_USART9_BRR	0x40007808	UASRT9 baud rate register	0x00000000
R32_USART9_CTLR1	0x4000780C	UASRT9 control register1	0x00000000
R32_USART9_CTLR2	0x40007810	UASRT9 control register2	0x00000000
R32_USART9_CTLR3	0x40007814	UASRT9 control register3	0x00000000
R32_USART9_GPR	0x40007818	UASRT9 guard time and prescaler register	0x00000000
R32_USART9_CTRL4	0x4000781C	UASRT9 control register4	0x00000000

Table 17-11 USART10-related registers

Name	Access address	Description	Reset value
R32_USART10_STATR	0x40007C00	USART10 status register	0x000000C0
R32_USART10_DATAR	0x40007C04	UASRT10 data register	0x00000000
R32_USART10_BRR	0x40007C08	UASRT10 baud rate register	0x00000000
R32_USART10_CTLR1	0x40007C0C	UASRT10 control register1	0x00000000
R32_USART10_CTLR2	0x40007C10	UASRT10 control register2	0x00000000
R32_USART10_CTLR3	0x40007C14	UASRT10 control register3	0x00000000
R32_USART10_GPR	0x40007C18	UASRT10 guard time and prescaler register	0x00000000
R32_USART10_CTRL4	0x40007C1C	UASRT10 control register4	0x00000000

17.10.1 USART Status Register (USARTx_STATR) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				MS_ERR	RX_BUSY	CTS	LBD	TXE	TC	RXNE	IDLE	ORE	NE	FE	PE

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
11	MS_ERR	RO	MARK or SPACE check error flag. In receive mode,	0

			if there is a MARK or SPACE check error, the hardware sets this bit. Reading this bit and then reading the data register will reset this bit. If MS_ERRIE has been set before, setting this bit will generate corresponding interrupt.	
10	RX_BUSY	RO	Receive status indicator bit: 1: Set when operating in receive mode; 0: Unset when not in receive mode.	0
9	CTS	RW0	CTS status change flag. If the CTSE bit is set, when the nCTS output state changes, the bit will be set high by the hardware. Zero is cleared by the software. If the CTSIE bit has been set, an interrupt occurs. 1: There is a change in the nCTS state line. 0: There is no change on the nCTS state line.	0
8	LBD	RW0	LIN Break detection flag. When LIN Break is detected, this bit will be set by hardware. It is cleared by the software. If the LBDIE bit has been set, an interrupt will be generated. 1: LIN Break detected; 0: No LIN Break detected.	0
7	TXE	RO	Transmission data register empty flag. When data in TDR register is transferred to shift register by hardware, this bit will be set by hardware. If TXEIE bit has been set, an interrupt will be generated, the data register will be written and this bit will be reset. 1: Data is transferred to the shift register; 0: Data is not transferred to the shift register.	1
6	TC	RW0	Transmission complete flag. When a frame containing data is sent and TXE bit is set, the hardware will set this bit. If TCIE is set, a corresponding interrupt will be generated. The software will read this bit and then write the data register to clear this bit. You can also directly write 0 to clear this bit. 1: Transmission completed; 0: Transmission not completed.	1
5	RXNE	RW0	Read data register not empty flag. When the data in the shift register is transferred to the data register, this bit will be set by the hardware. If the RXNEIE bit has been set, the corresponding interrupt will be generated. This bit can be cleared by the write operation of the data register. This bit can be also cleared by directly writing 0. 1: The data is received and can be read; 0: The data is not received.	0
4	IDLE	RO	Idle line flag. When an idle line is detected, the bit	0

			will be set by hardware. If IDLEIE bit has been set, the corresponding interrupt will be generated. This bit can be cleared by reading the status register and then reading the data register. 1: The bus is idle now; 0: Idle bus is not detected. *Note: This bit will not be set again until RXNE is set.*	
3	ORE	RO	Overrun error flag. When the receiving shift register has data that needs to be transferred to the data register, but this bit will be set when there is still data that has not been read in the receiving field of the data register. If the RXNEIE bit is set, the corresponding interrupt will be generated. 1: The overrun error has occurred; 0: No overrun error has occurred. *Note: When an overrun error occurs, the value of the data register will not be lost, but the value of the shift register will be overwritten. If the EIE bit is set, the ORE flag bit will generate an interrupt in the multi-buffer communication mode.*	0
2	NE	RO	Noise error flag. When the noise error flag is detected, it will be set by hardware. This bit can be reset by reading the status register and then reading the data register. 1: The noise is detected; 0: No noise is detected. *Note: This bit will not generate the interrupt. If the EIE bit has been set, the FE flag bit will generate an interrupt in the multi-buffer communication mode.*	0
1	FE	RO	Frame error flag. When a synchronization error, excessive noise or disconnection is detected, this bit will be set by hardware. This bit can be reset by reading the bit and then reading the data register. 1: A frame error is detected; 0: No frame error is detected. *Note: This bit will not generate interrupt. If the EIE bit has been set, the FE flag bit will generate an interrupt in the multi-buffer communication mode.*	0
0	PE	RO	Parity error flag. In the receiving mode, if a parity error occurs, this bit can be set by hardware. This bit can be reset by reading the bit and then reading the data register. Before this bit is cleared, the software must wait for the RXNE flag bit to be set. If PEIE bit has been set before, then the corresponding interrupt will be generated when this bit is set. 1: Parity check error occurs;	0

			0: No parity check error occurs.	

17.10.2 USART Data Register (USARTx_DATAR) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							DR[8:0]								

Bit	Name	Access	Description	Reset value
[31:9]	Reserved	RO	Reserved	0
[8:0]	DR[8:0]	RW	Data register. This register is actually composed of 2 registers: receive data register (RDR) and transmit data register (TDR). The start of the read and write operations of DR is to read the receive data register (RDR) and write to the transmit data register (TDR).	X

17.10.3 USART Baud Rate Register (USARTx_BRR) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIV_Mantissa[11:0]												DIV_Fraction[3:0]			

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:4]	DIV_Mantissa[11:0]	RW	These 12 bits define the integer portion of the divider division factor.	0
[3:0]	DIV_Fraction[3:0]	RW	These 4 bits define the decimal part of the divider division factor.	0

17.10.4 USART Control Register 1 (USARTx_CTLR1) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
M_EXT[1:0]		UE	M	WAKE	PCE	PS	PEIE	TXEIE	TCIE	RXNEIE	IDLEIE	TE	RE	RWU	SBK

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:14]	M_EXT	RW	Word length extension bit:	0

			00: Invalid, M bits determine the data length. 01: 7 data bits; 10: 6 data bits; 11: 5 data bits.	
13	UE	RW	USART enable. 1: Enable USART; 0: Disable USART. When this bit is cleared, the frequency divider and output of USART both stop working after the current byte transmission is completed.	0
12	M	RW	Word length. 1: 9 data bits; 0: 8 data bits.	0
11	WAKE	RW	Wake-up. This bit decides the method to wake up USART: 1: Address flag; 0: Idle line.	0
10	PCE	RW	Parity control enable. 1: Enable parity bit; 0: Disable parity bit. For the receiver, the parity of the data is performed; for the transmitter, the check bit is inserted. Once this bit is set, the parity control enable takes effect only after the current byte transmission is completed.	0
9	PS	RW	Parity selection. 1: Odd parity; 0: Even parity. After this bit is set, the parity control enable takes effect only after the current byte transmission is completed.	0
8	PEIE	RW	Parity check interrupt enable. 1: Allow generation of parity error interrupt; 0: Prohibit generation of parity error interrupt.	0
7	TXEIE	RW	Transmit buffer empty interrupt enable. 1: Allow generation of transmit buffer empty interrupt; 0: Prohibit generation of transmit buffer empty interrupt.	0
6	TCIE	RW	Transmission completion interrupt enable. 1: Allow generation of transmission complete interrupt; 0: Prohibit generation of transmission complete interrupt.	0
5	RXNEIE	RW	Receive buffer non-empty interrupt enable. 1: Allow generation of receive buffer not empty interrupt; 0: Prohibit generation of receive buffer not empty interrupt.	0
4	IDLEIE	RW	Idle line interrupt enable. 1: Allow generation of idle line interrupt; 0: Prohibit generation of idle line interrupt.	0
3	TE	RW	Transmit enable. 1: Enable transmitter; 0: Disable transmitter.	0

2	RE	RW	Receiver enable. 1: Enable receiver, and the receiver starts detecting the start bit on the RX pin. 0: Disable receiver.	0
1	RWU	RW	Receiver wake-up. This bit decides whether the USART is in mute mode: 1: The receiver is in mute mode; 0: The receiver is in active mode. *Note 1: Before the RWU bit is set, USART needs to receive a data byte firstly. Otherwise, it cannot be woken up by the idle bus in mute mode;* *Note 2: When configured to wake up from address flag, the RWU bit cannot be modified by software when RXNE is set.*	0
0	SBK	RW	Send break character control. This bit is set to transmit a frame break character. For the stop bit of break frame, the bit is set by hardware. 1: Break character transmitted; 0: No break character transmitted.	0

17.10.5 USART Control Register 2 (USARTx_CTLR2) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	LINEN	STOP		CLKEN	CPOL	CPHA	LBCL	Reserved	LBDIE	LBDL	Reserved	ADD[3:0]			

Bit	Name	Access	Description	Reset value
[31:15]	Reserved	RO	Reserved	0
14	LINEN	RW	LIN mode enable. When this bit is set, the LIN mode is enabled. In LIN mode, you can use the SBK bit to send the LIN synchronization disconnection symbol and detect the LIN synchronization disconnection symbol.	0
[13:12]	STOP[1:0]	RW	Stop bit setting. These bits are used to set the stop bits. 00: 1 stop bit; 01: 0.5 stop bit; 10: 2 stop bits; 11: 1.5 stop bit.	00b
11	CLKEN	RW	Clock enable. This bit is used to enable CK pin. 1: Enable; 0: Disable.	0
10	CPOL	RW	Clock polarity. In synchronous mode, this bit can be used to select the polarity of the clock output on the CK pin, and work with CPHA to generate the required clock/data sampling relationship. 1: High level is maintained on the CK pin when the bus is	0

			idle; 0: Low level is maintained on the CK pin when the bus is idle. *Note: This bit cannot be modified after enabling transmission.*	
9	CPHA	RW	Clock phase position setting. In the synchronization mode, you can use this bit to select the phase position of the clock output on the CK pin, and work with CPOL bit to generate the required clock/data sampling relationship. 1: Data capture is performed on the second edge of the clock; 0: Data capture is performed on the first edge of the clock. *Note: This bit cannot be modified after enabling transmission.*	0
8	LBCL	RW	Last bit clock pulse control. In synchronous mode, it is used to control whether to output the clock pulse corresponding to the last data byte sent on the CK pin; 1: The clock pulse of the last bit of data is not output from CK; 0: The clock pulse of the last bit of data is output from CK. *Note: This bit cannot be modified after enabling transmission.*	0
7	Reserved	RO	Reserved.	0
6	LBDIE	RW	The LIN break detection interrupt is enabled. Setting this bit will enable the interrupt caused by LBD.	0
5	LBDL	RW	LIN Break detection length, used to select 11-bit or 10-bit break character detection. 1: 11-bit break detection; 0: 10-bit break detection.	0
4	Reserved	RO	Reserved.	0
[3:0]	ADD[3:0]	RW	Address of the USART node, used to set the USART node address of the device. When the data is used during mute mode in multi-processor communication, the address flag is used to wake up a certain USART device.	0

17.10.6 USART Control Register 3 (USARTx_CTLR3) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					CTSIE	CTSE	RTSE	DMAT	DMAR	SCEN	NACK	HDSEL	IRLP	IREN	EIE

Bit	Name	Access	Description	Reset value

[31:11]	Reserved	RO	Reserved	0
10	CTSIE	RW	CTSIE interrupt enable. 1: An interrupt will be generated when CTS is set; 0: No interrupt is generated when CTS is set.	0
9	CTSE	RW	CTS enable. 1: Enable CTS flow control; 0: Disable CTS flow control.	0
8	RTSE	RW	RTS enable. 1: Enable RTS flow control; 0: Disable RTS flow control.	0
7	DMAT	RW	DMA transmission enable. 1: Use DMA when sending; 0: DMA is not used when sending.	0
6	DMAR	RW	DMA reception enable. 1: Use DMA when receiving; 0: DMA is not used when receiving.	0
5	SCEN	RW	Smart card mode enable. 1: Enable smartcard mode; 0: Disable smartcard mode.	0
4	NACK	RW	Smart card NACK enable. 1: NACK is transmitted when the check error occurs; 0: NACK is not transmitted when the check error occurs.	0
3	HDSEL	RW	Half-duplex mode selection bit: 1: Enable half-duplex mode; 0: Disable half-duplex mode.	0
2	IRLP	RW	Infrared low power selection bit: 1: Enable low-power mode when selecting infrared; 0: Disable low-power mode when selecting infrared.	0
1	IREN	RW	Infrared enable. 1: Enable infrared mode; 0: Disable infrared mode.	0
0	EIE	RW	Error interrupt bit enable: 1: Enable error interrupt; 0: Disable error interrupt. When this bit is set, and when DMAR is set, an interrupt is generated if the FE or ORE or NE bit is set.	0

17.10.7 USART Guard Time and Prescaler Register (USARTx_GPR) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

GT[7:0]	PSC[7:0]

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:8]	GT[7:0]	RW	Guard time domain. These bits specify the guard time in unit of baud rate clock. In smartcard mode, the transmission complete flag is set after the guard time has passed.	0
[7:0]	PSC[7:0]	RW	Prescaler domain. In infrared low-power mode, the source clock is divided by this value (all 8 bits are valid), and a value of 0 means reservation; In infrared normal mode, these bits can only be set to 1; In smartcard mode, the value (the lower 5 bits are valid) is multiplied by 2 to give the division factor of the source clock frequency, to provide the clock to the smart card. A value of 0 means reservation.	0

17.10.8 USART Control Register 4 (USARTx_CTLR4) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CHECK_SEL[1:0]		MS_ERRIE	Reserved

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
[3:2]	CHECK_SEL[1:0]	RW	Parity Function Selection Bit: 0x: Disable MARK and SPACE parity; 10: Enable MARK parity; 11: Enable SPACE parity.	0
1	MS_ERRIE	RW	SPACE or MARK Parity Enable Bit: 1: Enable SPACE or MARK parity error interrupts; 0: Disable SPACE or MARK parity error interrupts.	0
0	Reserved	RO	Reserved	0

Chapter 18 Inter-integrated Circuit (I2C) Interface

The internal integrated circuit bus (I2C) is widely used in the communication between microcontrollers, sensors and other off-chip modules. It supports multi-master and multi-slave mode, and can communicate at both 100kHz (standard) and 400kHz (fast) speeds using only 2 wires (SDA and SCL). I2C bus is also compatible with SMBus protocol. It not only supports I2C timing, but also supports arbitration, timing and DMA, and has CRC check function.

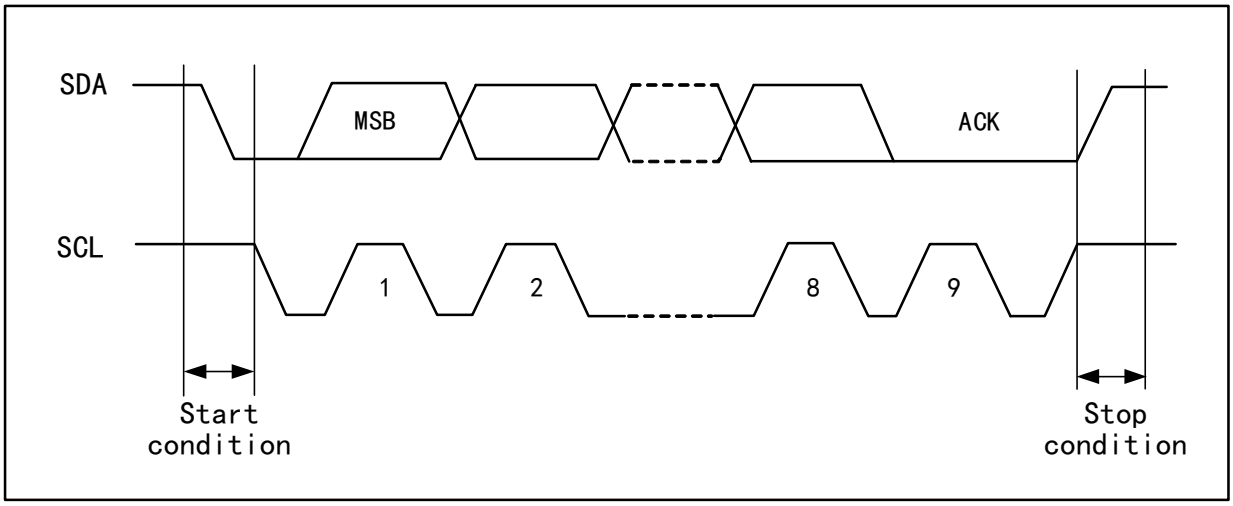
18.1 Main Features

- Master mode and slave mode
- 7-bit or 10-bit address
- Slave device supports dual 7-bit address.
- Two speed modes: 100kHz and 400kHz
- Multiple status modes, multiple error flags
- Support extended clock function
- 2 interrupt vectors
- Support DMA
- Support PEC
- Compatible with SMBus

18.2 Overview

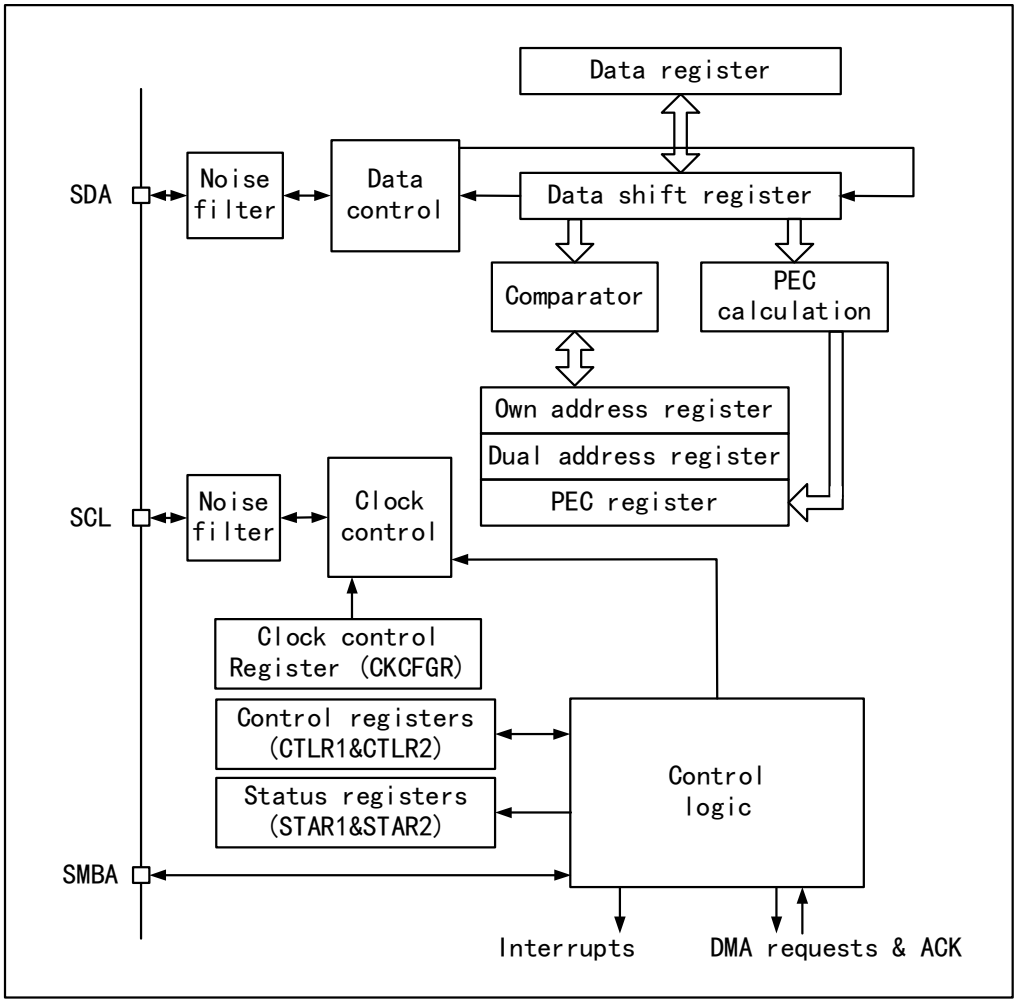
I2C is a half-duplex bus, which can only run in one of the following 4 modes: master device sending mode, master device receiving mode, slave device sending mode and slave device receiving mode. The I2C module works in slave mode by default. After generating the starting condition, it will automatically switch to the master mode. When the arbitration is lost or a stop signal is generated, it will switch to the slave mode. The I2C module supports multi-host functions. When working in main mode, the I2C module will actively send out data and addresses. Data and addresses are transmitted in 8-bit units, with the high bit in front and the low bit in the back. After the start event is a byte (7-bit address mode) or two-byte (10-bit address mode) address. Every time the host sends 8-bit data or address, the slave needs to reply a reply ACK, that is, pull down the SDA bus, as shown in figure 18-1.

Figure 18-1 I2C timing diagram



For normal use, the correct clock must be input to I2C. In the standard mode, the minimum input clock is 2MHz, while the minimum input clock is 4MHz in the fast mode.
Figure 18-2 shows the block diagram of I2C.

Figure 18-2 I2C block diagram



18.3 Master Mode

In master mode, the I2C module leads the data transmission and outputs the clock signal. The data transfer starts with a Start event and ends with a Stop event. The following is the required operations in master mode:

Set the correct clock in the control register2 (R16_I2Cx_CTLR2) and the clock control register (R16_I2Cx_CKCFGR).

Set a proper rising edge in the rising time register (R16_I2Cx_RTR).

Set the PE bit in the control register (R16_I2Cx_CTLR1) to start the peripheral.

Set the START bit in the control register (R16_I2Cx_CTLR1) to generate a start event. After the START bit is set, the I2C module automatically switches to master mode, the MSL bit is set, and a Start event is generated. After the start event is generated, the SB bit is set. If the ITEVTEN bit (in R16_I2Cx_CTLR2) is set, an interrupt is generated. In this case, it is needed to read the R16_I2Cx_STAR1 register. After the slave address is written to the data register, the SB bit is automatically cleared.

If the 10-bit address mode is enabled, then write the data register to send the header sequence (the header sequence is 11110xx0b, of which the xx bits are the highest 2 bits of the 10-bit address).

After the header sequence is transmitted, the ADD10 bit in the status register is set. If the ITEVTEN bit is already set, an interrupt will be generated. At this time, read the R16_I2Cx_STAR1 register and write the second address byte to the data register. Then, clear the ADD10 bit.

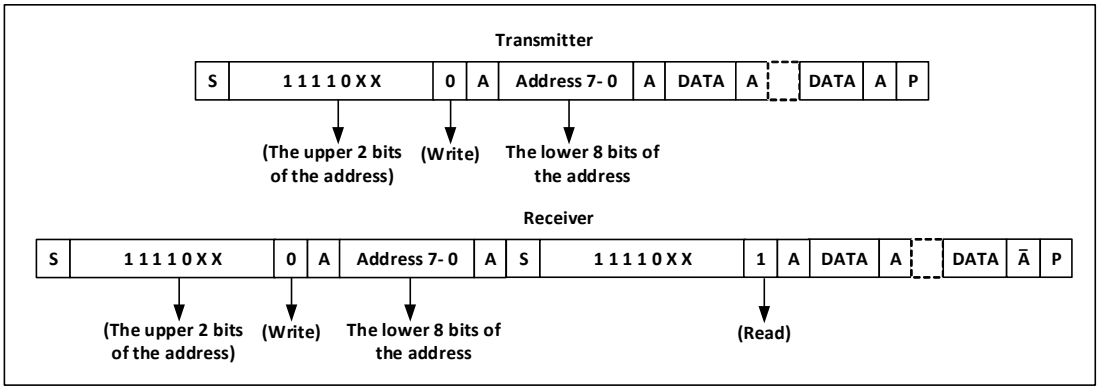
Then, write the data register to send the second address byte. After sending the second address byte, the ADDR bit in the status register is set. If the ITEVTEN bit has been set, an interrupt will be generated. Read the R16_I2Cx_STAR2 register at this time and then read R16_I2Cx_STAR1 register again to clear the ADDR bit;

If the 7-bit address mode is enabled, the write data register transmit the address byte. After the address byte is sent, the ADDR bit in the status register is set. If the ITEVTEN bit has been set, an interrupt will be generated. Read the R16_I2Cx_STAR1 register and then read the R16_I2Cx_STAR2 register again to clear the ADDR bit.

In the 7-bit address mode, the first byte sent is the address byte, the first 7 bits represent the address of the target slave device, the 8th bit determines the direction of the subsequent message, and '0' means the master device writes data to the slave device, '1' means that the master device reads information from the slave device.

In the 10-bit address mode, as shown in Figure 18-3, the first byte is 11110xx0, xx are the highest 2 bits of the 10-bit address, and the second byte is the lower 8 bits of the 10-bit address in the address transmission phase. If you subsequently enter the master transmitter mode, send data continuously. If the device enters the master receiver mode subsequently, a start event needs to be re-sent, and a byte of 11110xx1 will be sent together. Then, enter the master receiver mode.

Figure 18-3 Master receive/transmits data in 10-bit address mode



In transmit mode, the shift register inside the master device sends data from the data register to the SDA line. When the master device receives ACK, TxE of status register 1 (R16_I2C_STAR1) is set. If ITEVTEN and ITBUFEN are set, an interrupt will also be generated. Writing data to the data register will clear the TxE bit. If the TxE bit is set and no new data is written to the data register before the last data transmission, the BTF bit will be set. Before it is cleared, SCL will remain low. After reading R16_I2C_STAR1, writing data to the data register will clear the BTF bit.

In receive mode, the I2C module receives data from the SDA line and writes it into the data register through the shift register. After each byte, if the ACK bit is set, the I2C module will send a response low level, and the RxNE bit will be set. If ITEVTEN and ITBUFEN are set, an interrupt will also be generated. If RxNE is set and the original data has not been read before new data is received, the BTF bit will be set. Before clearing the BTF, SCL will remain low. After reading R16_I2C_STAR1, reading the data register will clear the BTF bit.

When the master device finishes sending data, it will actively send an end event, that is, set the STOP bit. In receive mode, the master device needs to NAK in the acknowledge position of the last data bit. Note that after a NAK is generated, the I2C module will switch to slave mode.

18.4 Slave Mode

In slave mode, the I2C module can recognize its own address and the general broadcast address. The software can control the identification of broadcast call addresses on or off. Once the start event is detected, the I2C module compares the SDA data through the shift register with its own address (the number of bits depends on ENDUAL and ADDMODE) or the broadcast address (when ENGC is set). If there is a mismatch, it will be ignored until a new start event is generated. If it matches the header sequence, an ACK signal is generated and waits for the address of the second byte; if the address of the second byte also matches, or if the address of the whole segment matches in the case of a 7-bit address, then:

First an ACK reply is generated; the ADDR bit is set, and if the ITEVTEN bit is set, there will be a corresponding interrupt.

If you are using dual-address mode (the ENDUAL bit is set), you also need to read the DUALF bit to determine which address the host is calling.

The slave mode defaults to receive mode, and when the last bit of the received header sequence is 1, or the last bit of the 7-bit address is 1 (depending on whether the header sequence is received for the first time or a normal 7-bit address), the I2C module will enter sender mode, and the TRA bit will indicate whether it is currently in receiver or sender mode.

In transmit mode, after clearing the ADDR bit, the I2C module sends the byte from the data register to the SDA line through the shift register. After receiving a response ACK, the TxE bit will be set, and if ITEVTEN and ITBUFEN are set, an interrupt will be generated. If TxE is set but no new data is written to the data register before the end of the next data transmission, the BTF bit will be set. Before clearing BTF, SCL will remain low. After reading status register 1 (R16_I2C_STAR1), writing data to the data register will clear the BTF bit.

In receive mode, after ADDR is cleared, the I2C module stores the data on SDA into the data register through the shift register. After each byte is received, the I2C module will set an ACK bit and set the RxNE bit. If ITEVTEN and ITBUFEN are set, an interrupt will be generated. If RxNE is set and old data is not read before new data is received, BTF will be set. SCL will remain low until the BTF bit is cleared. Reading status register 1 (R16_I2C_STAR1) and reading the data in the data register will clear the BTF bit.

When the I2C module detects a stop event, the STOPF bit is set and an interrupt is generated if the ITEVFEN bit is set. The user needs to read the status register (R16_I2C_STAR1) and then write the control register (such as the reset control word SWRST) to clear it.

18.5 Error

18.5.1 Bus Error (BERR)

When the I2C module detects an external start or stop event during address or data transmission, a bus error occurs. When a bus error is generated, the BERR bit is set, and an interrupt occurs if ITERREN is set. In slave mode, the data is discarded and the hardware releases the bus. If it is a start signal, the hardware will consider it as a restart signal and start waiting for the address or stop signal; if it is a stop signal, it will operate according

to the normal stop condition in advance. In main mode, the hardware does not release the bus and does not affect the current transmission, and it is up to the user code to decide whether to abort the transmission.

18.5.2 Acknowledge Failure (AF)

When the I2C module detects no response after a byte, a reply error occurs. When a reply error is generated: AF will be set, and an interrupt will be generated if ITERREN is set; if an AF error is encountered, if the I2C module works in slave mode, the hardware must release the bus, and if it is in master mode, the software must generate a stop event.

18.5.3 Arbitration Lost (ARLO)

When the I2C module detects the arbitration loss, an arbitration loss error occurs. When an arbitration loss error occurs: the ARLO bit is set and an interrupt is generated if ITERREN is set; the I2C module switches to slave mode and no longer responds to slave-initiated transfers against it unless a host initiates a new start event; the hardware releases the bus.

18.5.4 Overrun/Underrun Error (OVR)

- Overload error:

In slave mode, if clock extension is prohibited, the I2C module is receiving data. If one byte of data has been received, but the last received data has not been read out, an overload error will occur. When an overload error occurs, the last received byte is discarded and the sender should resend the last sent byte.

- Underload error:

In slave mode, if clock extension is prohibited, the I2C module is sending data, and if new data is not written to the data register before the next byte clock arrives, an underload error will occur. When an underload error occurs, the data in the previous data register will be sent twice. If an underload error occurs, the receiver should discard the repeatedly received data. In order not to generate underload errors, the I2C module should write data to the data register before the first rising edge of the next byte.

18.6 Clock Extension

If clock extension is prohibited, there is a possibility of overload/underload errors. But if the clock is extended when enabled:

- In transmit mode, if TxE is set and BTF is set, SCL will always be low, waiting for the user to read the status register and write the data to be sent to the data register.
- In receive mode, if RxNE is set and BTF is set, the SCL will remain low after receiving the data until the user reads the status register and reads the data register.

Thus it can be seen that the extension of the enable clock can avoid overload/underload errors.

18.7 SMBus

SMBus is also a 2-wire interface, which is generally used between the system and power management. SMBus and I2C have many similarities. For example, SMBus uses the same 7-bit address mode as I2C.

Similarities between SMBus and I2C:

1) Master-slave communication mode; the host provides the clock and supports multiple masters and multiple slaves;

2) 2-wire communication structure, of which a warning line can be selected for SMBus;
3) Support 7-bit address format.

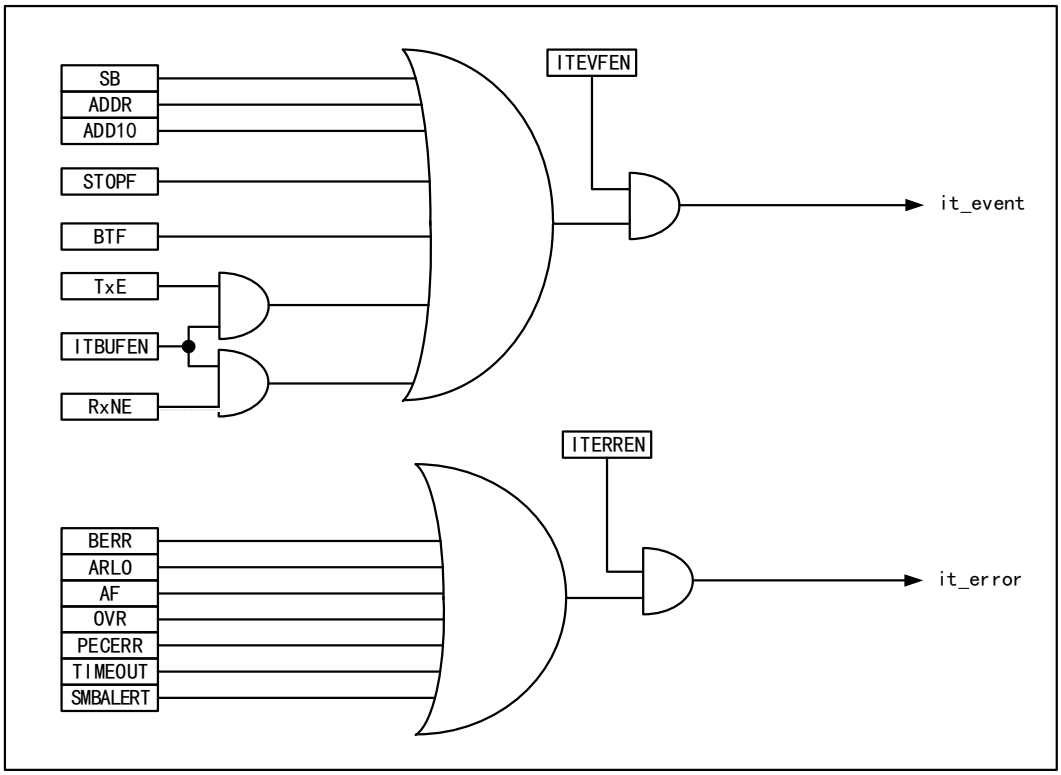
Differences between SMBus and I2C:
1) I2C supports the maximum speed of 400kHz, while SMBus supports the maximum speed of 100kHz, and SMBus has the minimum speed limit of 10kHz;
2) When the SMBus clock is lower than 35mS, it will report a timeout, but I2C has no such limitation;
3) SMBus has a fixed logic level, but I2C does not, depending on VDD;
4) SMBus has a bus protocol, but I2C does not.

SMBus also includes device identification, address resolution protocol, unique device identifier, SMBus reminder, and various bus protocols. For details, please refer to SMBus specification version 2.0. When SMBus is used, only the SMBus bit of the control register needs to be set, and the SMBTYPE and ENAARP bits need to be configured as needed.

18.8 Interrupt

Each I2C module has 2 interrupt vectors: event interrupt and error interrupt. 2 types of interrupts support the interrupt sources as shown in Figure 18-4.

Figure 18-4 I2C interrupt request



18.9 DMA

DMA can be used to receive/transmit bulk data. When DMA is used, the ITBUFEN bit of the control register cannot be set.

- **DMA is used for transmission**

The DMA mode can be activated by setting the DMAEN bit in the CTLR2 register. As long as the TxE bit is

set, the data can be loaded into the I2C data register from the set memory by DMA. The following settings are required to allocate channels for I2C.

1) Set the I2C_DATAR register address to the DMAy_PADDRx register, and set the memory address in the DMAy_MADDRx register, so that the data will be sent from the memory to the I2Cx_DATAR register after each TxE event.
2) Set the required number of transferred bytes in the DMAy_CNTRx register. After each TxE event, this value will be reduced progressively.
3) The channel priority is configured using the PL[0:1] bits in the DMAy_CFGRx register.
4) Set the DIR bit in the DMAy_CFGRx register, and it can be configured to issue an interrupt request according to application requirements when the entire transmission is half or wholly completed.
5) Activate the channel by setting the EN bit in the DMAy_CFGRx register.

When the number of data transfer set in the DMA controller has been completed, the DMA controller will send an EOT/EOT_1 signal indicating the end of the transmission to the I2C interface. When the interrupt is allowed, a DMA interrupt will be generated.

- **DMA is used for reception**

After the DMAEN bit in the CTLR2 register is set, DMA receiver mode can be started. When DMA is used for reception, DMA transfers the data in the data register to the preset memory area. The following steps are required to allocate channels for I2C.

1) Set the I2Cx_DATAR register address to the DMAy_PADDRx register, and set the memory address in the DMAy_MADDRx register, so that the data will be written into the memory from the I2C_DATAR register after each RxNE event.
2) Set the required number of transferred bytes in the DMAy_CNTRx register. After each RxNE event, this value will be reduced progressively.
3) The channel priority is configured by the PL[1:0] bits in the DMAy_CFGRx register.
4) Clear the DIR bit in the DMAy_CFGRx register, and it can be configured to issue an interrupt request according to application requirements when the data transmission is half or wholly completed.
5) Activate the channel by setting the EN bit in the DMAy_CFGRx register.

When the number of data transfer set in the DMA controller has been completed, the DMA controller will send an EOT/EOT_1 signal indicating the end of the transmission to the I2C interface. When the interrupt is allowed, a DMA interrupt will be generated.

18.10 Packet Error Checking

Packet error checking (PEC) is a CRC8 check step added to provide the transmission reliability. Each bit of serial data can be calculated through the following polynomial:

$$C=X^8+X^2+X+1$$

PEC calculation is activated by the ENPEC bit in the control register, and all information bytes are calculated, including address and read/write bits. During transmission, enabling PEC will add a byte of CRC8 calculation result after the last byte of data. While in receiver mode, the last byte is considered to be the CRC8 check result. If it does not match the internal calculation result, it will reply with a NAK. For the master receiver, it will reply with a NAK regardless of whether the check result is correct or not.

18.11 Debug Mode

After the system enters the debug mode, the DBG_I2C_SMBUS_TIMEOUT bit in the DEBUG module can be used to determine whether to continue operating or stop the time-out control of I2CSMBus.

18.12 Register Description

Table 18-1 I2C1 registers

Name	Access address	Description	Reset value
R16_I2C_CTLR1	0x40005400	I2C control register 1	0x0000
R16_I2C_CTLR2	0x40005404	I2C control register 2	0x0000
R16_I2C_OADDR1	0x40005408	I2C address register 1	0x0000
R16_I2C_OADDR2	0x4000540C	I2C address register 2	0x0000
R16_I2C_DATAR	0x40005410	I2C data register	0x0000
R16_I2C_STAR1	0x40005414	I2C status register 1	0x0000
R16_I2C_STAR2	0x40005418	I2C status register 2	0x0000
R16_I2C_CKCFGR	0x4000541C	I2C clock register	0x0000
R16_I2C_RTR	0x40005420	I2C rise time register	0x0002

18.12.1 I2C Control Register (I2C_CTLR1)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWRST	Reserved	ALERT	PEC	POS	ACK	STOP	START	NOSTRETCH	ENGC	ENPEC	ENARP	SMBTYPE	Reserved	SMBUS	PE

Bit	Name	Access	Description	Reset value
15	SWRST	RW	Software reset. Setting this bit by user code will reset the I2C peripheral. Before reset, make sure that the pins of the I2C bus are released and the bus is idle. *Note: This bit can reset the I2C module when no stop condition is detected on the bus but the busy bit is 1.*	0
14	Reserved	RO	Reserved.	0
13	ALERT	RW	SMBus alert. This bit can be set or cleared by the user code. When PE is set, this bit can be cleared by hardware. 1: Drive the SMBusALERT pin to make it low, and the response address header shall closely follow the ACK signal; 0: Release the SMBusALERT pin to make it high, and the response address header shall closely follow the NACK signal.	0
12	PEC	RW	Packet error check enable. Set this bit to enable data packet error detection. This bit can be set or cleared by the user code. When the PEC is transmitted, or a	0

			start or end signal is generated, or the PE bit is cleared to 0, the bit can be cleared by hardware. 1: Provided with PEC; 0: Not provided with PEC. *Note: PEC will fail when the arbitration is lost.*	
11	POS	RW	ACK and PEC position setting. This bit can be set and cleared by user code, and it can be cleared by hardware after PE is cleared; 1: The ACK bit controls the ACK or NAK of the next byte received in the shift register. The next byte received in the PEC shift register is PEC; 0: The ACK bit controls the ACK or NAK of the byte currently being received in the shift register. The PEC bit indicates that the byte of the shift register before the current bit is PEC. *Note: The usage of POS bit in 2-byte data reception is as follows: It must be configured before receiving. For the second byte of NACK, the ACK bit must be cleared immediately after the ADDR bit is cleared; in order to detect the PEC of the second byte, the PEC bit must be set after the ADDR event occurs following the POS bit.*	0
10	ACK	RW	Acknowledge enable. This bit can be set or cleared by user code. When PE bit is set, this bit can be cleared by hardware; 1: Acknowledge returned after a byte is received; 0: No acknowledge is returned.	0
9	STOP	RW	Stop event generation. It can be set or cleared by user code, or cleared by hardware when a stop event is detected, or set by hardware when a timeout error is detected. In master mode: 1: A stop event is generated after the current byte transfer or the current start condition is issued; 0: No stop event occurs. In slave mode: 1: Release the SCL and SDA lines after the current byte transfer; 0: No stop event occurs.	0
8	START	RW	Start event generation. This bit can be set or cleared by the user code. When the start condition is issued or PE is cleared, it can be cleared by hardware. In master mode: 1: A start event is generated repeatedly; 0: No start event is generated. In slave mode:	0

			1: When the bus is idle, a start event is generated; 0: No start event is generated.	
7	NOSTRETCH	RW	Clock stretching disable. This bit is used to disable clock stretching in slave mode when the ADDR or BTF flag bit is set until it is cleared by software. 1: Clock stretching disabled; 0: Clock stretching enabled.	0
6	ENGC	RW	General call enable. Set this bit to enable the general call, and respond to general address 00h.	0
5	ENPEC	RW	PEC enable. 1: Enable PEC calculation; 0: Disable PEC calculation.	0
4	ENARP	RW	ARP enable. 1: Enable ARP; 0: Disable ARP. If SMBTYPE=0, the default address of the SMBus device is used. If SMBTYPE=1, the main address of the SMBus is used.	0
3	SMBTYPE	RW	SMBus device type: 1: SMBus master device; 0: SMBus slave device.	0
2	Reserved	RO	Reserved.	0
1	SMBUS	RW	SMBus mode selection. 1: SMBus mode; 0: I2C mode.	0
0	PE	RW	I2C peripheral enable. 1: I2C module enabled; 0: I2C module disabled.	0

18.12.2 I2C Control Register 2 (I2C_CTLR2)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			LAST	DMA EN	ITBU FEN	ITEV TEN	ITER REN	Reserved		FREQ[5:0]					

Bit	Name	Access	Description	Reset value
[15:13]	Reserved	RO	Reserved	0
12	LAST	RW	Last transfer setting of DMA. 1: Next DMA EOT is the last transfer; 0: Next DMA EOT is not the last transfer. *Note: This bit is used in master receiver mode and can generate a NAK when the data is received at the last time.*	0
11	DMAEN	RW	DMA request enable. 1: Enable DMA request when TxE or RxNE bit is	0

			set; 0: Disable DMA request when TxE or RxNE bit is set.	
10	ITBUFEN	RW	Buffer interrupt enable. 1: When the TxE bit is set, or when the RxNE bit is set, an event interrupt is generated; 0: When the TxE bit is set, or when the RxNE bit is set, no interrupt is generated.	0
9	ITEVTEN	RW	Event interrupt enable. 1: Enable event interrupt; 0: Disable event interrupt. Under the following conditions, the interrupt can be generated: SB=1 (master mode); ADDR=1(master and slave modes); ADD10=1 (master mode); STOPF=1 (slave mode); BTF=1, but no TxE or RxNE event occurs; If ITBUFEN=1, TxE event is 1; If ITBUFEN=1, RxNE event is 1.	0
8	ITERREN	RW	Error interrupt enable. 1: The error interrupt is enabled; 0: The error interrupt is disabled Under the following conditions, the interrupt can be generated: BERR=1; ARLO=1; AF=1; OVR=1; PECERR=1; TIMEOUT=1; SMBAlert=1.	0
[7:6]	Reserved	RO	Reserved.	0
[5:0]	FREQ[5:0]	RW	I2C module clock frequency domain, must input the correct clock frequency to produce the correct timing, the allowable range is between 4-60MHz. Must be set between 000100b and 111100b in MHz.	0

18.12.3 I2C Address Register 1 (I2C_OADDR1)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDMODE	Reserved							ADD[9:8]		ADD[7:1]					ADD0

Bit	Name	Access	Description	Reset value
15	ADDMODE	RW	Addressing mode. 1: 10-bit slave address (7-bit address not acknowledged); 0: 7-bit slave address (10-bit address not acknowledged).	0
[14:10]	Reserved	RO	Reserved.	0

[9:8]	ADD[9:8]	RW	Interface address, which is 9-8 bits when using 10-bit addresses and ignored when 7-bit addresses are used.	0
[7:1]	ADD[7:1]	RW	Interface address, bit 7-1.	0
0	ADD0	RW	Interface address. Bit0 when a 10-bit address is used. It is ignored when a 7-bit address is used.	0

18.12.4 I2C Address Register 2 (I2C_OADDR2)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								ADD2[7:1]							ENDUAL

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:1]	ADD2[7:1]	RW	Interface address. Bit7 to bit1 of address in dual-address mode.	0
0	ENDUAL	RW	Dual addressing mode enable. When this bit is set, the ADD2 can be identified.	0

18.12.5 I2C Data Register (I2C_DATAR)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DR[7:0]							

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved.	0
[7:0]	DR[7:0]	RW	Data register. This domain is used to store received data or store data to be transmitted to the bus.	0

18.12.6 I2C Status Register 1 (I2C_STAR1)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMBALERT	TIMEOUT	Reserved	PECERR	OVR	AF	ARLO	BERR	TxE	RxNE	Reserved	STOPF	ADD10	BTF	ADDR	SB

Bit	Name	Access	Description	Reset value
15	SMBALERT	RW0	SMBus alert. It can be reset by user writing 0, or reset by hardware when PE becomes low. In master mode of SMBus: 1: SMBus alert is generated on the pin; 0: No SMBus alert. In slave mode of SMBus: 1: SMBAlert response address header to SMBAlert LOW received;	0

			0: No SMBAlert response address header.	
14	TIMEOUT	RW0	Timeout or Tlow error flag. It can be reset by user writing 0, or reset by hardware when PE becomes low. 1: SCL is low for 25ms, or the cumulative low-level clock expansion time of the host exceeds 10ms, or the cumulative low-level time of the slave device exceeds 25ms; 0: No timeout error. *Note: When this bit is set in slave mode, the slave device resets the communication and the hardware releases the bus. When this bit is set in master mode, hardware issues a stop condition.*	0
13	Reserved	RO	Reserved	0
12	PECERR	RW0	PEC error flag occurs during reception. This bit can be reset by user writing 0, or reset by hardware when PE becomes low. 1: PEC error. After PEC is received, NAK is returned; 0: No PEC error.	0
11	OVR	RW0	Overrun and underrun flag. 1: Overrun or underrun event occurs: In case of NOSTRETCH=1, when a new byte is received in the receiver mode and the content in the data register has not been read, the newly received byte will be lost. In the transmitter mode, no new data is written into the data register, and the same byte will be sent twice; 0: No overrun and underrun event.	0
10	AF	RW0	Acknowledge failure flag. This bit can be reset by user writing 0, or reset by hardware when PE becomes low. 1: Acknowledge error; 0: Normal acknowledge.	0
9	ARLO	RW0	Arbitration lost flag. It can be reset by user writing 0, or reset by hardware when PE becomes low. 1: Arbitration lost is detected and the module loses control of the bus; 0: Normal arbitration.	0
8	BERR	RW0	Bus error flag. It can be reset by user writing 0, or reset by hardware when PE becomes low. 1: Start or stop condition error; 0: Normal.	0
7	TxE	RO	Data register empty flag, which can be cleared by writing data to the data register, or it is automatically cleared by hardware after a start or	0

			stop bit is generated, or when PE is 0. 1: When the data is transmitted, the transmit data register is empty; 0: The data register is non-empty.	
6	RxNE	RO	The data register is not an empty log bit, which will be cleared by reading and writing to the data register, or by the hardware when PE is 0. 1: When receiving data, the data register is not empty. 0: Normal.	0
5	Reserved	RO	Reserved.	0
4	STOPF	RO	Stop event flag. After the user reads the status register1, writing to the control register1 will clear this bit, or when PE is 0, the hardware will clear this bit. 1: After the response, the slave device will detect a stop event on the bus; 0: No stop event is detected.	0
3	ADD10	RO	10-bit address header sent flag. After the user reads the status register1, writing to the control register1 will clear this bit, or when PE is 0, the hardware will clear this bit. 1: In 10-bit address mode, the master device has sent the first address byte; 0: None.	0
2	BTF	RO	Byte transmission end flag. After the user reads the status register1, reading and writing to the data register will clear this bit. During transmission, after a start or stop event is initiated, or when PE is 0, this bit will be cleared by hardware. 1: Byte transmission completed. In case of NOSTRETCH=0: when a new data is sent and the data register has not been written with new data during transmission; when a new byte is received but the data register has not been read; 0: None.	0
1	ADDR	RW0	Address transmitted/matched flag. After the user reads the status register 1, the read operation of the status register2 will clear this bit, or when PE is 0, the hardware will clear this bit. Master mode: 1: End of address transmission: In 10-bit address mode, it is set when the ACK of the second byte of the address is received; in 7-bit address mode, it is set when the ACK of the address is received; 0: Address transmission is not completed.	0

			Slave mode: 1: The received address matches; 0: The address does not match or no address is received.	
0	SB	RO	Start bit transmission flag. After reading the status register 1, the operation of writing the data register will clear this bit, or when PE is 0, the hardware will clear this bit. 1: The start bit has been transmitted; 0: The start bit has not been transmitted.	0

18.12.7 I2C Status Register 2 (I2C_STAR2)

Offset address: 0x18

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PEC[7:0]								DUALF	SMBHOST	SMBDEFAULT	GENCALL	Reserved	TRA	BUSY	MSL

Bit	Name	Access	Description	Reset value
[15:8]	PEC[7:0]	RO	Packet error checking. When PEC is enabled (ENPEC is set), this domain stores the value of PEC.	0
7	DUALF	RO	Matched detection flag. When the stop bit or start bit is generated, or when PE=0, the hardware will clear this bit. 1: The received address matched with OAR2; 0: The received address matched with OAR1.	0
6	SMBHOST	RO	SMBus host header flag. When the stop bit or start bit is generated, or when PE=0, the hardware will clear this bit. 1: When SMBTYPE=1 and ENARP=1, the SMBus host address will be received; 0: SMBus host address is not received.	0
5	SMBDEFAULT	RO	SMBus device default address flag. When the stop bit or start bit is generated, or when PE=0, the hardware will clear this bit. 1: When ENARP=1, the default address of the SMBus device is received; 0: No address is received.	0
4	GENCALL	RO	General call address flag. When the stop bit or start bit is generated, or when PE=0, the hardware will clear this bit. 1: When ENGC=1, the address of general call is received; 0: No general call address is received.	0
3	Reserved	RO	Reserved.	0

2	TRA	RO	Transmitter/receiver flag, cleared by hardware when a stop event (STOPF=1) is detected, repeated start condition or bus arbitration is lost (ARLO=1) or PE=0. 1: Data has been sent; 0: Data is received. This bit is determined by R/W bit of address byte.	0
1	BUSY	RO	Bus busy flag. This bit is cleared when a stop bit is detected. When the interface is disabled (PE=0), the information is still updated. 1: Busy bus: SDA or SCL has a low level; 0: The bus is idle and does not have communication.	0
0	MSL	RO	Master/slave mode indication. When the interface is in master mode (SB=1), the hardware will set this bit. When the bus detects a stop bit and the arbitration is lost, or PE=0, the hardware will clear this bit.	0

18.12.8 I2C Clock Register (I2C_CKCFGR)

Offset address: 0x1C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F/S	DUTY	Reserved		CCR[11:0]											

Bit	Name	Access	Description	Reset value
15	F/S	RW	Master mode selection. 1: Fast mode; 0: Standard mode.	0
14	DUTY	RW	Duty cycle in the fast mode: 1: $T_{Low\ level}/T_{High\ level}$=16/9; 0: $T_{Low\ level}/T_{High\ level}$=2.	0
[13:12]	Reserved	RO	Reserved.	0
[11:0]	CCR[11:0]	RW	Clock frequency division factor. These bits determine the frequency waveform of the SCL clock.	0

18.12.9 I2C Rise Time Register (I2C_RTR)

Offset address: 0x20

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										TRISE[5:0]					

Bit	Name	Access	Description	Reset value
[15:6]	Reserved	RO	Reserved	0
[5:0]	TRISE[5:0]	RW	Maximum rise time domain. The rise time of SCL in master mode is set at this bit. The maximum	000010b

			rising edge time is equal to TRISE-1 clock cycle. This bit can only be set when PE is cleared. If the input clock cycle of the IIC module is 125nS and the value of TRISE is 9, then the maximum rising edge time is (9-1)*125nS, i.e., 1000nS.	

Chapter 19 I3C Bus (I3C)

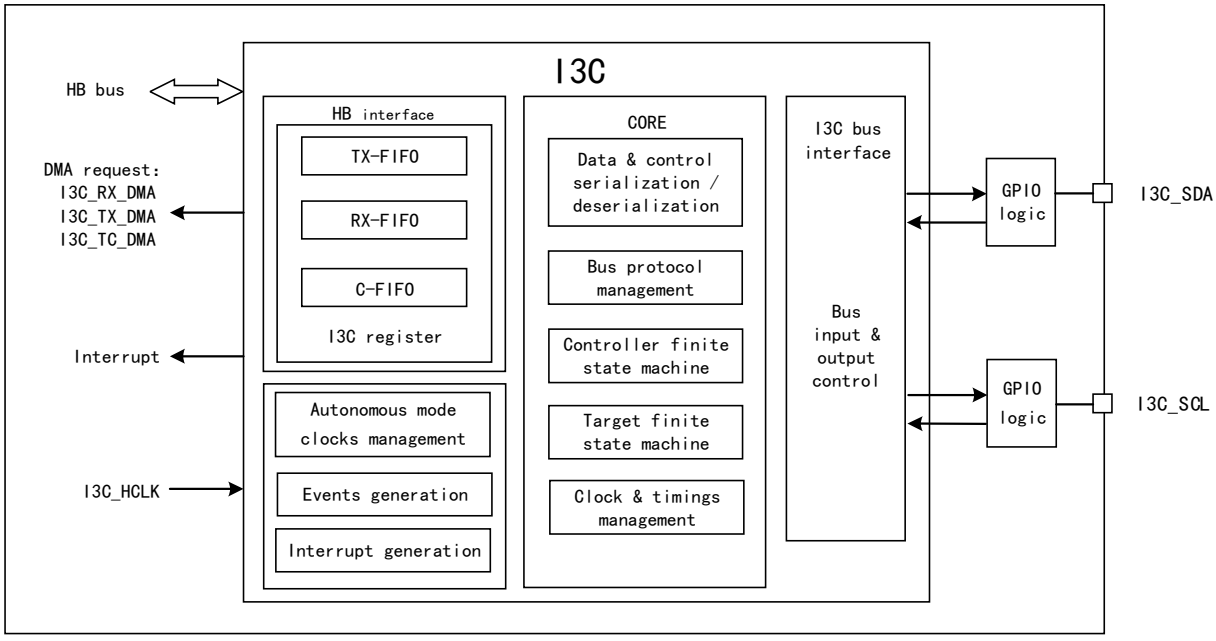
The I3C bus is a 2-wire serial single-ended multi-drop bus designed to enhance the traditional I2C bus. The I3C interface handles communication between this device and other devices connected to the I3C bus. It supports operation as both a master and a slave device. When functioning as a master, it extends the capabilities of the I2C interface while maintaining a degree of backward compatibility.

19.1 Main Features

- Support master and slave devices
- Support MIPI I3C specification v1.1
- Support multi-host functionality
- Support DMA
- In-band interrupt (IBI) capability
- Hot join (HJ) function
- Built-in error detection and recovery
- Support dynamic address allocation, direct and broadcast common command codes (CCC), and private read/write transfers

19.2 Overview

Figure 19-1 I3C timing diagram



19.3 Function Description

19.3.1 I3C Peripheral Status

I3C peripherals can be used as both I3C master devices and I3C slave devices. In any case, the peripheral is in one of the following states:

- **Forbidden state**

Operating Conditions: After the I3C peripheral is reset (by setting bit 1 of I3CRST in the RCC module), the peripheral enters a disabled state.

Transition from disabled to idle state: When the software sets bit 1 of the R32_I3C_CFGR register's EN field, the peripheral transitions from the disabled state to the idle state after completing verification of its internal configuration parameters.

- Idle state

In the idle state, the software can partially update the I3C peripheral configuration.

Transition from idle state to working state:

1) Software-controlled triggering:

In master mode, triggered when frame transmission starts via software write operation setting TSFSET bit 1 in the R32_I3C_CFGR register;

In slave mode, triggered when software issues an IBI/CR/HJ request;

2) Hardware Event Triggering:

The master device triggers upon receiving a start request and maximum TCAS time from the slave device during bus arbitration;

The slave device triggers upon detecting broadcast/direct CCC or private read/write operations via the SDA/SCL lines.

- **Working state**

In the working state, the peripheral performs transmission on the bus.

In the working state, when the requested transmission is completed, the software will be notified by the event in the R32_I3C_EVR register, and the corresponding interrupt will be enabled through the R32_I3C_INTENR register. After the peripheral is switched from the working state to the idle state, it can still communicate on the bus and can be partially reconfigured.

As master device: In the working state, the event/flag generated after the requested transmission is completed can be Frame Complete (FCF), IBI/Master Device Role/Hot Join Request Complete (IBIF/CRF/HJF) or Transmission Error (ERRF).

19.3.2 I3C Master Status and Programming Sequence

In this section, we provide a complete programming sequence when I3C peripheral is the master device, covering state transition, main subtasks and conditions.

Master device initialization

When the master device is in the forbidden state (the EN bit in R32_I3C_CFGR register is 0), the software must follow the following steps to initialize:

- Configure the R32_I3C_CFGR register, set CRINIT to 1 to mark the master device role
- Clear the HST_SIE_RST bit in the R32_I3C_RESET register
- Configure I3C bus timing:

1) Configure the R32_I3C_TIMINGR0 register to set the SCL clock high-level duration and low-level duration for the open-drain/push-pull stages of traditional I2C and I3C.
2) Configure the R32_I3C_TIMINGR1 register to set the SDA hold time (t_{HD_PP}) for the push-pull phase, the bus idle condition time (I3C t_{CAS}, traditional I2C t_{BUF}), the I3C repeat start bit timing (t_{CASr}, t_{CBSr}), I3C stop timing (t_{CBP}), and maximum SCL clock low stall time ($t_{STALLDAA}$ for ENTDAA CCC, ACK/NACK for traditional I2C, the address phase of I3C transfers, the parity bit of write data transfers,

the ACK/NACK data phase of traditional I2C write transfers, the transition bit of I3C read transfers, or the ACK/NACK phase of traditional I2C write transfers (t_{STALL}). Peripherals acting as master devices may adjust the SCL clock low-level stall time and $t_{NEWCRLock}$ according to their requirements to accommodate the master role transition process (following the GETACCCR CCC command).

3) Configure the R32_I3C_TIMINGR2 register to set the SCL clock low-level dwell time. Each phase can be individually enabled or disabled to adjust the SCL clock low-level dwell time (to meet the requirements of the addressed I3C slave device or traditional I2C slave device during SDA switching).

- Configure the DA[6:0] bits of the R32_I3C_DEVR0 register to set the device's dynamic address
- Configure the R32_I3C_DEVRx register to set management parameters for any slave device x (x = 1/2/3/4)
- Configure the R32_I3C_CFGR register to set the execution mode for frame transfer or target request transfer
- Configure the R32_I3C_INTENR register to set interrupt generation or polling mode for any event
- After initialization completes, set EN to 1 to enable the I3C peripheral

Start the frame transmission initiated by the master device

When the master device is in the enabled state (the EN bit in R32_I3C_CFGR register is 1), the software can start frame transmission by any of the following configuration methods:

1. Software trigger: set the TSFSET bit of R32_I3C_CFGR register to 1 by performing a write operation;
   - This operation will cause the hardware to set the CFNFF flag in the R32_I3C_EVR register to request writing the first control word to the R32_I3C_CTLR register.
2. No trigger: directly write the first control word to R32_I3C_CTLR register through software.

Regardless of the initialization method used, the I3C peripheral subsequently transitions to operational status. When the control word is not the final message in an I3C frame (i.e., the MEND bit in the R32_I3C_CTLR register is not set) and no transmission errors have occurred (i.e., the ERRF bit in the R32_I3C_EVR register is not set), the hardware will continue to request the next control word and proceed with frame transmission:
1) If the C-FIFO is in non-DMA mode (CDMAEN bit in the R32_I3C_CFGR register is 0), the software will write the next control word after the CFNFF flag in the R32_I3C_EVR register is set or the corresponding interrupt is triggered (CFNFIE bit in the R32_I3C_INTENR register is 1);
2) If the C-FIFO is in DMA mode (CDMAEN bit in the R32_I3C_CFGR register is set to 1), the corresponding I3C DMA request will be automatically processed by the assigned DMA channel, and the next control word will be written.

Initiate and receive transmissions initiated from devices

When the master device is in the enabled state (the EN bit in the R32_I3C_CFGR register is 1), the master device can initiate the transmission, and the slave device can also initiate the transmission by issuing a start request (driving SDA to a low level), but only if the master device has allowed the hot join request, IBI request or master role request through the R32_I3C_DEVR0 register.

Perform frame transmission initiated by the master device

On the bus, the master device continuously executes frame transfers until the final message is transmitted (R32_I3C_EVR register FCF bit is 1) or an error occurs during transmission (R32_I3C_EVR register ERRF bit is 1), triggering the corresponding interrupt (if enabled). This process relies on the R32_I3C_CTLR and R32_I3C_TDW/TDR registers (written explicitly by software or automatically pushed by assigned DMA channels) and the R32_I3C_RDW/RDR registers (read explicitly by software or by assigned DMA channels).

After completing the transmission, the I3C master device returns to idle state.

Update master device transmission configuration

After the software returns to the idle state, it can update the configuration of the I3C peripheral before the next transmission:

- Configure the R32_I3C_TIMINGR2 register to modify the SCL clock pause;
- Configure the R32_I3C_INTENR register to modify the interrupt/polling mode strategy;
- Configure the TXTHRES, RXTHRES, TMODE, SMODE, TXDMAEN, CDMAEN, RXDMAEN, SDMAEN, EXITPTRN, RSTPTRN, and NOARBH bits in the R32_I3C_CFGR register;

Modify/prepare the control word, status word and read/write data of the next frame transmission to be executed by software and/or DMA.

Master device receives (broadcast CCC, direct read/write CCC or private read/write) message

When the slave device is in the idle state (the EN bit of the R32_I3C_CFGR register is 1), it means that it is ready to receive communication messages from the master device on the I3C bus and is ready to switch to the working state.

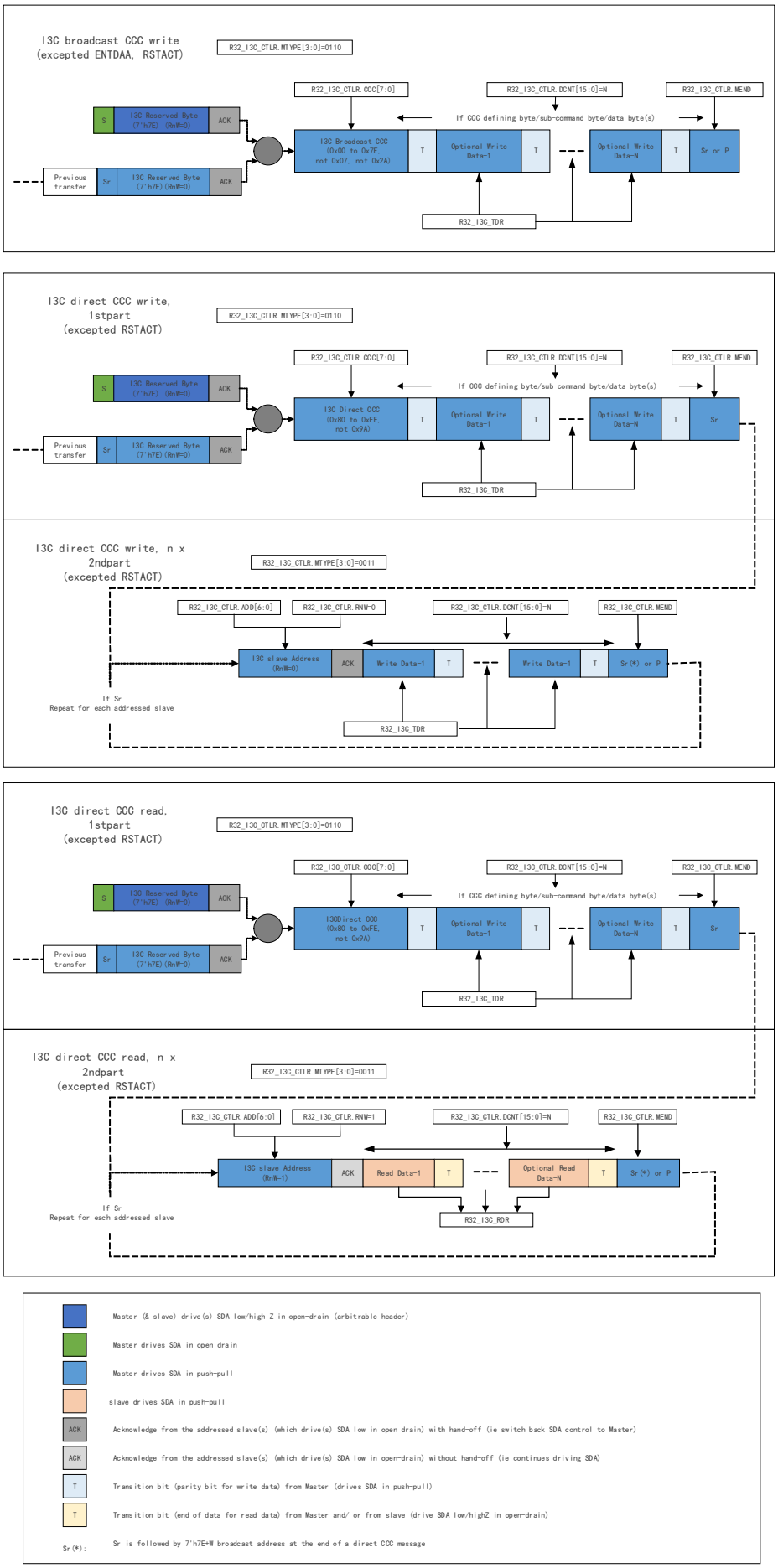
In the working state, the slave device will first receive the broadcast ENTDAACCC (possibly after receiving the optional broadcast ENEC/DISECCCC), and then be assigned a dynamic address. The event DAUPF in the R32_I3C_EVR register is set and the associated interrupt is generated (if enabled) and the slave device subsequently returns to the idle state.

The slave in the idle state is then ready to receive any other broadcast CCC messages, direct read/write CCC or private read/write messages from the master.

19.3.3 I3C Bus Transmission and Programming

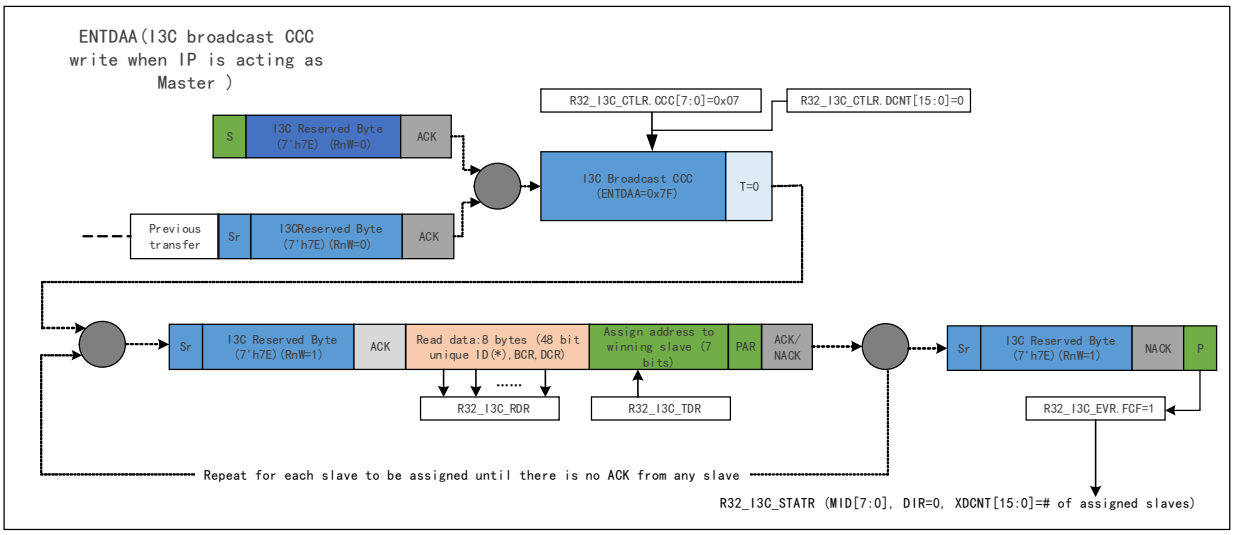
19.3.3.1 I3C Broadcast/Direct CCC Transfer (except ENTDAA and RSTACT) (as master device)

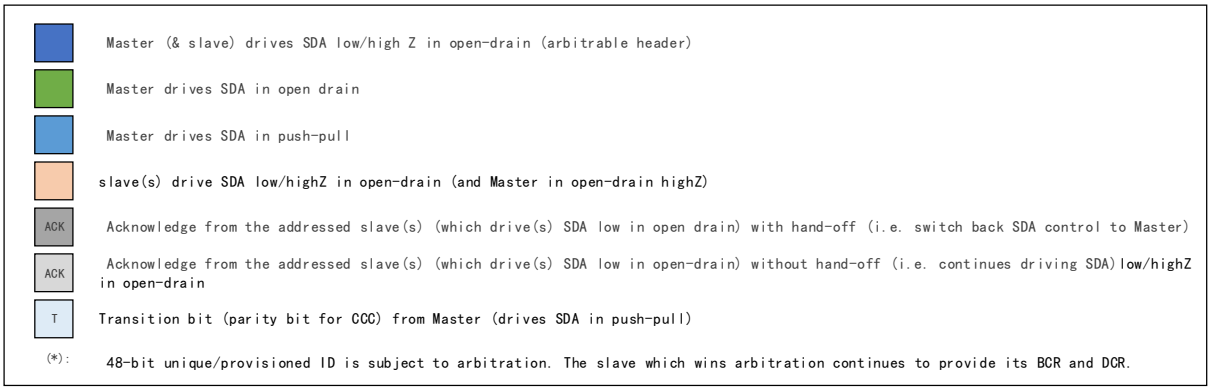
Table 19-2 I3C CCC message



19.3.3.2 I3C Broadcast ENTDAA CCC Transfer (as master device)

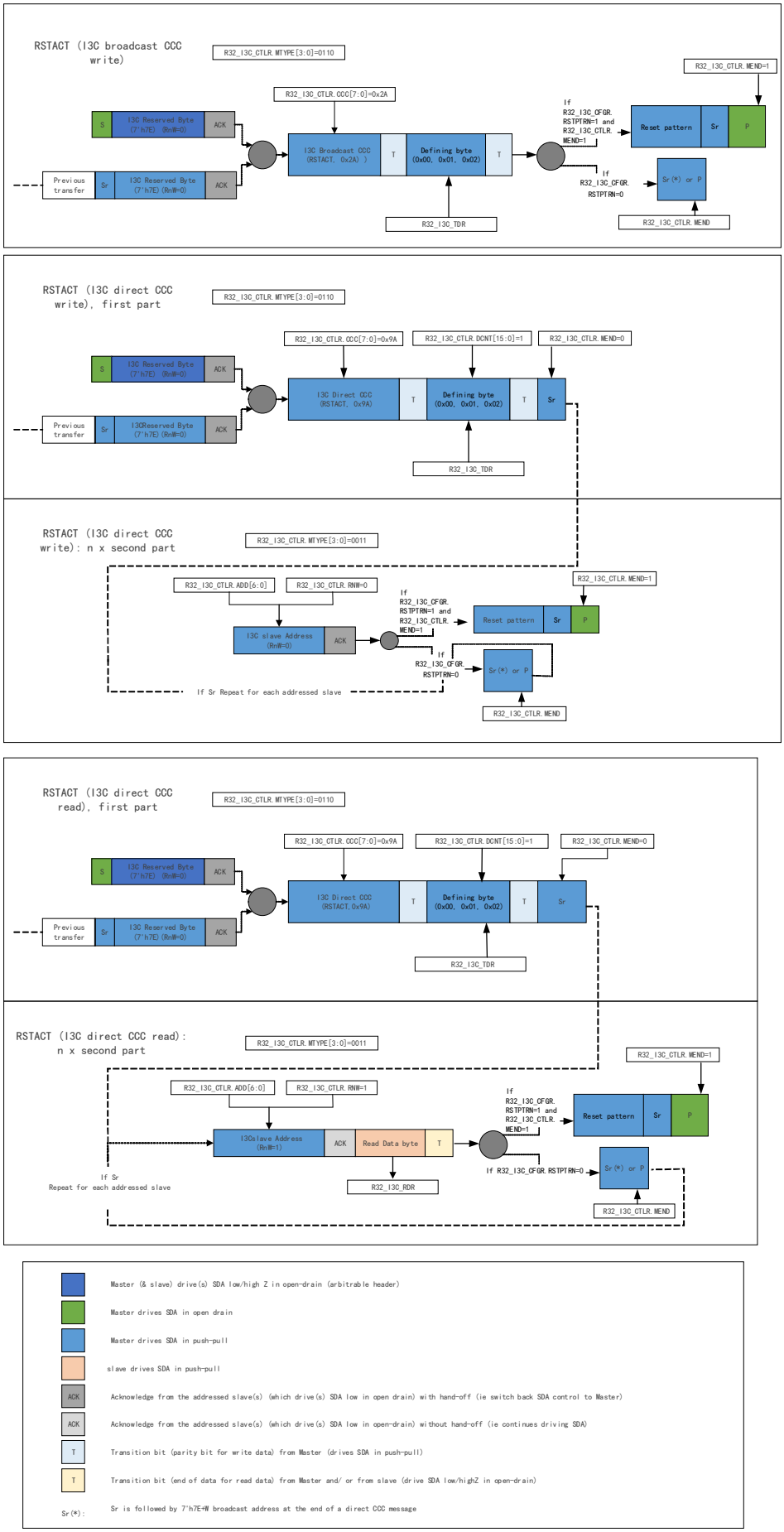
Table 19-3 I3C broadcast ENTDAA CCC transmission (as master device)





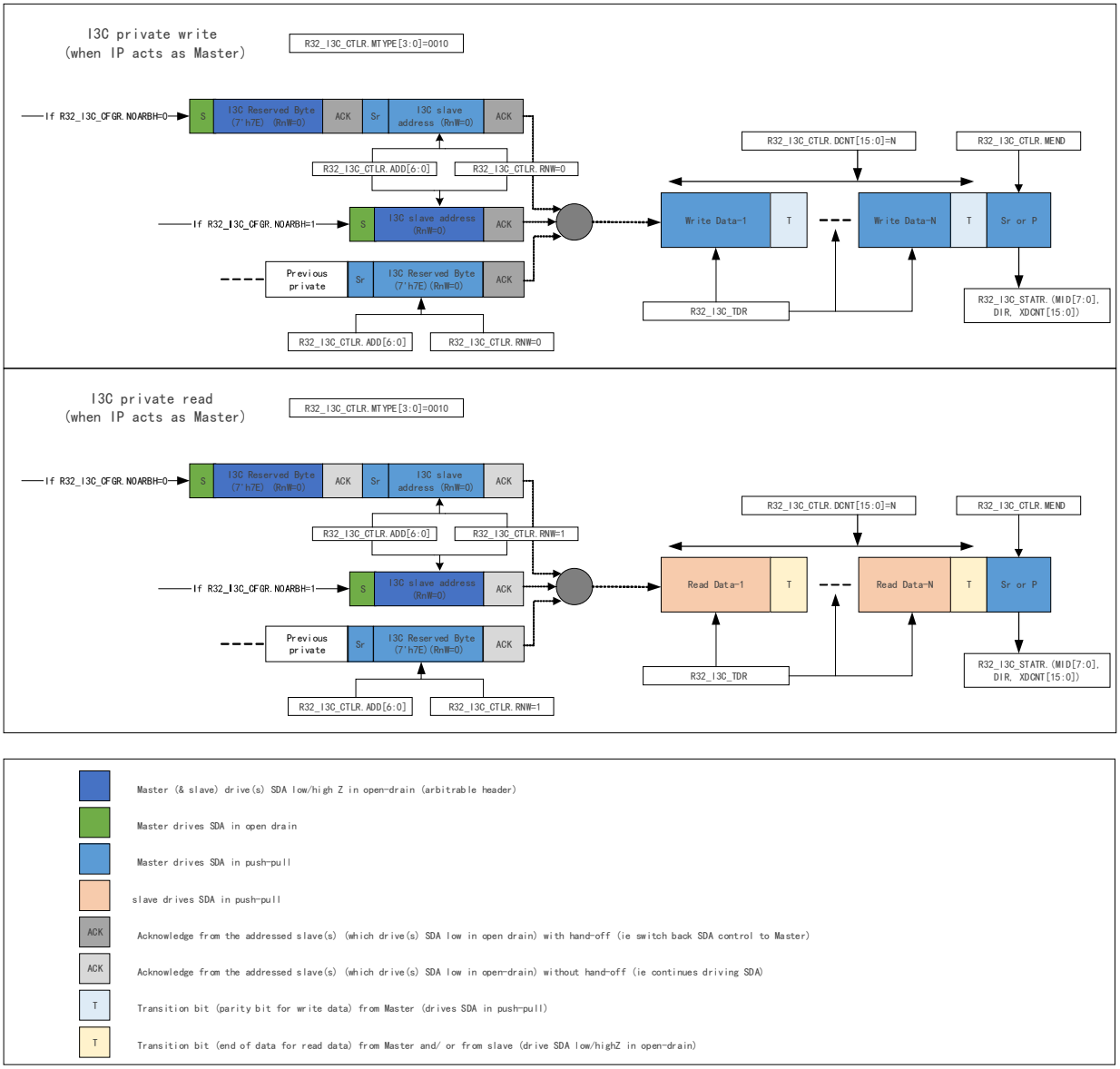
19.3.3.3 I3C Broadcast/Direct RSTACT CCC Transfer (as master device)

Table 19-4 I3C broadcast, direct read/write RSTACT CCC (as master device)



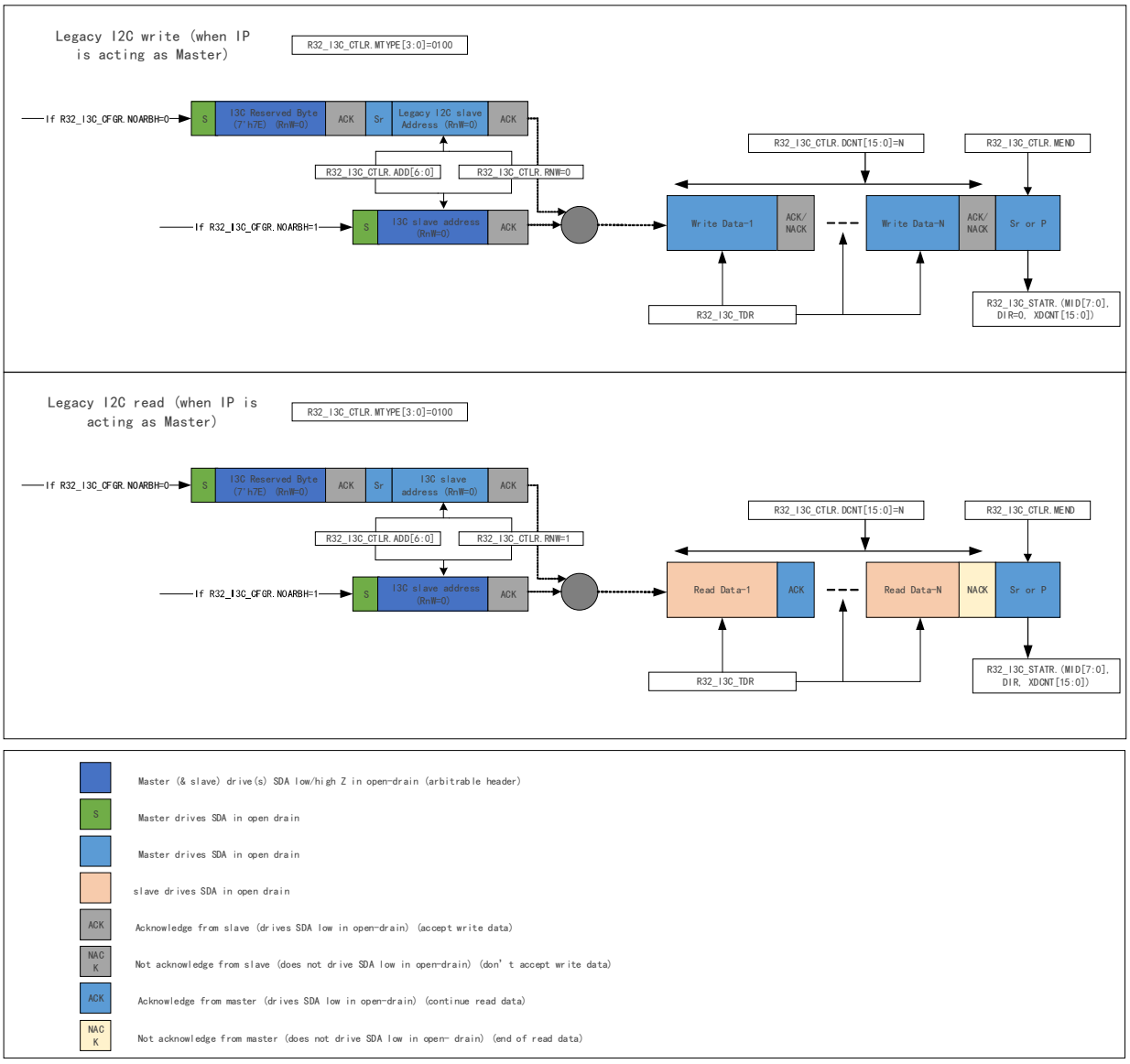
19.3.3.4 I3C Private Read/Write Transfer (as master devices)

Figure 19-5 I3C private read/write transfer (as master devices)



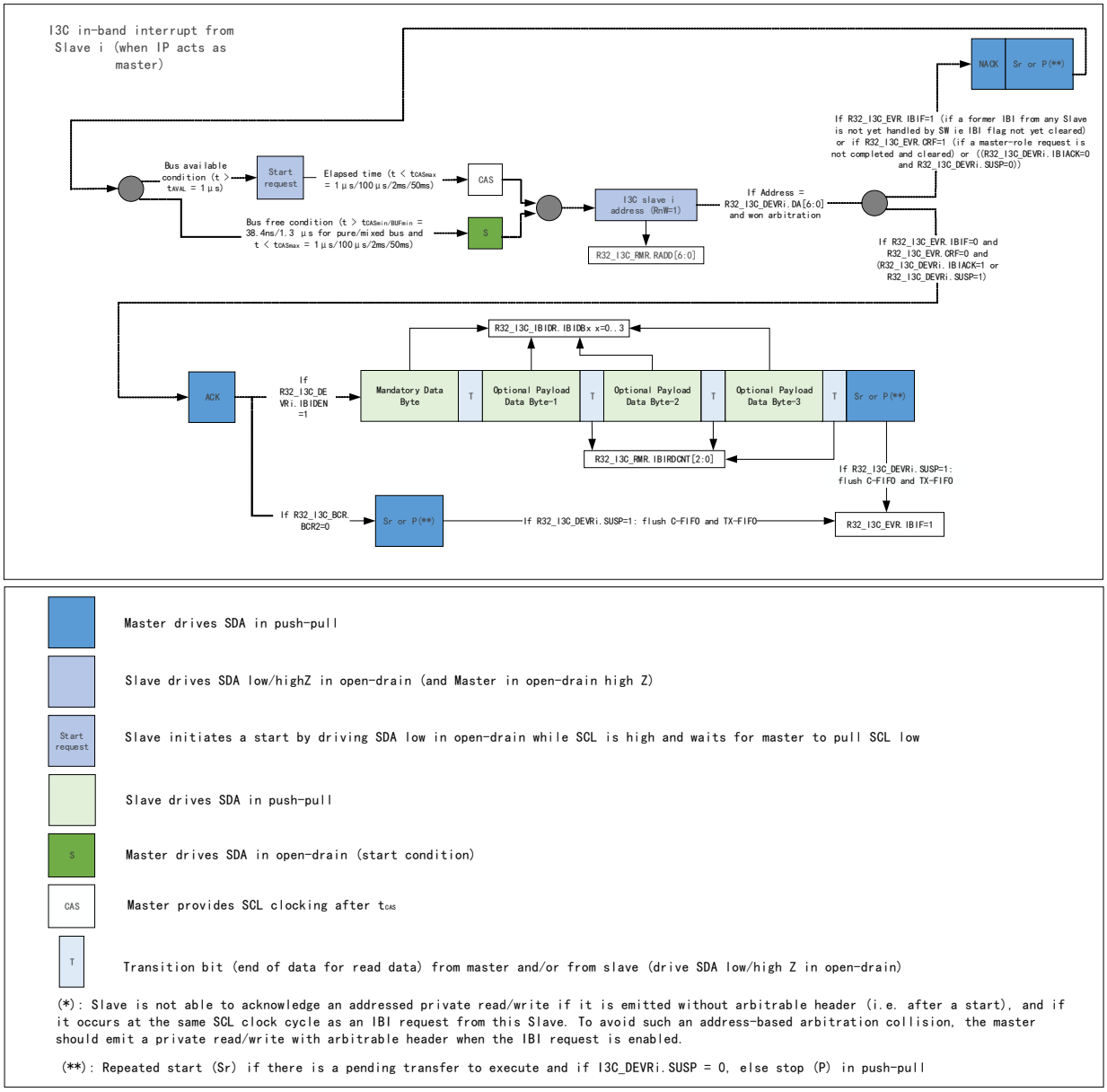
19.3.3.5 Traditional I2C Read/Write Transfer (as master devices)

Figure 19-6 Traditional I2C read/write transfer (as master devices)



19.3.3.6 I3C IBI Transfer (as master devices)

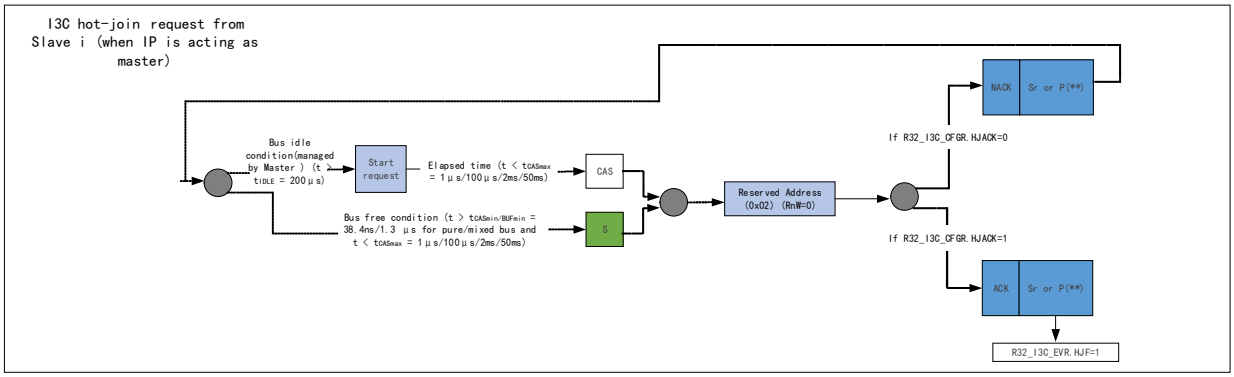
Figure 19-7 I3C IBI Transfer (as master devices)

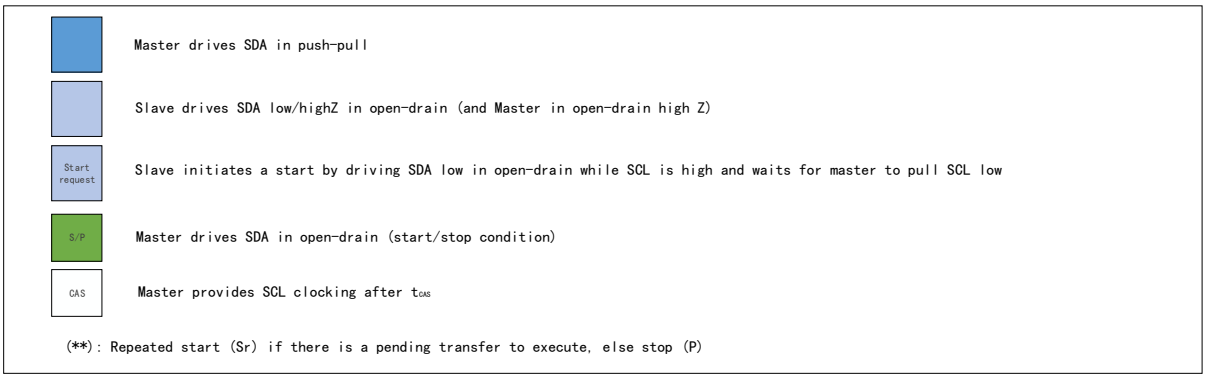


When the peripheral operates as the master device, the R32_I3C_IBIDR register is used to receive the IBI data payload. Therefore, IBI requests from the slave device must not exceed a 4-byte data payload. If more information needs to be exchanged within the context of this in-band interrupt, the master device software must issue a private read control word.

19.3.3.7 I3C Hot Join Request Transfer (as master devices)

Figure 19-8 I3C hot join request transfer (as master devices)





19.3.4 I3C FIFO Management

Table 19-1 I3C FIFO implementation

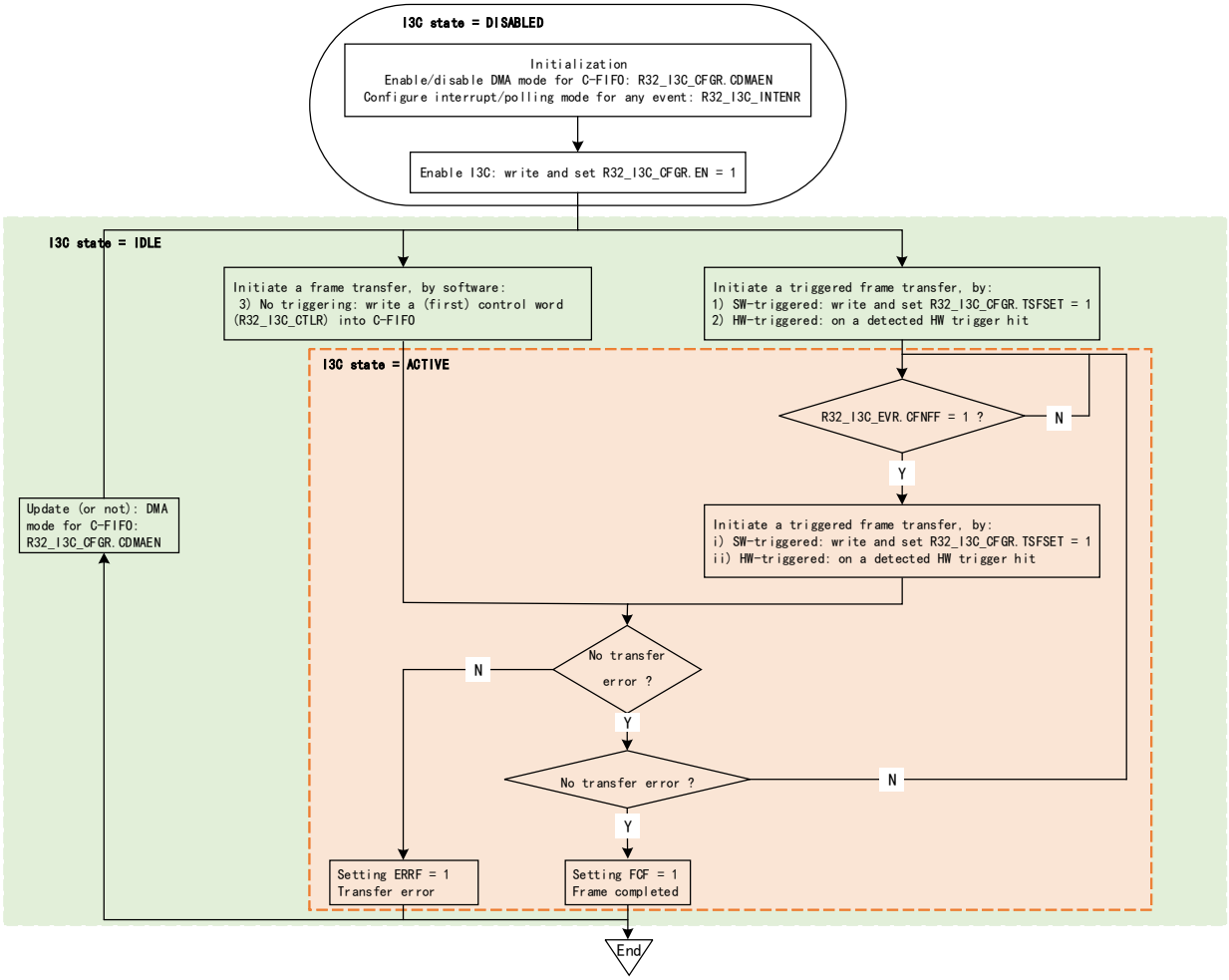
FIFO	Content	Unit	Size	Act as a master/slave (basic principle)
C-FIFO	32-bit control word	Word	2	Master device (a frame can be based on multiple control words; But not when used as a slave device)
TX-FIFO	Transmitted data	Byte	8	Master devices
RX-FIFO	Received data			

19.3.4.1 C-FIFO Management (as master)

When it is used as a master device, it can be used during C-FIFO broadcast CCC, direct read/write CCC, private read/write, traditional I2C read/write, and software-initiated error recovery.

Figure 19-9 illustrates how to manage the C-FIFO to queue the control words on the I3C bus when the I3C peripheral is used as the master device.

Figure 19-9 C-FIFO management (as master)



First, the software must initialize the C-FIFO management through the R32_I3C_CFGR register CDMAEN, and the specific steps are as follows:

- The software writes directly according to the control word (when CDMAEN=0):

- By polling mode (setting CFNFIE=0 in R32_I3C_INTENR register): Wait for the hardware to request the next control word (CFNFF=1 in R32_I3C_INTENR register) before explicitly writing into R32_I3C_CTLR register.
- Through enabled interrupt notification (when CFNFIE=1):

- Written by the assigned DMA channel (when CDMAEN=1) to enable the corresponding I3C peripheral DMA request:

- According to the configuration at the module level, DMA will automatically push/write the control word from its memory source buffer to the R32_I3C_CTLR register until the frame is completed (after the last message of the frame, a stop bit will be issued on the I3C bus), except when a transmission error occurs.

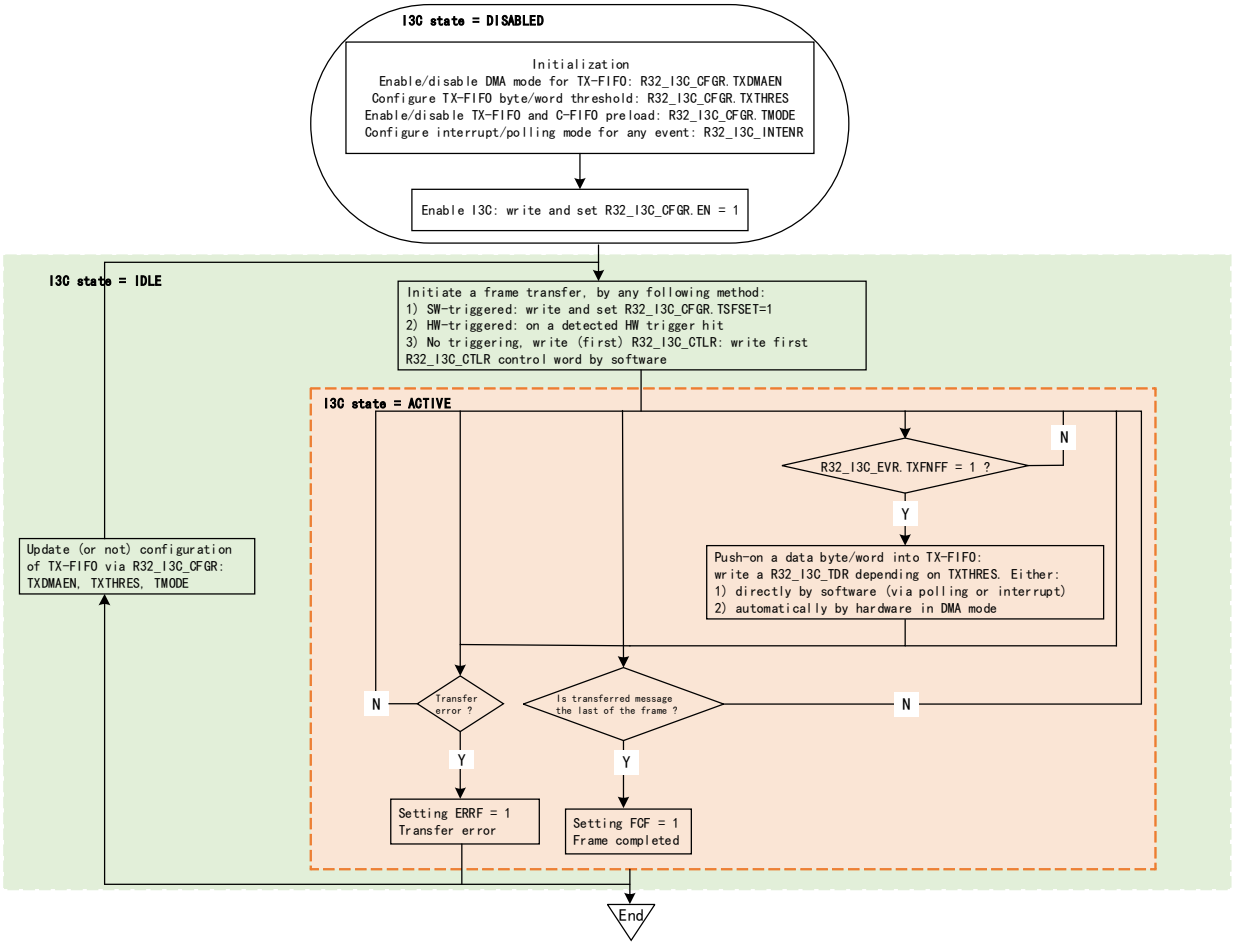
- In any case, as long as the C-FIFO is empty and the repeated start bit of a new control word must be issued, the C-FIFO underflow will be reported (ERRF=1 in the R32_I3C_EVR register and COVR=1 in the R32_I3C_STATR register). If the ERRIE bit in the R32_I3C_INTENR register is enabled, an interrupt is generated.
- When the I3C peripheral is not working, the DMA mode configuration of C-FIFO management can be modified.
- When used as a master device, if there is a transmission error (ERRF=1 in the R32_I3C_EVR register), the hardware will automatically empty the C-FIFO.

19.3.4.2 TX-FIFO Management (as master)

When operating as a master device, the TX-FIFO may be used during broadcasts or direct CCCs (including ENTDAA and RSTACT), when parsing bytes/subcommand bytes or data bytes, during private writes, or during traditional I2C writes.

Figure 19-10 illustrates how the TX-FIFO is managed to queue data bytes or words to be transmitted on the I²C bus when the I²C peripheral operates as a master device.

Figure 19-10 TX-FIFO management (as master)



First, TX-FIFO management must be initialized through the following fields in the R32_I3C_CFGR register:

- TXDMAEN: Enable/disable DMA mode for the TX-FIFO
- TXTHRES: Push data bytes or words into the TX-FIFO
- TMODE: Enable/disable TX-FIFO and C-FIFO preload

Then, according to the TXDMAEN bit in R32_I3C_CFGR register, there are two ways to write data into the TX-FIFO:

- Software writes directly by byte/word (when TXDMAEN=0):

- Through polling mode (R32_I3C_INTENR register TXFNFIE=0): Wait for the hardware to request the next data byte/word (R32_I3C_INTENR register TXFNFF = 1) before explicitly writing to R32_I3C_TDR or R32_I3C_TDWR register, depending on R32_I3C_CFGR register TXTHRES.

- Through enabled interrupt notification (when TXFNFIE = 1)

- Written by the assigned DMA channel (when TXDMAEN = 1) to enable the corresponding I3C peripheral DMA request:

- According to the configuration at DMA module level, DMA will automatically push/write data bytes/words from its memory source buffer into R32_I3C_TDR or R32_I3C_TDWR register (depending on the TXTHRES bit in R32_I3C_CFGR register) until the frame is completed (after the last message of the frame, a stop bit will be issued on I3C bus), except when a transmission error occurs.

I3C messages start with a start bit or a repeat start bit and end with a stop bit or a repeat start bit. At the message level, the last data byte/word to be sent is indicated by the R32_I3C_EVR register TXLASTF=1. When the I3C frame contains multiple messages (separated by repeated start bits), the software can use this event to

update the corresponding pointer to the buffer for storing the byte/word of the next message.

When the report frame completes (R32_I3C_EVR register FCF = 1 and the corresponding interrupt is enabled), the TX-FIFO is empty. If the TX-FIFO is empty and a data byte must be transmitted on the I3C bus, a TX-FIFO underflow is reported (R32_I3C_EVR register ERRF = 1 and R32_I3C_STATER register DOVR = 1). If the corresponding interrupt is enabled (R32_I3C_INTENR register ERRIE = 1), an interrupt is generated. When the I3C peripheral is not active, the configuration managing the TX-FIFO can be modified.

When operating as a master device, the hardware automatically clears the TX-FIFO upon a transmission error (ERRF = 1).

No C-FIFO/TX-FIFO preload

C-FIFO size is 2 words and TX-FIFO size is 8 bytes.

When the preload function of the C-FIFO/TX-FIFO is not enabled, that is, the R32_I3C_CFGR register TMODE=0, once the C-FIFO receives the first control word, the I3C peripheral will immediately send out the start bit on the I3C bus. Subsequently, the peripheral will decode the R32_I3C_CTLR register and write the next data byte or word as needed. Once it is detected that the hardware requires to send the next control word on the I3C bus (for example, it is necessary to send a repeated start bit on the I3C bus or the C-FIFO needs to obtain available space), the peripheral will immediately request to write the control word into the C-FIFO until the last message (that is, R32_I3C_CTLR register MEND=1). Similarly, once it is detected that the hardware requires another data byte or word to be sent on the I3C bus (for example, a repeated start bit needs to be sent on the I3C bus, the TX-FIFO is not full, or the data byte or word must be sent during the I3C message), the data byte/word must be written into the TX-FIFO immediately.

C-FIFO/TX-FIFO preload

When C-FIFO/TX-FIFO preload is configured (R32_I3C_CFGR register TMODE = 1), I3C peripherals need to wait for C-FIFO and TX-FIFO to load as much as possible before sending out the start bit on the bus, as follows:

Wait for the first control word to be written to the C-FIFO. Wait for data bytes/words to be written to the TX-FIFO (if any), as defined by the first control word (if RNW=0 in the R32_I3C_CTLR register and DCNT[15:0]=0), up to the size of the TX-FIFO. If TX-FIFO is not full and the first control word is not the final message in the frame (MEND=0 in the R32_I3C_CTLR register):

- Wait for the second control word to be written into the C-FIFO, after which the C-FIFO becomes full.
- If the TX-FIFO is not full, wait for data bytes/words to be written into the TX-FIFO (if any), as defined by the second control word (RNW=0 in the R32_I3C_CTLR register and DCNT[15:0]), up to the maximum size of the TX-FIFO.

Then, whenever the hardware detects a requirement to transmit the next control word on the I3C bus (if a duplicate start bit must be sent on the I3C bus), it immediately requests that control word be written into the C-FIFO, continuing until the final message (MEND=1 in the R32_I3C_CTLR register).

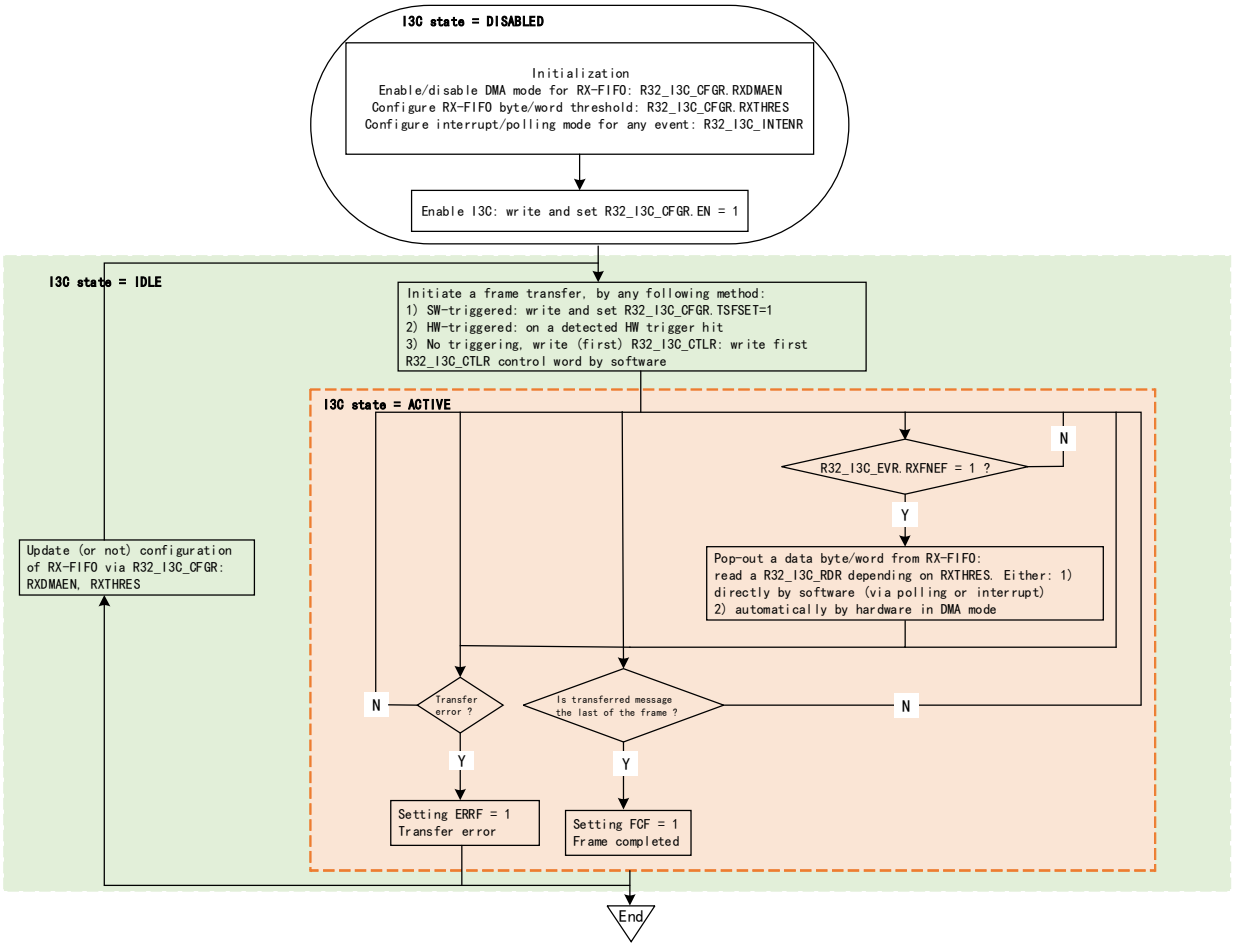
Similarly, whenever a hardware request is detected to send the next data byte/word on the I3C bus (if a repeat start bit must be issued on the I3C bus, the TX-FIFO is not full, or data bytes/words will be sent during that I3C message), that data byte/word must be written to the TX-FIFO immediately.

19.3.4.3 RX-FIFO Management (as master)

When used as a master device, RX-FIFO can be used during broadcasting ENTDAA CCC, direct CCC reading (including direct RSTACT CCC reading), private reading and traditional I2C reading.

Figure 19-11 illustrates how to manage the RX-FIFO to queue and eject data bytes or words to be received on the I3C bus when the I3C peripheral is used as the master device.

Figure 19-11 RX-FIFO management (as master)



First, the software must initialize the RX-FIFO management through the following fields in the R32_I3C_CFGR register:

- RXDMAEN: Enable/disable DMA mode for the RX-FIFO
- RXTHRES: Pop data bytes or words from the RX-FIFO

Then, read the RX-FIFO according to the RXDMAEN bit:

- Software reads directly by byte/word (RXDMAEN=0):

- Through polling mode (RXFNEIE=0 in R32_I3C_INTENR register): Wait for the hardware to request the next data byte/word (RXFNEF=1 in R32_I3C_INTENR register) before explicitly reading R32_I3C_RDR or R32_I3C_RDWR register, depending on R32_I3C_CFGR register.

- Notify via enabled interrupt (if R32_I3C_INTENR register RXFNEIE=1)

- Read by the assigned DMA channel (if RXDMAEN=1) to enable the corresponding I3C peripheral DMA request:

- According to the configuration at DMA module level, DMA will automatically pop up/read data bytes/words from R32_I3C_RDR or R32_I3C_RDWR register (depending on the RXTHRES bit) and write them into its memory slave buffer until the frame is completed (after the last message of the frame, a stop bit will be issued on the I3C bus), except when a transmission error occurs.

I3C messages start with a start bit or a repeat start bit and end with a stop bit or a repeat start bit. At the message level, the last data byte/word received from the I3C bus is indicated by IRXLASTF=1 in the I3C_EVR register. When the I3C frame contains multiple messages (separated by repeated start bits), the software can use this event to update the corresponding pointer to the buffer for storing the data byte/word of the next message.

If the RX-FIFO is full and data bytes are received on the I3C bus, the RX-FIFO overflow will be reported (ERRF=1 in the R32_I3C_EVR register and DOVR=1 in the R32_I3C_STATER register). If the corresponding interrupt is enabled (ERRIE=1 in the R32_I3C_INTENR register), an interrupt is generated.

When the I3C peripheral is not working, the configuration of RX-FIFO management can be modified.

Device early termination of read transfer

Disabling the S-FIFO (SMODE=0 in the R32_I3C_CFGR register) will notify the software of a read transfer premature termination via RXTGTENDF=1 in the R32_I3C_EVR register and the corresponding interrupt (if enabled). The software can then read the status register R32_I3C_STATR to examine information related to the last message and obtain the number of data bytes received during the prematurely terminated read transfer (XDCNT[15:0] in the R32_I3C_STATR register).

In any case, if the S-FIFO is enabled (SMODE=1 in the R32_I3C_CFGR register), the software or DMA (depending on SDMAEN in the R32_I3C_CFGR register) must read the status register R32_I3C_STATR once for each message. The number of valid data bytes received in a preemptively terminated read message is reported by XDCNT[15:0] in the R32_I3C_STATR register (followed by ABT=1).

19.3.5 I3C Master Device Error Management

Table 19-2 Master device error management

Error type	Description	Error detection	Master device operation	Report errors
CE0	Illegally formatted CCC (for example, fewer data bytes are returned)	When direct CCC read transmission, the slave device ends reading data ahead of schedule.	Hardware issue stop bit	ERRF=1, PERR=1 and CODERR[3:0]=0000b
CE1	Monitoring error	An incorrect ACK was detected at the end of the traditional I2C read transmission.	The hardware keeps SCL running for nine clock cycles, and another byte can be exchanged, and then a NACK is issued again, followed by a stop bit.	ERRF=1, PERR=1 and CODERR[3:0]=0001b
		When I3C SDR read transmission ends, repeated start bit or stop bit cannot be generated.	SCL remains running. Software can stop SCL by writing the control word MTYPE[3:0]=0000 (header message) to the R32_I3C_CTLR register, and then it usually has to wait at least 150μs before	
		Unable to generate start bit after CE1 error is detected.		

			sending another message.	
CE2	No response to broadcast address (1111110b)	In the process of message transmission (except for upgrade failure or reset timing message), it is detected that the header (1111110b+RnW=0) is NACK.	The hardware issues HDR exit timing followed by a stop bit.	ERRF=1, PERR=1 and CODERR[3:0]=0010b
CE3	Master device switching failed.	The new master device has not driven SCL low after the SDA signal went low (triggered by a test header or a start request from a slave device) and after the delay time defined in its operating state has elapsed.	The hardware issues the start bit +1111110b+RnW=0, followed by the ACK/NACK from the slave device, and finally the stop bit.	ERRF=1, PERR=1 and CODERR[3:0]=0011b
-	The 7-bit slave address with odd parity returned in GETACCCR CCC is incorrect.	An incorrect dynamic address and/or parity bit was detected.	The hardware cancels the GETACCCR CCC by issuing the restart bit +1111110b+RnW=0, followed by the ACK/NACK from the slave device, and finally the stop bit.	ERRF=1 and DERR=1
-	The addressed slave device was NACK for the first time in the direct CCC read transmission.	NACK detected	The hardware performs a single retry by issuing a duplicate start bit +7 bits of the same slave address +RnW=1.	-
-	The addressed slave device is NACK for the second time in direct CCC write transmission, I3C private read/write transmission, traditional I2C transmission or direct CCC read transmission.	NACK detected	The hardware issues a stop bit.	ERRF=1 and ANACK=1
-	The allocated/sent dynamic address with parity bit was	NACK detected	The hardware performs a single-retry allocation cycle by issuing a	-

	NACK in ENTDAA CCC for the first time.		repeated start bit +1111110b + RnW = 1, followed by an ACK from the (highest priority) slave device and 8 bytes of read data, followed by the allocated address + parity bit.	
-	The allocated/transmitted dynamic address with parity bit was NACK in ENTDAACCC for the second time.	NACK detected	The hardware issues a stop bit.	ERRF=1 and DNACK=1
-	Write data is NACK in traditional I2C write transmission.			
-	Not writing/reading control words, status words, sending data or reading data in time relative to I3C bus timing.	SCL pause timeout	The hardware issues a stop bit.	ERRF=1 and (COVR=1 or DOVR=1)

19.3.6 I3C Low-power Mode

I3C in low-power mode is not affected when it is in sleep mode, and I3C interrupt can make the device exit sleep mode.

When in low power mode, changes on the SCL and SDA buses wake up the system and exit low power mode.

19.3.7 I3C Interrupt

Table 19-3 I3C interrupt request

Interrupt event	Interrupt division	Interrupt enable control	Event flag	Event clearing control
Request control bit	Request	CFNFIE	CFNFF	-
Request data to be transmitted		TXFNFIE	TXFNFF	-
Request data to be received		RXFNEIE	RXFNEF	-
Master device: frame transfer completed		FCIE	FCF	CFCF
IBI request received		IBIIE	IBIF	CIBIF
Hot join request received		HJIE	HJF	CHJF
Error occurred	Error	ERRIE	ERRF	CERRF

19.4 Register Description

Table 19-4 I3C-related registers

Name	Access address	Description	Reset value
R32_I3C_CTLR	0x40024C00	I3C control register	0x00000000
R32_I3C_CFGR	0x40024C04	I3C configuration register	0x00000040
R32_I3C_RDR	0x40024C10	I3C receive data byte register	0x00000000
R32_I3C_TDR	0x40024C18	I3C transmit data byte register	0x00000000
R32_I3C_IBIDR	0x40024C20	I3C IBI payload data register	0x00000000
R32_I3C_RESET	0x40024C2C	I3C reset register	0x03000000
R32_I3C_STATR	0x40024C30	I3C status register	0x00000000
R32_I3C_STATER	0x40024C34	I3C status error register	0x00000000
R32_I3C_RMR	0x40024C40	I3C receive message register	0x00000000
R32_I3C_EVR	0x40024C50	I3C event register	0x00000003
R32_I3C_INTENR	0x40024C54	I3C interrupt enable register	0x00000000
R32_I3C_CEVR	0x40024C58	I3C clear event register	0x00000000
R32_I3C_DEVR0	0x40024C60	I3C device register 0	0x00000000
R32_I3C_DEVR1	0x40024C64	I3C device register 1	0x00000000
R32_I3C_DEVR2	0x40024C68	I3C device register 2	0x00000000
R32_I3C_DEVR3	0x40024C6C	I3C device register 3	0x00000000
R32_I3C_DEVR4	0x40024C70	I3C device register 4	0x00000000
R32_I3C_TIMINGR0	0x40024CA0	I3C timing register 0	0x00000000
R32_I3C_TIMINGR1	0x40024CA4	I3C timing register 1	0x00000000
R32_I3C_TIMINGR2	0x40024CA8	I3C timing register 2	0x00000000
R32_I3C_GETMDSR	0x40024CD0	I3C get maximum data speed register	0x00000000

19.4.1 I3C Control Register (R32_I3C_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MEND	MTYPE[3:0]				Reserved			ADD[6:0]							RNW

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DCNT[15:0]															

Bit	Name	Access	Description	Reset value
31	MEND	WO	Message end type/last message in frame: 1: This message from the master device ends with a stop bit and is the last message in the frame; 0: This message from the master device is followed by a repeat start bit before another message can be sent. *Note: When MTYPE[3:0] = 0100b, this bit cannot be 0.*	0

[30:27]	MTYPE[3:0]	WO	Message type When I3C is used as the master device: 0000: SCL clock forced stop until the next control word is executed; 0001: Header message; 0010: Private message; 0011: Direct message (second part of I3C SDR direct CCC commands); 0100: Traditional I2C message; Others: Reserved.	0
[26:24]	Reserved	RO	Reserved	0
[23:17]	ADD[6:0]	WO	7-bit I3C dynamic/I2C static slave device address. *Note: When I3C operates as a master device, this bit field is used if MTYPE[3:0] is 0010b, 0011b, or 0100b.*	0
16	RNW	WO	Read/Non-Write Message. 1: Read message; 0: Write message. *Note: When I3C operates as a master device, this bit field is used if MTYPE[3:0] is 0010b, 0011b, or 0100b.*	0
[15:0]	DCNT[15:0]	WO	Data count to be transmitted during read/write operations, in bytes: 0x0000: No data to transmit; 0x0001: 1 byte; 0x0002: 2 bytes; ... 0xFFFF: 64KB-1 byte. *When I3C operates as a master device, this bit field is used to set the number of data bytes exchanged on the bus if MTYPE[3:0] is 0010b, 0011b, or 0100b. For private or traditional I2C read/write messages, this bit field must be non-empty.*	0

19.4.2 I3C Control Register[multiplex] (R32_I3C_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MEND	MTYPE[3:0]				Reserved			CCC[7:0]							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DCNT[15:0]															

Bit	Name	Access	Description	Reset value
31	MEND	WO	Message end type/last message in frame:	0

			1: This message from the master device ends with a stop bit and is the last message in the frame; 0: This message from the master device is followed by a repeat start bit before another message can be sent. *Note: When MTYPE[3:0] = 0100b, this bit cannot be 0.*	
[30:27]	MTYPE[3:0]	WO	Message type. When I3C is used as master: 0110: Broadcast/direct CCC commands; Bit [23:16] (CCC[7:0]) is the transmitted 8-bit CCC code; Bit [15:0] (DCNT[15:0]) indicates the number of CCC interpretation bytes, CCC subcommand bytes, or CCC data bytes. If bit [23] = CCC[7] = 1: The message is a direct CCC command transmitted as the first part of an I3C SDR direct CCC command. The direct CCC command (first part) message is: – {S/S + 0b111_1110 +RnW = 0/Sr+*} + (Direct CCC + T) + (8-bit data + T)* + Sr; - After S (start byte), the NOARBH bit in the R32_I3C_CFGR register determines whether to insert an arbitration header (0b111_1110 + RnW = 0). - Sr+*: After Sr (repeat start byte), the hardware automatically inserts it (0b111_1110 + R/W). If bit [23] = CCC[7] = 0: The message is an I3C SDR broadcast CCC command; the transmitted broadcast CCC command message is: - {S/S + 0b111_1110 + RnW = 0/Sr + *} + (Broadcast CCC + T) + (8-bit data + T)* + Sr/P - After S (Start bit), the NOARBH bit determines whether to insert an arbitrable header (0b111_1110 + RnW = 0). - Sr+*: After Sr (Repeat Start bit), the hardware automatically inserts (0b111_1110 + R/W). Other values: Reserved.	0
[26:24]	Reserved	RO	Reserved	0
[23:16]	CCC[7:0]	WO	8-bit CCC code: If bit [23] = CCC[7] = 1, the message is the first part of an I3C SDR direct CCC command. If bit [23] = CCC[7] = 0, the message is an I3C SDR broadcast CCC command (including ENTDAA).	0
[15:0]	DCNT[15:0]	WO	The count (in bytes) of CCC command-related data to be transmitted as CCC interpretation bytes, CCC subcommand bytes, or CCC data bytes:	0

			Linear encoding up to 64KB - 1. 0x0000: No data to transmit; *Note: Mandatory value when issuing an ENTDAA broadcast CCC.* 0x0001: 1 byte; *Note: Mandatory value when issuing a direct/broadcast CCC for RSTACT.* 0x0002: 2 bytes; ... 0xFFFF: 64KB - 1 byte.	

19.4.3 I3C Configuration Register (R32_I3C_CFGR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	TSFSET	Reserved								CFLUSH	CDMAEN	TMODE	Reserved		

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	TXTHRES	TXFLUSH	TXDMAEN	Reserved	RXTHRES	RXFLUSH	RXDMAEN	HJACK	PULLUP_EN	Reserved	EXITPTRN	RSTPTRN	NOARBH	CRINIT	EN

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
30	TSFSET	WO	Frame transfer settings: 1: Hardware sets the CFNFF bit to initiate frame transfer; 0: No action is taken. *Note: If this bit is not set to 1, frame transfer can be initiated via software by directly writing the first control word register (R32_I3C_CTLR) when the C-FIFO is empty (R32_I3C_EVR register's CFEF bit is 1). Then, if the first control word written is not marked as message end (MEND bit in R32_I3C_CTLR register is 0), it will cause the hardware to set CFNFF to 1.*	0
[29:22]	Reserved	RO	Reserved	0
21	CFLUSH	WO	C-FIFO clear: 1: Clear C-FIFO; 0: Do not perform any operation.	0
20	CDMAEN	RW	C-FIFO DMA request enable: 1: Enables DMA mode for C-FIFO; 0: Disables DMA mode for C-FIFO.	0
19	TMODE	RW	Frames transmitted on the I3C bus manage the C-	0

			FIFO and TX-FIFO: 1: Preload the C-FIFO and TX-FIFO before initiating frame transmission. 0: Do not preload the C-FIFO and TX-FIFO before initiating frame transmission. Frame transmission begins immediately when the first control word appears in the C-FIFO.	
[18:15]	Reserved	RO	Reserved	0
14	TXTHRES	RW	TX-FIFO threshold: 1: 1-word/4-byte threshold; 0: 1-byte threshold.	0
13	TXFLUSH	WO	TX-FIFO clear: 1: Clear the TX-FIFO; 0: Do not perform any operation.	0
12	TXDMAEN	RW	TX-FIFO DMA request enable: 1: Enables DMA mode for the TX-FIFO; 0: Disables DMA mode for the TX-FIFO.	0
11	Reserved	RO	Reserved	0
10	RXTHRES	RW	RX-FIFO threshold: 1: 1-word/4-byte threshold; 0: 1-byte threshold.	0
9	RXFLUSH	WO	RX-FIFO clear: 1: Clear the RX-FIFO; 0: Do not perform any operation.	0
8	RXDMAEN	RW	RX-FIFO DMA request enable: 1: Enables DMA mode for the RX-FIFO; 0: Disables DMA mode for the RX-FIFO.	0
7	HJACK	RW	Hot join request acknowledgment: 1: Hot join request acknowledged; 0: Hot join request not acknowledged. *Note: This bit applies only to the master device.*	0
6	PULLUP_EN	RW	SDA pull-up enable: 1: Enable SDA pull-up; 0: In low-power mode, the pull-up can be turned off to reduce power consumption.	1
5	Reserved	RO	Reserved	0
4	EXITPTRN	RW	HDR exit timing enable: 1: After issuing the message header (R32_I3C_CTLR register MTYPE[3:0] set to 0001b), transmit the HDR exit timing. This is used for master device error detection and escalation when the slave device fails to respond to a private message or a direct CCC read. 0: Do not transmit HDR exit timing after issuing the message header (R32_I3C_CTLR register MTYPE[3:0] set to 0001b). This is used to send the header and test bus ownership if a problem is	0

			suspected after a master role switch (where the new master fails to enable its master role by accessing the previous master within the delay defined in the operating state).	
3	RSTPTRN	RW	HDR reset timing enable: 1: Insert HDR reset timing before the stop bit of any transmitted frame (including RSTACT CCC commands); 0: Transmit standard stop bit at frame end.	0
2	NOARBH	RW	No arbitration header after the start bit: 1: No arbitration header - For traditional I2C messages or I3C SDR proprietary read/write messages, the slave device address is sent immediately after the start bit. - This is a more efficient option (when sending the 0x7E arbitrable header is unnecessary), but it can only be used when the master is certain that the addressed slave device cannot simultaneously send an IBI or Master Role Request (to ensure that for IBIs or MRs, there is no misinterpretation or potential conflict between the address sent by the master in open-drain mode and the same address that the slave device could send after the start bit). 0: Arbitrated header (0b111_1110 + RnW = 0) transmitted after the start bit, preceding traditional I2C messages or I3C SDR proprietary read/write messages (default).	0
1	CRINIT	RW	Mark primary/secondary device roles: 1: Master device role; 0: Slave device role.	0
0	EN	RW	I3C enable: 1: Enable I3C; 0: Disable I3C.	0

19.4.4 I3C Receive Data Byte Register (R32_I3C_RDR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								RDB0[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	RDB0[7:0]	RO	8-bit receive data on the I3C bus.	0

19.4.5 I3C Transmit Data Byte Register (R32_I3C_TDR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								TDB0[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	TDB0[7:0]	RO	8-bit transmit data on the I3C bus.	0

19.4.6 I3C IBI Payload Data Register (R32_I3C_IBIDR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IBIDB3[7:0]								IBIDB2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IBIDB1[7:0]								IBIDB0[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	IBIDB3[7:0]	RW	8-bit IBI payload data (latest byte on I3C bus).	0
[23:16]	IBIDB2[7:0]	RW	8-bit IBI payload data (byte following IBIDB1[7:0] on I3C bus).	0
[15:8]	IBIDB1[7:0]	RW	8-bit IBI payload data (bytes following IBIDB0[7:0] on I3C bus).	0
[7:0]	IBIDB0[7:0]	RW	8-bit IBI payload data (the earliest byte on the I3C bus, MDB[7:0] forced data bytes).	0

19.4.7 I3C Reset Register (R32_I3C_RESET)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							HST_SIE_RST	Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															

Bit	Name	Access	Description	Reset value
[31:25]	Reserved	RO	Reserved	0x1
24	HST_SIE_RST	RW	Reset status of the master device: 1: The protocol layer of the master device is in reset state; 0: The protocol layer of the master device is in	1

			operational state;	
[23:0]	Reserved	RO	Reserved	0

19.4.8 I3C Status Register (R32_I3C_STATR)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MID[7:0]								Reserved					DIR	ABT	Reserved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XDCNT[15:0]															

Bit	Name	Access	Description	Reset value
[31:24]	MID[7:0]	RO	Message Identifier/Counter for a Given Frame: When I3C operates as a master device, this bit field identifies the control word message referenced by the R32_I3C_STATR status register (R32_I3C_CTLR). The first message in a frame is identified with MID[7:0] = 0. This bit field is incremented (by hardware) when a new message control word (R32_I3C_CTLR) on the I3C bus is completed. It is reset each time a new frame begins.	0
[23:19]	Reserved	RO	Reserved	0
18	DIR	RO	Message Direction: 1: Read; 0: Write. *Note: ENTDAA CCC is treated as a write command.*	0
17	ABT	RO	Early termination of private read messages from the slave device: 1: Slave device completed early; 0: Slave device did not complete early.	0
16	Reserved	RO	Reserved	0
[15:0]	XDCNT[15:0]	RO	Data counters: Condition: During dynamic address allocation (ENTDAA CCC) - When I3C is used as a master device: The number of slave devices detected on the bus; Condition: For other transfers during the message period - Regardless of whether I3C is used as a master device: The number of data bytes read from the I3C bus or transmitted on the I3C bus during the message period.	0

19.4.9 I3C Status Error Register (R32_I3C_STATER)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					DERR	DNACK	ANACK	COVR	DOVR	STALL	PERR	CODERR[3:0]			

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
10	DERR	RO	Data error: 1: The master detects a data error during the master role switch (GETACC RCCC, formerly GETACC MST) when the received slave address or/and parity bits do not match. The active master retains the master role. 0: No errors detected.	0
9	DNACK	RO	Data not confirmed: 1: The master device detected that the slave device did not acknowledge data bytes during the following processes: i) Traditional I2C write transfer; ii) Second attempt to send a dynamic address during the ENTDAA process. 0: No error detected.	0
8	ANACK	RO	Address not confirmed: 1: The master detects that the slave does not acknowledge static/dynamic addresses in the following procedure: i) Traditional I2C read/write transfer; ii) Direct CCC write transfer; iii) Try direct CCC read transfer for the second time; iv) Private read/write transfer; 0: No errors were detected.	0
7	COVR	RO	C-FIFO underflow or S-FIFO overflow: 1: Master device detected: i) C-FIFO underflow: Control FIFO is empty; a repeat start bit must be issued; ii) S-FIFO overflow: S-FIFO is full; new message ends; 0: No error detected.	0
6	DOVR	RO	RX-FIFO overflow or TX-FIFO underflow: 1: Regardless of whether acting as master or slave, the hardware detects:	0

			i) TX-FIFO underflow: TX-FIFO is empty and must transmit a write data byte; ii) RX-FIFO overflow: RX-FIFO is full and receives a new data byte; 0: No error detected.	
5	Reserved	RO	Reserved	0
4	PERR	RO	Protocol error: 1: Protocol error detected by hardware, whether acting as master or slave device; 0: No error detected.	0
[3:0]	CODERR[3:0]	RO	0000: CE0 Error (Transaction After CCC Transmission): The master device detected an illegally formatted CCC; 0001: CE1 Error (Monitoring Error): The master device detected data transmitted on the bus that did not match expectations; 0010: CE2 Error (No Response to Broadcast Address): The master device detected an unacknowledged broadcast address (0b111_1110); 0011: CE3 Error (Master Role Switch Failure): The master device detected that the new master device failed to drive the bus after the master role switch; Other values: Reserved.	0

19.4.10 I3C Receive Message Register (R32_I3C_RMR)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								RADD[6:0]							Reserved

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RCODE[7:0]								Reserved					IBIRDCNT[2:0]		

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:17]	RADD[6:0]	RO	Received Slave Address: When I3C is configured as a master device, this bit field records the dynamic address received from the slave device during an acknowledged IBI or master role request.	0
16	Reserved	RO	Reserved	0
[15:8]	RCODE[7:0]	RO	Received CCC Code:	0
[7:3]	Reserved	RO	Reserved	0
[2:0]	IBIRDCNT[2:0]	RO	IBI Received Payload Data Count: When I3C is configured as the master device, this bit	0x7

			field records the number of valid data bytes received in the R32_I3C_IBIDR register. *Note: For chips whose fifth digit from the bottom of the lot number is greater than 0, it becomes 0x7 after clearing the IBI request flag.*	

19.4.11 I3C Event Register (R32_I3C_EVR)

Offset address: 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												HJF	Reserved		

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IBIF	Reserved			ERRF	RXTGTENDF	FCF	Reserved	RXLASTF	TXLASTF	RXFNEF	TXFNFF	Reserved	CFNFF	TXFEF	CFEF

Bit	Name	Access	Description	Reset value
[31:20]	Reserved	RO	Reserved	0
19	HJF	RO	Hot-joined Flag: When I3C operates as the master device, this flag is set to 1 by hardware to indicate a received hot-joined request. This flag is cleared by writing a 1 to the CHJF bit of the R32_I3C_CEVR register via software.	0
[18:16]	Reserved	RO	Reserved	0
15	IBIF	RO	IBI Flag: When I3C operates as a master device, this flag is set to 1 by hardware to indicate an IBI request has been received. This flag is cleared by writing a 1 to the CIBIF bit in the R32_I3C_CEVR register via software.	0
[14:12]	Reserved	RO	Reserved	0
11	ERRF	RO	Flag: This flag is set by hardware to indicate an error has occurred. Subsequently, software must read the R32_I3C_STATR register to determine the error type. This flag is cleared by writing a 1 to the CERRF bit in the R32_I3C_CEVR register via software.	0
10	RXTGTENDF	RO	Read End Flag Initiated by Slave Device: When I3C operates as a master device, and only when the S-FIFO is disabled (SMODE bit in the R32_I3C_CFGR register is 0), this flag is set to 1 by hardware to indicate that the slave device has prematurely terminated the read transfer. Software	0

			must then read the status register R32_I3C_STATR to examine information related to the last message and retrieve the number of data bytes received during the prematurely terminated read transfer (XDCNT in the R32_I3C_STATR register). This flag is cleared by writing a 1 to the CRXTGTENDF bit in the R32_I3C_CEVR register via software.	
9	FCF	RO	Frame Complete Flag: When I3C is used as a master device, this flag is set to 1 by hardware to indicate that a frame on the I3C bus has completed (normally), such as when a stop bit is issued. This flag is cleared by writing a 1 to the CFCF bit of the R32_I3C_CEVR register via software.	0
8	Reserved	RO	Reserved	0
7	RXLASTF	RO	Last Data Byte/Word Flag: When I3C operates as a master device, this flag is set by hardware to indicate the last data byte/word of a message that must be read from the RX-FIFO (depending on the RXTHRES bit in the I3C_CFGR register). This flag is cleared by hardware when the last data byte/word of the message is read.	0
6	TXLASTF	RO	Last Data Byte/Word Written Flag: This flag is set by hardware to indicate the last data byte/word of a message that must be written to the TX-FIFO (depending on the TXTHRES bit in the R32_I3C_CFGR register). When the last data byte/word of the message is written, this flag is cleared by hardware.	0
5	RXFNEF	RO	RX-FIFO Not Empty Flag: This flag is set/cleared by hardware to indicate whether data bytes must/must not be read from the RX-FIFO. *Note: Before reading data bytes from the RX-FIFO (by executing R32_I3C_RDR or R32_I3C_RDWR depending on RXTHRES), the software must wait for RXFNEF = 1 (via polling or an enabled interrupt).*	0
4	TXFNFF	RO	TX-FIFO Not Empty Flag: This flag is set/cleared by hardware to indicate whether data bytes must/must not be written to the TX-FIFO. *Note: Before writing to the TX-FIFO (writing to R32_I3C_TDR depending on TXTHRES), software must wait for TXFNFF = 1 (via polling or an enabled interrupt).*	0

3	Reserved	RO	Reserved	0
2	CFNFF	RO	C-FIFO Not Full Flag: When I3C operates as a master device, this flag is set by hardware to indicate that a control word must be written to the C-FIFO. This flag is cleared by hardware to indicate that no control word is written to the C-FIFO. *Note: Before writing to the C-FIFO (writing to R32_I3C_CTLR), the software must wait for CFNFF = 1 (either through polling or via an enabled interrupt).*	0
1	TXFEF	RO	TX-FIFO Empty Flag: This flag is set to 1 by hardware to indicate the TX-FIFO is empty; This flag is cleared by hardware to indicate the TX-FIFO is not empty.	1
0	CFEF	RO	C-FIFO Empty Flag: When acting as the master device: This flag is set to 1 by hardware to indicate the C-FIFO is empty; This flag is cleared by hardware to indicate the C-FIFO is not empty. *Note: If C-FIFO and TX-FIFO preloading are configured (R32_I3C_CFGR register TMODE bit set to 1), the software must wait for TXFEF = 1 and CFEF = 1 before initiating a new frame transmission.*	1

19.4.12 I3C Interrupt Enable Register (R32_I3C_INTENR)

Offset address: 0x54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												HJIE	Reserved		

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IBIIE	Reserved			ERRIE	RXTGTENDIE	FCIE	Reserved			RXFNEIE	TXFNEIE	Reserved	CFNFIE	Reserved	

Bit	Name	Access	Description	Reset value
20	Reserved	RO	Reserved。	0
19	HJIE	RW	Hot join interrupt enable: 1: Enable; 0: Disable. *Note: This bit applies only to slave devices.*	0
18	CRUPDIE	RW	Master role update interrupt enable:	0

			1: Enable; 0: Disable. *Note: This bit applies only to slave devices.*	
17	CRIE	RW	Master role request interrupt enable: 1: Enable; 0: Disable. *Note: This bit applies only to master devices.*	0
16	IBIENDIE	RW	IBI end interrupt enable: 1: Enable; 0: Disable. *Note: This bit applies only to slave devices.*	0
15	IBIIE	RW	IBI request interrupt enable: 1: Enable; 0: Disable. *Note: This bit applies only to master devices.*	0
[14:12]	Reserved	RO	Reserved	0
11	ERRIE	RW	Error interrupt enable: 1: Enable; 0: Disable.	0
10	RXTGTENDIE	RW	Read end interrupt enable for slave device: 1: Enable; 0: Disable. *Note: This bit applies only to master devices.*	0
9	FCIE	RW	Frame completion interrupt enable: 1: Enable; 0: Disable.	0
[8:6]	Reserved	RO	Reserved	0
5	RXFNEIE	RW	RX-FIFO non-empty interrupt enable: 1: Enable; 0: Disable.	0
4	RXFNEIE	RW	TX-FIFO non-full interrupt enable: 1: Enable; 0: Disable.	0
3	SFNEIE	RW	S-FIFO non-empty interrupt enable: 1: Enable; 0: Disable.	0
2	CFNFIE	RW	C-FIFO non-full interrupt enable: 1: Enable; 0: Disable.	0
[1:0]	Reserved	RO	Reserved	0

19.4.13 I3C Clear Event Register (R32_I3C_CEVR)

Offset address: 0x58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												CH	Reserved		

												JF			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CIBIF	Reserved			CERRF	CRXTGTENDF	CFCF	Reserved								

Bit	Name	Access	Description	Reset value
[31:20]	Reserved	RO	Reserved	0
19	CHJF	WO	Clear hot join flag: 1: Clear HJF; 0: No effect.	0
[18:16]	Reserved	RO	Reserved	0
15	CIBIF	WO	Clear IBI request flag: 1: Clear IBIF; 0: No effect.	0
[14:12]	Reserved	RO	Reserved	0
11	CERRF	WO	Clear error flag: 1: Clear ERRF; 0: No effect.	0
10	CRXTGTENDF	WO	Clear read end flag for slave device: 1: Clear RXTGTENDF; 0: No effect.	0
9	CFCF	WO	Clear frame completion flag: 1: Clear FCF; 0: No effect.	0
[8:0]	Reserved	RO	Reserved	0

19.4.14 I3C Device Register 0 (R32_I3C_DEVR0)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DA[6:0]							DAVAL

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:1]	DA[6:0]	RW	7-bit dynamic address: When I3C operates as a master device, this bit field can be written by software to define its own dynamic address.	0
0	DAVAL	RW	Dynamic address valid:	0

			When I3C operates as a master device, this bit may be written by software to validate its own dynamic address.	

19.4.15 I3C Device Register x (R32_I3C_DEVRx) (x=1/2/3/4)

Offset address: 0x64, 0x68, 0x6C, 0x70

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DIS	Reserved											SUSP	IBIDEN	CRACK	IBIACK

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DA[6:0]							Reserved

Bit	Name	Access	Description	Reset value
31	DIS	RO	DA[6:0] write disable: 1: Disables/locks write operations to DA[7:0] and IBIDEN; 0: Enables write operations to the R32_I3C_DEVRx register's DA[7:0] and IBIDEN.	0
[30:20]	Reserved	RO	Reserved	0
19	SUSP	RW	Suspending/Stopping I3C transmission upon receiving IBI: 1: Upon receiving an IBI request from slave device x, stop the I3C transmission and clear the C-FIFO and TX-FIFO; 0: Upon acknowledging and completing the IBI request from slave device x, do not clear the C-FIFO and TX-FIFO, and issue a repeat start bit or stop bit based on whether the next control word exists.	0
18	IBIDEN	RW	IBI data enabled: 1: Confirmed IBI from slave device x followed by mandatory data bytes MDB[7:0]; 0: Confirmed IBI from slave device x with no data bytes.	0
17	Reserved	RO	Reserved	0
16	IBIACK	RW	IBI request acknowledgment: 1: Must acknowledge the IBI request originating from slave device x (via 7-bit dynamic address DA[6:0]) - The -DIS bit field is set to 1 by hardware to protect DA[6:0] from software modification, while the hardware internally stores the current DA[6:0] into the core clock domain. - After ACK, the master device records the IBI	0

			payload data (if present) based on I3C_DEVRx.IBIDEN. - Software is notified via the IBI flag (IBIF=1) and/or the corresponding interrupt (if enabled); -Regardless of the IBIACK configuration on this or other devices, all subsequent IBI requests will be NACKed until both the IBI request flag (IBIF) and the master role request flag (CRF) are cleared. 0: After the NACK for the IBI request from slave device x is received, message transmission resumes according to the initial programming (the slave device has received the NACK and may issue another IBI request later).	
[15:8]	Reserved	RO	Reserved	0
[7:1]	DA[6:0]	RW	I3C dynamic address assigned from slave device x: When I3C operates as the master device, this bit field must be written by software to store the 7-bit dynamic address sent by the master device via a broadcast ENTDAA acknowledged by slave device x or a direct SETNEWDA CCC command.	0
0	Reserved	RO	Reserved	0

19.4.16 I3C Timing Register 0 (R32_I3C_TIMINGR0)

Offset address: 0xA0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SCLH_I2C[7:0]								SCLL_OD[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SCLH_I3C[7:0]								SCLL_PP[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	SCLH_I2C[7:0]	RW	SCL high duration for traditional I2C messages, in microsecond units: $t_{SCLH_I2C} = (SCLH_I2C + 1) \times t_{HCLK}$	0
[23:16]	SCLL_OD[7:0]	RW	The duration of the low level of the SCL pin during the open-drain phase, measured in core clock cycles, for traditional I2C messages and I3C open-drain phases (address phase following the start bit, ACK phase during master-initiated messages, and T-bit phase during direct/private/IBI payloads): $t_{SCLL_OD} = (SCLL_OD + 1) \times t_{HCLK}$	
[15:8]	SCLH_I3C[7:0]	RW	SCL high level duration for I3C messages (in push-pull and open-drain phases), measured in core clock cycles: $t_{SCLH_I3C} = (SCLH_I3C + 1) \times t_{HCLK}$	

[7:0]	SCLL_PP[7:0]	RW	The low-level duration of SCL in the I3C push-pull stage, measured in core clock cycles: t_{SCLL_PP} = （SCLL_PP + 1）× t_{HCLK}	

19.4.17 I3C Timing Register 1 (R32_I3C_TIMINGR1)

Offset address: 0xA4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			SDA_HD	Reserved					FREE[6:0]						

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						ASNCR[1:0]		AVAL[7:0]							

Bit	Name	Access	Description	Reset value
[31:29]	Reserved	RO	Reserved	0
28	SDA_HD	RW	SDA hold time in core clock cycles. SDA hold time = (SDA_HD + 0.5)× t_{HCLK}	0
[27:23]	Reserved	RO	Reserved	0
[22:16]	FREE[6:0]	RW	The number of core clock cycles used to set certain MIPI timing parameters (e.g., bus idle condition time).	0
[15:10]	Reserved	RO	Reserved	0
[9:8]	ASNCR[1:0]	RW	Operating status of new master device: The time required to wait before accessing as a new slave device. Refer to AVAL[7:0].	0
[7:0]	AVAL[7:0]	RW	Number of core clock cycles used to set the 1μs time unit: The hardware then uses this time unit to construct several internal timers corresponding to the following MIPI I3C timing: When the I3C is used as a master, its clock pause time must not exceed the maximum pause time (pause SCL clock low), as follows: 1. For the first bit of the address allocated during dynamic address allocation: the pause clock shall not exceed the MIPI timing $t_{STALLDAA}$ = 15ms. The timing is defined as: $t_{STALLDAAmax}$ = (AVAL[7:0] + 1) × 15000 × t_{HCLK} 2. ACK/NACK phase for I3C/I2C transmission, parity bit for write data transmission, transition bit for I3C read transmission: pause clock time must not exceed MIPI timing t_{STALL} = 100μs.The timing is defined as: $t_{STALLmax}$ = (AVAL[7:0] + 1) × 100 × t_{HCLK}.	0

19.4.18 I3C Timing Register 2 (R32_I3C_TIMINGR2)

Offset address: 0xA8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STALL[7:0]								Reserved				STALLA	STALLC	STALLD	STALLT

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:8]	STALL[7:0]	RW	Master clock stall time in core clock cycles: t_{SCLL_STALL} = STALL × t_{HCLK}	0
[7:4]	Reserved	RO	Reserved	0
3	STALLA	RW	ACK phase master clock stall enable: 1: Enable stall; 0: Disable stall.	0
2	STALLC	RW	CCCPAR phase master clock stall enable: 1: Enable stall; 0: Disable stall.	0
1	STALLD	RW	Data PAR phase master clock stall enable: 1: Enable stall; 0: Disable stall.	0
0	STALLT	RW	Master clock stall enable for data T-bit phase (and ACK/NACK phase for data bytes of legacy I2C read messages): 1: Enable stall; 0: Disable stall.	0

19.4.19 I3C Get Maximum Data Speed Register (R32_I3C_GETMDSR)

Offset address: 0xD0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							TSCO	Reserved							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															

Bit	Name	Access	Description	Reset value
[31:25]	Reserved	RO	Reserved	0
24	TSCO	RW	Clock to data turn time (t_{SCO}): 1: t_{SCO} > 12ns; 0: t_{SCO} ≤ 12ns.	0
[23:0]	Reserved	RO	Reserved	0

Chapter 20 Serial Peripheral Interface (SPI/I2S)

SPI supports data interaction in 3-wire synchronous serial mode, and the chip select line supports hardware switching between master and slave modes, supporting communication with a single data line.

I2S is also a 3-wire synchronous serial interface communication protocol, which supports 4 audio standards, including Philips I2S standard, MSB alignment standard, LSB alignment standard and PCM standard.

20.1 Main Features

20.1.1 SPI Features

- Support full-duplex synchronous serial mode
- Support 1-wire half-duplex mode
- Support master mode and slave mode, multi-slave mode
- Support 8-bit or 16-bit data structures
- The highest clock frequency is supported to half of F_{pclk}
- Data ordering supports MSB or LSB first
- Support hardware or software control of the NSS pin
- Transceiver supports hardware CRC
- Transceiver buffer supports DMA transfer
- Support modifying clock phase and polarity

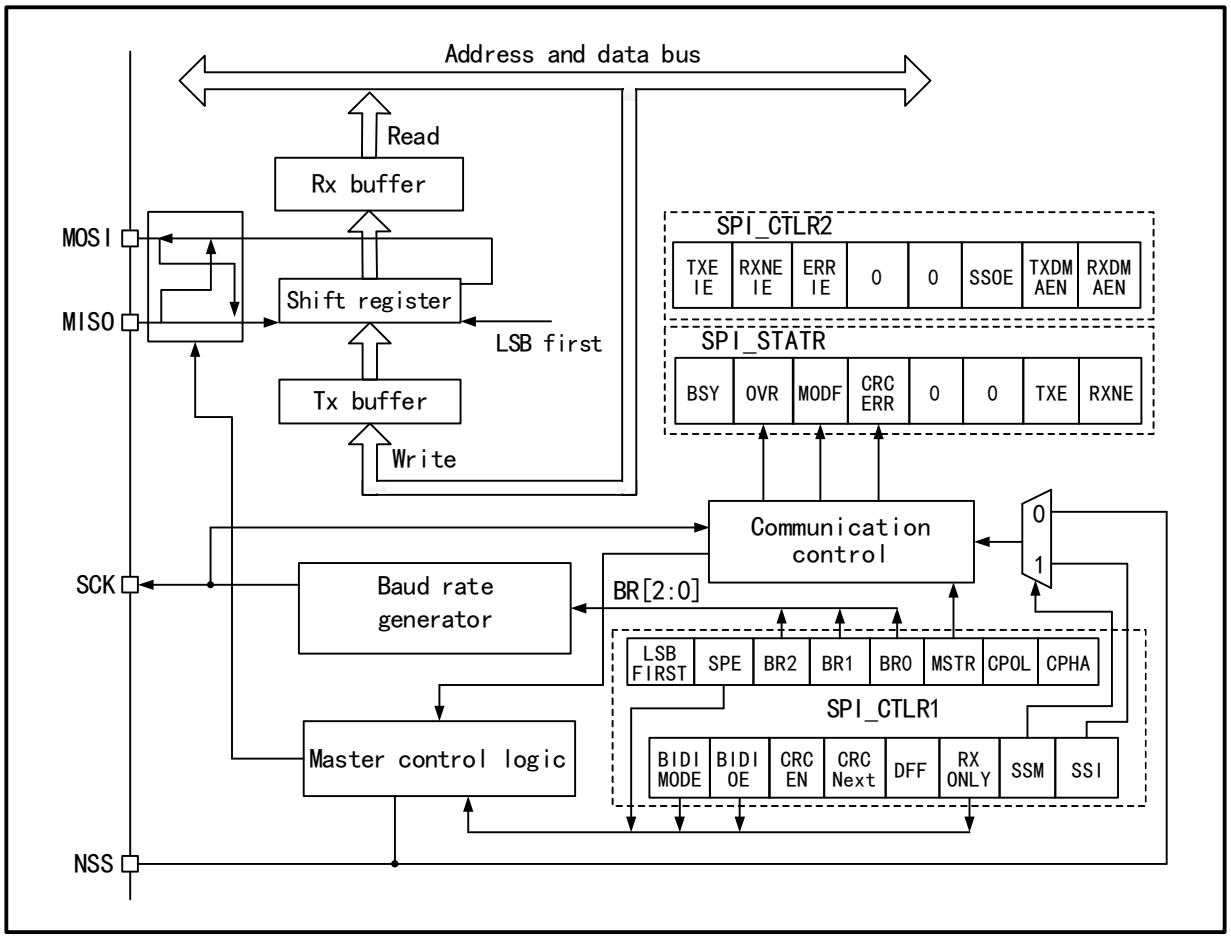
20.1.2 I2S Features

- Support simplex communication
- Support master mode and slave mode
- Support 16-bit, 24-bit and 32-bit data formats
- Audio sampling frequency support range 8kHz-562.2kHz
- Support clock polarity programmable
- Support common I2S protocols: Philips standard, MSB alignment standard, LSB alignment standard and PCM standard
- Transceiver buffer supports DMA transfer
- Support master clock output to external audio equipment

20.2 SPI functional Description

20.2.1 Overview

Figure 20-1 SPI structure diagram



As can be seen from Figure 20-1, the 4 pins related to SPI are MISO, M0SI, SCK and NSS. The MISO pin is the data input pin when the SPI module works in the master mode; it is the data output pin when it works in the slave mode. When the MOSI pin works in master mode, it is a data output pin; when it works in slave mode, it is a data input pin. SCK is the clock pin, the clock signal is always output by the master, and the slave receives the clock signal and synchronizes data transmission and reception. The NSS pin is a chip select pin and has the following uses:

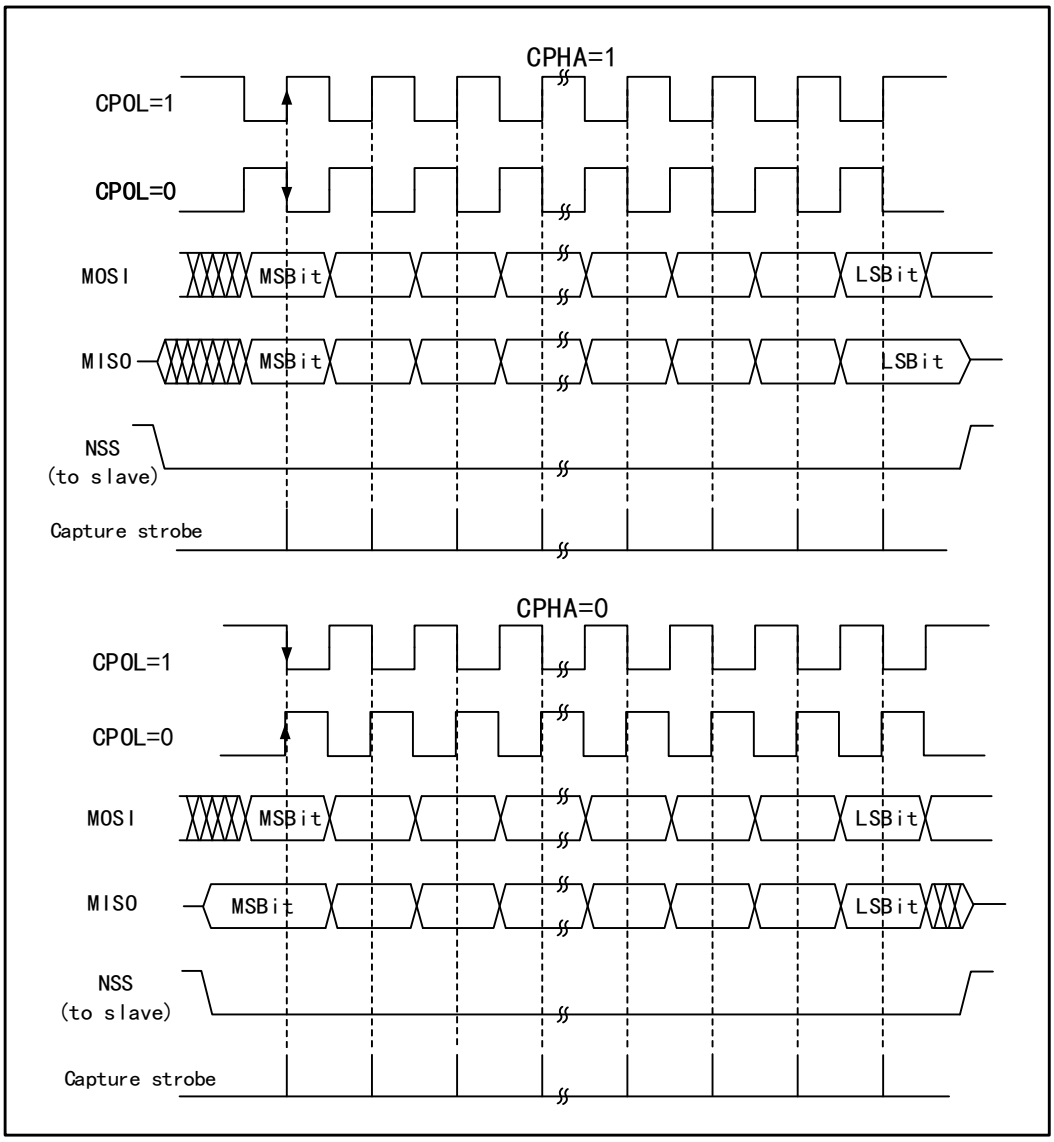
1） NSS is controlled by software: At this time, SSM is set, and the internal NSS signal is determined by SSI to output high or low. This situation is generally used in SPI master mode;

2） NSS is controlled by hardware: When the NSS output is enabled, i.e., when SSOE is set, the NSS pin will be actively pulled down when the SPI host sends the output outward, and a hardware error will be generated if the NSS pin cannot be successfully pulled down and pulled down, indicating that there are other host devices on the main line that are communicating; SSOE If it is not set, it can be used in multi-master mode. If it is pulled low, it will be forced into slave mode, and the MSTR bit will be automatically cleared.

The working mode of the SPI can be configured through CPHA and CPOL. If CPHA is set, it means that the module samples the data on the second edge of the clock, and the data is latched. If CPHA is not set, it means that the SPI module samples on the first edge of the clock, and the data is latched. CPOL indicates whether the

clock remains high or low when there is no data. See Figure 20-2 below for details.

Figure 20-2 SPI model



The host and device need to be set to the same SPI mode, and the SPE bit needs to be cleared before configuring the SPI mode. The DEF bit can decide whether the single data length of SP is 8 bits or 16 bits. LSBFIRST can control whether a single data word is big-endian or little-endian.

20.2.2 Master Mode

When the SPI module works in master mode, the serial clock is generated by SCK. To configure into master mode do the following steps:

Configure the BR[2:0] bits in the control register to determine the clock;

Configure the CPOL and CPHA bits to determine the SPI mode;

Configure DEF to determine the data word length;

Configure LSBFIRST to determine the frame format;

Configure the NSS pin, such as setting the SSOE bit to let the hardware reset NSS. You can also set the SSM bit and set the SSI bit high;

To set the MSTR bit and the SPE bit, it is necessary to ensure that NSS is already high at this time.

When you need to send data, you only need to write the data to be sent to the data register. SPI will send data from the transmit buffer to the shift register in parallel, and then send the data from the shift register according to the setting of LSBFIRST. When the data has reached the shift register, the TXE flag will be set. If it has been set TXEIE, then an interrupt will be generated. If the TXE flag bit is set, data needs to be filled into the data register to maintain a complete data flow.

When the receiver receives data, when the last sampling clock edge of the data word arrives, the data is transferred from the shift register to the receive buffer in parallel, the RXNE bit is set, and if the RXNEIE bit was previously set, it will also generate interrupt. At this time, the data register should be read as soon as possible to remove the data.

20.2.3 Slave Mode

When the SPI module works in slave mode, SCK is used to receive the clock sent by the host, and its own baud rate setting is invalid. The steps to configure into slave mode are as follows:
Configure the DEF bit to set the data bit length;
Configure the CPOL and CPHA bits to match the host mode;
Configure LSBFIRST to match the host data frame format;
In hardware management mode, the NSS pin needs to be kept at a low level. If NSS is set to software management (SSM is set), then please keep SSI not set;
Clear the MSTR bit and set the SPE bit to enable SPI mode.

When transmitting, when the first slave receive sampling edge occurs on SCK, the slave starts to transmit. The process of sending is to move the data in the send buffer to the send shift register. When the data in the send buffer is moved to the shift register, the TXE flag will be set. If the TXEIE bit was previously set, an interrupt will be generated.

When receiving, after the last clock sampling edge, the RXNE bit is set, the byte received by the shift register is transferred to the receive buffer, and the read operation of the read data register can obtain the data in the receive buffer. An interrupt will be generated if RXNEIE is set before RXNE is set.

20.2.4 Simplex Mode

The SPI interface can work in half-duplex mode, that is, the master device uses the MOSI pin, and the slave device uses the MISO pin for communication. When using half-duplex communication, BIDIMODE needs to be set, and BIDIOE is used to control the transmission direction.

Setting the RXONLY bit in normal full-duplex mode can set the SPI module to receive-only simplex mode. After RXONLY is set, a data pin will be released. The pins released in master mode and slave mode are not the same. It is also possible to ignore the received data and set the SPI to transmit-only mode.

20.2.5 CRC

The SPI module uses CRC to ensure the reliability of full-duplex communication, and separate CRC calculators are used for data transmission and reception. The polynomial of the CRC calculation is determined by the polynomial register. For 8-bit data width and 16-bit data width, different calculation methods are used respectively.

Setting the CRCEN bit enables CRC and resets the CRC calculator. After the last data byte is sent, setting the CRCNEXT bit will send the calculation result of the TXCRCR calculator after the current byte is sent. At the

same time, if the last received value of the receive shift register is not the same as the locally calculated value of RXCRCR, then the CRCERR bit will be set. To use the CRC, you need to set the polynomial calculator and set the CRCEN bit when configuring the SPI working mode, and set the CRCNEXT bit in the last word or half word to send the CRC and check the received CRC. Note that the CRC calculation polynomials of the sender and receiver should be unified.

20.2.6 DMA

The SPI module supports the use of DMA to speed up data communication. You can use DMA to fill in data into the send buffer, or use DMA to take data from the receive buffer in time. DMA will take or send data in time with RXNE and TXE as signals. DMA can also work in simplex or CRC mode.

20.2.7 Error

- Master mode failure error

When the SPI works in the NSS pin hardware management mode, an external operation of pulling down the NSS pin occurs; or in the NSS pin software management mode, the SSI bit is cleared; or the SPE bit is cleared, causing the SPI to be turned off; or the MSTR bit is cleared, the SPI enters slave mode. An interrupt will also be generated if the ERRIE bit has been set. Clear MODF bit steps: first perform a read or write operation to R16_SPI1_STATR, and then write to R16_SPI1_CTLR1.

- Overflow error

If the master sends data and there is unread data in the slave's receive buffer, an overrun error occurs, the OVR bit is set, and an interrupt is generated if ERRIE is set. Sending an overflow error should restart the current transfer. Reading the data register followed by the status register will clear this bit.

- CRC error

When the received CRC word does not match the value of RXCRCR, a CRC error will occur, and the CRCERR bit will be set.

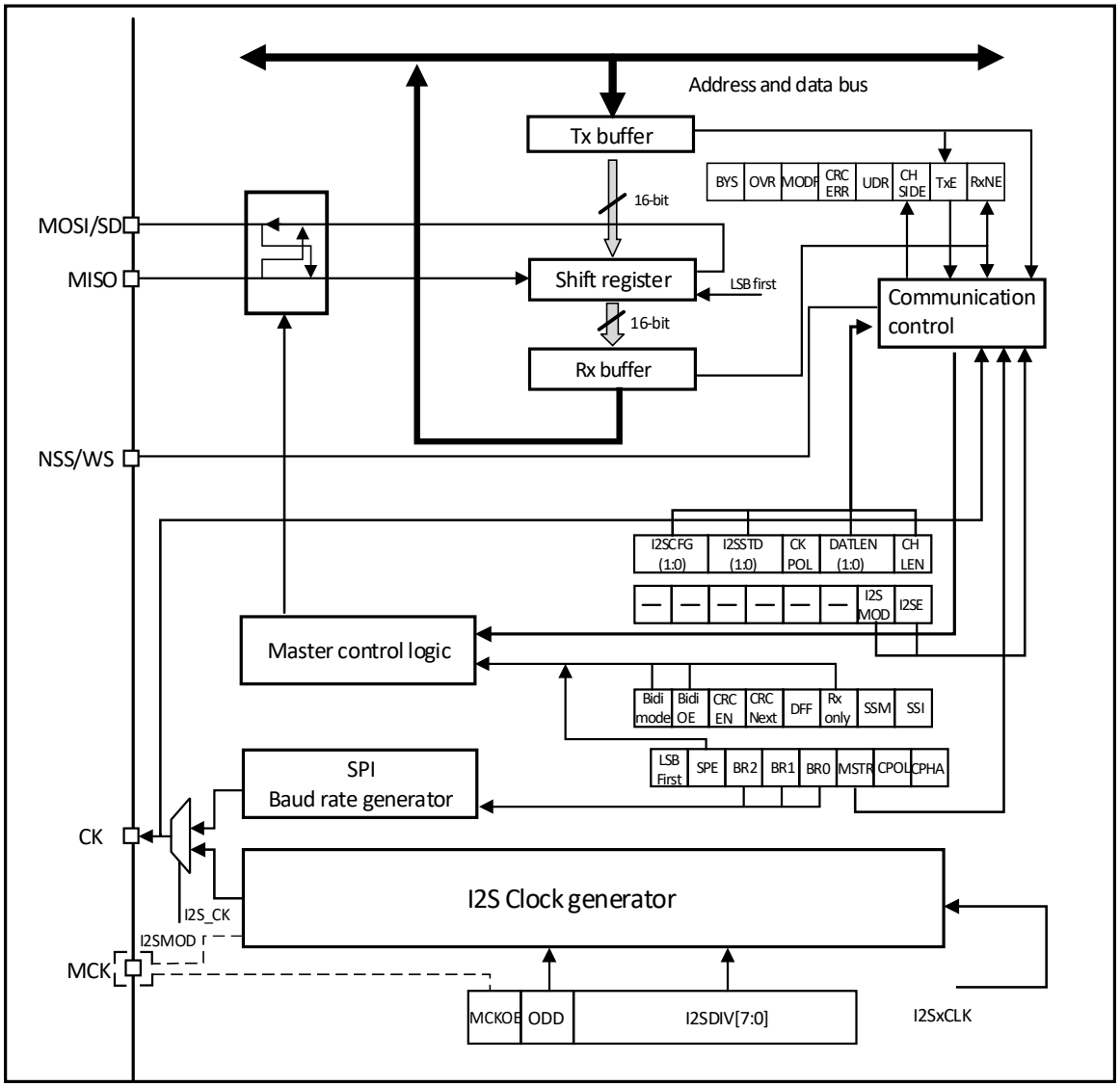
20.2.8 Interrupt

The interrupt of the SPI module supports 5 interrupt sources. The 2 events of the send buffer empty and the receive buffer non-empty will set TXE and RXNE respectively, and an interrupt will be generated when the TXEIE and RXNEIE bits are set respectively. In addition, the 3 errors mentioned above will also generate interrupts, namely MODF, OVR and CRCERR. After the ERRIE bit is enabled, these 3 errors will also generate error interrupts.

20.3 I2S Functional Description

20.3.1 I2S Overview

Figure 20-3 I2S structure diagram



The I2S function is enabled by setting the I2SMOD bit in register I2SCFGR. At this point, the SPI module can be used as an I2S audio interface. I2S and SPI share 3 pins:

- SD: Serial data (Mapped to MOSI pin), used to send and receive 2 time-multiplexed data channels;
- WS: word selection (Mapped to NSS pin), output as data control signal in master mode, and input in slave mode;
- CK: Serial clock (Mapped to the SCK pin), output as a clock signal in master mode, and input in slave mode.

When the main clock is required for some external audio devices, there is an additional pin to output the clock:

- MCK: main clock (Independent mapping), when the I2S is configured as the main mode and the MCKOE bit of the register I2SPR is 1, it is used as an output additional clock signal pin. The frequency of the output clock signal is preset to 256×Fs, where Fs is the sampling frequency of the audio signal.

When set to master mode, I2S uses its own clock generator to generate the clock signal for communication. This clock generator is also the clock source for the master clock output. There are 2 additional registers in I2S mode, one is the clock generator configuration related register I2SPR, the other is the I2S general

configuration register I2SCFGR (Can set the audio standard, slave/master mode, data format, packet frame, clock pole parameters, etc.). Register CTLR1 and all CRCR registers are not used in I2S mode. Similarly, the SSOE bit in the CTLR2 register, and the MODF bit and the CRCERR bit in the register STATR are not used in the I2S mode. I2S uses the same register DATAR as SPI for 16-bit wide mode data transfer.

20.3.2 Supported Audio Protocols

The 3-wire bus supports time division multiplexing of audio data on 2 channels: left and right, but only one 16-bit register is used for transmit or receive. Therefore, when writing data to the data register, the software must write the corresponding data according to the channel currently being transmitted; similarly, when reading the register data, check the CHSIDE bit of the register STATR to determine which channel the received data belongs to. The left channel always sends data before the right channel (The CHSIDE bit has no meaning under the PCM protocol). There are 4 available data and packet frame combinations. Data can be sent in the following 4 data formats:

- 16-bit data is packed into 16-bit frames
- 16-bit data is packed into 32-bit frames
- 24-bit data is packed into 32-bit frames
- 32-bit data is packed into 32-bit frames

When using 16-bit data to expand to a 32-bit frame, the first 16 bits (MSB) are meaningful data, and the last 16 bits (LSB) are forced to 0. This operation requires no software intervention and no DMA request (Only requires a read or write operation). 24-bit and 32-bit data frames require the CPU to perform 2 read or write operations on the register DATAR, and 2 DMA transfers are required when using DMA. For 24-bit data, after expansion to 32 bits, the lowest 8 bits are set to 0 by hardware. For all data formats and communication standards, the most significant bit (MSB) is always sent first. The I2S interface supports 4 audio standards, which can be selected by setting the I2SSTD[1:0] bits and PCMSYNC bits in the I2SCFGR register.

20.3.2.1 I2S Philips Standard

Under this standard, pin WS is used to indicate which channel the data being sent belongs to. The pin is valid 1 clock cycle before the first data bit (MSB) is sent. The sender changes the data on the falling edge of the clock signal (CK) and the receiver reads the data on the rising edge. The WS signal also changes on the falling edge of the clock signal.

Figure 20-4 Philips protocol waveform (16/32 full precision, CPOL=0)

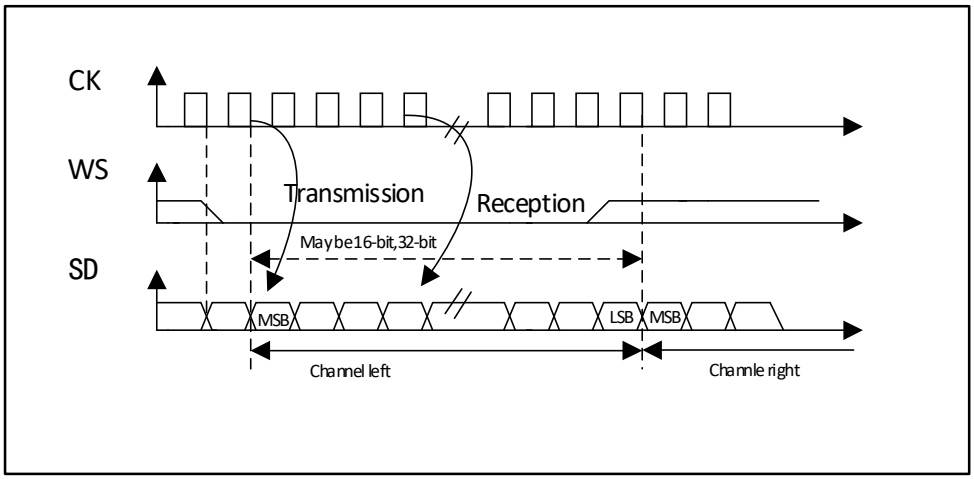
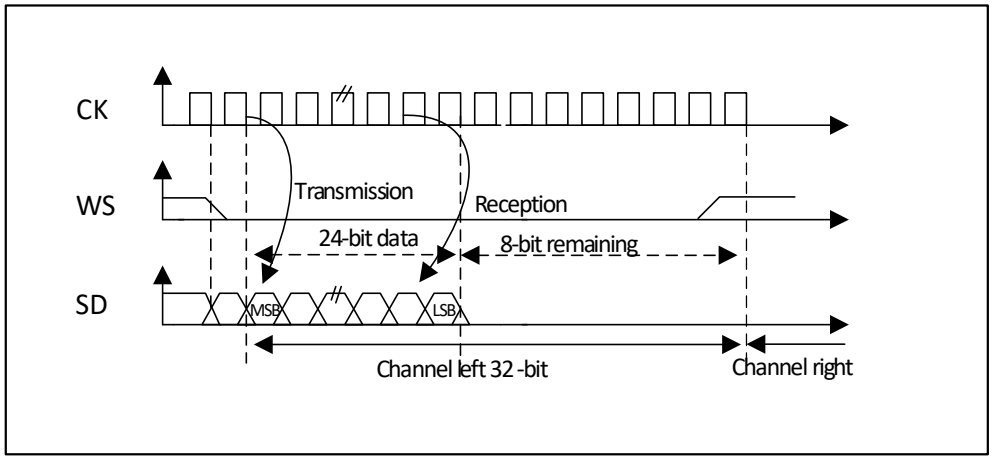
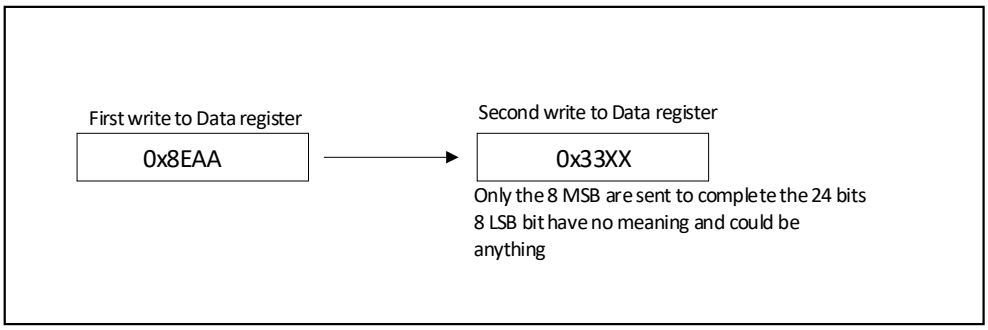


Figure 20-5 Philips protocol waveform (24-bit frame, CPOL=0)



This mode requires 2 read or write operations to the SPI_DATAR register. In transmit mode: if you need to send 0x8EAA33 (24-bits):



In receive mode: if 0x8EAA33 is received:

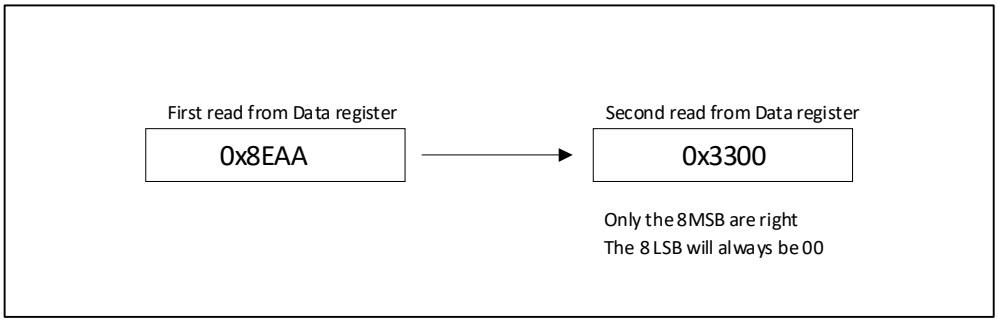
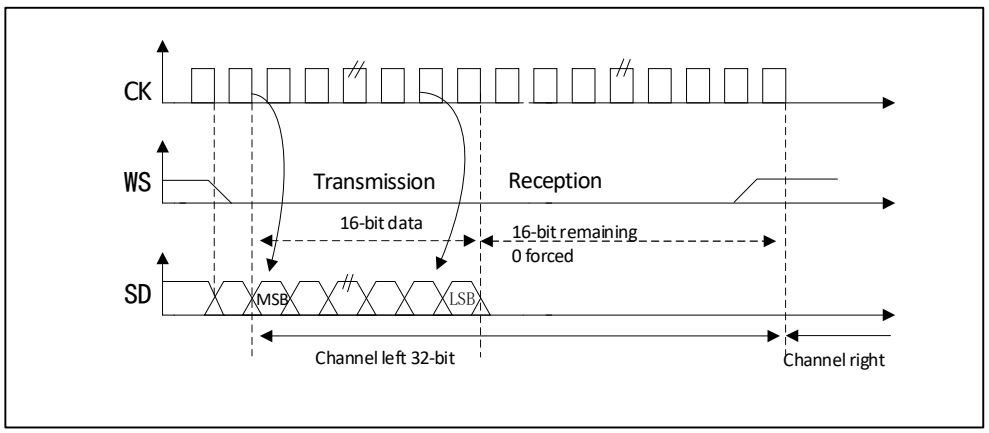


Figure 20-6 Philips protocol standard waveforms (16-bit extended to 32-bit packet frame, CPOL=0)



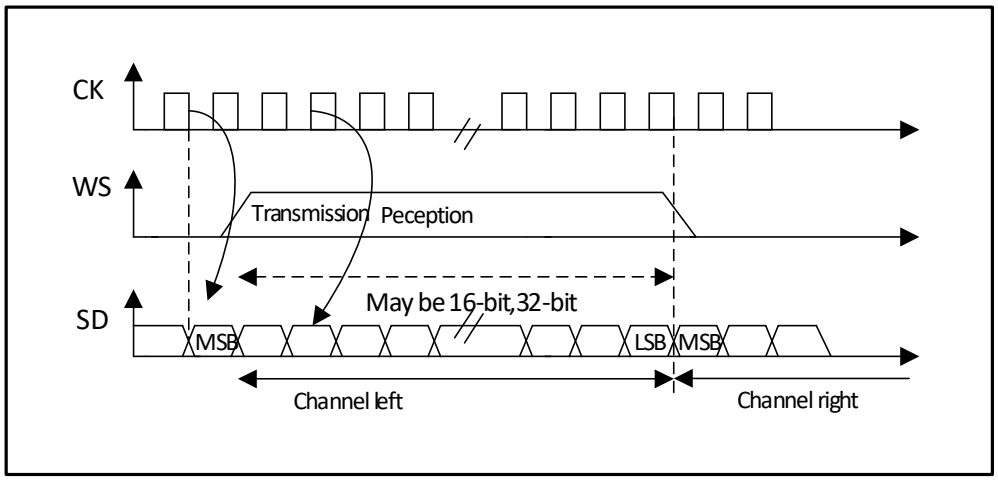
In the I2S configuration stage, if you choose to extend the 16-bit data to a 32-bit channel frame, you only need

to access the register DATAR once to extend to 32-bit. The lower 16 bits are set to 0x0000 by hardware. If the data to be transmitted or received is 0x76A3 (Extended to 32 bits is 0x76A30000), only one operation of DATAR is required. When sending, the MSB needs to be written into the register DATAR; if the flag bit TXE is 1, it means that new data can be written, and if the corresponding interrupt is allowed, an interrupt can be generated. The transmission is done by hardware, even if the last 16 bits of 0x0000 have not been sent, the TXE will be set and the corresponding interrupt will be generated. When receiving, each time the upper 16-bit half-word (MSB) is received, the flag bit RXNE is set to 1, and if the corresponding interrupt is enabled, an interrupt can be generated. This way, there is more time between the 2 reads and writes, preventing underflow or overflow conditions.

20.3.2.2 MSB alignment standard

Under this standard, the WS signal and the first data bit, the most significant bit (MSB), are generated simultaneously.

Figure 20-7 MSB aligned 16-bit or 32-bit full precision (CPOL = 0)



The sender changes data on the falling edge of the clock signal; the receiver reads data on the rising edge.

Figure 20-8 MSB aligned 24-bit data, CPOL = 0

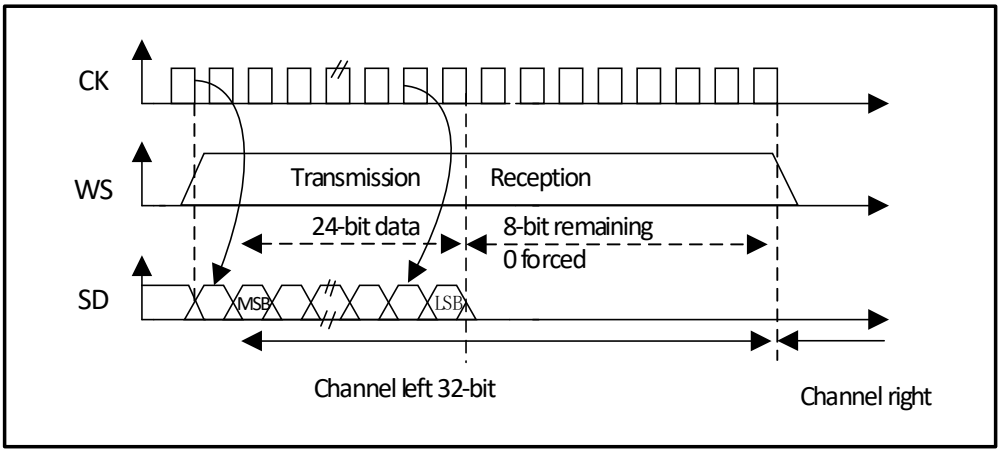
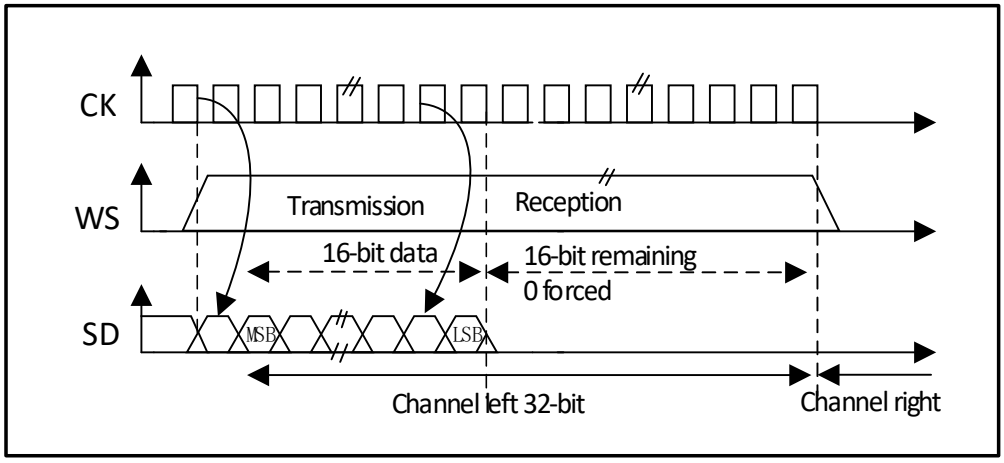


Figure 20-9 MSB-aligned 16-bit data extended to 32-bit packet frame, CPOL = 0



20.3.2.3 LSB alignment standard

This standard is similar to the MSB alignment standard (No difference in 16-bit or 32-bit full-precision frame format).

Figure 20-10 LSB-aligned 16- or 32-bit full precision, CPOL = 0

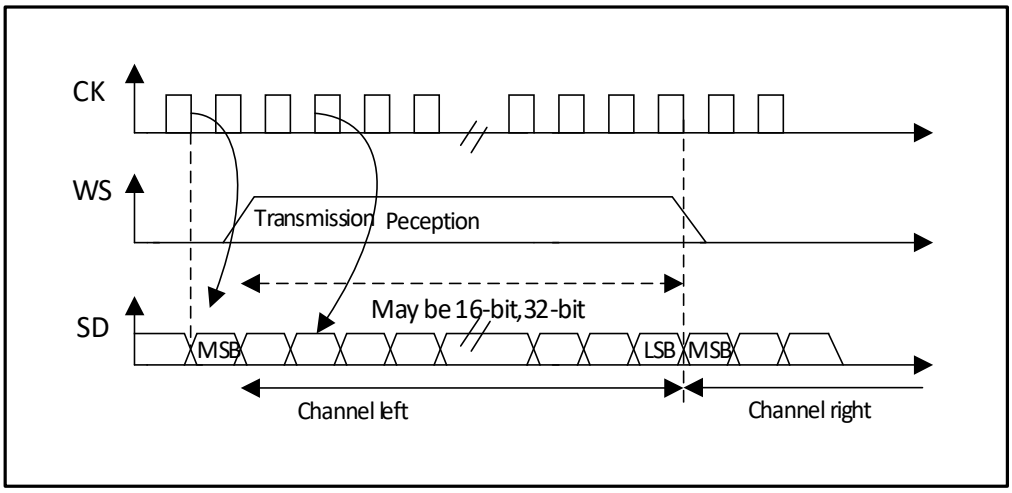
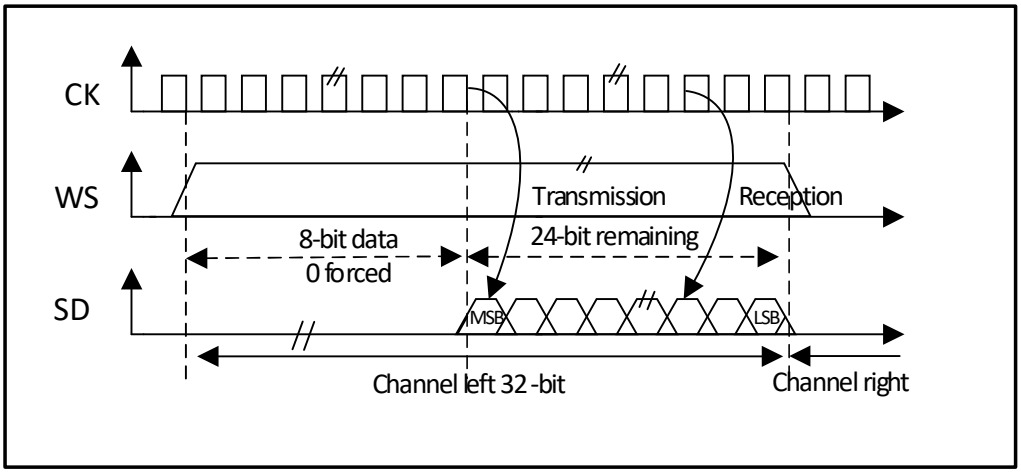
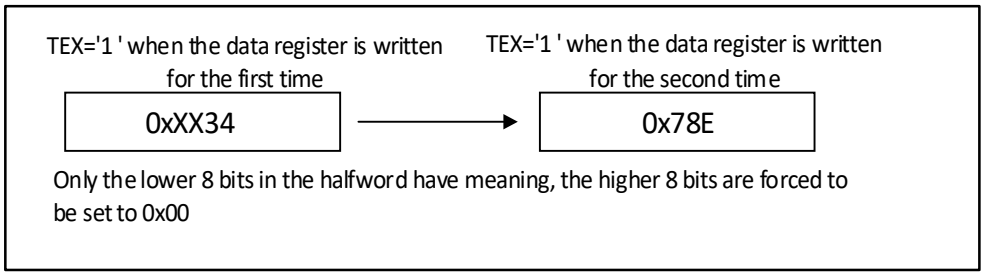


Figure 20-11 LSB-aligned 24-bit data, CPOL = 0



In transmit mode, if you want to send data 0x3478AE, you need to write the register DATAR twice by software or DMA.



In receive mode, if you want to receive data 0x3478AE, you need to read the register DATAR once when 2 consecutive RXNE events occur.

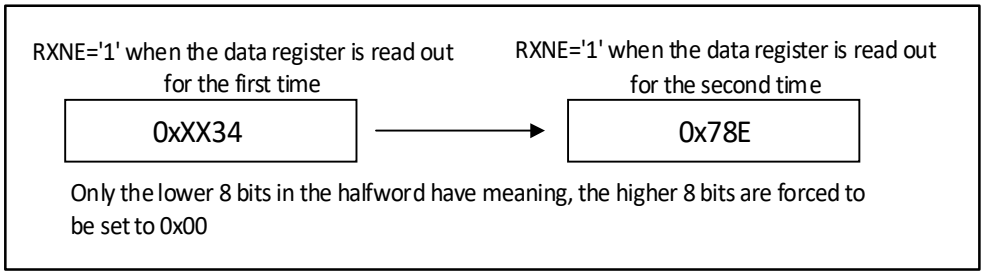
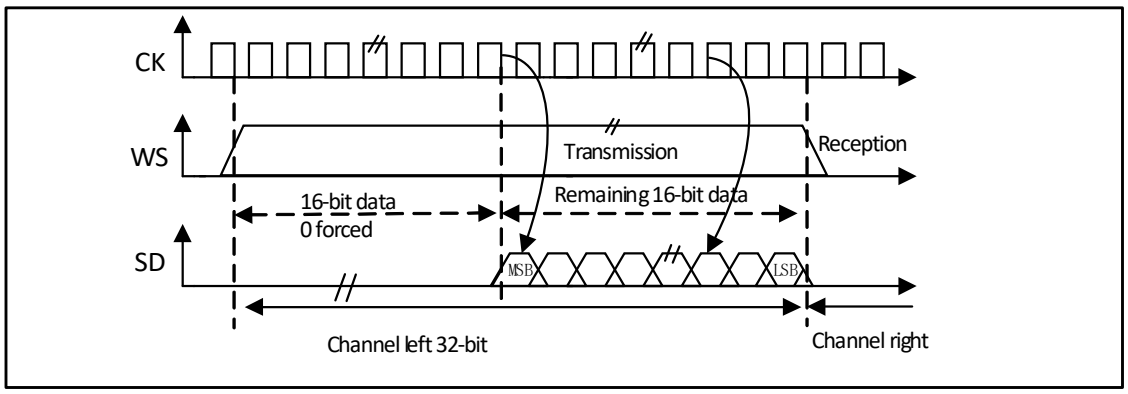


Figure 20-12 LSB aligned 16-bit data extended to 32-bit packet frame, CPOL = 0



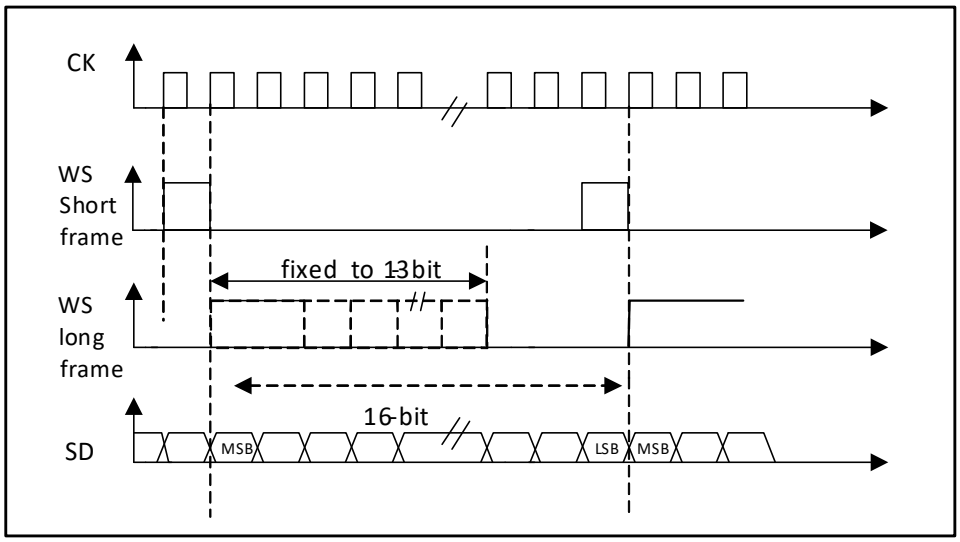
During the I2S configuration stage, if you choose to extend the 16-bit data to a 32-channel frame, you only need to access the DATAR register once. At this point, the high half word (16-bit MSB) after extending to 32 bits is set to 0x0000 by hardware.

If the data to be transmitted or received is 0x76A3 (Extended to 32 bits is 0x000076A3), only one operation of DATAR is required. When sending, if TXE is 1, the user needs to write the data to be sent (i.e. 0x76A3). The 0x0000 part, which is used to extend to 32 bits, is first sent by the hardware, and once the valid data starts to be sent from the SD pin, the next TXE event occurs. On reception, the RXNE event occurs as soon as valid data is received (Instead of the 0x0000 part). This way, there is more time between the 2 reads and writes, preventing underflow or overflow conditions.

20.3.2.4 PCM standard

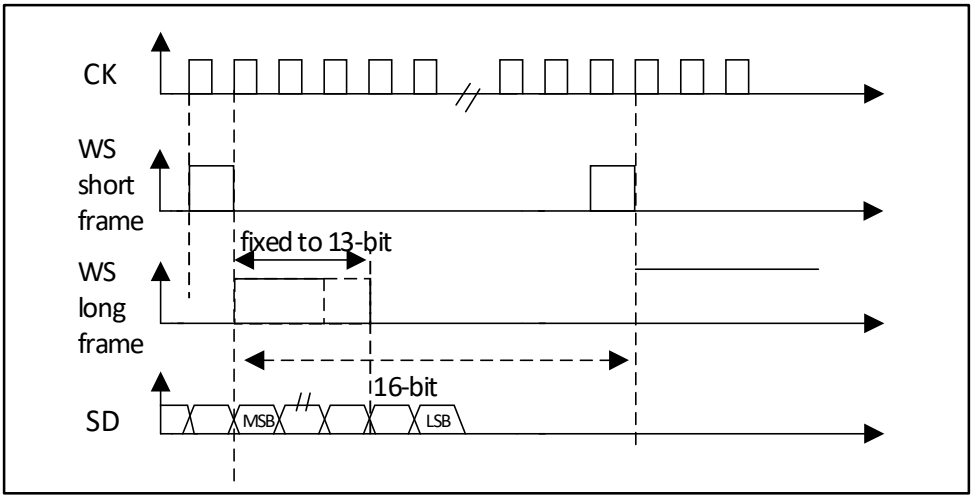
Under the PCM standard, there is no information for channel selection. The PCM standard has 2 available frame structures, short or long, which can be selected by setting the PCMSYNC bit in register I2SCFGR.

Figure 20-13 PCM standard waveform (16-bit)



For the long frame, in the main mode, the valid time of the WS signal used for synchronization is fixed to 13 bits. For short frames, the length of the WS signal used for synchronization is only 1 bit.

Figure 20-14 PCM standard waveform (16-bit)



No matter which mode (Master or slave), which synchronization method (Short frame or long frame), the time difference between 2 consecutive frames of data and between 2 synchronization signals, (Even in slave mode) needs to be set by setting the I2SCFGR register determined by the DATLEN bit and the CHLEN bit.

20.3.3 Clock Generator

The bit rate of I2S determines the data flow on the I2S data line and the frequency of the I2S clock signal. I2S bit rate = number of bits per channel × number of channels × audio sampling frequency.

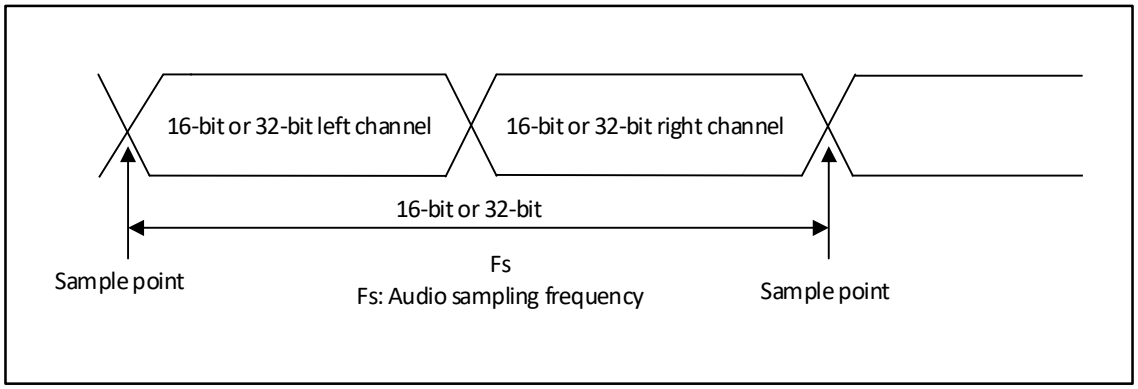
For a signal with left and right channels and 16-bit audio, the I2S bit rate is calculated as follows:

I2S bit rate = 16 × 2 × Fs

If the packet length is 32 bits, the I2S bit rate is calculated as follows:

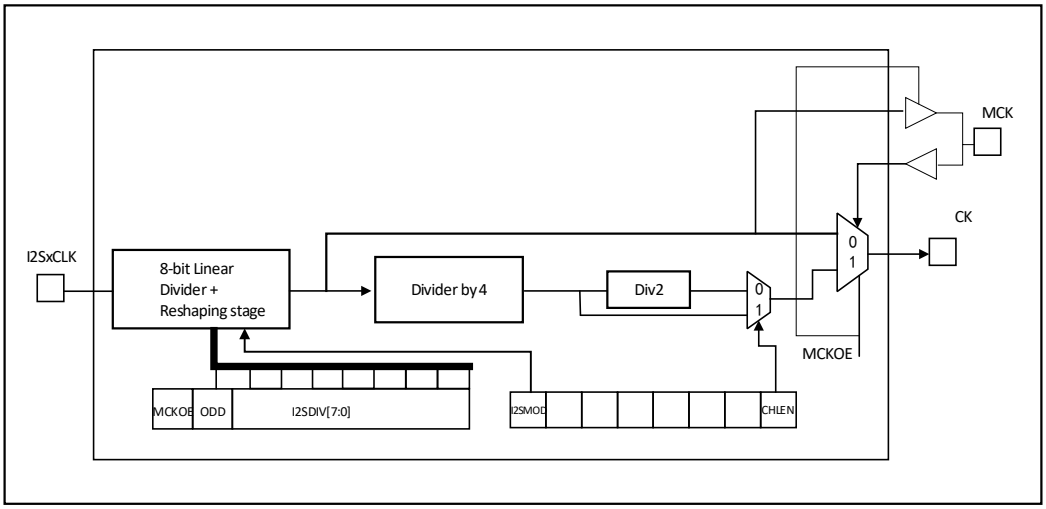
I2S bit rate = 32 × 2 × Fs

Figure 20-15 Audio sampling frequency definition



In master mode, the linear divider needs to be set correctly in order to obtain the desired audio frequency.

Figure 20-16 I2S clock generator structure



The clock source of I2SxCLK in the figure is the system clock (i.e., the HSI, HSE, or PLL that drives the PB1 clock). I2SxCLK can come from SYSCLK, or the PLL3 VCO (2xPLL3CLK) clock, which can be selected by the I2S2SRC and I2S3SRC bits in the RCC_CFGR2 register. Audio can be sampled at 96kHz, 48kHz, 44.1kHz, 32kHz, 22.05kHz, 16kHz, 11.025kHz, or 8kHz (Or any value within this range). To obtain the desired frequency, set the linear divider according to the following formula:

When the master clock needs to be generated (The MCKOE bit of the register SPI_I2SPR is 1):

When the frame length of the channel is 16 bits, Fs = I2SxCLK / [(16*2) * ((2*I2SDIV) + ODD)*8]

When the frame length of the channel is 32 bits, Fs = I2SxCLK / [(32*2) * ((2*I2SDIV) + ODD)*4]

When the main clock is turned off (MCKOE bit is '0'):

When the frame length of the channel is 16 bits, Fs = I2SxCLK / [(16*2) * ((2*I2SDIV) + ODD)]

When the frame length of the channel is 32 bits, Fs = I2SxCLK / [(32*2) * ((2*I2SDIV) + ODD)]

20.3.4 I2S Master Mode

Set I2S to work in master mode, the serial clock is output by pin CK, and the word selection signal is generated by pin WS. The master clock (MCK) can be selected to be output or not output by setting the MCKOE bit in register I2SPR.

20.3.4.1 Configuration Process

- Set I2SDIV[7:0] in the I2SPR register to define the serial clock baud rate corresponding to the audio sampling frequency. Also define the ODD bit of the register I2SPR.
- Set the CKPOL bit to define the level state of the communication clock when it is idle. If the main clock MCK needs to be provided to the external DAC/ADC audio device, the MCKOE position of the register I2SPR is required.
- Set the I2SMOD bit in the I2SCFGR register to 1 to activate the I2S function, set the I2SSTD[1:0] and PCMSYNC bits to select the I2S standard used, and set CHLEN to select the number of data bits per channel. Also set the I2SCFG[1:0] in the SPI_I2SCFGR register to select the I2S master mode and direction (transmitter or receiver).
- If required, the required interrupt function and DMA function can be turned on by setting register CR2.
- The I2SE bit in the I2SCFGR register must be set to 1.
- Pins WS and CK need to be configured for output mode. If the MCKOE bit in the SPI_I2SPR register is 1, the pin MCK is also configured to output mode.

20.3.4.2 Transmission Process

The transmission process starts when 1 half word (16 bits) of data is written to the transmit buffer. It is assumed that the first data written to the transmit buffer corresponds to the left channel data. When the data is moved from the transmit buffer to the shift register, the flag bit TXE is set to 1. At this time, the data corresponding to the right channel should be written into the transmit buffer. The flag bit CHSIDE indicates which channel the data to be transmitted currently corresponds to. The value of the flag bit CHSIDE is updated when TXE is 1, so it has meaning when TXE is 1. After the data of the left channel and then the right channel have been transmitted, it can be regarded as a complete data frame. It is not possible to transmit only part of the data frame, such as data only for the left channel.

When the first bit of data is sent out, the half-word data is transferred to the 16-bit shift register in parallel, and then the following bits are sent out from pins MOSI/SD in high-order first. Each time data is moved from the transmit buffer to the shift register, the flag bit TXE is set to 1, and an interrupt is generated if the TXEIE bit in the CR2 register is 1.

In order to ensure continuous audio data transmission, it is recommended to write the next data to be transmitted to the register DATAR before the current transmission is completed. It is recommended to wait for the flag bit TXE=1 and BSY=0 when the I2S function is to be turned off, and then clear the I2SE bit to '0'.

20.3.4.3 Reception Process

Except for point 3, the configuration steps of the reception process are consistent with the transmission process (See the aforementioned "transmission process"). It is necessary to select the main reception mode by configuring I2SCFG[1:0]. Regardless of the data and channel length, audio data is always received in 16-bit packets. That is, after the receive buffer is filled up each time, the flag bit RXNE is set to 1, and if the RXNEIE bit of the register CR2 is 1, an interrupt is generated. Depending on the configured data and channel length, receiving data from the left or right channel will require 1 or 2 times to transmit the data to the receive buffer. The RXNE flag is cleared by reading the DATAR register. CHSIDE is updated after each reception. Its value depends on the WS signal generated by the I2S unit. If the previous received data has not been read, and new data is received, that is, an overflow occurs, and the flag bit OVR is set to 1. If the ERRIE bit in the CR2 register is 1, an interrupt is generated, indicating that an error has occurred. To disable I2S, special operations need to be performed to ensure that the I2S module can normally complete the transfer cycle without starting a new data transfer. The operation process is related to the data configuration and channel length, and the mode

of the audio protocol:

● 16-bit data extended to 32-bit channel length (DATLEN=00 and CHLEN=1), using LSB (Low bit) alignment mode (I2SSTD=10)

a) Wait for the penultimate (n-1) RXNE=1;

b) Wait for 17 I2S clock cycles (Using software delay);

c) Disable I2S (I2SE=0).

● 16-bit data extended to 32-bit channel length (DATLEN=00 and CHLEN=1), using MSB (High bit) alignment, I2S or PCM mode (I2SSTD=00, I2SSTD=01 or I2SSTD=11 respectively)

a) Wait for the last RXNE=1;

b) Wait for 1 I2S clock cycle (Using software delay);

c) Disable I2S (I2SE=0).

● For all other combinations of DATLEN and CHLEN, any audio mode selected by I2SSTD, use the following method to turn off I2S:

a) Wait for the penultimate (n-1) RXNE=1;

b) Wait for one I2S clock cycle (Using software delay);

c) Disable I2S (I2SE=0).

Note: The BSY flag is always low during transmission.

20.3.5 I2S Slave Mode

In slave mode, I2S can be set to transmission and reception mode. The configuration method of the slave mode basically follows the same process as the configuration of the master mode. In slave mode, no clock is required from the I2S interface. Both the clock signal and the WS signal are provided by an external master I2S device connected to the corresponding pins. So the user does not need to configure the clock.

The configuration steps are as follows:

- Set the I2SMOD bit in the I2SCFGR register to activate the I2S function; set I2SSTD[1:0] to select the I2S standard used; set DATLEN[1:0] to select the number of data bits; set CHLEN to select the number of data bits per channel. Set the I2SCFG[1:0] in the I2SCFGR register to select the data direction (transmitter or receiver) of the I2S slave mode.
- As needed, set register CR2 to turn on the desired interrupt function and DMA function.
- The I2SE bit in the I2SCFGR register must be set to 1.

20.3.5.1 Transmission Process

The transmission process starts when the external master sends a clock signal, and when the NSS_WS signal requests data transmission. The slave device must be enabled and the I2S data register must be written before the external master device can start communication. For the MSB-aligned and LSB-aligned modes of I2S, the first data item written to the data register corresponds to the data of the left channel. When the communication starts, the data is transferred from the transmit buffer to the shift register, and then the flag bit TXE is set to 1; at this time, the data item corresponding to the right channel should be written into the I2S data register. The flag bit CHSIDE indicates which channel the data to be transmitted currently corresponds to. Compared to the transmit flow in master mode, in slave mode, CHSIDE depends on the WS signal from the external master I2S. This means that the slave I2S has to prepare the first data to be sent before receiving the clock signal generated by the master. The WS signal of 1 means that the left channel is sent first.

20.3.5.2 Reception Process

Except for point 1, the configuration steps are the same as the transmission process. The slave reception mode

needs to be selected by configuring I2SCFG[1:0]. Regardless of the data and channel length, audio data is always received in the form of 16-bit packets, that is, every time the receive buffer is filled, the flag bit RXNE is set to 1, and if the RXNEIE bit in the I2S_CTLR2 register is 1, an interrupt is generated. According to different data and channel length settings, receiving left channel or right channel data will require 1 or 2 times to transmit data to the receive buffer. CHSIDE is updated every time data is received (to be read out from DATAR), which corresponds to the WS signal generated by the I2S unit. Reading the SPI_DATAR register will clear the RXNE bit. When the previous received data has not been read out and new data is received, an overflow occurs, and the flag bit OVR is set to 1; if the ERRIE bit in the register I2S_CTLR2 is 1, an interrupt is generated, indicating that an error has occurred. To turn off the I2S function, it is necessary to clear the I2SE bit to 0 when the last RXNE=1 is received.

20.3.6 Status Flag

There are 3 status flags for the user to monitor the status of the I2S bus.

20.3.6.1 Busy Flag (BSY)

The BSY flag is set and cleared by hardware (writing this bit has no effect). This flag bit indicates the status of the I2S communication layer. When this bit is 1, it indicates that I2S communication is in progress, with one exception: in master receive mode (I2SCFG=11), the BSY flag is always low during reception. Before the software closes the SPI module, the BSY flag can be used to detect whether the transfer is over, so as to avoid destroying the last transfer, so it is necessary to strictly follow the procedure below. When a transfer starts, the BSY flag is set to 1, unless the I2S module is in master receive mode. This flag is cleared when the transfer ends or when the I2S module is turned off. When communication is continuous, in master transmit mode, the BSY flag is always high during the entire transfer; in slave mode, between each data item transmission, the BSY flag goes low within 1 I2S clock cycle.

20.3.6.2 TX Buffer Empty Flag (TXE)

When the flag bit is 1, it indicates that the sending buffer is empty, and new data to be sent can be written into the sending buffer. When there is data in the transmit buffer, the flag bit is cleared to 0. When I2S is turned off (I2SE bit is 0), this flag is also 0.

20.3.6.3 RX Buffer Not Empty Flag (RXNE)

Setting this flag bit indicates that there is valid data received in the receive buffer. This bit is cleared to 0 when the DATAR register is read.

20.3.6.4 Channel Side Flag (CHSIDE)

In transmit mode, this flag is refreshed when TXE is high, indicating which channel the data is sent from the SD pin. If an underflow error occurs in the slave sending mode, the value of this flag is invalid, and the I2S needs to be turned off and on before restarting the communication. In receive mode, this flag bit is refreshed when the register DATAR receives data, indicating the channel where the received data is located. Note that if an error occurs, this flag is meaningless, and I2S needs to be turned off and on again. Under the PCM standard, no matter the short frame format or the long frame format, this flag has no meaning. If the flag bit OVR or UDR in the STATR register is 1, and the ERRIE bit in the CR2 register is 1, an interrupt will be generated. (After the interrupt source has been cleared) The interrupt flag can be cleared by reading the STATR register.

20.3.7 Error Flag

20.3.7.1 Underrun Flag (UDR)

In the slave transmit mode, if the new data has not been written to the DATAR register when the first clock edge of the data transfer arrives, this flag will be set to 1. This flag is valid only after the I2SMOD bit of the register I2SCFGR is set. An interrupt will be generated if the ERRIE bit in register CR2 is 1. This flag is cleared by reading the STATR register.

20.3.7.2 Overrun Flag (OVR)

If the previous received data has not been read, and new data is received, that is, an overflow occurs, and the flag is set to 1. If the ERRIE bit of the register CTLR2 is 1, an interrupt is generated to indicate that an error has occurred. At this time, the content of the receiving buffer will not be refreshed with the new data sent from the sending device. A read of register DATAR returns the last correctly received data. All other 16-bit data sent by the sending device after the overflow occurs are lost. This flag is cleared by first reading the DATAR register and then reading the STATR register.

20.3.8 I2S Interrupt

I2S has 4 interrupt sources, of which the transmit buffer is empty and the receive buffer is not empty. These 2 events will set TXE and RXNE respectively, and an interrupt will be generated when the TXEIE and RXNEIE bits are set respectively. If the previous received data has not been read out, new data is received, that is, an overflow occurs. If ERRIE is set, an overflow interrupt will be generated; in slave transmit mode, if the first clock of data transmission When the edge arrives, the new data is still not written to the DATAR register. If ERRIE is set, an underflow interrupt will be generated.

20.3.9 DMA Features

The way DMA works in I2S mode is exactly the same as in SPI mode except that the CRC function is not available. Because there is no data transmission protection system in I2S mode.

20.4 Register Description

Table 20-1 SPI1 registers

Name	Access address	Description	Reset value
R16_SPI1_CTLR1	0x40013000	SPI1 control register1	0x0000
R16_SPI1_CTLR2	0x40013004	SPI1 control register2	0x0000
R16_SPI1_STATR	0x40013008	SPI1 status register	0x0002
R16_SPI1_DATAR	0x4001300C	SPI1 data register	0x0000
R16_SPI1_CRCR	0x40013010	SPI1 polynomial register	0x0007
R16_SPI1_RCRCR	0x40013014	SPI1 receive CRC register	0x0000
R16_SPI1_TCRCR	0x40013018	SPI1 transmit CRC register	0x0000
R16_SPI1_HSCR	0x40013024	SPI1 high-speed control register	0x0000

Table 20-2 SPI2 registers

Name	Access address	Description	Reset value
R16_SPI2_CTLR1	0x40003800	SPI2 control register1	0x0000
R16_SPI2_CTLR2	0x40003804	SPI2 control register2	0x0000
R16_SPI2_STATR	0x40003808	SPI2 status register	0x0002

R16_SPI2_DATAR	0x4000380C	SPI2 data register	0x0000
R16_SPI2_CRCR	0x40003810	SPI2 polynomial register	0x0007
R16_SPI2_RCRCR	0x40003814	SPI2 receive CRC register	0x0000
R16_SPI2_TCRCR	0x40003818	SPI2 transmit CRC register	0x0000
R16_SPI2_I2S_CFGR	0x4000381C	SPI2_I2S configuration register	0x0000
R16_SPI2_I2SPR	0x40003820	SPI2_I2S prescaler register	0x0000
R16_SPI2_HSCR	0x40003824	SPI2 high-speed control register	0x0000

Table 20-3 SPI3 registers

Name	Access address	Description	Reset value
R16_SPI3_CTLR1	0x40003C00	SPI3 control register1	0x0000
R16_SPI3_CTLR2	0x40003C04	SPI3 control register2	0x0000
R16_SPI3_STATR	0x40003C08	SPI3 status register	0x0002
R16_SPI3_DATAR	0x40003C0C	SPI3 data register	0x0000
R16_SPI3_CRCR	0x40003C10	SPI3 polynomial register	0x0007
R16_SPI3_RCRCR	0x40003C14	SPI3 receive CRC register	0x0000
R16_SPI3_TCRCR	0x40003C18	SPI3 transmit CRC register	0x0000
R16_SPI3_I2S_CFGR	0x40003C1C	SPI3_I2S configuration register	0x0000
R16_SPI3_I2SPR	0x40003C20	SPI3_I2S prescaler register	0x0000
R16_SPI3_HSCR	0x40003C24	SPI3 high-speed control register	0x0000

20.4.1 SPI Control Register 1 (SPIx_CTLR1) (x=1/2/3)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BIDIMODE	BIDIOE	CRCEN	CRCNEXT	DFF	RXONLY	SSM	SSI	LSBFIRST	SPE	BR[2:0]			MSTR	CPOL	CPHA

Bit	Name	Access	Description	Reset value
15	BIDIMODE	RW	Unidirectional data mode enable. 1: 1-wire bidirectional mode; 0: 2-wire bidirectional mode;	0
14	BIDIOE	RW	Single-line output enable, used in conjunction with BIDIMODE. 1: Enable output, only transmit; 0: Disable output, only receive.	0
13	CRCEN	RW	Hardware CRC check enable, this bit can only be written when SPE is 0, this bit can only be used in full duplex mode. 1: Enable CRC calculation; 0: Disable CRC calculation.	0
12	CRCNEXT	RW	Value of the transmit CRC register after next data transfer. This bit should be set immediately after writing the last data to the data register. 1: Transmit the CRC result;	0

			0: Continue to transmit the data of the data register.	
11	DFF	RW	Data frame length, this bit can only be written when SPE is 0. 1: Use 16-bit data length to transmit and receive; 0: Use 8-bit data length for transmission and reception.	0
10	RXONLY	RW	Rx only in 2-wire mode. It is used in conjunction with BIDIMODE. Setting this bit allows the device to only receive and not transmit. 1: Receive only, simplex mode; 0: Full-duplex mode.	0
9	SSM	RW	CS pin management, this bit determines whether the level of the NSS pin is controlled by hardware or software. 1: Software control NSS pin; 0: Hardware controls NSS pin.	0
8	SSI	RW	CS pin control. When SSM is set, this bit determines the level of the NSS pin. 1: NSS is high level; 0: NSS is low level.	0
7	LSBFIRST	RW	Frame format control. This bit cannot be modified during communication. 1: Transmit LSB first; 0: Transmit MSB first.	0
6	SPE	RW	SPI enable. 1: Enable SPI; 0: Disable SPI.	0
[5:3]	BR[2:0]	RW	Baud rate setting domain, this domain cannot be modified during communication. When the HSRXNE bit is 0, the SCK frequency is: 000: $F_{PCLK}/2$; 001: $F_{PCLK}/4$; 010: $F_{PCLK}/8$; 011: $F_{PCLK}/16$; 100: $F_{PCLK}/32$; 101: $F_{PCLK}/64$; 110: $F_{PCLK}/128$; 111: $F_{PCLK}/256$. When the HSRXNE bit is 1, the SCK frequency is: 000: $F_{PCLK}/2$; 001: $F_{PCLK}/3$; 010: $F_{PCLK}/4$; 011: $F_{PCLK}/5$; 100: $F_{PCLK}/6$; 101: $F_{PCLK}/7$; 110: $F_{PCLK}/8$; 111: $F_{PCLK}/9$.	000b
2	MSTR	RW	Master/slave setting. It cannot be modified during communication. 1: Configure as the master device; 0: Configured as the slave device.	0

1	CPOL	RW	Clock polarity selection. It cannot be modified during communication. 1: In idle state, SCK remains high; 0: In idle state, SCK remains low.	0
0	CPHA	RW	Clock phase setting. It cannot be modified during communication. 1: Data sampling starts from the second clock edge; 0: Data sampling starts from the first clock edge.	0

20.4.2 SPI Control Register 2 (SPIx_CTLR2) (x=1/2/3)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								TXEIE	RXNEIE	ERRIE	Reserved		SSOE	TXDMAEN	RXDMAEN

Control register 2

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
7	TXEIE	RW	Transmit buffer empty interrupt enable. 1: Enable interrupt generation when TXE is set; 0: Disable interrupt generation when TXE is set.	0
6	RXNEIE	RW	Receive buffer not empty interrupt enable. 1: Enable interrupt generation when RXNE is set; 0: Disable interrupt generation when RXNE is set.	0
5	ERRIE	RW	Error interrupt enable. 1: Enable interrupt generation when an error (CRCERR, OVR, MODF) is generated; 0: Disable interrupt generation when an error (CRCERR, OVR, MODF) is generated.	0
[4:3]	Reserved	RO	Reserved	0
2	SSOE	RW	SS output enable. Disable SS output to work in multi-master mode. 1: Enable SS output; 0: Disable SS output in master mode.	0
1	TXDMAEN	RW	Transmit buffer DMA enable. 1: Enable transmit buffer DMA; 0: Disable transmit buffer DMA.	0
0	RXDMAEN	RW	Receive buffer DMA enable. 1: Enable receive buffer DMA; 0: Disable receive buffer DMA.	0

20.4.3 SPI Status Register (SPIx_STATR) (x=1/2/3)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								BSY	OVR	MODF	CRCERR	UDR	CHSIDE	TXE	RXNE

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
7	BSY	RO	Busy flag. Set and reset by hardware. 1: SPI is communicating, or the transmit buffer is not empty; 0: SPI is not communicating.	0
6	OVR	RO	Overflow flag. Set by hardware, and reset by software. 1: An overflow error occurred; 0: No overflow error occurred.	0
5	MODF	RO	Mode error flag. Set by hardware, and reset by software. 1: A mode error occurred; 0: No mode error occurred.	0
4	CRCERR	RW0	CRC error flag. Set by hardware, and reset by software. 1: The received CRC value is inconsistent with the RCRCR value; 0: The received CRC value is the same as the RCRCR value.	0
3	UDR	RO	Underflow flag. Set by hardware, and reset by software. 1: Underflow occurs; 0: Underflow has not occurred.	0
2	CHSIDE	RO	Channel. Set by hardware and reset by software. 1: The right channel needs to be transmitted or received; 0: The left channel needs to be transmitted or received.	0
1	TXE	RO	Transmit buffer empty flag. 1: The transmit buffer is empty; 0: The transmit buffer is not empty.	1
0	RXNE	RO	Receive buffer not empty flag. 1: The receive buffer is not empty; 0: The receive buffer is empty. *Note: Read DATAR and auto-zero.*	0

20.4.4 SPI Data Register (SPIx_DATAR) (x=1/2/3)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DR[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	DR[15:0]	RW	Data register. It is used to store the received data or pre-store the data to be sent, so the read and write of the data register is actually a different area for operations, among which the read pair uses the receive buffer, and the write corresponds to the send buffer. Data reception and transmission can be 8-bit or 16-bit, and it is necessary to determine how many bits of data to use before transmission. When using 8 bits for data transmission, only the lower 8 bits of the data register are used, and the upper 8 bits are forced to 0 when receiving. Using the 16-bit data structure will cause all 16-bit data registers to be used.	0

20.4.5 SPI Polynomial Register (SPIx_CRCR) (x=1/2/3)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CRCPOLY[15:0]															

Bit	Name	Access	Description	Reset Value
[15:0]	CRCPOLY[15:0]	RW	CRC polynomial. This domain defines the polynomial used for the CRC calculation.	0x0007

20.4.6 SPI Receive CRC Register (SPIx_RCRCR) (x=1/2/3)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXCRC[15:0]															

Bit	Name	Access	Description	Reset Value
[15:0]	RXCRC[15:0]	RO	Rx CRC value. Stores the result of the calculated CRC check of the received bytes. Setting CRCEN will reset this register. The calculation method uses the polynomial used by CRCPOLY. In 8-bit mode, only the lower 8 bits participate in the calculation, and in 16-bit mode, all 16 bits will participate in the calculation. This register needs to be read when BSY is 0.	0

20.4.7 Transmit CRC Register (SPIx_TCRCR) (x=1/2/3)

Offset address: 0x18

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

TXCRC[15:0]

Bit	Name	Access	Description	Reset value
[15:0]	TXCRC[15:0]	RO	Tx CRC value. Stores the calculated result of the CRC check of the bytes that have been sent. Setting CRCEN will reset this register. The calculation method uses the polynomial used by CRCPOLY. In 8-bit mode, only the lower 8 bits participate in the calculation, and in 16-bit mode, all 16 bits will participate in the calculation. This register needs to be read when BSY is 0.	0

20.4.8 SPI_I2S Configuration Register (SPIx_I2S_CFGR) (x=2/3)

Offset address: 0x1C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				I2S MOD	I2SE	I2SCFG [1:0]		PCM SYNC	Reserved	I2SSTD [1:0]		CKPOL	DATLEN [1:0]		CHLEN

Bit	Name	Access	Description	Reset value
[15:12]	Reserved	RO	Reserved	0
11	I2SMOD	RW	I2S mode selection, this bit can only be set when SPI or I2S is disabled. 1: Select I2S mode; 0: Select SPI mode.	0
10	I2SE	RW	I2S enable, not used in SPI mode. 1: I2S enabled; 0: I2S disabled.	0
[9:8]	I2SCFG[1:0]	RW	I2S mode selection, this bit can only be set when I2S is disabled: 00: Slave device transmits; 01: Slave device receives; 10: Master device transmit; 11: Master device receives.	00b
7	PCMSYNC	RW	PCM frame synchronization. This bit is only meaningful when I2SSTD = 11 (Using the PCM standard). 1: Long frame synchronization; 0: Short frame synchronization.	0
6	Reserved	RO	Reserved	0
[5:4]	I2SSTD[1:0]	RW	I2S standard selection, this bit can only be set when I2S is disabled. 00: I2S Philips standard; 01: High byte alignment standard (Left alignment);	00b

			10: Low byte alignment standard (Right justified); 11: PCM standard.	
3	CKPOL	RW	Quiescent clock polarity, for correct operation, this bit should only be set when I2S is turned off. 0: I2S clock quiescent state is low; 1: I2S clock quiescent state is high level.	0
[2:1]	DATLEN[1:0]	RW	Length of the data to be transmitted. For correct operation, this bit can only be set when I2S is turned off. 00: 16-bit data length; 01: 24-bit data length; 10: 32-bit data length; 11: Not allowed.	00b
0	CHLEN	RW	Channel length. Only when DATLEN = 00, the write operation of this bit is meaningful, otherwise the channel length is fixed to 32 bits by hardware. 0: 16-bit wide; 1: 32-bit wide.	0

20.4.9 SPI_I2S Prescaler Register (SPIx_I2SPR) (x=2/3)

Offset Address: 0x20

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						MCKOE	ODD	I2SDIV[7:0]							

Bit	Name	Access	Description	Reset value
[15:10]	Reserved	RO	Reserved	0
9	MCKOE	RW	Master clock output enable, for proper operation, this bit can only be set when I2S is turned off. This bit is only used in I2S master mode. 1: Master device clock output enabled; 0: Master clock output disabled.	0
8	ODD	RW	Odd coefficient prescaler, for correct operation, this bit should only be set when I2S is turned off. This bit is only used in I2S master mode. 1: Actual division factor = (I2SDIV*2)+1; 0: Actual division factor = I2SDIV*2.	0
[7:0]	I2SDIV[7:0]	RW	I2S linear prescaler. For proper operation, this bit should only be set when I2S is turned off. This bit is only used in I2S master mode. Disable setting I2SDIV[7:0] = 0 or I2SDIV[7:0] = 1 See Section 20.3.3.	0

20.4.10 SPI High-speed Control Register (SPIx_HSCR) (x=1/2/3)

Offset address: 0x24

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												I2S_WS_SYNC	HSRXNE2	Reserved	HSRXEN

Bit	Name	Access	Description	Reset value
[15:4]	Reserved	RO	Reserved	0
3	I2S_WS_SYNC	RW	Strengthen the synchronization capability of I2S as a slave: 1: Enabled; 0: Disabled.	0
2	HSRXNE2	RW	SPI high-speed read mode 2 This mode is used only when HSRXNE is turned on and SPI PCLK is greater than or equal to 120M; this bit is not recommended for less than 120M. A higher host read speed can be obtained than if only HSRXNE is turned on. 1: Enable high-speed read mode 2; 0: Disables high-speed read mode 2.	0
1	Reserved	RO	Reserved	0
0	HSRXNE	W0	High-speed read mode enable bit: 1: High-speed read mode enabled; 0: High-speed read mode disabled. This mode is supported only when the clock is divided by 2 (that is, BR=000 in the CTLR1 register).	0

Chapter 21 Operational Amplifier (OPA)

Operational amplifier module (OPA), containing 1 independently configurable operational amplifier. The input and output of the operational amplifier are connected to the I/O port and are pin-selectable. The operational amplifier (OPA or PGA) supports gain selection.

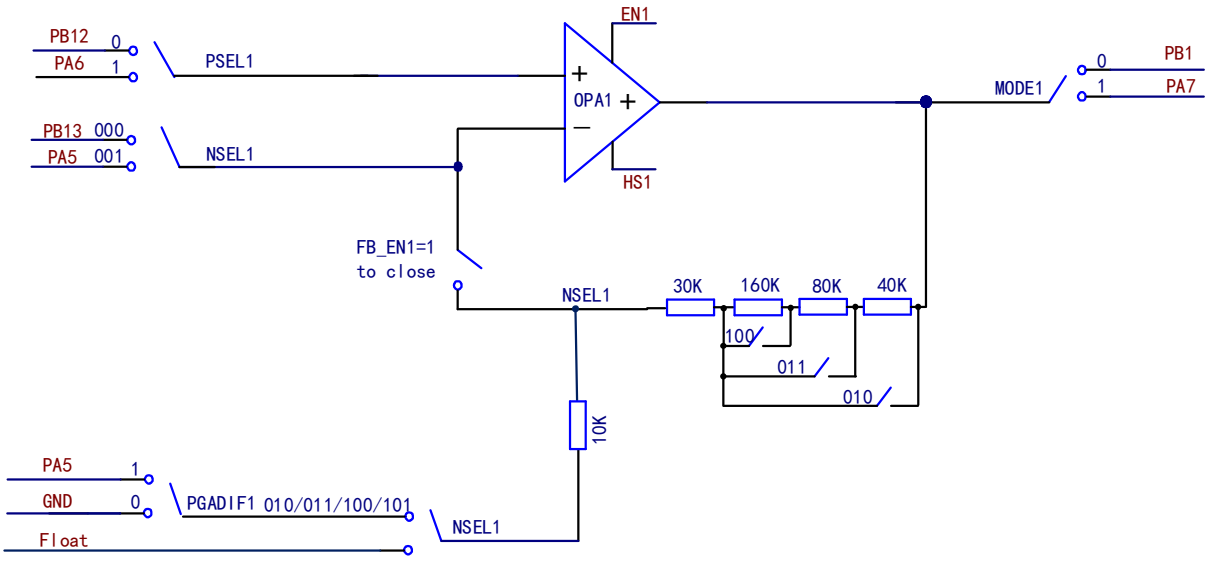
21.1 Main Features

- Input pin selectable
- Output pin selectable
- Support programmable gain amplifier (PGA)

21.2 Function Description

In the R32_OPA_CTLR register: set EN1 to enable the corresponding OPA1; configure MODE1 to select the output channel of OPA1; configure PSEL1 to select the forward input pin of OPA1; configure NSEL1, you can select the negative input channel of OPA1, or the gain when used as a PGA; set FB_EN1 to enable the PGA mode of OPA1; set PGADIF1 to enable the differential input PGA mode; set HS1 to enable the high-speed mode of OPA1.

Figure 21-1 OPA structure diagram



21.3 Register Description

Table 21-1 OPA-related registers

Name	Access address	Description	Reset value
R32_OPA_CTLR	0x40023804	OPA control register	0x00000000

21.3.1 OPA Control Register (R32_OPA_CTLR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							HS1	FB_EN1	PGADIF1	PSEL1	NSEL1[2:0]			MODE1	EN1

Bit	Name	Access	Description	Reset value
[31:9]	Reserved	RO	Reserved	0
8	HS1	RW	OPA1 high-speed mode enable: 1: Enabled; 0: Disabled.	0
7	FB_EN1	RW	OPA1 PGA mode feedback enable: 1: Enabled; 0: Disabled.	0
6	PGADIF1	RW	OPA1 is used with NSEL as a PGA, and the N terminal is connected to OPA1_CHN1 (PA5): 1: Turn on the differential input PGA mode; 0: Turn off the differential input PGA mode.	0
5	PSEL1	RW	OPA1 positive terminal channel selection: 1: OPA1_CHP1 (PA6); 0: OPA1_CHP0 (PB12).	0
[4:2]	NSEL1[2:0]	RW	OPA1 negative channel selection and PGA gain selection: 000: OPA1_CHN0 (PB13); 001: OPA1_CHN1 (PA5); 010: PGA mode, no negative input channel, internal gain is 4; 011: PGA mode, no negative input channel, internal gain is 8; 100: PGA mode, no negative input channel, internal gain is 16; 101: PGA mode, no negative input channel, internal gain is 32; Others: Off.	0
1	MODE1	RW	OPA1 output channel selection: 1: Output signal via PA7; 0: Output signal via PB1;	0
0	EN1	RW	OPA1 enable: 1: Enabled; 0: Disabled.	0

Chapter 22 USB High-speed Host/Device Controller (USBHS)

The chip has 2 built-in USB2.0 high-speed controllers, which have dual roles of host controller and device controller, and a built-in 480Mbps USB-PHY physical layer transceiver. When operating as a host controller, it supports low-speed, full-speed, and high-speed USB devices. When operating as a device controller, it can be flexibly configured for low-speed, full-speed, or high-speed modes to accommodate diverse applications.

22.1 USB Controller Introduction

- Support USB2.1, USB2.0, USB1.1, and USB1.0 protocol specifications
- Support USB Host function and USB Device function
- Support control transfer, bulk transfer, interrupt transfer and isochronous/real-time transfer
- Provide bus reset, suspend, wake-up and recovery functions
- Host supports USB HUB
- Non-zero endpoints all support the maximum packet size of 1024 bytes, with built-in FIFO, interrupt and DMA support.
- Support USART or I2C pin mapping, and is also used for 2-wire debugging.

22.2 Register Description

The USB registers are divided into 2 parts:

- USB high-speed device control registers
- USB high-speed host control registers

The base address for the USBHS1 module's peripherals is: 0x40024000;

The base address for the USBHS2 module's peripherals is: 0x40023400.

22.2.1 Device Register Description

Table 22-1 USBHS1 device-related registers

Name	Access address	Description	Reset value
R8_USB_CTRL	0x40024000	USBHS control register	0x07
R8_USB_BASE_MODE	0x40024001	USBHS mode control register	0x00
R8_USB_INT_EN	0x40024002	USBHS interrupt enable register	0x00
R8_USB_DEV_AD	0x40024003	USBHS device address register	0x00
R8_USB_WAKE_CTRL	0x40024004	USBHS remote wakeup register	0x00
R8_USB_TEST_MODE	0x40024005	USBHS test mode register	0x00
R16_USB_LPM_DATA	0x40024006	USBHS power management register	0x8000
R8_USB_INT_FG	0x40024008	USBHS interrupt flag register	0x00
R8_USB_INT_ST	0x40024009	USBHS interrupt status register	0x00
R8_USB_MIS_ST	0x4002400A	USBHS miscellaneous status register	0x00
R16_USB_FRAME_NO	0x4002400C	USBHS frame number register	0x0000
R16_USB_BUS	0x4002400E	USBHS bus status register	0x0000
R16_UEP_TX_EN	0x40024010	USBHS endpoint transmit enable register	0x0000

R16_UEP_RX_EN	0x40024012	USBHS endpoint receive enable register	0x0000
R16_UEP_T_TOG_AUTO	0x40024014	USBHS endpoint transmit auto-flip enable register	0x0000
R16_UEP_R_TOG_AUTO	0x40024016	USBHS endpoint receive auto-flip enable register	0x0000
R8_UEP_T_BURST	0x40024018	USBHS endpoint transmit burst register	0x00
R8_UEP_T_BURST_MODE	0x40024019	USBHS endpoint transmit burst mode register	0x00
R8_UEP_R_BURST	0x4002401A	USBHS endpoint receive burst register	0x00
R8_UEP_R_RES_MODE	0x4002401B	USBHS endpoint receive burst mode register	0x00
R32_UEP_AF_MODE	0x4002401C	USBHS endpoint alter Register	0x00000000
R32_UEP0_DMA	0x40024020	Start address register of endpoint 0 buffer	0x000XXXXX
R32_UEP1_RX_DMA	0x40024024	Start address register of endpoint 1 receive buffer	0x000XXXXX
R32_UEP2_RX_DMA	0x40024028	Start address register of endpoint 2 receive buffer	0x000XXXXX
R32_UEP3_RX_DMA	0x4002402C	Start address register of endpoint 3 receive buffer	0x000XXXXX
R32_UEP4_RX_DMA	0x40024030	Start address register of endpoint 4 receive buffer	0x000XXXXX
R32_UEP5_RX_DMA	0x40024034	Start address register of endpoint 5 receive buffer	0x000XXXXX
R32_UEP6_RX_DMA	0x40024038	Start address register of endpoint 6 receive buffer	0x000XXXXX
R32_UEP7_RX_DMA	0x4002403C	Start address register of endpoint 7 receive buffer	0x000XXXXX
R32_UEP1_TX_DMA	0x40024040	Start address register of endpoint 1 transmit buffer	0x000XXXXX
R32_UEP2_TX_DMA	0x40024044	Start address register of endpoint 2 transmit buffer	0x000XXXXX
R32_UEP3_TX_DMA	0x40024048	Start address register of endpoint 3 transmit buffer	0x000XXXXX
R32_UEP4_TX_DMA	0x4002404C	Start address register of endpoint 4 transmit buffer	0x000XXXXX
R32_UEP5_TX_DMA	0x40024050	Start address register of endpoint 5 transmit buffer	0x000XXXXX
R32_UEP6_TX_DMA	0x40024054	Start address register of endpoint 6 transmit buffer	0x000XXXXX
R32_UEP7_TX_DMA	0x40024058	Start address register of endpoint 7 transmit buffer	0x000XXXXX
R32_UEP0_MAX_LEN	0x4002405C	Endpoint 0 maximum length packet register	0x000000XX
R32_UEP1_MAX_LEN	0x40024060	Endpoint 1 maximum length packet register	0x00000XXX
R32_UEP2_MAX_LEN	0x40024064	Endpoint 2 maximum length packet register	0x00000XXX
R32_UEP3_MAX_LEN	0x40024068	Endpoint 3 maximum length packet register	0x00000XXX
R32_UEP4_MAX_LEN	0x4002406C	Endpoint 4 maximum length packet register	0x00000XXX
R32_UEP5_MAX_LEN	0x40024070	Endpoint 5 maximum length packet register	0x00000XXX

R32_UEP6_MAX_LEN	0x40024074	Endpoint 6 maximum length packet register	0x00000XXX
R32_UEP7_MAX_LEN	0x40024078	Endpoint 7 maximum length packet register	0x00000XXX
R16_UEP0_RX_LEN	0x4002407C	Endpoint 0 receive length register	0x00XX
R16_UEP1_RX_LEN	0x40024080	Endpoint 1 single receive length register	0xXXXX
R16_UEP1_R_SIZE	0x40024082	Endpoint 1 total receive data length register	0xXXXX
R16_UEP2_RX_LEN	0x40024084	Endpoint 2 single receive length register	0xXXXX
R16_UEP2_R_SIZE	0x40024086	Endpoint 2 total receive data length register	0xXXXX
R16_UEP3_RX_LEN	0x40024088	Endpoint 3 single receive length register	0xXXXX
R16_UEP3_R_SIZE	0x4002408A	Endpoint 3 total receive data length register	0xXXXX
R16_UEP4_RX_LEN	0x4002408C	Endpoint 4 single receive length register	0xXXXX
R16_UEP4_R_SIZE	0x4002408E	Endpoint 4 total receive data length register	0xXXXX
R16_UEP5_RX_LEN	0x40024090	Endpoint 5 single receive length register	0xXXXX
R16_UEP5_R_SIZE	0x40024092	Endpoint 5 total receive data length register	0xXXXX
R16_UEP6_RX_LEN	0x40024094	Endpoint 6 single receive length register	0xXXXX
R16_UEP6_R_SIZE	0x40024096	Endpoint 6 total receive data length register	0xXXXX
R16_UEP7_RX_LEN	0x40024098	Endpoint 7 single receive length register	0xXXXX
R16_UEP7_R_SIZE	0x4002409A	Endpoint 7 total receive data length register	0xXXXX
R16_UEP0_T_LEN	0x4002409C	Endpoint 0 transmit length register	0x00XX
R8_UEP0_TX_CTRL	0x4002409E	Endpoint 0 transmit control register	0x00
R8_UEP0_RX_CTRL	0x4002409F	Endpoint 0 receive control register	0x00
R16_UEP1_T_LEN	0x400240A0	Endpoint 1 transmit length register	0x0000
R8_UEP1_TX_CTRL	0x400240A2	Endpoint 1 transmit control register	0x00
R8_UEP1_RX_CTRL	0x400240A3	Endpoint 1 receive control register	0x00
R16_UEP2_T_LEN	0x400240A4	Endpoint 2 transmit length register	0x0000
R8_UEP2_TX_CTRL	0x400240A6	Endpoint 2 transmit control register	0x00
R8_UEP2_RX_CTRL	0x400240A7	Endpoint 2 receive control register	0x00
R16_UEP3_T_LEN	0x400240A8	Endpoint 3 transmit length register	0x0000
R8_UEP3_TX_CTRL	0x400240AA	Endpoint 3 transmit control register	0x00
R8_UEP3_RX_CTRL	0x400240AB	Endpoint 3 receive control register	0x00
R16_UEP4_T_LEN	0x400240AC	Endpoint 4 transmit length register	0x0000
R8_UEP4_TX_CTRL	0x400240AE	Endpoint 4 transmit control register	0x00
R8_UEP4_RX_CTRL	0x400240AF	Endpoint 4 receive control register	0x00
R16_UEP5_T_LEN	0x400240B0	Endpoint 5 transmit length register	0x0000
R8_UEP5_TX_CTRL	0x400240B2	Endpoint 5 transmit control register	0x00
R8_UEP5_RX_CTRL	0x400240B3	Endpoint 5 receive control register	0x00
R16_UEP6_T_LEN	0x400240B4	Endpoint 6 transmit length register	0x0000
R8_UEP6_TX_CTRL	0x400240B6	Endpoint 6 transmit control register	0x00
R8_UEP6_RX_CTRL	0x400240B7	Endpoint 6 receive control register	0x00
R16_UEP7_T_LEN	0x400240B8	Endpoint 7 transmit length register	0x0000
R8_UEP7_TX_CTRL	0x400240BA	Endpoint 7 transmit control register	0x00
R8_UEP7_RX_CTRL	0x400240BB	Endpoint 7 receive control register	0x00
R16_UEP_T_ISO	0x400240BC	USBHS endpoint transmit synchronization mode enable register	0x0000
R16_UEP_R_ISO	0x400240BE	USBHS endpoint receive synchronization mode enable register	0x0000

Table 22-2 USBHS2 device-related registers

Name	Access address	Description	Reset value
R8_USB_CTRL	0x40023400	USBHS control register	0x07
R8_USB_BASE_MODE	0x40023401	USBHS mode control register	0x00
R8_USB_INT_EN	0x40023402	USBHS interrupt enable register	0x00
R8_USB_DEV_AD	0x40023403	USBHS device address register	0x00
R8_USB_WAKE_CTRL	0x40023404	USBHS remote wakeup register	0x00
R8_USB_TEST_MODE	0x40023405	USBHS test mode register	0x00
R16_USB_LPM_DATA	0x40023406	USBHS power management register	0x8000
R8_USB_INT_FG	0x40023408	USBHS interrupt flag register	0x00
R8_USB_INT_ST	0x40023409	USBHS interrupt status register	0x00
R8_USB_MIS_ST	0x4002340A	USBHS miscellaneous status register	0x00
R16_USB_FRAME_NO	0x4002340C	USBHS frame number register	0x0000
R16_USB_BUS	0x4002340E	USBHS bus status register	0x0000
R16_UEP_TX_EN	0x40023410	USBHS endpoint transmit enable register	0x0000
R16_UEP_RX_EN	0x40023412	USBHS endpoint receive enable register	0x0000
R16_UEP_T_TOG_AUTO	0x40023414	USBHS endpoint transmit auto-flip enable register	0x0000
R16_UEP_R_TOG_AUTO	0x40023416	USBHS endpoint receive auto-flip enable register	0x0000
R8_UEP_T_BURST	0x40023418	USBHS endpoint transmit burst register	0x00
R8_UEP_T_BURST_MODE	0x40023419	USBHS endpoint transmit burst mode register	0x00
R8_UEP_R_BURST	0x4002341A	USBHS endpoint receive burst register	0x00
R8_UEP_R_RES_MODE	0x4002341B	USBHS endpoint receive burst mode register	0x00
R32_UEP_AF_MODE	0x4002341C	USBHS endpoint alter Register	0x00000000
R32_UEP0_DMA	0x40023420	Start address register of endpoint 0 buffer	0x000XXXXX
R32_UEP1_RX_DMA	0x40023424	Start address register of endpoint 1 receive buffer	0x000XXXXX
R32_UEP2_RX_DMA	0x40023428	Start address register of endpoint 2 receive buffer	0x000XXXXX
R32_UEP3_RX_DMA	0x4002342C	Start address register of endpoint 3 receive buffer	0x000XXXXX
R32_UEP4_RX_DMA	0x40023430	Start address register of endpoint 4 receive buffer	0x000XXXXX
R32_UEP5_RX_DMA	0x40023434	Start address register of endpoint 5 receive buffer	0x000XXXXX
R32_UEP6_RX_DMA	0x40023438	Start address register of endpoint 6 receive buffer	0x000XXXXX
R32_UEP7_RX_DMA	0x4002343C	Start address register of endpoint 7 receive buffer	0x000XXXXX
R32_UEP1_TX_DMA	0x40023440	Start address register of endpoint 1 transmit buffer	0x000XXXXX
R32_UEP2_TX_DMA	0x40023444	Start address register of endpoint 2 transmit	0x000XXXXX

		buffer	
R32_UEP3_TX_DMA	0x40023448	Start address register of endpoint 3 transmit buffer	0x000XXXXX
R32_UEP4_TX_DMA	0x4002344C	Start address register of endpoint 4 transmit buffer	0x000XXXXX
R32_UEP5_TX_DMA	0x40023450	Start address register of endpoint 5 transmit buffer	0x000XXXXX
R32_UEP6_TX_DMA	0x40023454	Start address register of endpoint 6 transmit buffer	0x000XXXXX
R32_UEP7_TX_DMA	0x40023458	Start address register of endpoint 7 transmit buffer	0x000XXXXX
R32_UEP0_MAX_LEN	0x4002345C	Endpoint 0 maximum length packet register	0x000000XX
R32_UEP1_MAX_LEN	0x40023460	Endpoint 1 maximum length packet register	0x00000XXX
R32_UEP2_MAX_LEN	0x40023464	Endpoint 2 maximum length packet register	0x00000XXX
R32_UEP3_MAX_LEN	0x40023468	Endpoint 3 maximum length packet register	0x00000XXX
R32_UEP4_MAX_LEN	0x4002346C	Endpoint 4 maximum length packet register	0x00000XXX
R32_UEP5_MAX_LEN	0x40023470	Endpoint 5 maximum length packet register	0x00000XXX
R32_UEP6_MAX_LEN	0x40023474	Endpoint 6 maximum length packet register	0x00000XXX
R32_UEP7_MAX_LEN	0x40023478	Endpoint 7 maximum length packet register	0x00000XXX
R16_UEP0_RX_LEN	0x4002347C	Endpoint 0 receive length register	0x00XX
R16_UEP1_RX_LEN	0x40023480	Endpoint 1 single receive length register	0xXXXX
R16_UEP1_R_SIZE	0x40023482	Endpoint 1 total receive data length register	0xXXXX
R16_UEP2_RX_LEN	0x40023484	Endpoint 2 single receive length register	0xXXXX
R16_UEP2_R_SIZE	0x40023486	Endpoint 2 total receive data length register	0xXXXX
R16_UEP3_RX_LEN	0x40023488	Endpoint 3 single receive length register	0xXXXX
R16_UEP3_R_SIZE	0x4002348A	Endpoint 3 total receive data length register	0xXXXX
R16_UEP4_RX_LEN	0x4002348C	Endpoint 4 single receive length register	0xXXXX
R16_UEP4_R_SIZE	0x4002348E	Endpoint 4 total receive data length register	0xXXXX
R16_UEP5_RX_LEN	0x40023490	Endpoint 5 single receive length register	0xXXXX
R16_UEP5_R_SIZE	0x40023492	Endpoint 5 total receive data length register	0xXXXX
R16_UEP6_RX_LEN	0x40023494	Endpoint 6 single receive length register	0xXXXX
R16_UEP6_R_SIZE	0x40023496	Endpoint 6 total receive data length register	0xXXXX
R16_UEP7_RX_LEN	0x40023498	Endpoint 7 single receive length register	0xXXXX
R16_UEP7_R_SIZE	0x4002349A	Endpoint 7 total receive data length register	0xXXXX
R16_UEP0_T_LEN	0x4002349C	Endpoint 0 transmit length register	0x00XX
R8_UEP0_TX_CTRL	0x4002349E	Endpoint 0 transmit control register	0x00
R8_UEP0_RX_CTRL	0x4002349F	Endpoint 0 receive control register	0x00
R16_UEP1_T_LEN	0x400234A0	Endpoint 1 transmit length register	0x0000
R8_UEP1_TX_CTRL	0x400234A2	Endpoint 1 transmit control register	0x00
R8_UEP1_RX_CTRL	0x400234A3	Endpoint 1 receive control register	0x00
R16_UEP2_T_LEN	0x400234A4	Endpoint 2 transmit length register	0x0000
R8_UEP2_TX_CTRL	0x400234A6	Endpoint 2 transmit control register	0x00
R8_UEP2_RX_CTRL	0x400234A7	Endpoint 2 receive control register	0x00
R16_UEP3_T_LEN	0x400234A8	Endpoint 3 transmit length register	0x0000
R8_UEP3_TX_CTRL	0x400234AA	Endpoint 3 transmit control register	0x00

R8_UEP3_RX_CTRL	0x400234AB	Endpoint 3 receive control register	0x00
R16_UEP4_T_LEN	0x400234AC	Endpoint 4 transmit length register	0x0000
R8_UEP4_TX_CTRL	0x400234AE	Endpoint 4 transmit control register	0x00
R8_UEP4_RX_CTRL	0x400234AF	Endpoint 4 receive control register	0x00
R16_UEP5_T_LEN	0x400234B0	Endpoint 5 transmit length register	0x0000
R8_UEP5_TX_CTRL	0x400234B2	Endpoint 5 transmit control register	0x00
R8_UEP5_RX_CTRL	0x400234B3	Endpoint 5 receive control register	0x00
R16_UEP6_T_LEN	0x400234B4	Endpoint 6 transmit length register	0x0000
R8_UEP6_TX_CTRL	0x400234B6	Endpoint 6 transmit control register	0x00
R8_UEP6_RX_CTRL	0x400234B7	Endpoint 6 receive control register	0x00
R16_UEP7_T_LEN	0x400234B8	Endpoint 7 transmit length register	0x0000
R8_UEP7_TX_CTRL	0x400234BA	Endpoint 7 transmit control register	0x00
R8_UEP7_RX_CTRL	0x400234BB	Endpoint 7 receive control register	0x00
R16_UEP_T_ISO	0x400234BC	USBHS endpoint transmit synchronization mode enable register	0x0000
R16_UEP_R_ISO	0x400234BE	USBHS endpoint receive synchronization mode enable register	0x0000

22.2.1.1 USBHS Control Register (R8_USB_CTRL)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
7	RB_UD_LPM_EN	RW	LPM enable: 1: Enable; 0: Disable.	0
6	Reserved	RO	Reserved	0
5	RB_UD_DEV_EN	RW	USB device enable: 1: Enable; 0: Disable.	0
4	RB_UD_DMA_EN	RW	DMA transfer enable: 1: Enable; 0: Disable.	0
3	RB_UD_PHY_SUSPENDM	RW	USB PHY suspended: 1: Normal operation; 0: Suspended.	0
2	RB_UD_CLR_ALL	RW	Clear all interrupt flags: 1: Clear USB interrupt flag and FIFO; requires software reset; 0: Do not clear.	1
1	RB_UD_RST_SIE	RW	USB protocol processor reset: 1: Force reset the USB Protocol Processor (SIE), including endpoint-related registers, requiring software clearing; 0: Do not reset.	1
0	RB_UD_RST_LINK	RW	LINK layer reset: 1: USB Link layer reset;	1

			0: No reset.	

22.2.1.2 USBHS Mode Control Register (R8_USB_BASE_MODE)

Offset address: 0x01

Bit	Name	Access	Description	Reset value
[7:2]	Reserved	RO	Reserved	0x0
[1:0]	RB_UD_SPEED_TYPE[1:0]	RW	Desired speed mode for device: 00: Full-Speed device; 01: High-Speed device; 10: Low-Speed device; 11: Reserved.	0

22.2.1.3 USBHS Interrupt Enable Register (R8_USB_INT_EN)

Offset address: 0x02

Bit	Name	Access	Description	Reset value
7	RB_UDIE_FIFO_OVER	RW	FIFO overflow interrupt. 1: Enable interrupt; 0: Disable interrupt.	0
6	RB_UDIE_LINK_RDY	RW	USB connection interrupt enabled. 1: Enable interrupt; 0: Disable interrupt.	0
5	RB_UDIE_SOF_ACT	RW	SOF packet interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0
4	RB_UDIE_TRANSFER	RW	USB transfer complete interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0
3	RB_UDIE_LPM_ACT	RW	LPM transfer end interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0
2	RB_UDIE_BUS_SLEEP	RW	USB bus sleep interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0
1	RB_UDIE_SUSPEND	RW	USB bus interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0
0	RB_UDIE_BUS_RST	RW	USB bus reset interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0

22.2.1.4 USBHS Device Address Register (R16_USB_DEV_AD)

Offset address: 0x03

Bit	Name	Access	Description	Reset value
7	Reserved	RO	Reserved	0

[6:0]	RB_UD_DEV_ADDR[6:0]	RW	USB device address.	0

22.2.1.5 USBHS Remote Wakeup Register (R8_USB_WAKE_CTRL)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[7:1]	Reserved	RO	Reserved	0
0	RB_UD_REMOTE_WKUP	RW1	Remote wake-up, hardware automatically reset.	0

22.2.1.6 USBHS Test Mode Register (R8_USB_TEST_MODE)

Offset address: 0x05

Bit	Name	Access	Description	Reset value
7	RB_UD_TEST_EN	RW	Test mode enable.	0
[6:4]	Reserved	RO	Reserved	0
3	RB_UD_TEST_SE0NAK	RW	Test mode, output SE0.	0
2	RB_UD_TEST_PKT	RW	Test mode, output one packet. DATA0, the packet uses the data address and length from endpoint 4, with TOG set to DATA0.	0
1	RB_UD_TEST_K	RW	Test mode, output K.	0
0	RB_UD_TEST_J	RW	Test mode, output J.	0

22.2.1.7 USBHS Power Management Register (R16_USB_LPM_DATA)

Offset address: 0x06

Bit	Name	Access	Description	Reset value
15	RB_UD_LPM_BUSY	RW	Power management busy.	1
[14:11]	Reserved	RO	Reserved	0
[10:0]	RB_UD_LPM_DATA[10:0]	RO	Power management data.	0

22.2.1.8 USB Interrupt Flag Register (R8_USB_INT_FG)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
7	RB_UDIF_FIFO_OV	RW1Z	FIFO overflow interrupt flag bit; write 1 to clear. 1: FIFO overflow triggered; 0: No event.	0
6	RB_UDIF_LINK_RDY	RW1Z	USB connection interrupt flag bit; write 1 to clear. 1: USB connection event triggered; 0: No event.	0

5	RB_UDIF_RX_SOF	RW1Z	Receive SOF packet interrupt flag bit; write 1 to clear. 1: SOF packet event triggered; 0: No event.	0
4	RB_UDIF_RTX_ACT	RO	USB transfer complete interrupt flag: Receive cleared by RB_UEP_R_DONE; Transmit cleared by RB_UEP_T_DONE.	0
3	RB_UDIF_LPM_ACT	RW1Z	LPM transmission end interrupt flag, write 1 to clear. 1: LPM transmission end event triggered; 0: No event.	0
2	RB_UDIF_BUS_SLEEP	RW1Z	USB Bus Sleep Interrupt Flag When LPM is enabled, entering L1 (Sleep) state triggers this flag, writing 1 to clear. 1: USB bus sleep event triggered; 0: No event.	0
1	RB_UDIF_SUSPEND	RW1Z	USB bus suspend interrupt flag, write 1 to clear. 1: USB suspend event triggered; 0: No event.	0
0	RB_UDIF_BUS_RST	RW1Z	USB bus reset interrupt flag, write 1 to clear. 1: USB bus reset event triggered; 0: No event.	0

22.2.1.9 USB Interrupt Status Register (R8_USB_INT_ST)

Offset address: 0x09

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved	0
4	RB_UDIS_EP_DIR	RO	Endpoint data transmission direction: 1: Endpoint IN data; 0: Endpoint OUT/SETUP data.	0
3	Reserved	RO	Reserved	0
[2:0]	RB_UDIS_EP_ID_MASK[2:0]	RO	The endpoint number where data transmission occurs.	0

22.2.1.10 USB Miscellaneous Status Register (R8_USB_MIS_ST)

Offset address: 0x0A

Bit	Name	Access	Description	Reset value
7	RB_UDMS_HS_MOD	RO	Whether the host is at high speed.	0
[6:5]	Reserved	RO	Reserved	0
4	RB_UDMS_SUSP_REQ	RO	USB suspend request.	0
3	RB_UDMS_SIE_FREE	RO	USB idle state.	0
2	RB_UDMS_SLEEP	RO	USB sleep state.	0
1	RB_UDMS_SUSPEND	RO	USB suspend state.	0

			1: The USB bus is in suspend state, with no USB activity for a period of time; 0: The USB bus is in non-suspend state.	
0	RB_UDMS_READY	RO	USB connection status.	0

22.2.1.11 USB Frame Register (R16_USB_FRAME_NO)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[15:13]	RB_UD_MFRAME_NO[2:0]	RO	The received micro-frame number.	0
[12:11]	Reserved	RO	Reserved	0
[10:0]	RB_UD_FRAME_NO[10:0]	RO	The received frame number.	0

22.2.1.12 USB Bus Status Register (R16_USB_BUS)

Offset address: 0x0E

Bit	Name	Access	Description	Reset value
[15:4]	Reserved	RO	Reserved	0
3	RB_USB_DM_ST	RO	UDM status.	0
2	RB_USB_DP_ST	RO	UDP status.	0
1	Reserved	RO	Reserved	0
0	RB_USB_WAKEUP	RO	USB wakeup (active high).	0

22.2.1.13 USBHS Endpoint Transmit Enable Register (R16_UEP_TX_EN)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[15:0]	RB_UEP_TX_EN[15:0]	RW	0~15 endpoint transmit enable register: 1: Enable; 0: Disable.	0

22.2.1.14 USBHS Endpoint Receive Enable Register (R16_UEP_RX_EN)

Offset address: 0x12

Bit	Name	Access	Description	Reset value
[15:0]	RB_UEP_RX_EN[15:0]	RW	0~15 endpoint receive enable register: 1: Enable; 0: Disable.	0

22.2.1.15 USBHS Endpoint Transmit Auto-flip Enable Register (R16_UEP_T_TOG_AUTO)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:1]	RB_UEP_T_TOG_AUTO[7:1]	RW	Auto flip enable for synchronous trigger bits on endpoints 1~7: 1: Automatically flips after successful data	0

			transmission; 0: Manually controls flipping.	
0	Reserved	RO	Reserved	0

22.2.1.16 USBHS Endpoint Transmit Auto-flip Enable Register (R16_UEP_R_TOG_AUTO)

Offset address: 0x16

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:1]	RB_UEP_R_TOG_AUTO[7:1]	RW	Auto flip enable for synchronous trigger bits on endpoints 1–7: 1: Automatically flips after successful data reception; 0: Manually controls flipping.	0
0	Reserved	RO	Reserved	0

22.2.1.17 USBHS Endpoint Transmit Burst Register (R8_UEP_T_BURST)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[7:1]	RB_UEP_T_BURST_EN[6:0]	RW	1~7 endpoint burst transmit enable: 1: Enable; 0: Disable.	0
0	Reserved	RO	Reserved	0

22.2.1.18 USBHS Endpoint Transmit Mode Register (R8_UEP_T_BURST_MODE)

Offset address: 0x19

Bit	Name	Access	Description	Reset value
[7:1]	RB_UEP_T_BURST_MODE[6:0]	RW	1~7 endpoint burst transmit mode: 1: Do not transmit zero-length data in burst mode; 0: Transmit zero-length data in burst mode.	0
0	Reserved	RO	Reserved	0

22.2.1.19 USBHS Endpoint Receive Burst Register (R8_UEP_R_BURST)

Offset address: 0x1A

Bit	Name	Access	Description	Reset value
[7:1]	RB_UEP_R_BURST_EN[6:0]	RW	1~7 endpoint burst receive enable: 1: Enable; 0: Disable.	0
0	Reserved	RO	Reserved	0

22.2.1.20 USBHS Endpoint Reply Mode Register (R8_UEP_R_RES_MODE)

Offset address: 0x1B

Bit	Name	Access	Description	Reset value

[7:1]	RB_UEP_R_RES_MODE[6:0]	RW	1~7 endpoint receive return mode: 1: NYET; 0: ACK.	0
0	Reserved	RO	Reserved	0

22.2.1.21 USBHS Endpoint Alternate Register (R32_UEP_AF_MODE)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								UEP_T_AF[6:0]							Reserved

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:1]	RB_UEP_T_AF[6:0]	RW	Endpoint multiplexing enable for 1~7: 1: Multiplex 9~15; 0: Multiplex 1~7.	0
0	Reserved	RO	Reserved	0

22.2.1.22 Start Address Register of Endpoint 0 Buffer (R32_UEP0_DMA)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UEP0_BUF_DMA[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UEP0_BUF_DMA[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	UEP0_BUF_DMA[31:0]	RW	Start address register of endpoint 0 buffer.	X

22.2.1.23 Start Address Register of Endpoint n Receive Buffer (R32_UEPn_RX_DMA) (n=1~7)

Offset address: 0x24 + 4*(n-1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UEPn_RX_DMA[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UEPn_RX_DMA[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	UEPn_RX_DMA[31:0]	RW	Start address register of endpoint n receive buffer. The address must be 4 bytes aligned.	X

22.2.1.24 Start Address Register of Endpoint n Transmit Buffer (R32_UEPn_TX_DMA) (n=1~7)

Offset address: 0x40 + 4*(n-1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UEPn_TX_DMA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UEPn_TX_DMA[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	UEPn_TX_DMA[31:0]	RW	The start address of endpoint n transmit buffer. The address must be 4-byte aligned.	X

22.2.1.25 Endpoint 0 Maximum Length Package Register (R32_UEPn_MAX_LEN)

Offset address: 0x5C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									UEP0_MAX_LEN[6:0]						

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RO	Reserved	0
[6:0]	UEP0_MAX_LEN[6:0]	RW	The maximum offset length allowed for endpoint 0 DMA.	X

22.2.1.26 Endpoint n Maximum Length Package Register (R32_UEPn_MAX_LEN) (n=1~7)

Offset address: 0x5C + 4n

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					UEPn_MAX_LEN[10:0]										

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[10:0]	UEPn_MAX_LEN[10:0]	RW	The maximum offset length allowed for endpoint n DMA.	X

22.2.1.27 Endpoint 0 Receive Length Register (R16_UEP0_RX_LEN)

Offset address: 0x7C

Bit	Name	Access	Description	Reset value
[15:7]	Reserved	RO	Reserved	0
[6:0]	UEP0_RX_LEN[6:0]	RW	Endpoint 0 receive data length.	X

22.2.1.28 Endpoint n Single Receive Length Register (R16_UEPn_RX_LEN) (n=1~7)

Offset address: 0x80 + 4*(n-1)

Bit	Name	Access	Description	Reset value
[15:0]	UEPn_RX_LEN[15:0]	RW	Endpoint n single receive data length.	X

22.2.1.29 Endpoint n Total Receive Length Register (R16_UEPn_R_SIZE) (n=1~7)

Offset address: 0x82 + 4*(n-1)

Bit	Name	Access	Description	Reset value
[15:0]	UEPn_R_SIZE[15:0]	RW	Endpoint n total receive data length.	X

22.2.1.30 Endpoint 0 Transmit Length Register (R16_UEP0_T_LEN)

Offset address: 0x9C

Bit	Name	Access	Description	Reset value
[15:7]	Reserved	RO	Reserved	0
[6:0]	UEP0_T_LEN[6:0]	RW	Endpoint 0 transmit data length.	X

22.2.1.31 Endpoint 0 Transmit Control Register (R8_UEP0_TX_CTRL)

Offset address: 0x9E

Bit	Name	Access	Description	Reset value
7	RB_UEP_T_DONE	RW0	Endpoint 0 transmits the end flag, writes 0 to clear it.	0
6	RB_UEP_T_NAK_ACT	RW0	Endpoint 0 transmits NAK end flag, writes 0 to clear it.	0
[5:3]	Reserved	RO	Reserved	0
2	RB_UEP_T_TOG_MASK	RW	Synchronous trigger bit expected to be sent by endpoint 0: 1: DATA1; 0: DATA0.	0
[1:0]	RB_UEP_T_RES_MASK[1:0]	RW	Endpoint 0 response control for transmission: 00: Respond with NAK or busy; 01: Respond with STALL or error; 10: Respond with ACK; 11: Reserved.	0

22.2.1.32 Endpoint 0 Receive Control Register (R8_UEP0_RX_CTRL)

Offset address: 0x9F

Bit	Name	Access	Description	Reset value
7	RB_UEP_R_DONE	RW0	Endpoint 0 receives the end flag and writes 0 to clear it.	0
6	RB_UEP_R_NAK_ACT	RW0	Endpoint 0 receives the NAK end flag and writes 0 to clear it.	0
5	RB_UEP_R_NAK_TOG	RO	Endpoint 0 For the received return NAK,	0

			packet type: 1: DATA1; 0: DATA0.	
4	RB_UEP_R_TOG_MATCH	RO	Received synchronous trigger bit matching status with expected synchronous trigger bit: 1: Synchronized; 0: Not synchronized.	0
3	RB_UEP_R_SETUP_IS	RO	Whether endpoint 0 receives a SETUP transaction.	0
2	RB_UEP_R_TOG_MASK	RW	Expected synchronous trigger bits for endpoint 0 reception: 1: DATA1; 0: DATA0.	0
[1:0]	RB_UEP_R_RES_MASK[1:0]	RW	Endpoint 0 response control for received responses: 00: Respond with NAK or busy; 01: Respond with STALL or error; 10: Respond with ACK; 11: Reserved.	0

22.2.1.33 Endpoint n Transmit Length Register (R16_UEPn_T_LEN) (n=1~7)

Offset address: 0xA0 + 4*(n-1)

Bit	Name	Access	Description	Reset value
[15:0]	UEPn_T_LEN[15:0]	RW	The length of data sent by endpoint n.	0

22.2.1.34 Endpoint n Transmit Control Register (R8_UEPn_TX_CTRL) (n=1~7)

Offset address: 0xA2 + 4*(n-1)

Bit	Name	Access	Description	Reset value
7	RB_UEP_T_DONE	RW0	Endpoint n sends the end flag and writes 0 to clear it.	0
6	RB_UEP_T_NAK_ACT	RW0	Endpoint n sends the NAK end flag and writes 0 to clear it.	0
[5:4]	Reserved	RO	Reserved	0
[3:2]	RB_UEP_T_TOG_MASK[1:0]	RW	Expected synchronization trigger bit for transmission of endpoint n: 00: DATA0; 01: DATA1; 10: DATA2; 11: MDATA. *Note: This bit can be configured in manual mode and not in automatic mode.*	0
[1:0]	RB_UEP_T_RES_MASK[1:0]	RW	Endpoint n response control for transmission: 00: Respond with NAK or busy;	0

			01: Respond with STALL or error; 10: Respond with ACK; 11: Reserved.	

22.2.1.35 Endpoint n Receive Control Register (R8_UEPn_RX_CTRL) (n=1~7)

Offset address: 0xA3 + 4*(n-1)

Bit	Name	Access	Description	Reset value
7	RB_UEP_R_DONE	RW0	Endpoint n receives the end flag and writes 0 to clear it.	0
6	RB_UEP_R_NAK_ACT	RW0	Endpoint n receives the NAK end flag and writes 0 to clear it.	0
5	RB_UEP_R_NAK_TOG	RO	Endpoint n for the received return NAK, packet type: 1: DATA1; 0: DATA0.	0
4	RB_UEP_R_TOG_MATCH	RO	The synchronization trigger bit received by endpoint n matches the expected synchronization trigger bit: 1: Synchronized; 0: Not synchronized.	0
[3:2]	RB_UEP_R_TOG_MASK[1:0]	RW	Receiving expected synchronization trigger bit of endpoint n: 00: DATA0; 01: DATA1; 10: DATA2; 11: MDATA. *Note: This bit can be configured in manual mode and not in automatic mode.*	0
[1:0]	RB_UEP_R_RES_MASK[1:0]	RW	Endpoint n controls responses received: 00: Responds with NAK or busy; 01: Responds with STALL or error; 10: Responds with ACK; 11: Responds with ACK (Interrupt download).	0

22.2.1.36 Endpoint Transmit Synchronization Mode Enable Register (R16_UEP_T_ISO)

Offset address: 0xBC

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:1]	RB_UEPn_T_ISO_EN[6:0]	RW	The upload endpoint (IN) synchronization mode is enabled.	0
0	Reserved	RO	Reserved	0

22.2.1.37 USBHS Endpoint Receive Synchronization Mode Enable Register (R16_UEP_R_ISO)

Offset address: 0xBE

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:1]	RB_UEPn_R_ISO_EN [6:0]	RW	Downstream endpoint (OUT) synchronization mode is enabled.	0
0	Reserved	RO	Reserved	0

22.2.2 Host Register Description

Table 22-2 USBHS host-related registers

Name	Access address	Description	Reset value
R8_UH_CFG	0x40023500	USBHS host configuration register	0x07
R8_UH_INT_EN	0x40023502	USBHS host interrupt enable register	0x00
R8_UH_DEV_AD	0x40023503	USBHS host device address register	0x00
R32_UH_CONTROL	0x40023504	USBHS host control register	0x00XXXXXX
R8_UH_INT_FLAG	0x40023508	USBHS host interrupt flag register	0x00
R8_UH_INT_ST	0x40023509	USBHS host interrupt status register	0xXX
R8_UH_MIS_ST	0x4002350A	USBHS host miscellaneous status register	0xXX
R32_UH_LPM_DATA	0x4002350C	USBHS host power management data register	0x00000XXX
R32_UH_SPLIT_DATA	0x40023510	USBHS host SPLIT data register	0x000XXXXX
R32_UH_FRAME	0x40023514	USBHS host frame register	0x00000000
R32_UH_TX_LEN	0x40023518	USBHS host transmit length register	0x00000XXX
R32_UH_RX_LEN	0x4002351C	USBHS host receive length register	0x00000XXX
R32_UH_RX_MAX_LEN	0x40023520	USBHS host receive maximum length register	0x00000XXX
R32_UH_RX_DMA	0x40023524	DMA receive address register	0x000XXXXX
R32_UH_TX_DMA	0x40023528	DMA transmit address register	0x000XXXXX
R32_UH_PORT_CTRL	0x4002352C	USBHS host port control register	0x0000X000
R8_UH_PORT_CFG	0x40023530	USBHS host port configuration register	0x00
R8_UH_PORT_INT_EN	0x40023532	USBHS host port interrupt enable register	0x00
R8_UH_PORT_TEST_CT	0x40023533	USBHS host port test mode register	0x00
R16_UH_PORT_ST	0x40023534	USBHS host port status register	0x0010
R8_UH_PORT_CHG	0x40023536	USBHS host port status change register	0x00
R32_UH_BC_CTRL	0x4002353C	USBHS host BC charging control register	0x00000000

Table 22-4 USBHS2 host-related registers

Name	Access address	Description	Reset value
R8_UH_CFG	0x40024100	USBHS host configuration register	0x07
R8_UH_INT_EN	0x40024102	USBHS host interrupt enable register	0x00
R8_UH_DEV_AD	0x40024103	USBHS host device address register	0x00
R32_UH_CONTROL	0x40024104	USBHS host control register	0x00XXXXXX
R8_UH_INT_FLAG	0x40024108	USBHS host interrupt flag register	0x00

R8_UH_INT_ST	0x40024109	USBHS host interrupt status register	0xXX
R8_UH_MIS_ST	0x4002410A	USBHS host miscellaneous status register	0xXX
R32_UH_LPM_DATA	0x4002410C	USBHS host power management data register	0x00000XXX
R32_UH_SPLIT_DATA	0x40024110	USBHS host SPLIT data register	0x000XXXXX
R32_UH_FRAME	0x40024114	USBHS host frame register	0x00000000
R32_UH_TX_LEN	0x40024118	USBHS host transmit length register	0x00000XXX
R32_UH_RX_LEN	0x4002411C	USBHS host receive length register	0x00000XXX
R32_UH_RX_MAX_LEN	0x40024120	USBHS host receive maximum length register	0x00000XXX
R32_UH_RX_DMA	0x40024124	DMA receive address register	0x000XXXXX
R32_UH_TX_DMA	0x40024128	DMA transmit address register	0x000XXXXX
R32_UH_PORT_CTRL	0x4002412C	USBHS host port control register	0x0000X000
R8_UH_PORT_CFG	0x40024130	USBHS host port configuration register	0x00
R8_UH_PORT_INT_EN	0x40024132	USBHS host port interrupt enable register	0x00
R8_UH_PORT_TEST_CT	0x40024133	USBHS host port test mode register	0x00
R16_UH_PORT_ST	0x40024134	USBHS host port status register	0x0010
R8_UH_PORT_CHG	0x40024136	USBHS host port status change register	0x00
R32_UH_BC_CTRL	0x4002413C	USBHS host BC charging control register	0x00000000

22.2.2.1 USBHS Host Control Register (R8_UH_CFG)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
7	RB_UH_LPM_EN	RW	LPM enable.	0
6	RB_UH_FORCE_FS	RW	Forced to use USB_FS.	0
5	RB_UH_SOF_EN	RW	Enable SOF package transmitted.	0
4	RB_UH_DMA_EN	RW	Enable DMA transmitted.	0
3	RB_UH_PHY_SUSPENDM	RW	PHY suspended.	0
2	RB_UH_CLR_ALL	RW	Clearing the interrupt flag and FIFO requires software clearing.	1
1	RB_UH_RST_SIE	RW	The USB protocol processor is reset, which requires software clearing.	1
0	RB_UH_RST_LINK	RW	The USB connection control module is reset.	1

22.2.2.2 USBHS Host Interrupt Enable Register (R8_UH_INT_EN)

Offset address: 0x02

Bit	Name	Access	Description	Reset value
7	RB_UHIE_FIFO_OVER	RW	FIFO overflow interrupt enable	0
6	RB_UHIE_TX_HALT	RW	Transmit pause interrupt enable	0
5	RB_UHIE_SOF_ACT	RW	SOF packet transmission interrupt enable	0
4	RB_UHIE_TRANSFER	RW	USB end or transfer complete interrupt enable	0

3	RB_UHIE_RESUME_ACT	RW	Bus recovery interrupt enable	0
2	RB_UHIE_WKUP_ACT	RW	Wake-up interrupt enable	0
[1:0]	Reserved	RO	Reserved	0

22.2.2.3 USBHS Host Device Address Register (R8_UH_DEV_AD)

Offset address: 0x03

Bit	Name	Access	Description	Reset value
[7:0]	RB_UH_DEV_ADDR[6:0]	RW	The address of the currently operating USB device.	0

22.2.2.4 USBHS Host Control Register (R32_UH_CONTROL)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
23	RB_UH_RX_NO_RES	RW	IN-DATA no response, used for synchronous transmission or high-speed SPLIT packets.	X
22	RB_UH_TX_NO_RES	RW	OUT/SETUP–DATA does not expect a response and is used for synchronous transmission or high-speed SPLIT packets.	X
21	RB_UH_RX_NO_DATA	RW	IN token packet followed by no expected data packet, used for high-speed SPLIT packets.	X
20	RB_UH_TX_NO_DATA	RW	No data packets follow the OUT/SETUP token packet, intended for high-speed SPLIT packets.	X
19	RB_UH_PRE_PID_EN	RW	Low-speed preamble packet PRE PID enable bit: When the port operates at full speed and requires transmission of low-speed packets (PRE), this bit must be enabled. 1: Enabled, used for communication with low-speed USB devices via an external full-speed HUB; 0: Disables low-speed preamble packets.	X
18	RB_UH_SPLIT_VALID	RW	Transmit SPLIT packet is valid.	X
17	RB_UH_LPM_VALID	RW	Transmit LPM packet is valid.	X
16	RB_UH_HOST_ACTION	RW	HOST performs transaction enabling, and this bit is cleared automatically after the transaction is completed.	0
[15:11]	Reserved	RO	Reserved	0
10	RB_UH_BUF_MODE	RW	Data cache control bit.	X

			1: Use R32_UH_RX_DMA for transmission and R32_UH_TX_DMA for reception; 0: Use R32_UH_TX_DMA for transmission and R32_UH_RX_DMA for reception.	
[9:8]	RB_UH_T_TOG_MASK [1:0]	RW	Transmit data PID: 00: PID_DATA0; 01: PID_DATA1; 10: PID_DATA2; 11: PID_MDATA。	X
[7:4]	RB_UH_T_ENDP_MASK [3:0]	RW	The endpoint number of the transaction token packet sent by the host.	X
[3:0]	RB_UH_T_TOKEN_MASK [3:0]	RW	The PID of the transaction token packet sent by the host.	X

22.2.2.5 USBHS Host Interrupt Flag Register (R8_UH_INT_FLAG)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
7	RB_UHIF_FIFO_OVER	RW1Z	FIFO overflow interrupt flag.	0
6	RB_UHIF_TX_HALT	RW1Z	The transmission stop interrupt flag is used in the host mode for reading data overflow during downloading, or when the transmission is not finished at the time point of EOF2, the transmission is stopped, and the write 1 is cleared.	0
5	RB_UHIF_SOF_ACT	RW1Z	SOF packet transmission completion interrupt flag, write 1 to clear.	0
4	RB_UHIF_TRANSFER	RW1Z	USB transaction transmission complete interrupt flag, write 1 to clear.	0
3	RB_UHIF_RESUME_ACT_IF	RW1Z	Bus recovery interrupt flag, write 1 to clear.	0
2	RB_UHIF_WKUP_ACT	RW1Z	Wake-up interrupt flag, write 1 to clear.	0
[1:0]	Reserved	RO	Reserved	0

22.2.2.6 USBHS Host Interrupt Status Register (R8_UH_INT_ST)

Offset address: 0x09

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved	X
4	RB_UHIS_PORT_RX_RESUME	RW	A bit of 1 indicates that the port has received a wake-up signal.	X
[3:0]	RB_UH_R_TOKEN_MASK [3:0]	RO	The received PID.	X

22.2.2.7 USBHS Host Miscellaneous Status Register (R8_UH_MIS_ST)

Offset address: 0x0A

Bit	Name	Access	Description	Reset value
7	RB_UHMS_BUS_SE0	RO	SE0 on USB bus.	X
6	RB_UHMS_BUS_J	RO	J on USB bus.	X
[5:4]	RB_UHMS_LINESTATE[1:0]	RO	PHY Linestate signal.	X
3	RB_UHMS_USB_WAKEUP	RO	USB bus wake status: 1: Bus wake in progress; 0: No bus wake in progress.	X
2	RB_UHMS_SOF_ACT	RO	USB bus SOF packet transmission status bit: 1: SOF packet being transmitted; 0: Transmission complete or idle.	X
1	RB_UHMS_SOF_PRE	RO	The USB bus SOF packet status indicates: 1: An SOF packet is about to be sent; 0: No SOF packet is being sent.	X
0	RB_UHMS_SOF_FREE	RO	Port enable status: 1: Port enabled; 0: Port disabled.	X

22.2.2.8 USBHS Host Power Management Data Register (R32_UH_LPM_DATA)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[10:0]	RB_UH_LPM_DATA[10:0]	RW	Link power management packet data content.	X

22.2.2.9 USBHS Host SPLIT Register (R32_UH_SPLIT_DATA)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[31:19]	Reserved	RO	Reserved	0
[18:0]	RB_UH_SPLIT_DATA[18:0]	RW	The data content of the SPLIT packet sent by the host endpoint, from high to low bits, is: ET, E, S, Port, SC, Hub Addr.	X

22.2.2.10 USBHS Host Frame Register (R32_UH_FRAME)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[31:26]	Reserved	RO	Reserved	0
25	RB_UH_SOF_CNT_CLR	RW	SOF count clear.	0

24	RB_UH_SOF_CNT_EN	RW	SOF count enable.	0
[23:19]	Reserved	RO	Reserved	0
[18:16]	RB_UH_MFRAME_NO[2:0]	RO	The micro-frame number of the upcoming SOF packet.	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	RB_UH_FRAME_NO[10:0]	RW	The frame number of the upcoming SOF packet.	0

22.2.2.11 USBHS Host Transmit Length Register (R32_UH_TX_LEN)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[10:0]	RB_UH_TX_LEN[10:0]	RW	Data transmission length.	X

22.2.2.12 USBHS Host Receive Length Register (R32_UH_RX_LEN)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					RX_LEN[10:0]										

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[10:0]	RB_UH_RX_LEN[10:0]	RW	Received data length.	X

22.2.2.13 USBHS Host Receive Maximum Length Register (R32_UH_RX_MAX_LEN)

Offset address: 0x20

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[10:0]	RB_UH_RX_MAX_LEN[10:0]	RW	Maximum length of reception.	X

22.2.2.14 DMA Receive Address Register (R32_UH_RX_DMA)

Offset address: 0x24

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
[16:0]	R32_UH_RX_DMA[16:0]	RW	Receiving address, the lower two digits are invalid, and 4 bytes alignment is required.	X

22.2.2.15 DMA Transmit Address Register (R32_UH_TX_DMA)

Offset address: 0x28

Bit	Name	Access	Description	Reset value
[31:0]	R32_UH_TX_DMA[31:0]	RW	Transmitting address.	X

22.2.2.16 USBHS Host Port Control Register (R32_UH_PORT_CTRL)

Offset address: 0x2C

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	RB_UH_BUS_RST_LONG	RW	Bus reset time selection: 1: Extended reset; 0: Normal reset.	0
[15:12]	RB_UH_PORT_SLEEP_BESL[3:0]	RW	Wake-up time control: 0000: 125us; 1000: 3ms; 0001: 150us; 1001: 4ms; 0010: 200us; 1010: 5ms; 0011: 300us; 1011: 6ms; 0100: 400us; 1100: 7ms; 0101: 500us; 1101: 8ms; 0110: 1ms; 1110: 9ms; 0111: 2ms; 1111: 10ms.	X
[11:9]	Reserved	RO	Reserved	0
8	RB_UH_CLR_PORT_SLEEP	WO	PORT exit SLEEP status (LPM).	0
[7:6]	Reserved	RO	Reserved	0
5	RB_UH_CLR_PORT_CONNECT	WO	Put the port into the port status.	0
4	RB_UH_CLR_PORT_EN	WO	PORT exits the enabled state and enters the DISABLED status.	0
3	RB_UH_SET_PORT_SLEEP	WO	PORT enter SLEEP status (LPM).	0
2	RB_UH_CLR_PORT_SUSP	WO	PORT exits the suspended status.	0
1	RB_UH_SET_PORT_SUSP	WO	PORT enters the suspended status.	0
0	RB_UH_SET_PORT_RESET	WO	Port transmit reset.	0

22.2.2.17 USBHS Host Port Configuration Register (R8_UH_PORT_CFG)

Offset address: 0x30

Bit	Name	Access	Description	Reset value
7	RB_UH_PD_EN	RW	15K resistor pull-down is enabled in host mode.	0
[6:1]	Reserved	RO	Reserved	0

0	RB_UH_HOST_EN	RW	USB port mode selection: 1: Port operates in host mode; 0: Port operates in device mode.	0

22.2.2.18 USBHS Host Port Interrupt Enable Register (R8_UH_PORT_INT_EN)

Offset address: 0x32

Bit	Name	Access	Description	Reset value
[7:6]	Reserved	RO	Reserved	0
5	RB_UHIE_PORT_SLP	RW	Port sleep state change interrupt enabled.	0
4	RB_UHIE_PORT_RESET	RW	Port reset state change interrupt enabled.	0
3	Reserved	RO	Reserved	0
2	RB_UHIE_PORT_SUSP	RW	Port suspended state change interrupt enabled.	0
1	RB_UHIE_PORT_EN	RW	Port enable state change interrupt enabled.	0
0	RB_UHIE_PORT_CONNECT	RW	Port connection state change interrupt enabled.	0

22.2.2.19 USBHS Host Port Test Mode Register (R8_UH_PORT_TEST_CT)

Offset address: 0x33

Bit	Name	Access	Description	Reset value
[7:3]	Reserved	RO	Reserved	0
2	RB_UH_TEST_FORCE_EN	RW	Test mode enabled.	0
1	RB_UH_TEST_K	RW	Test output K.	0
0	RB_UH_TEST_J	RW	Test output J.	0

22.2.2.20 USBHS Host Port Status Register (R16_UH_PORT_ST)

Offset address: 0x34

Bit	Name	Access	Description	Reset value
[15:12]	Reserved	RO	Reserved	0
11	RB_UHIS_PORT_TEST	RO	Whether the port is in test mode.	0
10	RB_UHIS_PORT_HS	RO	Whether the port connection speed is high.	0
9	RB_UHIS_PORT_LS	RO	Whether the port connection speed is low.	0
[8:6]	Reserved	RO	Reserved	0
5	RB_UHIS_PORT_SLP	RO	Port sleep.	0
4	RB_UHIS_PORT_RST	RO	Port reset status.	1
3	Reserved	RO	Reserved	0
2	RB_UHIS_PORT_SUSP	RO	Port suspended status.	0
1	RB_UHIS_PORT_EN	RO	Port enabled.	0
0	RB_UHIS_PORT_CONNECT	RO	Port connection status.	0

22.2.2.21 USBHS Host Port Status Change Register (R8_UH_PORT_CHG)

Offset address: 0x36

Bit	Name	Access	Description	Reset value
[7:6]	Reserved	RO	Reserved	0
5	RB_UHIF_PORT_SLP	RW1Z	Port sleep state changes, write 1 to clear.	0
4	RB_UHIF_PORT_RESET	RW1Z	Port reset state changes, write 1 to clear.	0
3	Reserved	RO	Reserved	0
2	RB_UHIF_PORT_SUSP	RW1Z	Port suspend state changes, write 1 to clear.	0
1	RB_UHIF_PORT_EN	RW1Z	Port enable state changes, write 1 to clear.	0
0	RB_UHIF_PORT_CONNECT	RW1Z	Port connection status changes, write 1 to clear.	0

22.2.2.22 USBHS Host BC Charging Control Register (R32_UH_BC_CTRL)

Offset address: 0x3C

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
10	RB_UDM_VSRC_ACT	RO	In automatic mode, this bit indicates UDM output V_{BC_SRC}; otherwise, it is controlled by RB_UDM_BC_CMPE. *Note: This bit is set to 1 when UDP is higher than V_{BC_SRC} but UDP is not high; otherwise, it is 0.*	0
9	RB_UDM_BC_VSRC	RW	UDM pin BC protocol source voltage enable: 1: UDM pin outputs BC protocol source voltage V_{BC_SRC}; 0: Output disabled.	0
8	RB_UDP_BC_VSRC	RW	UDP Pin BC Protocol Source Voltage Enable: 1: Outputs BC protocol source voltage V_{BC_SRC} via UDP pin; 0: Disables output.	0
7	Reserved	RO	Reserved	0
6	RB_BC_AUTO_MODE	RW	Auto mode enable: 1: Enabled; 0: Disabled.	0
5	RB_UDM_BC_CMPE	RW	UDM pin BC protocol comparator enable: 1: Enable; 0: Disable.	0
4	RB_UDP_BC_CMPE	RW	UDP pin BC protocol comparator enable: 1: Enable; 0: Disable.	0

[3:2]	Reserved	RO	Reserved	0
1	RB_UDM_BC_CMPO	RO	UDM pin BC protocol comparator state: 1: UDM pin voltage is higher than the BC protocol reference value V_{BC_REF}; 0: UDM pin voltage is lower than the BC protocol reference value V_{BC_REF}.	0
0	RB_UDP_BC_CMPO	RO	UDP pin BC protocol comparator state: 1: UDP pin voltage is higher than the BC protocol reference value V_{BC_REF}; 0: UDP pin voltage is lower than the BC protocol reference value V_{BC_REF}.	0

Chapter 23 Digital Video Port (DVP)

Digital Video Port (DVP) is used to connect the camera module to obtain the image data stream. 8/10/12-bit parallel interface communication is provided, which supports the maximum input frequency of 150MHz pixel clock. It supports image data organized in original line and frame formats, such as YUV and RGB, and also supports compressed image data in JPEG format, and can receive high-speed parallel data streams output by external 8-bit, 10-bit and 12-bit camera modules. When receiving, it mainly depends on the synchronization of VSYNC and HSYNC signals. Support image cropping function.

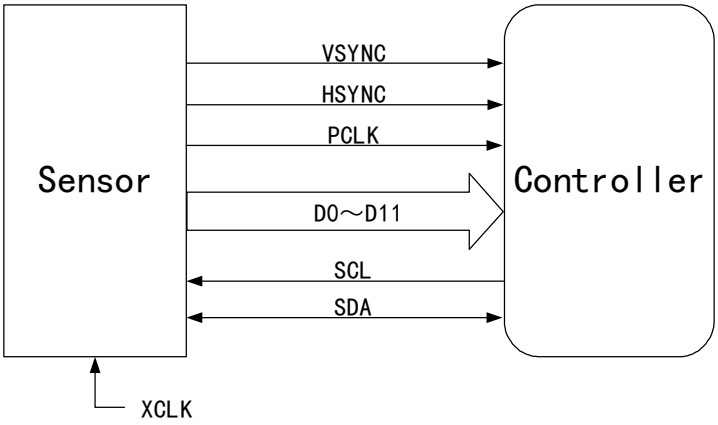
23.1 Main Features

- Configurable 8/10/12-bit data width modes
- Support YUV and RGB format data
- Support JPEG compressed format data
- Built-in FIFO with DMA transfer support
- Support dual-buffered reception
- Support cropping functionality
- Support continuous mode and snapshot mode

23.2 Register Description

23.2.1 Connected to the Sensor

Figure 23-1 DVP interface connection



- PLCK (Pixel clk): pixel clock, each clock corresponds to single pixel data (Uncompressed data). The PCLK clock output by the external DVP interface sensor supports a maximum of 150MHz.
- HSYNC (Horizontal synchronization): horizontal synchronization signal.
- VSYNC (Vertical synchronization): Frame synchronization signal.
- DATA: pixel data or compressed data, the bit width supports 8/10/12 bits.
- XCLK: The reference clock of the Sensor, which can be provided by the microcontroller or externally, generally using a crystal oscillator.
- I2C interface: The register used to configure the sensor, which can simulate the I2C timing communication through the controlled common GPIO port, or use the hardware I2C interface to operate.

Table 23-1 DVP pins

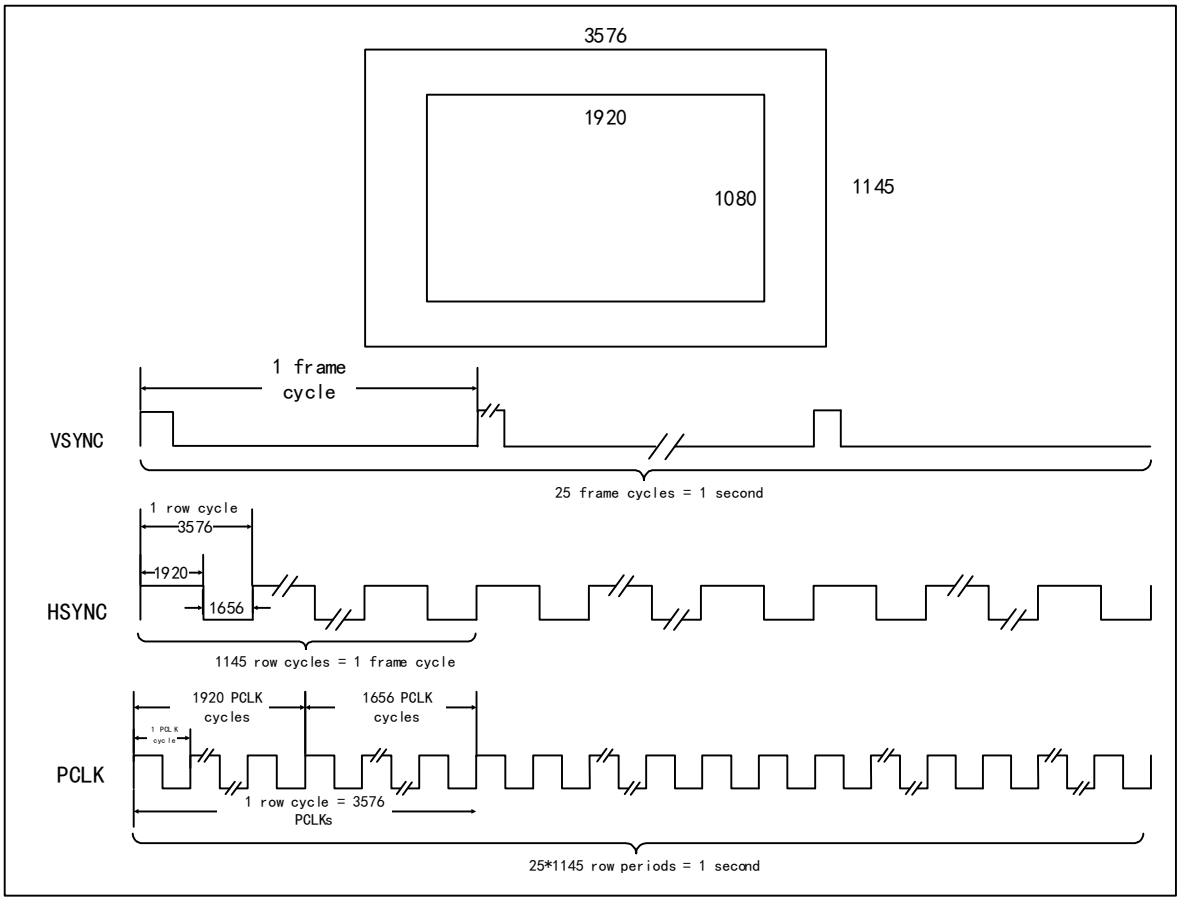
Name	Signal Type Description
PCLK	Pixel clock input
DATA[11:0]	Pixel data input
VSYNC	Frame sync signal input
HSYNC	Line sync signal input

23.2.2 Working Timing

There is a certain relationship between the digital signal data stream output by the commonly used DVP interface sensor module and the image size. The following is an example of image data.

As shown in Figure 23-2, the inside of the sensor is a complete image size of 3576*1145. After internal scaling process, the final image data size output from the interface is 1920*1080, and the image refresh rate is 25.

Figure 23-2 Timing description



PCLK is the transmission time of a pixel, so HSYNC is 3576 times more than PCLK. Among the 3576 pixels, only 1920 pixels are valid, and the sensor does not transmit data during the remaining 1656 pixels. VSYNC is a frame synchronization signal, so the VSYNC time is 3576*1145 times of PCLK. Similarly, the sensor is transmitting data only during the 1920*1080 effective pixel time. If the sensor is transmitting JPEG compressed data, the HSYNC signal may not be needed.

The relationship between DVP interface signal and image data is mainly based on the description of the selected sensor data sheet.

23.2.3 RGB/YUV/JPEG Compressed Data Format Description

- RGB
  3 primary colors: Red, green, blue
- YUV
  The luminance signal Y, the chrominance signal U and V, Pb and Pr, Cb and Cr.
- JPEG (Joint Photographic Experts Group)

There is lossy compression, but the lost part is the part that is not easily perceptible by human vision, and the human eye is insensitive to the high-frequency information part of the computer color. Remove the visual redundant information (Spatial redundancy) and remove the redundant information of the data itself (Structural redundancy).

23.3 Application of Digital Video Port

23.3.1 Digital Video Port Configuration Description

1） In the data reception of DVP, each frame of data is stored alternately by BUF0 and BUF1, starting from BUF0. For RGB and YUV data streams, the hardware resets and selects BUF0 every time the frame signal changes from an inactive level to an active level. When a line of data is full, BUF1 will be switched to realize alternate storage; for JPEG compressed data, the hardware will set the switching threshold of BUF0 and BUF1 according to the set DMA receive length.

2） When the data bus width is 10-bit or 12-bit, each time a data is received, the system will automatically unsigned-extend the data to 16-bit and then store it.

3） The R16_DVP_ROW_NUM and R16_DVP_COL_NUM registers must match the image size actually output by the sensor.

4） In video stream RGB mode, R16_DVP_COL_NUM represents the number of valid PCLK cycles of 1-line data, and R16_DVP_ROW_NUM represents the number of lines contained in a frame of image data; in image JPEG mode, R16_DVP_COL_NUM is used to configure the DMA length, and in this mode, the R16_DVP_ROW_NUM register cannot be used effectively.

23.3.2 Digital Video Port Application Note

When using the digital image interface to receive image data, the DVP control registers must be correctly configured to match the mode of the image sensor. The specific operation steps are as follows:

1） Clear RB_DVP_ALL_CLR and RB_DVP_RCV_CLR using the R8_DVP_CR1 register.

2） Configure the image mode, data bit width, PCLK polarity, HSYNC polarity and VSYNC polarity through the R8_DVP_CR0 register to match the output of the sensor.

3） According to the effective image pixels output by the configured image sensor, configure the R16_DVP_ROW_NUM and R16_DVP_COL_NUM registers to match the output of the sensor. In the image JPEG mode, only configure the R16_DVP_COL_NUM register.

4） Configure the DMA receive address through the R32_DVP_DMA_BUF0/1 register.

5） If the snapshot mode is used, the RB_DVP_CM field needs to be configured through the R8_DVP_CR1 register to enable the snapshot mode.

6） If the cropping mode is used, the RB_DVP_CROP and RB_DVP_FCRC fields need to be configured through the R8_DVP_CR1 register to enable the cropping function and control the frame capture rate. At the same time, configure the R16_DVP_HOFFCNT, R16_DVP_VST, R16_DVP_CAPCNT and R16_DVP_VLINE to set the size of the cropped image.

7） According to the requirements, enable the corresponding interrupt through the R8_DVP_IER register, configure the interrupt priority through the interrupt controller NVIC or PFIC, and enable the DVP interrupt.

8） The DMA is enabled through the R8_DVP_CR1 register, and the DVP interface is enabled through the R8_DVP_CR0 register.

9） Wait for the generation of the relevant receiving interrupt, and process the received data in time.

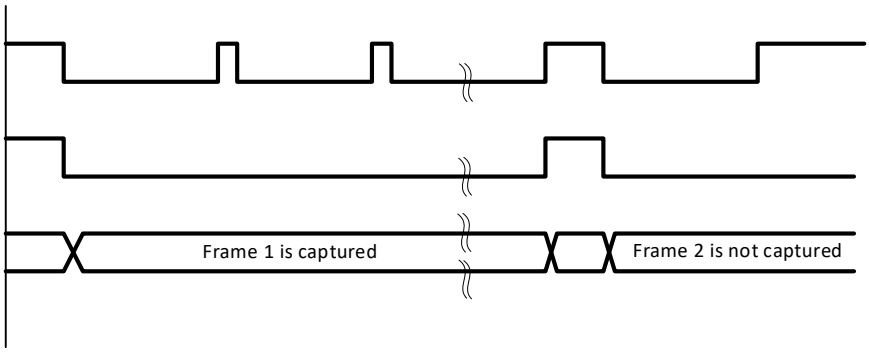
23.3.3 Capture Mode Description

The DVP interface supports 2 capture modes: snapshot (Single frame) mode and continuous mode.

23.3.3.1 Snapshot Mode

In this mode, only a single frame is captured (RB_DVP_CM is set to 1 in the R8_DVP_CR1 register). When the DVP interface is enabled, wait for the system to detect the start of the frame, and then start to sample the image data. After receiving a complete frame of data, the DVP interface will be turned off. (The RB_DVP_ENABLE field of the R8_DVP_CR0 register is cleared to 0). Figure 29-3 shows the single frame capture waveform in snapshot mode.

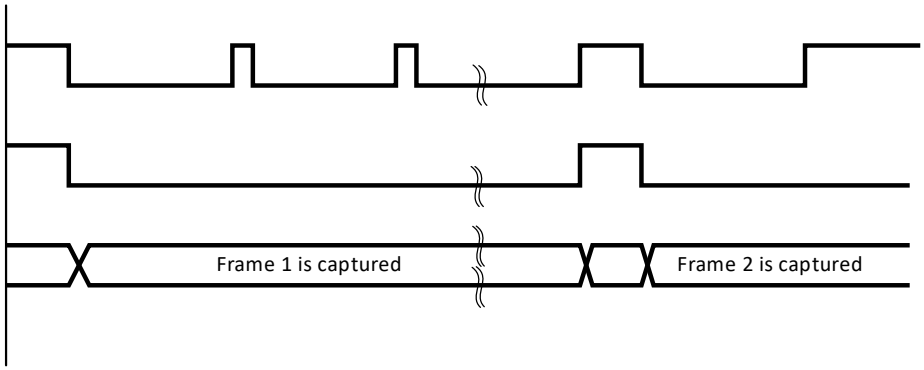
Figure 23-3 Single frame capture waveform in Snapshot mode



23.3.3.2 Continuous Mode

In this mode (RB_DVP_CM is cleared to 0 in the R8_DVP_CR1 register), after the DVP interface is enabled, each frame of data will be continuously sampled before the RB_DVP_ENABLE field of the R8_DVP_CR0 register is cleared to 0. The frame capture waveform in continuous mode is shown in Figure 23-4.

Figure 23-4 Frame capture waveform in continuous mode



23.3.4 Cropping Function Description

DVP can use the crop function to cut a rectangular window from the received image. The starting coordinates

of the rectangular window (X coordinate of the upper left corner of the rectangle R16_DVP_HOFFCNT, Y coordinate R16_DVP_VST) and window size (R16_DVP_CAPCNT represents the horizontal size, R16_DVP_VLINE represents the vertical size) are configurable. The coordinates and size of the cropping window are shown in Figure 23-5. The cropped window data capture waveform is shown in Figure 23-6.

Figure 23-5 Crop window coordinates and size

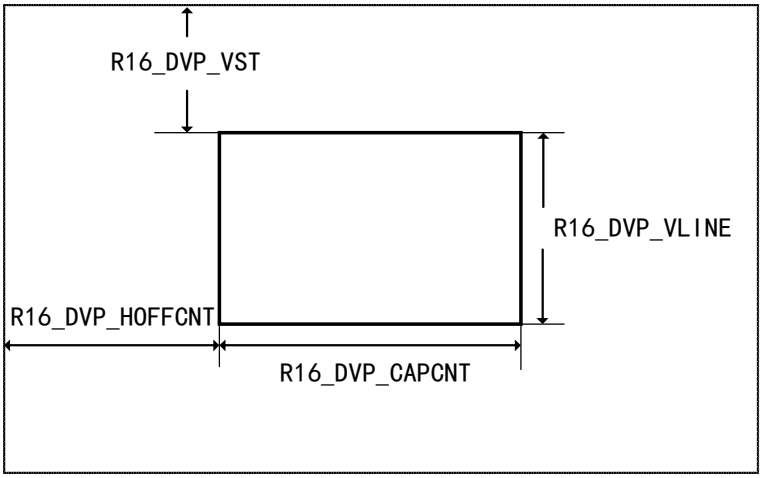
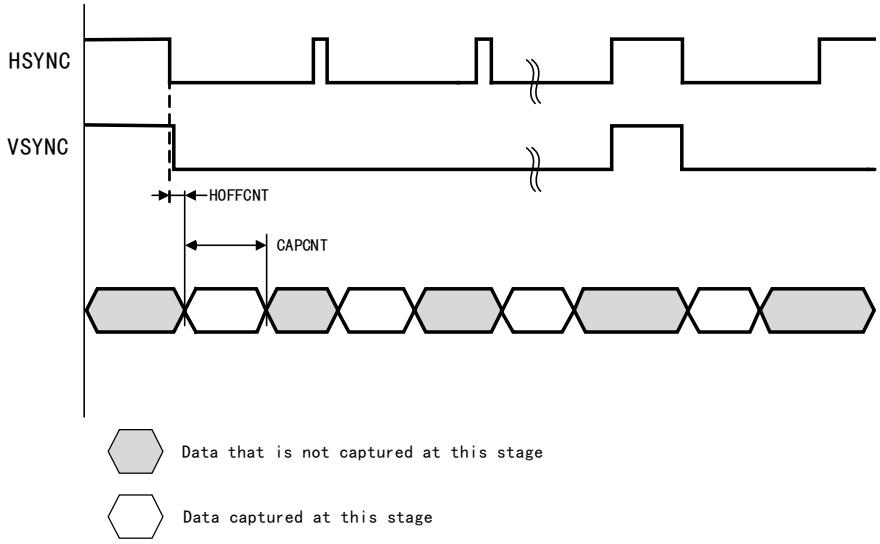


Figure 23-6 Crop window data capture waveform



23.4 Register Description

Table 23-2 DVP registers

Name	Access address	Description	Reset value
R8_DVP_CR0	0x50050000	DVP control register0	0x00
R8_DVP_CR1	0x50050001	DVP control register1	0x06
R8_DVP_IER	0x50050002	DVP interrupt enable register	0x00
R16_DVP_ROW_NUM	0x50050004	DVP row number configuration register	0x0000
R16_DVP_COL_NUM	0x50050006	DVP column number configuration register	0x0000
R32_DVP_DMA_BUF0	0x50050008	DVP DMA address 0 register	0x00000000
R32_DVP_DMA_BUF1	0x5005000C	DVP DMA address 1 register	0x00000000
R8_DVP_IFR	0x50050010	DVP interrupt flag register	0x00

R8_DVP_STATUS	0x50050011	DVP receive FIFO status register	0x00
R16_DVP_ROW_CNT	0x50050014	DVP row count register	0x0000
R16_DVP_HOFFCNT	0x50050018	Horizontal displacement register for the start of the window	0x0000
R16_DVP_VST	0x5005001A	Window start line number register	0x0000
R16_DVP_CAPCNT	0x5005001C	DVP capture count register	0x0000
R16_DVP_VLINE	0x5005001E	DVP vertical line count register	0x0000
R32_DVP_DR	0x50050020	DVP data register	0x00000000

23.4.1 DVP Control Register 0 (R8_DVP_CR0)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
7	Reserved	RO	Reserved	0
6	RB_DVP_JPEG	RW	JPEG mode enable. 0: Raw data format; 1: JPEG compression format.	0
[5:4]	RB_DVP_MSK_DAT_MOD[1:0]	RW	DVP data bit width configuration. 00: 8-bit mode; 01: 10-bit mode; 1x: 12-bit mode.	00b
3	RB_DVP_P_POLAR	RW	PCLK polarity configuration. 0: Sample data on the rising edge of PCLK; 1: Sample data on the falling edge of PCLK.	0
2	RB_DVP_H_POLAR	RW	HSYNC polarity configuration. 0: HSYNC high level data is valid; 1: HSYNC low level data is valid.	0
1	RB_DVP_V_POLAR	RW	VSYNC polarity configuration. 0: VSYNC low level data is valid; 1: VSYNC high level data is valid.	0
0	RB_DVP_ENABLE	RW	DVP enable. 0: Disable DVP; 1: Enable DVP.	0

23.4.2 DVP Control Register 1 (R8_DVP_CR1)

Offset address: 0x01

Bit	Name	Access	Description	Reset value
[7:6]	RB_DVP_FCRC[1:0]	RW	DVP frame capture rate control. 00: Capture all frames; 01: Capture every other frame; 10: Capture every third frame. 11: Reserved	00b
5	RB_DVP_CROP	RW	Crop function control. 0: Capture full image; 1: Capture only the data in the window specified by the clipping register.	0

4	RB_DVP_CM	RW	Capture mode. 0: Continuous mode; 1: Snapshot mode.	0
3	RB_DVP_BUF_TOG	RWT	Buffer address flag. Hardware control toggle, the software sets 1 to flip this bit, and writing 0 is invalid. 0: Data is stored at receive address 0; 1: Data is stored at receive address 1.	0
2	RB_DVP_RCV_CLR	RW	Receive logic reset control. 0: Cancel the reset operation; 1: Reset the receiving logic circuit.	1
1	RB_DVP_ALL_CLR	RW	Flag and FIFO clear control, write 1 or write 0 by software: 0: Cancel the reset operation; 1: Reset flag and FIFO.	1
0	RB_DVP_DMA_ENABLE	RW	DMA enable: 0: Disable DMA; 1: Enable DMA.	0

23.4.3 DVP Interrupt Enable Register (R8_DVP_IER)

Offset address: 0x02

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved	0
4	RB_DVP_IE_STP_FRM	RW	Frame stop interrupt enable. (Interrupt is generated when VSYNC changes from an active level to an inactive level.) 0: Disable frame stop interrupt; 1: Enable frame stop interrupt.	0
3	RB_DVP_IE_FIFO_OV	RW	Receive FIFO overflow interrupt enable. 0: Disable FIFO overflow interrupt; 1: Enable FIFO overflow interrupt.	0
2	RB_DVP_IE_FRM_DONE	RW	Frame reception done interrupt enable. (Interrupt is generated when the counter reaches the ROW/COL_NUM configuration value, indicating that the last data has been written to RAM) 0: Disable frame reception done interrupt; 1: Enable frame reception done interrupt.	0
1	RB_DVP_IE_ROW_DONE	RW	Row done interrupt enable. (Interrupt is generated when the counter reaches the COL_NUM configuration value) 0: Disable row done interrupt; 1: Enable row done interrupt.	0
0	RB_DVP_IE_STR_FRM	RW	New frame start interrupt enable. (Interrupt is generated when VSYNC changes from an inactive level to an active level, indicating that	0

			a new frame starts and data is coming.) 0: Disable new frame start interrupt; 1: Enable new frame start interrupt.	

23.4.4 DVP Row Number Configuration Register (R16_DVP_ROW_NUM)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_ROW_NUM[15:0]	RW	In RGB mode, it indicates the number of rows contained in a frame of image data. In JPEG mode, this register has no practical meaning.	0

23.4.5 DVP Column Number Configuration Register (R16_DVP_COL_NUM)

Offset address: 0x06

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_COL_NUM[15:0]	RW	In RGB mode, indicates the number of PCLK cycles contained within a line of data. In JPEG mode, it is used to configure the DMA receive length.	0

23.4.6 DVP DMA Address 0 Register (R32_DVP_DMA_BUF0)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
[31:0]	RB_DVP_DMA_BUF0[31:0]	RW	DMA receive address 0.	0

23.4.7 DVP DMA Address 1 Register (R32_DVP_DMA_BUF1)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[31:0]	RB_DVP_DMA_BUF1[31:0]	RW	DMA receive address 1.	0

23.4.8 DVP Interrupt Flag Register (R8_DVP_IFR)

Offset Address: 0x10

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved.	0
4	RB_DVP_IF_STP_FRM	RW1Z	Frame stop interrupt flag, active high, clear by writing 1.	0
3	RB_DVP_IF_FIFO_OV	RW1Z	Receive FIFO overflow interrupt flag, active high, clear by writing 1.	0
2	RB_DVP_IF_FRM_DONE	RW1Z	Frame receive completion interrupt flag, valid high, clear by writing 1.	0
1	RB_DVP_IF_ROW_DONE	RW1Z	Row done interrupt flag, active high, clear by writing 1.	0
0	RB_DVP_IF_STR_FRM	RW1Z	Frame start interrupt flag, active high, clear by writing 1.	0

23.4.9 DVP Receive FIFO Status Register (R8_DVP_STATUS)

Offset address: 0x11

Bit	Name	Access	Description	Reset value
7	Reserved	RO	Reserved.	0
[6:4]	RB_DVP_FIFO_CNT[2:0]	RO	FIFO counter.	0
3	Reserved	RO	Reserved.	0
2	RB_DVP_FIFO_OV	RO	FIFO overflow status. 0: FIFO not overflow; 1: FIFO overflow.	0
1	RB_DVP_FIFO_FULL	RO	FIFO full status. 0: The FIFO is not full; 1: The buffer is full.	0
0	RB_DVP_FIFO_RDY	RO	FIFO ready status. 0: No data in the FIFO; 1: There is data in the FIFO.	0

23.4.10 DVP Row Count Register (R16_DVP_ROW_CNT)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_ROW_CNT[15:0]	RO	In actual reception, the number of rows contained in a frame of image data, this register is updated at the end of the frame. In JPEG format, the value of this register has no meaning.	0

23.4.11 DVP Horizontal Shift Register for the Start of the Window (R16_DVP_HOFFCNT)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_HOFFCNT[15:0]	RW	Within the window line, each row needs to be blanked for the number of PCLK cycles before capturing data. The lower 14 bits are valid.	0

23.4.12 DVP Window Start Line Number Register (R16_DVP_VST)

Offset address: 0x1A

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_VST[15:0]	RW	The number of lines the image to start capturing, data before this line is not captured. The lower 13 bits are valid.	0

23.4.13 DVP Capture Count Register (R16_DVP_CAPCNT)

Offset address: 0x1C

Bit	Name	Access	Description	Reset value

[15:0]	RB_DVP_CAPCNT[15:0]	RW	The number of HCLK cycles to capture within the clipping window. The lower 14 bits are valid.	0

23.4.14 DVP Vertical Line Count Register (R16_DVP_VLINE)

Offset address: 0x1E

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_VLINE[15:0]	RW	The number of lines to capture within the crop window. The lower 14 bits are valid.	0

23.4.15 DVP Data Register (R32_DVP_DR)

Offset address: 0x20

Bit	Name	Access	Description	Reset value
[31:0]	RB_DVP_DR[31:0]	RO	Each time the DVP interface receives 4 bytes of data, a DMA request is triggered, and the 4-byte deep FIFO can leave sufficient time for DMA transfers. It can effectively prevent the occurrence of DMA overflow.	0

Chapter 24 SDIO Interface (SDIO)

The "SDIO" mentioned separately in this chapter refers to a communication interface on the microcontroller designed to operate external memory cards such as SD cards or other devices, and is a peripheral of the microcontroller. The SDIO of the microcontroller is directly mounted on the HB bus, and the clock is directly provided by HCLK, which can achieve higher communication speed. The SDIO of the microcontroller is used as the SDIO host, and the controlled devices are also collectively referred to as SDIO devices. In applications, SDIO is generally used to read and write SD cards, TF cards or eMMC particles, or to control other devices that use SDIO as a communication interface, such as Wi-Fi/4G modules.

24.1 Main Features

24.1.1 Features

- Support SD card, SDIO card and MMC card
- 1-bit, 4-bit and 8-bit bus support
- SDIO supports optional clock frequency, up to 80MHz in 4-wire mode and 50MHz in 8-wire mode
- Compatible with MMC Specification 4.5 (Forward compatibility)
- Compatible with SD card specification 3.0, SDIO card specification 2.0
- Not compatible with SPI

24.1.2 Overview

The SDIO of the microcontroller supports communication with memory such as SD cards or MMC cards. It needs to be clear that SDIO only provides a set of clocks, data and command control sequences required to realize the single command transmission of SD cards and MMC cards. The sequence and combination of commands need to be determined by the user through the program. In addition, for various memory cards, SDIO can only realize the function of reading and writing, and the function of the file system provided by the file system needs to be realized by the user to build the file system through the program.

SDIO is different from the QSPI interface. It has no chip select pin and has an extra CMD pin. CMD can be regarded as a special data line, which is specially used to transmit commands and responses; SDIO is available in 1-bit, 4-bit and 8-bit data line widths; the clock of SDIO is optional during configuration. When data transmission is officially carried out, the clock supports 4-wire mode up to 80MHz and 8-wire mode up to 50MHz. According to the protocol, when the clock received by the SDIO device is greater than a certain threshold, the waveform peak value of the clock line, data line and command line needs to be reduced to save the time-consuming rise and fall of the waveform.

Unlike SD cards, SDIO cards often refer to non-storage devices such as WIFI/Bluetooth modules and 4G modules that use SDIO interfaces. If there is no special instruction, the content described in this chapter must be applicable to SD card, and the content only applicable to SDIO card or MMC card will be specially pointed out. In the description of this chapter, the SD card is regarded as the potential operation object first, followed by the SDIO card, and finally the MMC card.

24.2 Interface and Clock

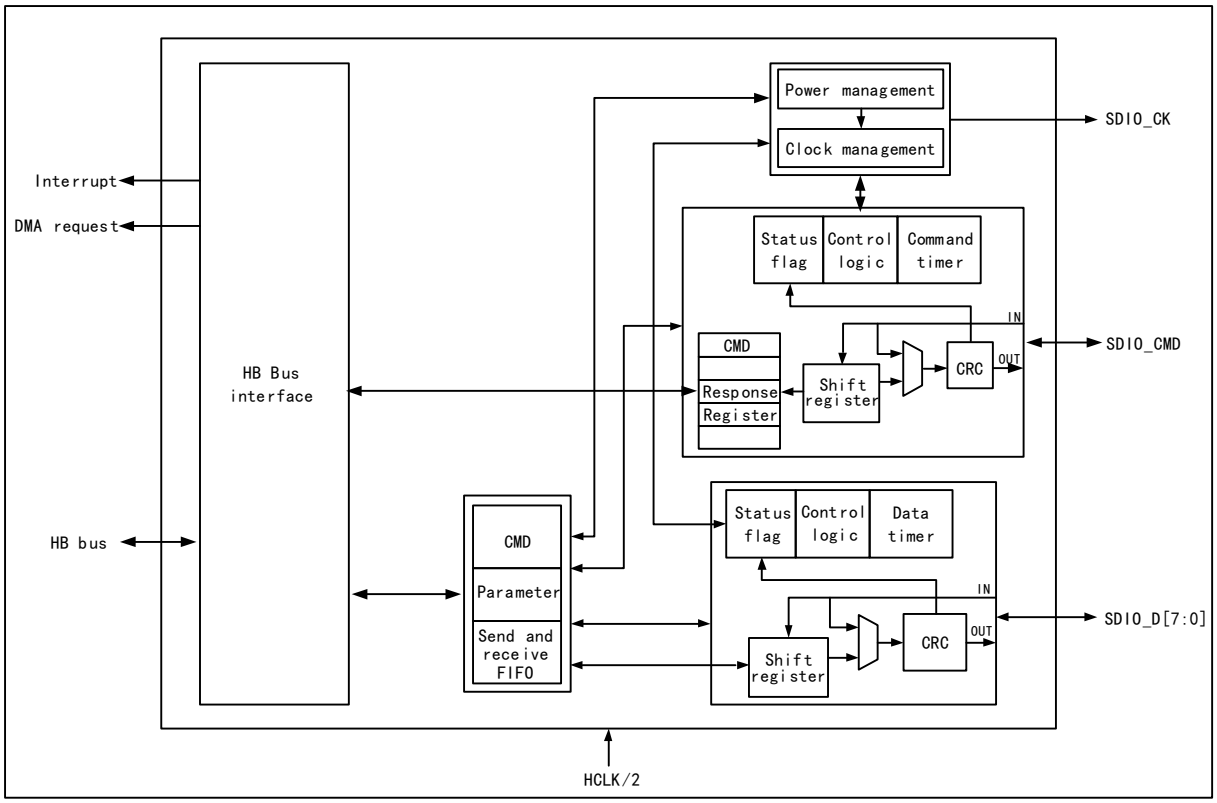
SDIO receives the control of the CPU through the HB bus. There is a FIFO interface in the SDIO register. The CPU or DMA obtains or generates data by reading and writing the FIFO. SDIO is directly clocked by HCLK, and has an interrupt entry that supports multiple interrupt sources. The pins directly controlled by SDIO are

SDIO_CK, SDIO_CMD, SDIO_D[7:0], which are connected to SDIO devices through several pins. SDIO is a host-dominated communication interface, and all transfers must be initiated by the microcontroller.

24.2.1 Peripheral Structure

The structure of SDIO is shown in Figure 24-1.

Figure 24-1 SDIO structure diagram



SDIO is a peripheral that the microcontroller directly operates the SDIO device as a host. It is roughly composed of 5 modules: HB interface, clock control part, CMD line control part, data line control part and control register part. SDIO is a half-duplex for peripherals, the CPU writes the commands and data to be sent to the control register, and the command line and data line control module is responsible for pushing the command or data to the IO and adding CRC. The data flow of SDIO is responsible for the transfer from RAM to SDIO FIFO by general-purpose DMA. SDIO FIFO has 32×32-bit sizes.

24.2.2 Pins and Configuration

The pins that need to be configured for SDIO and their modes are shown in Table 24-1.

Table 24-1 SDIO pin configuration

SDIO alternate function	The pin mode that needs to be configured	The pin speed that needs to be configured
SDIO_D[0:3]	Push-pull alternate output	50MHz
SDIO_CK	Push-pull alternate output	50MHz
SDIO_CMD	Push-pull alternate output	50MHz
SDIO_D4	Push-pull alternate output	50MHz
SDIO_D5	Push-pull alternate output	50MHz
SDIO_D6	Push-pull alternate output	50MHz

SDIO_D7	Push-pull alternate output	50MHz

Note: The SDIO_CK pin should be configured as a float input when in slave mode.

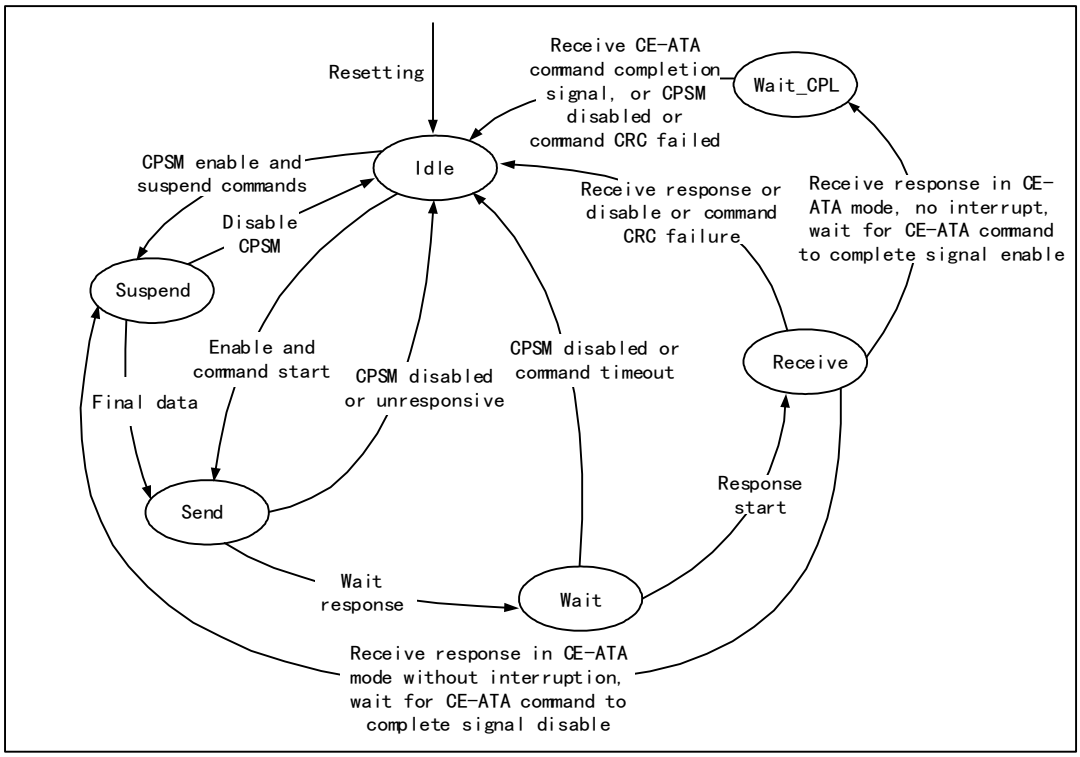
24.2.3 Clock

The clock pin of SDIO is SDIO_CK, and its output clock is obtained by dividing the frequency of HCLK, and the frequency dividing coefficient can be configured as any integer value between 2 and 261. SDIO devices generally only support a single bus mode with a clock of up to 400kHz during initialization. After initialization, the host generally initiates a switch to a low voltage operation, and at the same time increases the clock to the maximum clock that both the microcontroller and the SDIO device can accept. Different versions and speed grades of SD cards support different clock speeds and switching processes, and users need to understand for themselves.

24.2.4 Command Path State Machine

The command workflow of SDIO follows the state machine shown in the figure below.

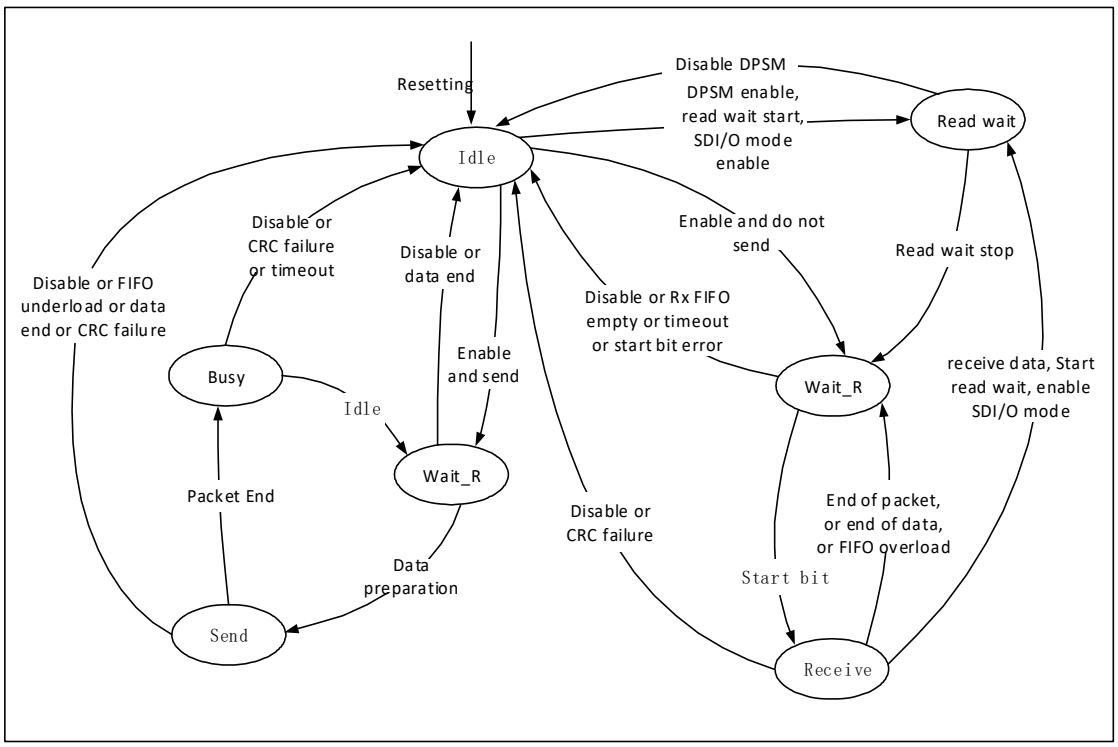
Figure 24-2 Command path state machine



24.2.5 Data Path State Machine

The data workflow of SDIO follows the state machine shown in the figure below.

Figure 24-3 Data path state machine



24.3 SDIO Protocol Description

The communication on SDIO is based on transmission as the smallest unit. Each transmission always starts with the host sending a command on the CMD line. After some commands are sent, the SDIO device will also send a piece of data on the CMD line to reply to the host, which is called "Response", sometimes accompanied by the transmission of data, the transmission of data is on the D line. The format of commands and responses is fixed, and the definition of each field is determined according to different commands or responses. Responses and data transfers need to be issued or stopped within a specified time after the end of the command or response, otherwise a timeout error will be generated.

The purpose of this section is to let users have a preliminary understanding of the details of some specifications necessary to use SDIO in a relatively small space, and it does not guarantee detailed and timely updates. The SDIO of the microcontroller only guarantees that it implements the hardware operation basis of the SD 2.0, SDIO 2.0 and MMC 4.5 specifications, and does not necessarily support the functions defined in the higher version specifications, such as double-edge sampling. Users should carry out SDIO interactive programming based on SD specification, SDIO specification and MMC specification when doing specific development.

24.3.1 Bus Timing

The transmission of the SD card is initiated by the host initiating the CMD. The SD card may not reply with a response, but may reply with a short response and a long response, and some responses will be accompanied by data transmission. In the communication of the SDIO card or MMC card, there may also be a situation where the SDIO device actively reports an interruption. The timing is shown in the group of figures below.

Figure 24-4 SDIO no response timing and no data timing (Taking SD card as an example)

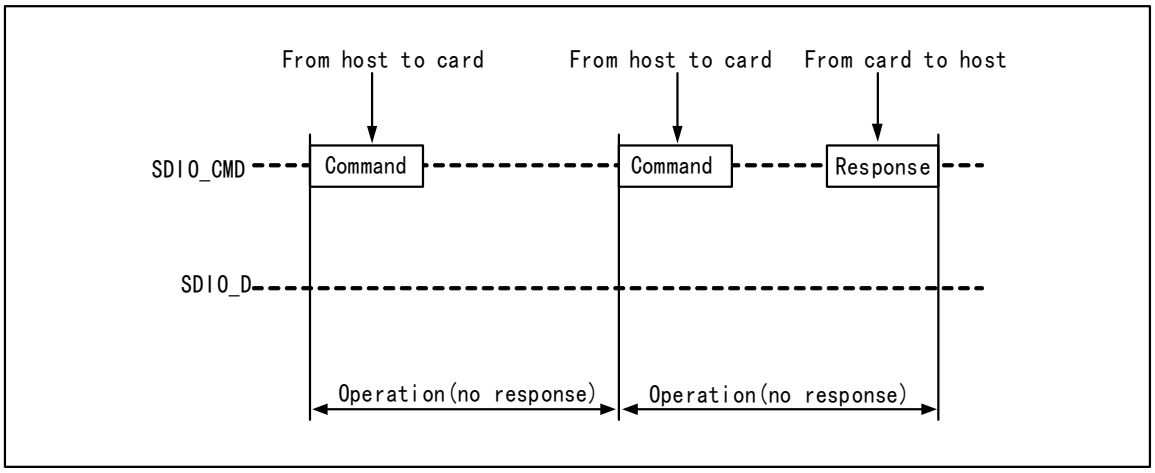


Figure 24-5 SDIO multi-block read timing (Taking SD card as an example)

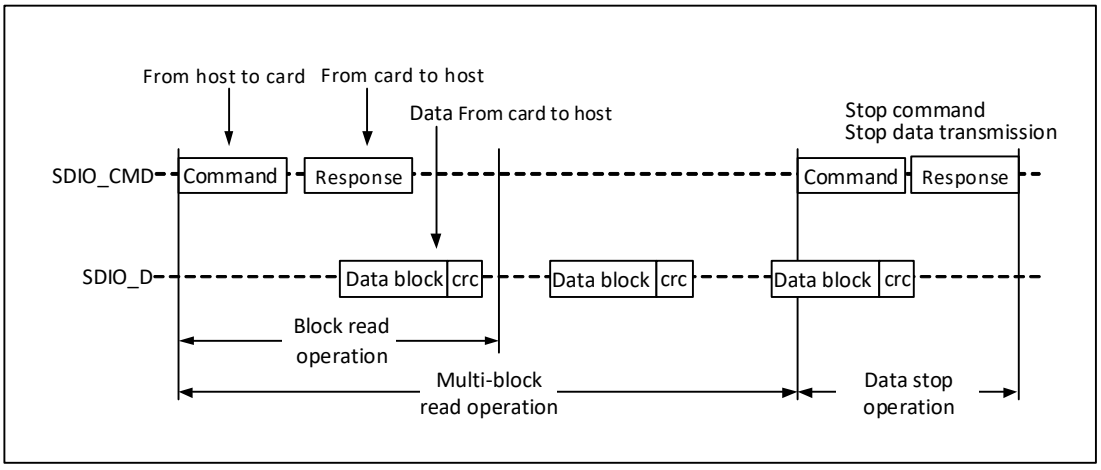


Figure 24-6 SDIO multi-block write timing (Taking SD card as an example)

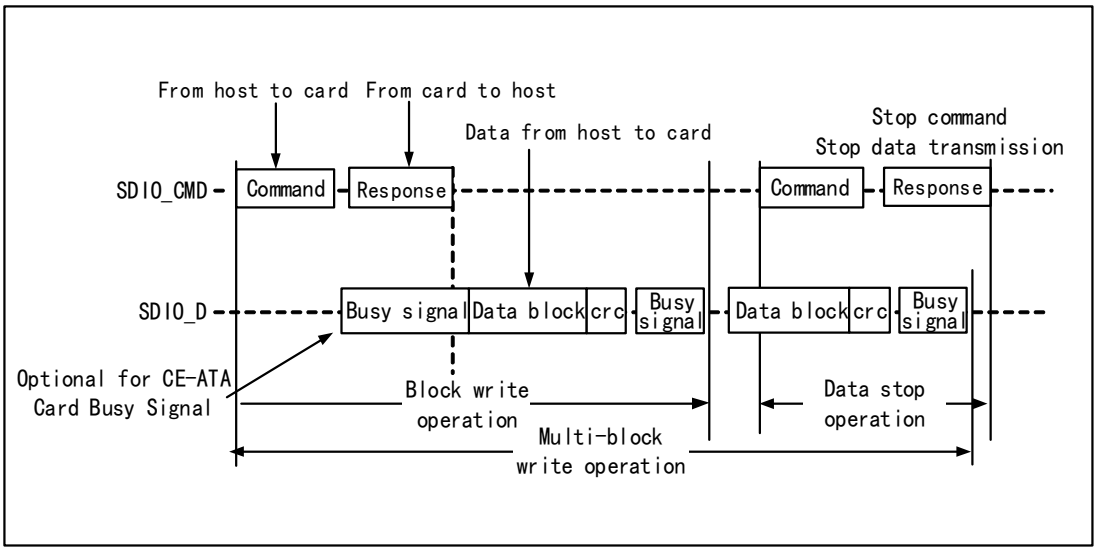


Figure 24-7 SDIO data stream read timing (Taking SDIO card as an example)

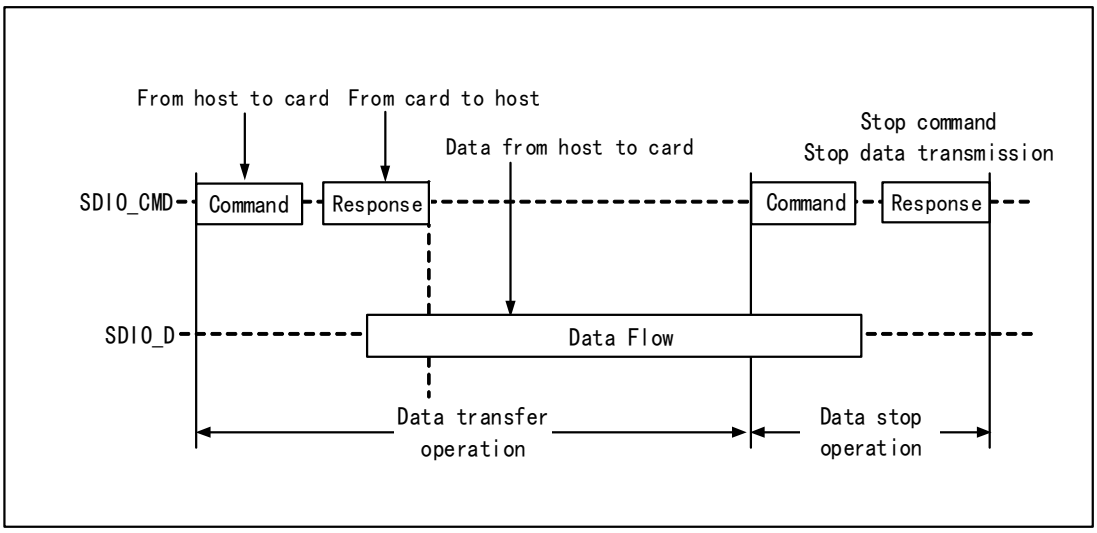
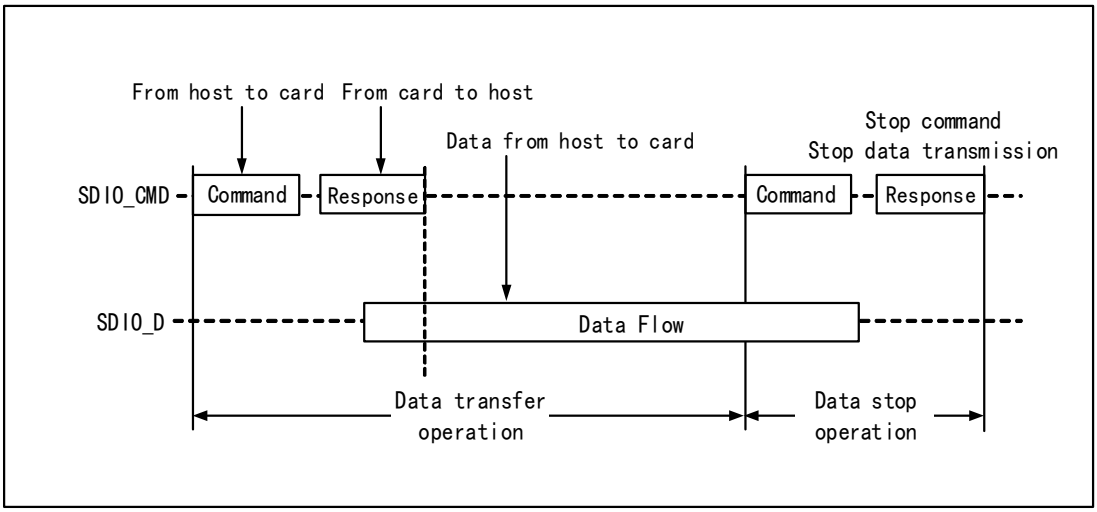


Figure 24-8 SDIO data stream write timing (Taking SDIO card as an example)



24.3.2 Command

Most SDIO transfers start with a command (CMD), through which the SDIO host informs the device of its intentions. The format of the command is as follows.

Table 24-2 Command format

Bit/Field name	Start bit	Transmission bit	Command index	Command parameters	CRC7	End bit
Rank/Width	47	46	[45:40]/6bits	[39:8]/32bits	[7:1]/7bits	0
Value	0b	1b	X	X	X	1b

4 types of commands:

1) Broadcast command (bc): sent to all cards on the bus, no response returned;
2) Broadcast command with response (bcr): sent to all cards on the bus, and a response is returned;
3) Point-to-point command (ac): issued to a specific card, but no data transmission;
4) Point-to-point command with data transmission (adtc): issued to a specific card with data transmission attached.

Below are some commonly used commands.

24.3.2.1 Basic Command

Basic commands are some of the more basic functions supported by the SD card.

Table 24-3 SD Card Basic Commands

Command index	Type	Parameter	Response format	Abbreviation	Description
CMD0	bc	[31:0] stuff bits are meaningless.	None	GO_IDLE_STATE	Reset all cards to IDLE state
CMD2	bcr	[31:0] stuff bits are meaningless.	R2	ALL_SEND_CID	All cards reply to CID
CMD3	bcr	[31:0] stuff bits are meaningless.	R6	SEND_RELATIVE_ADDR	Reply to new RCA
CMD7	ac	[31:16] RCA; [15:0] stuff bits are meaningless.	R1b Selected card	SELECT/DESELECT_CARD	Check or uncheck a card
CMD8	bcr	[31:12] reserved bits; [11:8] powered; [7:0] check mode.	R7	SEND_IF_COND	Send SD card interface conditions.
CMD9	ac	[31:16] RCA; [15:0] stuff bits are meaningless.	R2	SEND_CSD	Require CSD。
CMD10	ac	[31:16] RCA; [15:0] stuff bits are meaningless.	R2	SEND_CID	Require CID。
CMD11	ac	[31:0] stuff bits are meaningless.	R1	VOLTAGE_SWITCH	Switch to 1.8V.
CMD12	ac	[31:0] stuff bits are meaningless.	R1b	STOP_TRANSMISSION	Force the card to stop transmission;
CMD13	ac	[31:16] RCA; [15] stuff bits are meaningless; [14:0] stuff bits.	R1	SEND_STATUS/SEND_TASK_STATUS	Transmit Status or Task Status Register
CMD15	ac	[31:16] RCA, [15:0] stuff bits are meaningless.	None	GO_INACTIVE_STATE	The card is required to enter the INACTIVE mode;

24.3.2.2 Erase Command

The SD card is also a FLASH structure, and the FLASH needs to be erased before writing, but the SD card integrates erasing logic. If it is found that there is no erasure before executing the write command, it will automatically supplement the erasing operation. In many cases, especially before large lot writes, it can help to improve efficiency if the erase is actively performed.

Table 24-4 SD Card Erase Command [1]

Command index	Type	Parameter	Response format	Abbreviation	Description
CMD32	ac	[31:0] address to start erasing [2]	R1	ERASE_WR_BLK_START	Set the first address for erasing
CMD33	ac	[31:0] address to start erasing [2]	R1	ERASE_WR_BLK_END	Set the first address for erasing
CMD38	ac	[31:0] erase mode	R1b	ERASE	The parameter is 0, that is, ordinary erase

Note 1: The erase command here is the erase command of SD card defined by SD protocol specification, and the erase command of SDIO card and MMC card is different from this.
Note 2: Currently commonly used SDHC/SDXC (2GB to 2TB) level erase address written by the card must be a block address, that is, 512-byte alignment.

24.3.2.3 Block Read Command

Table 24-5 SD Card block read command [1]

Command index	Type	Parameter	Response format	Abbreviation	Description
CMD16	ac	[31:0] block length	R1	SET_BLOCKLEN	write block length, 512
CMD17	adtc	[31:0] address of a single block [1]	R1	READ_SINGLE_BLOCK	Set the start address of a single read
CMD18	adtc	[31:0] address of a single block [1]	R1	READ_MULTIPLE_BLOCK	Continuously transfer data blocks from the card to the host until interrupted by the STOP_TRANSMISSION command
CMD19	adtc	[31:0] reserved bits	R1	SEND_TUNING_BLOCK	Send a 64-byte sequence representing the mode change
CMD20	ac	[31:28] Speed control bit [27:0] See command description for details	R1b	SPEED_CLASS_CONTROL	Speed level control command: [27:0] Speed level [27:0] UHS speed level [27:0] Video speed level
CMD22	ac	[31:6] reserved	R1	ADDRESS_EXTENSION	SDUC will be used.

		bits [5:0] extended address			
CMD23	ac	[31:0] block counter	R1	SET_BLOCK_COUNT	block counter

Note 1: The erasing address written by the SDHC/SDXC (2GB to 2TB) level cards commonly used at present must be a block address, that is, 512-byte alignment.

24.3.2.4 Write Command for Block Transfer

Table 24-6 SD Card block write command [1]

Command index	Type	Parameter	Response format	Abbreviation	Description
CMD16	ac	[31:0] block length	R1	SET_BLOCKLEN	write block length, 512
CMD24	adtc	[31:0] block address [1]	R1	WRITE_BLOCK	Set the start address of a single block write
CMD25	adtc	[31:0] addresses of multiple blocks [1]	R1	WRITE_MULTIPLE_BLOCK	Set the start address of multiple blocks write
CMD27	adtc	[31:0] stuff bits	R1	PROGRAM_CSD	Programmable word programming for CSD

Note 1: The erasing address written by the SDHC/SDXC (2GB to 2TB) level cards commonly used at present must be a block address, that is, 512-byte alignment.

24.3.3 Response

As a necessary reply from the SDIO device to the host, the response is also transmitted on the CMD line and must be replied within a specified time. The response transmission is high-order first and low-order last. The length of the response and the definition of each bit and each field are determined by the type of response, but all responses start with a fixed 0 start bit, followed by a fixed 0 transmission Direction bit [1]. All responses have a stop bit at the end, fixed at 1[1]. There are roughly 7 kinds of responses, SD card supports R1/R1b/R2/R3/R6/R7, SDIO card also supports R4 / R5, see the format below.

24.3.3.1 R1 Response

Normal response, the total length is 48 bits, with CRC7, and the card status field is 32 bits. The format is as follows.

Table 24-7 R1 response

Bit/Field name	Start bit	Transmission bit	Response index	Card Status	CRC7	End bit
Rank/Width	47	46	[45:40] / 6 bits	[39:8]/32 bits	[7:1]/7 bits	0

value	0b	0b	Follow CMD Index	X	X	1b

24.3.3.2 R1b Response

The format of R1b is the same as that of R1, but a busy signal can be added after the response, that is, the data line D2 is clamped. The host needs to deal with it after receiving the busy signal (Detecting that SDIO_D2 is low).

24.3.3.3 R2 Response

Responses to specific commands, the total length is 136 bits, the CRC7 is included in the card status field, and the card status field is 128 bits. The card status field stores the value of the CID register or CSD register. The CID register is generally used as the reply to CMD2/CMD10, and the CSD register is generally used as the reply to CMD9. For the specific meaning of the CID/CSD register, see the section 32.4.2 Device Registers. It should be noted that R2 will only reply to the [127:1] segment of the CID/CSD register. The reserved bit [2] of the CID/CSD [0] fixed to 1 is occupied by the end bit, and the end bit is also fixed to 1. The format is as follows.

Table 24-8 R2 response

Bit/Field name	Start bit	Transmission bit	Command index	Card status	End bit
Rank/Width	135	134	[133:128]/6 bits	[127:1]/127 bits	0
Value	0b	0b	111111b	CID/CSD	1b

24.3.3.4 R3 Response

Reply to the dedicated response of the OCR register, the total length is 48 bits, and there is no CRC7. The OCR register is generally used as a reply to ACMD41. The format is as follows.

Table 24-9 R3 response

Bit/Field name	Start bit	Transmission bit	Command index	Card Status	Reserved	End bit
Rank/Width	47	46	[45:40]/6 bits	[39:8]/32 bits	[7:1]/7 bits	0
Value	0b	0b	111111b	OCR	1111111b	1b

24.3.3.5 R4 Response

In response to the response of CMD5, reply to the special response of OCR and related registers, with a total length of 48 bits and a CRC7. R4 is used in SDIO card, the format is as follows.

Table 24-10 R4 response

Bit/Field name	Bit position	Width (bit)	value
Start bit	47	1	0b
Transmission bit	46	1	0b
Reserve	[45:40]	6	111111b
Card ready	39	1	X
IO number of functions	[38:36]	3	X
Current register	[35]	1	X
Stuff bits	[34:33]	2	00b

S18A	32	1	X
IO OCR	[31:8]	24	OCR
CRC field	[7:1]	7	1111111b
End bit	0	1	1b

Note: The format of MMC card R4 is different from SDIO card.

24.3.3.6 R5 Response

A dedicated response to CMD5, with a total length of 48 bits and a CRC7. R5 is used in SDIO card, the format is as follows.

Table 24-11 R5 response

Bit/Field name	Start bit	Transmission bit	Command index	Stuff bit	Response format	Read and write data	CRC7	End bit
Rank	47	46	[45:40]	[39:24]	[23:16]	[15:8]	[7:1]	0
width	1	1	6	16	8	8	7	1
Value	0b	0b	110100b	0000h	X	X	X	1b

Note: The format of MMC card R5 is different from SDIO card.

24.3.3.7 R6 Response

Reply to the special response of RCA, with a total length of 48 bits and a CRC7. The format is as follows.

Table 24-12 R6 response

Bit/Field name	Start bit	Transmission bit	Command index	Card RCA	Card status bit	CRC7	End bit
rank	47	46	[45:40]	[39:24]	[23:8]	[7:1]	0
Width	1	1	6	16	16	7	1
Value	0b	0b	000011b	X	X	X	1b

24.3.3.8 R7 Response

In response to the dedicated response of CMD8, it indicates the information of the supported voltage, with a total length of 48 bits, with CRC7, and the format is as follows.

Table 24-13 R7 response

Bit/Field name	Start bit	Transmission bit	Command index	Reserved bit	PCIe 1V2 support	PCIe response	Accepted Voltage	Check Feedback	CRC7	End bit
Bit position	47	46	[45:40]	[39:22]	21	20	[19:16]	[15:8]	[7:1]	0
Width	1	1	6	18	1	1	4	8	7	1
Value	0b	0b	001000b	00000h	X	X	X	X	X	1b

24.3.4 Data Transfer

Data transfer is carried out on the data line SDIO_D, which has 3 widths of 1/4/8 bits. During data transfer, the start bit of a clock is generally at the front, the high bit of each byte is in the front, and the low bit is in the

back. For the block transfer mode only supported by the SD card, the data transfer of each block is followed by a CRC.

Enable the RANDOM_LEN_EN bit, configure the byte length indicated by DBLOCKSIZE2, configure the total length of the transferred data indicated by R32_SDIO_DLEN, enable the DTEN bit, and fill in the data to the R32_SDIO_FIFO, then you can complete the transfer of the block of any byte length with CRC checksum.

The MMC card also supports the data streaming transfer mode, at this time no CRC is attached. the following figure shows the format of the data transfer.

Figure 24-9 SD card single bus data transfer byte format

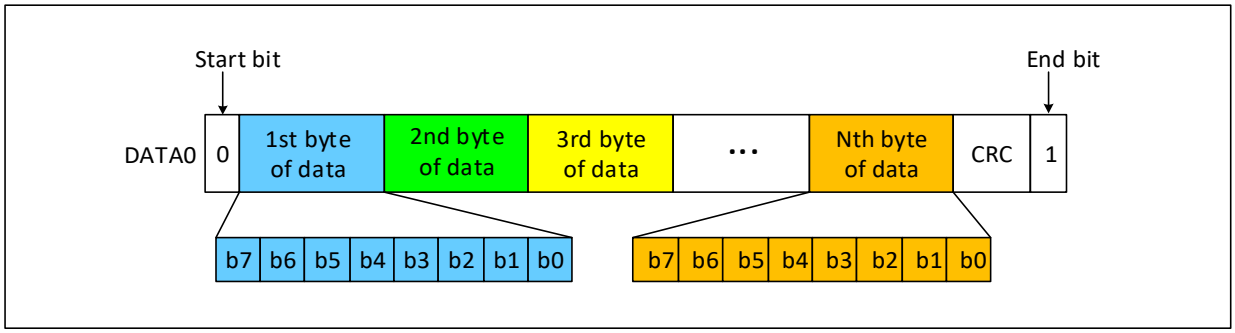


Figure 24-10 SD card 4 bus data transfer byte format

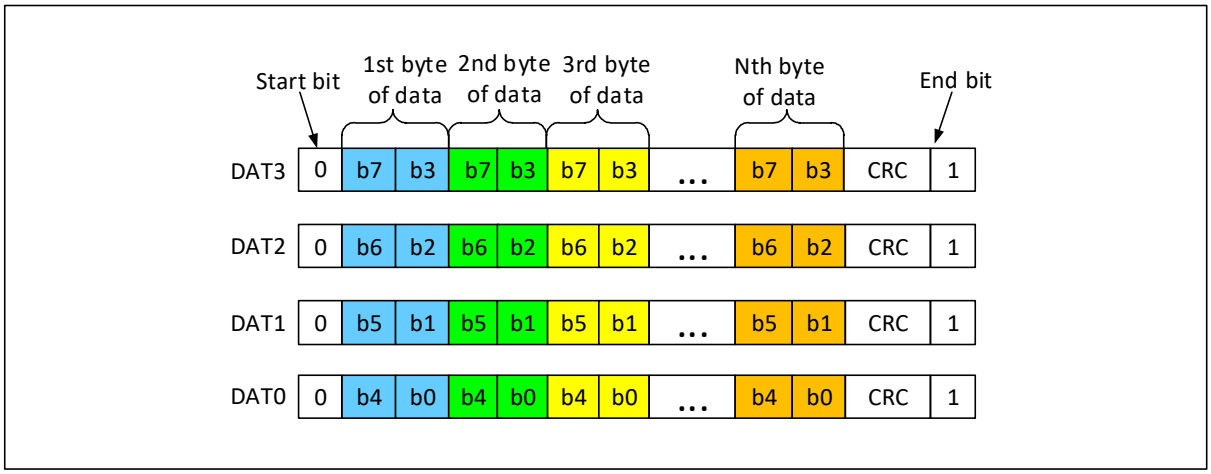


Figure 24-11 SD card single bus data transfer in a 512-bit word format

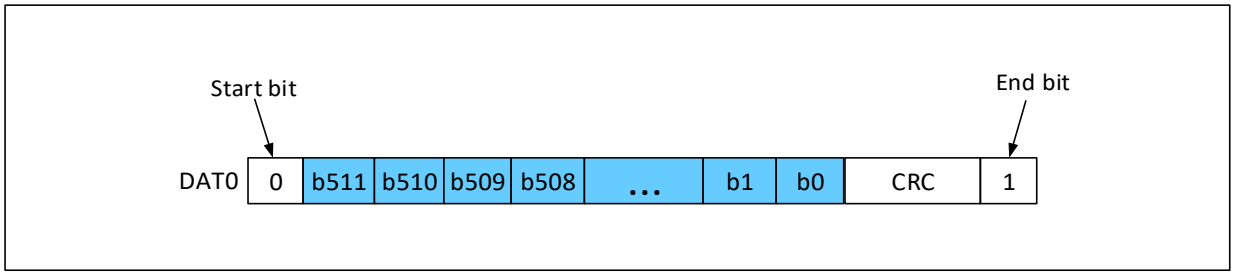
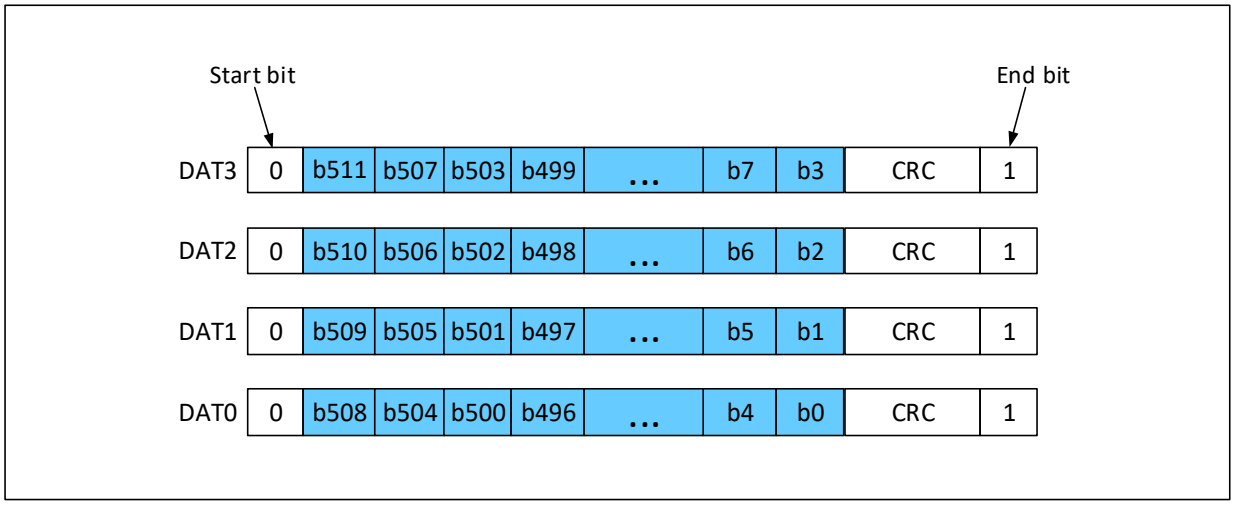


Figure 24-12 SD card 4 bus data transfer in a 512-bit word format



24.3.5 Slave Mode

Turning on the SLV_MODE bit means that it is in the state of waiting for commands from the host, with received command parameters placed in RESP1 and received command indexes placed in RESP2, while automatically answering to the host with an R1 type of answer, with the answer parameter being the value of CMDARG.

Data sent and received in slave mode refer to master mode, but the slave can use the SLV_FORCE_ERR bit to force a data block CRC error, thus indicating an undesired data read or write.

Note: The slave supports only the R1 type of answer, and the command indexes and parameter meanings used for master-slave communication are defined by the software;

24.4 Application

24.4.1 Device Initialization and Device Registers

24.4.1.1 OCR

The Operation Conditions Register stores some information about the power supply voltage and related status bits of the SD card. The relevant bits are defined in the following table.

Table 24-14 OCR bit definitions

Bit	Bit definition	Description
[0-3]	Reserved	Supported V_{DDA} voltage range in volts.
4	Reserved	
5	Reserved	
6	Reserved	
7	Reserved for low voltage range	
8	Reserved	
9	Reserved	
10	Reserved	
11	Reserved	
12	Reserved	

13	Reserved	
14	Reserved	
15	2.7-2.8	
16	2.8-2.9	
17	2.9-3.0	
18	3.0-3.1	
19	3.1-3.2	
20	3.2-3.3	
21	3.3-3.4	
22	3.4-3.5	
23	3.5-3.6	
24	accept switching to 1.8V	
[25:26]	Reserved	
27	Over 2TB support status bit	
28	Reserved	
29	UHS-II card status bits	This bit is set to indicate that the card supports the UHS-II interface.
30	Card capacity status (CCS)	This bit is set to indicate that the card capacity is greater than 2GB.
31	Card power-on status bit (busy)	This bit is set when the card is powered up. After power-on, other bits have meaning.

24.4.1.2 CID Register

CID stores some identifying information.

Table 24-15 CID bit definitions

Bit/Field name	Abbreviation	Width	Bit position
Manufacturer ID	MID	8	[127:120]
App ID	OID	16	[119:104]
Product Name	PNM	40	[103:64]
Product Revision	PRV	8	[63:56]
Product Serial Number	PSN	32	[55:24]
Reserved	None	4	[23:20]
Date of Manufacture	MDT	12	[19:8]
CRC7	CRC	7	[7:1]
Fixed bit, always 1	None	1	[0]

24.4.1.3 CSD Register

The CSD register stores the characteristic data of the SD card. Taking the most commonly used SDHC and SDXC cards of the second version of the CSD as an example, the definitions of each field are as follows.

Table 24-16 CSD bit definitions

Name	Abbreviation	Width	Value	Read and write	Bit position
CSD version	CSD_STRUCTURE	2	01b	RO	[127:126]
Reserved	None	6	00_0000b	RO	[125120
Reading access time	TAAC	8	0Eh	RO	[119:112]
Reading access time in clock cycles	NSAC	8	00h	RO	[111:104]
Maximum data transmission speed	TRAN_SPEED	8	32h5Ah0Bh2Bh	RO	[103:96]
Card command class	CCC	12	X1X1101101X1b	RO	[95:84]
Maximum length of read data block	READ_BL_LEN	4	9	RO	[83:80]
Allow block partial read	READ_BL_PARTIAL	1	0	RO	[79]
block write misalignment	WRITE_BLK_MISALIGN	1	0	RO	[78]
block read misaligned	READ_BLK_MISALIGN	1	0	RO	[77]
Executed DSR	DSR_IMP	1	X	RO	[76]
Reserved	None	6	00_0000b	RO	[75:70]
Device size	C_SIZE	22	XXXXXXh	RO	[69:48]
Reserved	None	1	0	RO	[47]
Single block erase enable	ERASE_BLK_EN	1	1	RO	[46]
Erase sector size	SECTOR_SIZE	7	7Fh	RO	[45:39]
Write-protect group size	WP_GRP_SIZE	7	0000000b	RO	[38:32]
Write Protect Group Enable	WP_GRP_ENABLE	1	0	RO	[31]
Reserved	None	2	00b	RO	[30:29]
Write speed factor	R2W_FACTOR	3	010b	RO	[28:26]
Maximum write data block length	WRITE_BL_LEN	4	9	RO	[25:22]
Allow block	WRITE_BL_PARTIAL	1	0	RO	[21]

partial writes					
Reserved	None	5	00000b	RO	[20:16]
file format group	FILE_FORMAT_GRP	1	0	RO	[15]
Copy sign	TMP_WRITE_PROTECT	1	X	RW OTP	[14]
Permanent write protection	PERM_WRITE_PROTECT	1	X	RW OTP	[13]
Temporary write protection	TMP_WRITE_PROTECT	1	X	RW	[12]
File format	FILE_FOMAT	2	00b	RO	[11:10]
Reservef	None	2	00b	RO	[9:8]
CRC	CRC	7	0000000b	RW	[7:1]
Not used, must use 1	None	1	1b	RO	[0]

24.4.1.4 RCA

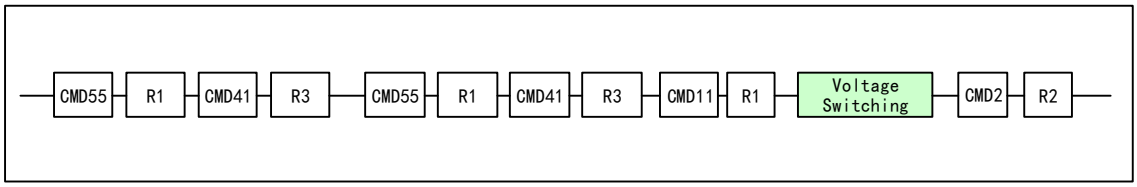
The relative card address register stores the address of the card, which is 16 bits, and the default value is 0.

24.4.2 Voltage Switch

In the later stage of SD card initialization, it is necessary to switch the interface level, and switch the IO level of the clock line, data line and command line of the SD card to the 1.8V level. For devices whose slew rate is not good enough, using a lower-level standard can help increase the frequency. However, the power supply voltage of SD does not necessarily change, and only SD cards with low voltage power supply appear in the newer version of the protocol.

The steps for switching the voltage are shown in the figure below.

Figure 24-13 Voltage switch sequence



24.4.3 Clock Switch

During initialization, the clock of the SD card is only 400kHz. After the voltage switch is completed, the clock can be increased to a higher level. For example, the first speed of the SDHC card UHS-I mode, the bus clock can reach 80MHz. In view of the IO output capability of the microcontroller, the clock should be limited to 50MHz.

24.5 Interrupt

24.5.1 SDIO Interrupt

SDIO supports a variety of interrupt sources. As shown in the interrupt enable register (R32_SDIO_IER), there

are 24 situations that can trigger interrupts, and users can enable them as appropriate.

24.5.2 SDIO Device Interrupt

It should be noted that not only SDIO peripherals can report interrupts to the CPU, but also external SDIO cards and MMC cards can report interrupts to SDIO peripherals. In 4-bit bus mode, the interrupt line is D1, and in 8-bit bus mode, the interrupt line is D7, active low. If SDIO detects that D1 or D7 is low level in the idle state, it should read the status register or interrupt flag register of the SDIO device to respond to the interrupt in time. The CPU can get whether the SDIO host receives an interrupt through the SDIOIT bit of the R32_SDIO_STA register.

24.6 Register Description

Table 24-17 SDIO registers

Name	Address	Description	Reset Value
R32_SDIO_POWER	0x40018000	Power register	0x00000000
R32_SDIO_CLKCR	0x40018004	Clock register	0x00000000
R32_SDIO_ARG	0x40018008	Command parameter register	0x00000000
R32_SDIO_CMD	0x4001800C	Command register	0x00000000
R32_SDIO_RESPCMD	0x40018010	Response register	0x00000000
R128_SDIO_RESPX	0x40018014	Response parameter register	0x00000000
R32_SDIO_DTIMER	0x40018024	Data timing register	0x00000000
R32_SDIO_DLEN	0x40018028	Data length register	0x00000000
R32_SDIO_DCTLR	0x4001802C	Data control register	0x00000000
R32_SDIO_DCOUNT	0x40018030	Data count register	0x00000000
R32_SDIO_STA	0x40018034	Status register	0x00000000
R32_SDIO_ICR	0x40018038	Interrupt clear register	0x00000000
R32_SDIO_MASK	0x4001803C	Interrupt enable register	0x00000000
R32_SDIO_ FIFOCNT	0x40018048	FIFO counter	0x00000000
R32_SDIO_DCTRL2	0x40018060	Data control register 2	0x00000000
R32_SDIO_FIFO	0x40018080	FIFO register	0x00000000

24.6.1 Power Register (R32_SDIO_POWER)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														PWRCTRL[1:0]	

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	RO	Reserved	0
[1:0]	PWRCTRL[1:0]	RW	Power control: 00: Power down, clock stops; 01: Reserved;	0

			10: Reserved power-on status; 11: Power-on, the card clock is turned on;	

24.6.2 Clock Register (R32_SDIO_CLKCR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	HWFC_EN	NEGEDGE	WIDBUS[1:0]		BYPASS	PWRSAV	CLKEN	CLKDIV[7:0]							

Bit	Name	Access	Description	Reset Value
[31:15]	Reserved	RO	Reserved	0
[14]	HWFC_EN	RW	Hardware flow control enable. TXFIFOE and RXFIFOF signals have effect after this bit is set. 0: Disable hardware flow control; 1: Enable hardware flow control.	0
[13]	NEGEDGE	RW	SDIO_CK phase selection. 0: SDIO_CK is generated on the rising edge of HCLK; 1: SDIO_CK is generated on the falling edge of HLCK.	0
[12:11]	WIDBUS[1:0]	RW	Bus width configuration. 00: 1-bit bus mode, use SDIO_D0; 01: 4-bit bus mode, use SDIO_D[3:0]; 10: 8-bit bus mode, use SDIO_D[7:0]; 11: Not used.	0
[10]	BYPASS	RW	Clock bypass enable. 0: SDIO_CK is obtained after divider; 1: SDIO_CK is directly connected to HCLK. *Note: The HWFC_EN bit needs to be enabled when using this bit.*	0
[9]	PWRSAV	RW	Free clock state configuration. After this bit is set, SDIO_CK output is disabled when bus is free, to save power. 0: SDIO_CK is always output; 1: SDIO_CK is only output when needed.	0
[8]	CLKEN	RW	Clock enable. 0: SDIO_CK output is disabled; 1: SDIO_CK output is enabled.	0
[7:0]	CLKDIV[7:0]	RW	Clock frequency division factor. This field represents the relationship between SDIO_CK and HCLK. SDIO_CK=HCLK/(CLKDIV+2). Note that SDIO_CK should be lower than 400kHz during initialization.	0

Note: (1) The clock configuration register is used to control SDIO_CK related parameters. It should be noted that this register cannot be changed during the period of reading and writing data to 7 HCLK cycles.

(2) No need to configure clock register in slave mode.

24.6.3 Command Parameter Register (R32_SDIO_ARG)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CMDARG[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMDARG[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	CMDARG	RW	Master mode: Command parameter field. This field stores the parameters in the command, which will be sent to the CMD line as part of the command. Slave mode: 32-bit response parameter register to which the slave replies to the answer.	0

24.6.4 Command Register (R32_SDIO_CMD)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	ATACMD	NIEN	ENCMDCompl	SDIOSuspend	CPSMEN	WAITPEND	WAITINT	WAITRESP[1:0]		CMDINDEX[5:0]					

Bit	Name	Access	Description	Reset Value
[31:15]	Reserved	RO	Reserved	0
[14]	ATACMD	RW	Carry out CE-ATA command. If this bit is set, CPSM will go to CMD61.	0
[13]	NIEN	RW	Not enable CE-ATA interrupt setting. If this bit is set, CE-ATA will not generate an interrupt.	0
[12]	ENCMDCompl	RW	Enable CMD completed signal enable. If this bit is set, command completion will be signaled.	0
[11]	SDIOSuspend	RW	Suspend command. If this bit is set, a suspend signal will be sent. Applies to SDIO cards only.	0
[10]	CPSMEN	RW	CPSM (Command Path State Machine) enable. If set, CPSM is enabled.	0
[9]	WAITPEND	RW	Command wait control. If this bit is set, the CPSM will wait for the data transfer to complete before sending the command.	0
[8]	WAITINT	RW	Command wait interrupt control. If this bit is set, the CPSM will turn off the timeout control and	0

			wait for an interrupt to be generated.	
[7:6]	WAITRESP[1:0]	RW	Response type. This field indicates the type of response the CPSM expects to receive. 00: No response, wait for the CMDSENT flag; 01: Short response, wait for CMDREND or CCRCFAIL flag; 10: No response, wait for the CMDSENT flag; 11: Long response, wait for CMDREND or CCRCFAIL flag;	0
[5:0]	CMDINDEX[5:0]	RW	Command index. This field indicates the specific command value.	0

24.6.5 Response Register (R32_SDIO_RESPCMD)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RESPCMD[5:0]					

Bit	Name	Access	Description	Reset Value
[31:6]	Reserved	RO	Reserved	0
[5:0]	RESPCMD[5:0]	RO	This field records the index value of the received response.	0

24.6.6 Response Parameter Register (R128_SDIO_RESPX)

Bit	Name	Access	Description	Reset Value
[127:0]	CARDSTATUSx	RO	Master mode: When the response is a long response, the whole 128 bits indicate the card status; when the response is a short response, the lower 32 bits indicate the card status. the SDIO peripheral receives the highest bit of the card status first and stores it starting from the lowest bit of R128_SDIO_RESPX.	0

24.6.6.1 Response Parameter Register High 32 Bits (R128_SDIO_RESP1[127:96])

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS1[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS1[111:96]															

Bit	Name	Access	Description	Reset Value
[31:0]	CARDSTATUS1	RO	Master mode:	0

			Long response: Card status [127:96]; Short response: Card status [31:0]. Slave mode: Command parameters received by the receive command.	

24.6.6.2 Response Parameter Register Second High 32 Bits (R128_SDIO_RESP2[95:64])

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS2[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS2[79:64]															

Bit	Name	Access	Description	Reset Value
[31:0]	CARDSTATUS2	RO	Master mode: Card status [95:64]; Slave mode: Index of the command received by the receive command.	0

24.6.6.3 Response Parameter Register Second Low 32 Bits (R128_SDIO_RESP3[63:32])

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS3[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS3[47:32]															

Bit	Name	Access	Description	Reset Value
[31:0]	CARDSTATUS3	RO	Master mode: Card status [63:32].	0

24.6.6.4 Response Parameter Register Low 32 Bits (R128_SDIO_RESP4[31:0])

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS4[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS4[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	CARDSTATUS4	RO	Master mode: Card status [31:0];	0

24.6.7 Data Timing Register (R32_SDIO_DTIMER)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATATIME[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATATIME[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	DATATIME	RW	Data timeout period. SDIO_CK cycle as the unit.	0

24.6.8 Data Length Register (R32_SDIO_DLEN)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							DATALENGTH[24:16]								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATALENGTH[15:0]															

Bit	Name	Access	Description	Reset Value
[31:25]	Reserved	RO	Reserved	0
[24:0]	DATALENGTH[24:0]	RW	Data length. The value of this field is loaded into the transfer counter when a transfer is started. For block transfers, the value of this field must be an integer multiple of the block size. The block size is defined by the SDIO device and stored in R32_SDIO_DCTLR[7:4]. Common values are 512B, etc.	0

24.6.9 Data Control Register (R32_SDIO_DCTRL)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SDIOEN	RWMOD	RWSTOP	RWSTART	DBLOCKSIZE[3:0]				DMAEN	DTMODE	DTDIR	DTEN

Bit	Name	Access	Description	Reset Value
[31:12]	Reserved	RO	Reserved	0
11	SDIOEN	RW	SDIO enable. When this bit is set, the DPSM can perform some operations specific to the SDIO card.	0
10	RWMOD	RW	Read wait mode. 0: Stop SDIO_CK control read wait; 1: Use SDIO_D2 to control read wait.	0
9	RWSTOP	RW	Read wait stop. If the RWSTART bit is set, the read wait will be stopped.	0
8	RWSTART	RW	Read wait start. Setting this bit will perform a read wait operation.	0
[7:4]	DBLOCKSIZE[3:0]	RW	Data block size. This field stores the size of the data block, and the block transfer length must be defined	0

			before using the block transfer. The value that can be written in this field is between 0 and 1110b, which indicates that the block transfer length is 2^{BLKLEN} (Between 0 and 16384 bytes).	
3	DMAEN	RW	DMA enable. 0: Disable DMA; 1: Enable DMA.	0
2	DTMODE	RW	Data mode. Set this bit to configure the transfer mode. 0: Block transfer; 1: Stream transfer. *Note: The 4-bit and 8-bit bus mode of streaming transmission only supports products with the last six digits of lot numbers of CH32F20x_D8, CH32F20x_D8C, CH32V30x_D8C, and CH32V31x_D8C not being 0.*	0
1	DTDIR	RW	Data direction. Set this bit to configure the transfer direction. 0: From controller to card; 1: From card to controller.	0
0	DTEN	RW	Data enable. Set this bit to start data transfer. Specific process: after setting this bit, DPSM enters the process of Wait_S or Wait_R (Depending on the data direction).	0

24.6.10 Data Count Register (R32_SDIO_DCOUNT)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							DATACOUNT[24:16]								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATACOUNT[15:0]															

Bit	Name	Access	Description	Reset Value
[31:25]	Reserved	RO	Reserved	0
[24:0]	DATACOUNT[24:0]	RO	Data count. The value of the transmission length register will be loaded into this counter when a transfer is initiated and decremented as the transfer progresses.	0

24.6.11 Status Register (R32_SDIO_STA)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CEATAEND	SDIOIT	RXDAVL	TXDAVL	RXFIFOE	TXFIFOE	RXFIFOF	TXFIFOF
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

RXFIFOHF	TXFIFOHE	RXACT	TXACT	CMDACT	DBCKEND	STBITERR	DATAEND	CMDSENT	CMDREND	RXOVERR	TXUNDERR	DTIMEOUT	CTIMEOUT	DCRCFAIL	CCRCFAIL

Bit	Name	Access	Description	Reset Value
[31:24]	Reserved	RO	Reserved	0
23	CEATAEND	RO	When this bit is set, CMD61 receives the CE-ATA completed signal.	0
22	SDIOIT	RO	When this bit is set, SDIO receives a device interrupt.	0
21	RXDAVL	RO	When this bit is set, data in the Rx FIFO is valid.	0
20	TXDAVL	RO	When this bit is set, data in the Tx FIFO is valid.	0
19	RXFIFOE	RO	When this bit is set, the Rx FIFO is empty.	0
18	TXFIFOE	RO	When this bit is set, the Tx FIFO is empty.	0
17	RXFIFOF	RO	When this bit is set, the Rx FIFO is full.	0
16	TXFIFOF	RO	When this bit is set, the Tx FIFO is full.	0
15	RXFIFOHF	RO	When this bit is set, the Rx FIFO is half full.	0
14	TXFIFOHE	RO	When this bit is set, the Tx FIFO is half empty.	0
13	RXACT	RO	When this bit is set, data is being received.	0
12	TXACT	RO	When this bit is set, data is being transmitted.	0
11	CMDACT	RO	When this bit is set, a command is being transmitted.	0
10	DBCKEND	RO	When this bit is set, the data block has been sent or received and the CRC has passed.	0
9	STBITERR	RO	When this bit is set, no start signal is detected on all data lines in wide bus mode.	0
8	DATAEND	RO	When this bit is set, the data transfer is complete (The transfer counter is zero).	0
7	CMDSENT	RO	When this bit is set, the command has been sent.	0
6	CMDREND	RO	When this bit is set, the response has been received and the CRC is successful.	0
5	RXOVERR	RO	When this bit is set, the Rx FIFO overflows.	0
4	TXUNDERR	RO	When this bit is set, the transmit FIFO underflows.	0
3	DTIMEOUT	RO	When this bit is set, the data times out.	0
2	CTIMEOUT	RO	When this bit is set, the command timeout exceeds 64 SDIO_CK cycles.	0
1	DCRCFAIL	RO	When this bit is set, a data block has been sent or received but the CRC failed.	0
0	CCRCFAIL	RO	When this bit is set, the response has been received but the CRC has failed.	0

24.6.12 Interrupt Clear Register (R32_SDIO_ICR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CEATAENDC	SDIOITC	Reserved					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Reserved	DBCKENDC	STBITERRC	DATAENDC	CMDSENTC	CMDRENDC	RXOVERRC	TXUNDERRC	DTIMEOUTC	CTIMEOUTC	DCRCFAILC	CCRCFAILC

Bit	Name	Access	Description	Reset Value
[31:24]	Reserved	RO	Reserved	0
23	CEATAENDC	RW	Setting this bit clears the CEATAEND bit in the status register.	0
22	SDIOITC	RW	Setting this bit clears the SDIOIT bit in the status register.	0
[21:11]	Reserved	RW	Reserved	0
10	DBCKENDC	RW	Setting this bit clears the DBCKEND bit in the status register.	0
9	STBITERRC	RW	Setting this bit clears the STBITERR bit in the status register.	0
8	DATAENDC	RW	Setting this bit clears the DATAEND bit in the status register.	0
7	CMDSENTC	RW	Setting this bit clears the CMDSENT bit in the status register.	0
6	CMDRENDC	RW	Setting this bit clears the CMDREND bit in the status register.	0
5	RXOVERRC	RW	Setting this bit clears the RXOVERR bit in the status register.	0
4	TXUNDERRC	RW	Setting this bit clears the TXUNDERR bit in the status register.	0
3	DTIMEOUTC	RW	Setting this bit clears the DTIMEOUT bit in the status register.	0
2	CTIMEOUTC	RW	Setting this bit clears the CTIMEOUT bit in the status register.	0
1	DCRCFAILC	RW	Setting this bit clears the DCRCFAIL bit in the status register.	0
0	CCRCFAILC	RW	Setting this bit clears the CCRCFAIL bit in the status register.	0

24.6.13 Interrupt Enable Register (R32_SDIO_MASK)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CEATAENDIE	SDIOITIE	RXDAVLIE	TXDAVLIE	RXFIFOEIE	TXFIFOEIE	RXFIFOFIE	TXFIFOFIE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXFIFOHFIE	TXFIFOHEIE	RXACTIE	TXACTIE	CMDACTIE	DBCKENDIE	STBITERRIE	DATAENDIE	CMDSENTIE	CMDRENDIE	RXOVERRIE	TXUNDERRIE	DTIMEOUTIE	CTIMEOUTIE	DCRCFAILIE	CCRCFAILIE

Bit	Name	Access	Description	Reset

				Value
[31:24]	Reserved	RO	Reserved	0
23	CEATAENDIE	RW	After this bit is set, the status register will generate an interrupt when the CEATAEND bit is set.	0
22	SDIOITIE	RW	After this bit is set, the status register will generate an interrupt when the SDIOIT bit is set.	0
21	RXDAVLIE	RW	After this bit is set, the status register will generate an interrupt when the RXDAVL bit is set.	0
20	TXDAVLIE	RW	After this bit is set, the status register will generate an interrupt when the TXDAVL bit is set.	0
19	RXFIFOEIE	RW	After this bit is set, the status register will generate an interrupt when the RXFIFOE bit is set.	0
18	TXFIFOEIE	RW	After this bit is set, the status register will generate an interrupt when the TXFIFOE bit is set.	0
17	RXFIFOFIE	RW	After this bit is set, the status register will generate an interrupt when the RXFIFOF bit is set.	0
16	TXFIFOFIE	RW	After this bit is set, the status register will generate an interrupt when the TXFIFOF bit is set.	0
15	RXFIFOHFIE	RW	After this bit is set, the status register will generate an interrupt when the RXFIFOHF bit is set.	0
14	TXFIFOHEIE	RW	After this bit is set, the status register will generate an interrupt when the TXFIFOHE bit is set.	0
13	RXACTIE	RW	After this bit is set, the status register will generate an interrupt when the RXACT bit is set.	0
12	TXACTIE	RW	After this bit is set, the status register will generate an interrupt when the TXACT bit is set.	0
11	CMDACTIE	RW	After this bit is set, the status register will generate an interrupt when the CMDACT bit is set.	0
10	DBCKENDIE	RW	After this bit is set, the status register will generate an interrupt when the DBCKEND bit is set.	0
9	STBITERRIE	RW	After this bit is set, the status register will generate an interrupt when the STBITERR bit is set.	0
8	DATAENDIE	RW	After this bit is set, the status register will generate an interrupt when the DATAEND bit is set.	0
7	CMDSENTIE	RW	After this bit is set, the status register will generate an interrupt when the CMDSENT bit is set.	0
6	CMDRENDIE	RW	After this bit is set, the status register will generate an interrupt when the CMDREND bit is set.	0
5	RXOVERRIE	RW	After this bit is set, the status register will generate an interrupt when the RXOVERR bit is set.	0
4	TXUNDERRIE	RW	After this bit is set, the status register will generate an interrupt when the TXUNDERR bit is set.	0
3	DTIMEOUTIE	RW	After this bit is set, the status register will generate an interrupt when the DTIMEOUT bit is set.	0
2	CTIMEOUTIE	RW	After this bit is set, the status register will generate an	0

			interrupt when the CTIMEOUT bit is set.	
1	DCRCFAILIE	RW	After this bit is set, the status register will generate an interrupt when the DCRCFAIL bit is set.	0
0	CCRCFAILIE	RW	After this bit is set, the status register will generate an interrupt when the CCRCFAIL bit is set.	0

24.6.14 FIFO Count Register (R32_SDIO_FIFOCNT)

Offset address: 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FIFOCOUNT[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FIFOCOUNT[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	FIFOCOUNT[31:0]	RO	The FIFO contains the number of words (32 bits) that have not been written to or read from the FIFO. When setting R32_SDIO_DCTLR: R32_SDIO_DCTLR, if the DPSM is idle, the FIFO counter will load the transfer length value from R32_SDIO_TLEN, if this value is not divisible by 4, the last 1 to 3 bytes will be treated as a word.	0

24.6.15 FIFO Data Control Register 2 (R32_SDIO_DCTRL2)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					SLV_CK_PHASE	SLV_FORCE_ERR	SLV_MODE	Reserved							RANDOM_LEN_EN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DBLOCKSIZE2											

Bit	Name	Access	Description	Reset Value
[31:27]	Reserved	R0	Reserved	0
26	SLV_CK_PHASE	RW	Phase selection bit when outputting DATA from mode when SLV_MODE is 1: 0: Outputs DATA on the rising edge of the internal SDCK; 1: outputs DATA on the falling edge of the internal SDCK. When SLV_MODE is 0: 0: DTEN bit writes 0 to clear the FIFO; 1: DTEN bit writes 0 without clearing the FIFO.	0

[25]	SLV_FORCE_ERR	RW	Software forced data block CRC error in slave mode 0: Use hardware correct CRC value; 1: Software forced data block CRC is the wrong value.	0
[24]	SLV_MODE	RW	Slave mode enable bit: 0: The controller operates in host mode; 1: The controller CPSM and DPSM work in slave mode.	0
[23:17]	Reserved	R0	Reserved	0
[16]	RANDOM_LEN_EN	RW	Data block arbitrary byte length enable bit: 0: The data block can only be $2^{DBLOCKSIZE}$ bytes; 1: The data block can be any length from 0 to 2048 in DBLOCKSIZE2.	0
[15:12]	Reserved	R0	Reserved	0
[11:0]	DBLOCKSIZE2	RW	The data block length field for arbitrary byte length mode.	0

24.6.16 FIFO Register (R32_SDIO_FIFO)

Offset address: 0x80

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FIFODATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FIFODATA[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	FIFODATA[31:0]	RW	FIFO data field. Reading and writing this field will read the received data or send the pending data. The SDIO FIFO has a total of 32 words (One word is 32 bits).	0

Chapter 25 Controller Area Network (CAN)

Controller Area Network is a high-performance communication protocol for serial data communication. The CAN controller provides a complete implementation of the CAN protocol, supporting CAN protocols 2.0A and 2.0B. The CAN controller can be used to construct a powerful local area network to realize safe distributed real-time control, process a large number of data packets with a small CPU load, and has a wide range of applications in the industrial and automotive fields.

25.1 Main Features

- Compatible with CAN specification 2.0A and 2.0B
- Programmable transfer rate up to 1Mbit/s
- Support time-triggered communication function to avoid low-priority message blocking
- Support 3 sending mailboxes, the priority of sending messages can be determined by the message identifier or the order of sending requests, and can record the time stamp of the SOF moment of sending the message
- Each CAN group supports 2 receive FIFOs with 3 levels of mailbox depth. A total of 14 filter groups are available for configuration across three CANs. Each filter group can be configured in 32-bit or 16-bit mode, using mask bits or identifier list mode. This minimizes software intervention in message filtering. FIFO overflow handling is flexible and can record timestamps for the SOF moment of received messages.
- Occupies 4 interrupt vectors, each interrupt source can be configured independently

25.2 CAN Controller Operating Mode

The CAN controller can operate the SLEEP or INRQ bits in the register CAN_CTLR to switch between the initialization mode, sleep mode and normal mode.

25.2.1 Initialization Mode

After reset, CAN works in sleep mode by default to reduce power consumption. At this time, the sending and receiving of messages is prohibited, the internal pull-up resistor of the TX pin is enabled, and the TX pin outputs a recessive bit. The INRQ bit in the register CAN1_CTLR is set to 1, and the CAN controller is requested to enter the initialization mode. When the INAK bit of the register CAN1_STATR is automatically set to 1, the initialization state is successfully entered. Similarly, clear the INRQ bit in the register CAN1_CTLR to request the CAN controller to exit the initialization mode. When the INAK bit of the register CAN1_STATR is automatically cleared to 0, the initialization state is successfully exited.

The filter group can be initialized in the non-initialization mode, but the FINIT bit of the register CAN1_FCTLR must be set to 1, and the reception of the message is prohibited at this time.

25.2.2 Sleep Mode

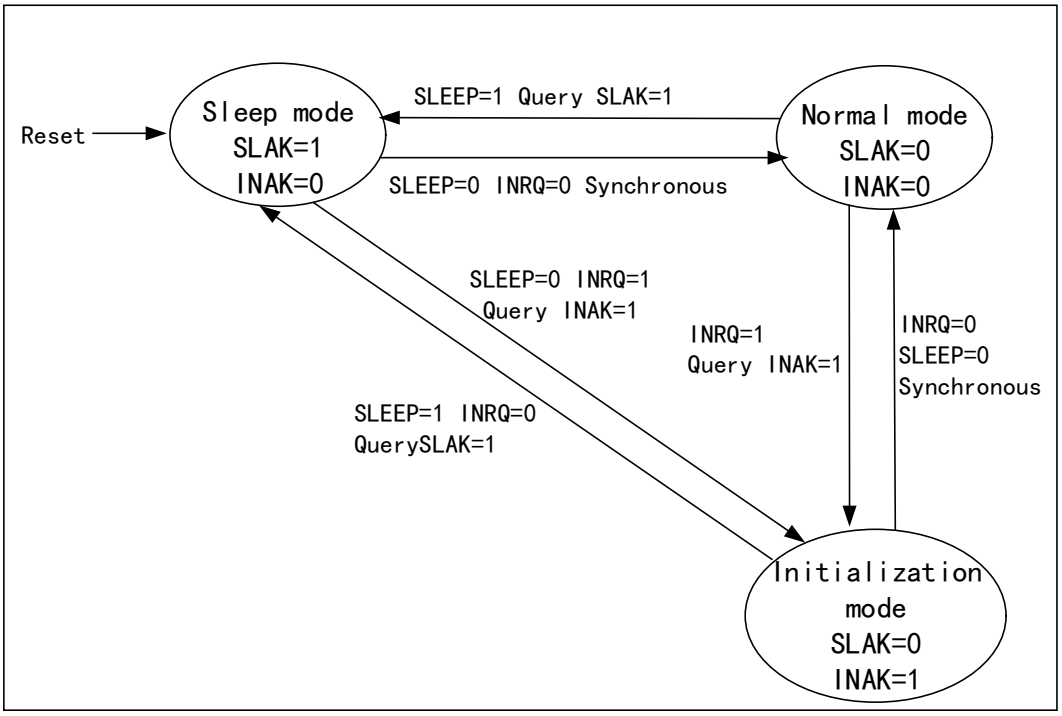
For the SLEEP position 1 in the register CAN1_CTLR, the CAN controller is requested to enter the sleep mode. When the SNAK bit of the register CAN1_STATR is automatically set to 1, the CAN successfully enters the sleep mode, and the clock of the CAN controller stops, but the mailbox register can still be accessed.

From sleep mode to initialization mode, the SLEEP bit of CAN1_CTLR must be cleared to position 1, and when the INAK bit of register CANx1_STATR is automatically set to 1, it will be switched to initialization

state.

To enter the normal mode from sleep mode, the SLEEP bit of CAN1_CTLR must be cleared 0, and when the SNAK bit of register CAN1_STATR is automatically cleared 0, it will enter normal mode.

Figure 25-1 CAN operating mode switch



25.3 CAN Controller Test Mode

In the initialization mode, operate the SILM and LBKM bits of the register CAN1_BTIMR to select a test mode, and then exit the initialization mode and enter the test mode by clearing the INRQ bit of the register CAN1_CTLR. There are 3 test modes: silent mode, loopback mode and silent loopback mode.

25.3.1 Silent Mode

Setting the SILM bit in register CAN1_BTIMR to 1 can optionally enter silent mode. In this mode, the CAN controller can receive, but cannot send messages to the outside world. It is always in a recessive bit to the outside world, which can avoid affecting the bus, but the message can be received by the controller of the node where it is located. Usually, silent mode is used for the status analysis of the CAN bus.

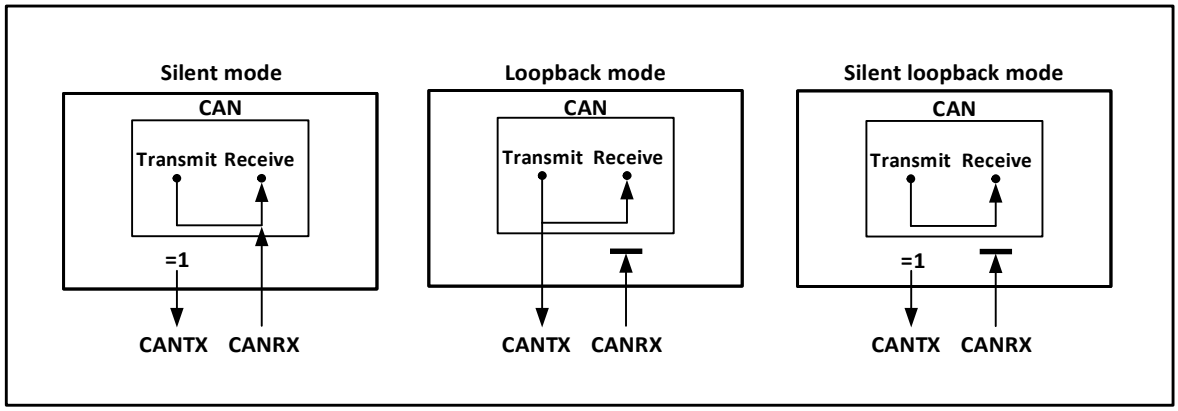
25.3.2 Loopback Mode

By setting the LBKM bit of the register CAN1_BTIMR to 1, the loopback mode can be selected. In this mode, the CAN controller can send external messages, but cannot receive external messages, but the sent messages can be received by the controller of the node where it is located, and the reception filtering mechanism is effective. Usually, loopback mode is used for transceiver testing of CAN controllers.

25.3.3 Silent Loopback Mode

Setting the SILM and LBKM bits in register CAN1_BTIMR to 1 can optionally enter silent loopback mode. This mode is usually used for the closed self-test of the CAN controller. In this mode, it has no effect on the CAN bus, the RX pin is disconnected from the bus, and the TX pin is set to a recessive bit.

Figure 25-2 CAN 3 test modes of the bus



25.4 MCU Operating State of CAN Controller in Debug Mode

When the MCU enters the debug mode, the core is in a suspended state, but it can be determined whether the CAN controller is in a normal operation or a stop state through the configuration bits in the debug module.

25.5 CAN Controller Functional Description

25.5.1 Transmit Processing Flow

The transmit processing flow is as follows: If there are vacant mailboxes among the 3 sending mailboxes, the application layer software only has write access to the registers of the vacant mailboxes, and operates the registers CAN1_TXMIyR, CAN1_TXMDTyR, CAN1_TXMDLyR and CAN1_TXMDHyR, and can set the message identifier, message length, time stamp, and message data. After the data is ready, the TXRQ bit of the register CAN1_TXMIyR is set to 1 to request transmission, the mailbox enters the registered state, and the priority is queued; once it becomes the highest priority mailbox, it becomes the scheduled transmission state and waits for the CAN bus to be idle; when the CAN bus is idle When the message is scheduled to be sent, the message of the mailbox will enter the sending state immediately; after the message is sent, the mailbox will become a vacant mailbox again, and the RQCP and TXOK bits of the register CAN1_TSTATR are set to 1 to indicate that the sending is successful; if the arbitration fails during sending, the ALST of the register CAN1_TSTATR Set to 1, TERR set to 1 if an error is sent.

25.5.2 Transmit Priority

The transmit priority can be determined by the identifier or the order of sending requests. The TXFP position of the register CAN1_CTLR is 1 and sent according to the order of sending requests. According to the order of sending requests, it is mainly used for segmented sending; the TXFP bit is cleared to 0 and sent according to the priority of the identifier. In order, the smaller the identifier is, the higher the priority is. In the case of the same identifier, the mailbox with the lower number has a higher priority.

25.5.3 Transmit Abort Processing

If the ABRQ bit in the register CAN1_TSTATR is set, the transmission request can be aborted. When the mailbox status is registered or scheduled to send, the sending request is directly aborted; when the mailbox is in the sending state, the abort request may succeed (stop sending) or fail (sending complete), and the result can be queried by the TXOK bit in the CAN1_TSTATR register.

25.5.4 Time-based Trigger Mode

When the traditional CAN communication bus is busy, it is easy to cause low-priority messages to be blocked for a long time, and even cannot meet the requirements of its time limit. In order to solve the bottleneck, related protocols based on time-triggered mode have been introduced. Such protocols have a certain scale of application in the industry, and the functions based on time-triggered mode are the application of such protocols.

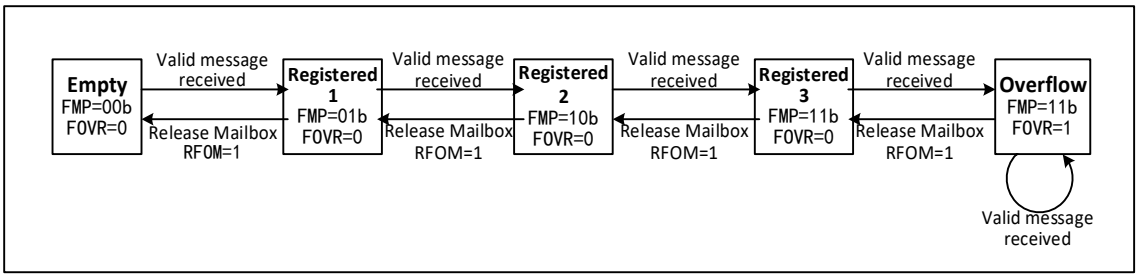
There are 2 modes to choose from in the time-triggered mode. To use this mode, the automatic retransmission function needs to be turned off. The default mode and enhanced mode are selected by configuring the MODE bit of the CAN1_TTCTLR register. Set the TTCM and NART bits of the register CAN1_CTLR to 1, enable the time-triggered mode and disable automatic retransmission. The MODE bit of the CAN1_TTCTLR register defaults to 0. At this time, it works in the default mode, and the internal timer is activated to generate timestamps of the transmit and receive mailboxes. The timer accumulates at the CAN bit time, and the internal timer is sampled and generates a timestamp at the sample point position of the received and transmitted frame start bits. If the enhanced mode is used, the MODE bit of the CAN1_TTCTLR register needs to be set to 1 to enable the enhanced mode. Using this mode, there must be 3 or more nodes in the entire CAN network, one of which sends the time reference, and the other nodes receive the time stamp of the reference node, they reset the internal counter by writing 1 to the TIMRST bit of the CAN1_TTCTLR register to synchronize the internal counter. So that in addition to the node that sends the time reference, the rest of the CAN nodes achieve time synchronization. Afterwards, write the data to be sent to the sending mailbox, configure the time trigger count value (TIMCNT of the CAN1_TTCNT register) and the internal counter count end value (TIMCMV of the CAN1_TTCTLR register) of each node in turn. The time-triggered count value and the final count value of the internal counter are determined by the CAN nodes, the CAN communication rate and the number of data bits in a frame. After the configuration is completed, each node waits for the internal counter to count to the time-triggered count value, and then triggers the transmitting action.

25.5.5 Receive Processing Flow

The reception of CAN bus messages is completed by the controller hardware without the intervention of the MCU, which reduces the processing load of the MCU. The received messages are stored in 2 FIFOs with 3-level mailbox depth according to the setting of the register CAN1_FAFIFOR. If the application layer needs to obtain the message, it can only read the valid received message through the receiving FIFO mailbox.

Initially, the receiving FIFO is empty, and the value of FMPx[1:0] in the receiving FIFO register CAN1_RFIFOx is binary 00b. After receiving a valid receiving message, it becomes the registered 1 state, and the controller automatically sets the FMPx[1:0] in the receiving FIFO register CAN1_RFIFOx to binary 01b; if the mailbox data registers CAN1_RXMDLRx and CAN1_RXMDHRx are read at this time, the mailbox is released by setting the RFOM bit of the receiving FIFO register CAN1_RFIFOx to 1, and the receiving FIFO state becomes empty again; if the mailbox is not released in the registered 1 state, after the next valid receiving message is received, the receiving FIFO state switches to the registered 2 state; at this time, the FMPx[1:0] of the receiving FIFO register CAN1_RFIFOx is automatically set to binary 10b. If the mailbox data register is read and the mailbox is released, then the state returns to registered 1; if the mailbox is not released in the registered 2 state, the receiving FIFO enters the registered 3 state; also in the registered 3 state, the message is read and the mailbox is released, then the registered 2 state is returned; if the registered 3 state is not If the mailbox is released, when the next valid packet is received, packet loss will inevitably occur.

Figure 25-3 Receive FIFO state switching diagram



In the case of message loss above, that is, the receiving FIFO is full, and the message overflow causes the message to be lost. The FOVR bit of the receiving FIFO register CAN1_RFIFOx will be automatically set to 1 by hardware for overflow query. When the RFLM bit of the register CAN1_CTLR is set to 1, the receiving FIFO locking function is enabled, and the discarded message is a new received message; when the RFLM bit of the register CAN1_CTLR is cleared to 0, the receiving FIFO locking function is disabled. Among the 3 original messages of the receiving FIFO, the last received message will be overwritten by the new message.

When the relevant bit of the register CAN1_INTENR is set, an interrupt can be generated when the state of the receiving FIFO is switched, so as to process the received message more efficiently, see section 25.6 CAN interrupt for details.

25.5.6 Received Message Identifier Filtering

There are as many as 14 filter groups in the module. By setting the filter groups, each CAN node can receive messages that meet the filtering rules, and messages that do not meet the filtering rules are discarded by hardware without software intervention.

Each filter group consists of two 32-bit registers: CANx_FxR0 and CANx_FxR1. The bit width of the filter group can be independently configured as either one 32-bit filter or two 16-bit filters by setting individual bits in the CAN1_FSCFGR register. Each filter group can be configured as a mask bit or identifier list mode by setting individual bits in the CAN1_FMCFGR register. Individual filter groups can be enabled or disabled by setting corresponding bits in the CAN1_FWR register. Bits in the CAN1_FAFIFOR register determine which receive FIFO stores messages passing through the filter.

As shown in Table 25-1, in mask bit mode, the two registers are the identifier register and the mask register, which must be used in conjunction. Each bit in the identifier register indicates whether the corresponding bit should be set to a dominant or recessive state. Each bit in the mask register indicates whether the corresponding bit should match the desired state specified by the identifier register.

Table 25-1 32-bit mask bit patterns

Identifier register	CANx_FxR1[31:24]		CANx_FxR1[23:16]		CANx_FxR1[15:8]		CANx_FxR1[7:0]		
Mask bit register	CANx_FxR2[31:24]		CANx_FxR2[23:16]		CANx_FxR2[15:8]		CANx_FxR2[7:0]		
Mapping	STID[10:3]		STID[2:0]	EXID[17:13]	EXID[12:5]		EXID[4:0]	IDE	RTR	0

In identifier list mode, both registers are used as identifier registers, and the identifier of the received message must be consistent with one of the registers before it can pass the screening.

Table 25-2 32-bit identifier list mode

Identifier	CANx_FxR1[31:24]	CANx_FxR1[23:16]	CANx_FxR1[15:8]	CANx_FxR1[7:0]

register									
Mask bit register	CANx_FxR2[31:24]		CANx_FxR2[23:16]		CANx_FxR2[15:8]		CANx_FxR2[7:0]		
Mapping	STID[10:3]	STID[2:0]	EXID[17:13]		EXID[12:5]		EXID[4:0]	IDE	RTR	0

In the 16-bit mode, the register group is divided into four registers, and the mask bit mode of each group of filters can have two filters, each of which contains a 16-bit identifier register and a 16-bit mask register; All four registers in identifier list mode are used as identifier registers.

Table 25-3 16-bit mask bit patterns

Identifier register n	CANx_FxR1[15:8]	CANx_FxR1[7:0]			
Mask bit register n	CANx_FxR1[31:24]	CANx_FxR1[23:16]			
Identifier register n+1	CANx_FxR2[15:8]	CANx_FxR2[7:0]			
Mask bit register n+1	CANx_FxR2[31:24]	CANx_FxR2[23:16]			
Mapping	STID[10:3]	STID[2:0]	RTR	IDE	EXID[17:15]

Table 25-4 16-bit identifier list mode

Identifier register n	CANx_FxR1[15:8]	CANx_FxR1[7:0]			
Mask bit register n	CANx_FxR1[31:24]	CANx_FxR1[23:16]			
Identifier register n+1	CANx_FxR2[15:8]	CANx_FxR2[7:0]			
Mask bit register n+1	CANx_FxR2[31:24]	CANx_FxR2[23:16]			
Mapping	STID[10:3]	STID[2:0]	RTR	IDE	EXID[17:15]

Messages entering the FIFO mailbox are read and stored by the application, which typically distinguishes message data based on message identifiers. The CAN controller provides a filter number for messages filtered through different filters in the receive FIFO. This number is stored in the FMI[7:0] field of the CANx_RXMDTyR register, regardless of whether the filter group is enabled. Refer to Figure 25-4 for the numbering rules.

When a message passes multiple filters, the stored filter number in the receive buffer is determined by the filter priority rules. The filter priority rules are as follows:

- All 32-bit filters have higher priority than 16-bit filters.
- For filters of the same width, those using an identifier list have higher priority than those using a mask bit pattern.
- For filters with identical width and pattern, the one with the lower number has higher priority.

As shown in Figure 25-5: When receiving a message, the identifier is first matched against the 32-bit identifier list pattern filter. If no match is found, it is then matched against the 32-bit mask pattern filter. If still no match occurs, it proceeds to match against the 16-bit identifier list pattern filter. If no match is found, it finally matches against the 16-bit mask pattern filter. If no match is found at any stage, the message is discarded. If a match occurs, the message is stored in the mailbox of the receive FIFO, and the identifier number is stored in the FMI field of the CAN1_RXMDTRx register.

Figure 25-4 Example of Filter Number

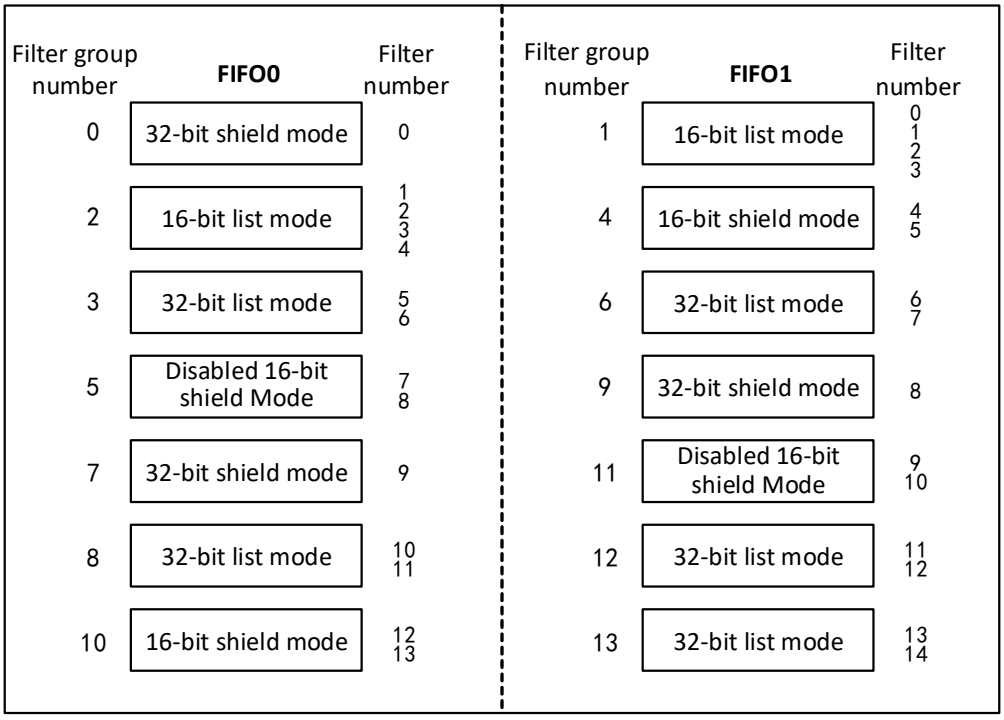
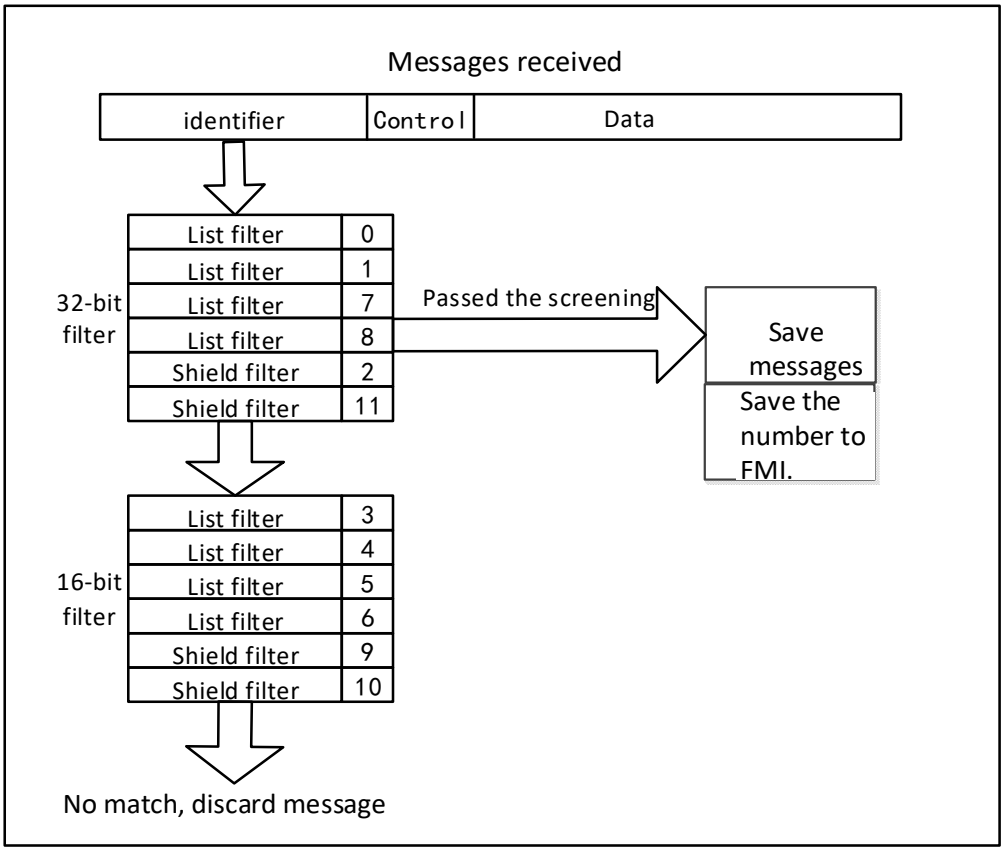


Figure 25-5 Filter filtering example



25.5.7 Error Handling

The CAN controller relies on the state error register CAN1_ERRSR to manage errors on the bus. The TEC and REC in the state error register CAN1_ERRSR represent the sending and receiving error count values

respectively. According to the increase of the sending and receiving errors and the decrease of the sending and receiving success, the stability of the CAN bus can be judged according to their values.

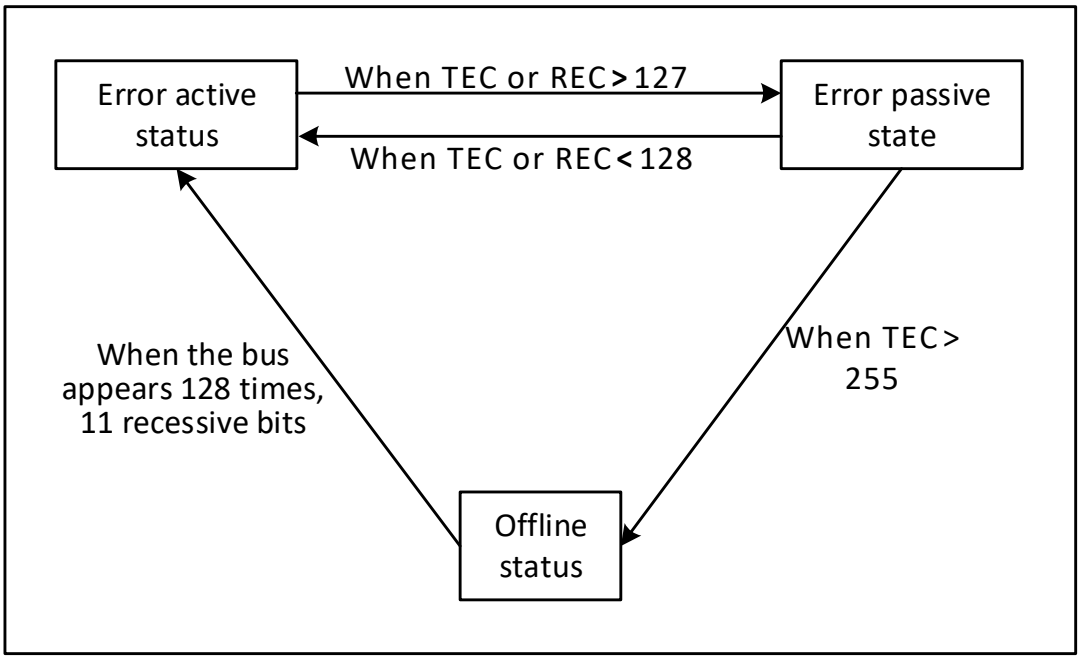
When the TEC and REC in the state error register CAN1_ERRSR are less than 128, the current CAN node is in an error active state, can normally participate in bus communication, and issue an active error flag when an error is detected.

When the TEC and REC in the state error register CAN1_ERRSR are greater than 127, the current CAN node is in an error passive state, and when an error is detected, it is not allowed to issue an active error flag, but can only issue a passive error flag.

When the TEC in the status error register CAN1_ERRSR is greater than 255, the current CAN node goes offline.

When the bus monitors 11 consecutive hidden bits for 128 times, it returns to the error active state, and the recovery mode is affected by the ABOM bit in the main control register CAN1_CTLR. If ABOM is set to 1, the hardware automatically exits the offline state. If ABOM is 0, the software needs to operate the INRQ bit to enter the initialization mode, and then exit the initialization before exiting the offline state.

Figure 25-6 CAN error state switch diagram



25.5.8 Bit Timing

According to the standard of CAN bus, each bit time is divided into 4 segments: synchronization segment, propagation time segment, phase buffer segment 1 and phase buffer segment 2. These segments consist of the minimum time unit Tq. The CAN controller monitors the CAN bus change by sampling and synchronizes through the edge of the start bit of the frame.

The CAN controller re-divides the above 4 segments into 3 segments, which are:

- Synchronization segment (SS): the synchronization segment in the CAN standard is fixed as a minimum time unit, and the expected bit jump normally occurs within this period of time.
- Time period 1 (BS1): contains the propagation time period and phase buffer section 1 in the CAN standard, which can be set to contain 1 to 16 minimum time units and can be automatically extended to compensate for the positive phase drift caused by frequency accuracy errors of different nodes on the CAN bus. The

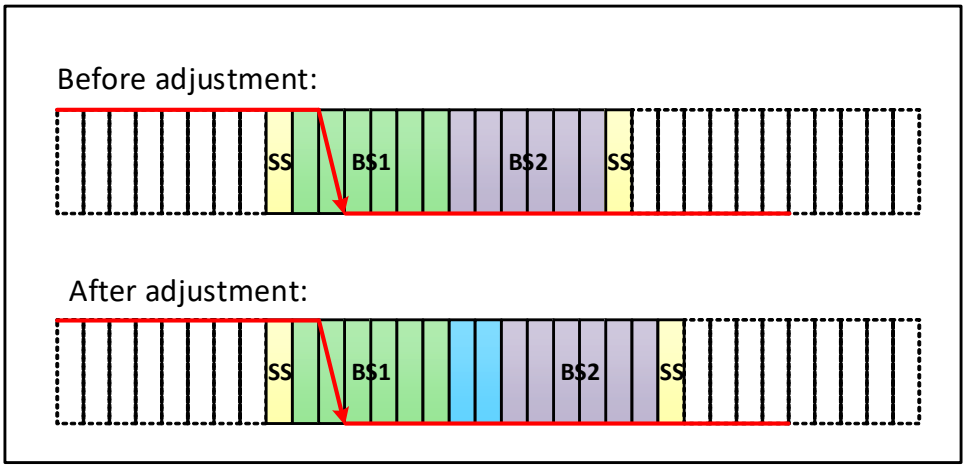
end of the time period is the location of the sampling point.

- Time period 2 (BS2): phase buffer section 2 in the CAN standard can be set to 1 to 8 minimum time units and can be automatically shortened to compensate for the negative phase drift caused by frequency accuracy errors of different nodes on the CAN bus.

Resynchronization jump width (SJW) is the upper limit of the minimum number of time units that can be extended and reduced per person, and the range can be set to 1 to 4 minimum time units.

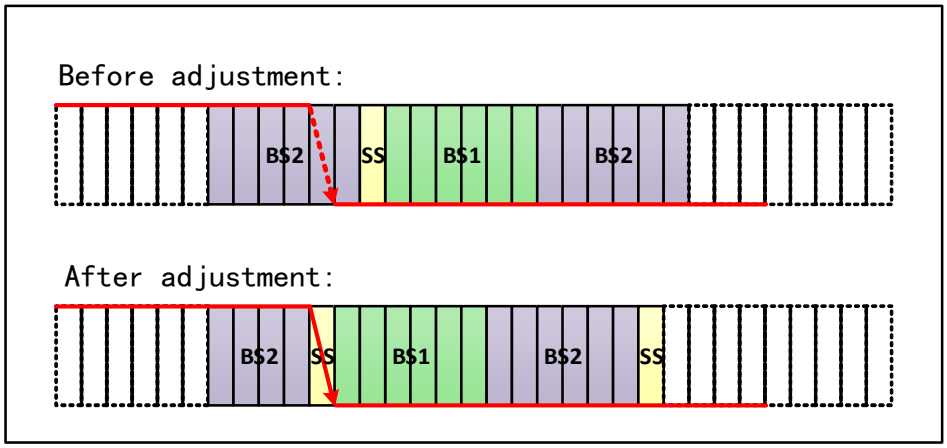
The above parameters can be configured in the CAN bus timing register CAN1_BTIMR.

Figure 25-7 The jump appears in BS1



If the SJW of figure 25-7 is 2, and the bus level jump is detected in time period 1, the length of time period 1 needs to be extended, with a maximum extension of SJW, thus delaying the position of the sampling point.

Figure 25-8 The jump appears in BS2



If the SJW of figure 25-8 is 2 and the bus level jump is detected in time period 2, it is necessary to reduce the length of time period 2 and the maximum SJW, so as to advance the position of the sampling point.

25.6 CAN Interrupt

The CAN controller has 4 interrupt vectors, which are send interrupt, FIFO_0 interrupt, FIFO_1 interrupt, error and state change interrupt.

Set the CAN interrupt allow register CAN1_INTENR to allow or disable each interrupt source.

The sending interrupt is caused by the sending mailbox empty event. After the interrupt is generated, the

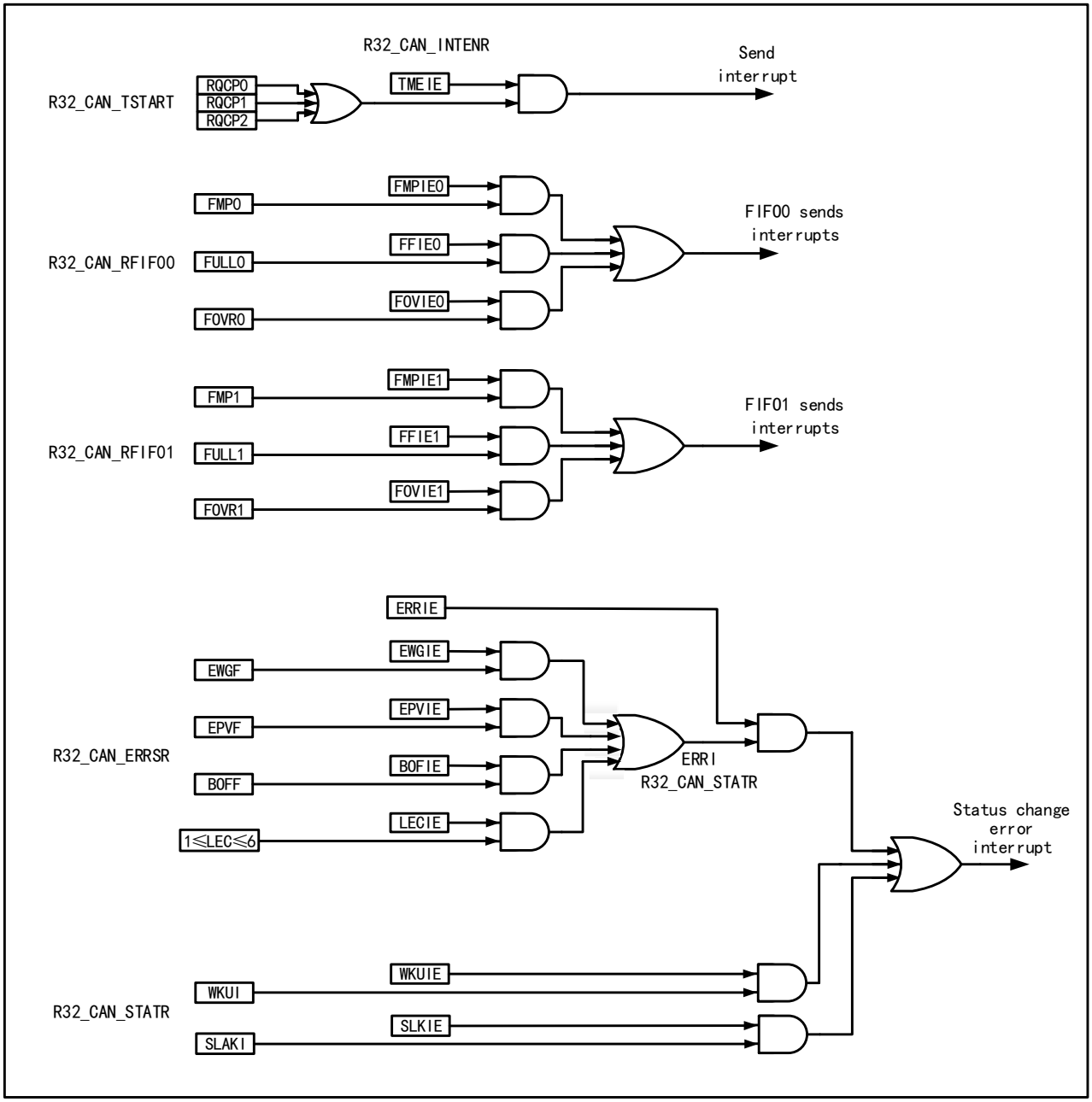
RQCP0, RQCP1 and RQCP2 bits of the register CAN1_TSTATR are queried to determine which mailbox empty event is generated.

FIFO0 interrupts are caused by receiving new messages, receiving mailbox fullness and overflow events. After the interrupts are generated, the FMP0, FULL0 and FOVER0 bits of register CAN1_RFIFO0 are queried to determine which mailbox emptiness event is generated.

FIFO1 interrupts are caused by receiving new messages, receiving mailbox fullness and overflow events. After the interrupts are generated, the FMP1, FULL1 and FOVER1 bits of register CAN1_RFIFO1 are queried to determine which mailbox emptiness event is generated.

Errors and state change interruptions are caused by errors, arousal, and sleep events.

Figure 25-9 CAN interrupt logic diagram



25.7 Register Description

The registers related to the CAN controller must be manipulated in 32-bit words. In order to avoid the influence of the current node on the entire CAN bus, the application software can only modify the bit timing register CAN1_BTIMR in the initialization mode.

Table 25-5 CAN-related registers

Name	Access address	Description	Reset value
R32_CAN1_CTLR	0x40006400	CAN1 main control register	0x00010002
R32_CAN1_STATR	0x40006404	CAN1 main status register	0x00000x02
R32_CAN1_TSTATR	0x40006408	CAN1 transmit status register	0x1C000000
R32_CAN1_RFIFO0	0x4000640C	CAN1 receive FIFO0 control and status registers	0x00000000
R32_CAN1_RFIFO1	0x40006410	CAN1 receive FIFO1 control and status registers	0x00000000
R32_CAN1_INTENR	0x40006414	CAN1 interrupt enable register	0x00000000
R32_CAN1_ERRSR	0x40006418	CAN1 error status register	0x00000000
R32_CAN1_BTIMR	0x4000641C	CAN1 bit timing register	0x01230000
R32_CAN1_TTCTLR	0x40006420	CAN1 time trigger control register	0x0000FFFF
R32_CAN1_TTCNT	0x40006424	CAN1 time trigger count value register	0x00000000
R32_CAN1_TERR_CNT	0x40006428	CAN1 offline recovery error counter	0x00000000

Table 25-6 CAN1 mailbox-related registers

Name	Access address	Description	Reset value
R32_CAN1_TXMI0R	0x40006580	CAN1 Tx mailbox 0 identifier register	X
R32_CAN1_TXMDT0R	0x40006584	CAN1 Tx mailbox 0 data length and timestamp register	X
R32_CAN1_TXMDL0R	0x40006588	CAN1 Tx mailbox 0 data low register	X
R32_CAN1_TXMDH0R	0x4000658C	CAN1 Tx mailbox 0 data high register	X
R32_CAN1_TXMI1R	0x40006590	CAN1 Tx mailbox 1 identifier register	X
R32_CAN1_TXMDT1R	0x40006594	CAN1 Tx mailbox 1 data length and timestamp register	X
R32_CAN1_TXMDL1R	0x40006598	CAN1 Tx mailbox 1 data low register	X
R32_CAN1_TXMDH1R	0x4000659C	CAN1 Tx mailbox 1 data high register	X
R32_CAN1_TXMI2R	0x400065A0	CAN1 Tx mailbox 2 identifier register	X
R32_CAN1_TXMDT2R	0x400065A4	CAN1 Tx mailbox 2 data length and timestamp register	X
R32_CAN1_TXMDL2R	0x400065A8	CAN1 Tx mailbox 2 data low register	X
R32_CAN1_TXMDH2R	0x400065AC	CAN1 Tx mailbox 2 data high register	X
R32_CAN1_RXMI0R	0x400065B0	CAN1 Rx FIFO0 mailbox identifier register	X
R32_CAN1_RXMDT0R	0x400065B4	CAN1 Rx FIFO0 mailbox data length and timestamp register	X
R32_CAN1_RXMDL0R	0x400065B8	CAN1 Rx FIFO0 mailbox data low register	X

R32_CAN1_RXMDH0R	0x400065BC	CAN1 Rx FIFO0 mailbox data high register	X
R32_CAN1_RXMI1R	0x400065C0	CAN1 Rx FIFO1 mailbox identifier register	X
R32_CAN1_RXMDT1R	0x400065C4	CAN1 Rx FIFO1 mailbox data length and timestamp register	X
R32_CAN1_RXMDL1R	0x400065C8	CAN1 Rx FIFO1 mailbox data low register	X
R32_CAN1_RXMDH1R	0x400065CC	CAN1 Rx FIFO1 mailbox data high register	X

Table 25-7 CAN filter-related registers

Name	Access address	Description	Reset value
R32_CAN1_FCTLR	0x40006600	CAN1 filter main control register	0x2A1C0E01
R32_CAN1_FMCFGR	0x40006604	CAN1 filter mode register	0x00000000
R32_CAN1_FSCFGR	0x4000660C	CAN1 filter bit width register	0x00000000
R32_CAN1_FAFIFOR	0x40006614	CAN1 filter FIFO association register	0x00000000
R32_CAN1_FWR	0x4000661C	CAN1 filter activation register	0x00000000
R32_CAN1_F0R1	0x40006640	CAN1 filter group 0 register 1	X
R32_CAN1_F0R2	0x40006644	CAN1 filter group 0 register 2	X
R32_CAN1_F1R1	0x40006648	CAN1 filter group 1 register 1	X
R32_CAN1_F1R2	0x4000664C	CAN1 filter group 1 register 2	X
R32_CAN1_F2R1	0x40006650	CAN1 filter group 2 register 1	X
R32_CAN1_F2R2	0x40006654	CAN1 filter group 2 register 2	X
R32_CAN1_F3R1	0x40006658	CAN1 filter group 3 register 1	X
R32_CAN1_F3R2	0x4000665C	CAN1 filter group 3 register 2	X
R32_CAN1_F4R1	0x40006660	CAN1 filter group 4 register 1	X
R32_CAN1_F4R2	0x40006664	CAN1 filter group 4 register 2	X
R32_CAN1_F5R1	0x40006668	CAN1 filter group 5 register 1	X
R32_CAN1_F5R2	0x4000666C	CAN1 filter group 5 register 2	X
R32_CAN1_F6R1	0x40006670	CAN1 filter group 6 register 1	X
R32_CAN1_F6R2	0x40006674	CAN1 filter group 6 register 2	X
R32_CAN1_F7R1	0x40006678	CAN1 filter group 7 register 1	X
R32_CAN1_F7R2	0x4000667C	CAN1 filter group 7 register 2	X
R32_CAN1_F8R1	0x40006680	CAN1 filter group 8 register 1	X
R32_CAN1_F8R2	0x40006684	CAN1 filter group 8 register 2	X
R32_CAN1_F9R1	0x40006688	CAN1 filter group 9 register 1	X
R32_CAN1_F9R2	0x4000668C	CAN1 filter group 9 register 2	X
R32_CAN1_F10R1	0x40006690	CAN1 filter group 10 register 1	X
R32_CAN1_F10R2	0x40006694	CAN1 filter group 10 register 2	X
R32_CAN1_F11R1	0x40006698	CAN1 filter group 11 register 1	X
R32_CAN1_F11R2	0x4000669C	CAN1 filter group 11 register 2	X
R32_CAN1_F12R1	0x400066A0	CAN1 filter group 12 register 1	X
R32_CAN1_F12R2	0x400066A4	CAN1 filter group 12 register 2	X
R32_CAN1_F13R1	0x400066A8	CAN1 filter group 13 register 1	X
R32_CAN1_F13R2	0x400066AC	CAN1 filter group 13 register 2	X

25.7.1 CAN1 Main Control Register (CAN1_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															DBF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RST	Reserved							TTCM	ABOM	AWUM	NART	RFLM	TXFP	SLEEP	INRQ

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	DBF	RW	Debug whether to disable CAN bus from working 1: During debugging, the CAN transceiver is prohibited, but the control and read and write operations of the receiving FIFO are normal; 0: When debugging, the CAN controller works normally.	1
15	RST	RW1	CAN controller software reset request. Writing 0 to this bit is invalid 1: Reset the CAN controller. After reset, the controller enters sleep mode, and then the hardware automatically clears it to 0; 0: The CAN controller is in normal state.	0
[14:8]	Reserved	RO	Reserved	0
7	TTCM	RW	Whether to allow time trigger mode 1: Enable time trigger mode; 0: Disable time trigger mode. The time trigger mode is mainly used with the TTCAN protocol.	0
6	ABOM	RW	Offline automatic exit control 1: The hardware detects 11 consecutive implicit bits 128 times and automatically exits the offline state; 0: The INRQ bit of the software operation register CAN_CTLR is required to be set to 1 and then cleared to 0. After 11 consecutive implicit bits are detected 128 times, the offline state is exited.	0
5	AWUM	RW	CAN controller automatic wake-up enable 1: When a message is detected, the hardware automatically wakes up, and the SLEEP and SLAK bits of the register CAN_STATR are automatically cleared to 0; 0: Software operation is required to clear the SLEEP bit of the CAN_CTLR register to wake up the CAN controller.	0
4	NART	RW	The automatic packet retransmission function is disabled	0

			1: No matter whether the transmission is successful or not, the message can only be sent once; 0: The CAN controller keeps retransmitting until the transmission is successful.	
3	RFLM	RW	Receive FIFO message lock mode enable. 1: When the received FIFO overflows, the received mailbox message is not read, and the mailbox is not released, the newly received message is discarded. 0: When the received FIFO overflows, the received mailbox message is not read, and when the mailbox is not released, the newly received message will overwrite the original message. *Note: this bit is only used for traditional CAN.*	0
2	TXFP	RW	Transmit mailbox priority method selection 1: The priority is determined by the order in which the requests are sent; 0: The priority is determined by the message identifier.	0
1	SLEEP	RW	Sleep mode request 1: Set to 1 to request the CAN controller to enter sleep mode. After the current activity is completed, the controller enters sleep mode. If the AWUM bit is set to 1, the controller clears the SLEEP bit to 0 when a message is received; 0: After the software clears to 0, the controller exits the sleep mode.	1
0	INRQ	RW	Initialize mode request 1: Set to 1 to request the CAN controller to enter the initialization mode. After the current activity is completed, the controller enters the initialization mode, and the hardware sets the INAK bit of the register CAN_STATR to 1; 0: Set to 0 to request the CAN controller to exit the initialization mode and enter the normal mode, and the hardware clears the INAK bit of the CAN_STATR register to 0.	0

25.7.2 CAN1 Main Status Register (CAN1_STATR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				RX	SAMP	RXM	TXM	Reserved			SLAKI	WKUI	ERRI	SLAK	INAK

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
11	RX	RO	Current actual level of CAN controller Rx pin.	1
10	SAMP	RO	Level of a receive bit on the CAN controller RX pin	1
9	RXM	RO	Rx mode query 1: CAN controller is in Rx mode currently; 0: CAN controller is not in Rx mode currently.	0
8	TXM	RO	Tx mode query 1: CAN controller is in Tx mode currently; 0: CAN controller is not in Tx mode currently.	0
[7:5]	Reserved	RO	Reserved	0
4	SLAKI	RW1	Sleep interrupt enable, that is, interrupt generation flag when the SLKIE bit in the CAN_INTENR register is set to 1, write 1 to clear it, and writing 0 is invalid. 1: When entering sleep mode, an interrupt is generated and the hardware is set to 1; 0: When exiting sleep mode, it can be cleared by hardware or by software.	0
3	WKUI	RW1	Wake-up interrupt flag. When the WKUI bit in the CAN_INTENR register is set to 1, if the SOF bit is detected when the CAN controller is in sleep mode, the hardware will set it to 1. Set to 1 by software to clear to 0, and set to 0 is invalid.	0
2	ERRI	RW1	Error interrupt. When the ERRIE bit in the CAN_INTENR register is set to 1, an error and status change interrupt is generated. This bit is set to 1 and cleared to 0 by software, and set to 0 is invalid.	0
1	SLAK	RO	Sleep mode indication. 1: CAN controller is in sleep mode; 0: CAN controller is not in sleep mode.	1
0	INAK	RO	Initialization mode indication. 1: CAN controller is in initialization mode; 0: CAN controller is not in initialization mode.	0

25.7.3 CAN1 Transmit Status Register (CAN1_TSTATR)

Offset Address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOW2	LOW1	LOW0	TME2	TME1	TME0	CODE[1:0]		ABRQ2	Reserved			TERR2	ALST2	TXOK2	RQCP2

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ABRQ1	Reserved			TERR1	ALST1	TXOK1	RQCP1	ABRQ0	Reserved			TERR0	ALST0	TXOK0	RQCP0

Bit	Name	Access	Description	Reset value
31	LOW2	RO	Lowest priority flag for Tx mailbox2 1: The priority of Tx mailbox2 is the lowest; 0: The priority of Tx mailbox2 is not the lowest.	0
30	LOW1	RO	Lowest priority flag for Tx mailbox1 1: The priority of Tx mailbox1 is the lowest; 0: The priority of Tx mailbox1 is not the lowest.	0
29	LOW0	RO	Lowest priority flag for Tx mailbox0 1: The priority of Tx mailbox0 is the lowest; 0: The priority of Tx mailbox0 is not the lowest.	0
28	TME2	RO	Indicates the empty flag bit of sending mailbox 2. 1: Indicates that mailbox 2 does not have a message waiting to be sent. 0: Indicates that mailbox 2 is waiting to send a message.	1
27	TME1	RO	Indicates the empty flag bit of sending mailbox 1. 1: Indicates that mailbox 1 does not have a message waiting to be sent. 0: Indicates that mailbox 1 is waiting to send a message.	1
26	TME0	RO	Indicates the empty flag bit of sending mailbox 0. 1: Indicates that mailbox 0 does not have a message waiting to be sent. 0: Indicates that mailbox 0 is waiting to send a message.	1
[25:24]	CODE[1:0]	RO	Mailbox number. When more than one mailbox is empty, the next mailbox number is empty; when the mailbox is empty, it means the mailbox number with the lowest priority.	0
23	ABRQ2	RW1	Transmit a send abort request for mailbox 2. Software setting 1 can abort the transmitting request of mailbox 2, and the hardware is cleared when transmitting messages are cleared. If mailbox 2 is empty, software setting 1 is invalid.	0
[22:20]	Reserved	RO	Reserved	0
19	TERR2	RW1	Tx mailbox2 transmit error. When Tx mailbox2 fails, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0
18	ALST2	RW1	Tx mailbox 2 arbitration failure flag. When the Tx mailbox 2 has a low arbitration priority and fails to send, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0
17	TXOK2	RW1	Tx mailbox 2 transmit OK.	0

			1: The last transmission was successful; 0: The last transmission failed. Set to 1 by software to clear, software write 0 is invalid.	
16	RQCP2	RW1	Tx mailbox 2 request completion flag, this bit is automatically set to 1 when the sending or aborting request of sending mailbox 2 is completed. Set to 1 by software to clear, software write 0 is invalid.	0
15	ABRQ1	RW1	Tx a transmit abort request for mailbox 1. When the software is set to 1, the transmitting request of mailbox 1 can be aborted and the hardware will clear 0 when the message is cleared. The software write 0 is invalid.	0
[14:12]	Reserved	RO	Reserved	0
11	TERR1	RW1	Tx mailbox1 transmit error. When Tx mailbox1 fails, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0
10	ALST1	RW1	Tx mailbox1 arbitration failure flag. When the Tx mailbox1 has a low arbitration priority and fails to send, this bit is automatically set to 1.	0
9	TXOK1	RW1	Tx mailbox1 transmit OK. 1: The last transmission was successful; 0: The last transmission failed. Set to 1 by software to clear, software write 0 is invalid.	0
8	RQCP1	RW1	Tx mailbox1 request completion flag, this bit is automatically set to 1 when the sending or aborting request of sending mailbox 1 is completed. Set to 1 by software to clear, software write 0 is invalid.	0
7	ABRQ0	RW1	Tx a transmit abort request for mailbox 0. Set to 1 by software to abort the transmitting request of mailbox 0, and reset to 0 by hardware when the sent message is cleared. Software write 0 is invalid.	0
[6:4]	Reserved	RO	Reserved	0
3	TERR0	RW1	Tx mailbox0 transmit error. When Tx mailbox0 fails, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0
2	ALST0	RW1	Tx mailbox0 arbitration failure flag. When Tx mailbox0 has a low arbitration priority and fails to send, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0
1	TXOK0	RW1	Tx mailbox0 transmit OK.	0

			1: The last transmission was successful; 0: The last transmission failed. Set to 1 by software to clear, software write 0 is invalid.	
0	RQCP0	RW1	Tx mailbox0 request completion flag. When the send or abort request of Tx mailbox0 is completed, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0

25.7.4 CAN1 Receive FIFO 0 Status Register (CAN1_RFIFO0)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RFOM0	FOVR0	FULL0	Reserved	FMP0[1:0]	

Bit	Name	Access	Description	Reset value
[31:6]	Reserved	RO	Reserved	0
5	RFOM0	RW1	When the software sets this bit to 1, it releases the current mailbox message of the receiving FIFO_0, and automatically clears it to 0 after the release, and software write 0 is invalid.	0
4	FOVR0	RW1	Receive FIFO_0 overflow flag. When there are 3 messages in FIFO_0, a new message is received, and the hardware is set to 1. This bit needs software to be set to 1 and cleared to 0, and software write 0 is invalid.	0
3	FULL0	RW1	Receive FIFO_0 full flag. Set by hardware when there are 3 messages in FIFO_0. This bit needs software to be set to 1 and cleared to 0, and software write 0 is invalid.	0
2	Reserved	RO	Reserved	0
[1:0]	FMP0[1:0]	RO	Number of received FIFO_0 messages.	0

25.7.5 CAN1 Receive FIFO 1 Status Register (CAN1_RFIFO1)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RFOM1	FOVR1	FULL1	Reserved	FMP1[1:0]	

Bit	Name	Access	Description	Reset value

[31:6]	Reserved	RO	Reserved	0
5	RFOM1	RW1	When the software sets this bit to 1, it releases the current mailbox message of the receiving FIFO_1, and automatically clears it to 0 after the release, and software write 0 is invalid.	0
4	FOVR1	RW1	Receive FIFO_1 overflow flag. When there are 3 messages in FIFO_1, a new message is received, and the hardware is set to 1. This bit needs software to be set to 1 and cleared to 0, and software write 0 is invalid.	0
3	FULL1	RW1	Receive FIFO_1 full flag. Set by hardware when there are 3 messages in FIFO_1. This bit needs software to be set to 1 and cleared to 0, and software write 0 is invalid.	0
2	Reserved	RF	Reserved	0
[1:0]	FMP1[1:0]	RO	Number of received FIFO_1 messages.	0

25.7.6 CAN1 Interrupt Enable Register (CAN1_INTENR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														SLKIE	WKUIE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ERRIE	Reserved			LECIE	BOFIE	EPVIE	EWGIE	Reserved	FOVIE1	FFIE1	FMPIE1	FOVIE0	FFIE0	FMPIE0	TMEIE

Bit	Name	Access	Description	Reset value
[31:18]	Reserved	RO	Reserved	0
17	SLKIE	RW	Sleep interrupt enable. 1: When entering the sleep state, an interrupt is generated; 0: No interrupt is generated when entering sleep state.	0
16	WKUIE	RW	Wake-up interrupt enable. 1: When the CAN controller is woken up, an interrupt is generated; 0: No interrupt is generated when the CAN controller wakes up.	0
15	ERRIE	RW	Error interrupt enable, CAN error interrupt always enable. 1: When the CAN controller generates an error, an interrupt is generated; 0: No interrupt is generated when the CAN controller generates an error.	0
[14:12]	Reserved	RF	Reserved.	0

11	LECIE	RW	Last error number interrupt enable. 1: When an error is detected, the hardware updates LEC[2:0], updates the ERRI bit to 1, and triggers an error interrupt; 0: When an error is detected, the hardware updates LEC[2:0], does not update the ERRI bit, and does not trigger an error interrupt.	0
10	BOFIE	RW	Offline interrupt enable. 1: When entering the offline state, update the ERRI bit to 1, triggering an error interrupt; 0: When entering the offline state, the ERRI bit will not be updated, and the error interrupt will not be triggered.	0
9	EPVIE	RW	Error passive interrupt enable. 1: When entering the error passive state, update the ERRI bit to 1, triggering an error interrupt; 0: When entering the error passive state, the ERRI bit is not updated and the error interrupt is not triggered.	0
8	EWGIE	RW	Error warning interrupt enable. 1: When the number of errors reaches the warning threshold, update the ERRI bit to 1, triggering an error interrupt; 0: When the number of errors reaches the warning threshold, the ERRI bit will not be updated, and the error interrupt will not be triggered.	0
7	Reserved	RF	Reserved	0
6	FOVIE1	RW	Receive FIFO_1 overflow interrupt enable. 1: When FIFO_1 overflows, trigger FIFO_1 interrupt; 0: When FIFO_1 overflows, not trigger FIFO_1 interrupt.	0
5	FFIE1	RW	Receive FIFO_1 full interrupt enable. 1: When FIFO_1 is full, trigger FIFO_1 interrupt; 0: When FIFO_1 is full, not trigger FIFO_1 interrupt.	0
3	FOVIE0	RW	Receive FIFO_0 overflow interrupt enable. 1: When FIFO_0 overflows, trigger FIFO_0 interrupt; 0: When FIFO_0 overflows, not trigger FIFO_0 interrupt.	0
2	FFIE0	RW	Receive FIFO_0 full interrupt enable. 1: When FIFO_0 is full, trigger FIFO_0 interrupt; 0: When FIFO_0 is full, not trigger FIFO_0 interrupt.	0
1	FMPIE0	RW	Receive FIFO_0 message registration interrupt	0

			enable. 1: When FIFO_0 updates the FMP bit and is not 0, trigger FIFO_0 interrupt; 0: When FIFO_0 updates the FMP bit, and it is not 0, the FIFO_0 interrupt is not triggered.	
0	TMEIE	RW	Tx mailbox empty interrupt. 1: When the Tx mailbox is empty, an interrupt is generated; 0: No interrupt is generated when the Tx mailbox is empty.	0

25.7.7 CAN1 Error Status Register (CAN1_ERRSR)

Offset Address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REC[7:0]								TEC[7:0]							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									LEC[2:0]			Reserved	BOFF	EPVF	EWGF

Bit	Name	Access	Description	Reset value
[31:24]	REC[7:0]	RO	Receive error counter. When the CAN receives an error, according to the error condition, the counter is incremented by 1 or 8; after successful reception, the counter is decreased by 1 or set to 120 (the error count value is greater than 127). When the counter value exceeds 127, the CAN enters the error passive state.	0
[23:16]	TEC[7:0]	RO	Transmit error counter. When the CAN sends an error, according to the error condition, the counter is incremented by 1 or 8; after the transmission is successful, the counter is decremented by 1 or set to 120 (the error count value is greater than 127). When the counter value exceeds 127, the CAN enters the error passive state.	0
[15:7]	Reserved	RO	Reserved	0
[6:4]	LEC{2:0}	RW	Last error code. When an error is detected on the CAN bus, the controller sets this bit to 000b, depending on the nature of the error; when messages are transmitted and received correctly, it is set to 000b. 000: no error; 001: Bit stuffing error; 010: FORM format error; 011: ACK confirmation error; 100: recessive bit error;	0

			101: Dominant bit error; 110: CRC error; 111: Software settings. Usually when the application software reads the error, the code name is set to 111b, and the code name update can be detected.	
3	Reserved	RO	Reserved	0
2	BOFF	RO	Offline status flag. When the CAN controller enters the offline state, the hardware automatically sets it to 1; when it exits the offline state, the hardware automatically clears it to 0.	0
1	EPVF	RO	Error passive flag. When the transceiver error counter reaches the error passive threshold, that is, greater than 127, the hardware is set to 1.	0
0	EWGF	RO	Error warning flag bit. When the sending and receiving error counter reaches the warning threshold, that is, greater than or equal to 96, the hardware is set to 1.	0

25.7.8 CAN1 Bit Timing Register (CAN1_BTIMR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SILM	LBKM	Reserved		SJW[3:0]				TS2[3:0]				TS1[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BTR_TS1_T				Reserved		BRP[9:0]									

Bit	Name	Access	Description	Reset value
31	SILM	RW	Silent mode setting. 1: Enter silent mode; 0: Exit silent mode.	0
30	LBKM	RW	Loopback mode setting. 1: Enter loopback mode; 0: Exit loopback mode.	0
[29:28]	Reserved	RO	Reserved	0
[27:24]	SJW[3:0]	RW	Defines the resync jump width setting value. When implementing resynchronization, the upper limit of the minimum number of time units that can be extended and reduced in the bit, the actual value is (SJW[1:0]+1), and the range can be set to 1 to 4 minimum time units.	0001b
[23:20]	TS2[3:0]	RW	Time period 2 set value. It defines how many minimum time units are	0010b

			occupied by time period 2, and the actual value is (TS2[1:0]+1).	
[19:16]	TS1[3:0]	RW	Time period 1 set value. It defines how many minimum time units are occupied by time period 1, and the actual value is (TS1[1:0]+1).	0011b
[15:12]	BTR_TS1_T	RW	For legacy CAN, CLAS_LONG_TS1=0, then TS1 is TS[3:0] (4bit); CLAS_LONG_TS1=1, then TS1 is (TS[1:0]<<4)+BTR_TS1_T[15:12] (6bit).	0
[11:10]	Reserved	RO	Reserved	0
[9:0]	BRP[9:0]	RW	Minimum time unit length setting value Tq = (BRP[9:0]+1) ×t_{HCLKS}	0

25.7.9 CAN1 Time Trigger Control Register (CAN1_TTCTLR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														MODE	TIMRST

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIMCMV [15:0]															

Bit	Name	Access	Description	Reset value
[31:18]	Reserved	RO	Reserved	0
17	MODE	RW	Time-triggered mode selection. 1: Enhanced mode; 0: Default mode.	0
16	TIMRST	WZ	Internal counter reset control. Write 1 to reset the internal counter, the hardware will automatically clear 0	0
[15:0]	TIMCMV[15:0]	RW	Internal counter count end value	ffffh

25.7.10 CAN1 Time Trigger Count Value Register (CAN1_TTCNT)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIMCNT[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	TIMCNT[15:0]	RW	Time-triggered count value	0

25.7.11 CAN1 Offline Recovery Error Counter (CAN1_TERR_CNT)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							TX_ERR_CNT								

Bit	Name	Access	Description	Reset value
[31:9]	Reserved	RO	Reserved	0
[8:0]	TX_ERR_CNT	RW	Currently, the error count value is recovered offline. If you modify the count value, you can recover it immediately from offline.	0

25.7.12 CAN1 Tx Mailbox Identifier Register (CAN1_TXMIyR) (y=0/1/2)

Offset address: 0x180,0x190,0x1A0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STID[10:0]/EXID[28:18]											EXID[17:13]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXID[12:0]													IDE	RTR	TXRQ

Bit	Name	Access	Description	Reset value
[31:21]	STID[10:0] /EXID[28:18]	RW	The upper 11 bits of a standard or extended identifier.	x
[20:3]	EXID[17:0]	RW	The lower 18 bits of the extended identifier.	x
2	IDE	RW	Identifier selection flag. 1: Extended identifier; 0: Standard identifiers.	x
1	RTR	RW	Remote frame selection flag. 1: Remote frame; 0: Data frame.	x
0	TXRQ	RW	Data transmission request flag. When the software is set to 1, the data in the mailbox is requested to be sent. When the mailbox is empty after sending, the hardware is cleared to 0.	0

25.7.13 CAN1 Tx Mailbox Data Length and Timestamp Register (CAN1_TXMDTyR) (y=0/1/2)

Offset address: 0x184,0x194,0x1A4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TIME[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Reserved	TGT	Reserved	DLC[3:0]

Bit	Name	Access	Description	Reset value
[31:16]	TIME[15:0]	RW	The 16-bit timer value used to send the message SOF time.	x
[15:9]	Reserved	RO	Reserved	0
8	TGT	RW	Message timestamp transmission selection flag. This bit is valid when TTCM is set to 1 and the message length is 8. 1: Transmit timestamp, the value is the immediate value of TIME[15:0], replacing the last 2 bytes of the 8-byte message; 0: No timestamp is transmitted.	x
[7:4]	Reserved	RO	Reserved	0
[3:0]	DLC[3:0]	RW	The data length of the data frame or the data length of the remote frame request. The data length can be set from 0 to 64.	0

25.7.14 CAN1 Tx Mailbox Data Low Register (CAN1_TXMDLyR) (y=0/1/2)

Offset address: 0x188,0x198,0x1A8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA3[7:0]								DATA2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA1[7:0]								DATA0[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	DATA3[7:0]	RW	Tx the content of data byte 3.	x
[23:16]	DATA2[7:0]	RW	Tx the content of data byte 2.	x
[15:8]	DATA1[7:0]	RW	Tx the content of data byte 1.	x
[7:0]	DATA0[7:0]	RW	Tx the contents of data byte 0.	x

25.7.15 CAN1 Tx Mailbox Data High Register (CAN1_TXMDHyR) (y=0/1/2)

Offset address: 0x18C,0x19C,0x1AC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA7[7:0]								DATA6[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA5[7:0]								DATA4[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	DATA7[7:0]	RW	Tx the content of data byte 7.	x
[23:16]	DATA6[7:0]	RW	Tx the content of data byte 6.	x
[15:8]	DATA5[7:0]	RW	Tx the content of data byte 5.	x
[7:0]	DATA4[7:0]	RW	Tx the contents of data byte 4.	x

25.7.16 CAN1 Rx Mailbox Identifier Register (CAN1_RXMIyR) (y=0/1)

Offset address: 0x1B0, 0x1C0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STID[10:0]/EXID[28:18]											EXID[17:13]				

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXID[12:0]													IDE	RTR	FDF

Bit	Name	Access	Description	Reset value
[31:21]	STID[10:0] /EXIDH[28:18]	RO	The upper 11 bits of a standard or extended identifier.	x
[20:3]	EXIDL[17:0]	RO	The lower 18 bits of the extended identifier.	x
2	IDE	RO	Identifier selection flag. 1: Select extended identifier; 0: Use standard identifiers.	x
1	RTR	RO	Remote frame selection flag. 1: Remote frame; 0: Data frame.	x
0	FDF	RO	Reserved	0

25.7.17 CAN1 Rx Mailbox Data Length and Timestamp Register (CAN1_RXMDTyR) (y=0/1)

Offset address: 0x1B4, 0x1C4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TIME[15:0]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FMI[7:0]								Reserved	RES	ESI	BRS	DLC[3:0]			

Bit	Name	Access	Description	Reset value
[31:16]	TIME[15:0]	RO	16-bit timer value used to receive the SOF time of the message.	0
[15:7]	FMI[7:0]	RO	The filter number to which the message matches.	x
6	RES	RO	The RES bit of the currently received frame	0
5	ESI	RO	The ESI bit of the currently received frame	0
4	BRS	RO	The BRS bit of the currently received frame	0
[3:0]	DLC[3:0]	RO	Received message data length. DLC=0-8: Indicating data frame lengths of 0 through 8 and 0 for remote frames.	x

25.7.18 CAN1 Rx Mailbox Data Low Register (CAN1_RXMDLyR) (y=0/1)

Offset address: 0x1B8, 0x1C8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA3[7:0]								DATA2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA1[7:0]								DATA0[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	DATA3[7:0]	RO	The data byte 3 that receives the message.	x
[23:16]	DATA2[7:0]	RO	The data byte 2 that receives the message.	x
[15:8]	DATA1[7:0]	RO	The data byte 1 that receives the message.	x
[7:0]	DATA0[7:0]	RO	The data byte 0 that receives the message.	x

25.7.19 CAN1 Rx Mailbox Data High Register (CAN1_RXMDHyR) (y=0/1)

Offset address: 0x1BC, 0x1CC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA7[7:0]								DATA6[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA5[7:0]								DATA4[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	DATA7[7:0]	RO	The data byte 7 that receives the message.	x
[23:16]	DATA6[7:0]	RO	The data byte 6 that receives the message.	x
[15:8]	DATA5[7:0]	RO	The data byte 5 that receives the message.	x
[7:0]	DATA4[7:0]	RO	The data byte 4 that receives the message.	x

25.7.20 CAN1 Filter Main Control Register (CAN1_FCTLR)

Offset address: 0x200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															FINIT

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0x2A1C
[15:1]	Reserved	RO	Reserved	0
0	FINIT	RW	Filter initialization mode enable flag. 1: Filter group is in initialization mode; 0: Filter group is in normal mode.	1

25.7.21 CAN1 Filter Mode Register (CAN1_FMCFGR)

Offset address: 0x204

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		FBM13	FBM12	FBM11	FBM10	FBM9	FBM8	FBM7	FBM6	FBM5	FBM4	FBM3	FBM2	FBM1	FBM0

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:0]	FBMx	RW	The FINIT bit must be set to 1 to write to the operating mode control bit of filter group x. 1: Filter group x registers operate in identifier list mode; 0: Filter group x registers operate in mask bit mode.	0

25.7.22 CAN1 Filter Bit Width Register (CAN1_FSCFGR)

Offset address: 0x20C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		FSC13	FSC12	FSC11	FSC10	FSC9	FSC8	FSC7	FSC6	FSC5	FSC4	FSC3	FSC2	FSC1	FSC0

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:0]	FSCx	RW	Bit width control bit for filter group x; FINIT must be 1 to write. 1: Filter group x registers are a single 32-bit register; 0: Filter group x registers are two 16-bit registers.	0

25.7.23 CAN1 Filter FIFO Association Register (CAN1_FAFIFOR)

Offset address: 0x214

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		FFA13	FFA12	FFA11	FFA10	FFA9	FFA8	FFA7	FFA6	FFA5	FFA4	FFA3	FFA2	FFA1	FFA0

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0

[13:0]	FFAx	RW	The associated FIFO control bit for filter group x must be FINIT=1 to enable writing. 1: Filter group x is associated with FIFO_1; 0: Filter group x is associated with FIFO_0.	0

25.7.24 CAN1 Filter Activation Register (CAN1_FWR)

Offset address: 0x21C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		FACT13	FACT12	FACT11	FACT10	FACT9	FACT8	FACT7	FACT6	FACT5	FACT4	FACT3	FACT2	FACT1	FACT0

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:0]	FACTx	RW	Activation control bit for filter group x. 1: Filter group x activated; 0: Filter group x disabled.	0

25.7.25 CAN1 Filter Group Register (CAN1_FiRx) (i=0-13, x=1/2)

Offset address: 0x240-0x2AC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FB31	FB30	FB29	FB28	FB27	FB26	FB25	FB24	FB23	FB22	FB21	FB20	FB19	FB18	FB17	FB16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FB15	FB14	FB13	FB12	FB11	FB10	FB9	FB8	FB7	FB6	FB5	FB4	FB3	FB2	FB1	FB0

Bit	Name	Access	Description	Reset value
[31:0]	FB	RW	The FINIT flag in the filter group register must be set to 1 before writing. Identifier mode: 1: The corresponding bit expects a low level; 0: The corresponding bit expects a high level. Mask bit mode: 1: Must match the corresponding identifier register bit; 0: Does not need to match the corresponding identifier register bit.	0

Chapter 26 Flexible Static Memory Controller (FSMC)

The flexible static memory controller (FSMC) supports a variety of static memory types and rich storage operation methods, and can expand different types of large-capacity static memory according to the needs of system applications.

26.1 Main Features

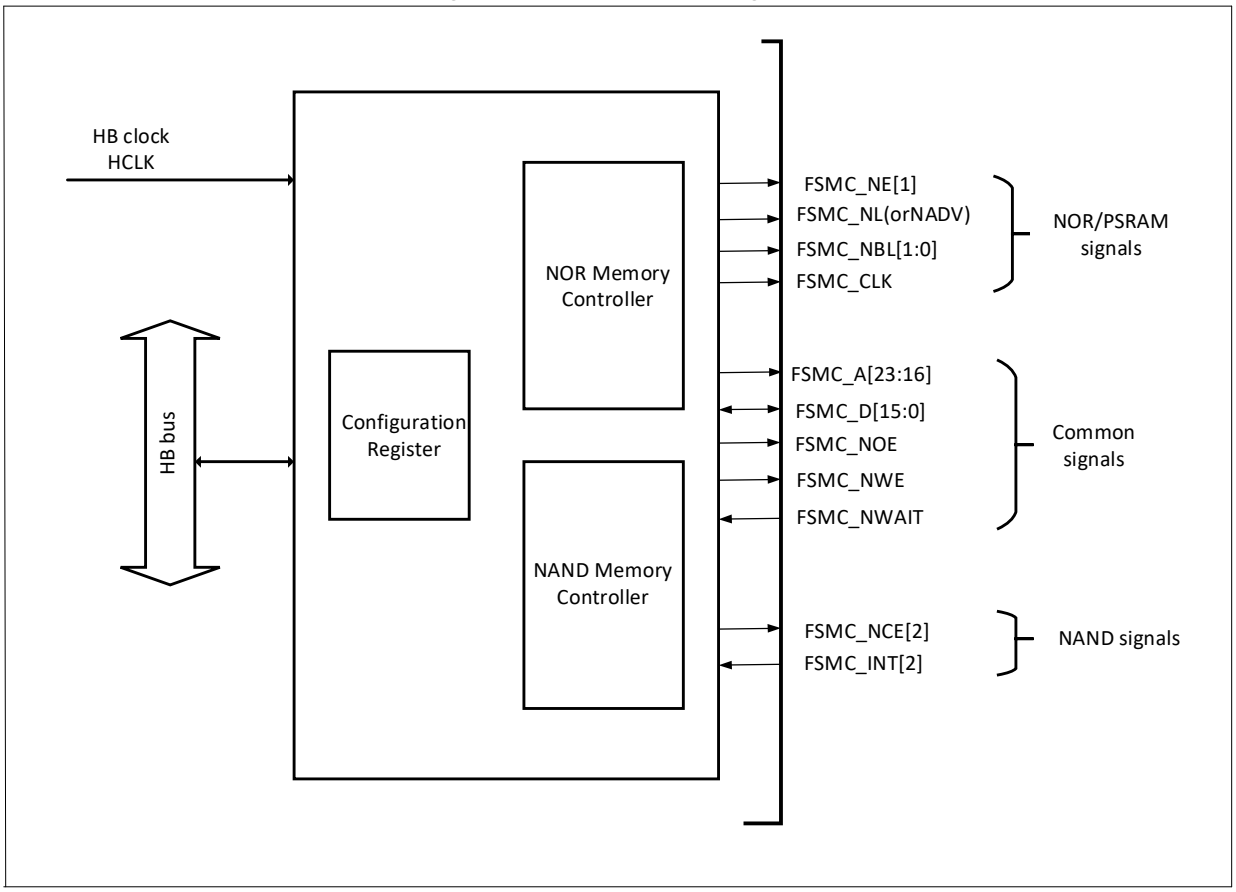
- Support to operate SRAM, ROM, NOR FLASH and PSRAM
- Support NAND FLASH operation, and built-in hardware ECC, can detect up to 8KByte data
- Support for synchronous device operation, such as PSRAM
- Support 8bit or 16bit data bus width
- Timing signals can be programmed by software

26.2 Functional Description

26.2.1 Module Structure

FSMC mainly includes HB bus interface, NOR memory controller, NAND memory controller and external device interface.

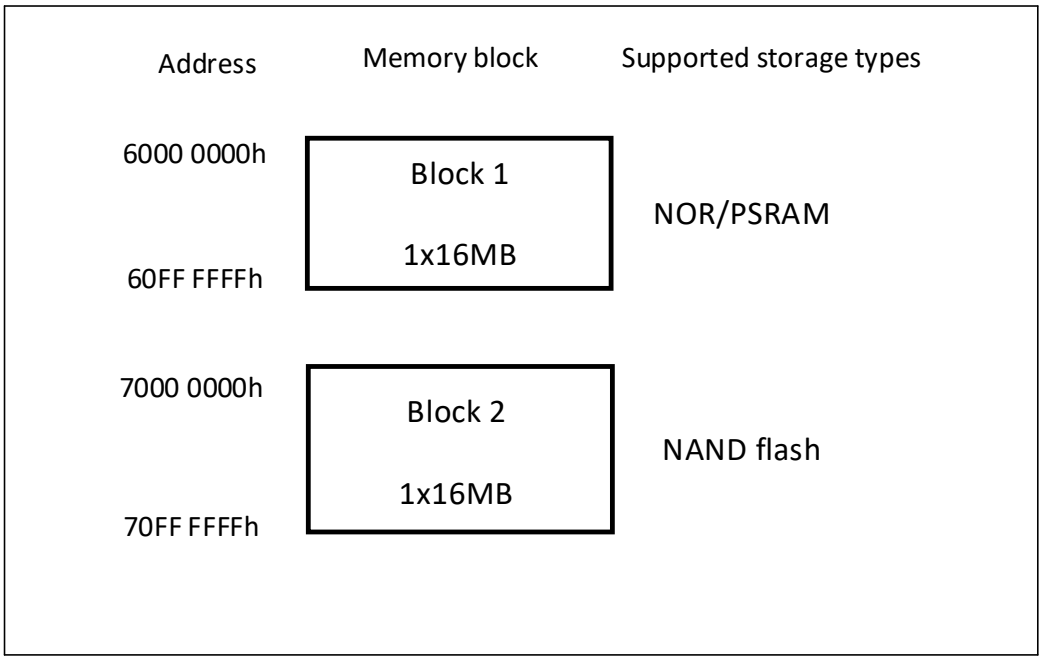
Figure 26-1 FSMC block diagram



26.2.2 External Device Address Mapping

The FSMC divides the memory block into 2 memory blocks with a fixed size of 16MByte according to the address line, as shown in Figure 26-2.

Figure 26-2 FSMC memory block



26.2.2.1 NOR/PSRAM Address Mapping

Table 26-1 External memory address

Data width	Address lines connected to memory	Maximum access memory space (bit)
8bit	HADDR[23:0] is correspondingly connected to FSMC_A[23:0]	16MB×8=128Mbit
16bit	HADDR[23:1] is correspondingly connected to FSMC_A[22:0], HADDR[0] is not connected	16MB/2×16=128Mbit

Note: Those with 100 pins support FSMC, and only support address and data line multiplexing mode.

NOR FLASH and PSRAM support unaligned access. For asynchronous mode, each operation requires an accurate address; for synchronous mode, only single address signal is issued, and then batches of data are sequentially processed through CLK. For NOR FLASH that supports unaligned batch access, the unaligned access mode of the memory can be set to the same mode as HB. If it cannot be set, the unaligned access mode is disabled, and the unaligned access request is divided into 2 consecutive access operations.

26.2.2.2 NAND Address Mapping

NAND FLASH image and timing registers, see Table 26-2.

Table 26-2 Memory map and timing registers

Start address	End address	FSMC memory block	Storage	Timing register
0x78000000	0x78FFFFFF	Block 2-NAND	Attribute	FSMC_PATT2(0x6C)

0x70000000	0x70FFFFFF	FLASH	General	FSMC_PMEM2(0x68)

The general and attribute space can be divided into address area (second 128KB area), command area (second 64KB area) and data area (first 64KB area) in the lower 256KB, see Table 26-3.
The software accesses the specific process of NAND FLASH by operating these 3 areas:

- Send read and write operation commands to NAND FLASH: the software can operate any address in the command area to send commands;
- Send the address to be operated to the NAND FLASH: the software can operate any address in the address area to send commands;
- Read and write data from NAND FLASH: software can operate any address in the data area to write or read data;

Table 26-3 NAND memory block selection

Area	HADDR[17:16]	Range
Address area	1X	0x020000 to 0x03FFFF
Command area	01	0x010000 to 0x01FFFF
Data area	00	0x000000 to 0x00FFFF

26.2.3 NOR/PSRAM Controller

FSMC supports 8bit, 16bit and 32bit asynchronous operation SRAM and ROM, supports asynchronous mode and burst mode operation PSRAM, supports asynchronous mode and burst mode operation NOR FLASH. The output signals of all controllers change on the rising edge of the internal clock HCLK. For synchronous write mode (PSRAM), the output data changes on the falling edge of the memory clock (CLK). For details, refer to the synchronous transfer and asynchronous transfer timing diagrams. The read and write parameters of the memory can be configured by software, see Table 26-4.

Table 26-4 Software controllable NOR/PSRAM read and write parameters

Parameters	Read and write	Parameter value range
Address setup time	Asynchronous	1<< T <<16 (HB HCLK)
Address hold time	Asynchronous	1<< T <<16 (HB HCLK)
Data setup time	Asynchronous	2<< T <<256 (HB HCLK)
Bus recovery time	Asynchronous or synchronous read	1<< T <<16 (HB HCLK)
Clock divider	Synchronous	2<< T <<16 (HB HCLK)
Data generation time	Synchronous	2<< T <<17 (Memory CLK)

26.2.3.1 External Memory Multiplexing Interface Signal

For NOR FLASH and PSRAM interfaces, see Table 26-5 and Table 26-6.

Table 26-5 Multiplexed NOR FLASH interface

FSMC Pins	Direction	Description
CLK	Output	Clock line (For synchronous burst mode only)
A[23:16]	Output	Address line
AD[15:0]	Input/Output	16bit address/data line (Multiplexed)
NE[1]	Output	Chip select line

NOE	Output	Output enable
NWE	Output	Write enable
NL(NADV)	Output	Latch Enable
NWAIT	Input	Wait for signal line (Only for NOR FLASH)

Table 26-6 Multiplexed PSRAM interface

FSMC Pins	Direction	Description
CLK	Output	Clock line (For synchronous burst mode only)
A[23:16]	Output	Address line
AD[15:0]	Input / Output	16bit address/data line (Alternate)
NE[1]	Output	Chip select line
NOE	Output	Output enable
NWE	Output	write enable
NL(NADV)	Output	Latch enable
NWAIT	Input	Wait for signal line
NBL[1]	Output	High byte enable (NUB)
NBL[0]	Output	Low byte enable (NLB)

Note: The signal with the prefix 'N' means that it is active low.

26.2.3.2 Supported Memory and Operation Methods

See Table 26-7 for supported memories and operation methods.

Table 26-7 Supported memories and operation methods.

Memory	Mode	HB data width	Memory width	Description
NOR FLASH	Asynchronous read	8	16	
	Asynchronous read	16	16	
	Asynchronous write	16	16	
	Asynchronous read	32	16	Divided into 2 FSMC access
	Asynchronous write	32	16	Divided into 2 FSMC access
	Synchronous read	16	16	
	Synchronous read	32	16	Divided into 2 FSMC access
PSRAM	Asynchronous read	8	16	
	Asynchronous write	8	16	Use byte letter NBL[1:0]
	Asynchronous read	16	16	
	Asynchronous write	16	16	
	Asynchronous read	32	16	Divided into 2 FSMC access
	Asynchronous write	32	16	Divided into 2 FSMC access
	Synchronous read	16	16	
	Synchronous read	32	16	Divided into 2 FSMC access
	Synchronous write	8	16	Use byte letter NBL[1:0]
	Synchronous write	16/32	16	
SRAM and ROM	Asynchronous read	8/16/32	8/16	Use byte letter NBL[1:0]
	Asynchronous write	8/16/32	8/16	Use byte letter NBL[1:0]

26.2.3.3 NOR/PSRAM Timing Diagram of Asynchronous Transfer Address/Data Multiplexing

Note the following for asynchronous static memory (NOR FLASH and PSRAM) operations:

- When extended mode is enabled, read and write operations to memory using modes A, B, C, and D can be mixed.

Figure 26-3 Mode 1 read operation (Multiplexed)

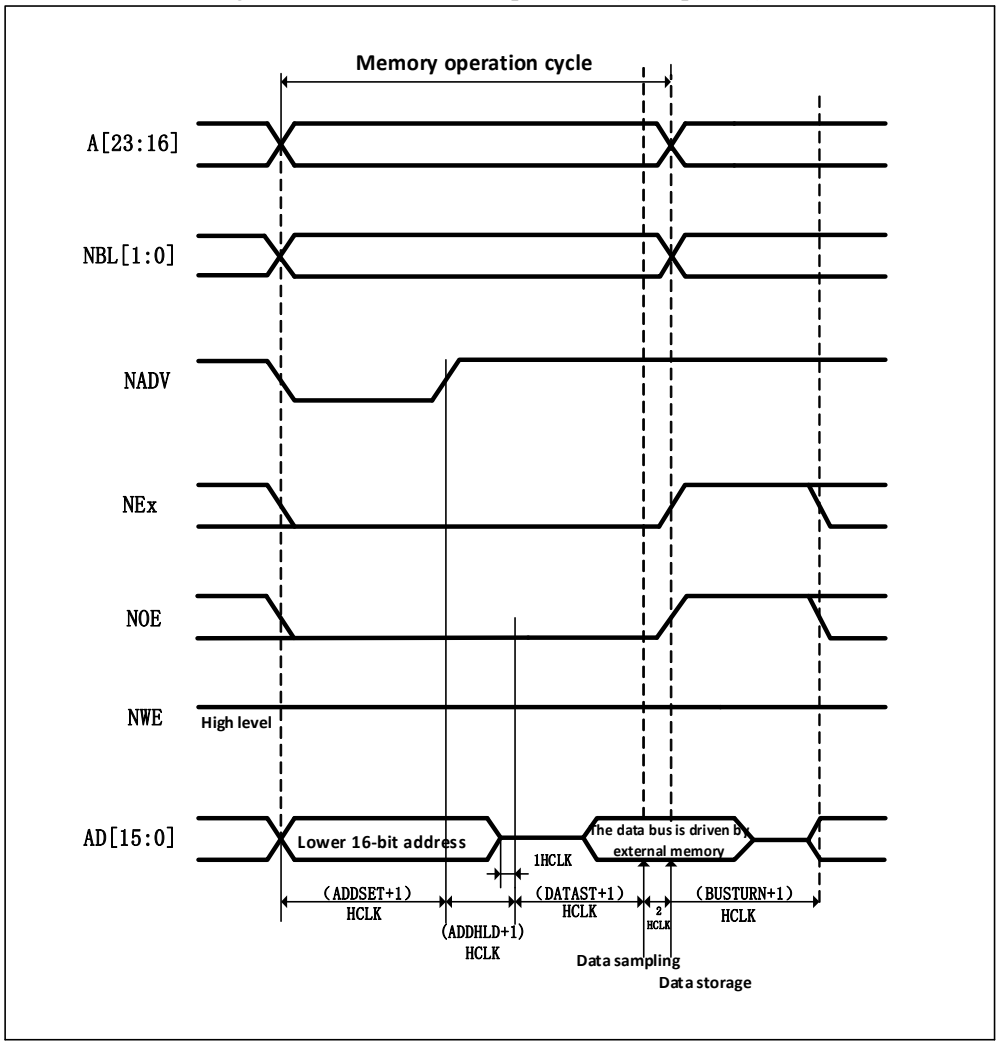
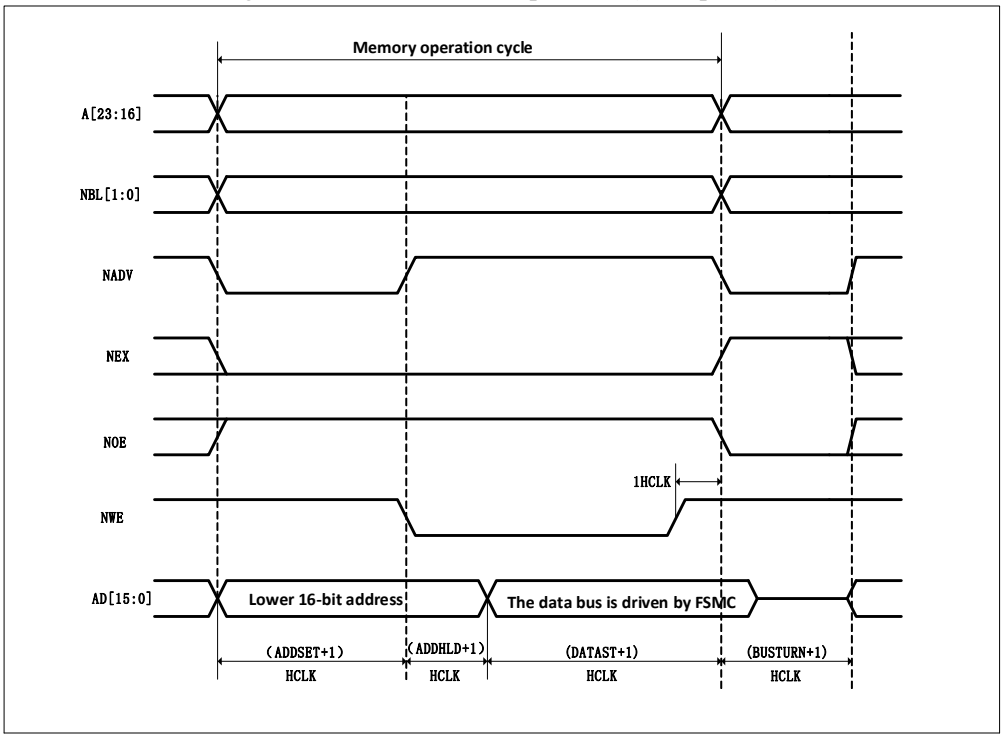


Figure 26-4 Mode 1 write operation (Multiplexed)



Note: The last HCLK cycle of a write operation guarantees the hold time for address and data after the rising edge of NWE. Due to this HCLK cycle, the value of DATAST must be greater than zero (DATAST > 0).

Table 26-8 Mode 1 FSMC_BCR1 bit field

Bitx	Name	Configuration Value
[31:20]	Reserved	0x000
19	CBURSTRW	0x0
[18:16]	Reserved	0x0
15	ASYNCWAIT	It is 1 if the memory supports this function, otherwise it is 0.
14	EXTMOD	0x0
13	WAITEN	0x0
12	WREN	Set as needed
11	WAITCFG	No effect
10	Reserved	0x0
9	WAITPOL	When bit15 is 1, it is meaningful
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	No effect
[5:4]	MWID	Set as needed
[3:2]	MTYP	Set as needed, excluding 0x2 (NOR FLASH)
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-9 Mode 1 FSMC_BTR1 bit field

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0

[29:28]	ACCMOD	No effect
[27:24]	DATLAT	No effect
[23:20]	CLKDIV	No effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. The write operation is (DATAST+1 HCLK), and the read operation is (DATAST+3 HCLK). The minimum value of this bit is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. (ADDSET+1 HCLK).

Figure 26-5 Mode A read operation (Multiplexed)

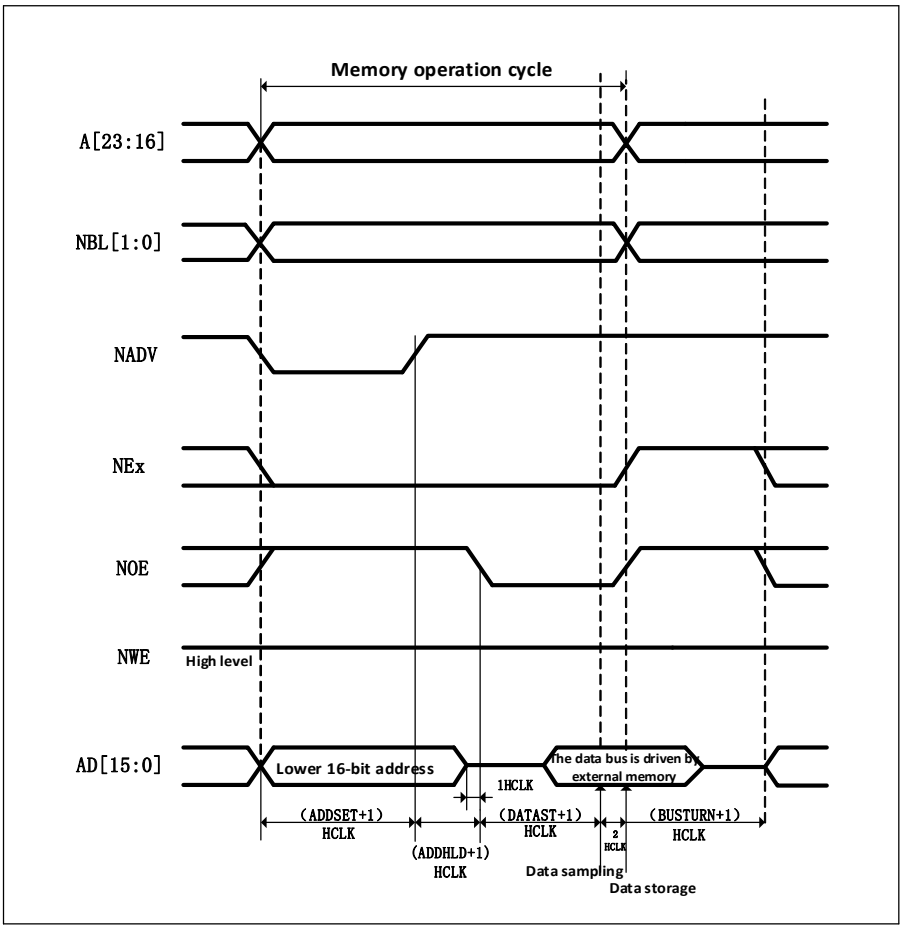


Figure 26-6 Mode A write operation (Multiplexed)

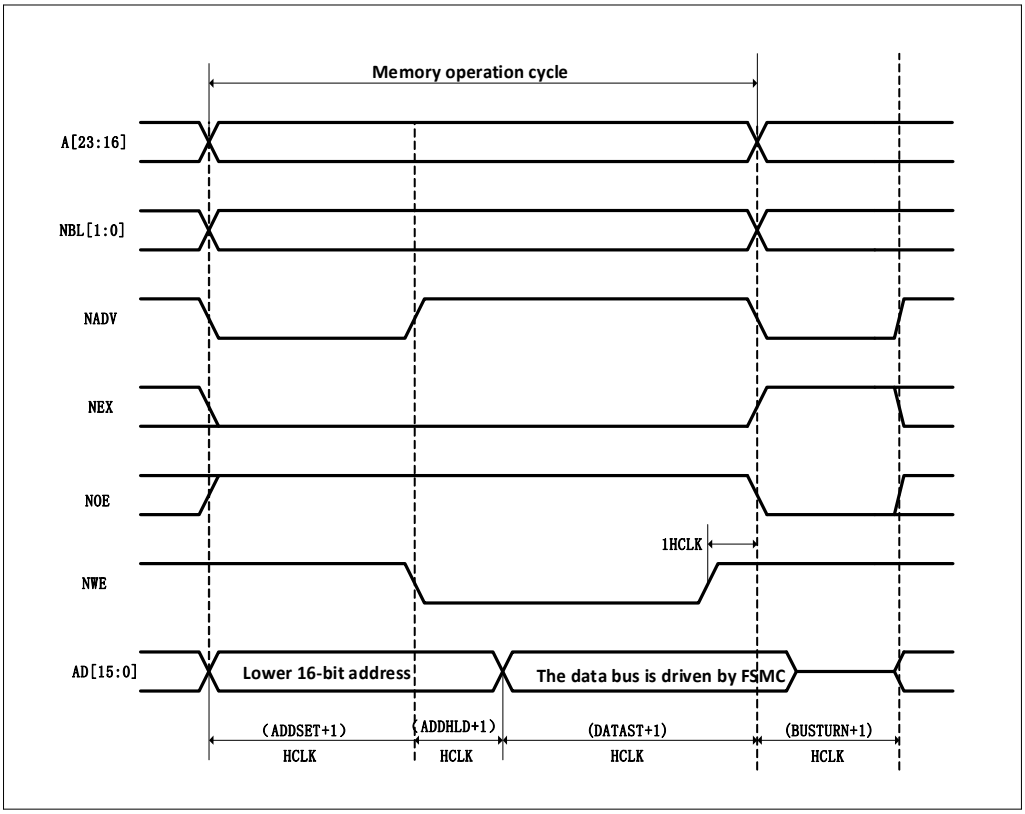


Table 26-10 Mode A FSMC_BCR1 bit field

Bitx	Name	Configuration Value
[31:20]	Reserved	0x000
19	CBURSTRW	0x0
[18:16]	Reserved	0x0
15	ASYNCWAIT	It is 1 if the memory supports this function, otherwise it is 0.
14	EXTMOD	0x1
13	WAITEN	0x0
12	WREN	Set as needed
11	WAITCFG	No effect
10	Reserved	0x0
9	WAITPOL	When bit15 is 1, it is meaningful
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	No effect
[5:4]	MWID	Set as needed
[3:2]	MTYP	Set as needed, excluding 0x2 (NOR FLASH)
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-11 Mode A FSMC_BTR1 bit field

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	No effect
[23:20]	CLKDIV	No effect
[19:16]	BUSTYRN	0x0
[15:8]	DATAST	Data setup time. The read operation is (DATAST+3 HCLK). The minimum value of this bit is 1.
[7:4]	ADDHLD	No effect
[3:0]	ADDSET	Address setup time. The read operation is (ADDSET+1 HCLK).

Table 26-12 Mode A FSMC_BWTR1 bit field

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	No effect
[23:20]	CLKDIV	No effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. The write operation is (DATAST+1 HCLK). The minimum value of this bit is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. The write operation is (ADDSET+1 HCLK).

Figure 26-7 Mode B read operation (multiplexed)

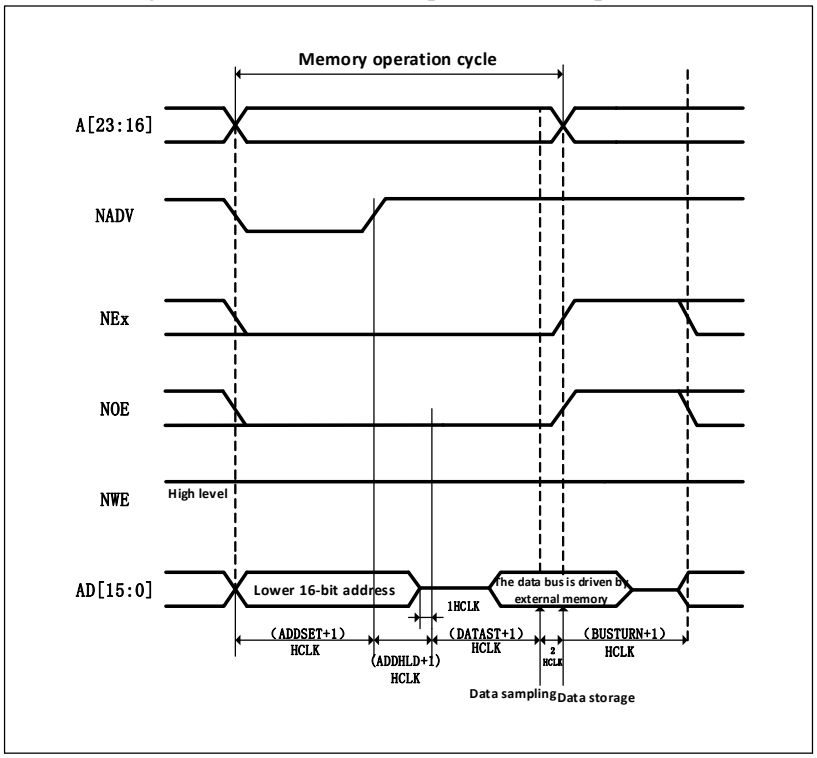


Figure 26-8 Mode B write operation (Multiplexed)

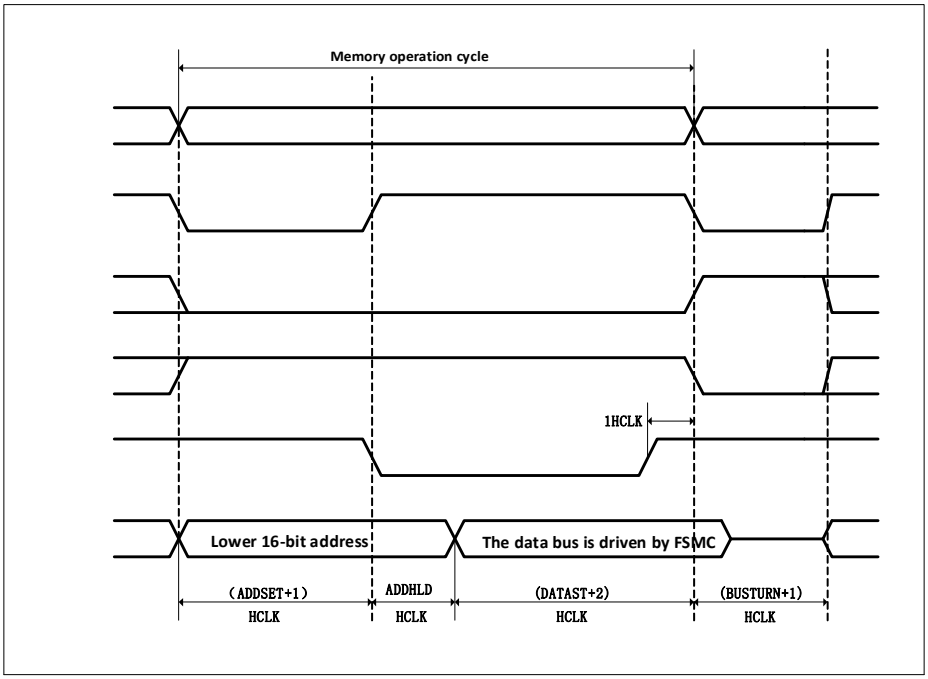


Table 26-13 Mode B FSMC_BCR1 bit field

Bitx	Name	Configuration Value
[31:20]	Reserved	0x000
19	CBURSTRW	0x0
[18:16]	Reserved	0x0
15	ASYNCWAIT	It is 1 if the memory supports this function,

		otherwise it is 0.
14	EXTMOD	0x1
13	WAITEN	0x0
12	WREN	Set as needed
11	WAITCFG	No effect
10	Reserved	0x0
9	WAITPOL	When bit15 is 1, it is meaningful.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
[5:4]	MWID	Set as needed
[3:2]	MTYP	0x2(NOR FLASH)
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-14 Mode B FSMC_BTR1 bit field

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x1
[27:24]	DATLAT	No effect
[23:20]	CLKDIV	No effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. The read operation is (DATAST+3 HCLK). The minimum value of this bit is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. The read operation is (ADDSET+1 HCLK).

Table 26-15 Mode B FSMC_BWTR1 bit field

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x1
[27:24]	DATLAT	No effect
[23:20]	CLKDIV	No effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. The write operation is (DATAST+1 HCLK). The minimum value of this bit is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. The write operation is (ADDSET+1 HCLK).

Figure 26-9 Mode C read operation (Multiplexed)

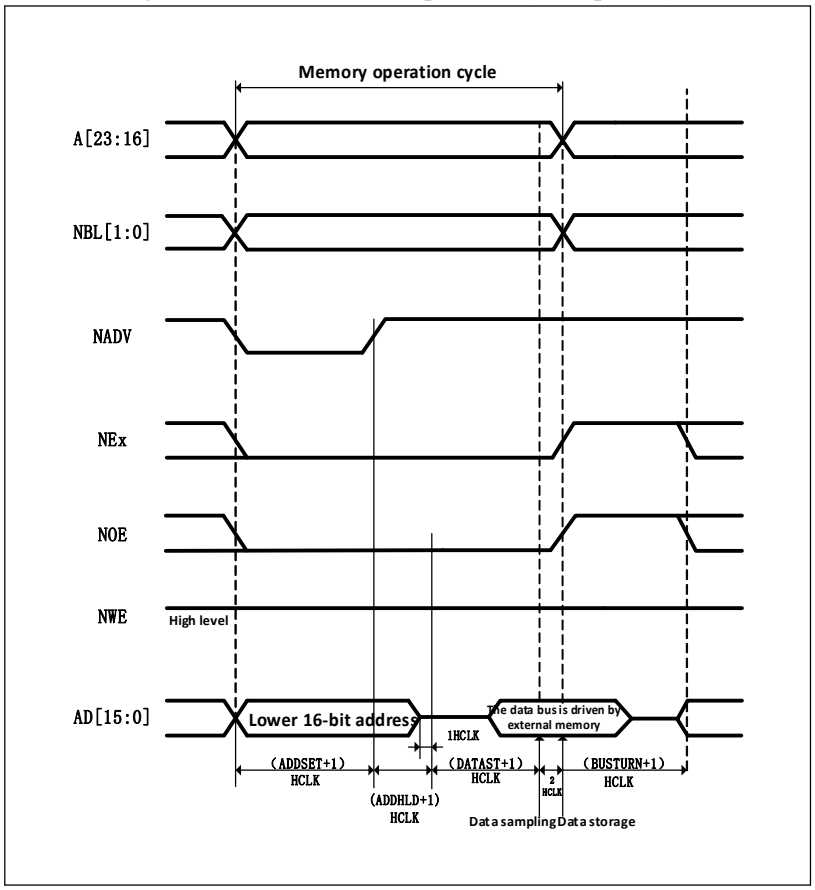


Figure 26-10 Mode C write operation (Multiplexed)

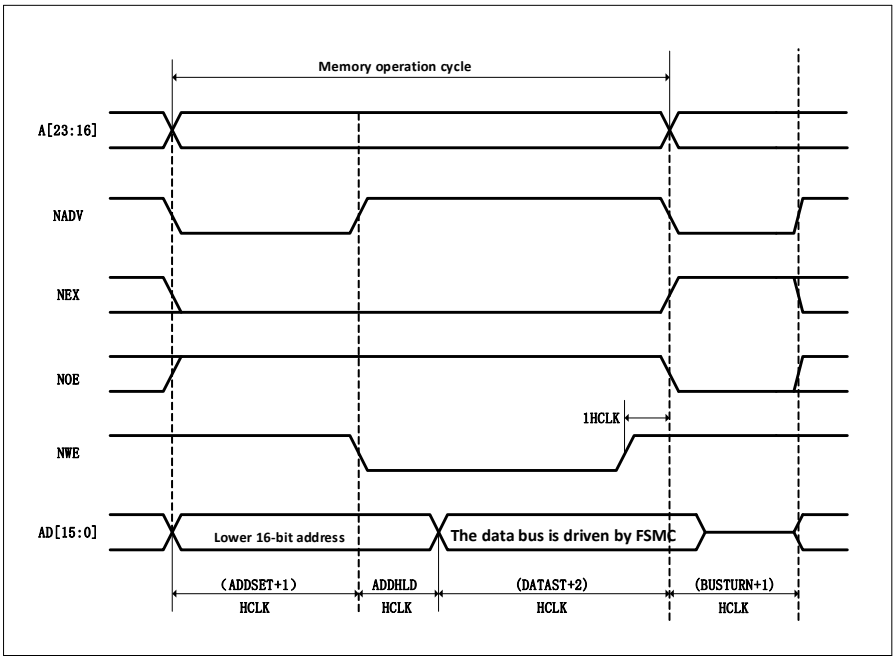


Table 26-16 Mode C FSMC_BCR1 bit field

Bitx	Name	Configuration Value
[31:20]	Reserved	0x000
19	CBURSTRW	0x0
[18:16]	Reserved	0x0
15	ASYNCWAIT	It is 1 if the memory supports this function,

		otherwise it is 0.
14	EXTMOD	0x1
13	WAITEN	0x0
12	WREN	Set as needed
11	WAITCFG	No effect
10	Reserved	0x0
9	WAITPOL	When bit15 is 1, it is meaningful.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
[5:4]	MWID	Set as needed
[3:2]	MTYP	0x2(NOR FLASH)
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-17 Mode C FSMC_BTR1 bit field

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x2
[27:24]	DATLAT	No effect
[23:20]	CLKDIV	No effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. The read operation is (DATAST+3 HCLK). The minimum value of this bit is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. The read operation is (ADDSET+1 HCLK).

Table 26-18 Mode C FSMC_BWTR1 bit field

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x2
[27:24]	DATLAT	No effect
[23:20]	CLKDIV	No effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. The write operation is (DATAST+1 HCLK). The minimum value of this bit is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. The write operation is (ADDSET+1 HCLK).

Figure 26-11 Mode D read operation (Multiplexed)

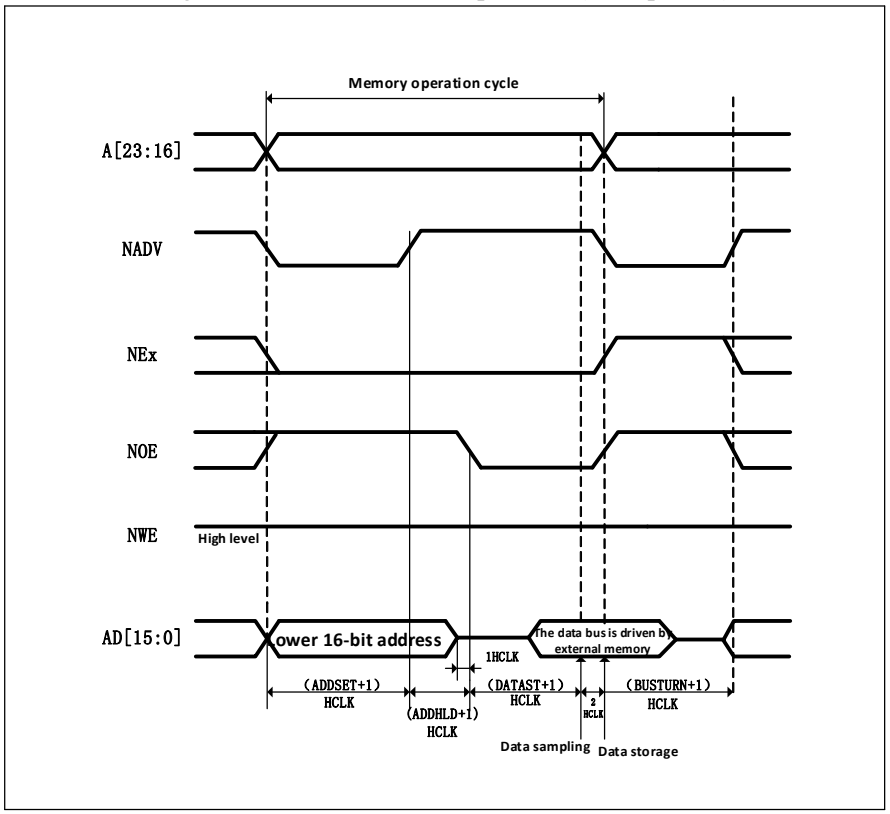


Figure 26-12 Mode D write operation (Multiplexed)

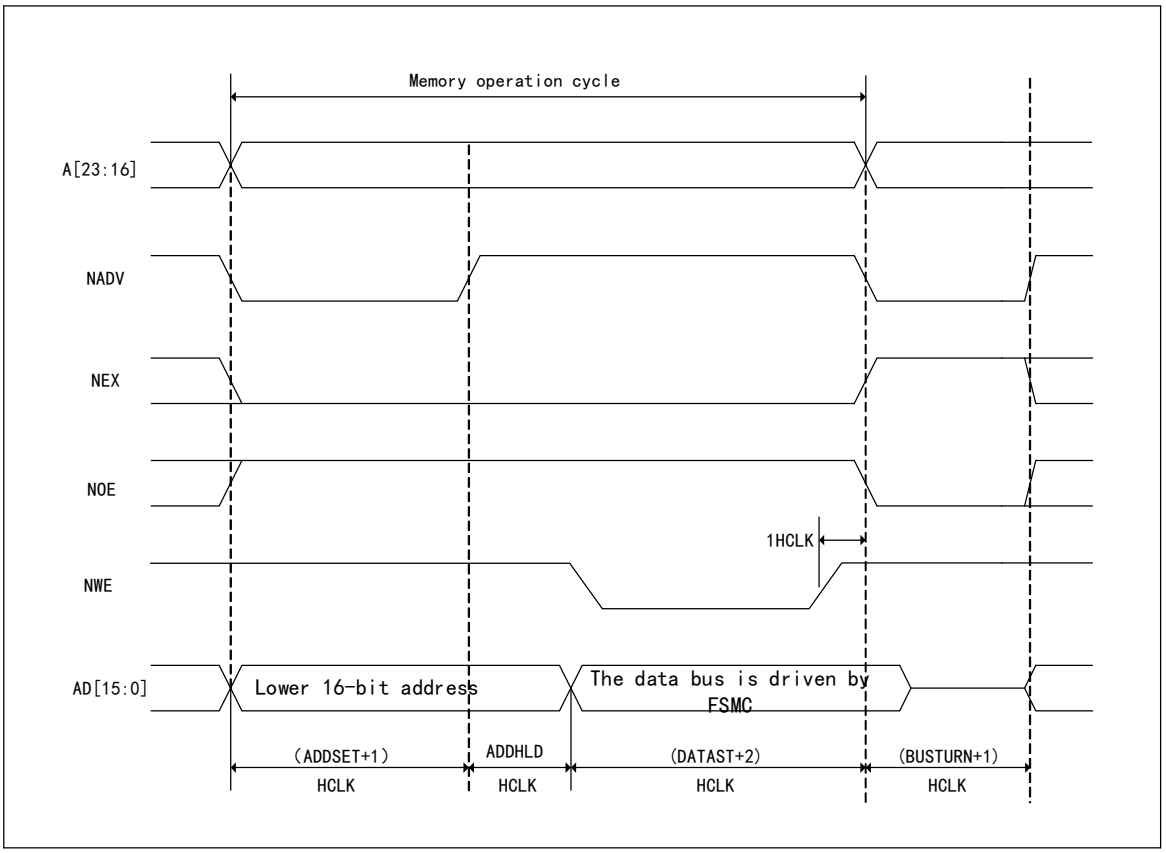


Figure 26-19 Mode D FSMC_BCR1 bit field

Bitx	Name	Configuration Value

[31:20]	Reserved	0x000
19	CBURSTRW	0x0
[18:16]	Reserved	0x0
15	ASYNCWAIT	It is 1 if the memory supports this function, otherwise it is 0.
14	EXTMOD	0x1
13	WAITEN	0x0
12	WREN	Set as needed
11	WAITCFG	No effect
10	Reserved	0x0
9	WAITPOL	When bit15 is 1, it is meaningful.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	Set as needed
[5:4]	MWID	Set as needed
[3:2]	MTYP	Set as needed
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-20 Mode D FSMC_BTR1 bit field

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x3
[27:24]	DATLAT	No effect
[23:20]	CLKDIV	No effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. The read operation is (DATAST+3 HCLK). The minimum value of this bit is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK).
[3:0]	ADDSET	Address setup time. The read operation is (ADDSET+1 HCLK).

Table 26-21 Mode D FSMC_BWTR1 bit field

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x3
[27:24]	DATLAT	0x0
[23:20]	CLKDIV	0x0
[19:16]	BUSTYRN	0x0
[15:8]	DATAST	Data setup time. The write operation is (DATAST+1 HCLK). The minimum value of this bit is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK).
[3:0]	ADDSET	Address set time. The write operation is

		(ADDSET+1 HCLK).

26.2.3.4 Timing Diagram of NOR/PSRAM Synchronous Transfer Address/Data Multiplexing

Data delay and NOR FLASH delay should be noted that the DATLAT value must be consistent with the definition in the NOR FLASH configuration register. The clock period when the NADV signal is low is not counted in the delay parameter. The DATLAT parameter of the FSMC can be DATLAT+2 or DATLAT+3. For special memories, the NWAIT signal will be generated during the data retention period, and DATLAT needs to be set to the minimum value. For other memories, the NWAIT signal will not be generated during the data hold time phase, the data hold time of FSMC and memory must be set consistently.

For a single batch transmission, it should be noted that the memory configuration bit is synchronized in batch mode. If 16bit data is transmitted, FSMC will perform a batch transmission with a length of 1. If 32bit data is transmitted, the FSMC is divided into 2 times of 16bit transmission, and the length of an execution is 1. 2 batch transfers.

In NOR FLASH synchronous batch mode access, after the hold time (DATLAT+1 CLK), if the NWAIT signal is detected to be low, the FSMC needs to insert a wait period before NWAIT becomes high. When NWAIT becomes high, FSMC considers the data valid. During the wait state insertion period controlled by the NWAIT signal, the controller will continue to send clock pulses to the memory, hold the chip select signal and output the valid signal, while ignoring the invalid data signal. In bulk transfer mode, the NWAIT signal of NOR FLASH selects 2 timing configurations by configuring the WAITCFG bit.

Figure 26-13 Synchronous bus multiplexing read operations (Multiplexed)

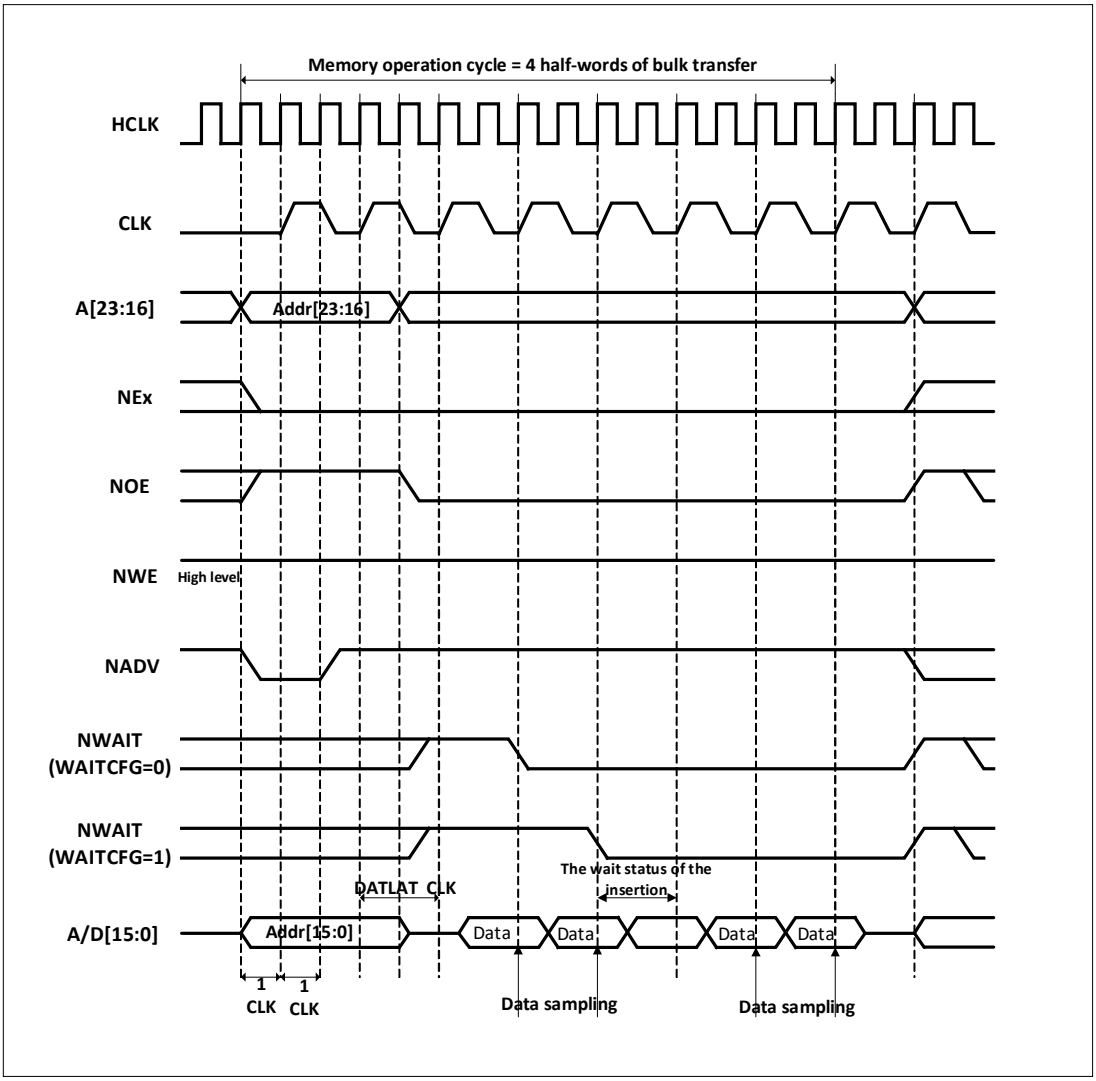


Table 26-22 FSMC_BCR1 bit field (Synchronous read mode)

Bitx	Name	Configuration Value
[31:20]	Reserved	0x000
19	CBURSTRW	No effect
[18:16]	Reserved	0x0
15	ASYNCWAIT	0x0
14	EXTMOD	0x0
13	WAITEN	0x0
12	WREN	No effect
11	WAITCFG	Set as needed
10	Reserved	0x0
9	WAITPOL	Set as needed
8	BURSTEN	0x1
7	Reserved	0x1
6	FACCEN	Set as needed
[5:4]	MWID	Set as needed
[3:2]	MTYP	0x1or 0x2

1	MUXEN	0x1
0	MBKEN	0x1

Table 26-23 FSMC_BTR1 bit field (Synchronous read mode)

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	Data latch time
[23:20]	CLKDIV	0x0: Reserved 0x1: CLK=2 HCLK
[19:16]	BUSTURN	No effect
[15:8]	DATAST	No effect
[7:4]	ADDHLD	No effect
[3:0]	ADDSET	No effect

Figure 26-14 Synchronous bus multiplexing write operations (Multiplexed)

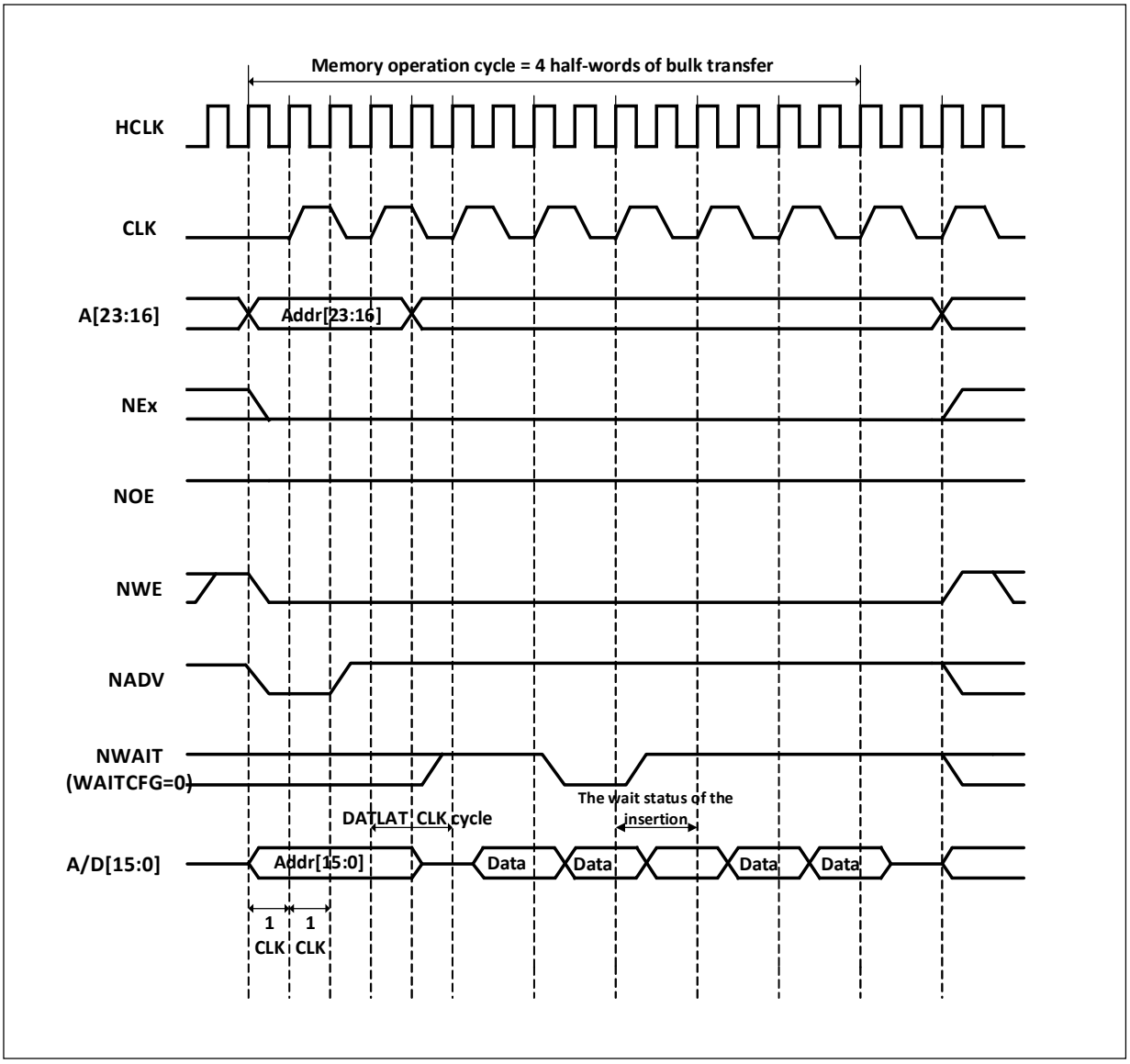


Table 26-24 FSMC_BCR1 bit field (Synchronous write mode)

Bitx	Name	Configuration Value
[31:20]	Reserved	0x000
19	CBURSTRW	0x1
[18:16]	Reserved	0x0
15	ASYNCWAIT	0x0
14	EXTMOD	0x0
13	WAITEN	It is 1 if the memory supports this function, 0 otherwise it is 0.
12	WREN	0x1
11	WAITCFG	0x0
10	Reserved	0x0
9	WAITPOL	Set as needed
8	BURSTEN	No effect
7	Reserved	0x1
6	FACCEN	Set as needed
[5:4]	MWID	Set as needed
[3:2]	MTYP	0x1
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-25 FSMC_BTR1 bit field (Synchronous write mode)

Bitx	Name	Configuration Value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	Data latch time
[23:20]	CLKDIV	0x0: Reserved 0x1: CLK=2 HCLK
[19:16]	BUSTURN	No effect
[15:8]	DATAST	No effect
[7:4]	ADDHLD	No effect
[3:0]	ADDSET	No effect

26.2.4 NAND FLASH controller

FSMC supports 8bit and 16bit operation of NAND FLASH. FSMC can generate multiple address cycles as needed. In theory, FSMC has no limit to the capacity of NAND FLASH. The read and write parameters of NAND FLASH can be configured by software, see Table 26-26.

Table26-26 Software controllable NAND read and write parameters

Parameter	Function	Read and write	Parameter value range
Memory setup time	Time to setup the address before sending command	read/write	1<< T <<256 (HB HCLK)
Memory wait	Minimum duration to	read/write	1<< T <<256 (HB HCLK)

time	send a command		
Memory hold time	Time to hold the address at the end of command transmission, also the data hold time during write operation	read/write	1<< T <<255 (HB HCLK)
Memory data bus high resistance state time	The data bus remains high resistance state time after initiating a write operation	write	0<< T <<255 (HB HCLK)

26.2.4.1 External Memory Interface Signals

8bit and 16bit NAND FLASH interface, see Table 26-27, Table 26-28.

Table 26-27 8-bit NAND FLASH

FSMC Pin	Direction	Description
A[17]	Output	Address latch enable signal (ALE)
A[16]	Output	Command latch enable signal (CLE)
D[7:0]	Input/output	8-bit bidirectional address/data multiplexing bus
NCE[2]	Output	Chip select line
NOE	Output	Output enable
NWE	Output	Write enable
NWAIT	Input	Wait signal line

Table 26-28 16-bit NAND FLASH

FSMC Pins	Direction	Description
A[17]	Output	Address latch enable signal (ALE)
A[16]	Output	Command latch enable signal (CLE)
D[15:0]	Input/output	16-bit bidirectional address/data multiplexing bus
NCE[2]	Output	Chip select line
NOE	Output	Output enable
NWE	Output	Write enable
NWAIT	Input	Wait signal line

26.2.4.2 Supported Memory and Operation Methods

See Table 26-29 for supported memories and operation methods.

Table 26-29 Supported memory and operation modes

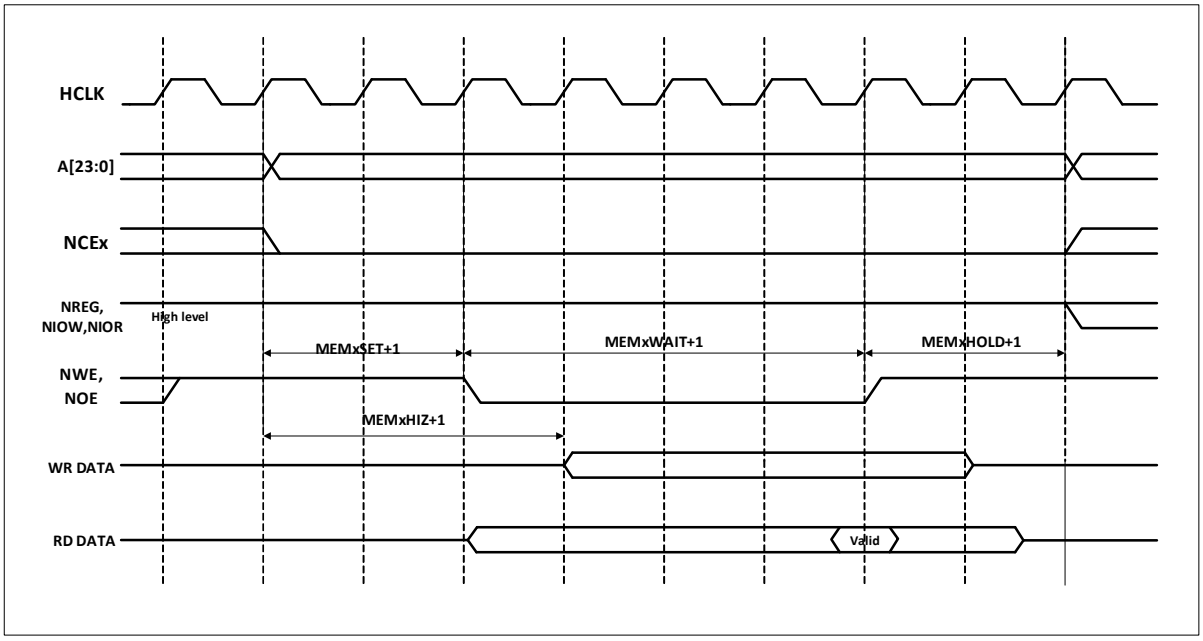
Memory	Mode	HB data width	Memory width	Description
8bit NAND	Asynchronous read	8	8	
	Asynchronous write	8	8	
	Asynchronous read	16	8	Divided into 2 FSMC access
	Asynchronous write	16	8	Divided into 2 FSMC access
	Asynchronous read	32	8	Divided into 4 FSMC access

	Asynchronous write	32	8	Divided into 4 FSMC access
16bit NAND	Asynchronous read	8	16	
	Asynchronous read	16	16	
	Asynchronous write	16	16	
	Asynchronous read	32	16	Divided into 2 FSMC access
	Asynchronous write	32	16	Divided into 2 FSMC access

26.2.4.3 NAND FLASH Timing Diagram (Including pre-wait function)

NAND FLASH operation timing control, involving FSMC_PMEM2 and FSMC_PATT2 timing registers, each register contains 4 parameters. The 3 parameters MEMHOLD, MEMWAIT and ATTSET correspond to the number of HCLK cycles in the 3 stages of operating the NAND FLASH, and the MEMHIZ parameter corresponds to the timing when the FSMC starts to drive the data bus. NAND FLASH controller general memory space access timing, see Figure 26-15.

Figure 26-15 NAND FLASH controller general purpose memory access timing



26.2.4.4 NAND FLASH Operating Procedure

The command latch enables signal (CLE) and address latch enable signal (ALE) of NAND FLASH are driven by FSMC address lines A16 and A17 respectively. Therefore, when sending commands or addresses to NAND FLASH operations, it is necessary to perform a write operation on the specified address of the storage space. The read operation process of NAND FLASH is as follows:

1） Configure the PWID, PTYP, PWAITEN and PBKEN of the FSMC_PCR2, FSMC_PMEM2 and FSMC_PATT2 registers according to the NAND FLASH characteristics to be operated.

2） Write a command byte in the general storage space. During the period when NWE is low, CLE outputs a high level. At this time, the command byte is recognized as a command by NAND FLASH, and the command is latched. The subsequent page read operation does not need to send the command repeatedly.

3） Then, by writing 4 bytes to the memory space, it is used as the starting address of the read operation. During the period when NWE is low, ALE outputs high level, and 4 bytes are recognized by NAND FLASH as the starting address of the read operation at this time. When operating the attribute storage

space, the FSMC can be made to generate different timings and realize some NAND FLASH pre-wait functions.

4） Before the FSMC controller starts a new operation, it needs to wait for the R/NB signal of the NAND FLASH to become a high level. During the wait period, the FSMC controller needs to keep the NCE signal at a low level.

5） The FSMC controller can read the memory page byte by byte from the NAND FLASH by operating the general memory space.

26.2.4.5 NAND FLASH Pre-wait Function

For some special NAND FLASHs, after the last address byte is input, the R/NB signal changes to a low level, see Figure 26-16.

(1) in the figure shows the CPU writing the byte 0x00 to 0x70010000;

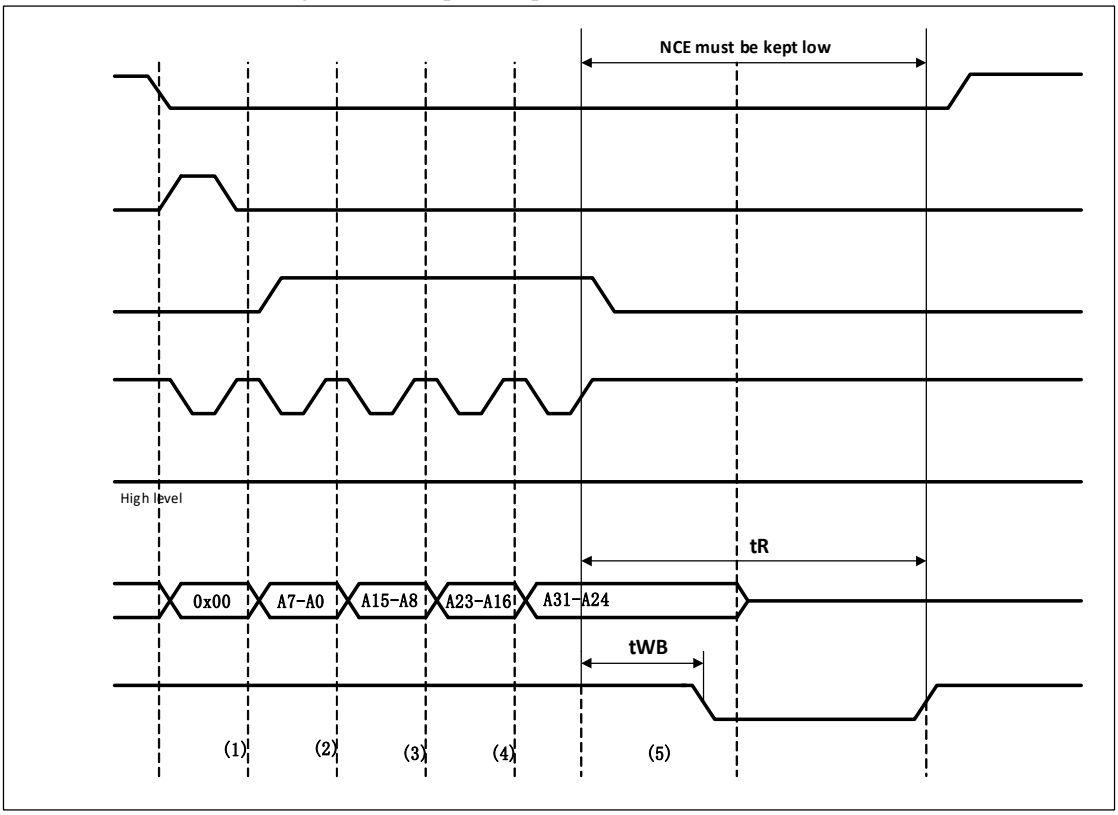
(2) in the figure shows the CPU writing to NAND FLASH addresses A7–A0 at 0x70020000;

(3) in the figure shows the CPU writing to NAND FLASH addresses A15–A8 at 0x70020000;

(4) in the figure shows the CPU writing to NAND FLASH addresses A23–A16 up to 0x70020000;

(5) in the figure shows the CPU writing to NAND FLASH addresses A31–A24 up to 0x70020000;

Figure 26-16 Special operation on NAND FLASH



When using this function, the timing of tWB is guaranteed by configuring the MEMHOLD bit in the FSMC_PMEM2 register. After reading and writing operations to NAND FLASH, the FSMC controller will insert (MEMHOLD+1) HCLK between the rising edge of the NEW signal and the next operation. Keep the delay. In order to solve this problem, the attribute space is used to configure the ATTHOLD value so that it conforms to the timing of tWB, and at the same time, it is necessary to keep MEMHOLD at a minimum value. At this time, only when writing the last byte of the NAND FLASH address, the FSMC controller needs to

write the attribute storage space, and the rest of the NAND FLASH read and write operations use the general storage space.

26.2.4.6 NAND FLASH ECC Features

FSMC's NAND FLASH controller includes an error correction code calculation hardware module, which can effectively reduce the software workload of the CPU when processing error correction codes. When the ECC module reads and writes NAND FLASH, it supports correction of 1 bit error and detection of 2-bit errors in every 256, 512, 1024, 2048, 4096 or 8192 bytes. The page size corresponds to the valid bits of the ECC result, see Table 26-30. After the ECC calculation circuit is enabled, the ECC module monitors the data bus and read/write signals (NCE and NWE) of the NAND FLASH. Pay attention to the following when using ECC:

- When accessing the NAND FLASH, the data appearing on the D[15:0] bus is latched and used for ECC calculations.
- After the specified number of bytes have been written into or read from the NAND FLASH, the software reads the ECC value from the FSMC_ECCR2 register. To calculate ECC again, first clear ECCEN to 0, and then write 1 to re-enable ECC calculation.

Table 26-30 The page size corresponds to the valid bits of the ECC result

ECCPS[2:0]	Page size (bytes)	ECC significant bit
000	256	ECC[21:0]
001	512	ECC[23:0]
010	1024	ECC[25:0]
011	2048	ECC[27:0]
100	4096	ECC[29:0]
101	8192	ECC[31:0]

26.3 Register Description

Table 26-31 FSMC registers

Name	Access address	Description	Reset value
R32_FSMC_BCR1	0xA0000000	SRAM/NOR-Flash chip select control register 1	0x000030DX
R32_FSMC_BTR1	0xA0000004	SRAM/NOR-Flash chip select timing register 1	0x0FFFFFFF
R32_FSMC_PCR2	0xA0000060	NAND-Flash control register 2	0x00000018
R32_FSMC_SR2	0xA0000064	FIFO status and interrupt register 2	0x00000040
R32_FSMC_PMEM2	0xA0000068	General purpose memory timing register 2	0xFCFCFCFC
R32_FSMC_PATT2	0xA000006C	Attribute memory timing register 2	0xFCFCFCFC
R32_FSMC_ECCR2	0xA0000074	ECC result register 2	0x00000000
R32_FSMC_BWTR1	0xA0000104	SRAM/NOR-Flash write timing register 1	0x0FFFFFFF

26.3.1 SRAM/NOR-Flash Chip Select Control Register 1 (FSMC_BCR1)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												CBU RSTR	Reserved		

												W			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ASYNCWAIT	EXTMOD	WAITEN	WREN	WAITCFG	Reserved	WAITPOL	BURSTEN	Reserved	FACCEN	MWID[1:0]		MTYP[1:0]		MUXEN	MBKEN

Bit	Name	Access	Description	Reset value
[31:20]	Reserved	RO	Reserved	0
19	CBURSTRW	RW	Synchronous write enable. 0: Write operation is in asynchronous mode; 1: Write operation is in synchronous mode.	0
[18:16]	Reserved	RO	Reserved	0
15	ASYNCWAIT	RW	Wait signal during asynchronous transfer. 0: Disable wait signal during asynchronous transfer; 1: Enable wait signal during asynchronous transfer.	0
14	EXTMOD	RW	Extended mode enable. 0: Disable writing to use FSMC_BWTR; 1: Enable writing to use FSMC_BWTR.	0
13	WAITEN	RW	Wait enable. When the flash memory is in bulk transfer mode, this bit controls whether the wait signal is inserted according to the NWAIT signal. 0: Disable NWAIT signal; 1: Enable NWAIT signal.	1
12	WREN	RW	Write enable. 0: Disable FSMC to write to the memory; 1: Enable FSMC to write to the memory.	1
11	WAITCFG	RW	Wait timing configuration. 0: NWAIT signal is valid one data cycle before the wait state; 1: NWAIT signal is valid during the wait state.	0
10	Reserved	RO	Reserved	0
9	WAITPOL	RW	Wait signal polarity. 0: NWAIT is active low; 1: NWAIT is active high.	0
8	BURSTEN	RW	Synchronous read enable bit: 0: Synchronous reading mode is prohibited. Read access in asynchronous mode; 1: Enable synchronous reading mode. Read access in synchronous mode.	0
7	Reserved	RO	Reserved	1
6	FACCEN	RW	NOR FLASH access enable. 0: Disable access to NOR FLASH; 1: Enable access to NOR FLASH.	1

[5:4]	MWID[1:0]	RW	Data bus width. 11: Reserved; 10: Reserved; 01: 16 bits; 00: 8 bits.	01b
[3:2]	MTYP[1:0]	RW	Memory type. 11: Reserved; 10: NOR FLASH; 01: PSRAM; 00: SRAM, ROM.	xxb
1	MUXEN	RW	Address/data multiplexing enable. 0: Address/data are not multiplexed; 1: Address/data multiplexing.	x
0	MBKEN	RW	Memory block enable. 0: Disable the corresponding memory block; 1: Enable the corresponding memory block.	x

26.3.2 SRAM/NOR-Flash Chip Select Timing Register 1 (FSMC_BTR1)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		ACCMOD[1:0]		DATLAT[3:0]				CLKDIV[3:0]				BUSTURN[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATAST[7:0]								ADDHLD[3:0]				ADDSET[3:0]			

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:28]	ACCMOD[1:0]	RW	Asynchronous access mode. 00: Mode A; 01: Mode B; 10: Mode C; 11: Mode D.	00b
[27:24]	DATLAT[3:0]	RW	Data latch time. For synchronous bulk mode (only for NOR FLASH) 0000: The first data latch time is 2 CLK clocks; ... 1110: The first data latch time is 16 CLK clocks; 1111: The first data latch time is 17 CLK clocks.	1111b
[23:20]	CLKDIV[3:0]	RW	Clock divider ratio (CLK signal). This parameter has no effect when accessing asynchronous NOR FLASH, SRAM and ROM. 0000: Reserved; 0001: 1 CLK cycle = 2 HCLK cycles;	1111b

			... 1110: 1 CLK cycle = 15 HCLK cycles; 1111: 1 CLK cycle = 16 HCLK cycles.	
[19:16]	BUSTURN[3:0]	RW	Bus recovery time. (only for NOR FLASH) 0000: Bus recovery time = 1 HCLK cycle; ... 1110: Bus recovery time = 15 HCLK cycles; 1111: Bus recovery time = 16 HCLK cycles.	1111b
[15:8]	DATAST[7:0]	RW	Data hold time. 00000000: Data hold time = 1 HCLK cycles; 00000001: Data hold time = 2 HCLK cycles; ... 11111110: Data hold time = 255 HCLK cycles; 11111111: Data hold time = 256 HCLK cycles. For the data retention time of each memory type and access method, refer to the corresponding charts (see Figures 26-3 to 26-12). For example: For Mode 1 and Mode A, when DATAST=1 during a read operation: Data retention time = DATAST + 3 = 4 HCLK clock cycles.	0xFF
[7:4]	ADDHLD[3:0]	RW	Address hold time. (Only for asynchronous operations) 0000: Address hold time = 1 HCLK cycle; ... 1110: Address hold time = 15 HCLK cycles; 1111: Address hold time = 16 HCLK cycles.	1111b
[3:0]	ADDSET[3:0]	RW	Address setup time. (Only for asynchronous operations) 0000: Address setup time = 1 HCLK cycle; ... 1110: Address setup time = 15 HCLK cycles; 1111: Address setup time = 16 HCLK cycles. For the data retention time of each memory type and access method, refer to the corresponding charts (see Figures 26-3 to 26-12). For example: For Mode 1 and Mode A, when DATAST=1 during a read operation: Data retention time = DATAST + 3 = 4 HCLK clock cycles.	1111b

26.3.3 NAND-Flash Control Register 2 (FSMC_PCR2)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												ECCPS[2:0]			TAR[3]

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TAR[2:0]			TCLR[3:0]				Reserved		ECC EN	PWID[1:0]		PTYP	PBK EN	PWA ITEN	Reser ved

Bit	Name	Access	Description	Reset value
[31:20]	Reserved	RO	Reserved	0
[19:17]	ECCPS[2:0]	RW	ECC page size. 000: 256Byte; 001: 512Byte; 010: 1024Byte; 011: 2048Byte; 100: 4096Byte; 101: 8192Byte; 111: Reserved.	000b
[16:13]	TAR[2:0]	RW	ALE to RE delay. 0000: 1 HCLK cycle; ... 1110: 15 HCLK cycles; 1111: 16 HCLK cycles.	0000b
[12:9]	TCLR[3:0]	RW	CLE to RE delay. 0000: 1 HCLK cycle; ... 1110: 15 HCLK cycles; 1111: 16 HCLK cycles.	0000b
[8:7]	Reserved	RO	Reserved	0
6	ECCEN	RW	ECC enable. 0: Disable ECC; 1: Enable ECC.	0
[5:4]	PWID[1:0]	RW	Data bus width. 00: 8 bits; 01: 16 bits; 10: Reserved; 11: Reserved.	01b
3	PTYP	RW	Memory Type. 0: Reserved; 1: NAND flash memory.	1
2	PBKEN	RW	NAND memory enable. 0: Disable the corresponding memory block; 1: Enable the corresponding memory block.	0
1	PWAITEN	RW	Wait function enable. 0: Disable; 1: Enable.	0
0	Reserved	RO	Reserved	0

26.3.4 FIFO Status and Interrupt Registers 2 (FSMC_SR2)

Offset address: 0x64

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									FEMPT	Reserved					

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RO	Reserved	0
6	FEMPT	RO	FIFO empty flag. 0: FIFO not empty; 1: FIFO empty.	1
[5:0]	Reserved	RO	Reserved	0

26.3.5 General-purpose Memory Timing Registers 2 (FSMC_PMEM2)

Offset address: 0x68

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MEMHIZx[7:0]								MEMHOLDx[7:0]							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MEMWAITx[7:0]								MEMSETx[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	MEMHIZx[7:0]	RW	High-impedance time of the general memory data bus. 00000000: NAND FLASH is 0 HCLK cycles; ... 11111110: NAND FLASH is 254 HCLK cycles; 11111111: NAND FLASH is 255 HCLK cycles.	0xFC
[23:16]	MEMHOLDx	RW	General memory hold time. 00000000: Reserved; 00000001: 1 HCLK cycle; ... 11111110: 254 HCLK cycles; 11111111: 255 HCLK cycles.	0xFC
[15:8]	MEMWAITx[7:0]	RW	General memory wait time. (Need to add the wait period introduced by the NWAIT signal going low) 00000000: Reserved; 00000001: 2 HCLK cycles; ... 11111110: 255 HCLK cycles;	0xFC

			11111111: 256 HCLK cycles.	
[7:0]	MEMSETx[7:0]	RW	General memory setup time. 00000000: NAND FLASH 2 HCLK cycles; ... 11111110: NAND FLASH 256 HCLK cycles; 11111111: NAND FLASH 257 HCLK cycles.	0xFC

26.3.6 Attribute Memory Timing Register 2 (FSMC_PATT2)

Offset address: 0x6C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ATTHIZx[7:0]								ATTHOLDx[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ATTWAITx[7:0]								ATTSETx[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	ATTHIZx[7:0]	RW	High-impedance time of the attribute memory data bus. 00000000: 0 HCLK cycles; ... 11111110: 254 HCLK cycles; 11111111: 255 HCLK cycles.	0xFC
[23:16]	ATTHOLDx[7:0]	RW	Attribute memory hold time. 00000000: Reserved; 00000001: 1 HCLK cycle; ... 11111110: 254 HCLK cycles; 11111111: 255 HCLK cycles.	0xFC
[15:8]	ATTWAITx[7:0]	RW	Attribute memory wait time. (Need to add the wait period introduced by the NWAIT signal going low) 00000000: 1 HCLK cycle; 00000001: 2 HCLK cycles; …… 11111110: 255 HCLK cycles; 11111111: 256 HCLK cycles.	0xFC
[7:0]	ATTSETx[7:0]	RW	Attribute memory setup time. 00000000: 1 HCLK cycle; ... 11111110: 255 HCLK cycles; 11111111: 256 HCLK cycles.	0xFC

26.3.7 ECC Result Register2 (FSMC_ ECCR2)

Offset address: 0x74

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

ECC[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ECC[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	ECC[31:0]	RO	ECC calculation result	0

26.3.8 SRAM/NOR-Flash Write Timing Register1 (FSMC_BWTR1)

Offset address: 0x104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		ACCMOD[1:0]		Reserved								BUSTYRN[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATAST[7:0]								ADDHLD[3:0]				ADDSET[3:0]			

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:28]	ACCMOD[1:0]	RW	Access mode. (Valid when EXTMOD is 1) 00: Access mode A; 01: Access mode B; 10: Access mode C; 11: Access mode D.	00b
[27:20]	Reserved	RO	Reserved	0
[19:16]	BUSTYRN[3:0]	RW	Bus recovery time. 0000: 1 HCLK cycle; 0001: 2 HCLK cycles; ..... 1110: 15 HCLK cycles; 1111: 16 HCLK cycles.	1111b
[15:8]	DATAST[7:0]	RW	Data retention time. 00000000: Reserved; 00000001: 2 HCLK cycles; ... 11111110: 255 HCLK cycles; 11111111: 256 HCLK cycles.	0xFF
[7:4]	ADDHLD[3:0]	RW	Address hold time. 0000: Reserved; 0001: 2 HCLK cycles; ... 1110: 15 HCLK cycles; 1111: 16 HCLK cycles.	1111b
[3:0]	ADDSET[3:0]	RW	Address setup time. 0000: 1 HCLK cycle;	1111b

			0001: 2 HCLK cycles; ... 1110: 15 HCLK cycles; 1111: 16 HCLK cycles.	

Chapter 27 LCD-TFT Display Controller (LTDC)

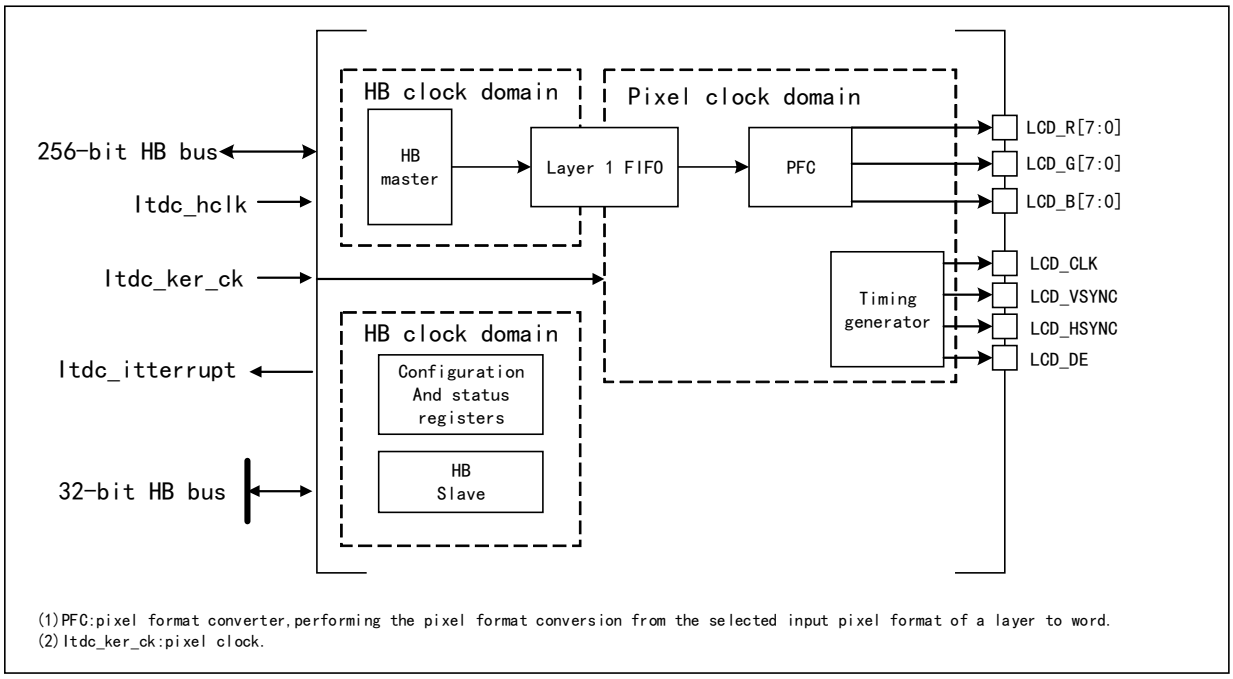
LCD-TFT (Thin Film Transistor-Liquid Crystal Display) Display Controller (LTDC) mainly provides parallel digital RGB, horizontal synchronization, vertical synchronization, pixel clock and data enable signals, which can be used as output signals to interface with different LCD and TFT panels.

27.1 Main Features

- Provide 1 display layers, each containing a proprietary 8*32 bits FIFO
- Support color lookup tables (CLUT) with up to 256 colors per layer
- Support programmable timing for different display panels
- Support programmable background color
- Support programmable polarity for HSYNC, VSYNC, and data enable signals
- Each display layer supports up to 8 input colors
- Support programmable window position and size
- Support thin-film transistor (TFT) color displays
- Support up to 3 programmable interrupt timers

27.2 Overview

Figure 27-1 Structural block diagram of LTDC



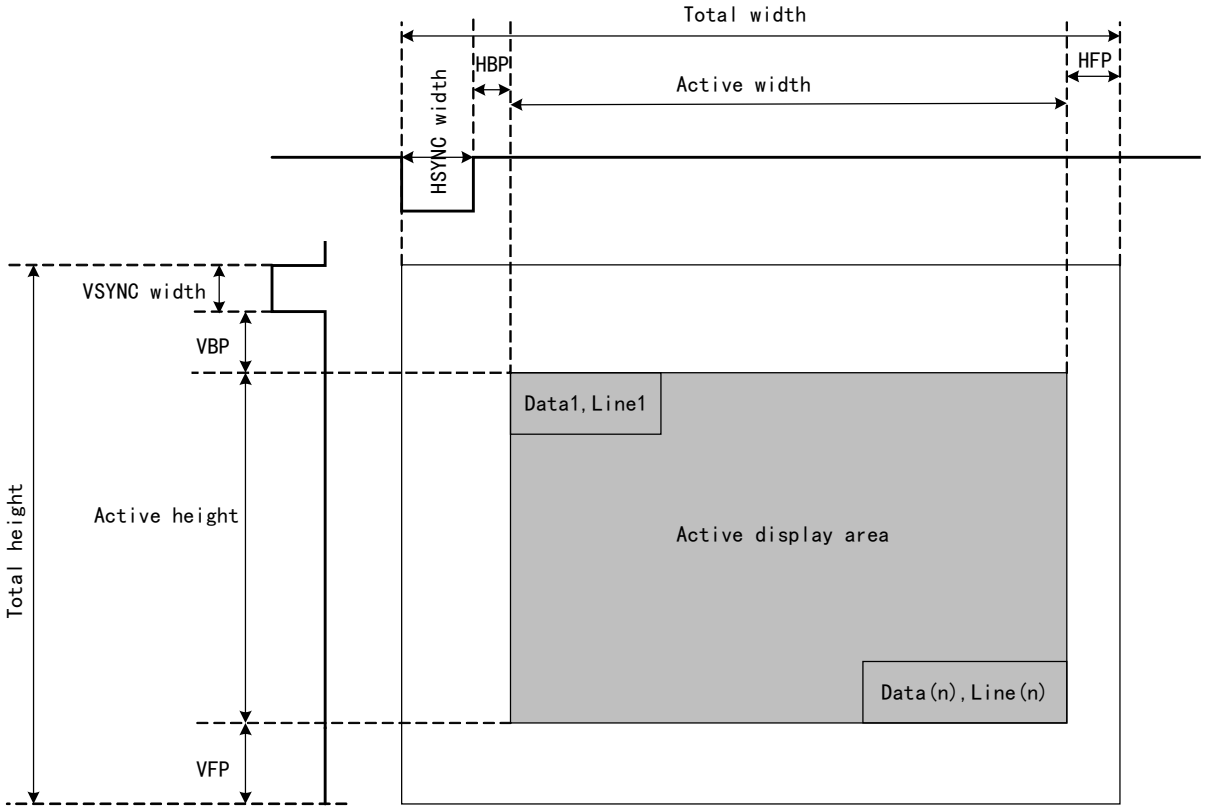
27.3 Function Description

27.3.1 LTDC Synchronization Timing

The LTDC interface is a synchronous data interface comprising pixel clock, pixel data, and horizontal and vertical sync signals. Figure 27-2 illustrates the configurable timing parameters generated by the LTDC synchronous timing

generator module.

Figure 27-2 LTDC synchronization timing



Note: (1) HSYNC and VSYNC denote horizontal and vertical sync widths, respectively; HBP and HFP denote horizontal back-edge period and horizontal front-edge period, respectively; VBP and VFP denote vertical back-edge period and vertical front-edge period, respectively.

(2) When LTDC is enabled, the generated timing uses the X/Y=0/0 position as the first horizontal sync pixel within the vertical sync region, followed by the back edge, active data display region, and front edge.

(3) When LTDC is disabled, the timing generator module resets to X=total width-1 and Y=total height-1, holding the previous pixel during the vertical sync phase and before FIFO refresh. Consequently, only blanking data is continuously output.

An example of synchronization timing configuration is as follows:

1) LCD-TFT timing (to be extracted from the panel datasheet);
2) Horizontal and vertical sync width: 0xA pixels, 0x2 lines;
3) Horizontal and vertical trailing edge: 0x14 pixels, 0x2 lines;
4) Effective width and effective height: 0x140 pixels, 0xF0 lines (320×240);
5) Horizontal leading edge: 0xA pixels;
6) Vertical leading edge: 0x4 lines;

The value programmed in the LTDC timing register will be:

1) R32_LTDC_SSCR register: Program to 0x00090001. (HSW[11:0] is 0x9 and VSH[10:0] is 0x1);
2) R32_LTDC_BPCR register: Program to 0x001D0003. (AHBP[11:0] is 0x1D (0xA + 0x13), and AVBP[10:0] is 0x3 (0x2 + 0x1));
3) R32_LTDC_AWCR register: Program to 0x015D00F3; (AAW[11:0] is 0x15D (0xA + 0x14 + 0x13F), and

AAH[10:0] is 0xF3 (0x2 + 0x2 + 0xEF));
4) R32_LTDC_TWCR register: Program to 0x00000167. (TOTALW[11:0] is 0x167 (0xA + 0x14 + 0x140 + 0x9));
5) R32_LTDC_THCR register: Program to 0x000000F7. (TOTALW[10:0] is 0xF7 (0x2 + 0x2 + 0xF0 + 3)).

27.3.2 Pixel Format

Programmable pixel format is used for data stored in the frame buffer of the layer. Up to eight input pixel format can be configured for each layer through the R32_LTDC_LxPFCR register.

The ARGB8888 format requires 8 bits of data per channel (Alpha, Red, Green and Blue). However, some channels in ARGB1555 and ARGB4444 formats are less than 8 bits. The pixel processing unit converts it into ARGB8888 by copying the high bits and filling them into the low bits. When processing RGB888 and RGB565 formats, the pixel processing unit assumes Alpha=255, and if the number of bits of the channel is less than 8, the copy high bits are also padded to the low bits.

27.3.3 Color Look-up Table (CLUT)

CLUT can be enabled for each layer at runtime through the CLUTEN bit in the R32_LTDC_LCR register. CLUT is only suitable for index colors when using L8, AL44 and AL88 input pixel format.

First, CLUT must load R, G and B values to replace the original R, G and B values of the corresponding pixels (index colors). Each color (RGB value) has its own corresponding address in CLUT. The values of r, g and b and their respective addresses are programmed by R32_LTDC_LCLUTWR register.

When using L8 and AL88 to input pixel format, CLUT must be loaded with 256 colors. The address of each color is configured in the CLUTADD bit of R32_LTDC_LCLUTWR register.

When using AL44 to input pixel format, CLUT must only load 16 colors. The address of each color must be filled by repeating the 4-bit L channel into 8 bits, as follows:
- L0 (Index Color 0), at address 0x00;
- L1, at address 0x11;
- L2, at address 0x22;
- .....;
- L15, at address 0xFF.

27.3.4 Color Frame Buffer

27.3.4.1 Color Frame Buffer Address

The color frame buffer of each layer has a start address, which is configured by R32_LTDC_LCFBAR register. When a layer is enabled, the data will be obtained from the color frame buffer.

27.3.4.2 Color Frame Buffer Length

Each layer is set with the total length (in bytes) and the number of lines of the color frame buffer, which can be configured by the R32_LTDC_LCFBLR and R32_LTDC_LCFBLNR registers respectively.

The length and number of rows are set to prevent data beyond the end of the frame buffer from being prefetched into the FIFO corresponding to the layer.

If it is set below the required bytes, a FIFO underflow interrupt will be generated (if previously enabled).

If it is set to more bytes than actually needed, useless data read from FIFO will be discarded and will not be displayed.

27.3.4.3 Color Frame Buffer Spacing

The color frame buffer of each layer has a configurable spacing, which is the distance (in bytes) from the beginning of one line to the beginning of the next line. It is configured through the R32_LTDC_LCFBLR register.

27.3.5 Chroma Key

Chroma key (RGB) configurable for pixel replacement. At runtime, pixel values can be replaced by matching pixels using color key values.

When chroma key is enabled, the current pixel (pixel after format conversion and before CLUT mixing) will be compared with the chroma key. If the current pixel matches the programmed RGB value, that pixel's RGB channel is configured as the default color.

The chroma key is configured through the LTDC_LCKCR register. The programmed value depends on the pixel format as it is compared to the current pixel after the pixel format is converted to RGB888.

For example, if you use medium yellow (50% red + 50% green) as the replacement chroma key, then:
- In RGB565, medium yellow is 0x8400. Set LTDC_LCKCR to 0x848200.
- In ARGB8888, medium yellow is 0x808000; set LTDC_LCKCR to 0x808000.
- In all CLUT-based color modes (L8, AL88, AL44), set one of the palette entries to medium yellow (0x808000) and set LTDC_LCKCR to 0x808000.

27.3.6 Reloaded Shadow Register

Some configuration registers perform shadow operations. When writing to the active register, or at the beginning of the vertical blanking period after the configuration phase of R32_LTDC_SRCR register, the shadow register value can be reloaded into the active register immediately. If the immediate overload configuration is selected, overloading should only be activated after all new registers have finished writing.

The shadow register should not be modified again before the overload is completed. Reading the shadow register will return the actual effective value. The newly written value can only be read after the overload occurs.

If enabled in the R32_LTDC_IER register, a register overload interrupt can be generated.

Shadow registers are Layer 1 and Layer 2 registers, except R32_LTDC_LCLUTWR register.

27.3.7 Interrupt

The LTDC provides 3 maskable interrupts, which are logically ORed into a single interrupt vector.

Interrupt sources can be individually enabled or disabled via the R32_LTDC_IER register; corresponding interrupt flags can be read through the R32_LTDC_ISR register; and the corresponding interrupt flag bits can be cleared by setting the R32_LTDC_ICR register to 1.

Interrupts are generated when the following events occur:

- Line interrupt: Generated upon reaching a programmed line. The interrupt position is programmed in the R32_LTDC_LIPCR register;
- Register reload interrupt: Generated during shadow register reload execution within the vertical blanking

interval;

- FIFO underflow interrupt: Generated when requesting pixels from an empty layer FIFO.

27.4 Register Description

Table 27-1 LTDC-related registers

Name	Access address	Description	Reset value
R32_LTDC_SSCR	0x40024800	LTDC synchronous size configuration register	0x00000000
R32_LTDC_BPCR	0x40024804	LTDC trailing-edge configuration register	0x00000000
R32_LTDC_AWCR	0x40024808	LTDC effective width configuration register	0x00000000
R32_LTDC_TWCR	0x4002480C	LTDC total width configuration register	0x00000000
R32_LTDC_GCR	0x40024810	LTDC global control register	0x00002220
R32_LTDC_SRCR	0x40024814	LTDC shadow reload configuration register	0x00000000
R32_LTDC_BCCR	0x40024818	LTDC background color configuration register	0x00000000
R32_LTDC_IER	0x4002481C	LTDC interrupt enable register	0x00000000
R32_LTDC_ISR	0x40024820	LTDC interrupt status register	0x00000000
R32_LTDC_ICR	0x40024824	LTDC interrupt clear register	0x00000000
R32_LTDC_LIPCR	0x40024828	LTDC line interrupt position configuration register	0x00000000
R32_LTDC_CPSR	0x4002482C	LTDC current position status register	0x00000000
R32_LTDC_CDSR	0x40024830	LTDC current display status register	0x0000000F
R32_LTDC_LCR	0x40024834	LTDC layer control register	0x00000000
R32_LTDC_LWHPCR	0x40024838	LTDC layer window horizontal position configuration register	0x00000000
R32_LTDC_LWVPCR	0x4002483C	LTDC layer window vertical position configuration register	0x00000000
R32_LTDC_LCKCR	0x40024840	LTDC layer chroma key configuration register	0x00000000
R32_LTDC_LPFCR	0x40024844	LTDC layer pixel format configuration register	0x00000000
R32_LTDC_LDCCR	0x4002484C	LTDC layer default color configuration Register	0x00000000
R32_LTDC_LCFBAR	0x40024854	LTDC layer color frame buffer address register	0x00000000
R32_LTDC_LCFBLR	0x40024858	LTDC layer color frame buffer length register	0x00000000
R32_LTDC_LCFBLNR	0x4002485C	LTDC layer color frame buffer line number register	0x00000000
R32_LTDC_LCLUTWR	0x40024860	LTDC layer CLUT write register	0x00000000

27.4.1 LTDC Synchronous Size Configuration Register (R32_LTDC_SSCR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				HSW[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				VSH[10:0]											

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	HSW[11:0]	RW	Horizontal synchronization width (in pixel clock period): This field defines the number of horizontal synchronization pixels minus 1.	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	VSH[10:0]	RW	Vertical synchronization height (in horizontal scanning lines): This field defines the vertical synchronization height minus 1. It represents the number of horizontal synchronization lines.	0

27.4.2 LTDC Trailing-Edge Configuration Register (R32_LTDC_BPCR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				AHBP[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				AVBP[10:0]											

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	AHBP[11:0]	RW	Accumulation level trailing edge (in pixel clock cycles): This field defines the width of the accumulation level trailing edge (horizontal sync pixels plus trailing edge pixels minus one). The trailing edge is the interval between when the horizontal sync signal becomes inactive and the start of the next scan line's active display.	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	AVBP[10:0]	RW	Accumulated vertical Trailing Edge (Per Horizontal Scan Line): This field defines the accumulated vertical trailing edge width (vertical sync line plus vertical trailing edge line minus 1). The vertical trailing edge is the number of horizontal scan lines contained between the start of a frame and the start of the first valid scan line of the next frame.	0

27.4.3 LTDC Effective Width Configuration Register (R32_LTDC_AWCR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				AAW[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				AAH[10:0]											

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	AAW[11:0]	RW	Cumulative effective width (in pixel clock cycles): This field defines the cumulative effective width (horizontal sync pixels plus horizontal trailing edge pixels minus one effective pixel). The effective width represents the number of pixels within the active display area of the panel scan line.	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	AAH[10:0]	RW	Cumulative effective height (per horizontal scan line): This field defines the cumulative height (vertical sync line plus vertical trailing edge line plus effective height line minus 1). Effective height represents the number of valid lines within the panel.	0

27.4.4 LTDC Total Width Configuration Register (R32_LTDC_TWCR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				TOTALW[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				TOTALH[10:0]											

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	TOTALW[11:0]	RW	Total width (in pixel clock cycles): This field defines the accumulated total width (horizontal sync pixel plus horizontal trailing edge pixel plus active width plus horizontal leading edge pixel minus 1).	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	TOTALH[10:0]	RW	Total height (per horizontal scan line): This field defines the cumulative height (vertical sync line plus vertical trailing edge line plus effective height plus vertical leading edge line minus 1).	0

27.4.5 LTDC Global Control Register (R32_LTDC_GCR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HSPOL	VSPOL	DEPOL	PCPOL	Reserved											

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															LTDCEN

Bit	Name	Access	Description	Reset value
31	HSPOL	RW	Horizontal sync polarity: 1: Horizontal sync active low; 0: Horizontal sync active high.	0
30	VSPOL	RW	Vertical sync polarity: 1: Vertical sync active high; 0: Vertical sync active low.	0
29	DEPOL	RW	Non-data enable polarity: 1: Non-data enable polarity active high; 0: Non-data enable polarity active low.	0
28	PCPOL	RW	Pixel clock polarity: 1: Pixel clock active high; 0: Pixel clock active-low.	0
[27:1]	Reserved	RO	Reserved	0x444
0	LTDCEN	RW	LCD-TFT controller enable bit: 1: Enable; 0: Disable.	0

27.4.6 LTDC Shadow Reload Configuration Register (R32_LTDC_SRCR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														VBR	IMR

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	RO	Reserved	0
1	VBR	RW	Vertical blanking Reload: 1: Shadow register reloads during the vertical blanking period (starting at the first line after the active display area); 0: No effect. *Note: This bit is set to 1 by software and cleared to 0 by hardware only after reloading.*	0
0	IMR	RW	Immediate reload:	0

			1: Shadow register immediate reload; 0: No effect. *Note: This bit is set to 1 by software and is only cleared by hardware after reload.*	

27.4.7 LTDC Background Color Configuration Register (R32_LTDC_BCCR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								BCRED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BCGREEN[7:0]								BCBLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:16]	BCRED[7:0]	RW	Background red value.	0
[15:8]	BCGREEN[7:0]	RW	Background green value.	0
[7:0]	BCBLUE[7:0]	RW	Background blue value.	0

27.4.8 LTDC Interrupt Enable Register (R32_LTDC_IER)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												RRIE	DMA_DISABLE	FUIE	LIE

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	RRIE	RW	Register reload interrupt enable: 1: Enable; 0: Disable.	0
2	DMA_DISABLE	RW	DMA disable function bit: 1: DMA is disabled. When this bit is set to 1, the current data address is forced to switch to the CFBADD after reloading, and the internal FIFO is cleared. It is recommended to use this bit to switch to the address during line blanking or field blanking (configure the address, immediately reload the address and then pull this bit low to switch to the destination address);	0

			0: DMA Enable. *Note: Only applicable to chips whose fifth digit from the bottom of the lot number is greater than 0.*	
1	FUIE	RW	FIFO underflow interrupt enable: 1: Enable; 0: Disable.	0
0	LIE	RW	Line interrupt enable: 1: Enable; 0: Disable.	0

27.4.9 LTDC Interrupt Status Register (R32_LTDC_ISR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												RRIF	Reserved	FUIF	LIF

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	RRIF	RO	Register overload interrupt flag: 1: Generates a register overload interrupt during vertical blanking reload (and on the first line after reaching the active area); 0: Does not generate a register overload interrupt.	0
2	Reserved	RO	Reserved	0
1	FUIF	RO	FIFO underflow interrupt flag: 1: A FIFO underflow interrupt occurs when one of the layer FIFOs is empty and pixel data is read from the FIFO. 0: No FIFO underflow interrupt has occurred.	0
0	LIF	RO	Line interrupt flag: 1: Generates a line interrupt when reaching the programmed line; 0: No line interrupt generated.	0

27.4.10 LTDC Interrupt Clear Register (R32_LTDC_ICR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CRRI	Reser	CFUI	CLIF

	F	ved	F	

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	CRRIF	RW1Z	Register reload interrupt clear flag, write 1 to clear 0, write 0 is invalid.	0
2	Reserved	RO	Reserved	0
1	CFUIF	RW1Z	FIFO underflow interrupt clear flag, write 1 to clear 0, write 0 is invalid.	0
0	CLIF	RW1Z	Line interrupt clear flag, write 1 to clear 0, write 0 is invalid.	0

27.4.11 LTDC Line Interrupt Position Configuration Register (R32_LTDC_LIPCR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					LIPOS[10:0]										

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
[10:0]	LIPOS[10:0]	RW	Line interrupt position	0

27.4.12 LTDC Current Position Status Register (R32_LTDC_CPSR)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CXPOS[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CYPOS[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	CXPOS[15:0]	RW	Current X position.	0
[15:0]	CYPOS[15:0]	RW	Current Y position.	0

27.4.13 LTDC Current Display Status Register (R32_LTDC_CDSR)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												HSYN	VSYN	HDE	VDES

	CS	CS	S	

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	HSYNCS	RO	Horizontal synchronization display status: 1: Active high; 0: Active low.	1
2	VSYNCS	RO	Vertical synchronization display status: 1: Active high; 0: Active low.	1
1	HDES	RO	Horizontal data enabled display status: 1: Active high; 0: Active low.	1
0	VDES	RO	Vertical data enable display status: 1: Active high; 0: Active low.	1

27.4.14 LTDC Layer Control Register (R32_LTDC_LCR)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											CLUTEN	Reserved		COLKEN	LEN

Bit	Name	Access	Description	Reset value
[31:5]	Reserved	RO	Reserved	0
4	CLUTEN	RW	Color lookup table enable: 1: Enable; 0: Disable. *Note: CLUT is only relevant for L8, AL44, and AL88 pixel formats.*	0
[3:2]	Reserved	RO	Reserved	0
1	COLKEN	RW	Color keyboard enable: 1: Enable; 0: Disable.	0
0	LEN	RW	Layer enable: 1: Enable; 0: Disable.	0

27.4.15 LTDC Layer Window Horizontal Position Configuration Register (R32_LTDC_LWHPCR)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				WHSPPOS[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				WHSTPOS[11:0]											

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	WHSPPOS[11:0]	RW	Window horizontal end position: WHSPPOS[11:0] >= AHBP[11:0] + 1.	0
[15:12]	Reserved	RO	Reserved	0
[11:0]	WHSTPOS[11:0]	RW	Window horizontal start position: WHSTPOS[11:0] <= AAW[11:0].	0

For example, the R32_LTDC_BPCR register is configured as 0x000E0005 (AHBP[11:0] is 0xE), and the R32_LTDC_AWCR register is configured as 0x028E01E5 (AAW[11:0] is 0x28E). To configure the horizontal position of a 630x460 window (with a horizontal starting offset of 5 pixels within the valid data area).

1) First pixel of the layer window: WHSTPOS[11:0] should be programmed to 0x14 (0xE + 1 + 0x5);

2) Last pixel of the layer window: WHSPPOS[11:0] should be programmed to 0x28A.

27.4.16 LTDC Layer Window Vertical Position Configuration Register (R32_LTDC_LWVPCR)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					WVSPPOS[10:0]										
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					WVSTPOS[10:0]										

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
[26:16]	WVSPPOS[10:0]	RW	Window vertical end position: WVSPPOS[10:0] >= AVBP[10:0] + 1。	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	WVSTPOS[10:0]	RW	Window vertical start position: WVSTPOS[10:0] <= AAH[10:0]。	0

For example, the R32_LTDC_BPCR register is configured as 0x000E0005 (AVBP[10:0] is 0x5), and the R32_LTDC_AWCR register is configured as 0x028E01E5 (AAH[10:0] is 0x1E5). To configure the vertical position of a 630x460 window (with a vertical starting offset of 8 lines in the valid data area):

1) First line of the layer window: WVSTPOS[10:0] should be programmed to 0xE (0x5 + 1 + 0x8);

2) Last line of the layer window: WVSPPOS[10:0] should be programmed to 0x1DA.

27.4.17 LTDC Layer Chroma Key Configuration Register (R32_LTDC_LCKCR)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CKRED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CKGREEN[7:0]								CKBLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:16]	CKRED[7:0]	RW	Chroma key red value.	0
[15:8]	CKGREEN[7:0]	RW	Chroma key green value.	0
[7:0]	CKBLUE[7:0]	RW	Chroma key blue value.	0

27.4.18 LTDC Layer Pixel Format Configuration Register (R32_LTDC_LPFCR)

Offset address: 0x44

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													PF[2:0]		

Bit	Name	Access	Description	Reset value
[31:3]	Reserved	RO	Reserved	0
[2:0]	PF[2:0]	RW	Pixel format: 000: ARGB8888; 001: RGB888; 010: RGB565; 011: ARGB1555; 100: ARGB4444; 101: L8 (8-bit Luminance); 110: AL44 (4-bit Alpha, 4-bit Luminance); 111: AL88 (8-bit Alpha, 8-bit Luminance).	0

27.4.19 LTDC Layer Default Color Configuration Register (R32_LTDC_LDCCR)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								DCRED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DCGREEN[7:0]								DCBLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:16]	DCRED[7:0]	RW	Default color red.	0

[15:8]	DCGREEN[7:0]	RW	Default color green.	0
[7:0]	DCBLUE[7:0]	RW	Default color blue.	0

27.4.20 LTDC Layer Color Frame Buffer Address Register (R32_LTDC_LCFBAR)

Offset address: 0x54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CFBADD[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CFBADD[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	CFBADD[31:0]	RW	Start address of color frame buffer.	0

27.4.21 LTDC Layer Color Frame Buffer Length Register (R32_LTDC_LCFBLR)

Offset address: 0x58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			CFBP[12:0]												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CFBLL[12:0]												

Bit	Name	Access	Description	Reset value
[31:29]	Reserved	RO	Reserved	0
[28:16]	CFBP[12:0]	RW	Color frame buffer interval (in bytes): This field defines the increment from the start of one row of pixels to the start of the next row.	0
[15:13]	Reserved	RO	Reserved	0
[12:0]	CFBLL[12:0]	RW	Color frame buffer line length: Defines the length of a row of pixels (in bytes) + 31. The row length is calculated as: Effective Width × Bytes per Pixel + 31.	0

27.4.22 LTDC Layer Color Frame Buffer Line Number Register (R32_LTDC_LCFBLNR)

Offset address: 0x5C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					CFBLNBR[10:0]										

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0

[10:0]	CFBLNBR[10:0]	RW	Number of lines in frame buffer: This field defines the number of lines in the buffer, corresponding to the effective height.	0

Note: The line number and line length settings define the amount of data obtained from each frame for each layer. If configured below the required bytes, a FIFO underflow interrupt will be generated (if enabled). On the other hand, the starting address and spacing settings define the correct starting position of each line in the memory.

27.4.23 LTDC Layer CLUT Write Register (R32_LTDC_LCLUTWR)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CLUTADD[7:0]								RED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GREEN[7:0]								BLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	CLUTADD[7:0]	WO	Configure the CLUT address of each RGB value.	0
[23:16]	RED[7:0]	WO	Red value.	0
[15:8]	GREEN[7:0]	WO	Green value.	0
[7:0]	BLUE[7:0]	WO	Blue value.	0

Note: (1) The CLUT write register should only be configured during the blanking period or when the layer is disabled.

(2) The CLUT can be enabled or disabled via the R32_LTDC_LCR register.

(3) The CLUT is only relevant for the L8, AL44, and AL88 pixel formats.

Chapter 28 Addressable RGB (ARGB)

The chip has a built-in set of ARGB (Addressable RGB) modules that support the configurable output of code 0 and code 1 high- and low-level time and period, and supports DMA.

The ARGB module can realize timing control of single data line RGB light strips and supports single line RGB cascade. Compared with traditional RGB, it can support driving more different types of ARGB light control ICs to produce more light effects.

28.1 Main Features

- Support configurable output of code 0 and code 1 high- and low-level time and period
- Support DMA

28.2 Function Description

28.2.1 Polling and Interrupts

When using polling or interrupt mode to drive this peripheral, it is important to note that the ARGB internal buffer is only a single byte. During a data transmission cycle, the hardware will not detect whether new data is written to the R32_ARGB_DATAR register; when the 1-byte data transmission is completed, if the new data cannot be filled in time, the hardware will repeatedly send the last byte that has been sent.

Therefore, in order to ensure that the data sent is correct, the real-time nature of data writing must be ensured, and the writing operation of the next byte needs to be completed within the time window when TXEIF is set and BYTEIF is set.

Configuration phase

- Configure the MODE and LSB bits of the R32_ARGB_CTLR register to select the data frame structure.
- Configure the R32_ARGB_CCRL, R32_ARGB_CCRH, R32_ARGB_DAT_ARR, and R32_ARGB_RST_ARR registers to set the physical protocol parameters.
- Configure the R32_ARGB_DAT_CYC register to set the data frame transmission length.
- Enable the TE bit in the R32_ARGB_CTLR register.

If using the interrupt method: Since the status of the TXEIF flag must be monitored during data transmission, you must additionally set the TXEIE bit in the R32_ARGB_CTLR register and enable the ARGB interrupt.

Data transmitting phase

- Check the STATUS bit in the R32_ARGB_STATR register to confirm that the controller is currently in the IDLE state.
- Write data to the R32_ARGB_DATAR register to trigger a new frame transmission.
- Wait for the TXEIF bit to be set, then write the next set of data within the valid time window. When using the interrupt method, the next set of data must be written within the interrupt service routine upon the TXE

interrupt being triggered.

- Check the TCIF flag or use a software counter to determine whether the current data frame has been fully transmitted.
- If you need to update the display content of the lighting control IC, simply repeat the configuration and transmission steps described above.

28.2.2 DMA

The DMA configuration process is different from the polling method. The main difference is: in DMA mode, the detection and data filling of TXEIE during the data transmission phase are completed by the ARGB peripheral and the DMA controller hardware. As long as the bus is not continuously occupied by a very high priority, the hardware can always complete data filling in time. Therefore, it is recommended to use DMA to drive external devices, which can improve the reliability of data transmission.

Configuration phase

- Configure the MODE and LSB bits of the R32_ARGB_CTLR register to select the data frame structure, and enable the DMA_EN bit.
- Configure the R32_ARGB_CCRL, R32_ARGB_CCRH, R32_ARGB_DAT_ARR, and R32_ARGB_RST_ARR registers to set the physical protocol parameters.
- Configure the R32_ARGB_DAT_CYC register to set the data frame transmission length.
- Configure the DMA-related parameters and set the DMA transfer width to 1 byte.

Data transmitting phase

- Enable the TE bit in the R32_ARGB_CTLR register to initiate data transmission.
- Check the TCIF flag, or wait for the TCIF and RSTIF flags according to the frame protocol, to determine the current transmission status.
- If you need to update the display content of the lighting control IC, simply repeat the configuration and transmission steps described above.

28.3 Register Description

Table 28-1 ARGB related register list

Name	Access address	Description	Reset value
R32_ARGB_CTLR	0x40025000	ARGB control	0x00000000
R32_ARGB_DAT_ARR	0x40025004	ARGB 0/1 code cycle configuration register	0x000003FF
R32_ARGB_RST_ARR	0x40025008	ARGB RESET code width configuration register	0x0001FFFF
R32_ARGB_CCRL	0x4002500C	ARGB code 0 high level T0H width configuration register	0x00000000
R32_ARGB_CCRH	0x40025010	ARGB code 1 high level T1H width configuration register	0x00000000

R32_ARGB_DAT_CYC	0x40025014	ARGB data refresh cycle length configuration register	0x00000000
R32_ARGB_DATAR	0x40025018	ARGB data register	0x00000000
R32_ARGB_STATR	0x4002501C	ARGB status register	0x00000020
R32_ARGB_CNTR	0x40025020	ARGB counter register	0x00000000

28.3.1 ARGB Control Register (R32_ARGB_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		TXEIE	HALFIE	BYTEIE	TCIE	RSTIE	FRHIE	Reserved				LSB	MODE	DMAEN	TE

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
13	TXEIE	RW	Transmit buffer empty interrupt enable: 1: Enabled; 0: Disabled.	0
12	HALFIE	RW	Enable interruption when more than half of the data refresh cycle has elapsed: 1: Enabled; 0: Disabled.	0
11	BYTEIE	RW	Byte transmission completion interrupt enable: 1: Enabled; 0: Disabled.	0
10	TCIE	RW	Transmission completion interrupt enable: 1: Enabled; 0: Disabled.	0
9	RSTIE	RW	RESET code transmission completion interrupt enable: 1: Enabled; 0: Disabled.	0
8	FRHIE	RW	Interrupt enable at the end of data refresh cycle: 1: Enabled; 0: Disabled.	0
[7:4]	Reserved	RO	Reserved	0
3	LSB	RW	Data transmission mode enable bit: 1: Low-order data is transmitted first; 0: High-order data is transmitted first.	0
2	MODE	RW	Transmission mode configuration bits: 1: Transmit the data code first, then the RESET	0

			code; 0: Transmit the RESET code first, then the data code.	
1	DMAEN	RW	DMA enable: 1: Enabled; 0: Disabled.	0
0	TE	RW	Transmit enable: 1: Enabled; 0: Disabled.	0

28.3.2 ARGB 0/1 Code Cycle Configuration Register (R32_ARGB_DAT_ARR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						DAT_ARR[9:0]									

Bit	Name	Access	Description	Reset value
[31:10]	Reserved	RO	Reserved	0
[9:0]	DAT_ARR[9:0]	RW	0/1 code cycle, unit: system clock cycle.	0x3FF

28.3.3 ARGB RESET Code Width Configuration Register (R32_ARGB_RST_ARR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														RST_ARR[16]	

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RST_ARR[15:0]															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
[16:0]	RST_ARR[16:0]	RW	Reset code period, unit: system clock cycle.	0x1FFFF

28.3.4 ARGB Code 0 High Level T0H Width Configuration Register (R32_ARGB_CCRL)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CCRL[9:0]									

Bit	Name	Access	Description	Reset value
[31:10]	Reserved	RO	Reserved	0
[9:0]	CCRL[9:0]	RW	Code 0 high level T0H width, unit: system clock cycle.	0

28.3.5 ARGB Code 1 High Level T1H Width Configuration Register (R32_ARGB_CCRH)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CCRH[9:0]									

Bit	Name	Access	Description	Reset value
[31:10]	Reserved	RO	Reserved	0
[9:0]	CCRH[9:0]	RW	Code 1 high level T1H width, unit: system clock cycle.	0

28.3.6 ARGB Data Refresh Cycle Length Configuration Register (R32_ARGB_DAT_CYC)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DAT_CYC[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	DAT_CYC[15:0]	RW	ARGB data refresh cycle length, the actual value is the value of this field + 1, unit: byte.	0

28.3.7 ARGB Data Register (R32_ARGB_DATAR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DATA[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0

[7:0]	DATA[7:0]	RW	ARGB transmit data.	0

28.3.8 ARGB Status Register (R32_ARGB_STATR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BYTE_CNT[15:0]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					BIT_CNT[2:0]			STATE [1:0]		TX EIF	HA LFI F	BY TEI F	TCI F	RS TIF	FR HIF

Bit	Name	Access	Description	Reset value
[31:16]	BYTE_CNT[15:0]	RO	Transmit byte counter.	0
[15:11]	Reserved	RO	Reserved	0
[10:8]	BIT_CNT[2:0]	RO	Transmit bit counter.	0
[7:6]	STATE[1:0]	RO	State machine indicator signals: 00: Idle; 01: Data transmission phase; 10: RESET code transmission phase; 11: Reserved.	0
5	TXEIF	RO	Transmit buffer empty interrupt flag, writing DATAR will clear this flag. *Note: When this bit is set, it means that the contents in DATAR have been fetched into the shift register, the DATAR register has been released, and new data can be written.*	1
4	HALFIF	RW1Z	The interrupt flag is sent halfway through the data refresh cycle and cleared by writing 1. *Note: When this bit is set, it means that the number of bytes sent in this round of data sending cycle has reached half of DAT_CYC.*	0
3	BYTEIF	RW1Z	Byte transmission completion interrupt flag, clear by writing 1. *Note: When this bit is set, it means that the contents of the shift register have been sent.*	0
2	TCIF	RW1Z	Transmit completion interrupt flag, clear by writing 1. *Note: When this bit is set, it means that the number of bytes sent in this round of data sending cycle has reached the number set by DAT_CYC.*	0
1	RSTIF	RW1Z	RESET code transmission completion interrupt	0

			flag, clear by writing 1. *Note: When this bit is set, it means that the completion of the RST phase of the current cycle.*	
0	FRHIF	RW1Z	The data refresh cycle transmission completion interrupt flag is cleared by writing 1. *Note: When this bit is set, it means that the end of the current cycle.*	0

28.3.9 ARGB Counter Register (R32_ARGB_CNTR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															CNT[16]

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
[16:0]	CNT[16:0]	RO	ARGB base counter.	0

Chapter 29 Ethernet Controller (ETH)

The product provides an Ethernet controller (MAC) that complies with the IEEE802.3-2002 standard, acting as the data link layer. It also has built-in 10Mbps and 100Mbps Ethernet PHY physical layer transceivers, supports speed adaptation, and implements Ethernet communication on a single chip. When applying, combine the TCP/IP protocol stack interface to realize the development of network products.

29.1 Main Features

29.1.1 MAC Features

- Support 10M/100M Ethernet
- Support full-duplex and half-duplex
- Support automatic insertion of frame header sequence and SFD
- Support automatic insertion and check CRC
- Support automatic calculation and verification of the checksum of IP/ICMP/TCP/UDP protocol packets
- Support transmitting frame length control
- Support transmitting gap adjustment
- Support VLAN frame
- Support perfect address filtering and HSAH filtering of frame receiving addresses
- Support frame sending address filtering
- Support multicast broadcast frame reception control
- Support promiscuous mode
- Support magic frame and custom wake-up frame to wake up microcontroller
- Support dedicated Ethernet wake-up interrupt entry
- Support data loopback at the link layer

29.1.2 DMA Features

- MAC dedicated DMA, minimize CPU operations
- Support access to RAM in a byte-aligned manner
- Manage send and receive buffers in the form of descriptors
- Part of the status feedback sent by the receiver is in the descriptor
- Can dynamically modify descriptors and buffers that are not in use
- Support manual stop or start
- Support chained or ringed formal connection descriptors
- Support multiple interrupt sources such as transceiver completion interrupt

29.1.3 MMC Module Features

- Support manual reset, stop or freeze
- Support multiple interrupts
- Multiple counters supporting multiple counting modes for sending and receiving

29.1.4 PTP Module Features

- Support IEEE 1588 protocol

- Support to automatically save the current moment when sending and receiving
- Come with a 32-bit second timer and a 32-bit signed sub-second timer
- Support to adjust the time in 2 ways: fine adjustment and coarse adjustment
- Single interrupt source

29.1.5 Internal 10BASE-T/100BASE-TX Physical Layer Features

- Management via SMI interface
- Support 10BASE-T and 100BASE-TX and auto-negotiation
- Support power down mode
- Support Auto-MDIX switching TX/RX and automatic identification of positive and negative signal lines
- Support half-duplex, full-duplex
- Support UTP CAT5E and CAT6 twisted-pair cables, with a transmission distance of 120 meters
- Support WOL network wake-up
- Support interrupt function

29.1.6 Frame Filtering

Use the MAC frame filter to perform perfect filtering or HASH filtering on the destination MAC address and source MAC address of the received frame, and can set the frame that does not pass the filtering to be discarded, received, or marked in the receive descriptor. Please read the description of the MAC frame filter register (R32_ETH_MACFFR) for this function. The MAC has 4 built-in MAC address registers, of which MAC address register 0 is used by default to store its own MAC address, and the remaining 3 MAC address registers can be used for perfect filtering, compared with the source address or destination address of the received frame. After setting R32_ETH_MACFFR: RA, all received frames will be received, but the corresponding status will be marked in the status field of the first word of the descriptor after that, if R32_ETH_MACFFR:PM is set, there will have a similar effect, but the state will not be marked. After setting R32_ETH_MACFFR: DAIF/SAIF, the result will be flipped. Those that pass the filtering will be discarded or marked, and those that do not pass the filtering will be transferred to the buffer in RAM.

29.1.6.1 Unicast Filtering

The MAC uses the HPF bit and the HU bit to determine whether the unicast frame is HASH filtering or perfect address filtering.

29.1.6.2 Multicast Filtering

The MAC uses the HPF bit and the HM bit to determine whether the multicast frame is HASH filtered or perfect address filtering. When the PAM bit is set, all multicast packets can pass through the filter.

29.1.6.3 Broadcast Filtering

By setting the BFD bit, the MAC can block all broadcast packets.

29.1.6.4 Source Address Filtering Selection

The MAC address register can be enabled by setting the AE bit in the MAC address register, while setting the SA bit determines whether the MAC address register is to be compared as a source or target address sample.

29.1.6.5 Summary

The filter settings for target address and source address are shown in Table 29-1 and Table 29-2 respectively.

Table 29-1 Acceptance of the setting of each bit of R32_ETH_MACFFR to the target MAC address of the received frame

Frame type	PM	HPF	HU	DAIF	HM	PAM	Effect
Broadcast frame	1	-	-	-	-	-	Pass
	0	-	-	-	-	-	Fail (BFD set)
Unicast frame	1	-	-	-	-	-	All frames pass
	0	-	0	0	-	-	Pass when perfect filter matches
	0	-	0	1	-	-	Fail when perfect filter matches
	0	0	1	0	-	-	Pass when HSAH filter matches
	0	0	1	1	-	-	Fail when HSAH filter matches
	0	1	1	0	-	-	Pass when perfect filter or HSAH filter matches
	0	1	1	1	-	-	Fail when perfect filter or HSAH filter matches
Multicast frame	1	-	-	-	-	-	Pass
	-	-	-	-	-	1	Pass
	0	-	-	0	0	0	Pass when perfect filter matches
	0	0	-	0	1	0	Pass when HSAH filter matches
	0	1	-	0	1	0	Pass when perfect filter or HSAH filter matches
	0	-	-	1	0	0	Fail when perfect filter matches
	0	0	-	1	1	0	Fail when HSAH filter matches
	0	1	-	1	1	0	Fail when perfect filter or HSAH filter matches

Table 29-2 Acceptance of the setting of each bit of R32_ETH_MACFFR to the source MAC address of the received frame

Frame Type	RA	SAIF	SAF	Effect
Unicast frame	1	-	-	All frames pass
	0	0	0	Passes on perfect filter match, marks but does not discard failed frames
	0	1	0	Do not pass on perfect filter match, mark but do not discard failed frames
	0	0	1	Pass when perfect filter matches, discard failed frames
	0	1	1	Fail when perfect filter matches, discard failed frames

Note: '-' indicates that the setting of this bit is of no concern.

29.1.7 MMC

The function of the management counter (MMC) is mainly to count the receiving and sending status of various instruction frames and the running status of the MAC. It can generate settings and generate interrupts. In general, we can get the number of currently received good frames by receiving the "good" frame counter (MMCRGUFCR),

get the number of successfully sent frames by sending the "good" frame counter (MMCTGFCR), and get the number of frames successfully sent by receiving the CRC error frame counter (MMCRFCECR) to check whether there is a frame with CRC error received, in general, if the data is correct but the CRC is wrong, it can be considered that there is a problem with the RGMII line layout.

Users need to be clear about a concept: what kind of frame is a "good" frame. Under normal circumstances, as long as a frame starts to send, its frame length meets the requirements (Automatic padding is turned on), and automatic CRC calculation is turned on, then it will be a "good" frame. And a frame is received, as long as its CRC is correct, the frame length is within the Ethernet frame length range, the frame length and the value of the length field are the same (If the length type field indicates the length), or there is no misalignment, then it will be considered a "good" received frame.

29.1.8 PMT

29.1.8.1 Overview

The function of the PMT (Power management) part is to wake up the microcontroller from a low-power mode via Ethernet. The Ethernet controller supports 2 types of frames to wake up the system, magic frames and (Remote) wake-up frames. When Ethernet receives these 2 kinds of frames, because the microcontroller is in low-power mode, it does not necessarily generate a receive interrupt, even if it is legal, but if it is recognized by the MAC's magic frame and wake-up frame, it will generate PMT interrupt, PMT interrupt is a separate interrupt from the Ethernet interrupt. By querying the PMT control and status registers, you can find out which Ethernet frame generated the interrupt.

29.1.8.2 Magic Frame

The magic frame (Magic package, also called magic data packet) defined by AMD is a commonly used Ethernet frame to wake up the microcontroller. It has a fixed and special frame format, that is, the target network card can receive it through frame filtering. It can be encapsulated into a broadcast frame, and there are 6 consecutive bytes of full height (OXFF) in the frame, followed by the MAC address of the target network card repeated 16 times. This combination can exist anywhere in the frame, so magic frames can be encapsulated into Mac frames or IP packets or even UDP packets. Below is the format of the magic frame.

xx xx xx xx xx xx xx xx(Unlimited previous data, even no previous data)—ff ff ff ff ff ff 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 xx xx xx xx xx xx (Some network cards require the magic frame to be attached with a password at the end).

29.1.8.3 Wakeup Frame

Due to the limitation of the format of the magic frame, the format of the (Remote) wakeup frame can be defined by the user. As long as the Ethernet frame passes through the remote wakeup frame filter register (ETH_MACRWUFFR) group, it is regarded as a wakeup frame. Users can define the wakeup frame that meets their needs by setting the remote wakeup frame filter register group. It should be noted that the remote wakeup frame filter register group has only single address entry, the user can write data to it through 8 consecutive write operations, and then read the data originally written into the remote wakeup frame filter register by reading data 8 consecutive times.

The structure of the remote wakeup frame filter register group is as follows:

Table 29-3 Structure of the remote wakeup frame filtering register bank

Filter register 0	Filter byte mask of register 0							
Filter register 1	Filter byte mask of register 1							
Filter register 2	Filter byte mask of register 2							
Filter register 3	Filter byte mask of register 3							
Filter register 4	Reserved	Filter 3 command	Reserved	Filter 2 command	Reserved	Filter 1 command	Reserved	Filter 0 command
Filter register 5	Filter 3 offset		Filter 2 offset		Filter 1 offset		Filter 0 offset	
Filter register 6	Filter 1 CRC-16				Filter 0 CRC-16			
Filter register 7	Filter 3 CRC-16				Filter 2 CRC-16			

As can be seen from the above table, the four fields of the filter register, byte mask, command, offset, and filter, work together to determine whether a frame is a remote wake-up frame. In fact, four different frames that meet the requirements can be set.

The highest bit of the 32-bit byte mask must be 0. Bit[30:0] corresponds to the first 31 bytes of data defined by the offset field. If any bit is set to 1, it indicates that the corresponding byte participates in CRC- 16 parity, and up to 31 bytes participate in calculation;

The highest bit of the 4-bit command field indicates what kind of frame it works on. 1 is only valid for multicast addresses, 0 is only valid for unicast addresses; the 2nd and 1st bits of the command field are reserved, and bit 0 is Enable bit, set high to enable this set of filters;

The offset field indicates how many bytes are offset from the beginning of the frame to calculate the CRC16 value. The minimum value is 12. If the offset field is 12, the CRC16 value is calculated starting from the 13th byte at the beginning. (Parametric model using CRC-16-IBM with default parameters)

The filter stores the value of the CRC result that the user expects to calculate. The MAC will compare the value of the CRC16 calculated by itself with the value in this field. If they are consistent, it will be regarded as a (remote) wake-up frame. If used If the wake-up frame interrupt and PMT interrupt are disabled, a PMT interrupt will also be generated.

In addition, according to the bit definition of the PMT control status register, if the GU bit is set, the unicast frame that passes the frame filter will also be considered a wake-up frame.

29.1.9 IEEE1588 PTP

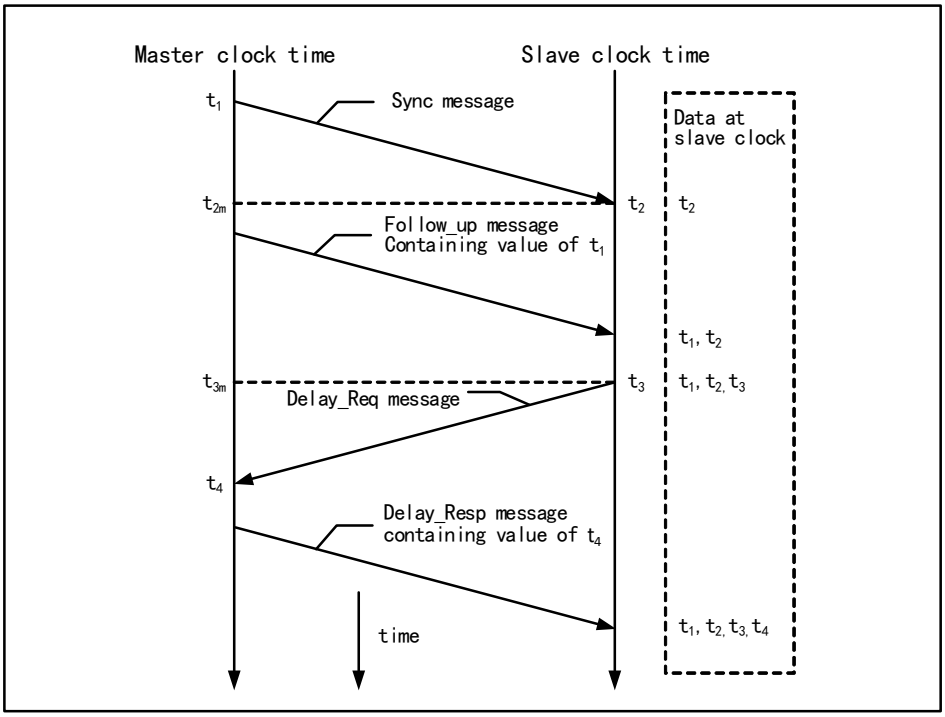
29.1.9.1 PTP Principle and Implementation

The IEEE1588 standard defines a set of precise protocols for acquiring time, and its goal is to achieve time synchronization within 10 microseconds of error. The original NTP protocol can already achieve time synchronization at the level of 200 microseconds, but in fact this level cannot meet the time synchronization requirements in the field of industrial automation, so the Network Precision Clock Synchronization Committee drafted the PTP protocol, which was approved by the IEEE Standards Committee at the end of 2002 as the IEEE1588 standard.

The implementation of the PTP protocol requires both the host and the slave to accurately record the time when the MAC receives and sends Ethernet frames, which requires both the host and the slave to have their own set of high-precision time counters. Then, the master and slave supporting PTP perform time synchronization through a set of

procedures, and the slave can obtain the time difference between itself and the master and correct it. The following figure shows the process of master-slave time synchronization:

Figure 29-1 IEEE1588 PTP protocol synchronization message timing diagram



Step-by-step description:

- The master sends a syn message to the slave, and the slave receives this message and records the local time t_2 when the syn message is received;
- The master sends a follow_up message to the slave, including the master time t_1 when the master sends the syn message;
- The slave sends a delay_req message to the master, and the slave records the sending time t_3;
- The master sends a delay_resq message to the slave, including the reception time t_4 of the delay_req message;

In actual use, the host sends out a syn message every 2 seconds, and every other syn message can be regarded as the follow_up message of the previous syn message, and they will all be accompanied by the sending time of the last syn message. Through this process, the slave can know the delay time Tdelay from the master network to the slave network, and then calculate the time offset between the master time and the slave time.

$$Tdelay = \frac{(t_2-t_1)+(t_4-t_3)}{2}$$

The time sent by either host minus the offset is the host's time.

The synchronization process of PTP is generally implemented through the UDP protocol, of course, the user can also implement a custom protocol. PTP is highly dependent on the latency stability of the intranet.

29.1.9.2 Local Time Update and Correction

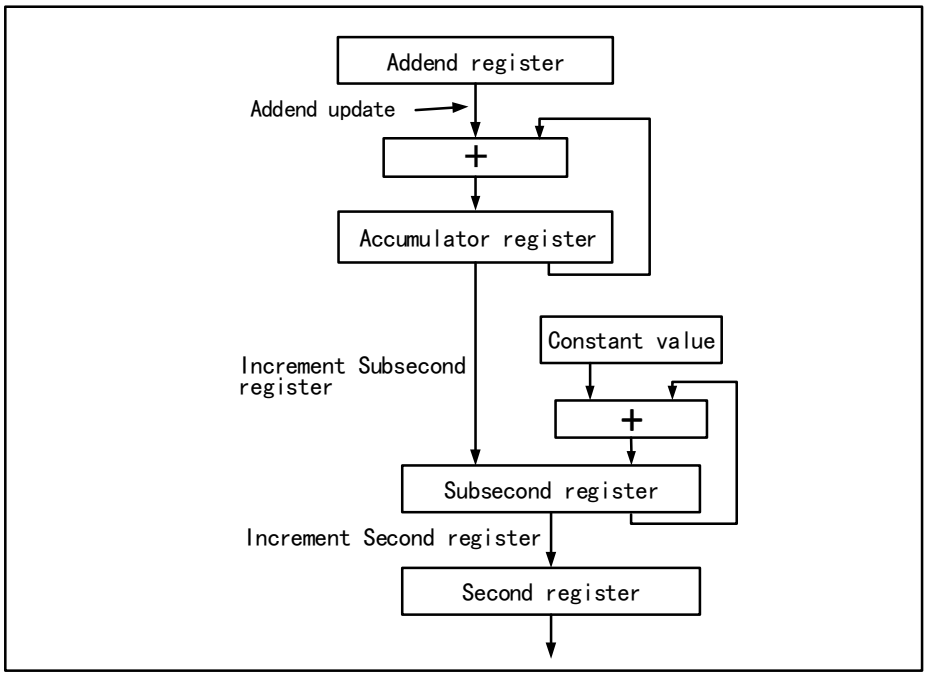
In order to implement the PTP host, the device needs to have a high-precision time counter locally, at least to the nanosecond level. The PTP module of the microcontroller's Gigabit Ethernet counter has a 32-bit counter in seconds, and a 31-bit sub-second counter. When the sub-second counter overflows, it will cause the second counter to

increment automatically, so the resolution of the local time. The rate can be done around 0.46 nanoseconds.

There are 2 ways to update the local time, namely coarse correction and fine correction. The update timing of the coarse correction mode is determined externally. When the time update is required, the time update bit (PTPTSCR: TSSTU) of the time stamp control register is set, and the PTP module of the microcontroller subtracts or adds the value of the timestamp update registers (TSHUR, TSLUR) to the second counter and sub-second counter. Coarse correction is a simpler and more convenient time update mechanism, but less precise.

The fine correction time update method is more commonly used. The update process is as follows:

Figure 29-2 Process for updating time using fine correction



The way to use the fine correction mode requires a clear understanding of the system frequency and the fine correction process. Different from the coarse correction mode, the timing of the update event in the fine correction mode is the overflow of the accumulator (32 bits, not listed in the register list. The Accumulator register in the figure) overflows, and the accumulator will automatically increment the number register in each system clock cycle. The value of (PTPTSAR, Addend register in the figure) will generate a time update event once it overflows, that is, the value of the sub-second auto-increment register (PTPSSIR, Constant Value in the figure) will be added to the sub-second counter (Sub-second register), complete time update. The second counter is incremented when the sub-second counter overflows. In fact, the time for the sub-second counter to increment by one bit is 1/(2^31)=0.46566128730...nanoseconds. The user needs to ensure that the time spent on the accumulator overflow is exactly equal to the value of the sub-second auto-increment register multiplied by the sub-second the time for the counter to increment by one bit.

The correction of local time is relatively simple. Set the time correction bit (PTPTSCR: TSSTI) of the time stamp control register, and the value of the second counter and sub-second counter will be replaced by the value of the time stamp update counter (TSHUR, TSLUR).

29.1.10 DMA Operation

29.1.10.1 Overview

The advantages of Ethernet transceivers are fast speed and large throughput. In order to maintain this advantage, it is necessary to minimize the intervention of the CPU in the process of receiving and sending Ethernet frames. Here, a dedicated DMA for Ethernet transceivers is used.

The DMA used by Ethernet is 32-bit, and it is managed by the CPU through 2 data structures: traditional control and status registers, and receive and transmit descriptors. Since the DMA is 32 bits wide, the base address of the transceiver descriptor queue in RAM is required to be aligned with 4 bytes. There is no alignment requirement for the receive/transmit buffer.

29.1.10.2 DMA Descriptor

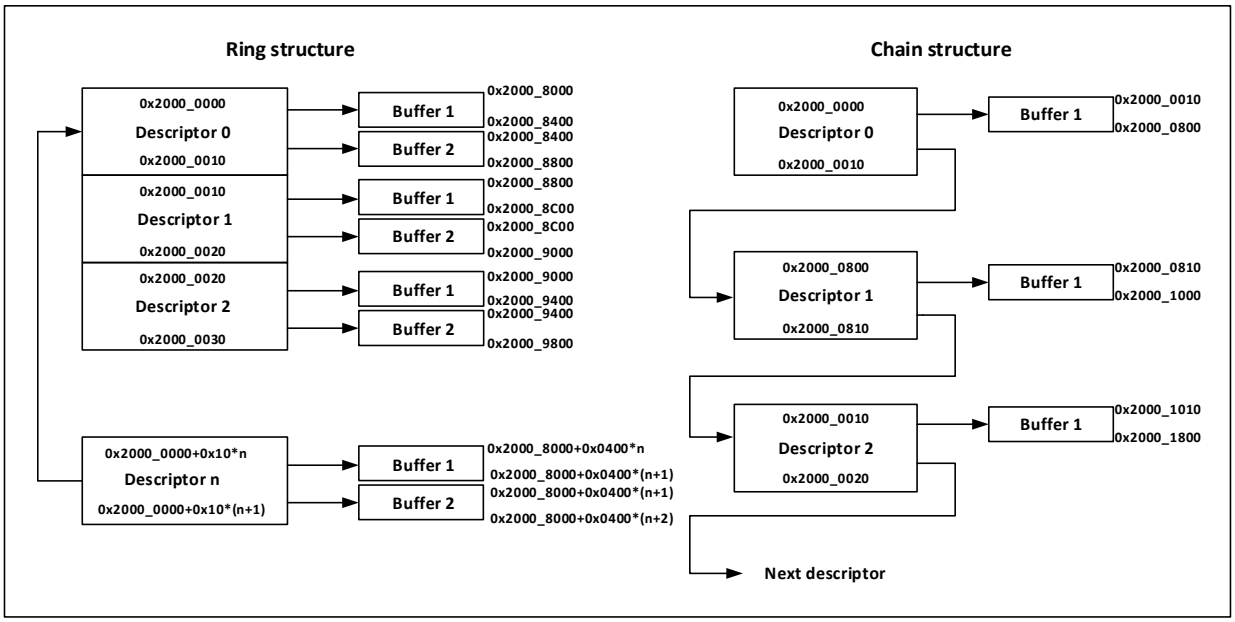
Users use traditional peripherals mainly by writing the control bits of the register, and by reading the status register to obtain the status and return information of the peripheral. These registers are independent of the core, space outside of SRAM and non-volatile memory, and actually exist, and can be called "hardware registers". Traditional communication peripherals, such as USART or SPI, have a data register to temporarily store the data sent and received, and a set of DMA to store all received data in a specific address space.

The Ethernet transceiver is different from the traditional communication transceiver because of its extremely high data transmission speed, great data throughput and unique data frame organization. The Ethernet transceiver receives a large amount of data as densely as possible. It must store the received data stream in a separate memory space in units of frames, and complete the receiving and sending operations by itself, so that the CPU can read it with a minimum of operations. And dispose of these data, release the corresponding memory space, and ensure that the DMA controller and the CPU will not conflict with the memory usage rights. The single size of the buffer that is opened up to temporarily store the Ethernet frame in the memory is generally set to the maximum packet length of the Ethernet frame. IEEE802.3 stipulates that it is 1518 bytes (Including the source-end hardware address, length or type field and CRC32 calibration). The number of buffers for sending and receiving Ethernet frames is determined by the user according to the frequency of actual interaction and the memory resources of the microcontroller.

The buffer of the Ethernet frame is organized in the form of a queue. In the receiving direction, the Ethernet frame is written into the memory by DMA through the MAC, and is read and dequeued by the CPU; in the sending direction, the Ethernet frame is written to the memory and queued by the CPU, read by DMA and pushed into the transmit FIFO and out of the queue. The depth of the queue is the number of buffers. Different from ordinary communication peripherals, the starting address and number of the buffer of the Ethernet frame are not fixed in a certain register, but are managed by a special type of data structure, which is stored in the memory. A single such data structure unit manages a buffer, which holds the start address, length of the buffer, the settings that the DMA controller needs to make when calling this buffer, the send status of the write-back when the DMA send is completed, and the next such data structure the address of. This data structure plays the role of the traditional communication peripheral control register and status register, but the actual location is in the memory, so it can be called "software register", and the official name is "DMA descriptor".

DMA descriptors are divided into 2 types: sending and receiving, and each format is fixed. The storage structure of DMA descriptors is divided into 2 types, one is linked list form, that is, the fourth word of each descriptor is the address of the next descriptor, and the DMA controller will directly read the next descriptor from TDes3/RDes3. The other is a ring structure, where all descriptors must be closely arranged, DMA control takes a descriptor from the end of the current descriptor, when the DMA controller detects the TER/RER bit of TDes4/RDes4, i.e., from the beginning of the descriptor list. The address of the descriptor array must be 4-byte aligned, and the size of a single descriptor is 16 bytes.

Figure 29-3 A buffer allocation scheme which describes the ring structure and the chain structure for reference when users allocate space.



29.1.10.3 Transmit DMA Descriptor

Table 29-4 shows the structure of the transmit descriptor.

Table 29-4 Structure of the transmit descriptor

	31							0
TDes0	OWM (31)	CTRL (30:26)	TTSE (25)	Reserved	Control （23:20）	Reserved (19:18)	TTSS (17)	State （16:0）
TDes1	Reserved （31:29）	Buffer 2-byte count （28:16）			Reserved （15:13）	Buffer 1-byte count （12:0）		
TDes2	Buffer 1 address, timestamp low							
TDes3	Buffer 2 address, address of next descriptor, timestamp high							

As can be seen from the above table, the send descriptor consists of four 32-bit words, namely TDes0, TDes1, TDes2 and TDes3. TDes0 is used to control and return the sending status, TDes1 is used to indicate the sending length, and TDes2 is used to indicate the sending The position of the buffer or the low bit of the sending timestamp is returned. TDes3 is used to indicate the address of the second sending buffer (when TCH is not set) or the address of the next descriptor (when TCH is set). When the timestamp is enabled, the high bit of the IEEE1588 timestamp is returned after sending. Each 32-bit word is described below.

Table 29-5 Definitions of TDes0 bits

31	30	29	28	27	26	25	24	23:22	21	19	19:18	17	16
OWN	IC	LS	FS	DC	DP	TTE	Res	CIC	TER	TCH	Res	TSS	IHE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

ES	JT	FF	IPE	LCA	NC	LCO	EC	VF	CC	ED	UF	DB

Bit	Name	Description
31	OWN	Descriptor own This bit indicates who occupies this descriptor. 0: This descriptor is owned by the CPU, and the CPU can modify the value of this descriptor; 1: This descriptor is owned by the DMA, and the CPU cannot modify this descriptor. When this bit is 0, it is set to 1 by the CPU after the CPU completes the operation on the descriptor and buffer. When this bit is 1, 0 is automatically written by the DMA after the DMA completes the operation on the descriptor and the buffer. In this way, the user software and hardware complete the operation handover of the descriptor and the receive/transmit buffer.
30	IC	Transmission completed interrupt enable When this bit is set, the transmission interrupt bit flag (ETH_DMASR: TS) will be set after the current frame is sent.
29	LS	Last segment This bit is set to indicate that the buffer indicated by this descriptor contains the end of the frame.
28	FS	First segment This bit is set to indicate that the buffer indicated by this descriptor contains the beginning of the frame.
27	DC	Disable automatic CRC calculation When this bit is set, the DMA controller will not calculate the CRC32 value of the Ethernet frame, and no value will be appended to the end of the frame. This bit is only valid when the FS bit is set. Also, the setting of the DP bit takes precedence over this bit.
26	DP	Disable auto-padding When this bit is set, the DMA controller will not add auto-padding for Ethernet frames less than 64 bytes. When this bit is 0, DMA will automatically add padding and CRC value to Ethernet frames less than 64 bytes, regardless of whether DC is set.
25	TTE	Timestamp transmission enable On the premise that ETH_PTPTSCR: TSE is set, the DMA controller will enable the IEEE1588 timestamp function in the Ethernet frame indicated by the face descriptor after this bit is set. This bit is only valid when FS is set.
24	Reserved	Reserved
[23:22]	CIC	Checksum and insert control 00: Prohibit insertion of checksum;

		01: Only enable the calculation and insertion of IP header checksum; 10: Reserved; 11: Enable calculation and insertion of IP header checksum and payload checksum, enable calculation of pseudo-header checksum.
21	TER	Transmit descriptor end flag setting Users set this bit to indicate to the DMA controller that the current transmit descriptor is the last descriptor of the transmit descriptor array. The DMA controller will next read the first descriptor in the transmit descriptor array.
20	TCH	Next descriptor addresses valid This bit is set to indicate that the second address is the address of the next descriptor and not the address of the next buffer. When this bit is set, the value of TBS2 has no effect. This bit is only valid when the FS bit is set. The TER bit has higher priority than this bit.
[19:18]	Reserved	Reserved
17	TSS	Transmit timestamp status DMA controller sets this bit to indicate that the transmit timestamp is captured and stored in TDes2 and TDes3.
16	IHE	IP header error DMA controller checks the IP header according to the received data: for IPv4, the DMA controller checks whether the length field of the header is correct; for IPv6, the DMA controller checks whether the header is 40 bytes. In addition, the IP protocol type must be the same as the type/length field in the Ethernet frame.
15	ES	Error summary Set by the DMA controller if an error occurs while sending a frame. The ES bit is set when one of the following bits is set: UF[TDES0:1] data underflow error bits; IPE[TDES0:12] IP data error bits; FF[TDES0:13] frame empty bits; JT[TDES0:14] Jabber timeout bits; IHE[TDES0:16] IP header error bits.
14	JT	Jabber timeout When this bit is set, a jabber timeout error occurs at the MAC transmitter. This bit is only set when the JD bit (ETH_MACCR:22) is not set.
13	FF	Frame flushed This bit is set to indicate that the DMA controller clears the frame from FIFO due to the command issued by the CPU.
12	IPE	IP payload error The MAC compares the total packet length in the IPv4 or IPv6

		header of the received TCP/UDP/ICMP packet with the actual packet length, and set this bit if they are inconsistent.
11	LCA	Lose carrier This bit is set to indicate that a carrier loss occurs when the frame is sent, that is, the CSR signal is invalid. This bit only works in half-duplex mode.
10	NC	No carrier This bit indicates that the carrier sense signal of the physical layer is invalid when the frame is sent. This bit only works in half-duplex mode.
9	LCO	Late collision This bit indicates that the frame collided after sending the preamble. This bit only works in half-duplex mode.
8	EC	Excessive collisions This bit indicates that a collision of more than 16 bits occurs when the frame is sent. If the RD bit in MACCR is set, this bit indicates that only one collision is sent. This bit only works in half-duplex mode.
7	VF	VLAN frame. Set when sending VLAN frames.
[6:3]	CC	Collision counter These bits indicate how many collisions the frame had when it was sent. No effect when EC is set. These bits only works in half-duplex mode.
2	EC	Excessive deferral This bit is set to indicate that when the DC in MACCR is set, the transmission of the frame fails due to the delay of more than 24288 bits. This bit only works in half-duplex mode.
1	UF	UF data underflow error When the DMA controller fetches data from the specified RAM and finds that the buffer is empty, the transmission enters the suspended state, and sets this bit and the relevant bit of the DMASR register.
0	DB	Deferred bit This bit is set to indicate that the transmission fails due to carrier occupancy. This bit only works in half-duplex mode.

Table 29-6 Definitions of TDes1 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			TBS2												

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			TBS1												

Bit	Name	Description
31:29	Reserved	Reserved
28:16	TBS2	Size of the transmit buffer 2.
15:13	Reserved	Reserved
12:0	TBS1	Size of the transmit buffer 1.

Table 29-7 Definitions of TDes2 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TBAD1/TTSL															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TBAD1/TTSL															

Bit	Name	Description
31:0	TBAD1/TTSL	All 32 bits of TDes2 are used as a whole unit to store the address of transmit buffer 1. In the IEEE1588 mode, after completing the transmission, TDes2 is also used to store the timestamp returned by the MAC, and the MAC will clear the OWN bit (TDes0:31). This field stores the lower 32 bits of the timestamp.

Table 29-8 Definitions of TDes3 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	20	17	16
TDAD2/TTSH															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TDAD2/TTSH															

Bit	Name	Description
31:0	TDAD2/TTSH	This field is used to store the address of the next descriptor. When TCH is not set, this field stores the address of transmit buffer 2. In the IEEE1588 mode, after completing the transmission, TDes3 is used to store the timestamp returned by the MAC, and the MAC will clear the OWN bit (TDes0:31). This field stores the high 32 bits of the timestamp.

29.1.10.4 Receive DMA Descriptor

Table 29-9 Receive descriptor structure

	31					0
RDes0	OWM(31)	Status (30:0)				
RDes1	Control (31)	Reserved (30:29)	Buffer 2-byte count (28:16)	RER (15:14)	Reserved 13	Buffer 1 byte count (12:0)
RDes2	Buffer 1 address, timestamp low					
RDes3	Buffer 2 address, address of next descriptor, timestamp high					

As can be seen from the above figure, the receive descriptor is also composed of four 32-bit words. The first 32-bit word mainly returns the status when receiving. The second 32-bit word contains the received data length. The third 32-bit word defines the address of the receiving buffer or returns the low bit of the timestamp. The fourth 32-bit word serves as the address of the second buffer (RCH is not set), the address of the next buffer (RCH is set) or returns the high bit of the timestamp.

The meanings of bits of the receive descriptor are as follows:

Table 29-10 Definitions of RDes0

31	30	29:16
OWN	AFM	FL

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ES	DE	SAF	LE	OE	VLAN	FS	LS	IPHCE	LCO	PT	RWT	RE	DE	CE	PCE

Bit	Name	Description
31	OWN	Descriptor ownership bit This bit indicates who is occupying this descriptor. 0: This descriptor is owned by the CPU, and the CPU can modify the value of this descriptor; 1: This descriptor is owned by the DMA, and the CPU cannot modify this descriptor. When this bit is 0, it is set to 1 by the CPU after the CPU completes the operations on the descriptor and buffer. When this bit is 1, 0 is automatically written by the DMA after the DMA completes the operations on the descriptor and buffer. In this way, the operation handover of the descriptor and the receive/transmit buffer between the user software and the hardware is completed.
30	AFM	Target address failed flag This flag is set if the received frame does not pass the MAC's target address filter.
[29:16]	FL	Frame length This field is valid when ES (RDes0:15) is 0. When LS (RDes0:8) is 1, this field

		indicates the length of the frame received by the DMA controller, including the CRC. When LS is 0, this field indicates the cumulative length sent to the memory by the DMA controller so far, the unit is in bytes.
15	ES	Error summary Set when the MAC detects any of the following errors: CE[RDes0:1]: CRC error; RE[RDes0:3]: Receive error; RWT[RDes0:4]: Watchdog timeout; IPHCE[RDes0:7]: Jumbo frame (note that it is necessary to distinguish whether it is a jumbo frame or an IP header error when reporting IPHCE); OE[RDes0:11]: Overflow error; DE[RDes0:14]: Descriptor error.
14	DE	Descriptor error This bit is set to indicate that the buffer indicated by the descriptor is cut off as the buffer indicated by the descriptor cannot fit the current frame, and the DMA does not occupy the next descriptor. This bit is only valid when the LS bit (RDes0 [8]) is set.
13	SAF	Source address filter fail This bit is set to indicate that the frame does not pass the source address filter of the MAC.
12	LE	Length error Set to indicate that the actual received frame length does not match the length indicated in the Ethernet Type/Length field. This bit is only valid when FT (RDes0 [5]) is set.
11	OE	Overflow error Set to indicate that the received frame is corrupted due to an overflow of the receive FIFO.
10	VLAN	VLAN tag Set to indicate that a VLAN frame is received.
9	FS	First descriptor This bit indicates that this descriptor contains the beginning of the frame.
8	LS	Last descriptor It indicates that this descriptor contains the end of the frame.
7	IPHCE	IP header checksum error This bit is set to indicate that there is an error in the IPv4 or IPv6 header. The specific reasons may be: 1. The protocol indicated by the Ethernet frame type/length field is inconsistent with the actual IP version; 2. The IP header checksum is incorrect; 3. The length indicated by the IP header is incorrect.
6	LCO	Late collision Set to indicate that a late collision occurs. This bit only works in half-duplex mode.
5	FT	Frame type indication bit.

		1 indicates that the received frame is an Ethernet type encapsulated frame (RFC 894). 0 indicates that the received frame is an IEEE802.3 type encapsulated frame (RFC1042). This bit is invalid when the frame length is less than 14 bytes. *Note: Refer to Table 29-11 for the special meaning of FT.*
4	RWT	Receive watchdog timeout Set to indicate that when the current frame is received, the watchdog times out and the current frame is truncated.
3	RE	Receive error Set to indicate that the RX_ERR signal is valid when RX_DV is valid in the process of receiving a frame.
2	DE	Dribble bit error Set to indicate that the length of the frame received by the MAC is not multiples of the 8-bit frame, and the period may be missed.
1	CE	CRC error This bit is set to indicate that the received frame has a CRC check error. This bit is valid only when the LS bit (RDes0 [8]) is set.
0	PCE	Payload checksum error This bit is set to indicate that the TCP/UDP/ICMP packet received by the MAC does not match the value indicated in the checksum field.

It can be seen that the verification mechanism is implemented in the receiving process of the MAC. In fact, the data length of this layer is described at the Ethernet frame layer (Data link layer), network layer (IPv4/IPv6) and transport layer (TCP/UDP/SCTP), and some check measures are taken to ensure the correctness of the data content. The 0th, 5th and 7th bits of Des0 are related to the checksum of the data, which is summarized in the following table.

Table 29-11 Relationship between the value of Des0:7/5/0 and the status of the received frame

RDes0:5	RDes0:7	RDes0:0	Frame status
Frame Type Indicator Bit (FT)	IP header checksum error flag bit (IPHCE)	Load checksum error (PCE)	
1	0	0	IP type frame, IP header not detected and load checksum error
1	0	1	IP type frame, load checksum error
1	1	0	IP type frame, IP header checksum error
1	1	1	IP type frame, IP header checksum error, load checksum error
0	0	0	IEEE802.3 Type Encapsulated Frame (RFC1042)
0	0	1	IP type frame, IP header checksum error not detected, not detected because load checksum is not supported
0	1	1	Not an IP type frame (e.g. ARP frame)
0	1	0	Reserved

Table 29-12 Definitions of RDes1 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DIC	Reserved		RBS2												

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RER	RCH	Res	RBS1												

Bit	Name	Description
31	DIC	Disable reception completion interrupt setting bit.
[30:29]	Reserved	Unused.
[28:16]	RBS2	Receive buffer 2 size.
15	RER	End receive descriptor flag. It indicates that the current descriptor is the last descriptor. The DMA controller will go back to the Descriptor Pair Column Base Address Register (ETH_DMARDLAR) to fetch the next descriptor.
14	RCH	Next receive descriptor address valid bit. This bit is set to indicate that in the last 32-bit word is the address of the next receive descriptor, otherwise the address of the second buffer.
13	Reserved	Unused.
[12:0]	RBS1	Receive buffer 1 size.

Table 29-13 Definitions of RDes2 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RBAD1/RTSL															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RBAD1/RTSL															

Bit	Name	Description
[31:0]	RBAD/RTSL	All 32 bits of RDes2 are used as a whole unit to store the address of the receive buffer. In the IEEE1588 mode, after completion of reception, RDes2 is also used to store the timestamp returned by the MAC, and the MAC will clear the OWN bit [RDes0:31].

Table 29-14 Definitions of RDes3 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDAD2/RTSH															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDAD2/RTSH															

Bit	Name	Description
[31:0]	RDAD/RTSH	When RCR is set, it is used to store the address of the next buffer, otherwise it is used to store the address of the second buffer. In the IEEE1588 mode, after completion of reception, RDes3 is used to store the timestamp returned by the MAC, and the MAC will clear the OWN bit [RDes0:31] to zero. This field stores the high 32 bits of the timestamp.

29.1.10.5 Data Cache Alignment

Since both the transceiver buffer and the transceiver descriptor are called by the DMA controller and actually exist in the RAM space, and the DMA is 32-bit, setting the transceiver descriptor queue requires ensuring that their starting address is in a 4-byte aligned position, but there is no such mandatory requirement for setting the buffer.

In order to improve efficiency, it is most appropriate to receive an Ethernet frame in a buffer. For example, setting the buffer to 1518 bytes can accommodate the largest ordinary Ethernet frame, including the source and destination address fields, length type fields, data padding fields, and CRC check fields. It should be noted that the maximum frame length of tagged VLAN frames is 4 bytes longer than ordinary Ethernet frames, that is, 1522 bytes. If the user wants to achieve 4-byte alignment, a buffer can be 1524 bytes.

29.1.10.6 DMA Transceiver Configuration

The data received and sent is automatically dumped and pushed by the DMA, but if a fatal error is encountered, the DMA will stop and update the DMA status register. The user needs to manually start the DMA after initializing the DMA to continue running.

29.1.10.6.1 Transmit DMA Configuration

The steps for the DMA controller to establish the operating mechanism are as follows:

1） Set up the transmit buffer queue and fill the transmit buffer with the content to be sent. Set up the transmit descriptor queue, fill in the fields of the transmit descriptor, set OWN, and deliver the management right of the transmit descriptor to the DMA controller; register the initial address of the descriptor in the DMATDLAR register;

2） Set the ST bit (DMAOMR: 13) to enable DMA;

3） In Run mode, the DMA descriptor will automatically read the content of the transmit descriptor, and push the data to the transmit FIFO according to the address and length indicated by it. After finishing the DMA, it will read the next transmit descriptor immediately following the chain structure for the next transmission. When the DMA controller detects that the OWM bit is not set, resulting in no access to the transmit descriptor or other normal errors, it will terminate the transfer and set the TBUS bit (DMASR: 2) or other bits (Other than errors caused by OWN being 0) and the NIS bit (DMASR:16) is set.

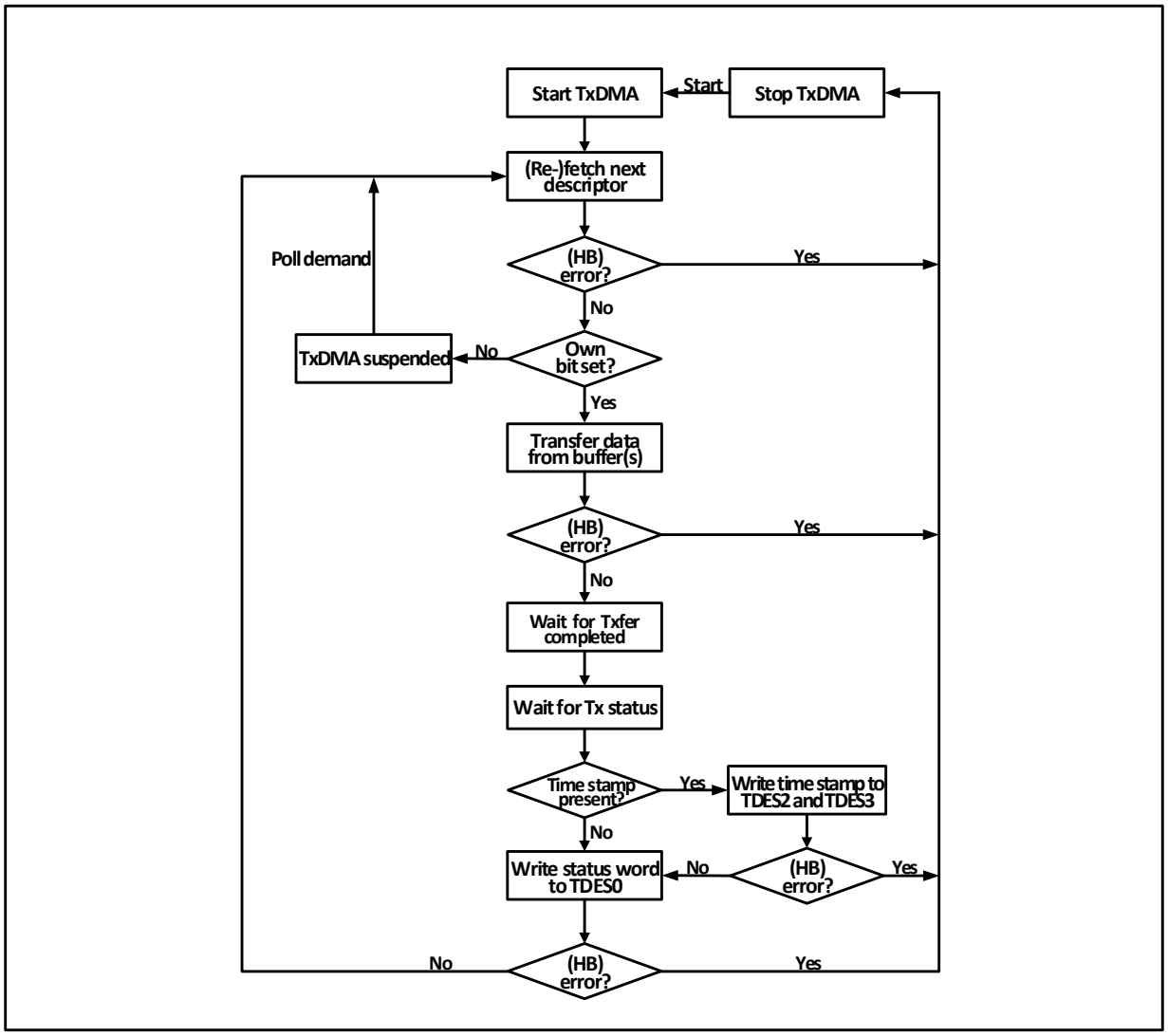
4） A single frame is not allowed to span multiple descriptors; a frame must be clearly associated with a descriptor and a buffer.

5） If the MAC turns on the IEEE1588 PTP mode, after the DMA controller pushes the data to the FIFO, the MAC will write the transmit timestamp to TDes2 and TDes3, and reset the OWN bit.

6） After sending a frame, if the transmit descriptor enables the transmit completion interrupt (Set TDes1:31), the DMA controller will set the transmit completion interrupt flag (DMASR:0), and then continue to fetch the next transmit descriptor.

The following figure shows the default process of transmission.

Figure 29-4 Process of transmission



29.1.10.6.2 Receive DMA Configuration

The steps to establish a DMA receive flow mechanism are as follows:

1) Set the transmit buffer queue and descriptor queue, and set up the bits of each field of the receive descriptor; register the initial address of the descriptor in the DMARDLAR register; set the OWN bit to give the descriptor use permission to the DMA controller;
2) Set the SR bit to start the receiving process;
3) When the receive flow mechanism is running, the DMA controller obtains the next descriptor, checks the receiving descriptor configuration, and when the FIFO receives the next frame, performs multiple checks such as filtering and identifying the frame content, and forwards the frame data to the buffer specified by the symbol, writes the descriptor status field and obtains the next receiving descriptor, and reports the receiving completion interrupt. If the frame does not pass the filtering, it will be marked or discarded. If the frame has a checksum error, CRC error or the frame is too short, it will be marked. If the frame is too long, it may be cut off or receive a watchdog timeout error. The DMA controller will stop the receiving process when it encounters a fatal error

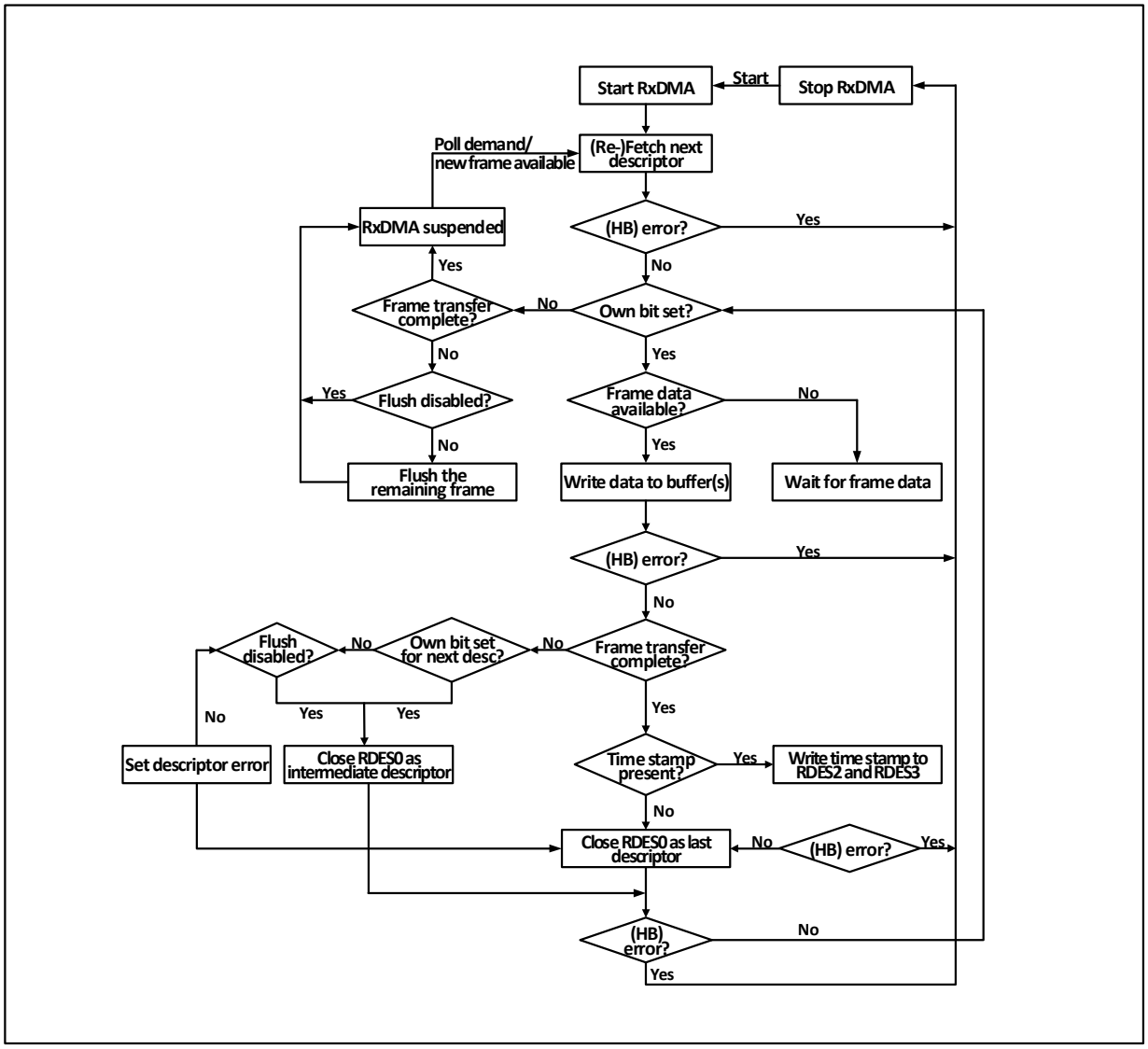
such as the unavailability of the receive descriptor, and the user needs to pay special attention;

4) The user needs to enable at least one receive completion interrupt, and restore the used receive descriptor to the standby state in the Ethernet interrupt function to ensure that the receive process can run uninterrupted. The user can pass the address of the data buffer to be processed in the interrupt function, or handle some abnormal events that interrupt the receiving process.

5) If the MAC opens the IEEE1588PTP mode and enables the enhanced descriptor, when the DMA controller dumps data and writes the descriptor status field, the current timestamp will also be written into the last two words of the descriptor.

The following figure shows the default flow mechanism of reception:

Figure 29-5 Process of reception



29.1.11 Interrupt

The Ethernet transceiver has 2 interrupt vectors, one is the Ethernet wake-up event and the other is the normal transceiver event. When a wake-up frame or magic frame is detected, an Ethernet wake-up event is triggered. Common Ethernet transceiver interrupt events include DMA interrupt and ETH interrupt.

29.2 Register Description

MAC control-related registers address mapping

The physical starting address of MAC control-related register is: 0x4002 8000

Table 29-15 Ethernet transceiver related register list

Name	Offset Address	Description	Reset Value
R32_ETH_MACCR	0x40028000	MAC control register	0x00000000
R32_ETH_MACFFR	0x40028004	MAC Frame filter register	0x00000000
R32_ETH_MACHTHR	0x40028008	Hash value list register high	0x00000000
R32_ETH_MACHTLR	0x4002800C	Hash value list register low	0x00000000
R32_ETH_MACMIIAR	0x40028010	MII address register	0x00000000
R32_ETH_MACMIIDR	0x40028014	MII data register	0x00000000
R32_ETH_MACFCR	0x40028018	MAC flow control register	0x00000000
R32_ETH_MACVLAN	0x4002801C	VLAN tag register	0x00000000
R32_ETH_MACRWUFFR	0x40028028	Wake-up frame filter register	0x00000000
R32_ETH_MACPMTCSR	0x4002802C	PMT control and status Register	0x00000000
R32_ETH_MACSR	0x40028038	MAC interrupt status register	0x00000000
R32_ETH_MACIMR	0x4002803C	MAC interrupt mask register	0x00000000
R32_ETH_MACA0HR	0x40028040	MAC address register 0 high 32 bits	0x8000FFFF
R32_ETH_MACA0LR	0x40028044	MAC address register 0 low 32 bits	0xFFFFFFFF
R32_ETH_MACA1HR	0x40028048	MAC address register 1 high 32 bits	0x0000FFFF
R32_ETH_MACA1LR	0x4002804C	MAC address register 1 low 32 bits	0xFFFFFFFF
R32_ETH_MACA2HR	0x40028050	MAC address register 2 high 32 bits	0x0000FFFF
R32_ETH_MACA2LR	0x40028054	MAC address register 2 low 32 bits	0xFFFFFFFF
R32_ETH_MACA3HR	0x40028058	MAC address register 3 high 32 bits	0x0000FFFF
R32_ETH_MACA3LR	0x4002805C	MAC address register 3 low 32 bits	0xFFFFFFFF
R32_ETH_PHY_CR	0x40028080	PHY configuration register	0x40000001

MMC control register address mapping, note: the addresses are not consecutive

The physical starting address of MMC control-related register is: 0x4002 8100

Name	Address	Description	Reset Value
R32_ETH_MMCCR	0x40028100	MMC control register	0x00000000
R32_ETH_MMCRIR	0x40028104	MMC receive register	0x00000000
R32_ETH_MMCTIR	0x40028108	MMC transmit interrupt register	0x00000000
R32_ETH_MMCRIMR	0x4002810C	MMC receive interrupt mask register	0x00000000
R32_ETH_MMCTIMR	0x40028110	MMC transmit interrupt mask register	0x00000000
R32_ETH_MMCTGFSCCR	0x4002814C	MMC transmit good frame counter after single conflict	0x00000000
R32_ETH_MMCTGFMSCCR	0x40028150	MMC transmit good frame counter after multiple conflicts	0x00000000
R32_ETH_MMCTGFCR	0x40028168	MMC transmit good frame count register	0x00000000
R32_ETH_MMCRFCECR	0x40028194	MMC receive CRC error frame count	0x00000000

		register	
R32_ETH_MMCRFAECR	0x40028198	MMC receive alignment error frame counter register	0x00000000
R32_ETH_STATE	0x400281A0	Ethernet state machine register	0x00000000
R32_ETH_MMCRGUFCR	0x400281C4	MMC receive good unicast frame count register	0x00000000

IEEE1588 (PTP) register address mapping

The physical starting address of IEEE1588 (PTP) register is: 0x4002 8700

Name	Address	Description	Reset Value
R32_ETH_PTPTSCR	0x40028700	PTP timestamp control register	0x00000000
R32_ETH_PTPSSIR	0x40028704	PTP subsecond increment register	0x00000000
R32_ETH_PTPTSHR	0x40028708	PTP timestamp register high	0x00000000
R32_ETH_PTPTSLR	0x4002870C	PTP timestamp register low	0x00000000
R32_ETH_PTPTSHUR	0x40028710	PTP timestamp update register high	0x00000000
R32_ETH_PTPTSLUR	0x40028714	PTP timestamp update register low	0x00000000
R32_ETH_PTPTSAR	0x40028718	PTP timestamp adder register	0x00000000
R32_ETH_PTPTTHR	0x4002871C	PTP target time register high 32-bit	0x00000000
R32_ETH_PTPTTLR	0x40028720	PTP target time register low 32-bit	0x00000000

DMA related register address mapping. Note: The addresses are not consecutive

The physical starting address of DMA related register is: 0x4002 9000

Name	Address	Description	Reset Value
R32_ETH_DMABMR	0x40029000	DMA bus mode register	0x00000001
R32_ETH_DMATPDR	0x40029004	DMA transmit query register	0x00000000
R32_ETH_DMARPDR	0x40029008	DMA receive query register	0x00000000
R32_ETH_DMARDLAR	0x4002900C	DMA receive descriptor address register	0x00000000
R32_ETH_DMATDLAR	0x40029010	DMA transmit descriptor address register	0x00000000
R32_ETH_DMASR	0x40029014	DMA status register	0x00000000
R32_ETH_DMAOMR	0x40029018	DMA operation mode register	0x00000000
R32_ETH_DMAIER	0x4002901C	DMA interrupt enable register	0x00000000
R32_ETH_DMAMFBOCR	0x40029020	DMA lost frame register	0x00000000
R32_ETH_DMACHTDR	0x40029048	DMA current transmit descriptor register	0x00000000
R32_ETH_DMACHRDR	0x4002904C	DMA current receive descriptor register	0x00000000
R32_ETH_DMACHTBAR	0x40029050	DMA current transmit buffer register	0x00000000
R32_ETH_DMACHRBAR	0x40029054	DMA current receive buffer register	0x00000000

The offset address of the internal physical layer register is used in the SMI interface

Configure all relevant PHY registers using the R32_ETH_MACMIIAR and R32_ETH_MACMIIDR registers.

Internal 10M/100M physical layer register address

Name	Offset Address	Description	Reset Value
R16_BMCR	0x00	Basic control register	0x3100

R16_BMSR	0x01	Basic status register	0x7849
R16_PHYIDR1	0x02	PHY identification register 1	0x7371
R16_PHYIDR2	0x03	PHY identification register 2	0x9011
R16_ETH_ANAR	0x04	Auto-negotiation message register	0x01E1
R16_ETH_ANALPAR	0x05	Auto-negotiation linker capability register	0x0001
R16_ANER	0x06	Auto-negotiation extension register	0x0004
R16_PAGE_SEL	0x1F	Page selection register	0x0000

PAGE0-related register address

Name	Offset Address	Description	Reset Value
R16_PHY_STATUS1	0x1A	PHY status register	0x0020
R16_INTERRUPT_IND	0x1E	Interrupt indication register	0x0000

PAGE7-related register address

Name	Offset Address	Description	Reset Value
R16_INTERRUPT_MASK	0x13	Interrupt/WOL/LED function register	0x0030

29.2.1 MAC Control Related Registers

29.2.1.1 MAC Control Register (R32_ETH_MACCR)

Offset Address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								WD	JD	Reserved		IFG[2:0]			Reserved

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FES	Reserved	LM	DM	IPCO	Reserved		APCS	Reserved			TE	RE	Reserved	

Name	Name	Access	Description	Reset Value
[31:24]	Reserved	RO	Reserved	0
23	WD	RW	Watchdog setting. 0: The MAC opens the watchdog, and can only accept Ethernet frames with a maximum size of 2048 bytes, and the super-long part will be cut off; 1: MAC closes the watchdog, and can receive Ethernet frames with a maximum size of 16384 bytes.	0
22	JD	RW	Jabber setting. 0: If the user tries to send an Ethernet frame with a length of more than 2048 bytes, the MAC will close the transmitter; 1: The MAC closes the Jabber timer, and can send up	0

			to 16384 bytes of Ethernet frames.	
[21:20]	Reserved	RO	Reserved	0
[19:17]	IFG[2:0]	RW	Frame gap setting, used to set the shortest time gap between sending 2 frames. 000: 96-bit time; 001: 88-bit time; 010: 80-bit time; 011: 72-bit time; 100: 64-bit time; 101: 56-bit time; 110: 48-bit time; 111: 40-bit time.	0
[16:15]	Reserved	RO	Reserved	0
14	FES	RW	Fast Ethernet, this bit indicates the speed of Fast Ethernet (MII) mode: 0: 10Mbit/s; 1: 100Mbit/s;	0
[13:12]	Reserved	RO	Reserved	0
11	DM	RW	Duplex mode enable. Setting this bit enables full duplex mode.	0
10	IPCO	RW	IPv4 checksum check enable. 0: Disable the checksum verification function of IPv4 at the receiving end, and the corresponding PCE and PHCE flags are always 0. See the definition of each bit of the receive descriptor; 1: Enable IPv4 checksum verification, the MAC controller will check the TCP, UDP and ICMP headers.	0
9	RD	RW	Retry disabled.	0
8	Reserved	RO	Reserved	0
7	APCS	RW	Padding & CRC auto-stripping enable. 0: The MAC does not modify the contents of the frame; 1: The MAC removes the Pad/FCS field from the frame only if the length/type field value is less than or equal to 1500 bytes. If the Length/Type field is greater than or equal to 1501 bytes, the frame contents are not changed.	0
[6:4]	Reserved	RO	Unused	0
3	TE	RW	Transmit enable. 0: Disable the transmitter, and no frame will be transmitted after the MAC finishes sending the current frame; 1: Enable the MAC transmitter.	0
2	RE	RW	Receive enable. 0: Disable the receiver, and no frame will be received	0

			after the MAC finishes receiving the current frame; 1: Enable the MAC receiver.	
[1:0]	Reserved	RO	Reserved	0

29.2.1.2 MAC Frame Filter Register (R32_ETH_MACFFR)

Offset Address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RA	Reserved														

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					HPF	SAF	SAIF	PCF[1:0]		BFD	PAM	DAIF	HM	HU	PM

Bit	Name	Access	Description	Reset Value
31	RA	RW	Receive all 0: MAC only forwards frames that have passed the filter to the receive queue; 1: The MAC forwards all received frames to the receive queue, regardless of whether they pass the filter or not.	0
[30:11]	Reserved	RO	Reserved	0
10	HPF	RW	HASH filter or perfect filter 0: On the premise that HM or HU is set, as long as it meets the HASH filter, it can pass the address filter; 1: Determine which filtering method is used in unicast/multicast mode according to the value of HM or HU.	0
9	SAF	RW	Source address filter 0: MAC marks frames that do not pass the source MAC address filtering; 1: MAC will directly discard frames that do not pass the source MAC address filtering.	0
8	SAIF	RW	Source address filter result inverted. 0: If the source address of the received frame is inconsistent with the source address enabled in the MAC address register, it is considered to have failed the source address filter; 1: If the source address of the received frame is consistent with the source address enabled in the MAC address register, it is considered to have failed the source address filter.	0
[7:6]	PCF[1:0]	RW	Flow control frames pass control. 00/01: The MAC does not forward any flow control frames to the application; 10: MAC forwards all flow control frames to the	0

			application, including flow control frames that fail the address filter; 11: MAC only forwards flow control frames that pass the address filter.	
5	BFD	RW	Broadcast frame reception control 0: Receive all broadcast frames; 1: Discard all broadcast frames.	0
4	PAM	RW	Pass all multicast frame control. 0: Whether the multicast frame can pass filtering depends on the value of HM; 1: All multicast frames can be filtered by address.	0
3	DAIF	RW	Destination MAC address filter results inverted control. 0: The filter result takes effect normally; 1: For multicast frames and unicast frames, whether the result of the filter passing is reversed before it takes effect.	0
2	HM	RW	Multicast frame filter mode 0: Perfect address filtering; 1: HSAH address filtering.	0
1	HU	RW	Unicast frame filter mode 0: Perfect address filtering; 1: HSAH address filtering.	0
0	PM	RW	Promiscuous mode enable All frames pass the address filter without marking the result of the filter.	0

29.2.1.3 Hash List Register (R32_ETH_MACHTHR, R32_ETH_MACHTLR)

Offset Address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTH[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTH[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	HTH[31:0]	RW	Hash list high 32 bits	0

Offset Address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTL[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTL[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	HTL[31:0]	RW	Hash list low 32 bits	0

29.2.1.4 MII Address Register (R32_ETH_MACMIIAR)

Offset Address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PA[4:0]					MR[4:0]					Reserved	CR[2:0]			MW	MB

Bit	Name	Access	Description	Reset Value
[31:16]	Reserved	RO	Unused.	0
[15:11]	PA[4:0]	RW	Physical layer address. The user writes the physical layer address to be operated into this field.	0
[10:6]	MR[4:0]	RW	Physical layer register address field. The user writes the register address to be operated into this field.	0
5	Reserved	RO	Unused.	0
[4:2]	CR[2:0]	RW	Clock range setting field. The CR clock range option determines the HCLK frequency and is used to determine the MDC clock frequency. 000: HCLK (60-90MHz), MDC clock (HCLK/42); 001: HCLK (90-120MHz), MDC clock (HCLK/62); 010: HCLK (20-35MHz), MDC clock (HCLK/16); 011: HCLK (35-62MHz), MDC clock (HCLK/26); 100: HCLK (150-200MHz), MDC clock (HCLK/102); 101: HCLK (120-150MHz), MDC clock (HCLK/74); Others: Reserved.	0
1	MW	RW	Read and write setting. 0: Read the physical layer; 1: Write to the physical layer.	0
0	MB	RW1Z	MII busy flag. This bit is set by the user, indicating that the hardware is instructed to start reading or writing. The physical address, register address and data field should remain unchanged during reading and writing. After the hardware clears this bit, the operation is completed.	0

29.2.1.5 MII Data Register (R32_ETH_MACMIIDR)

Offset Address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MD[15:0]															

Bit	Name	Access	Description	Reset Value
[31:16]	Reserved	RO	Reserved.	0
[15:0]	MD[15:0]	RW	MII operation data. This field is used to store the data to be read from the physical layer through the MII interface, or to store the data written to the physical layer.	0

29.2.1.6 MAC Flow Control Register (R32_ETH_MACFCR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PT[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												UPFD	RFCE	TFCE	FCB

Bit	Name	Access	Description	Reset Value
[31:16]	PT[15:0]	RW	Pause interval. This field is used to control the value of the Pause time field, and the unit is the time it takes for the current MII interface to send 64 bytes.	0
[15:4]	Reserved	RO	Reserved	0
3	UPFD	RW	Unicast Pause frame detection. 0: The MAC only receives Pause frames with the unique address defined by the protocol specification; 1: The MAC also detects whether the Pause frame is a unicast address defined in the MAC address register 0.	0
2	RFCE	RW	Receive flow control enable. 0: The MAC does not parse the Pause frame; 1: The MAC parses the Pause frame and turns off the transmitter for a while.	0
1	TFCE	RW	Transmit flow control enable. 0: MAC closes sending flow control and does not send Pause frames; 1: MAC enables sending flow control and can send Pause frames.	0
0	FCB	W1	Flow control busy flag. This bit can be set to send a Pause frame, which is cleared by hardware after the transmission is completed. When operating on the	0

			entire MACFCR register, it is necessary to ensure that the FCB bit is 0.	

29.2.1.7 VLAN Tag Register (R32_ETH_MACVLAN)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															VLANT

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VLANTI[15:0]															

Bit	Name	Access	Description	Reset Value
[31:17]	Reserved	RO	Reserved	0
16	VLANT	RW	Tag compare control. 0: Compare all 16-bit data of the 15th and 16th bytes of the VLAN frame with the VLANTI field; 1: Only use the [11:0] bits of the 15th and 16th bytes of the VLAN frame and compare them with the corresponding bits of the VLANTI field.	0
[15:0]	VLANTI[15:0]	RW	Label vs. sample domain. According to the IEEE 802.1 protocol, bits [15:13] of the VLAN frame are the user priority, [12] is the canonical format indicator, and bits [11:0] of the VLAN identifier field. If VLANTI is all 0, then the MAC will no longer care about the 15th and 16th bytes of the VLAN frame, and when the 13th and 14th bytes are 0x8100 (pay attention to the size end), it will be judged as a VLAN frame.	0

29.2.1.8 Wake-up Frame Filter Register (R32_ETH_MACRWUFFR)

Offset Address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RWUFFR[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RWUFFR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RWUFFR[31:0]	RW	The wake-up frame filter register is actually 8 different registers, and 8 consecutive read operations can read out all registers, and 8 consecutive write operations can be written into all 8 registers. The description of each bit of these eight registers is detailed in the table below.	0

Filter register 0	Byte mask of filter register 0							
Filter register 1	Byte mask of filter register 1							
Filter register 2	Byte mask of filter register 2							
Filter register 3	Byte mask of filter register 3							
Filter register 4	Reserved	Filter 3 command	Reserved	Filter 2 command	Reserved	Filter 1 command	Reserved	Filter 0 command
Filter register 5	Filter 3 offset		Filter 2 offset		Filter 1 offset		Filter 0 offset	
Filter register 6	Filter 1 CRC-16				Filter 0 CRC-16			
Filter register 7	Filter 3 CRC-16				Filter 2 CRC-16			

29.2.1.9 PMT Control and Status Register (R32_ETH_MACPMTCSR)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
WFFRP	Reserved														

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						GU	Reserved	PHYR	WFR	MPR	Reserved	PHYE	WFE	MPE	PD

Bit	Name	Access	Description	Reset Value
31	WFFRP	W	Wake-up frame filter register pointer reset. Setting this bit clears the MACRWUFFR register all the way to zero. This bit is automatically cleared on a clock cycle.	0
[30:10]	Reserved	RO	Reserved.	0
9	GU	RW	Global unicast. Setting this bit causes the MAC to consider all unicast frames that pass the filter to be wake-up frames.	0
8	Reserved	RO	Reserved.	0
7	PHYR	RZ	PHY wake-up frame flag bit. When a PHY wake-up frame is received, this bit will be set. Reading is automatically cleared.	0
6	WFR	RC	Wake-up frame received flag. This bit is set when a wake-up frame is received. Read it and it will be cleared automatically.	0

5	MPR	RC	Magic frame received flag. This bit is set when a magic frame is received. Read it and it will be cleared automatically.	0
4	Reserved	RO	Reserved.	0
3	PHY_PMEB_EN	RW	PHYPMEB enable bit. Setting this bit allows the PHY to generate a PMT event upon receiving a wake-up frame.	0
2	WFE	RW	Wake-up frame enable. Setting this bit allows a PMT event to be generated when a wake-up frame is received.	0
1	MPE	RW	Magic frame enable. Setting this bit allows a PMT event to be generated when a magic frame is received.	0
0	PD	W	Power-down control. Setting this bit will put the MAC into power-down mode: all other frames are discarded until it receives a wake-up frame or magic frame, and the MAC is automatically cleared after wake-up. Before setting this bit, WFE or MPE needs to be set.	0

29.2.1.10 MAC Interrupt Status Register (R32_ETH_MACSR)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						TSTS	Reserved		MMCTS	MMCRS	MMCS	PMTS	Reserved		

Bit	Name	Access	Description	Reset Value
[31:10]	Reserved	RO	Reserved	0
9	TSTS	RZ	Timestamp trigger interrupt flag. When the time set by the PTP system clock is reached, this flag will be set.	0
[8:7]	Reserved	RO	Reserved	0
6	MMCTS	R	MMC transmit interrupt flag. This bit is set when any interrupt is generated in the MMCTIR register in the MMC register bank. This bit is also cleared when all the MMCTIR registers in the MMC register bank are cleared.	0
5	MMCRS	R	MMC receive interrupt flag. This bit is set when any interrupt is generated in the MMCRIR register in the MMC register bank. This bit is also cleared when the MMCRIR registers in the MMC register bank are all cleared.	0
4	MMCS	R	MMC status flag. When MMCTS or MMCRS is set, this bit is triggered. When MMCTS and MMCRS are	0

			all cleared, this bit is cleared.	
3	PMTS	R	PMT status flag. In power-down mode, if the MAC receives a magic frame or wake-up frame wakes up the MAC, this bit will be set. After clearing the WFR and MPR bits, this bit is cleared.	0
[2:0]	Reserved	RO	Reserved	0

29.2.1.11 MAC Interrupt Mask Register (R32_ETH_MACIMR)

Offset Address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						TSTIM	Reserved					PMTIM	Reserved		

Bit	Name	Access	Description	Reset Value
[31:10]	Reserved	RO	Reserved	0
9	TSTIM	RW	Timestamp interrupt mask. Setting this bit disables generation of timestamp interrupts.	0
[8:4]	Reserved	RO	Reserved	0
3	PMTIM	RW	PMT interrupt mask. Setting this bit will mask the PMT interrupt.	0
[2:0]	Reserved	RO	Reserved	0

29.2.1.12 MAC Address Register 0 High 32 Bits (R32_ETH_MACA0HR)

Offset Address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MO	Reserved														

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA0H[15:0]															

Bit	Name	Access	Description	Reset Value
31	MO	RO	Always 1.	1
[30:16]	Reserved	RO	Reserved	0
[15:0]	MACA0H[15:0]	RW	High 16 bits of the MAC address, that is, 47:32 bits.	0xFFFF

29.2.1.13 MAC Address Register 0 Low 32 Bits (R32_ETH_MACA0LR)

Offset Address: 0x44

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

MACA0L[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA0L[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	MACA0L[31:0]	RW	Low 32 bits of the MAC address, namely [31:0] bits. In general, the address of the MAC address register 0 is the address of the MAC itself.	0xFFFFFFFF

29.2.1.14 MAC Address Register 1 High 32 Bits (R32_ETH_MACA1HR)

Offset address: 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AE	SA	MBC[5:0]						Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA1H[15:0]															

Bit	Name	Access	Description	Reset Value
31	AE	RW	Address filter enable. 0: The address filter ignores MAC address 1. 1: The address filter uses MAC address 1 for perfect filtering;	0
30	SA	RW	Address role selection. 0: MAC address 1 is used to compare with the destination address of the received frame; 1: MAC address 1 is used to compare with the source address of the received frame.	0
[29:24]	MBC[5:0]	RW	Mask word control. Each bit of the MBC is used to correspond to a certain byte of the masked MAC address 1, and is used to prohibit a certain byte of the MAC address 1 from participating in the comparison of the source address or the destination address of the received frame. R32_ETH_MACA1HR[29] is set to mask MACA1H[15:8]; R32_ETH_MACA1HR[28] is set to mask MACA1H[7:0]; R32_ETH_MACA1HR[27] is set to mask MACA1L[31:24]; R32_ETH_MACA1HR[26] is set to mask MACA1L[23:16]; R32_ETH_MACA1HR[25] is set to mask	0

			MACA1L[15:8]; R32_ETH_MACA1HR[24] is set to mask MACA1L[7:0].	
[23:16]	Reserved	RO	Reserved	0
[15:0]	MACA1H	RW	High 16 bits of the MAC address, that is, 47:32 bits.	0xFFFF

29.2.1.15 MAC Address Register 1 Low 32 Bits (R32_ETH_MACA1LR)

Offset Address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MACA1L[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA1L[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	MACA1L[31:0]	RW	Low 32 bits of the MAC address, namely [31:0] bits.	0xFFFFFFFF

29.2.1.16 MAC Address Register 2 High 32 Bits (R32_ETH_MACA2HR)

Offset Address: 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AE	SA	MBC[5:0]						Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA2H[15:0]															

Bit	Name	Access	Description	Reset value
31	AE	RW	Address filter enable. 0: The address filter ignores the MAC address 2. 1: The address filter uses MAC address 2 for perfect filtering.	0
30	SA	RW	Address role selection. 0: MAC address 2 is used to compare with the destination address of the received frame; 1: MAC address 2 is used to compare with the source address of the received frame.	0
[29:24]	MBC[5:0]	RW	Mask word control. Each bit of the MBC is used to correspond to a certain byte of the masked MAC address 2, and is used to prohibit a certain byte of the MAC address 2 from participating in the comparison of the source address or the destination address of the received frame. R32_ETH_MACA2HR[29] is set to mask	0

			MACA2H[15:8]; R32_ETH_MACA2HR[28] is set to mask MACA2H[7:0]; R32_ETH_MACA2HR[27] is set to mask MACA2L[31:24]; R32_ETH_MACA2HR[26] is set to mask MACA2L[23:16]; R32_ETH_MACA2HR[25] is set to mask MACA2L[15:8]; R32_ETH_MACA2HR[24] is set to mask MACA2L[7:0].	
[23:16]	Reserved	RO	Reserved	0
[15:0]	MACA2H[15:0]	RW	High 16 bits of the MAC address, namely [47:32] bits.	0xFFFF

29.2.1.17 MAC Address Register 2 Low 32 Bits (R32_ETH_MACA2LR)

Offset address: 0x54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MACA2L[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA2L[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	MACA2L[31:0]	RW	Low 32 bits of the MAC address, namely [31:0] bits.	0xFFFFFFFF

29.2.1.18 MAC Address Register 3 High 32 Bits (R32_ETH_MACA3HR)

Offset address: 0x58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AE	SA	MBC[5:0]						Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA3H[15:0]															

Bit	Name	Access	Description	Reset Value
31	AE	RW	Address filter enable. 0: The address filter ignores the MAC address 3; 1: The address filter uses MAC address 3 for perfect filtering.	0
30	SA	RW	Address role selection. 0: MAC address 3 is used to compare with the destination address of the received frame; 1: MAC address 3 is used to compare with the source address of the received frame.	0

[29:24]	MBC[5:0]	RW	Mask word control. Each bit of the MBC is used to correspond to a certain byte of the masked MAC address 3, and is used to prohibit a certain byte of the MAC address 3 from participating in the comparison of the source address or the destination address of the received frame. R32_ETH_MACA3HR[29] is set to mask MACA3H[15:8]; R32_ETH_MACA3HR[28] is set to mask MACA3H[7:0]; R32_ETH_MACA3HR[27] is set to mask MACA3L[31:24]; R32_ETH_MACA3HR[26] is set to mask MACA3L[23:16]; R32_ETH_MACA3HR[25] is set to mask MACA3L[15:8]; R32_ETH_MACA3HR[24] is set to mask MACA3L[7:0].	0
[23:16]	Reserved	RO	Reserved	0
[15:0]	MACA3H[15:0]	RW	High 16 bits of the MAC address, namely [47:32] bits.	0xFFFF

29.2.1.19 MAC Address Register 3 Low 32 Bits (R32_ETH_MACA3LR)

Offset address: 0x5C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MACA3L[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA3L[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	MACA3L[31:0]	RW	Low 32 bits of the MAC address, namely [31:0] bits.	0xFFFFFFFF

29.2.1.20 PHY Configuration Register (R32_ETH_PHY_CR)

Offset address: 0x80

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PHY_RSTN	PHY_PD	Reserved													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					DUPlEX	SPEED	Reserved	PHYADDR_	Reserved		REPHYADDR[4:0]				

				EN		

Bit	Name	Access	Description	Reset Value
31	PHY_RSTN	RW	Ethernet PHY Global Reset Signal: 1: Normal operating mode; 0: Reset the Ethernet PHY.	0
30	PHY_PD	RW	In power-down mode, the MAC can access the PHY's registers via the MDIO interface: 1: Power-down mode; 0: Normal operating mode.	1
[29:11]	Reserved	RO	Reserved	0
10	DUPlEX	RO	1: Connection is full-duplex; 0: Connection is half-duplex.	0
9	SPEED	RO	1: Connection is 100 Mbps; 0: Connection is 10 Mbps.	0
8	Reserved	RO	Reserved	0
7	PHYADDR_EN	RW	Reconfigure PHY Address: 1: Bits 4-0 serve as the PHY address. 0: PHY address defaults to 1.	0
[6:5]	Reserved	RO	Reserved	0
[4:0]	REPHYADDR[4:0]	RW	When bit7 is 1, bit 4-0 is used as the address of PHY.	0x01

Note: (1) When using Ethernet, you must first clear the PHY reset and PD signals.

(2) The bits in the R32_ETH_PHY_CR register are always read as 0.

29.2.2 MMC registers

29.2.2.1 MMC Control Register (R32_ETH_MMCCR)

Offset address: 0x0100

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												MCF	ROR	CSR	CR

Bit	Name	Access	Description	Reset Value
[31:4]	Reserved	RO	Reserved	0
3	MCF	RW	Counter freeze control. Setting this bit will freeze all MMC counter values. Resetting this bit will restore the count of each counter. Setting ROR while frozen and then reading any counter will cause that counter to be cleared.	0
2	ROR	RW	Reset control on read. Setting this bit will clear the value of the counter after reading either counter.	0

1	CSR	RW	Counter roll stop. Setting this bit causes the counter to increment to its maximum value and stop without auto-zeroing.	0
0	CR	RW1Z	Counter reset control. Setting this bit will reset all counters. This bit is automatically cleared after single system cycle.	0

29.2.2.2 MMC Receive Interrupt Register (R32_ETH_MMCRIR)

Offset address: 0x0104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														RGUFS	Reserved

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RFCES	Reserved				

Bit	Name	Access	Name	Reset Value
[31:18]	Reserved	RO	Reserved	0
17	RGUFS	RZ	This bit is set when more than half of the good frames are received.	0
[16:6]	Reserved	RO	Reserved	0
5	RFCES	RZ	This bit is set when more than half of the frames with CRC errors are received.	0
4:0	Reserved	RO	Reserved	0

29.2.2.3 MMC Transmit Interrupt Register (R32_ETH_MMCTIR)

Offset address: 0x0108

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										TGFS	Reserved				

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGFMSCS	TGFSCS	Reserved													

Bit	Name	Access	Description	Reset Value
[31:22]	Reserved	RO	Reserved	0
21	TGFS	RZ	This bit is set when more than half of the frames are sent.	0
[20:16]	Reserved	RO	Reserved	0
15	TGFMSCS	RW	In half-duplex mode, the counter is set when more than one collision is encountered when a frame reaches half.	0
14	TGFSCS	RW	In half-duplex mode, only one collision is encountered	0

			when a frame occurs and the counter is set when it reaches half.	
[13:0]	Reserved	RO	Reserved	0

29.2.2.4 MMC Receive Interrupt Mask Register (R32_ETH_MMCRIMR)

Offset address: 0x010C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														RGUFM	Reserved

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									RFAEM	RFCEM	Reserved				

Bit	Name	Access	Description	Reset Value
[31:18]	Reserved	RO	Reserved	0
17	RGUFM	RW	Receive a good frame half interrupt mask. Setting this bit will mask the interrupt generated when the received frame counter reaches half the value.	0
[16:7]	Reserved	RO	Reserved	0
6	RFAEM	RW	Receive non-alignment error frame half-way interrupt mask bit. Setting this bit will mask the interrupt generated when the frame counter reaches half the value when a non-alignment error is received.	0
5	RFCEM	RW	Receive CRC error frame half interrupt mask. Setting this bit will mask the interrupt that occurs when the frame counter value reaches half the value of the frame counter that receives a CRC error.	0
[4:0]	Reserved	RO	Reserved	0

29.2.2.5 MMC Transmit Interrupt Mask Register (R32_ETH_MMCTIMR)

Offset address: 0x0110

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										TGFM	Reserved				

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGFMSCM	TGFSCM	Reserved													

Bit	Name	Access	Description	Reset Value
[31:22]	Reserved	RO	Reserved	0

21	TGFM	RW	Transmit good frame half interrupt mask. Setting this bit will mask the interrupt generated when the sent frame counter reaches half the value.	0
[20:16]	Reserved	RO	Reserved	0
15	TGFMSCM	RW	Transmit a good frame with multiple collision flags.	0
14	TGFSCM	RW	Transmit a good frame with a single collision flag.	0
[13:0]	Reserved	RO	Reserved	0

29.2.2.6 MMC Transmit Good Frame after Single Conflict Counter (R32_ETH_MMCTGFSCCR)

Offset address: 0x014C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TGFSCCR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGFSCCR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TGFSCCR[31:0]	RO	This field contains the number of frames successfully sent after a single conflict in half-duplex mode, i.e., the counter of good frames sent after a single conflict.	0

29.2.2.7 MMC Transmit Good Frame Counter after Multiple Conflicts (R32_ETH_MMCTGFMSCCR)

Offset address: 0x0150

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TGFMSCCR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGFMSCCR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TGFMSCCR[31:0]	RO	This field contains the number of frames successfully sent after more than one conflict in half-duplex mode, i.e., a counter of good frames sent after multiple conflicts.	0

29.2.2.8 MMC Transmit Good Frame Count Register (R32_ETH_MMCTGFCR)

Offset address: 0x0168

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TGFC[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGFC[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TGFC[31:0]	RO	Count of correct frames sent by the MAC.	0

29.2.2.9 MMC Receive Frame with CRC Error Count Register (R32_ETH_MMCRFCECR)

Offset address: 0x0194

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RFCECR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RFCECR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RFCECR[31:0]	RO	Counter of frames received by the MAC with CRC errors.	0

29.2.2.10 MMC Receive Frame with Alignment Error Count Register (R32_ETH_MMCRFAECR)

Offset address: 0x0198

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RFAECR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RFAECR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RFAECR[31:0]	RO	Counter of frames received by the MAC with alignment errors.	0

29.2.2.11 Ethernet State Machine Register (R32_ETH_STATE)

Offset address: 0x01A0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					ERX_STATE[2:0]			Reserved					ETX_STATE[2:0]		

Bit	Name	Access	Description	Reset Value
[31:11]	Reserved	RO	Reserved	0
[10:8]	ERX_STATE[2:0]	RO	Ethernet receiving state machine: 000: IDLE; 001: MAC_ADDR;	0

			010: RCV_LEN_TYPE; 011: RCV_DATA; 100: RCV_FCS_PAD; 101: RCV_ZZZ; Others: Reserved.	
[7:3]	Reserved	RO	Reserved	0
[2:0]	ETX_STATE[2:0]	RO	Ethernet transmit state machine: 000: SHED; 001: IDLE; 010: PREAM; 011: SEND_DATA; 100: SEND_PAD; 101: SEND_FCS; 110: SEND_JAM; 111: SEND_EEE.	0

29.2.2.12 MMC Receive Good Unicast Frame Count Register (R32_ETH_MMCRGUFCR)

Offset address: 0x01C4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RGUFCR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RGUFCR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RGUFCR[31:0]	RO	The number of good unicast frames received by the MAC.	0

29.2.3 IEEE 1588 (PTP) Registers

29.2.3.1 PTP Timestamp Control Register (R32_ETH_PTPTSCR)

Offset address: 0x0700

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										TSARU	TSITE	TSSTU	TSSTI	TSFCU	TSE

Bit	Name	Access	Description	Reset Value
[31:6]	Reserved	RO	Reserved	0
5	TSARU	RW	Addend register update control. When this bit is set, the value of the addend register will be added to the accumulator. This bit is used in fine tuning mode.	0

			This bit is automatically cleared after the accumulator is incremented. This bit can only be set when it is 0.	
4	TSITE	RW	Timestamp interrupt trigger enable. When this bit is set, an interrupt will be generated when the PTP system time reaches the value set in the target time register.	0
3	TSSTU	RW	System time update control. When this bit is set, the PTP system time will be added to the value in the update register. This bit is automatically cleared after the update is complete.	0
2	TSSTI	RW	Timestamp initialization control. When this bit is set, the system time of the PTP will be replaced with the value in the update register. This bit is automatically cleared after the update is complete.	0
1	TSFCU	RW	Update mode selection. 0: Coarse mode; 1: Fine adjustment mode.	0
0	TSE	RW	Additional timestamp enable. 0: No timestamp is added to the descriptor; 1: Add a timestamp to the descriptor at the end of reception or transmission.	0

29.2.3.2 PTP Subsecond Increment Register (R32_ETH_PTPSSIR)

Offset address: 0x0704

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								STSSI[7:0]							

Bit	Name	Access	Description	Reset Value
[31:8]	Reserved	RO	Reserved	0
[7:0]	STSSI[7:0]	RW	Sub-second step value. In coarse mode, the PTP system time will automatically increase by this value every main frequency cycle. In the fine adjustment mode, when the accumulator overflows, the PTP system time will automatically increase by this value.	0

29.2.3.3 PTP Timestamp Register High (R32_ETH_PTPTSHR)

Offset address: 0x0708

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STS[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STS[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TSHR[31:0]	RO	PTP system time value, real-time value, in seconds.	0

29.2.3.4 PTP Timestamp Register Low (R32_ETH_PTPTSLR)

Offset address: 0x070C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STPNS	STSS[30:16]														

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STSS[15:0]															

Bit	Name	Access	Description	Reset Value
31	STPNS	RO	System time positive/negative flag. 1: Negative; 0: Positive.	0
[30:0]	STSS[30:0]	RO	PTP system time value, real-time value, in sub-seconds, i.e., about 0.46ns. When STSS overflows, STS increments by 1s.	0

29.2.3.5 PTP Timestamp Update Register High 32 Bits (R32_ETH_PTPTSHUR)

Offset address: 0x0710

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TSUS[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSUS[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TSUS	RW	Timestamp update second value, which is used to replace the high bit of the PTP system time or the second value representing its addition or subtraction to the system time.	0

29.2.3.6 PTP Timestamp Update Register Low 32 Bits (R32_ETH_PTPTSLUR)

Offset address: 0x0714

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TSUSS[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSUSS[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TSUSS[31:0]	RW	Timestamp update sub-second value, which is used to replace the high bit of the PTP system time or a sub-second value that represents its addition or subtraction from the system time.	0

29.2.3.7 PTP Timestamp Adder Register (R32_ETH_PTPTSAR)

Offset address: 0x0718

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TSA[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSA[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TSA[31:0]	RW	Timestamp adder value. This register is only used in fine tuning mode. The value of TSA is added to the accumulator every system cycle, and if the accumulator overflows, it will trigger a system time update in fine tuning mode.	0

29.2.3.8 PTP Target Time Register High 32 Bits (R32_ETH_PTPTTHR)

Offset address: 0x071C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TTSH[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TTSH[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TTSH[31:0]	RW	Target time in seconds. If the PTP system time reaches or exceeds this value and the associated interrupt is enabled, an interrupt will be generated.	0

29.2.3.9 PTP Target Time Register Low 32 Bits (R32_ETH_PTPTTLR)

Offset address: 0x0720

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TTSL[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TTSL[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TTSL[31:0]	RW	Target time sub-second value. An interrupt is generated if the PTP system time reaches or exceeds this value and the associated interrupt is enabled.	0

29.2.4 DMA Control Registers

29.2.4.1 DMA Bus Mode Register (R32_ETH_DMABMR)

Offset address: 0x1000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									DSL[4:0]					Reserved	SR

Bit	Name	Access	Description	Reset Value
[31:7]	Reserved	RO	Reserved	0
[6:2]	DSL[4:0]	RW	Descriptor hop length These bits define the jump distance in words (32 bits) between 2 descriptors that are not connected in a chained structure. The address hop is the address difference from the end of the current descriptor to the beginning of the next descriptor. When the DSL field is 0, the DMA considers the descriptors to be contiguous under a ring structure.	0
1	Reserved	RO	Reserved	0
0	SR	RW	Software Reset When set to 1, the MAC's DMA controller resets the internal registers and logic circuits of all subsystems of the MAC. This bit is automatically cleared after the reset operation is completed for the different clock domain modules inside the MAC. Before rewriting the MAC's registers, you should ensure that this bit is 0.	1

29.2.4.2 DMA Transmit Query Register (R32_ETH_DMATPDR)

Offset address: 0x1004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

TPDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TPDR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TPDR[31:0]	RW	Transmit query commands. The user initiates a suspended transmit process by writing any value to this register. After restarting the transmission process, this register will be automatically cleared.	0

29.2.4.3 DMA Receive Query Register (R32_ETH_DMARPDR)

Offset address: 0x1008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RPDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RPDR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RPDR[31:0]	RW	Receive query commands. The receiving process may be interrupted by various accidents, and the user needs to write any value to this register to restart the receiving process. After restarting the receive process, this register will be automatically cleared.	0

29.2.4.4 DMA Receive Descriptor Address Register (R32_ETH_DMARDLAR)

Offset address: 0x100C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDLAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDLAR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RDLAR[31:0]	RW	This register is used to store the address of the first received DMA descriptor. Note that descriptors require 4-byte alignment.	0

29.2.4.5 DMA Transmit Descriptor Address Register (R32_ETH_DMATDLAR)

Offset address: 0x1010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TDLAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TDLAR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TDLAR[31:0]	RW	This register is used to store the address of the first transmit DMA descriptor. Note that descriptors require 4-byte alignment.	0

29.2.4.6 DMA Status Register (R32_ETH_DMASR)

Offset address: 0x1014

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		TSTS	PMTS	MMCS	Reserved	EBS[2:0]			TPS[2:0]			RPS[2:0]			NIS
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AIS	ERS	Reserved		PHYLINK_SR	ETS	RWTS	RPSS	RBUS	RS	TUS	ROS	TJTS	TBUS	TPSS	TS

Bit	Name	Access	Description	Reset Value
[31:30]	Reserved	RO	Reserved	0
29	TSTS	RO	Timestamp trigger status. This bit will be set when an interrupt event occurs in the PTP section, or an interrupt will be generated if the PTP interrupt is enabled. This bit is automatically cleared after clearing all flag bits in the PTP section.	0
28	PMTS	RO	PMT trigger status. This bit will be set when an interrupt event is generated in the PMT part, and an interrupt will be generated if the PMT interrupt is enabled. This bit is automatically cleared after clearing all flag bits in the PMT section.	0
27	MMCS	RO	MMC trigger status. This bit will be set when an interrupt event is generated in the MMC part, and an interrupt will be generated if the MMC interrupt is enabled. This bit is automatically cleared after clearing all flag bits in the MMC section.	0
26	Reserved	RO	Reserved	0
[25:23]	EBS[2:0]	RO	Error status. This field indicates the type of bus error that caused it. This field is only valid when the DMASR[13] bit is set.	0

			DMASR[23]:0: Error sending DMA forwarding data; 1: Error receiving DMA forwarding data; DMASR[24]:0: Error reading data forwarding; 1: Error writing data forwarding; DMASR[25]:0: Error accessing descriptor; 1: Error accessing data cache.	
[22:20]	TPS[2:0]	RO	Transmit process status. This field is used to indicate the status of the current transmit DMA. 000: Stop. Receive reset or stop sending command; 001: Run. Fetching the send descriptor; 010: Run. Waiting for status information; 011: Run. Reading the transmit buffer data and pushing it into the FIFO; 100, 101: Reserved; 110: Pause. The transmit descriptor is not available or the transmit buffer data underflows; 111: Run. Closing the transmit descriptor.	0
[19:17]	RPS[2:0]	RO	Receive process status. This field is used to indicate the current state of receiving DMA. 000: Stop. Receive reset or stop sending command; 001: Run. Fetching the receive descriptor; 010: Reserved; 011: Run. Waiting to receive data packets; 100: Pause. The receive descriptor is not available; 101: Run. Closing the receive descriptor; 110: Reserved; 111: Run. Pushing the received data from the FIFO into the memory.	0
16	NIS	RW1	Normal interrupt summary. With interrupts enabled in the DMAIER register, if any of the following bits are set, the NIS bit will also be set. -DMASR[0]: Send interrupt; -DMASR[2]: The transmit buffer is not available; -DMASR[6]: Receive interrupt; -DMASR[11]: Ethernet physical layer connection status changes; -DMASR[14]: Early receive interrupt; The NIS bit is sticky, when the corresponding interrupt status bit that causes NIS location 1 is cleared, the NIS bit must also be cleared by writing 1.	0
15	AIS	RW1	Abort interrupt summary. With interrupts enabled in the DMAIER register, the AIS bit will also be set if any of the following bits are set. -DMASR[1]: The transmission is stopped; -DMASR[3]: Transmit Jabber timeout;	0

			-DMASR[4]: Receive FIFO overflow; -DMASR[5]: Transmit data underflow; -DMASR[7]: The receive buffer is not available; -DMASR[8]: The reception is stopped; -DMASR[9]: Receive watchdog timeout; -DMASR[10]: Transmit early; The AIS bit is a sticky bit. When the corresponding interrupt status bit that causes the AIS bit to be set to 1 is cleared, the AIS bit must also be cleared by writing 1.	
14	ERS	RW1	Early receive status. When this bit is set, it means that the DMA has filled the first buffer when the data frame is received, but the complete frame has not been received yet. After RS is set, the ERS bit is automatically cleared.	0
[13:12]	Reserved	RO	Reserved	0
11	PHYLINK_SR	RW1Z	The PHY has generated an interrupt; you need to read the relevant PHY registers to determine the source of the interrupt. Setting this bit to 1 clears the interrupt.	0
10	ETS	RW1	Early transmit status. When set, it indicates that the transmit frame has been fully pushed into the FIFO.	0
9	RWTS	RW1	Watchdog time-out status. When set, it indicates that the frame length has exceeded 2048 bytes.	0
8	RPSS	RW1	Receive process stop status. This bit is set to indicate that the receive process has stopped.	0
7	RBUS	RW1	Receive buffer unavailable status. This bit is set to indicate that the permission to receive the descriptor belongs to the CPU, the DMA cannot be obtained, and the receiving process has been suspended. The user needs to release the descriptor and fill the RPDR register with a value to resume the receive flow. The DMA will retry to acquire the descriptor when the next frame is received.	0
6	RS	RW1	Receive completed status. This bit is set to indicate that the reception of a frame has been completed, and the frame information is also updated into the descriptor.	0
5	TUS	RW1	Transmit data underflow status. This bit is set to indicate that the transmit buffer under flowed the transmit data while transmitting the frame. At this point the sending process is halted and the data underflow error bit is set.	0
4	ROS	RW1	Receive status overflow. This bit is set to indicate that a data overflow has occurred in the receive buffer.	0
3	TJTS	RW1	Transmit Jabber timer timeout status. This bit is set to	0

			indicate that the transmitter is too busy, transmission has stopped, and the descriptor's Jabber timeout bit has been set.	
2	TBUS	RW1	Transmit buffer unavailable status bit. This bit is set to indicate that the transmit descriptor is occupied by the CPU, the DMA cannot be acquired, and the transmit process has been suspended. Bits [22:20] show the current transmit status. The application needs to release the send descriptor.	0
1	TPSS	RW1	Transmit process stop status. This bit is set to indicate that the transmission process has stopped.	0
0	TS	RW1	Transmit completed status. This bit is set to indicate that the transmission of a frame has been completed, and the ownership of the descriptor has been returned to the CPU.	0

29.2.4.7 DMA Operation Mode Register (R32_ETH_DMAOMR)

Offset address: 0x1018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					DTCEFD	Reserved				TSF	FTF	Reserved			

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		ST	Reserved					FEF	FUGF	Reserved				SR	Res

Bit	Name	Access	Description	Reset Value
[31:27]	Reserved	RO	Reserved	0
26	DTCEFD	RW	Discard TCP/IP checksum error frame disable 0: If the FEF bit is 0, the MAC will discard all frames with errors; 1: When an error is found in the checksum of protocols such as TCP/IP/ICMP/UDP, the frame will not be discarded.	0
[25:22]	Reserved	RO	Reserved	0
21	TSF	RW	Transmit store and forward. 0: After the data written into the FIFO in the sending process reaches a certain value, the sending will be started; 1: The sending process will start sending after the complete frame is written into the FIFO.	0
20	FTF	RW	Transmit FIFO empty control. Setting this bit will reset the transmit FIFO.	0
[19:14]	Reserved	RO	Reserved	0

13	ST	RW	Start or stop transmission. 0: After sending the current frame, the sending process enters the stop state; 1: Put the sending process in the running state.	0
[12:8]	Reserved	RO	Reserved	0
7	FEF	RW	Forward error frame. 0: Receive FIFO discards frames with errors. 1: All frames are forwarded to the DMA except for frames that are too short.	0
6	FUGF	RW	Forward too short frame. 0: The receive FIFO discards all frames less than 64 bytes in length; 1: The receive FIFO forwards a frame whose length is too short.	0
[5:2]	Reserved	RO	Reserved	0
1	SR	RW	Start or stop reception. 0: After forwarding the currently received frame, the receiving DMA enters the stop mode, and the next transmission starts from the current receiving descriptor position; 1: Open the receiving process, DMA fetches the receiving descriptor from the current position or fetches the receiving descriptor from the identifier header position.	0
0	Reserved	RO	Reserved	0

29.2.4.8 DMA Interrupt Enable Register (R32_ETH_DMAIER)

Offset address: 0x101C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															NISE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AISE	ERS	Reserved		PHYLINK_IER	ETIE	RWTIE	RPSIE	RBUIE	RIE	TUIE	ROIE	TJTIE	TBUIE	TPSIE	TIE

Bit	Name	Access	Description	Reset Value
[31:17]	Reserved	RO	Reserved	0
16	NISE	RW	Normal interrupt enable. Enabling this bit will enable the following interrupts. -DMASR[0]: Transmit interrupt; -DMASR[2]: Transmit buffer unavailable; -DMASR[6]: Receive interrupt;	0

			-DMASR[14]: Early receive interrupt.	
15	AISE	RW	Abnormal interrupt. Enabling this bit will enable the following interrupts. -DMASR[1]: The sending process is stopped; -DMASR[3]: Send Jabber timeout; -DMASR[4]: Receive FIFO overflow; -DMASR[5]: transmit data underflow; -DMASR[7]: Send buffer unavailable; -DMASR[8]: The receiving process is stopped; -DMASR[9]: Receive watchdog timeout; -DMASR[10]: Early transmit;	0
14	ERS	RW	Early receive interrupt enable. Enable this bit to generate an early receive interrupt. NISE must be set. AISE must be set.	0
[13:12]	Reserved	RO	Reserved	0
11	PHYLINK_IER	RW	Transceiver connection interrupt enable bit. Enabling this bit generates a transceiver connection interrupt.	0
10	ETIE	RW	Early transmit interrupt enable. Enable this bit to generate an early transmit interrupt. AISE must be set.	0
9	RWTIE	RW	Receive watchdog interrupt enable. Enable this bit to generate a receive watchdog timeout interrupt. AISE must be set.	0
8	RPSIE	RW	Receive process stop interrupt enable. Enable this bit to generate a receive process stop interrupt.	0
7	RBUIE	RW	Receive buffer unavailable interrupt enable. Enable this bit to generate a receive buffer unavailable interrupt. AISE must be set.	0
6	RIE	RW	Receive completed interrupt enable bit. Enable this bit to generate receive completed interrupt. NISE must be set.	0
5	TUIE	RW	Transmit underflow interrupt enable. Enable this bit to generate a transmit underflow interrupt. AISE must be set.	0
4	ROIE	RW	Receive overflow interrupt. Enable this bit to generate a receive overflow interrupt. AISE must be set.	0
3	TJTIE	RW	Transmit Jabber timeout interrupt enable. Enable this bit to generate transmit Jabber timeout interrupt. AISE must be set.	0
2	TBUIE	RW	Transmit buffer unavailable interrupt enable. Enable this bit to generate transmit buffer unavailable interrupt. NISE must be set	0
1	TPSIE	RW	Transmit process stop interrupt enable. Enable this bit to generate transmit process stop interrupt. AISE must be set.	0

0	TIE	RW	Transmit completed interrupt enable. Enable this bit to generate transmit completed interrupt. NISE must be set.	0

29.2.4.9 DMA Lost Frame Register (R32_ETH_DMAMFBOCR)

Offset address: 0x1020

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			OFOC	MFA[10:0]											OMFC

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MFC[15:0]															

Bit	Name	Access	Description	Reset Value
[31:29]	Reserved	RO	Reserved	0
28	OFOC	RZ	The overflow bit of the FIFO overflow counter.	0
[27:17]	MFA[10:0]	RZ	Frames lost by the application.	0
16	OMFC	RZ	Missing frame counter overflow.	0
[15:0]	MFC[15:0]	RZ	Lost frame counter. This field indicates the number of frames lost due to unavailability of receive buffers.	0

29.2.4.10 DMA Current Transmit Descriptor Register (R32_ETH_DMACHTDR)

Offset address: 0x1048

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTDAR[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTDAR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	HTDAR[31:0]	RO	This register value points to the transmit descriptor currently in use. This register is updated in real time by DMA.	0

29.2.4.11 DMA Current Receive Descriptor Register (R32_ETH_DMACHRDR)

Offset address: 0x104C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HRDAR[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HRDAR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	HRDAR[31:0]	RO	This register value points to the receive descriptor currently in use. This register is updated in real time by DMA.	0

29.2.4.12 DMA Current Transmit Buffer Register (R32_ETH_DMACHTBAR)

Offset address: 0x1050

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTBAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTBAR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	HTBAR[31:0]	RO	This register value points to the transmit buffer currently in use. This register is updated in real time by DMA.	0

29.2.4.13 DMA Current Receive Buffer Register (R32_ETH_DMACHRBAR)

Offset address: 0x1054

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HRBAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HRBAR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	HRBAR[31:0]	RO	This register value points to the receive buffer currently in use. This register is updated in real time by DMA.	0

29.2.5 Internal 10M/100M Physical Layer Register Address

29.2.5.1 Basic Control Register (R16_BMCR)

Offset address: 0x00

Bit	Name	Access	Description	Reset Value
15	RST	RW/SC	1: PHY reset; 0: Normal operation.	0
14	Loopback	RW	1: Enable loopback; 0: Normal operation.	0
13	Speed Selection	RW	1: 100Mb/s; 0: 10Mb/s; After auto-negotiation, the status of negotiation working speed is displayed.	1

12	Auto-Negotiation	RW	0: Disable auto-negotiation; 1: Enable auto-negotiation.	1
11	Power Down	RW	1: Turn off the power supply; 0: Normal operation.	0
10	Isolate	RW	1. MII/RMII interface is isolated from PHY, and PHY can still respond to MDC/MDIO; 0: It works normally.	0
9	Restart Auto-Negotiation	RW/SC	0: Normal operation; 1: Restart auto-negotiation.	0
8	Duplex Mode	RW	0: Half-duplex; 1: Full-duplex.	1
7	Collision Test	RW	Collision test. 1: Conflict test enabled; 0: Normal operation.	0
[6:0]	Reserved	RO	Reserved	0

29.2.5.2 Basic Status Register (R16_BMSR)

Offset address: 0x01

Bit	Name	Access	Description	Reset Value
15	100BASE-T4	RO	1: Supports enabling 100BASE-T4; 0: Does not support 100BASE-T4.	0
14	100BASE-TX Full Duplex	RO	1: Supports enabling 100BASE-TX full-duplex; 0: Does not support 100BASE-TX full-duplex.	1
13	100BASE-TX Half Duplex	RO	1: Supports enabling 100BASE-TX half-duplex; 0: Does not support 100BASE-TX half-duplex.	1
12	10BASE-T Full Duplex	RO	1: Supports enabling 10BASE-TX full-duplex; 0: Does not support 10BASE-TX full-duplex.	1
11	10BASE-T Half Duplex	RO	1: Supports enabling 10BASE-TX half-duplex; 0: Does not support 10BASE-TX half-duplex.	1
[10:7]	Reserved	RO	Reserved	0
6	MF Preamble Suppression	RO	Management frames with preamble suppression are allowed to be received.	1
5	Auto-Negotiation Complete	RO	1: Auto-negotiation is completed; 0: Auto-negotiation is not completed.	0
4	Remote Fault	RO	1: Remote error condition detected (read clear); 0: No remote error condition was detected.	0
3	Auto-Negotiation Ability	RO	1: PHY can perform self-negotiation; 0: PHY cannot perform self-negotiation.	1
2	Link Status	RO	1: A valid link has been established; 0: No valid link has been established.	0
1	Jabber Detect	RO	1: Jabber condition detected; 0: Jabber condition not detected.	0
0	Extended Capability	RO	1: Extended register functionality; 0: No extended register functionality.	1

29.2.5.3 Auto-Negotiation Message Register(R16_ETH_ANAR)

Offset address: 0x04

Bit	Name	Access	Description	Reset Value
15	NP	RW	Next page position: 1: Send protocol rule data page; 0: Send primary capability data page.	0
14	ACK	RO	1: Acknowledged receipt of the link partner capability data word; 0: No acknowledgment signal received.	0
13	RF	RW	1: Notify remote error detection capabilities; 0: Do not notify remote error detection capabilities.	0
12	Reserved	RO	Reserved	0
11	ASYPAUSE	RW	1: Support asymmetric flow control; 0: Does not support asymmetric flow control.	0
10	PAUSE	RW	1: Support flow control; 0: Does not support flow control.	0
9	100BASE_T4	RO	1: Local node supports 100Base-T4; 0: Local node does not support 100Base-T4.	0
8	100BASE_TX_FD	RW	1: Local node supports 100Base-TX full-duplex; 0: Local node supports 100Base-TX full-duplex.	1
7	100BASE_TX	RW	1: Local node supports 100Base-TX; 0: Local node supports 100Base-TX.	1
6	10BASE_T_FD	RW	1: Local node supports 10Base-TX full-duplex; 0: Local node supports 10Base-TX full-duplex.	1
5	10BASE_T	RW	1: Local node supports 10Base-TX half-duplex; 0: Local node supports 10Base-TX half-duplex.	1
[4:0]	SELECT[4:0]	RO	The binary code selector supported by this node is currently only CSMA/CD00001, and no other protocols support it.	1

29.2.5.4 Auto-Negotiation Linker Capability Register(R16_ETH_ANLPAR)

Offset address: 0x05

Bit	Name	Access	Description	Reset Value
15	NP	RW	Next page bit. 1: Transmit protocol details data page; 0: Transmit basic capability data page;	0
14	ACK	RO	1: Acknowledgment received for peer auto-negotiation	0

			feature register; 0: No acknowledgment received.	
13	RF	RW	1: Enable remote error detection notification; 0: Disable remote error detection notification.	0
12	Reserved	RO	Reserved	0
11	ASYPAUSE	RO	1: Supports asymmetric pause notifications; 0: Does not support asymmetric pause. When auto-negotiation is enabled, this bit reflects the link partner's capability.	0
10	PAUSE	RO	1: Supports symmetric notification suspension; 0: Does not support symmetric suspension.	0
9	100BASE_T4	RO	1: PHY supports 100BASE-T4; 0: PHY does not support 100BASE-T4.	0
8	100BASE_TX_FD	RO	1: PHY supports 100BASE-T full-duplex; 0: PHY does not support 10BASE-T full-duplex.	0
7	100BASE_TX	RO	1: PHY supports 100BASE-T; 0: PHY does not support 10BASE-T.	0
6	10BASE_T_FD	RO	1: PHY supports 10BASE-T full-duplex; 0: PHY does not support 10BASE-T full-duplex.	0
5	10BASE_T	RO	1: PHY supports 10BASE-T; 0: PHY does not support 10BASE-T.	0
[4:0]	SELECT[4:0]	RO	The PHY supports a binary encoding selector. Currently, only CSMA/CD 00001 is special; no other protocols are supported.	0x01

29.2.5.5 Auto-negotiation Extension Register (R16_ANER)

Offset address: 0x06

Bit	Name	Access	Description	Reset Value
[15:7]	Reserved	RO	Reserved	0
6	RX_NPLA	RO	Whether the storage location of the next page sent by the peer is determined based on bit5.	0
5	RX_NPSL	RO	1: Indicates whether the next page of the peer is stored in register 8; 0: Indicates whether the next page of the peer is stored in register 5.	0
4	Parallel Detection Fault	RZ	1: An error was detected during the parallel detection function; 0: No error was detected during the parallel detection	0

			function.	
3	Link Partner Next Page Ability	RO	1: The link partner has the next page capability; 0: The link partner does not have the next page capability.	0
2	Local Next Page Ability	RO	1: The local device has the ability to go to the next page; 0: The local device does not have the ability to go to the next page.	1
1	Page Received	RZ	1: A new page was received; 0: A new page was not received.	0
0	Link Partner Auto-Negotiation Ability	RO	If the local device auto-negotiation function is enabled, this bit means: 1: The link partner has auto-negotiation capability; 0: The link partner does not have auto-negotiation capability.	0

29.2.5.6 Page Selection Register (R16_PAGE_SEL)

Offset address: 0x1F

Bit	Name	Access	Description	Reset Value
[15:8]	Reserved	RO	Reserved	0
[7:0]	SEL_PAGE_ADDR[7:0]	RW	Select page address: 00000000: 11111111.	0

29.2.6 PAGE0-related Registers

29.2.6.1 PHY Status Register (R16_PHY_STATUS1)

Offset address: 0x1A

Bit	Name	Access	Description	Reset Value
15	incompatible_link	RO	After auto-negotiation priority resolution, the compatibility status between the local device and the peer device's capabilities: 1: Incompatible; 0: Compatible.	0
[14:7]	Reserved	RO	Reserved	0
6	signal_status	RO	100M signal status: 1: Signal status is good; 0: No signal.	0
5	mdi_mdix_status	RO	MDI/MDIX status: 1: MDI; 0: MDIX.	1
4	polarity_status	RO	Polarity status:	0

			1: Polarity reversed; 0: Polarity normal.	
[3:0]	phy_sepd_dup[3:0]	RO	PHY operating rate and full/half-duplex operating mode.	0

phy_sepd_dup detail table				
3	phy_100bt	RO	Operation rate 100M.	0
2	phy10_indicate	RO	Operation rate 10M.	0
1	full_indicat	RO	Full-duplex.	0
0	hfull_indicat	RO	Half-duplex.	0

29.2.6.2 Interrupt Indicator Register (R16_INTERRUPT_IND)

Offset address: 0x1E

Bit	Name	Access	Description	Reset Value
[15:12]	Reserved	RO	Reserved	0
11	linkstatus_chg	RZ	Link state change notification.	0
[10:0]	Reserved	RO	Reserved	0

29.2.7 PAGE7-related Registers

29.2.7.1 Interrupt/WOL/LED Flag Register (R16_INTERRUPT_MASK)

Offset address: 0x13

Bit	Name	Access	Description	Reset Value
[15:14]	Reserved	RO	Reserved	0
13	int_linkchg	RW	Link change interrupt Mask.	0
[12:10]	Reserved	RO	Reserved	0
9	led_enable	RW	LED function enabled.	0
[8:0]	Reserved	RO	Reserved	0x30

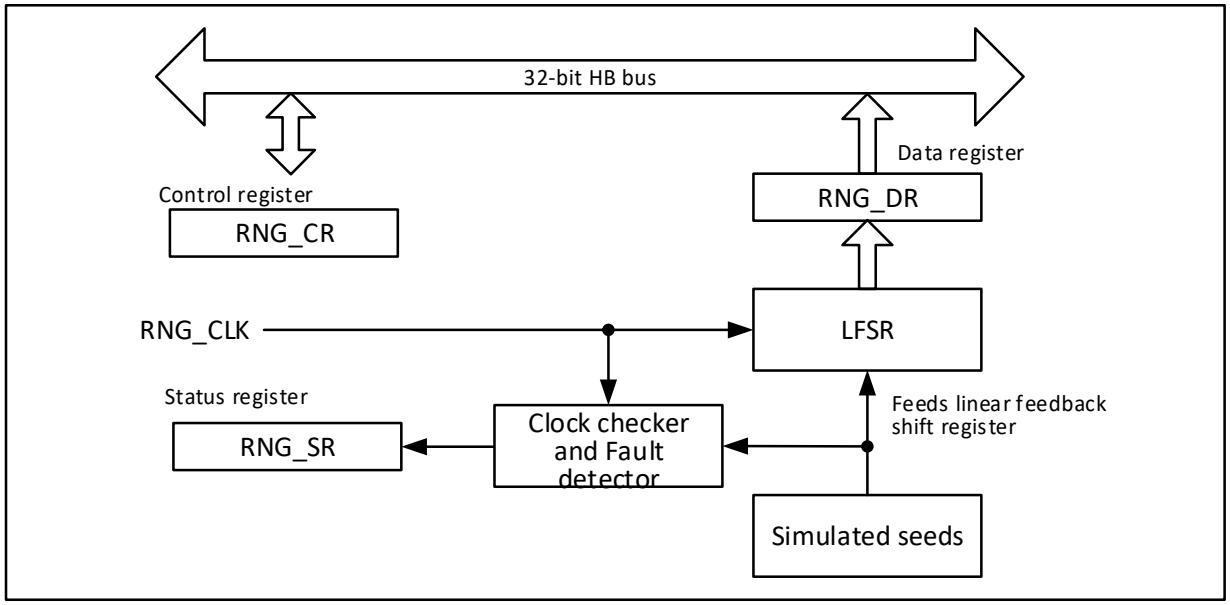
Chapter 30 Random Number Generator (RNG)

The chip has a built-in hardware random number generator, which provides a 32-bit random number through an internal analog circuit based on continuous analog noise.

30.1 Main Features

- Can generate 32-bit random numbers
- Error management is possible
- Can be disabled individually to reduce power consumption

Figure 30-1 RNG block diagram



30.2 Functional Description

The random transmitter is implemented using an analog circuit that generates a seed for a linear feedback shift register (RNG_LFSR) for generating 32-bit random numbers. RNG_LFSR provides clock information at a constant frequency by a dedicated clock (PLL48CLK), so the quality of random numbers is related to the HCLK clock. When a large number of seeds are introduced into RNG_LFSR, the content of RNG_LFSR will be transferred to the data register (RNG_DR).

30.2.1 RNG Operations

RNG operation procedure:

1） If the interrupt is enabled, by setting the IE bit to 1 in the RNG_CR register (The interrupt is generated when a random number is ready or an error occurs).

2） Random number generation is enabled by configuring the RNGEN bit in the RNG_CR register, while the analog section, RNG_LFSR and error detector are activated.

3） If the interrupt is enabled, every time an interrupt is generated, the SEIS and CEIS bits in the R32_RNG_SR register are 0 to confirm that no error occurs and that DRDY is 1 to confirm that the random number is ready.

The contents of the RNG_DR register can then be read.

30.2.2 Error Management

RNG errors include clock errors and seed errors. When a clock error occurs, the RNG can no longer generate random numbers. At this time, it is necessary to check whether the clock controller is configured correctly and whether the RNG clock can be provided, and then clear the CEIS bit. When the CECS bit is 0, the RNG can work normally. When a clock error occurs, it has no effect on the last random number and can be used normally. When a seed error occurs, the value in the RNG_DR register cannot use the random number at this time. If the RNG is to be reused, the SEIS bit needs to be cleared first, then the RNGEN bit is cleared and set to 1, and the RNG is reinitialized and restarted.

30.3 Register Description

Table 30-1 RNG registers

Name	Address	Description	Reset value
R32_RNG_CR	0x40023C00	RNG control register	0x00000000
R32_RNG_SR	0x40023C04	RNG status register	0x00000000
R32_RNG_DR	0x40023C08	RNG data register	0x00000000

30.3.1 RNG Control Register (R32_RNG_CR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												IE	RNG EN	Reserved	

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	IE	RW	Interrupt enable: 0: Disable RNG interrupt. 1: Enable RNG interrupt.	0
2	RNGEN	RW	Random number generator enable: 0: Disable random number generator. 1: Enable random number generator.	0
[1:0]	Reserved	RO	Reserved	0

30.3.2 RNG Status Register (R32_RNG_SR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									SEIS	CEIS	Reserved		SECS	CECS	DRDY

Bit	Name	Access	Description	Reset Value
[31:7]	Reserved	RO	Reserved	0
6	SEIS	RW	Seed error interrupt status (This bit is set at the same time as SECS) 1: One of the following error sequences is detected: - more than 64 identical consecutive bits; - more than 32 consecutive alternating 0 and 1. 0: No inverted error sequence detected.	0
5	CEIS	RW	Clock error interrupt status (This bit is set at the same time as CECS) 1: PLL48CLK clock not detected correctly; 0: PLL48CLK clock detected correctly.	0
[4:3]	MODE4	RW	Reserved	0
2	SECS	RO	Seed error current status 1: One of the following error sequences was detected: - more than 64 identical consecutive bits; - more than 32 consecutive alternating 0 and 1. 0: No error sequence detected.	0
1	CECS	RO	Clock error current status 1: PLL48CLK clock not detected correctly; 0: PLL48CLK clock detected correctly.	0
0	DRDY	RO	Data ready (This bit is cleared after reading R32_RNG_DR) 1: R32_RNG_DR register is valid, this random number is available; 0: R32_RNG_DR register is invalid, this random number is not available.	0

30.3.3 RNG Data Register (R32_RNG_DR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RNDATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RNDATA[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RNDATA	RO	32-bit random number	0

Chapter 31 Electronic Signature (ESIG)

Electronic Signature contains chip identification information: Flash memory area capacity and unique identification. It is burned into the system storage area of the memory module by the manufacturer when leaving the factory, and can be read through SWD (SDI) or application code.

31.1 Function Description

Flash memory area capacity: Indicates the size that the current chip user application can use.

Unique identification: 96-bit binary code, unique to any microcontroller, users can only read and access it and cannot modify it. This unique identification information can be used as the security password, decryption key, product serial number, etc. of the microcontroller (product) to improve the system security mechanism or indicate identity information.

Users of the above content can perform read access in 8/16/32 bits.

31.2 Register Description

Table 31-1 ESIG-related register

Name	Access address	Description	Reset value
R16_ESIG_FLACAP	0x1FFFF7E0	Flash capacity register	0xXXXX
R32_ESIG_UNIID1	0x1FFFF7E8	UID register 1	0xXXXXXXXX
R32_ESIG_UNIID2	0x1FFFF7EC	UID register 2	0xXXXXXXXX
R32_ESIG_UNIID3	0x1FFFF7F0	UID register 3	0xXXXXXXXX

31.2.1 Flash Capacity Register (ESIG_FLACAP)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F_SIZE[15:0]															

Bit	Name	Access	Description	Reset value
[15:0]	F_SIZE[15:0]	RO	Flash memory capacity in Kbyte. Example: 0x0080 = 128 Kbytes	X

31.2.2 UID Register (ESIG_UNIID1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
U_ID[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U_ID[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	U_ID[31:0]	RO	Bits 0-31 of the UID.	X

31.2.3 UID Register (ESIG_UNIID2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
U_ID[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U_ID[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	U_ID[63:32]	RO	Bits 32-63 of the UID.	X

31.2.4 UID Register (ESIG_UNIID3)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
U_ID[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U_ID[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	U_ID[95:64]	RO	Bits 64-95 of the UID.	X

Chapter 32 Flash Memory and User Option Bytes (FLASH)

32.1 Flash Memory Organization

The flash memory organization inside the chip is as follows:

Table 32-1 Flash organizational structure

Block	Name	Address range	Size (byte)
Main memory	Page 0	0x08000000 – 0x080000FF	256
	Page 1	0x08000100 – 0x080001FF	256
	Page 2	0x08000200 – 0x080002FF	256
	Page 3	0x08000300 – 0x080003FF	256
	Page 4	0x08000400 – 0x080004FF	256
	Page 5	0x08000500 – 0x080005FF	256
	Page 6	0x08000600 – 0x080006FF	256
	Page 7	0x08000700 – 0x080007FF	256
	…	…	…
	Page 3967	0x080F7F00 – 0x080F7FFF	256
Information block	Boot program code	0x1FFF8000 – 0x1FFFEFFF	28K
	User option bytes	0x1FFFF800 – 0x1FFFF87F	128

Note:

1) The above main memory area is used for the user's application storage, and the write protection is divided in 4K bytes (16 pages) units; except that the "manufacturer configuration word" area is factory locked, the user is inaccessible. Other areas can be operated by users under certain conditions.

2) For chips whose fifth digit from the bottom of the lot number is greater than 0, their non-zero waiting areas support two 64-bit accesses, RVV instructions and DMA.

3) For the CH32V407KEU6 chip, the main memory (Code FLASH) has a capacity of 480KB.

32.2 Flash Memory Programming and Safety

32.2.1 Two Program/Erase Methods

- Standard programming: This mode is the default programming mode (compatibility mode). In this mode, the CPU performs programming in a single 2-byte manner, and performs erase and whole-chip erase operations in a single 4K-byte manner.
- Fast programming: This method uses page operation mode (recommended). After a specific sequence is unlocked, a single 256-byte programming, 32K-byte erasing, and entire chip erasing are performed.

Note: Functions related to FLASH erasure and programming may only be placed in the zero-wait zone of the FLASH.

32.2.2 Security-preventing against Illegal Access (read, write and erase)

- Page write protection
- Read protection

When the chip is in read protection:

1) Main memory pages 0-15 (4K bytes) are automatically write-protected and are not controlled by the

FLASH_WPR register; when the read-protected state is released, all main memory pages are controlled by the FLASH_WPR register.

2) The system boot code area, SWD or SDI mode, and RAM area cannot erase or program the main memory, except for full chip erasure. User selectable word area can be erased or programmed. If you try to remove read protection (program user words), the chip will automatically erase the entire user area.

32.3 FLASH Enhanced Read Mode

FLASH Enhanced Read Mode is suitable for user programs to run in FLASH, which can improve the access efficiency of FLASH. To enable this mode, set bit 1 of the EHMOD field in the FLASH_ACTLR register. Then, read the ENHANCE_STATUS bit; if bit 1 of ENHANCE_STATUS is set, the enhanced read mode is active. To disable this mode, first clear bit 1 of EHMOD, then set RSENACT to 1.

Note: When using FLASH enhanced reading mode, you should pay attention to the following points:

1) Before performing any erase or program operations on the FLASH (including user programming such as read protection removal), you must first exit Enhanced Read Mode. Failure to do so will cause erase and program operations to fail.

2) Before entering Stop Mode, you must first exit Enhanced Read Mode. Failure to do so may cause Stop Mode to malfunction.

3) After a power reset or system reset completes, the chip automatically exits Enhanced Read Mode under hardware control.

32.4 Register Description

Table 32-2 FLASH-related registers

Name	Access address	Description	Reset value
R32_FLASH_KEYR	0x40022004	FPEC key register	0xXXXXXXXX
R32_FLASH_OBKEYR	0x40022008	OBKEY register	0xXXXXXXXX
R32_FLASH_STATR	0x4002200C	Status register	0x00000000
R32_FLASH_CTLR	0x40022010	Control register	0x02008080
R32_FLASH_ADDR	0x40022014	Address register	0x00000000
R32_FLASH_OBR	0x4002201C	Option byte register	0xXXXXXXXX
R32_FLASH_WPR	0x40022020	Write protection register	0xXXXXXXXX
R32_FLASH_MODEKEYR	0x40022024	Extension key register	0xXXXXXXXX

32.4.1 FPEC Key Register (FLASH_KEYR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
KEYR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KEYR[15:0]															

Bit	Name	Access	Description	Reset value

[31:0]	KEYR[31:0]	WO	FPEC key, the unlock key used to enter FPEC includes: RDPRT key = 0x000000A5; KEY1 = 0x45670123; KEY2 = 0xCDEF89AB.	X

32.4.2 OBKEY Register (FLASH_OBKEYR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OBKEYR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OBKEYR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	OBKEYR[31:0]	WO	Selection word key for entering a selection word key to disarm OBWRE.	X

32.4.3 Status Register (FLASH_STATR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								EHMODS	Reserved	EOP	WRPRTERR	Reserved		WRBSY	BSY

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
7	EHMODS	RO	FLASH enhanced read mode enable status bit: 1: FLASH enhanced read mode is enabled; 0: FLASH enhanced read mode is not enabled.	0
6	Reserved	RO	Reserved	0
5	EOP	RW1	Indicates the end of the operation, write 1 to clear. The hardware is set every time it is successfully erased or programmed.	0
4	WRPRTERR	RW1	Indicates write protection error, write 1 to clear. If you program a write-protected address, the hardware will set it.	0
[3:2]	Reserved	RO	Reserved	0
1	WRBSY	RO	This bit is used during fast page programming to indicate that programming data is being written. During page programming, when writing data, this bit is set to '1' and the hardware automatically clears '0'; If this bit is '0', it means that writing the next data is allowed.	0
0	BSY	RO	Indicates busy status:	0

			1: Indicates that a flash operation is in progress; 0: The operation is finished.	

Note: When performing programming operations, you need to ensure that the STRT bit of the FLASH_CTLR register is 0.

32.4.4 Control Register (FLASH_CTLR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					SCKMOD [1:0]		EHM OD	Reser ved	RSE NAC T	PGST RT	Reserved		BER3 2	Reser ved	FTPG

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLO CK	Reserved		EOPI E	Reser ved	ERRI E	OBW RE	Reser ved	LOC K	STR T	OBER	OBP G	Reser ved	MER	PER	PG

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
[26:25]	SCKMOD[1:0]	RW	FLASH Access Clock Configuration: 00: FLASH access clock frequency = HCLK; 01: FLASH access clock frequency = (HCLK/2); 10: FLASH access clock frequency = (HCLK/4); 11: Reserved. *Note: The FLASH access clock frequency does not exceed 100 MHz;* *This bit enables standard unlock + fast unlock.*	01b
24	EHMOD	RW	FLASH enhanced read mode: This mode improves access efficiency when the program is running in FLASH. 1: Enables FLASH Enhanced Read Mode. 0: Disables FLASH Enhanced Read Mode; must be used in conjunction with the RSENACT bit. To disable this mode, first clear the EHMOD bit to 0, then set the RSENACT bit to 1. *This bit enables standard unlock + fast unlock.*	0
23	Reserved	RO	Reserved	0
22	RSENACT	WO	To exit Enhanced Read Mode, the hardware automatically clears the mode. This operation requires coordination with the EHMOD bit. The exit procedure involves first clearing the EHMOD bit to 0, then setting RSENACT to 1. *This bit enables standard unlocking.*	0
21	PGSTRT	WO	Start. Set to 1 to start a page programming, and	0

			the hardware will be cleared automatically. *This bit enables standard unlocking.*	
[20:19]	Reserved	RO	Reserved	0
18	BER32	RW	Perform a 32KB erase. *This bit enables standard unlock + fast unlock.*	0
17	Reserved	RO	Reserved	0
16	FTPG	RW	Perform fast page programming operation. *This bit enables standard unlocking + quick unlocking.*	0
15	FLOCK	RW1	Fast Program Lock. Writes only '1'. When this bit is '1', quick program/erase mode is disabled. Hardware clears this bit to '0' upon detecting the correct unlock sequence. Software sets to '1' to re-lock. *This bit enables standard unlocking.*	1
[14:13]	Reserved	RO	Reserved	0
12	EOPIE	RW	Interrupt control after operation completion (EOP bit set in FLASH_STATR register): 1: Enable interrupt generation; 0: Disable interrupt generation. *This bit enables standard unlocking.*	0
11	Reserved	RO	Reserved	0
10	ERRIE	RW	Error status interrupt control (PGERR/WRPRTERR set in FLASH_STATR register): 1: Allows interrupt generation; 0: Disables interrupt generation. *This bit enables standard unlocking.*	0
9	OBWRE	RW0	User option word lock, software reset: 1: Indicates the user option byte is programmable. Requires writing the correct sequence to the R32_FLASH_OBKEYR register followed by hardware setting; 0: User option byte relocks after software reset. *This bit enables standard unlocking.*	0
8	Reserved	RO	Reserved	0
7	LOCK	RW1	Lock. Writable only as '1'. When this bit is '1', it indicates that FPEC and R32_FLASH_CTLR are locked and cannot be written. After detecting the correct unlock sequence, the hardware clears this bit to '0'. Following an unsuccessful unlock operation, this bit remains unchanged until the next system reset.	1

			This bit enables standard unlocking.	
6	STRT	RW1	Start. Set to 1 to start an erase operation, and the hardware automatically clears 0(BSY changes to '0'). *This bit enables standard unlocking.*	0
5	OBER	RW	User option word erasure is performed. *This bit enables standard unlocking.*	0
4	OBPG	RW	Perform user option byte programming. *This bit enables standard unlocking.*	0
3	Reserved	RO	Reserved	0
2	MER	RW	Perform a full erase operation (erase the entire user area). *This bit enables standard unlocking.*	0
1	PER	RW	Perform a sector (4KB) erase operation. *This bit enables standard unlocking.*	0
0	PG	RW	Perform standard programming operations. *This bit enables standard unlocking.*	0

32.4.5 Address Register (FLASH_ADDR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FAR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	FAR[31:0]	WO	The flash address, which is the programming address for programming and the starting address for erasure. The register cannot be written when the BSY bit in the FLASH_STATR register is '1'.	0

32.4.6 Option Byte Register (FLASH_OBR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved						DATA1[7:0]								DATA0[7:6]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA0[5:0]						Reserved	RAM_CODE_	Reserved			STANDYRST	STOPRST	IWDGSW	RDPRT	OBERR

		MOD							

Bit	Name		Access	Description	Reset value
[31:26]	Reserved		RO	Reserved	0
[25:18]	DATA1[7:0]		RO	Data byte 1.	X
[17:10]	DATA0[7:0]		RO	Data byte 0.	X
9	USER	Reserved	RO	Reserved	x
8		RAM_CODE_MOD	RO	1: CODE-576KB + RAM-136KB; 0: CODE-512KB + RAM-200KB. *Note: For the CH32V407KEU6 chip, it is recommended that this bit be set to 0.*	x
[7:5]		Reserved	RO	Reserved	x
4		STANDYRST	RO	System reset control in Standby mode.	x
3		STOPRST	RO	System reset control in Stop mode.	x
2		IWDGSW	RO	Independent watchdog (IWDG) hardware enable bit.	1
1	RDPRT		RO	Read protection status. 1: Indicate that the current read protection of the flash memory is valid; 0: Indicate that the current read protection of the flash memory is invalid.	1
0	OBERR		RO	Option byte error. 1: Indicates that the selected word does not match its complement; 0: Indicates that the selected word matches its complement.	0

Note: USER and RDPRT are loaded from the user selection word area after system reset.

32.4.7 Write Protection Register (FLASH_WPR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
WRP[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WRP[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	WRP[31:0]	RO	Flash memory write protection status. 1: Write protection disabled; 0: Write protection enabled. Each bit represents 4K bytes (16 pages) of storage write protection status.	X

Note: WPR is loaded from the user-selected word area after system reset.

32.4.8 Extension Key Register (FLASH_MODEKEYR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODEKEYR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODEKEYR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	MODEKEYR[31:0]	WO	Enter the following sequence to unlock the Quick Program/Erase mode: KEY1 = 0x45670123; KEY2 = 0xCDEF89AB.	X

32.5 Flash Operation Procedure

32.5.1 Read Operation

Direct addressing is in the general address space, and the user can access the content of the flash memory module and get the corresponding data through any read operation of 8/16/32-bit data.

32.5.2 Flash Memory Unlock

After the system reset, the flash memory controller (FPEC) and FLASH_CTLR register will be locked and cannot be accessed. The flash memory controller module can be unlocked by writing the sequence to the FLASH_KEYR register.

Unlock sequence:

1) Write KEY1 = 0x45670123 to the FLASH_KEYR register (must operate KEY1 first);

2) Write KEY2 = 0xCDEF89AB to the FLASH_KEYR register (must operate KEY2 secondly).

The above operations must be performed sequentially and continuously. Otherwise, it is an error operation, which will lock the FPEC module and FLASH_CTLR register and generate a bus error until the next system reset.

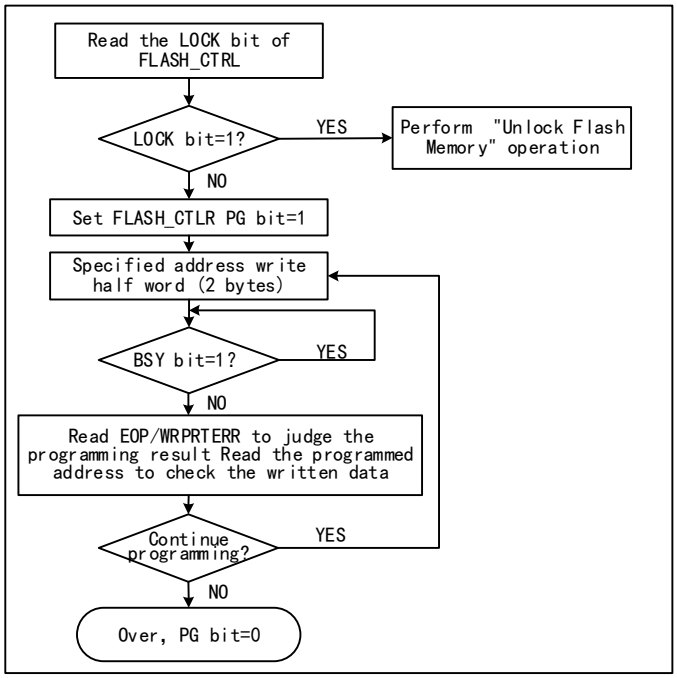
The flash memory controller (FPEC) and the FLASH_CTLR register can be locked again by setting the "LOCK" bit in the FLASH_CTLR register to 1.

32.5.3 Main Memory Standard Program

Standard programming writes 2 bytes per operation. When the PG bit in the R32_FLASH_CTLR register is set to '1', each write of a half-word (2 bytes) to a flash address initiates one programming cycle. Writing any non-half-word data causes FPEC to generate a bus error. During programming, the BSY bit is '1'; upon completion, BSY becomes '0' and EOP becomes '1'.

Note: When the BSY bit is '1', write operations to any register are prohibited.

Figure 32-1 FLASH program



1) Check the LOCK bit in the R32_FLASH_CTLR register. If it is 1, you need to perform the "Release Flash Memory Lock" operation.

2) Set the PEG bit in the FLASH_CTLR register to '1' to enable the standard page erasure mode.

3) Write the half-word to be programmed to the specified flash memory address (even address).

4) Wait for the BSY bit to become '0' or for the EOP bit in the R32_FLASH_STATR register to become '1', indicating programming completion. Then clear the EOP bit to '0'.

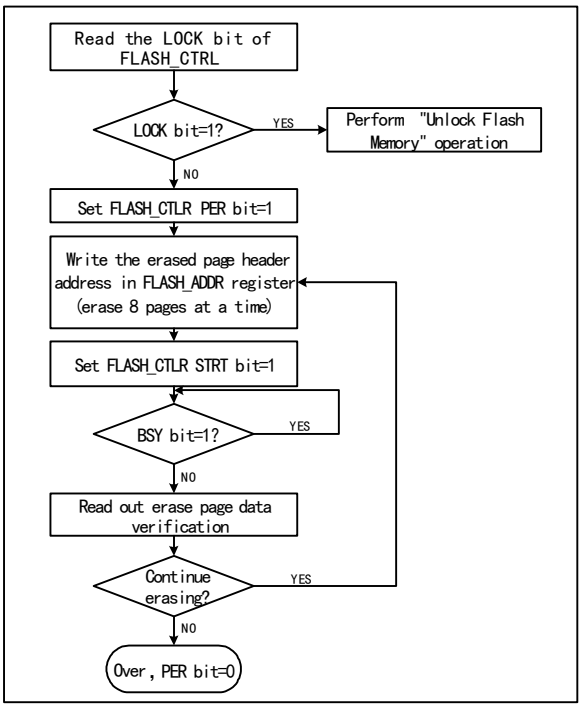
5) Check the R32_FLASH_STATR register for errors, or read the programmed address data for verification.

6) To continue programming, repeat steps 3-5. To end programming, clear the PG bit.

32.5.4 Main Memory Standard Erase

Flash memory can be erased by sectors (4K bytes) or in its entirety.

Figure 32-2 FLASH page erasure



1) Check the LOCK bit in the R32_FLASH_CTLR register. If it is 1, you need to perform the "Release Flash Memory Lock" operation.

2) Set the PEG bit in the FLASH_CTLR register to '1' to enable the sector erasure mode.

3) Write the page heading address of the page to be erased to the FLASH_ADDR register.

4) Set the STAT bit in the FLASH_CTLR register to '1' to start an erase action.

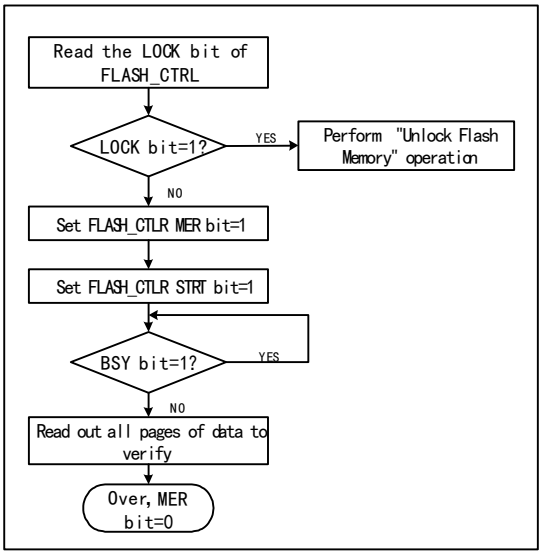
5) When the BYS bit changes to '0' or the EOP bit in the FLASH_STATR register to be '1', it indicates the end of erasure. Clear the EOP bit to 0.

6) Read the page of erasure page for verification.

7) To erase the sector continuously, you can repeat steps 3-5 to end erasing and clear the PEG bit to 0.

Note: After successful erasure, words read -0xe339e339, half words read -0xe339, even address bytes read -0x39, and odd addresses read 0xe3.

Figure 32-3 FLASH full chip erasure



1) Check the LOCK bit of the FLASH_CTLR register. If it is 1, you need to perform the "unlock flash lock" operation.
2) Set the MER bit of the FLASH_CTLR register to '1' to enable full chip erase mode.
3) Set the STRT bit of the FLASH_CTLR register to '1' to start the erase action.
4) Wait for the BSY bit to become '0' or the EOP bit of the FLASH_STATR register to be '1' to indicate the end of erasure, and clear the EOP bit to 0.
5) Read the data of the erased page for verification.
6) Clear the MER bit to 0.

32.5.5 Fast Programming Mode Unlock

Fast programming mode operation can be unlocked by writing a sequence to the FLASH_MODEKEYR register. After unlocking, the block bit of the FLASH_CTLR register will be cleared to 0, indicating that quick erasure and programming operations can be performed. Lock again by setting the "FLOCK" bit of the FLASH_CTLR register to 1.

Unlock sequence:
1) Write KEY1=0x45670123 to the FLASH_MODEKEYR register.
2) Write KEY2=0xCDEF89AB to the FLASH_MODEKEYR register.
The above operations must be performed sequentially and continuously, otherwise the error operation will be locked and cannot be unlocked until the next system reset.
Note: Fast programming requires unlocking the "LOCK" and "FLOCK" layers.

32.5.6 Main Memory Fast Programming

The fast programming (256 bytes) is made according to the page.

1) Check the LOCK bit of the FLASH_CTLR register. If it is '1', you need to perform the "unlock flash" operation.
2) Check the FLASH_CTLR register FLOCK bit, if it is '1', you need to perform a "quick programming mode unlock" operation.
3) Check the BSY bit of the FLASH_STATR register to confirm that there are no other programming operations in progress.
4) Set the FTPG bit of the FLASH_CTLR register to '1' to enable fast page programming mode.

5) Write data to a FLASH address in a 32-bit manner, such as.
* (uint32_t*) 0x80000000=0x12345678.
6) Wait for WRBSY of FLASH_STATR register to be '1', and write the next data.
7) Repeat steps 5-6 64 times
8) Set the FTPG bit in FLASH_CTLR register to' 1' to start fast page programming.
9) Waiting for the BSY bit to change to '0' or the EOP bit of the FLASH_STATR register to '1' means that a quick page programming is completed and the EOP bit is cleared to 0.
10) Query the FLASH_STATR register to see if there are any errors, or read the programming address data check.
11) Continue fast page programming to repeat steps 5-10 and end the programming to clear the FTPG bit 0.

32.5.7 Main Memory Fast Erasure

Fast erase erases in blocks (32K bytes).
1) Check the FLASH_CTLR register LOCK bit, if it is '1', you need to perform the "unlock flash memory" operation.
2) Check the FLASH_CTLR register FLOCK bit, if it is '1', you need to perform a "fast programming mode unlock" operation.
3) Check the BSY bit of the FLASH_STATR register to confirm that there are no other programming operations in progress.
4) Set the BER32 bit of the FLASH_CTLR register to '1' to enable the fast block erase (32K bytes) mode function.
5) Write the first address of the quick erase page to the FLASH_ADDR register.
6) Set the STRT bit of the FLASH_CTLR register to '1' to start a fast block erase (32K bytes) action.
7) Wait for the BSY bit to change to '0' or the EOP bit of the FLASH_STATR register to '1' means the erasure is over, and the EOP bit is cleared to 0.
8) Query the FLASH_STATR register to see if there are any errors, or read the erase page address data check.
9) You can repeat steps 5-8 to continue fast page erasure and clear the BER32 bit to 0 when erasing ends.
Note: After successful erasure, the byte reads -0xe339e339, the half-byte reads -0xe339, the even-address byte reads -0x39, and the odd-address byte reads 0xe3.

32.6 User Option Bytes

The User Option Bytes are solidified in FLASH and will be reloaded into the corresponding register after the system reset, and the user can erase and program at will. The user option word information block has a total of 8 bytes (4 bytes for write protection, 1 byte for read protection, 1 byte for configuration options, 2 bytes for user data storage), and each bit has the inverted code bit for checking during loading. The structure and meaning of the selected word information are described below.

Table 32-3 32-bit option bytes format division

[31:24]	[23:16]	[15:8]	[7:0]
Inverse code of option bytes 1	Option bytes 1	Inverse code of option bytes 0	Option bytes 0

Table 32-4 User option bytes information structure

Address / Bit	[31:24]	[23:16]	[15:8]	[7:0]
0x1FFFF800	nUSER	USER	nRDPR	RDPR

0x1FFFF804	nData1	Data1	nData0	Data0
0x1FFFF808	nWRPR1	WRPR1	nWRPR0	WRPR0
0x1FFFF80C	nWRPR3	WRPR3	nWRPR2	WRPR2

Name/Byte			Description	Reset value
RDPR			Read protection control. It configures whether the code in the flash memory can be read. 0xA5: If this byte is 0xA5 (nRDP must be 0x5A), it means that the current code is in a non-read protected state and can be read; Other values: Code read protection status, unreadable; pages 0-31 pages (4K) will be automatically write-protected and not controlled by WRPR0.	0xA5
USER	7	Reserved	Reserved	1
	6	RAM_CODE_MOD	1: CODE-576KB + RAM-136KB; 0: CODE-512KB + RAM-200KB. *Note: For the CH32V407KEU6 chip, it is recommended that this bit be set to 0.*	x
	[5:3]	Reserved	Reserved	111b
	2	STANDYRST	System reset control in Standby mode: 1: Disabled, the system will not reset when entering standby mode; 0: Enabled, system reset will occur when entering standby mode.	1
	1	STOPRST	System reset control in Stop mode: 1: Disabled, entering stop mode without resetting the system; 0: Enabled, entering stop mode to generate a system reset.	1
	0	IWDGSW	Independent watchdog (IWDG) hardware enable bit: 1: IWDG function is enabled by software, but hardware is not allowed. 0: IWDG function is turned on by the hardware (determined by the LSI clock).	1
Data0–Data1			Stores 2 bytes of user data.	0xFFFF
WRPR0 - WRPR3			Write protection control bit. Each bit is used to control the write protection status of 1 sector (4K bytes/sector) in main memory: 1: Disable write protection. 0: Enable write protection. 4 bytes are used to protect main memory. WRP0: Sector 0-7 storage write protection control; WRP1: Sector 8-15 storage write protection control; WRP2: Sector 16-23 storage write protection control;	0xFFFFFFFF

	WRP3: Bits 0-6 provide write protection for sectors 24-30; Bit 7 provides write protection for sectors 31-119.	

32.6.1 User Option Bytes Unlock

The user option bytes operation can be unlocked by writing the sequence to the FLASH_OBKEYR register. After unlocking, the OBWRE bit in the FLASH_CTLR register will be set to 1, indicating that user option bytes can be erased and programmed. By setting the OBWRE bit in the FLASH_CTLR register, it will be cleared to 0 by software to lock again.

Unlocking sequence

1) Write KEY1 = 0x45670123 to the FLASH_OBKEYR register;

2) Write KEY2 = 0xCDEF89AB to the FLASH_OBKEYR register.

Note: The user needs to unlock the 2 layers: "LOCK" and "OBWRE" for word selection.

32.6.2 User Option Bytes Programming

Only standard programming mode is supported, and half word (2 bytes) is written at a time. In the actual process, when programming the user-selected word, FPEC only uses the low byte in the half word, and automatically calculates the high byte (the high byte is the anti-code of the low byte), and then starts the programming operation, which will ensure that the byte in the user option byte and its anti-code are always correct.

1) Check the LOCK bit in the FLASH_CTLR register. If it is set to 1, perform the “Unlock Flash” operation.

2) Check the BSY bit in the FLASH_STATR register to confirm no other programming operations are in progress.

3) Check the OBWRE bit in the FLASH_CTLR register. If it is 0, perform the “User Option Byte Unlock” operation.

4) Set the OBPG bit in the FLASH_CTLR register to '1'.

5) Write the half-word (2 bytes) to be programmed to the specified address.

6) Wait for the BSY bit to become '0' or for the EOP bit in the FLASH_STATR register to become '1', indicating programming completion. Then clear the EOP bit.

7) Read the programming address data for verification.

8) To continue programming, repeat steps 5-7. To end programming, clear the OBPG bit.

Note: when "read protected" in the modified selection word is changed to "unprotected" state, an entire erase main storage area operation is automatically performed. If you modify a selection other than read Protection, the whole erase operation will not occur.

32.6.3 User Option Bytes Erasure

Erase the entire 128-byte user option bytes area directly.

1) Check the LOCK bit in the FLASH_CTLR register. If it is set to 1, perform the "Unlock Flash" operation.

2) Check the BSY bit in the FLASH_STATR register to confirm no programming operation is in progress.

3) Check the OBWRE bit in the FLASH_CTLR register. If it is 0, perform the "User Option Byte Unlock" operation.

4) Set the OBER bit in the FLASH_CTLR register to '1'. Subsequently, set the STRT bit in the FLASH_CTLR register to '1' to initiate user-selected word erase.

5) Wait for the BSY bit to become '0' or for the EOP bit in the FLASH_STATR register to become '1' indicating erase completion, then clear the EOP bit to 0.

6) Read the erased address data for verification.

7) Clear the OBER bit to 0 to conclude the operation.

Note: After successful erasure, the character reads -0xe339e339, the half-character reads -0xe339, and the byte

reads -0x39.

32.6.4 Read Protection Release

Whether the flash memory is read-protected or not is determined by the user option byte. Read the FLASH_OBR register. When the RDPRT bit is '1', it indicates that the current flash memory is in read-protected state, and the flash operation is protected by a series of read-protected states. The process of unprotecting read is as follows:

1) Erase the entire user option byte area, read the protection field RDPR, and the read protection is still valid.

2) The user chooses words to program and writes the correct RDPR code 0xA5 to remove the read protection of the flash memory. (This step will first cause the system to automatically erase the entire piece of flash memory.).

3) A power-on reset is performed to reload the selected bytes (including the new RDPR code), and the read protection is removed.

32.6.5 Write Protection Release

Whether flash memory is write-protected is determined by the user option byte. Reading the FLASH_WPR register reveals that each bit represents a sector of flash memory space. A bit value of '1' indicates an unprotected state, while '0' indicates write protection. The process to remove write protection is as follows:

1) Erase the entire user option byte area.

2) Write the correct RDPR code 0xA5 to allow read access.

3) Perform a system reset, reload the selected bytes (including the new WRPR[3:0] bytes), and write protection is removed.

Chapter 33 Extended Configuration (EXTEN)

33.1 Extended Configuration

The system provides the EXTEN extension configuration unit (EXTEN_CTR register). This unit uses the HB clock and only performs reset actions during system reset. It mainly includes the following extended control bit functions:

1) Adjust the built-in voltage: Select the default value for the ULLDOTRIM field, and you can modify its value when adjusting performance and power consumption.
2) Lock-up function monitoring: If the LKUPEN field is enabled, the system's Lock-up situation monitoring will be turned on. Once a Lock-up situation occurs, the system will perform a software reset and set the LKUPRST field to 1. After reading, you can write 1 to clear this flag.
3) Adjustment of V_{DD18}: The voltage of V_{DD18} can be configured.

33.2 Register Description

Table 33-1 EXTEN related register list

Name	Access address	Description	Reset value
R32_EXTEN_CTR	0x40023800	Configure extended control register	0x00000240
R32_FEATURE_SIGN	0x1FFFF7D0	Feature information indication register	0xE339XXXX
R32_FEATURE_SIGN2	0x1FFFF7F4	Feature information indication register 2	0xE339XXXX

33.2.1 Configure Extended Control Register (EXTEN_CTR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			LDO18_ADJ	Reserved		ULLDOTRIM[1:0]		LKUPRST	LKUPEN	Reserved					

Bit	Name	Access	Description	Reset value
[31:13]	Reserved	RO	Reserved	0
12	LDO18_ADJ	RW	Adjustment bit of V_{DD18}. 1: 1.9V; 0: 1.8V.	0
[11:10]	Reserved	RO	Reserved	0
[9:8]	ULLDOTRIM[1:0]	RW	Adjust the ULLDO voltage value in low power mode	10b
7	LKUPRST	RW1	LOCKUP reset flag: 1: LOCKUP occurs and causes system reset, write 1 to clear;	0

			0: Normal.	
6	LKUPEN	RW	LOCKUP monitoring function: 1: Enabled, when the system lock-up occurs, it performs a reset and sets LOCKUP_RST; 0: Disabled.	1
[5:0]	Reserved	RO	Reserved	0

33.2.2 Feature Information Indication Register (FEATURE_SIGN)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								Reserved							ETH_LED_EN

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0xE339
[15:8]	Reserved	RO	The reset value of bit[15:8] is the inversion of the reset value of bit[7:0].	X
[7:1]	Reserved	RO	Reserved	0x7F
0	ETH_LED_EN	RO	ETH LED function indication bit: 1: The ETHLED function is enabled, providing LINK and ACT functions; 0: The ETHLED function is not enabled, only LINK function is provided.	x

33.2.3 Feature Information Indication Register 2 (FEATURE_SIGN2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															PSRAM_CLK_IND

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0xE339
[15:8]	Reserved	RO	The reset value for bit[15:8] is the complement of the reset value for bit[7:0].	X
[7:1]	Reserved	RO	Reserved	0x7F
0	PSRAM_CLK_IND	RO	PSRAM Clock Frequency Indicator Bits: 1: 300MHz;	X

			0: 200MHz. *Note: This value applies to the product of the reset values of bits[7:0] and the inverted reset values of bits[15:8] in this register. If the values are not inverted, the maximum clock frequency of the PSRAM is 300MHz.*	

Chapter 34 Pseudo-static Random Access Memory (PSRAM)

Chapter module description is only applicable to CH32V467 microcontroller product.

Based on the CH32V407, the CH32V467 chip also has built-in 4M or 8M bytes of extended PSRAM memory, which supports data storage through addresses and read and write instructions, and supports DMA.

34.1 Main Features

- 8-bit hardware width, supports 8-bit, 16-bit or 32-bit random or block read and write
- Support a maximum clock rate of 300MHz, supports double edges, and a maximum data transfer rate of 600MB/s
- Support write data masking
- Support high-speed read operations with data strobes
- Support register-configurable write and read initial delays
- Support half sleep mode and deep sleep mode

34.2 Register Description

34.2.1 PSRAM Related Registers

Table 34-1 PSRAM related register list

Name	Access address	Description	Reset value
R32_PSRAM_CTLR	0x40024400	PSRAM control register	0x00000300
R32_PSRAM_TIMING	0x40024404	PSRAM timing register	0x070C0014
R32_PSRAM_DATA	0x40024408	PSRAM data register	0x00000000
R32_PSRAM_ADDR	0x4002440C	PSRAM address register	0x00000000
R32_PSRAM_CMD0_CFG	0x40024410	PSRAM command word 0 configuration register	0xA0208000
R32_PSRAM_CMD1_CFG	0x40024414	PSRAM command word 1 configuration register	0xB0FFC040
R32_PSRAM_DMACR	0x40024418	PSRAM DMA control register	0x00000000
R32_PSRAM_DATA_NUM	0x4002441C	PSRAM data transfer quantity register	0x00000000
R32_DMA_MEM_ADDR	0x40024420	Memory starting address register during DMA transfer	0x00000000
R32_DMA_PSRAM_ADDR	0x40024424	PSRAM starting address register during DMA transfer	0x00000000
R32_PSRAM_ISR	0x40024428	PSRAM interrupt status register	0x00000000
R32_PSRAM_IER	0x4002442C	PSRAM interrupt error enable register	0x00000000
R32_PSRAM_STATUS	0x40024430	PSRAM controller status machine register	0x00000000
R32_PSRAM_CECFG	0x40024434	PSRAM CE pulse configuration register	0x000004B0
R32_PSRAM_LATENCY	0x40024438	PSRAM read and write delay register	0x00000505

34.2.1.1 PSRAM Control Register (R32_PSRAM_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															EXIT_LPMD

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					CAP_CFG[2:0]			Reserved		DFIFO	CFIFO	Reserved			

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	EXIT_LPMD	RW	PSRAM exits the current low-power state enable: 1: Make PSRAM exit the current low power consumption; 0: No behavior is generated.	0
[15:11]	Reserved	RW	Reserved	0
[10:8]	CAP_CFG[2:0]	RW	PSRAM capacity configuration: 001: 32Mb; 011: 64Mb; 101: 128Mb; 111: 256Mb; Others: Reserved.	0x3
[7:6]	Reserved	RW	Reserved	0
5	DFIFO	RW	DMA write operation clears FIFO method selection: 1: A DMA write operation clears the data in the read cache of the HB slave interface at the same time; 0: A DMA write operation does not clear the data in the read cache of the HB slave interface.	0
4	CFIFO	RW	Clear read FIFO method selection: 1: Writing to any address in PSRAM will clear the read FIFO; 0: The read FIFO will be cleared only when the write address matches the address in the read buf.	0
[3:0]	Reserved	RW	Reserved	0

34.2.1.2 PSRAM Timing Register (R32_PSRAM_TIMING)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TXLPD[7:0]								TCPH[7:0]							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								TRC[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	TXLPD[7:0]	RW	PSRAM_CE_n sustained pulse width when exiting low power (half sleep and deep sleep): Tpulse= TXLPD[7:0] * HSI clock period.	0x07
[23:16]	TCPH[7:0]	RW	The minimum interval between two consecutive bursts. t_{CPH} = TCPH[7:0] * PSRAM clock period. For example: If PSRAM_CLK is 333M, it is 32n time to meet t_{CPH}, and the minimum writing position should be 12; if PSRAM_CLK is 166M, it is 22n time to meet t_{CPH}, and the minimum writing position should be 4.	0x0C
[15:8]	Reserved	RW	Reserved	0x00
[7:0]	TRC[7:0]	RW	Minimum time for read and write cycles: T= TRC[7:0] * PSRAM clock period. Example: To meet the 60n timing requirements for t_{RC_read} and t_{RC_write}, if PSRAM_CLK is 333M, the minimum value to write is 0x14; if PSRAM_CLK is 166M, the minimum value to write is 0x0A.	0x14

34.2.1.3 PSRAM Data Register (R32_PSRAM_DATA)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RW_HDATA[7:0]								RW_LDATA[7:0]							

Bit	Name	Access	Description	Reset value
[15:8]	RW_HDATA[7:0]	RW	High byte data written or read.	0
[7:0]	RW_LDATA[7:0]	RW	Low byte data written or read.	0

34.2.1.4 PSRAM Address Register (R32_PSRAM_ADDR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								MR_ADDR0[7:0]							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMD[7:0]								Reserved						MW	BUSY

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:16]	MR_ADDR0[7:0]	RW	The address of register 0 to be operated on.	0
[15:8]	CMD[7:0]	RW	Command executed:	0

			Read register: 0x40； Write register: 0xC0。	
[7:2]	Reserved	RO	Reserved	0
1	MW	RW	1: Inform PSRAM that a write operation to the PSRAM register will be initiated; 0: Inform PSRAM that a read operation will be initiated and the data will be placed in the R32_PSRAM_DATA data register.	0
0	BUSY	RW	This bit should read logic 0 before writing to R32_PSRAM_DATA and R32_PSRAM_ADDR. During a PSRAM register access, this bit is set to 1 by the application to indicate that a read or write access is in progress. During a write operation to the PSRAM register, R32_PSRAM_DATA should remain valid until the controller clears this bit. During a read operation to the PSRAM register, R32_PSRAM_DATA is always invalid until the controller clears this bit.	0

34.2.1.5 PSRAM Command Word 0 Configuration Register (R32_PSRAM_CMD0_CFG)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SW_LB[7:0]								SR_LB[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SW[7:0]								SR[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	SW_LB[7:0]	RW	Configurable synchronous write (linear burst) command.	0xA0
[23:16]	SR_LB[7:0]	RW	Configurable synchronous read (linear burst) command.	0x20
[15:8]	SW[7:0]	RW	Configurable synchronous write commands.	0x80
[7:0]	SR[7:0]	RW	Configurable synchronous read command.	0

34.2.1.6 PSRAM Command Word 1 Configuration Register (R32_PSRAM_CMD1_CFG)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REFRESH[7:0]								GLRST[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MREGW[7:0]								MREGR[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	REFRESH[7:0]	RW	Configurable refresh command.	0xB0
[23:16]	GLRST[7:0]	RW	Configurable global reset command.	0xFF
[15:8]	MREGW[7:0]	RW	Configurable register write command.	0xC0
[7:0]	MREGR[7:0]	RW	Configurable register read command.	0x40

34.2.1.7 PSRAM DMA Control Register (R32_PSRAM_DMACR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								TOWBL_TOUT[3:0]				DMA_ONEBL[3:0]			

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				MEMORY_SIZE[1:0]		Reserved					DAT_DIR	Reserved			DMA EN

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:20]	TOWBL_TOUT[3:0]	RW	Pause time between bursts, in HCLK units: 0000: Indicates that the transmission is not paused; 0001: Pauses for 1 cycle; 0010: Pauses for 2 cycles; ... 1110: Pauses for 14 cycles; 1111: Pauses for 15 cycles.	0
[19:16]	DMA_ONEBL[3:0]	RW	DMA burst length: 0000: 2; 0001: 4; 0010: 6; 0011: 8; Others: 16.	0
[15:12]	Reserved	RO	Reserved	0
[11:10]	MEMORY_SIZE[1:0]	RW	Memory data width: 00: Reserved; 01: 16-bit; 10: 32-bit; 11: Reserved. This bit is set and cleared by software.	0
[9:5]	Reserved	RO	Reserved	0
4	DAT_DIR	RW	Data transfer direction: 1: MEM to PSRAM; 0: PSRAM to MEM. This bit is set and cleared by software. *Note: This bit needs to be cleared at the same time when DMA is enabled and turned off.*	0
[3:1]	Reserved	RO	Reserved	0

0	DMAEN	RW	DMA enable: 1: Enable; 0: Disable.	0

34.2.1.8 PSRAM Data Transfer Number Register (R32_PSRAM_DATA_NUM)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA_NUM[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	DATA_NUM[15:0]	RW	Number of data transfers: Write when the DMAEN bit of the R32_PSRAM_DMACR register is 0, and read only when the DMAEN bit of the R32_PSRAM_DMACR register is 1, indicating the remaining number to be transferred.	0

34.2.1.9 Memory Starting Address Register During DMA Transfer (R32_DMA_MEM_ADDR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMA_MEM_ADDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMA_MEM_ADDR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	DMA_MEM_ADDR[31:0]	RW	The starting address of the memory during DMA transfer (single byte alignment).	0

34.2.1.10 PSRAM Starting Address Register During DMA Transfer (R32_DMA_PSRAM_ADDR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMA_PSRAM_ADDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMA_PSRAM_ADDR[15:0]															

Bit	Name	Access	Description	Reset value

[31:0]	DMA_PSRAM_ADDR[31:0]	RW	The starting address of PSRAM during DMA transfer (4-byte alignment in 32 mode, 2-byte alignment in 16 mode).	0

34.2.1.11 PSRAM Interrupt Status Register (R32_PSRAM_ISR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												AEIF	CENEIF	DMATF	Reserved

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	AEIF	RO	Address error interrupt flag, the address exceeds the MEM capacity.	0
2	CENEIF	RW1Z	CEM error interrupt flag: 1: CEM error interrupt is generated; 0: CEM error interrupt is not generated.	0
1	DMATF	RW1Z	DMA bus transfer completion flag: 1: DMA bus transfer is completed; 0: DMA bus transfer is not completed.	0
0	Reserved	RO	Reserved	0

34.2.1.12 PSRAM Interrupt Error Enable Register (R32_PSRAM_IER)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												AEIEN	CENEIEN	DMATEN	Reserved

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	AEIEN	RW	Address error interrupt enable: 1: Enable; 0: Disable.	0
2	CENEIEN	RW	CEM error interrupt flag enable: 1: Enable; 0: Disable.	0

1	DMATEN	RW	DMA bus transfer completion enable: 1: Enable; 0: Disable.	0
0	Reserved	RO	Reserved	0

34.2.1.13 PSRAM Controller Status Machine Register (R32_PSRAM_STATUS)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											PSRAM_STATE[4:0]				

Bit	Name	Access	Description	Reset value
[31:5]	Reserved	RO	Reserved	0
[4:0]	PSRAM_STATE[4:0]	RO	PSRAM controller status machine: 0000: Idle status; 0001: Transmit command address status; 0010: Write register status; 0011: Read register status; 0100: Write delay status; 0101: Reserved 0110: Write data status; 0111: Read delay status; 1111: Wait for the shortest CEM completion status.	0

34.2.1.14 PSRAM CE Pulse Configuration Register (R32_PSRAM_CECFG)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															CEP WEN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				CELPW[11:0]											

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	CEPWEN	RW	CE pulse width detection function enable: 1: Disable CE pulse width detection function; 0: Enable CE pulse width detection function.	0
[15:12]	Reserved	RO	Reserved	0
[11:0]	CELPW[11:0]	RW	The maximum duration of PSRAM_CE_n low pulse	0x4B0

			width (excluding half-sleep): T=CELPW[11:0]*PSRAM clock period.	

34.2.1.15 PSRAM Read and Write Delay Register (R32_PSRAM_LATENCY)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WLATENCY[7:0]								RLATENCY[7:0]							

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:8]	WLATENCY[7:0]	RW	Write delay, the delay from when the write command is issued to when the written data becomes valid.	0x05
[7:0]	RLATENCY[7:0]	RW	Read delay, the delay after issuing the read command until the read data is valid.	0x05

Chapter 35 Debug Support (DBG)

35.1 Main Features

This register allows configuration of the MCU in debug state. include:

- Counters that support an independent watchdog (IWDG)
- Counters that support a window watchdog (WWDG)
- Counters that support timers
- Support I2C SMBus timeout control
- Support CAN communication

35.2 Register Description

35.2.1 Debug MCU Configuration Register (DBGMCU_CR)

Offset address: 0x7C0(CSR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved											CAN1_STOP	TIM8_STOP	TIM7_STOP	TIM6_STOP	TIM5_STOP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIM4_STOP	TIM3_STOP	TIM2_STOP	TIM1_STOP	Reserved	I2C_SMBUS_TIMEOUT	WWDG_STOP	IWDG_STOP	Reserved					STANDBY	STOP	SLEEP

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
20	CAN1_STOP	RW	CAN debug stop bit. CAN1 stops running when the core enters the debug state. 1: The CAN receiving register does not continue to receive data; 0: CAN is still running normally.	0
19	TIM8_STOP	RW	Timer 8 debug stop bit. The counter stops working when the core enters debug state. 1: The counter of timer 8 stops working; 0: The counter of timer 8 still works normally.	0
18	TIM7_STOP	RW	Timer 7 debug stop bit. The counter stops working when the core enters debug state. 1: The counter of timer 7 stops working; 0: The counter of timer 7 still works normally.	0
17	TIM6_STOP	RW	Timer 6 debug stop bit. The counter stops working when the core enters debug state. 1: The counter of timer 6 stops working;	0

			0: The counter of timer 6 still works normally.	
16	TIM5_STOP	RW	Timer 5 debug stop bit. The counter stops working when the core enters debug state. 1: The counter of timer 5 stops working; 0: The counter of timer 5 still works normally.	0
15	TIM4_STOP	RW	Timer 4 debug stop bit. The counter stops working when the core enters debug state. 1: The counter of timer 4 stops working; 0: The counter of timer 4 still works normally.	0
14	TIM3_STOP	RW	Timer 3 debug stop bit. The counter stops working when the core enters debug state. 1: The counter of timer 3 stops working; 0: The counter of timer 3 still works normally.	0
13	TIM2_STOP	RW	Timer 2 debug stop bit. The counter stops working when the core enters debug state. 1: The counter of timer 2 stops working; 0: The counter of timer 2 still works normally.	0
12	TIM1_STOP	RW	Timer 1 debug stop bit. The counter stops working when the core enters debug state. 1: The counter of timer 1 stops working; 0: The counter of timer 1 still works normally.	0
11	Reserved	RO	Reserved	0
10	I2C_SMBUS_TIMEOUT	RW	SMBUS timeout mode debug stop bit. Stop SMBUS timeout mode when the core enters debug state. 1: Freeze SMBUS timeout control; 0: Same as normal mode operation.	0
9	WWDG_STOP	RW	Window watchdog debugging stop bit. The debug window watchdog stops working when the kernel enters debug state. 1: The window watchdog counter stops working; 0: The window watchdog counter still works normally.	0
8	IWDG_STOP	RW	Independent watchdog debug stop bit. The watchdog stops working when the core enters debug state. 1: The watchdog counter stops working; 0: The watchdog counter still works normally.	0
[7:3]	Reserved	RW	Reserved	0
2	STANDBY	RW	Debug standby mode bit. 1: (FCLK is on, HCLK is on) The digital circuit part is not powered off, and the FCLK and HCLK clocks are provided by the internal RC oscillator. In addition, the microcontroller exiting STANDBY mode by generating a system reset is the same as resetting; 0: (FCLK off, HCLK off) The entire digital circuit part is powered off. From a software point of view, exiting	0

			STANDBY mode is the same as resetting (except that some status bits indicate that the microcontroller has just exited STANDBY state).	
1	STOP	RW	Debug stop mode bit. 1: (FCLK on, HCLK on) In Stop mode, the FCLK and HCLK clocks are provided by the internal RC oscillator. When exiting stop mode, the software must reconfigure the clock system to start the PLL, crystal oscillator, etc. (the same operation as when configuring this bit to 0); 0: (FCLK off, HCLK off) In Stop mode, the clock controller disables all clocks (including HCLK and FCLK). Upon exiting Stop mode, the clock configuration is the same as it was after a reset (the microcontroller is clocked by the 20 MHz internal RC oscillator (HSI)). Therefore, the software must reconfigure the clock control system to enable the PLL, crystal oscillator, and other components.	0
0	SLEEP	RW	Debug sleep mode bit. 1: (FCLK on, HCLK on) In Sleep mode, both FCLK and HCLK clocks are provided by the originally configured system clock; 0: (FCLK on, HCLK off) In Sleep mode, FCLK is provided by the originally configured system clock, and HCLK is off. Since sleep mode does not reset the configured clock system, software does not need to reconfigure the clock system when exiting from sleep mode.	0